Ответы и официальные разборы

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IMO2024
65th International
Mathematical Olympiad
SHORTLISTED
PROBLEMS
WITH SOLUTIONS

65th International Mathematical Olympiad

Bath, United Kingdom, 10th–22nd July 2024



SHORTLISTED PROBLEMS

WITH SOLUTIONS

Note of Confidentiality

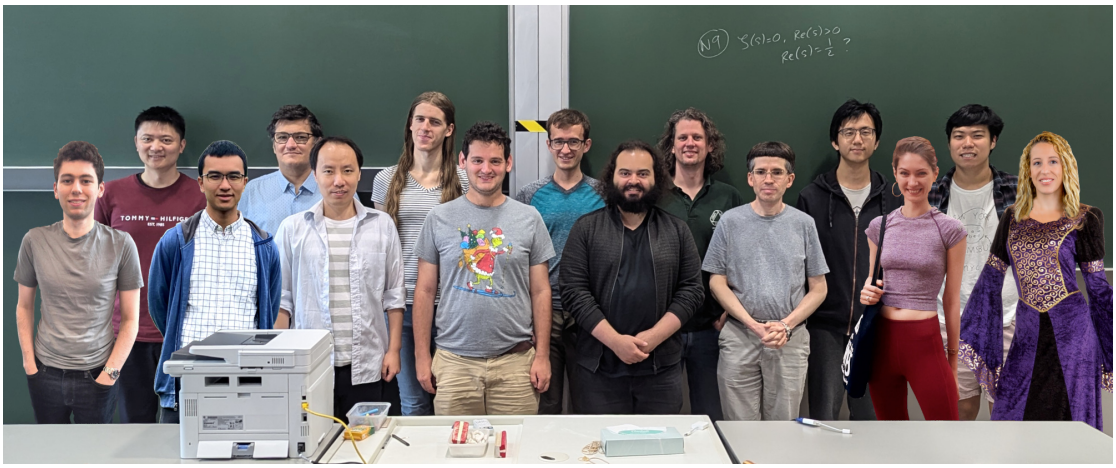
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Contributing Countries

The Organising Committee and the Problem Selection Committee of IMO 2024 thank the following 63 countries for contributing 229 problem proposals:

Algeria, Australia, Azerbaijan, Bangladesh, Belarus, Brazil, Bulgaria, Canada, China, Colombia, Croatia, Cyprus, Czech Republic, Denmark, Dominican Republic, Ecuador, Estonia, France, Georgia, Germany, Ghana, Greece, Hong Kong, India, Indonesia, Ireland, Iran, Israel, Japan, Kazakhstan, Kosovo, Latvia, Lithuania, Luxembourg, Malaysia, Mexico, Moldova, Netherlands, New Zealand, Norway, Peru, Poland, Portugal, Romania, Senegal, Serbia, Singapore, Slovakia, Slovenia, South Africa, South Korea, Spain, Sweden, Switzerland, Syria, Taiwan, Thailand, Tunisia, Türkiye, Uganda, Ukraine, U.S.A., Uzbekistan.

Problem Selection Committee



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Problems

Algebra

A1. Determine all real numbers α such that the number

$$\lfloor \alpha \rfloor + \lfloor 2\alpha \rfloor + \cdots + \lfloor n\alpha \rfloor$$

is a multiple of n for every positive integer n. (Here $\lfloor z \rfloor$ denotes the greatest integer less than or equal to z.)

(Colombia)

A2. Let n be a positive integer. Find the minimum possible value of

$$S = 2^0 x_0^2 + 2^1 x_1^2 + \cdots + 2^n x_n^2,$$

where $x_0, x_1, \ldots, x_n$ are nonnegative integers such that $x_0 + x_1 + \cdots + x_n = n$.

(China)

A3. Decide whether for every sequence (a_n) of positive real numbers,

$$\frac{3^{a_1} + 3^{a_2} + \cdots + 3^{a_n}}{(2^{a_1} + 2^{a_2} + \cdots + 2^{a_n})^2} < \frac{1}{2024}$$

is true for at least one positive integer n.

(China)

A4. Let $\mathbb{Z}_{>0}$ be the set of all positive integers. Determine all subsets $\mathcal{S}$ of $\{2^0, 2^1, 2^2, \ldots\}$ for which there exists a function $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that

$$\mathcal{S} = \{f(a+b) - f(a) - f(b) \mid a, b \in \mathbb{Z}_{>0}\}.$$

(Thailand)

A5. Find all periodic sequences $a_1, a_2, \ldots$ of real numbers such that the following conditions hold for all $n \geqslant 1$:

$$a_{n+2} + a_n^2 = a_n + a_{n+1}^2 \qquad \text{and} \qquad |a_{n+1} - a_n| \leqslant 1.$$

(Kosovo)

A6. Let $a_0, a_1, a_2, \ldots$ be an infinite strictly increasing sequence of positive integers such that for each $n \geqslant 1$ we have

$$a_n \in \left\{ \frac{a_{n-1} + a_{n+1}}{2}, \sqrt{a_{n-1} \cdot a_{n+1}} \right\}.$$

Let $b_1, b_2, \ldots$ be an infinite sequence of letters defined as

$$b_n = \begin{cases} A, & \text{if } a_n = \frac{1}{2}(a_{n-1} + a_{n+1}); \\ G, & \text{otherwise.} \end{cases}$$

Prove that there exist positive integers n_0 and d such that for all $n \geqslant n_0$ we have $b_{n+d} = b_n$.

(Czech Republic)

A7. Let $\mathbb{Q}$ be the set of rational numbers. Let $f\colon \mathbb{Q} \to \mathbb{Q}$ be a function such that the following property holds: for all $x, y \in \mathbb{Q}$,

$$f(x + f(y)) = f(x) + y \qquad \text{or} \qquad f(f(x) + y) = x + f(y).$$

Determine the maximum possible number of elements of $\{f(x) + f(-x) \mid x \in \mathbb{Q}\}$.

(Japan)

A8. Let $p \neq q$ be coprime positive integers. Determine all infinite sequences $a_1, a_2, \ldots$ of positive integers such that the following conditions hold for all $n \geqslant 1$:

$$\max(a_n, a_{n+1}, \ldots, a_{n+p}) - \min(a_n, a_{n+1}, \ldots, a_{n+p}) = p \qquad \text{and}$$
$$\max(a_n, a_{n+1}, \ldots, a_{n+q}) - \min(a_n, a_{n+1}, \ldots, a_{n+q}) = q.$$

(Japan)

Combinatorics

C1. Let n be a positive integer. A class of n students run n races, in each of which they are ranked with no draws. A student is eligible for a rating (a, b) for positive integers a and b if they come in the top b places in at least a of the races. Their final score is the maximum possible value of $a - b$ across all ratings for which they are eligible.

Find the maximum possible sum of all the scores of the n students.

(Australia)

C2. Let n be a positive integer. The integers 1, 2, 3, ..., n^2 are to be written in the cells of an $n \times n$ board such that each integer is written in exactly one cell and each cell contains exactly one integer. For every integer d with $d \mid n$, the *d-division* of the board is the division of the board into $(n/d)^2$ nonoverlapping sub-boards, each of size $d \times d$, such that each cell is contained in exactly one $d \times d$ sub-board.

We say that n is a *cool number* if the integers can be written on the $n \times n$ board such that, for each integer d with $d \mid n$ and $1 < d < n$, in the d-division of the board, the sum of the integers written in each $d \times d$ sub-board is not a multiple of d.

Determine all even cool numbers.

(Türkiye)

C3. Let n be a positive integer. There are $2n$ knights sitting at a round table. They consist of n pairs of partners, each pair of which wishes to shake hands. A pair can shake hands only when next to each other. Every minute, one pair of adjacent knights swaps places.

Find the minimum number of exchanges of adjacent knights such that, regardless of the initial arrangement, every knight can meet her partner and shake hands at some time.

(Belarus)

C4. On a board with 2024 rows and 2023 columns, Turbo the snail tries to move from the first row to the last row. On each attempt, he chooses to start on any cell in the first row, then moves one step at a time to an adjacent cell sharing a common side. He wins if he reaches any cell in the last row. However, there are 2022 predetermined, hidden monsters in 2022 of the cells, one in each row except the first and last rows, such that no two monsters share the same column. If Turbo unfortunately reaches a cell with a monster, his attempt ends and he is transported back to the first row to start a new attempt. The monsters do not move.

Suppose Turbo is allowed to take n attempts. Determine the minimum value of n for which he has a strategy that guarantees reaching the last row, regardless of the locations of the monsters.

(Hong Kong)

C5. Let N be a positive integer. Geoff and Ceri play a game in which they start by writing the numbers 1, 2, ..., N on a board. They then take turns to make a move, starting with Geoff. Each move consists of choosing a pair of integers (k, n), where $k \geqslant 0$ and n is one of the integers on the board, and then erasing every integer s on the board such that $2^k \mid n - s$. The game continues until the board is empty. The player who erases the last integer on the board loses.

Determine all values of N for which Geoff can ensure that he wins, no matter how Ceri plays.

(Indonesia)

C6. Let n and T be positive integers. James has $4n$ marbles with weights $1, 2, \ldots, 4n$. He places them on a balance scale, so that both sides have equal weight. Andrew may move a marble from one side of the scale to the other, so that the absolute difference in weights of the two sides remains at most T.

Find, in terms of n, the minimum positive integer T such that Andrew may make a sequence of moves such that each marble ends up on the opposite side of the scale, regardless of how James initially placed the marbles.

(Ghana)

C7. Let N be a positive integer and let $a_1, a_2, \ldots$ be an infinite sequence of positive integers. Suppose that, for each $n > N$, a_n is equal to the number of times a_{n-1} appears in the list $a_1, a_2, \ldots, a_{n-1}$.

Prove that at least one of the sequences $a_1, a_3, a_5, \ldots$ and $a_2, a_4, a_6, \ldots$ is eventually periodic.

(Australia)

C8. Let n be a positive integer. Given an $n \times n$ board, the unit cell in the top left corner is initially coloured black, and the other cells are coloured white. We then apply a series of colouring operations to the board. In each operation, we choose a 2×2 square with exactly one cell coloured black and we colour the remaining three cells of that 2×2 square black.

Determine all values of n such that we can colour the whole board black.

(Peru)

Geometry

G1. Let $ABCD$ be a cyclic quadrilateral such that $AC < BD < AD$ and $\angle DBA < 90^\circ$. Point E lies on the line through D parallel to AB such that E and C lie on opposite sides of line AD, and $AC = DE$. Point F lies on the line through A parallel to CD such that F and C lie on opposite sides of line AD, and $BD = AF$.

Prove that the perpendicular bisectors of segments BC and EF intersect on the circumcircle of $ABCD$.

(Ukraine)

G2. Let ABC be a triangle with $AB < AC < BC$, incentre I and incircle ω. Let X be the point in the interior of side BC such that the line through X parallel to AC is tangent to ω. Similarly, let Y be the point in the interior of side BC such that the line through Y parallel to AB is tangent to ω. Let AI intersect the circumcircle of triangle ABC again at $P \neq A$. Let K and L be the midpoints of AB and AC, respectively.

Prove that $\angle KIL + \angle YPX = 180^\circ$.

(Poland)

G3. Let $ABCDE$ be a convex pentagon and let M be the midpoint of AB. Suppose that segment AB is tangent to the circumcircle of triangle CME at M and that D lies on the circumcircles of triangles AME and BMC. Lines AD and ME intersect at K, and lines BD and MC intersect at L. Points P and Q lie on line EC so that $\angle PDC = \angle EDQ = \angle ADB$.

Prove that lines KP, LQ, and MD are concurrent.

(Belarus)

G4. Let $ABCD$ be a quadrilateral with AB parallel to CD and $AB < CD$. Lines AD and BC intersect at a point P. Point $X \neq C$ on the circumcircle of triangle ABC is such that $PC = PX$. Point $Y \neq D$ on the circumcircle of triangle ABD is such that $PD = PY$. Lines AX and BY intersect at Q.

Prove that PQ is parallel to AB.

(Ukraine)

G5. Let ABC be a triangle with incentre I, and let Ω be the circumcircle of triangle BIC. Let K be a point in the interior of segment BC such that $\angle BAK < \angle KAC$. The angle bisector of $\angle BKA$ intersects Ω at points W and X such that A and W lie on the same side of BC, and the angle bisector of $\angle CKA$ intersects Ω at points Y and Z such that A and Y lie on the same side of BC.

Prove that $\angle WAY = \angle ZAX$.

(Uzbekistan)

G6. Let ABC be an acute triangle with $AB < AC$, and let Γ be the circumcircle of ABC. Points X and Y lie on Γ so that XY and BC intersect on the external angle bisector of $\angle BAC$. Suppose that the tangents to Γ at X and Y intersect at a point T on the same side of BC as A, and that TX and TY intersect BC at U and V, respectively. Let J be the centre of the excircle of triangle TUV opposite the vertex T.

Prove that AJ bisects $\angle BAC$.

(Poland)

G7. Let ABC be a triangle with incentre I such that $AB < AC < BC$. The second intersections of AI, BI, and CI with the circumcircle of triangle ABC are M_A, M_B, and M_C, respectively. Lines AI and BC intersect at D and lines BM_C and CM_B intersect at X. Suppose the circumcircles of triangles XM_BM_C and XBC intersect again at $S \neq X$. Lines BX and CX intersect the circumcircle of triangle SXM_A again at $P \neq X$ and $Q \neq X$, respectively.

Prove that the circumcentre of triangle SID lies on PQ.

(Thailand)

G8. Let ABC be a triangle with $AB < AC < BC$, and let D be a point in the interior of segment BC. Let E be a point on the circumcircle of triangle ABC such that A and E lie on opposite sides of line BC and $\angle BAD = \angle EAC$. Let I, I_B, I_C, J_B, and J_C be the incentres of triangles ABC, ABD, ADC, ABE, and AEC, respectively.

Prove that I_B, I_C, J_B, and J_C are concyclic if and only if AI, I_BJ_C, and J_BI_C concur.

(Canada)

Number Theory

N1. Find all positive integers n with the following property: for all positive divisors d of n, we have that $d+1 \mid n$ or $d+1$ is prime.

(Ghana)

N2. Determine all finite, nonempty sets $\mathcal{S}$ of positive integers such that for every $a, b \in \mathcal{S}$ there exists $c \in \mathcal{S}$ with $a \mid b + 2c$.

(Netherlands)

N3. Determine all sequences $a_1, a_2, \ldots$ of positive integers such that, for any pair of positive integers $m \leqslant n$, the arithmetic and geometric means

$$\frac{a_m + a_{m+1} + \cdots + a_n}{n-m+1} \quad \text{and} \quad (a_m a_{m+1} \cdots a_n)^{\frac{1}{n-m+1}}$$

are both integers.

(Singapore)

N4. Determine all positive integers a and b such that there exists a positive integer g such that $\gcd(a^n + b, b^n + a) = g$ for all sufficiently large n.

(Indonesia)

N5. Let $\mathcal{S}$ be a finite nonempty set of prime numbers. Let $1 = b_1 < b_2 < \cdots$ be the sequence of all positive integers whose prime divisors all belong to $\mathcal{S}$. Prove that, for all but finitely many positive integers n, there exist positive integers $a_1, a_2, \ldots, a_n$ such that

$$\frac{a_1}{b_1} + \frac{a_2}{b_2} + \cdots + \frac{a_n}{b_n} = \left\lfloor \frac{1}{b_1} + \frac{1}{b_2} + \cdots + \frac{1}{b_n} \right\rfloor.$$

(Croatia)

N6. Let n be a positive integer. We say that a polynomial P with integer coefficients is *n-good* if there exists a polynomial Q of degree 2 with integer coefficients such that $Q(k)(P(k) + Q(k))$ is never divisible by n for any integer k.

Determine all integers n such that every polynomial with integer coefficients is an n-good polynomial.

(France)

N7. Let $\mathbb{Z}_{>0}$ denote the set of positive integers. Let $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ be a function satisfying the following property: for $m, n \in \mathbb{Z}_{>0}$, the equation

$$f(mn)^2 = f(m^2) f(f(n)) f(mf(n))$$

holds if and only if m and n are coprime.

For each positive integer n, determine all the possible values of $f(n)$.

(Japan)

Solutions

Algebra

A1. Determine all real numbers α such that the number

$$\lfloor \alpha \rfloor + \lfloor 2\alpha \rfloor + \cdots + \lfloor n\alpha \rfloor$$

is a multiple of n for every positive integer n. (Here $\lfloor z \rfloor$ denotes the greatest integer less than or equal to z.)

(Colombia)

Answer: All even integers satisfy the condition of the problem and no other real number α does so.

Solution 1. First we will show that even integers satisfy the condition. If $\alpha = 2m$ where m is an integer then

$$\lfloor \alpha \rfloor + \lfloor 2\alpha \rfloor + \cdots + \lfloor n\alpha \rfloor = 2m + 4m + \cdots + 2mn = mn(n+1)$$

which is a multiple of n.

Now we will show that they are the only real numbers satisfying the conditions of the problem. Let $\alpha = k + \epsilon$ where k is an integer and $0 \leqslant \epsilon < 1$. Then the number

$$\begin{aligned} \lfloor \alpha \rfloor + \lfloor 2\alpha \rfloor + \cdots + \lfloor n\alpha \rfloor &= k + \lfloor \epsilon \rfloor + 2k + \lfloor 2\epsilon \rfloor + \cdots + nk + \lfloor n\epsilon \rfloor \\ &= \frac{kn(n+1)}{2} + \lfloor \epsilon \rfloor + \lfloor 2\epsilon \rfloor + \cdots + \lfloor n\epsilon \rfloor \end{aligned}$$

has to be a multiple of n. We consider two cases based on the parity of k.

Case 1: k is even.

Then $\frac{kn(n+1)}{2}$ is always a multiple of n. Thus

$$\lfloor \epsilon \rfloor + \lfloor 2\epsilon \rfloor + \cdots + \lfloor n\epsilon \rfloor$$

also has to be a multiple of n.

We will prove that $\lfloor n\epsilon \rfloor = 0$ for every positive integer n by strong induction. The base case $n = 1$ follows from the fact that $0 \leqslant \epsilon < 1$. Let us suppose that $\lfloor m\epsilon \rfloor = 0$ for every $1 \leqslant m < n$. Then the number

$$\lfloor \epsilon \rfloor + \lfloor 2\epsilon \rfloor + \cdots + \lfloor n\epsilon \rfloor = \lfloor n\epsilon \rfloor$$

has to be a multiple of n. As $0 \leqslant \epsilon < 1$ then $0 \leqslant n\epsilon < n$, which means that the number $\lfloor n\epsilon \rfloor$ has to be equal to 0.

The equality $\lfloor n\epsilon \rfloor = 0$ implies $0 \leqslant \epsilon < 1/n$. Since this has to happen for all n, we conclude that $\epsilon = 0$ and then α is an even integer.

Case 2: k is odd.

We will prove that $\lfloor n\epsilon \rfloor = n-1$ for every natural number n by strong induction. The base case $n=1$ again follows from the fact that $0 \leqslant \epsilon < 1$. Let us suppose that $\lfloor m\epsilon \rfloor = m-1$ for every $1 \leqslant m < n$. We need the number

$$\begin{aligned}\frac{kn(n+1)}{2} + \lfloor \epsilon \rfloor + \lfloor 2\epsilon \rfloor + \cdots + \lfloor n\epsilon \rfloor &= \frac{kn(n+1)}{2} + 0 + 1 + \cdots + (n-2) + \lfloor n\epsilon \rfloor \\ &= \frac{kn(n+1)}{2} + \frac{(n-2)(n-1)}{2} + \lfloor n\epsilon \rfloor \\ &= \frac{k+1}{2}n^2 + \frac{k-3}{2}n + 1 + \lfloor n\epsilon \rfloor\end{aligned}$$

to be a multiple of n. As k is odd, we need $1 + \lfloor n\epsilon \rfloor$ to be a multiple of n. Again, as $0 \leqslant \epsilon < 1$ then $0 \leqslant n\epsilon < n$, so $\lfloor n\epsilon \rfloor = n-1$ as we wanted.

This implies that $1 - \frac{1}{n} \leqslant \epsilon < 1$ for all n which is absurd. So there are no other solutions in this case.

Solution 2. As in Solution 1 we check that for even integers the condition is satisfied. Then, without loss of generality we can assume $0 \leqslant \alpha < 2$. We set $S_n = \lfloor \alpha \rfloor + \lfloor 2\alpha \rfloor + \cdots + \lfloor n\alpha \rfloor$.

Notice that

$$S_n \equiv 0 \pmod{n} \tag{1}$$

$$S_n \equiv S_n - S_{n-1} = \lfloor n\alpha \rfloor \pmod{n-1} \tag{2}$$

Since $\gcd(n, n-1) = 1$, (1) and (2) imply that

$$S_n \equiv n\lfloor n\alpha \rfloor \pmod{n(n-1)}. \tag{3}$$

In addition,

$$0 \leqslant n\lfloor n\alpha \rfloor - S_n = \sum_{k=1}^{n} \Big(\lfloor n\alpha \rfloor - \lfloor k\alpha \rfloor \Big) < \sum_{k=1}^{n} \Big(n\alpha - k\alpha + 1 \Big) = \frac{n(n-1)}{2}\alpha + n. \tag{4}$$

For n large enough, the RHS of (4) is less than $n(n-1)$. Then (3) forces

$$0 = S_n - n\lfloor n\alpha \rfloor = \sum_{k=1}^{n} \Big(\lfloor n\alpha \rfloor - \lfloor k\alpha \rfloor \Big) \tag{5}$$

for n large enough.

Since $\lfloor n\alpha \rfloor - \lfloor k\alpha \rfloor \geqslant 0$ for $1 \leqslant k \leqslant n$, we get from (5) that, for all n large enough, all these inequalities are equalities. In particular $\lfloor \alpha \rfloor = \lfloor n\alpha \rfloor$ for all n large enough, which is absurd unless $\alpha = 0$.

Comment. An alternative ending to the previous solution is as follows.

By definition we have $S_n \leqslant \alpha \frac{n(n+1)}{2}$, on the other hand (5) implies $S_n \geqslant \alpha n^2 - n$ for all n large enough, so $\alpha = 0$.

Solution 3. As in other solutions, without loss of generality we may assume that $0 \leqslant \alpha < 2$. Even integers satisfy the condition, so we assume $0 < \alpha < 2$ and we will derive a contradiction.

By induction on n, we will simultaneously show that

$$\lfloor \alpha \rfloor + \lfloor 2\alpha \rfloor + \cdots + \lfloor n\alpha \rfloor = n^2, \tag{6}$$

$$\text{and} \qquad \frac{2n-1}{n} \leqslant \alpha < 2. \tag{7}$$

The base case is $n = 1$: If $\alpha < 1$, consider $m = \left\lceil \frac{1}{\alpha} \right\rceil > 1$, then

$$\lfloor \alpha \rfloor + \lfloor 2\alpha \rfloor + \cdots + \lfloor m\alpha \rfloor = 1$$

is not a multiple of m, so we deduce (7). Hence, $\lfloor \alpha \rfloor = 1$ and (6) follows.

For the induction step: assume the induction hypothesis to be true for n, then by (7)

$$2n + 1 - \frac{1}{n} \leqslant (n+1)\alpha < 2n + 2.$$

Hence,

$$n^2 + 2n \leqslant \lfloor \alpha \rfloor + \lfloor 2\alpha \rfloor + \cdots + \lfloor n\alpha \rfloor + \lfloor (n+1)\alpha \rfloor = n^2 + \lfloor (n+1)\alpha \rfloor < n^2 + 2n + 2.$$

So, necessarily $\lfloor (n+1)\alpha \rfloor = 2n + 1$ and

$$\lfloor \alpha \rfloor + \lfloor 2\alpha \rfloor + \cdots + \lfloor n\alpha \rfloor + \lfloor (n+1)\alpha \rfloor = (n+1)^2$$

in order to obtain a multiple of $n + 1$. These two equalities give (6) and (7) respectively.

Finally, we notice that condition (7) being true for all n gives a contradiction.

Solution 4. As in other solutions without loss of generality we will assume that $0 < \alpha < 2$ and derive a contradiction. For each n, we define

$$b_n = \frac{\lfloor \alpha \rfloor + \lfloor 2\alpha \rfloor + \cdots + \lfloor n\alpha \rfloor}{n},$$

which is a nonnegative integer by the problem condition and our assumption. Note that

$$\lfloor (n+1)\alpha \rfloor \geqslant \lfloor \alpha \rfloor, \lfloor 2\alpha \rfloor, \ldots, \lfloor n\alpha \rfloor \quad \text{and} \quad \lfloor (n+1)\alpha \rfloor > \lfloor \alpha \rfloor$$

for all $n > \frac{1}{\alpha}$. It follows that $b_{n+1} > b_n \implies b_{n+1} \geqslant b_n + 1$ for $n > \frac{1}{\alpha}$. Thus, for all such n,

$$b_n \geqslant n + C$$

where C is a fixed integer. On the other hand, the definition of b_n gives

$$b_n = \frac{\lfloor \alpha \rfloor + \lfloor 2\alpha \rfloor + \cdots + \lfloor n\alpha \rfloor}{n} \leqslant \frac{\alpha + 2\alpha + \cdots + n\alpha}{n} = \frac{\alpha}{2}(n+1),$$

which is a contradiction for sufficiently large n.

A2. Let n be a positive integer. Find the minimum possible value of

$$S = 2^0 x_0^2 + 2^1 x_1^2 + \cdots + 2^n x_n^2,$$

where $x_0, x_1, \ldots, x_n$ are nonnegative integers such that $x_0 + x_1 + \cdots + x_n = n$.

(China)

Answer: The minimum value is $\frac{n(n+1)}{2}$.

Solution 1. For a fixed n, let $f(n)$ denote the minimum possible value of S. Consider the following variant: among all infinite sequences of nonnegative integers $x_0, x_1, \ldots$, only finitely many of which are nonzero, satisfying $x_0 + x_1 + \cdots = n$, let $g(n)$ denote the minimum possible value of

$$T = 2^0 x_0^2 + 2^1 x_1^2 + 2^2 x_2^2 + \cdots .$$

It is clear that $g(n) \leqslant f(n)$. Conversely, it is easy to see that if a sequence $x_0, x_1, \ldots$ achieves the minimum of $g(n)$, then $x_0 \geqslant x_1 \geqslant \cdots$ and thus $x_{n+1} = x_{n+2} = \cdots = 0$. In particular, $f(n) = g(n)$.

Now, we hope to get an inductive formula for $g(n)$.

Note that, in order to minimise T for $n \geqslant 1$, we must have $x_0 \geqslant 1$ since the sequence (x_i) is nonincreasing. Note that the minimal value of

$$2^1 x_1^2 + 2^2 x_2^2 + \cdots = 2(2^0 x_1^2 + 2^1 x_2^2 + \cdots)$$

over all infinite sequences of nonnegative integers with $x_1 + x_2 + \cdots = m$ is exactly $2g(m)$. As a result, for $n \geqslant 1$ we have

$$g(n) = \min_{x_0 \in \{1,2,\ldots,n\}} \left(x_0^2 + 2g(n - x_0)\right).$$

We now prove $g(n) = \frac{n(n+1)}{2}$ by induction. It is clear that $g(0) = 0$. Assume that this has been proved for $n = 0, 1, \ldots, N-1$. Then,

$$\begin{aligned} x_0^2 + 2g(N - x_0) &= x_0^2 + (N - x_0)(N - x_0 + 1) \\ &= 2x_0^2 - (2N+1)x_0 + N(N+1) \\ &= \frac{1}{2}\left[(2x_0 - N)(2x_0 - N - 1) + N^2 + N\right]. \end{aligned} \tag{1}$$

The product of two consecutive integers $(2x_0 - N)(2x_0 - N - 1)$ is always nonnegative, and it is zero precisely when $2x_0$ is the even number in $\{N, N+1\}$. Thus the minimum of the final expression in equation (1) is $\frac{1}{2}(N^2 + N)$, so $g(N) = \frac{N(N+1)}{2}$, completing the inductive proof.

Solution 2. Consider the following table of numbers, where the row and column indices start from 0, and $a_{i,j} = 2^i(2j+1)$ for $i,\, j \geqslant 0$.

	$j=0$	1	2	3	4	5	$\cdots$
$i=0$	1	3	5	7	9	11	
1	2	6	10	14	18	22	
2	4	12	20	28	36	44	
3	8	24	40	56	72	88	
4	16	48	80	112	144	176	
$\vdots$							

Every number can be written uniquely as a product of a power of 2 and an odd number so every positive integer appears exactly once in the table above. It is easy to see that numbers in each row and each column are strictly increasing. Since the sum of the first x odd positive integers is x^2, the sum of the first x_k numbers in the kth row is $2^k x_k^2$, the kth term appearing in S.

Thus, the sum S can be interpreted as the result of taking a total of n numbers from the first n rows of the table such that we take the leftmost x_k numbers from row k (where $\sum_{k=1}^{n} x_k = n$), and then computing the sum of these n numbers. In particular, the minimum possible value of S is the same as the sum of the smallest n numbers in this table, since every row and every column of the table is strictly increasing.

Moreover, the smallest n numbers, namely 1, 2, …, n, appear in the first n rows, so the minimum of S is

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Comment. As can be seen from the table in Solution 2, the equality case of the problem is given by

$$x_i = \left\lfloor \frac{n}{2^{i+1}} + \frac{1}{2} \right\rfloor.$$

So x_i is the result of rounding $\frac{n}{2^{i+1}}$ to the nearest integer. This also gives a proof of the identity

$$n = \sum_{i=0}^{\infty} \left\lfloor \frac{n}{2^{i+1}} + \frac{1}{2} \right\rfloor,$$

which can be separately proven by induction on n: when n is incremented by 1, exactly one term on the right hand side, namely the one corresponding to $i = \nu_2(n)$, increases by 1 while the others remain the same.

Comment. If the condition that the x_i are nonnegative integers is relaxed to the x_i being nonnegative *reals*, the problem can be solved by an application of the Cauchy-Schwarz inequality:

$$(2^0 + 2^{-1} + \cdots + 2^{-n})(2^0 x_0^2 + 2^1 x_1^2 + \cdots + 2^n x_n^2) \geqslant (x_0 + \cdots + x_n)^2 = n^2$$
$$\implies 2^0 x_0^2 + 2^1 x_1^2 + \cdots + 2^n x_n^2 \geqslant \frac{n^2}{2 - 2^{-n}}.$$

The equality case for this relaxed problem is given by

$$x_i = \frac{2^{-i} n}{2 - 2^{-n}} \approx \left\lfloor \frac{n}{2^{i+1}} + \frac{1}{2} \right\rfloor.$$

In fact, when the terms in the optimal sequence for the real case are all rounded to the nearest integer, we obtain the optimal sequence for the original problem. While thinking about the real case may guide one towards the equality case of the original problem, it does not seem like it can be easily continued into a full solution.

A3. Decide whether for every sequence (a_n) of positive real numbers,

$$\frac{3^{a_1}+3^{a_2}+\cdots+3^{a_n}}{(2^{a_1}+2^{a_2}+\cdots+2^{a_n})^2}<\frac{1}{2024} \tag{1}$$

is true for at least one positive integer n.

(China)

Comment. The question can be asked in several forms, as follows:

(i) students could be asked, as above, to show the existence of such an n;

(ii) students could be asked to show that this happens for all sufficiently large n;

(iii) students could be given a concrete positive integer N and asked to show it for all $n>N$.

The solutions below provide varying bounds for N.

Answer: The answer is "yes": there is always such an n.

Common remarks. We write $\varepsilon=\frac{1}{2024}$.

Solution 1. For every positive integer n, let $M_n=\max(a_1,a_2,\ldots,a_n)$. We first prove that

$$\frac{3^{a_1}+3^{a_2}+\cdots+3^{a_n}}{(2^{a_1}+2^{a_2}+\cdots+2^{a_n})^2}\leqslant\left(\frac{3}{4}\right)^{M_n}. \tag{2}$$

For $i=1,2,\ldots,n$, from $\left(\frac{3}{2}\right)^{a_i}\leqslant\left(\frac{3}{2}\right)^{M_n}$ we can obtain $3^{a_i}\leqslant\left(\frac{3}{4}\right)^{M_n}\cdot 2^{M_n}\cdot 2^{a_i}$. By summing up over all i,

$$\sum_{i=1}^{n}3^{a_i}\leqslant\left(\frac{3}{4}\right)^{M_n}\cdot 2^{M_n}\cdot\sum_{i=1}^{n}2^{a_i}\leqslant\left(\frac{3}{4}\right)^{M_n}\cdot\left(\sum_{i=1}^{n}2^{a_i}\right)^2,$$

which is equivalent to (2).

Now let $\mu=\log_{4/3}\frac{1}{\varepsilon}$, so that μ is the positive real number with $\left(\frac{3}{4}\right)^{\mu}=\varepsilon$. If there is an index n such that $a_n>\mu$, then $M_n\geqslant a_n>\mu$, and hence

$$\frac{3^{a_1}+3^{a_2}+\cdots+3^{a_n}}{(2^{a_1}+2^{a_2}+\cdots+2^{a_n})^2}\leqslant\left(\frac{3}{4}\right)^{M_n}<\left(\frac{3}{4}\right)^{\mu}=\varepsilon.$$

Otherwise we have $0<a_i\leqslant\mu$ for all positive integers i, so

$$\frac{3^{a_1}+3^{a_2}+\cdots+3^{a_n}}{(2^{a_1}+2^{a_2}+\cdots+2^{a_n})^2}\leqslant\frac{n\cdot 3^{\mu}}{(n\cdot 1)^2}=\frac{3^{\mu}}{n}.$$

If $n>\left\lfloor\frac{3^{\mu}}{\varepsilon}\right\rfloor$, this is less than ε.

Comment. It is also possible to prove (2) by induction on n. The base case $n=1$ is clear. For the induction step, after ordering $a_1, a_2, \ldots, a_n$ in increasing order as $b_1\leqslant b_2\leqslant\cdots\leqslant b_n$, it suffices, for example, to verify that

$$\frac{3^{b_1}+3^{b_2}+\cdots+3^{b_n}}{(2^{b_1}+2^{b_2}+\cdots+2^{b_n})^2}\leqslant\frac{3^{b_1}+3^{b_2}+\cdots+3^{b_n}}{(2^{b_1}+2^{b_2}+\cdots+2^{b_n})(2^{b_2}+\cdots+2^{b_n})}\leqslant\frac{3^{b_2}+\cdots+3^{b_n}}{(2^{b_2}+\cdots+2^{b_n})^2}.$$

The second inequality is equivalent to $3^{b_1}\sum_{i=2}^{n}2^{b_i}\leqslant 2^{b_1}\sum_{i=2}^{n}3^{b_i}$, which follows from $\left(\frac{3}{2}\right)^{b_1}\leqslant\left(\frac{3}{2}\right)^{b_i}$.

Solution 2. We will combine two upper bounds.

First, start with the trivial estimate

$$\frac{3^{a_1}+\cdots+3^{a_n}}{(2^{a_1}+\cdots+2^{a_n})^2} \leqslant \frac{3^{a_1}+\cdots+3^{a_n}}{4^{a_1}+\cdots+4^{a_n}}.$$

By applying Jensen's inequality to the convex function $x^{\log_3 4}$ we get

$$\frac{4^{a_1}+\cdots+4^{a_n}}{n} = \frac{\left(3^{a_1}\right)^{\log_3 4}+\cdots+\left(3^{a_n}\right)^{\log_3 4}}{n} \geqslant \left(\frac{3^{a_1}+\cdots+3^{a_n}}{n}\right)^{\log_3 4},$$

so

$$\frac{3^{a_1}+\cdots+3^{a_n}}{(2^{a_1}+\cdots+2^{a_n})^2} \leqslant \frac{3^{a_1}+\cdots+3^{a_n}}{4^{a_1}+\cdots+4^{a_n}} \leqslant \left(\frac{n}{3^{a_1}+\cdots+3^{a_n}}\right)^{\log_3 4-1}.$$

Hence, (1) holds true whenever

$$3^{a_1}+\cdots+3^{a_n} > \left(\frac{1}{\varepsilon}\right)^{\frac{1}{\log_3 4-1}} \cdot n. \tag{3}$$

Second, trivially

$$\frac{3^{a_1}+\cdots+3^{a_n}}{(2^{a_1}+\cdots+2^{a_n})^2} \leqslant \frac{3^{a_1}+\cdots+3^{a_n}}{n^2},$$

so (1) is satisfied if

$$3^{a_1}+\cdots+3^{a_n} < \varepsilon \cdot n^2. \tag{4}$$

If $n > \left(\frac{1}{\varepsilon}\right)^{1+\frac{1}{\log_3 4-1}}$ then $\left(\frac{1}{\varepsilon}\right)^{\frac{1}{\log_3 4-1}} \cdot n < \varepsilon \cdot n^2$, and therefore at least one of (3) and (4) is satisfied.

Solution 3. Define $C = \log_{4/3} \frac{2}{\varepsilon}$, so that if $a_i > C$ then $3^{a_i} < \frac{\varepsilon}{2} \cdot 4^{a_i}$. We divide the sequence into "small" and "large" terms by how they compare to C: let

$$\mathcal{S}_n = \{i \leqslant n \mid a_i \leqslant C\} \quad \text{and} \quad \mathcal{L}_n = \{i \leqslant n \mid a_i > C\}.$$

Then (1) is equivalent to

$$\frac{\sum_{i\in\mathcal{S}_n} 3^{a_i}}{\left(\sum_{i\in\mathcal{S}_n} 2^{a_i}+\sum_{i\in\mathcal{L}_n} 2^{a_i}\right)^2} + \frac{\sum_{i\in\mathcal{L}_n} 3^{a_i}}{\left(\sum_{i\in\mathcal{S}_n} 2^{a_i}+\sum_{i\in\mathcal{L}_n} 2^{a_i}\right)^2} < \frac{\varepsilon}{2}+\frac{\varepsilon}{2}.$$

If $\mathcal{L}_n$ is nonempty, we have

$$\frac{\sum_{i\in\mathcal{L}_n} 3^{a_i}}{\left(\sum_{i\in\mathcal{S}_n} 2^{a_i}+\sum_{i\in\mathcal{L}_n} 2^{a_i}\right)^2} < \frac{\varepsilon}{2} \cdot \frac{\sum_{i\in\mathcal{L}_n} 4^{a_i}}{\left(\sum_{i\in\mathcal{L}_n} 2^{a_i}\right)^2} \leqslant \frac{\varepsilon}{2},$$

and this also trivially holds when $\mathcal{L}_n$ is empty (in which case the LHS is zero).

Now suppose that $n \geqslant \frac{2}{\varepsilon}\left(\frac{3}{2}\right)^C$. Note that $3^{a_i} \leqslant \left(\frac{3}{2}\right)^C 2^{a_i}$ for $i \in \mathcal{S}_n$, so we have

$$\frac{\sum_{i\in\mathcal{S}_n} 3^{a_i}}{\left(\sum_{i\in\mathcal{S}_n} 2^{a_i}+\sum_{i\in\mathcal{L}_n} 2^{a_i}\right)^2} \leqslant \frac{\left(\frac{3}{2}\right)^C \sum_{i\in\mathcal{S}_n} 2^{a_i}}{\left(\sum_{i\in\mathcal{S}_n} 2^{a_i}+\sum_{i\in\mathcal{L}_n} 2^{a_i}\right)^2} \leqslant \frac{\left(\frac{3}{2}\right)^C}{\sum_{i\in\mathcal{S}_n} 2^{a_i}+\sum_{i\in\mathcal{L}_n} 2^{a_i}} < \frac{\left(\frac{3}{2}\right)^C}{n} \leqslant \frac{\varepsilon}{2},$$

so we have (1).

Solution 4. For every index $i = 1, 2, \ldots, n$, apply the weighted AM-GM inequality to numbers 2^{a_i} and $(n-1)$ with weights $\log_2 \frac{3}{2} \approx 0.585$ and $\log_2 \frac{4}{3} \approx 0.415$ as

$$\begin{aligned} 2^{a_1} + 2^{a_2} + \cdots + 2^{a_n} &\geqslant 2^{a_i} + (n-1) \\ > \log_2 \frac{3}{2} \cdot 2^{a_i} + \log_2 \frac{4}{3} \cdot (n-1) &\geqslant \left(2^{a_i}\right)^{\log_2 \frac{3}{2}} \cdot (n-1)^{\log_2 \frac{4}{3}} \\ = \left(\frac{3}{2}\right)^{a_i} \cdot (n-1)^{\log_2 \frac{4}{3}} &> \left(\frac{3}{2}\right)^{a_i} \cdot (n-1)^{2/5}. \end{aligned}$$

By summing up for $i = 1, 2, \ldots, n$,

$$(2^{a_1} + \cdots + 2^{a_n})^2 = \sum_{i=1}^{n} 2^{a_i} \left(2^{a_1} + 2^{a_2} + \cdots + 2^{a_n}\right) > (n-1)^{2/5} \sum_{i=1}^{n} 3^{a_i}$$

so

$$\frac{3^{a_1} + 3^{a_2} + \cdots + 3^{a_n}}{(2^{a_1} + 2^{a_2} + \cdots + 2^{a_n})^2} < \frac{1}{(n-1)^{2/5}}.$$

If $n \geqslant \left(\frac{1}{\varepsilon}\right)^{5/2} + 1$ then $\frac{1}{(n-1)^{2/5}} < \varepsilon$.

A4. Let $\mathbb{Z}_{>0}$ be the set of all positive integers. Determine all subsets $\mathcal{S}$ of $\{2^0, 2^1, 2^2, \dots\}$ for which there exists a function $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that

$$\mathcal{S} = \{f(a+b) - f(a) - f(b) \mid a, b \in \mathbb{Z}_{>0}\}.$$

(Thailand)

Answer: $\mathcal{S}$ can be any subset of size 1 or 2.

Common remarks. For this problem, it is convenient to use notation such as $\{a, b, c\}$ for multisets rather than sets, and the subset relation is likewise that for multisets. Both solutions use the following property of powers of 2: if $2^a + 2^b = 2^c + 2^d$, then $\{a, b\}$ and $\{c, d\}$ are the same multiset. Define $e(a, b) = \log_2(f(a+b) - f(a) - f(b)) = e(b, a)$. Thus,

$$f(a+b) = 2^{e(a,b)} + f(a) + f(b).$$

Solution 1. Clearly $\mathcal{S}$ must be nonempty. We start with constructions when $1 \leqslant |\mathcal{S}| \leqslant 2$.

- If $\mathcal{S} = \{2^k\}$, then take $f(x) = cx - 2^k$ for any integer $c > 2^k$.
- If $\mathcal{S} = \{2^k, 2^\ell\}$ where $k > \ell$, then take $f(x) = (2^k - 2^\ell)\lfloor \alpha x \rfloor - 2^\ell$, where $\alpha > 2$ is not an integer. This works because $\lfloor \alpha(x+y) \rfloor - (\lfloor \alpha x \rfloor + \lfloor \alpha y \rfloor) \in \{0, 1\}$ for all x and y, and takes both values; the lower bound on α ensures the values of f are positive.

Observe that, inductively,

$$f(n) = 2^{e(1,1)} + 2^{e(2,1)} + \cdots + 2^{e(n-1,1)} + nf(1).$$

Lemma 1. For any positive integers n and k,

$$\{e(1,1), e(2,1), \dots, e(k-1,1)\} \subset \{e(n,1), e(n+1,1), \dots, e(n+k-1,1)\}.$$

Proof. We work by induction on k; in the case $k = 1$, the first multiset is empty, which provides our base case.

For the induction step, suppose $k \geqslant 2$ and we know that

$$\{e(1,1), e(2,1), \dots, e(k-2,1)\} \subset \{e(n,1), e(n+1,1), \dots, e(n+k-2,1)\}.$$

By definition, $f(n+k) - f(n) - f(k) = 2^{e(n,k)}$, and using the first observation we see that

$$f(n+k) - f(n) - f(k) = \left(2^{e(n,1)} + 2^{e(n+1,1)} + \cdots 2^{e(n+k-1,1)}\right) - \left(2^{e(1,1)} + 2^{e(2,1)} + \cdots + 2^{e(k-1,1)}\right).$$

From the induction hypothesis, we may write

$$\{e(n,1), e(n+1,1), \dots, e(n+k-2,1)\} = \{e(1,1), e(2,1), \dots, e(k-2,1)\} \cup \{a\}$$

for some a. Thus

$$2^{e(n,k)} = 2^a + 2^{e(n+k-1,1)} - 2^{e(k-1,1)}.$$

So $\{e(n,k), e(k-1,1)\} = \{a, e(n+k-1,1)\}$. Thus $e(k-1,1) = a$ or $e(k-1,1) = e(n+k-1,1)$, and in either case we have our result. □

Lemma 2. The sequence $e(1,1)$, $e(2,1)$, $e(3,1)$, … takes at most two different values.

Proof. Suppose for a contradiction that $k \geqslant 2$ is the least index with $e(k,1) \neq e(1,1)$, and that some $\ell > k$ has $e(\ell,1) \notin \{e(k,1), e(1,1)\}$. By Lemma 1, any block of k consecutive values of the sequence has at least $k-1$ values equal to $e(1,1)$. This forces

$$e(\ell-1,1) = e(\ell-2,1) = \cdots = e(\ell-(k-1),1) = e(1,1)$$

and

$$e(\ell+1,1) = e(\ell+2,1) = \cdots = e(\ell+(k-1),1) = e(1,1).$$

But then the block $e(\ell-1,1)$, $e(\ell,1)$, $e(\ell+1,1)$, $e(\ell+2,1)$, …, $e(\ell+(k-1),1)$ has length $k+1$ and does not contain $e(k,1)$, a contradiction. □

Finally, for any a and b we have

$$\begin{aligned} f(a+b) - f(a) - f(b) &= \left(2^{e(a,1)} + 2^{e(a+1,1)} + \cdots 2^{e(a+b-1,1)}\right) - \left(2^{e(1,1)} + 2^{e(2,1)} + \cdots + 2^{e(b-1,1)}\right) \\ &= 2^{e(i,1)} \end{aligned}$$

for some $a \leqslant i \leqslant a+b-1$. So $|\mathcal{S}| \leqslant 2$.

Comment. In the construction of functions, $\alpha > 2$ is only necessary if $k = \ell+1$, to make sure $f(1) \neq 0$. Otherwise, any nonintegral $\alpha > 1$ suffices.

Solution 2. Subsets of size 1 or 2 can be achieved as in Solution 1, and $\mathcal{S}$ must be nonempty. We consider such a set $\mathcal{S}$ with $|\mathcal{S}| \geqslant 3$ and a corresponding function f in order to achieve a contradiction. We will relate the $e(a,b)$ to values of $e(c,1)$ with $c+1 < a+b$, leading to a proof of Lemma 2 from Solution 1 that does not depend on Lemma 1 from that solution.

Suppose $a > 1$. We have $f(a+b) - f(a) - f(b) = 2^{e(a,b)}$ and also $f(a) - f(a-1) - f(1) = 2^{e(a-1,1)}$, so

$$f(a+b) - f(a-1) - f(1) - f(b) = 2^{e(a,b)} + 2^{e(a-1,1)}.$$

Similarly, $f(a+b) - f(a-1) - f(b+1) = 2^{e(a-1,b+1)}$ and $f(b+1) - f(1) - f(b) = 2^{e(b,1)}$, so

$$f(a+b) - f(a-1) - f(1) - f(b) = 2^{e(a-1,b+1)} + 2^{e(b,1)}.$$

Thus either

$$e(a,b) = e(a-1,b+1) \quad \text{and} \quad e(a-1,1) = e(b,1)$$

or

$$e(a,b) = e(b,1) \quad \text{and} \quad e(a-1,b+1) = e(a-1,1).$$

For $n \geqslant 4$, we consider these possibilities as (a,b) ranges over all pairs with $a+b = n$. If the first case holds for every such pair (that is, if $e(c,1) = e(d,1)$ for all $c+d = n-1$), then all the $e(a,b)$ for $a+b = n$ are equal (and the above equations do not constrain whether or not the value is the same as any $e(c,1)$ with $c+1 < n$). Otherwise, the values of $e(a,b)$ with $a+b = n$ are fully determined by the values of $e(c,1)$ for which $e(c,1) \neq e(n-1-c,1)$, and are not all equal.

Specifically, if $e(c,1) = j$ and $e(n-1-c,1) = k$ with $j \neq k$, we have $e(c,n-c) = j = e(n-c,c)$ and $e(c+1,n-c-1) = k = e(n-c-1,c+1)$. Every other value of $e(a,b)$ with $a+b = n$ is then determined by the rule that $e(a,b) = e(a-1,b+1)$ if $e(a-1,1) = e(b,1)$: if we have $e(c,1) \neq e(n-1-c,1)$, and $e(c',1) \neq e(n-1-c',1)$, but $e(d,1) = e(n-1-d,1)$ for all $c < d < c'$, then if $c < c'-1$ we have $e(c'-1,n-(c'-1)) = e(c',n-c')$, then if $c < c'-2$ we have $e(c'-2,n-(c'-2)) = e(c'-1,n-(c'-1)) = e(c',n-c')$, and so on until $e(c+1,n-c-1) = e(c',n-c')$ (yielding a contradiction if $e(n-c-1,1) \neq e(c',1)$; such a contradiction also arises trivially if $c+1 = c'$ and $e(n-c-1,1) \neq e(c',1)$). If c is the least integer such that $e(c,1) \neq e(n-1-c,1)$, the values of $e(a,b)$ with $a < c$ are similarly determined to be equal to $e(c,n-c)$ (and likewise for $a > n-c$).

In other words, if we list the values in ascending order of a from 1 to $n-1$, any gaps between the pairs of adjacent values determined when $e(c,1) \neq e(n-1-c,1)$ are filled with copies of the previously determined adjacent values, and if the values on either side of such a gap are different, we have a contradiction (including in the degenerate cases where the pairs are adjacent or overlap, if $c+1=c'$). Note in particular that every value of $e(a,b)$ is a value of $e(c,1)$ for some c with $c+1 \leqslant a+b$.

If $|\mathcal{S}| \geqslant 3$, that means that $e(c,1)$ takes at least three different values. Let m be such that $e(m,1)$ does not equal any $e(c,1)$ for $c<m$, and there are exactly two different values of $e(c,1)$ for $c<m$ (and thus $m \geqslant 3$).

Because $e(m,1)$ does not equal any $e(c,1)$ for $c<m$, we have that all $e(a,b)$ for $a+b=m+1$ are equal, and $e(c,1)=e(d,1)$ for all $c+d=m$. We now consider the values of $e(a,b)$ for $a+b=m+2$ determined by the above rules. Since $e(m,1) \neq e(1,1)$, we have $e(1,m+1)=e(1,1)$ and $e(2,m)=e(m,1)$. If there were any other $e(d,1) \neq e(m+1-d,1)$, consider the one with minimal $d>1$; because $e(m,1) \neq e(d,1)$, we arrive at a contradiction. So every $e(c,1)=e(d,1)$ for $c+d=m+1$ except for $e(m,1) \neq e(1,1)$. But these equalities form a path connecting all $e(c,1)$ for $c<m$:

$$e(1,1)=e(m-1,1)=e(2,1)=e(m-2,1)=e(3,1)=\cdots$$

which contradicts the assumption we made that there were exactly two different values of $e(c,1)$ for $c<m$.

Solution 3. Constructions for $1 \leqslant |\mathcal{S}| \leqslant 2$ are shown in Solution 1, and $\mathcal{S}$ must be nonempty. We suppose $|\mathcal{S}| \geqslant 3$ to derive a contradiction.

Claim 1. $e(a,b)$, $e(b,c)$, and $e(a,c)$ can take at most two different values.

Proof. By expanding $f(a+b+c)$ in three different ways, we get

$$2^{e(a,b)}+2^{e(c,a+b)}=2^{e(b,c)}+2^{e(a,b+c)}=2^{e(a,c)}+2^{e(b,a+c)}.$$

The result follows from the equality of the three multisets of exponents. □

For Claims 2 to 4, we fix k and let N be the smallest integer such that $e(a,N-a+1)=k$ for some $a \leqslant N$.

Claim 2. For any b with $b \leqslant N$, we must have $e(b,N-b+1)=k$.

Proof. Suppose that $e(a,N-a+1)=k$ and $a<b$. Expanding $f(a+(b-a)+N-b+1)$ in two different ways, we see that

$$2^{e(a,b-a)}+2^{e(b,N-b+1)}=2^{e(N-b+1,b-a)}+2^{e(N-a+1,a)}.$$

By the minimality of N, we must have $e(b,N-b+1)=e(N-a+1,a)$. The case of $a>b$ follows by replacing a and b with $N-a+1$ and $N-b+1$. □

Claim 3. $e(a,1)=e(N-a+1,1)$ for any a satisfying $1<a<N$.

Proof. By Claim 2, $e(a,N-a+1)=k$. Then by Claim 1, $e(a,N-a+1)$, $e(a,1)$, and $e(N-a+1,1)$ can take at most two different values. But by the minimality of N, we must have $e(a,N-a+1) \neq e(a,1)=e(N-a+1,1)$. □

Claim 4. $e(a,1)=e(N-a,1)$ for any a satisfying $1 \leqslant a<N$.

Proof. By Claim 2, $e(a,N-a+1)=e(a+1,N-a)=k$. Expanding $f(1+a+(N-a))$ in two different ways, we see that

$$2^{e(1,a)}+2^{e(a+1,N-a)}=2^{e(1,N-a)}+2^{e(N-a+1,a)}.$$

Therefore $e(1,a)=e(1,N-a)$, as required. □

If $|\mathcal{S}| \geqslant 3$, then there exist $1<N_k<N_\ell$ where N_k and N_ℓ are the minimal values corresponding to k and ℓ. But Claims 3 and 4 imply that $e(a,1)$ is constant for all $1 \leqslant a<N_\ell$, which is a contradiction.

A5. Find all periodic sequences $a_1, a_2, \ldots$ of real numbers such that the following conditions hold for all $n \geqslant 1$:

$$a_{n+2} + a_n^2 = a_n + a_{n+1}^2 \qquad \text{and} \qquad |a_{n+1} - a_n| \leqslant 1.$$

(Kosovo)

Answer: The sequences satisfying the conditions of the problem are:

$$c, -c, c, -c, \ldots,$$
$$d, d, d, d, \ldots,$$

where $c \in [-\frac{1}{2}, \frac{1}{2}]$ and d is any real number.

Solution 1. We rewrite the first condition as

$$a_{n+2} + a_{n+1} = (a_{n+1} + a_n)(a_{n+1} - a_n + 1). \tag{1}$$

If there exists a positive integer m such that $a_{m+1} + a_m = 0$, then from equation (1) we have $a_{n+1} + a_n = 0$ for all positive integers $n \geqslant m$. By the fact that the sequence $(a_{i+1} + a_i)$ is periodic, we get $a_{i+1} + a_i = 0$ for every positive integer i. Thus the sequence (a_i) is of the form $c, -c, c, -c, \ldots$ for some $|c| \leqslant \frac{1}{2}$.

Now suppose that $a_{n+1} + a_n \neq 0$ for every positive integer n. Let T be the period of the sequence. From equation (1) we have

$$1 = \prod_{i=1}^{T} \frac{a_{i+2} + a_{i+1}}{a_{i+1} + a_i} = \prod_{i=1}^{T} (a_{i+1} - a_i + 1).$$

Combining with the second condition $|a_{i+1} - a_i| \leqslant 1$, we have $a_{i+1} - a_i + 1 > 0$. Using the AM-GM inequality we get

$$1 = \prod_{i=1}^{T} (a_{i+1} - a_i + 1) \leqslant \left(\frac{\sum_{i=1}^{T} (a_{i+1} - a_i + 1)}{T} \right)^T = 1.$$

So the equality holds, and thus we get

$$a_2 - a_1 = a_3 - a_2 = \cdots = a_{T+1} - a_T,$$

which means that (a_i) is a constant sequence.

So all sequences satisfying the conditions of the problem are those listed above.

Solution 2. Define $s_n = a_{n+1} + a_n$ and $d_n = a_{n+1} - a_n$, so the original sequence is periodic if and only if both these sequences are periodic. Rearranging the given conditions yields $s_{n+1} = s_n(1 + d_n)$ and $d_{n+1} = d_n(s_n - 1)$, with $|d_n| \leqslant 1$ for all n.

If $s_n = 0$ for some n then $s_i = 0$ for all $i \geqslant n$, and for the sequence to be periodic we must have all $s_i = 0$ and the sequence c, $-c$, c, $-c$, $\ldots$, for some $|c| \leqslant \frac{1}{2}$. Similarly, if $d_n = 0$ for some n and the sequence is periodic, then all $d_i = 0$ and the sequence is c, c, c, $\ldots$.

We claim those are the only periodic sequences, so suppose for contradiction that we have a periodic sequence where no s_i or d_i is 0. Under this hypothesis, we will prove that (s_n), (d_n) have the following properties.

1. All s_n are positive numbers. As $|d_n| \leqslant 1$ and $s_{n+1} = s_n(1+d_n) \neq 0$, it follows that $d_n > -1$ and that all s_n have the same sign (all positive or all negative). If all s_n are negative, then $|d_{n+1}| = |d_n(s_n - 1)| > |d_n|$, so $|d_i|$ is a strictly increasing sequence, contradicting periodicity.

2. Whenever $d_n > 0$ we have $0 < s_n < 1$. Suppose for contradiction that we have $d_n > 0$ and $s_n \geqslant 1$ for some n. Since $d_{n+1} \neq 0$ we have $s_n > 1$, and then $d_{n+1} > 0$, $s_{n+1} > s_n > 1$. Inductively, all $d_i > 0$ for $i \geqslant n$, and $s_{i+1} > s_i$ for $i \geqslant n$, contradicting periodicity.

Now we can get the desired contraction as follows. Suppose that the period of (a_i) is T, then $\sum_{i=1}^{T} d_i = a_{T+1} - a_1 = 0$, hence there is an n such that $d_n > 0$. By property 2 we get $0 < s_n < 1$, and in particular $s_n < 2$. Suppose that we have $s_i < 2$. If $d_i < 0$, then $s_{i+1} = s_i(1+d_i) < s_i < 2$; if $d_i > 0$, then by property 2 we have $0 < s_i < 1$, and then $s_{i+1} = s_i(1 + d_i) \leqslant 2s_i < 2$. In both cases we get $s_{i+1} < 2$, and then by induction we get $s_k < 2$ for all $k \geqslant n$. But then we have $|d_{k+1}| = |d_k(s_k - 1)| < |d_k|$, which contradicts the fact that (d_k) is periodic.

So the only periodic sequences are the two listed above.

Solution 3. Note that if $a_{n+1} = -a_n$ for any n, then $a_{n+2} = a_n = -a_{n+1}$, yielding the first answer by periodicity. Also, if $a_{n+1} = a_n$ for any n, then $a_{n+2} = a_n = a_{n+1}$, yielding the second answer by periodicity. If $a_{n+2} = a_n$ for any n, then $a_n^2 = a_{n+1}^2$ so $a_{n+1} = \pm a_n$ and one of those two cases applies. Henceforth, we will assume that the sequence is neither one of the answers and $a_n \neq a_{n+1}$, $-a_{n+1}$, a_{n+2} for all n for the rest of the solution. Note that the recursion rearranges to

$$a_{n+2} - a_{n+1} = (a_{n+1}^2 - a_{n+1}) - (a_n^2 - a_n) = (a_{n+1} - a_n)(a_{n+1} + a_n - 1). \tag{2}$$

Claim 1. We have that $a_n \leqslant \frac{1}{2}$ for all n.

Proof. First, we cannot have $a_n > \frac{1}{2}$ for all n. Otherwise, $a_{n+1} + a_n - 1$ is positive for all n, so (2) implies that $a_{n+2} - a_{n+1}$ has the same sign as $a_{n+1} - a_n$ for all n. This would mean that the sequence is monotonic, contradicting periodicity.

On the other hand, if $a_{n+1} \leqslant \frac{1}{2}$ and $a_{n+2} > \frac{1}{2}$, then

$$a_{n+1}^2 = a_{n+2} + a_n^2 - a_n \geqslant a_{n+2} - \frac{1}{4} > \frac{1}{4} \implies |a_{n+1}| > \frac{1}{2},$$

where we use the fact that $t^2 - t \geqslant -\frac{1}{4}$ for all $t \in \mathbb{R}$. As $a_{n+1} \leqslant \frac{1}{2}$, this means that $a_{n+1} < -\frac{1}{2}$ so $|a_{n+1} - a_{n+2}| > 1$, a contradiction. □

The identity (2) now implies that $a_{n+2} - a_{n+1}$ and $a_{n+1} - a_n$ are of opposite signs for all n, so that $a_n < a_{n+1} \iff a_{n+1} > a_{n+2}$.

Claim 2. We have that $a_n > 0 \iff a_{n+1} \leqslant 0$: that is, the signs of the sequence are alternating.

Proof. First, it cannot be the case that $a_n > 0$ for all n. Indeed, then we would have from Claim 1 that $|a_{n+1} + a_n - 1| < 1$ for all n, which by (2) means that $|a_{n+1} - a_n|$ is strictly decreasing in n, a contradiction of the sequence's periodicity. It also cannot be the case that $a_n \leqslant 0$ for all n, as then we would have that $|a_{n+1} + a_n - 1| > 1$ for all n (noting that by the nonconstant assumption we will never have $a_n = a_{n+1} = 0$) so $|a_{n+1} - a_n|$ is strictly increasing in n.

Hence, if the signs of a_n are not alternating, then by periodicity there exists n with $a_n > 0$ and $a_{n+1}, a_{n+2} \leqslant 0$ or $a_n, a_{n+1} > 0$ and $a_{n+2} \leqslant 0$. In either scenario, we have that

$$a_n^2 - a_n = a_{n+1}^2 - a_{n+2} \geqslant a_{n+1}^2 \geqslant 0 \implies a_n > 1$$

as a_n is positive and a_{n+2} is nonpositive.

In the former case, we have that $a_n - a_{n+1} > 1$, a contradiction. In the latter case, as $a_{n+1} > a_{n+2}$, we must have that $a_n < a_{n+1}$. But then we have that $a_{n+1} - a_{n+2} > a_n - a_{n+2} > 1$, a contradiction. □

Note now that we cannot have $a_{n+2} > -a_{n+1} > a_n$ for any n, as we would then have

$$a_{n+1}^2 - a_n^2 = a_{n+2} - a_n > -a_{n+1} - a_n > 0 \implies a_n - a_{n+1} > 1,$$

a contradiction. Similarly, we cannot have $a_n > -a_{n+1} > a_{n+2}$ for any n, as we would then have

$$a_n^2 - a_{n+1}^2 = a_n - a_{n+2} > a_n + a_{n+1} > 0 \implies a_n - a_{n+1} > 1.$$

Having ruled out these scenarios, we may conclude that $|a_{n+1}|$ is not between $|a_n|$ and $|a_{n+2}|$ for any n.

Let k be an index such that $|a_k|$ is maximal. Note that we cannot have $|a_{k-2}| = |a_k|$, as that would imply that $a_{k-2} = a_k$ by Claim 2. We also cannot have $|a_{k-2}| \leqslant |a_{k-1}|$, as that would imply that $|a_{k-1}|$ is between $|a_{k-2}|$ and $|a_k|$. Hence, we must have that $|a_{k-1}| < |a_{k-2}| < |a_k|$. As $|a_k|$ is maximal, we cannot have $a_k = 0$. If $a_k > 0$, then we have that $a_k - a_{k-2} = a_{k-1}^2 - a_{k-2}^2 < 0$, a contradiction. If $a_k < 0$, then we have that $a_k - a_{k+2} = a_k^2 - a_{k+1}^2 > 0$, a contradiction.

A6. Let a_0, a_1, a_2, ... be an infinite strictly increasing sequence of positive integers such that for each $n \geqslant 1$ we have

$$a_n \in \left\{ \frac{a_{n-1} + a_{n+1}}{2}, \sqrt{a_{n-1} \cdot a_{n+1}} \right\}.$$

Let b_1, b_2, ... be an infinite sequence of letters defined as

$$b_n = \begin{cases} A, & \text{if } a_n = \frac{1}{2}(a_{n-1} + a_{n+1}); \\ G, & \text{otherwise.} \end{cases}$$

Prove that there exist positive integers n_0 and d such that for all $n \geqslant n_0$ we have $b_{n+d} = b_n$.
(Czech Republic)

Common remarks. In fact, all known proofs proceed by showing that the eventual period of the sequence (b_n) always consists of some number of occurrences of G (possibly zero) followed by an A.

Such sequences of any period $p \geqslant 1$ exist. Indeed, consider the sequence

$$\ldots, k^p, k^{p-1}(k+1), \ldots, k(k+1)^{p-1}, (k+1)^p, (k+1)^{p-1}(k+2), \ldots.$$

The Tournament of the Towns, in Spring 2009 (Junior A-Level Paper, problem 4), considered sequences satisfying exactly this fairly natural criterion. However, it asked a vastly easier question about them: in the language of this problem, it asked whether every such sequence had (b_n) eventually constant. The answer to that problem is "no", as heavily hinted by the statement of this problem. Thus, at least so far as the Problem Selection Committee knows, this is a novel problem about a family of sequences which has been previously considered.

Solution 1. We will show that the eventual period of sequence (b_n) consists of any fixed number of occurrences of G (possibly zero) followed by a single A.

We look at the ratios of consecutive terms of the sequence (a_n). Let C and D be coprime positive integers such that $a_1/a_0 = (C+D)/C$. If $b_n = G$ then $a_n/a_{n-1} = a_{n+1}/a_n$. If $b_n = A$ and $a_n/a_{n-1} = (C+kD)/(C+(k-1)D)$ for some positive integer k then

$$\frac{a_{n+1}}{a_n} = \frac{2a_n - a_{n-1}}{a_n} = \frac{C + (k+1)D}{C + kD}.$$

Thus, by induction, there is a sequence of positive integers (k_n) for $n \geqslant 1$ which satisfies $a_n/a_{n-1} = (C + k_n D)/(C + (k_n - 1)D)$ for all positive integers n. Moreover, we have $k_1 = 1$ and

$$k_{n+1} = \begin{cases} k_n, & \text{if } b_n = G; \\ k_n + 1, & \text{if } b_n = A. \end{cases}$$

If there are only finitely many values of n such that $b_n = A$ then the problem statement obviously holds (we can choose $d = 1$). Thus, we may assume that $b_n = A$ for infinitely many n. This means that the sequence (k_n) attains all positive integer values. Given a value $q \geqslant 1$, denote by m_q the last index where value q occurs, that is, the index such that $k_{m_q} = q$ and $k_{m_q+1} = q + 1$.

Our aim is to prove that the sequence of differences $(m_{q+1} - m_q)$ is eventually constant. We first show that it is bounded above. To that end, fix $t \geqslant 1$ (we will choose a suitably large t later on) and consider a sequence $s(t)_0$, $s(t)_1$, ... defined for $q \geqslant 1$ by $s(t)_q = a_{m_q}/(C + qD)^t$.

We note two properties of $s(t)_q$. First, simple algebra gives

$$s(t)_{q+1} = \frac{a_{m_{q+1}}}{(C+(q+1)D)^t} = \frac{a_{m_q}}{(C+(q+1)D)^t}\left(\frac{C+(q+1)D}{C+qD}\right)^{m_{q+1}-m_q}$$
$$= \frac{a_{m_q}}{(C+qD)^t}\left(\frac{C+(q+1)D}{C+qD}\right)^{m_{q+1}-m_q-t} = s(t)_q\left(\frac{C+(q+1)D}{C+qD}\right)^{m_{q+1}-m_q-t}.$$

It follows that

$$\left.\begin{aligned} s(t)_q &> s(t)_{q+1} \\ s(t)_q &= s(t)_{q+1} \\ s(t)_q &< s(t)_{q+1} \end{aligned}\right\} \quad \text{if and only if} \quad \begin{cases} m_{q+1}-m_q < t, \\ m_{q+1}-m_q = t, \\ m_{q+1}-m_q > t. \end{cases}$$

Second, suppose that $m_{q+1}-m_q \geqslant t$ for some positive integer q. We claim that in that case $s(t)_q$ is a positive integer. Indeed, we have

$$a_{m_q+t} = a_{m_q}\left(\frac{C+(q+1)D}{C+qD}\right)^t,$$

because $k_{m_q+1} = k_{m_q+2} = \cdots = k_{m_q+t} = q+1$. Since $C+(q+1)D$ and $C+qD$ are coprime we have that

$$s(t)_q = \frac{a_{m_q}}{(C+qD)^t}$$

is an integer.

We choose $T \geqslant 1$ such that $s(T)_1 < 1$ (which exists since $C+D>1$). Then, by induction we can show that $s(T)_q < 1$ for all q. Indeed, since $s(T)_q < 1$, it is not a positive integer; this means that $m_{q+1}-m_q < T$ by the second property above. Hence by the first property above we have $s(T)_{q+1} < s(T)_q < 1$, as needed.

This means that $m_{q+1}-m_q < T$ for all q. Thus there is a largest integer $T' \leqslant T$ with the property that an equality $m_{q+1}-m_q = T'$ holds for infinitely many values of q.

Therefore, for all sufficiently large values of q we have the inequality $m_{q+1}-m_q \leqslant T'$, which by the first property implies that the sequence $s(T')$ is decreasing from some point on. Moreover, we know that the sequence attains infinitely many integer values since there are infinitely many values of q for which we have the equality $m_{q+1}-m_q = T'$. As a consequence, the sequence $s(T')$ is constant from some sufficiently large index Q onwards.

This in turn means that the equality $m_{q+1}-m_q = T'$ holds for all $q \geqslant Q$. Note that $b_n = A$ is equivalent to the fact that $n = m_q$ for some integer q. Thus, the sequence (b_n) is periodic for $n \geqslant Q$ with period T', and the proof is complete.

Solution 2. First, observe that the statement holds immediately if $b_n = G$ for all n; otherwise, there must be some n for which $b_n = A$. Without loss of generality, we may assume that $n = 1$, as we can translate the sequence without affecting the statement.

We define an arithmetic sequence (p_n) by taking $p_0 = a_0/\gcd(a_0, a_1)$ and $p_1 = a_1/\gcd(a_0, a_1)$. Note that $p_0 < p_1$, and hence that (p_n) is an increasing sequence of positive integers, and also that $p_2 = a_2/\gcd(a_0, a_1)$.

We also define a sequence of positive integers $d_n = a_n - a_{n-1}$ and a sequence of positive rational numbers $q_n = a_n/a_{n-1}$.

Then the following facts are immediate consequences of the definitions:

- if $b_n = G$, then $q_{n+1} = q_n$ and $d_{n+1} = d_n q_n$;
- if $b_n = A$, then $d_{n+1} = d_n$;

- $q_1 = p_1/p_0$;
- if $b_n = A$ and $q_n = p_i/p_{i-1}$, then $q_{n+1} = p_{i+1}/p_i$.

Now, let k_i be the number of integers n for which $b_n = G$ and $q_n = p_i/p_{i-1}$. If some k_i is infinite then b_n is eventually always G; otherwise, all values of k_i are nonnegative integers.

The sequence of values for d_n can be written as

$$d_0, d_0\frac{p_1}{p_0}, \ldots, d_0\left(\frac{p_1}{p_0}\right)^{k_1}, d_0\left(\frac{p_1}{p_0}\right)^{k_1}\frac{p_2}{p_1}, \ldots, d_0\left(\frac{p_1}{p_0}\right)^{k_1}\left(\frac{p_2}{p_1}\right)^{k_2}, \ldots$$

and in particular all terms in this sequence are positive integers. Furthermore, p_i and p_{i+1} are coprime for all i, so the following sequence consists entirely of positive integers:

$$\begin{aligned} u_0 &= d_0 p_0^{-k_1}, \\ u_1 &= d_0 p_0^{-k_1} p_1^{k_1-k_2}, \\ u_2 &= d_0 p_0^{-k_1} p_1^{k_1-k_2} p_2^{k_2-k_3}, \\ &\vdots\ . \end{aligned}$$

We will prove that k_i is eventually constant, which implies that the sequence of b_n is eventually periodic with period consisting of k copies of G followed by an A (where k is that constant value).

Observe that either k_i is unbounded, or is bounded with eventual maximum k for some constant k. In the second case, let r_0 be minimal such that $k_{r_0} = k$; in the first case let $r_0 = 0$. We will construct an infinite sequence of integers as follows:

- If $k_{r_i+1} \geqslant k_{r_i}$, then $r_{i+1} = r_i + 1$
- If $k_{r_i+1} < k_{r_i}$, then r_{i+1} is the minimal positive integer greater than r_i such that $k_{r_{i+1}} \geqslant k_{r_i}$.

Observe that in the second case, such an r_{i+1} must exist by our construction of r_0.

We claim that $u_{r_{i+1}} \leqslant u_{r_i}$ with equality only if $k_{r_i+1} = k_{r_i}$ (so $r_{i+1} = r_i + 1$). Indeed, if $k_{r_i+1} \geqslant k_{r_i}$ then

$$u_{r_{i+1}} = u_{r_i+1} = u_{r_i} p_{r_i}^{k_{r_i}-k_{r_i+1}} \leqslant u_{r_i},$$

with equality if and only if $k_{r_i} = k_{r_i+1}$.

Otherwise, we have

$$\frac{u_{r_{i+1}}}{u_{r_i}} = p_{r_i}^{k_{r_i}-k_{r_i+1}} p_{r_i+1}^{k_{r_i+1}-k_{r_i+2}} \cdots p_{r_{i+1}-1}^{k_{r_{i+1}-1}-k_{r_{i+1}}},$$

so we just need to show that the right hand side is strictly less than 1. But this follows because

$$\begin{aligned} p_{r_i}^{k_{r_i}-k_{r_i+1}} p_{r_i+1}^{k_{r_i+1}-k_{r_i+2}} \cdots p_{r_{i+1}-1}^{k_{r_{i+1}-1}-k_{r_{i+1}}} &< p_{r_i+1}^{k_{r_i}-k_{r_i+2}} p_{r_i+2}^{k_{r_i+2}-k_{r_i+3}} \cdots p_{r_{i+1}-1}^{k_{r_{i+1}-1}-k_{r_{i+1}}} \\ &< p_{r_i+2}^{k_{r_i}-k_{r_i+3}} p_{r_i+3}^{k_{r_i+3}-k_{r_i+4}} \cdots p_{r_{i+1}-1}^{k_{r_{i+1}-1}-k_{r_{i+1}}} \\ &\vdots \\ &< p_{r_{i+1}-1}^{k_{r_i}-k_{r_{i+1}}} \\ &\leqslant 1, \end{aligned}$$

where each inequality besides the last follows from the fact that $p_j < p_{j+1}$ and $k_{r_i} > k_j$ for $j < r_{i+1}$, and the last follows from the fact that $k_{r_i} \leqslant k_{r_{i+1}}$.

Finally, the sequence u_{r_i} is an infinite nonincreasing sequence of positive integers so must eventually be constant, yielding the claim.

Comment. The two solutions above differ in approach, but have some overlap in the structure they reveal. Indeed, the $C + nD$ of Solution 1 is the p_n of Solution 2, while the $m_{r+1} - m_r$ of Solution 1 turns out to be equal to the k_r of Solution 2.

A7. Let $\mathbb{Q}$ be the set of rational numbers. Let $f\colon \mathbb{Q} \to \mathbb{Q}$ be a function such that the following property holds: for all x, $y \in \mathbb{Q}$,

$$f(x + f(y)) = f(x) + y \qquad \text{or} \qquad f(f(x) + y) = x + f(y).$$

Determine the maximum possible number of elements of $\{f(x) + f(-x) \mid x \in \mathbb{Q}\}$.

(Japan)

Answer: 2 is the maximum number of elements.

Common remarks. Suppose that f is a function satisfying the condition of the problem. We will use the following throughout all solutions.

- $a \sim b$ if either $f(a) = b$ or $f(b) = a$,
- $a \to b$ if $f(a) = b$,
- $P(x, y)$ to denote the proposition that either $f(x + f(y)) = f(x) + y$ or $f(f(x) + y) = x + f(y)$,
- $g(x) = f(x) + f(-x)$.

With this, the condition $P(x, y)$ could be rephrased as saying that $x + f(y) \sim f(x) + y$, and we are asked to determine the maximum possible number of elements of $\{g(x) \mid x \in \mathbb{Q}\}$.

Solution 1. We begin by providing an example of a function f for which there are two values of $g(x)$. We take the function $f(x) = \lfloor x \rfloor - \{x\}$, where $\lfloor x \rfloor$ denotes the floor of x (that is, the largest integer less than or equal to x) and $\{x\} = x - \lfloor x \rfloor$ denotes the fractional part of x.

First, we show that f satisfies $P(x, y)$. Given $x, y \in \mathbb{Q}$, we have

$$\begin{aligned} f(x) + y &= \lfloor x \rfloor - \{x\} + \lfloor y \rfloor + \{y\} = (\lfloor x \rfloor + \lfloor y \rfloor) + (\{y\} - \{x\}); \\ x + f(y) &= \lfloor x \rfloor + \{x\} + \lfloor y \rfloor - \{y\} = (\lfloor x \rfloor + \lfloor y \rfloor) + (\{x\} - \{y\}). \end{aligned}$$

If $\{x\} < \{y\}$, then we have that the fractional part of $f(x) + y$ is $\{y\} - \{x\}$ and the floor is $\lfloor x \rfloor + \lfloor y \rfloor$, so $f(x) + y \to x + f(y)$. Likewise, if $\{x\} > \{y\}$, then $x + f(y) \to f(x) + y$. Finally, if $\{x\} = \{y\}$, then $f(x) + y = x + f(y) = \lfloor x \rfloor + \lfloor y \rfloor$ is an integer. In all cases, the relation P is satisfied.

Finally, we observe that if x is an integer then $g(x) = 0$, and if x is not an integer then $g(x) = -2$, so there are two values for $g(x)$ as required.

Now, we prove that there cannot be more than two values of $g(x)$. $P(x, x)$ tells us that $x + f(x) \sim x + f(x)$, or in other words, for all x,

$$f(x + f(x)) = x + f(x). \tag{1}$$

We begin with the following lemma.

Lemma 1. f is a bijection, and satisfies

$$f(-f(-x)) = x. \tag{2}$$

Proof. We first prove that f is injective. Suppose that $f(x_1) = f(x_2)$; then $P(x_1, x_2)$ tells us that $f(x_1) + x_2 \sim f(x_2) + x_1$. Without loss of generality, suppose that $f(x_1) + x_2 \to f(x_2) + x_1$.

But $f(x_1) = f(x_2)$, so $f(f(x_1) + x_2) = f(f(x_2) + x_2) = f(x_2) + x_2$ by (1). Therefore, $f(x_2) + x_1 = f(x_2) + x_2$, as required.

Now, (1) with $x = 0$ tells us that $f(f(0)) = f(0)$ and so by injectivity $f(0) = 0$.

Applying $P(x, -f(x))$ tells us that $0 \sim x + f(-f(x))$, so either $0 = f(0) = x + f(-f(x))$ or $f(x + f(-f(x))) = 0$ which implies that $x + f(-f(x)) = 0$ by injectivity. Either way, we deduce that $x = -f(-f(x))$, or $x = f(-f(-x))$ by replacing x with $-x$.

Finally, note that bijectivity follows immediately from (2). □

Since f is bijective, it has an inverse, which we denote f^{-1}. Rearranging (2) (after replacing x with $-x$) gives that $f(-x) = -f^{-1}(x)$. We have $g(x) = f(x) + f(-x) = f(x) - f^{-1}(x)$.

Suppose $g(x) = u$ and $g(y) = v$, where $u \neq v$ are both nonzero. Define $x' = f^{-1}(x)$ and $y' = f^{-1}(y)$; by definition, we have

$$x' \to x \to x' + u$$
$$y' \to y \to y' + v.$$

Putting in $P(x', y)$ gives $x + y \sim x' + y' + v$, and putting in $P(x, y')$ gives $x + y \sim x' + y' + u$. These are not equal since $u \neq v$, and $x + y$ may have only one incoming and outgoing arrow because f is a bijection, so we must have either $x' + y' + u \to x + y \to x' + y' + v$ or the same with the arrows reversed. Swapping (x, u) and (y, v) if necessary, we may assume without loss of generality that this is the correct direction for the arrows.

Also, we have $-x' - u \to -x \to -x'$ by Lemma 1. Putting in $P(x + y, -x' - u)$ gives $y \sim y' + v - u$, and so $y' + v - u$ must be either $y' + v$ or y'. This means u must be either 0 or v, and this contradicts our assumption about u and v.

Comment. Lemma 1 can also be proven as follows. We start by proving that f must be surjective. Suppose not; then, there must be some t which does not appear in the output of f. $P(x, t - f(x))$ tells us that $t \sim x + f(t - f(x))$, and so by assumption $f(t) = x + f(t - f(x))$ for all x. But setting $x = f(t) - t$ gives $t = f(t - f(f(t) - t))$, contradicting our assumption about t.

Now, choose some t such that $f(t) = 0$; such a t must exist by surjectivity. $P(t, t)$ tells us that $f(t) = t$, or in other words $t = 0$ and $f(0) = 0$. The remainder of the proof is the same as the proof given in Solution 1.

Solution 2. We again start with Lemma 1, and note $f(0) = 0$ as in the proof of that lemma.

$P(x, -f(y))$ gives $x + f(-f(y)) \sim f(x) - f(y)$, and using (2) this becomes $x - y \sim f(x) - f(y)$. In other words, either $f(x - y) = f(x) - f(y)$ or $x - y = f(f(x) - f(y))$. In the latter case, we deduce that

$$\begin{aligned} f(-(x - y)) &= f(-f(f(x) - f(y))) \\ f(y - x) &= f(-f(f(x) - f(y))) \\ &= f(y) - f(x). \end{aligned}$$

Thus, $f(y) - f(x)$ is equal to either $f(y - x)$ or $-f(x - y)$. Replacing y with $x + d$, we deduce that $f(x + d) - f(x) \in \{f(d), -f(-d)\}$.

Now, we prove the following claim.

Claim. For any $n \in \mathbb{Z}_{>0}$ and $d \in \mathbb{Q}$, we have that either $g(d) = 0$ or $g(d) = \pm g(d/n)$.

In particular, if $g(d/n) = 0$ then $g(d) = 0$.

Proof. We first prove that if $g(d/n) = 0$ then $g(d) = 0$. Suppose that $g(d/n) = 0$. Then $f(d/n) = -f(-d/n)$ and so $f(x + d/n) - f(x) = f(d/n)$ for any x. Applying this repeatedly, we deduce that $f(x + d) - f(x) = nf(d/n)$ for any x. Applying this with $x = 0$ and $x = -d$ and adding gives $f(d) + f(-d) = 0$, so $g(d) = 0$, and in particular the claim is true whenever $g(d) = 0$.

Now, select $n \in \mathbb{Z}_{>0}$ and $d \in \mathbb{Q}$ such that $g(d) \neq 0$, and observe that we must have $g(d/n) \neq 0$. Observe that for any $k \in \mathbb{Z}$ we have that $f(kd/n) - f((k-1)d/n) \in \{f(d/n), -f(-d/n)\}$. Let A_i be the number of $k \in \mathbb{Z}$ with $i - n < k \leqslant i$ such that this difference equals $f(d/n)$.

We deduce that for any $i \in \mathbb{Z}$,

$$\begin{aligned} f(id/n) - f(id/n - d) &= \sum_{i-n<k\leqslant i} f(kd/n) - f((k-1)d/n) \\ &= A_i f(d/n) - (n - A_i) f(-d/n) \\ &= -nf(-d/n) + A_i g(d/n). \end{aligned}$$

Since $g(d/n)$ is nonzero, this is a nonconstant linear function of A_i. However, there are only two possible values for $f(id/n) - f(id/n - d)$, so there must be at most two possible values for A_i as i varies. And since $A_{i+1} - A_i \in \{-1, 0, 1\}$, those two values must differ by 1 (if there are two values).

Now, we have

$$\begin{aligned} f(d) - f(0) &= -nf(-d/n) + A_n g(d/n), \qquad \text{and} \\ f(0) - f(-d) &= -nf(-d/n) + A_0 g(d/n). \end{aligned}$$

Subtracting these (using the fact that $f(0) = 0$) we obtain

$$\begin{aligned} f(d) + f(-d) &= (A_n - A_0) g(d/n) \\ &= \pm g(d/n), \end{aligned}$$

where the last line follows from the fact that $g(d)$ is nonzero. □

It immediately follows that there can only be one nonzero number of the form $g(x)$ up to sign; to see why, if $g(d)$ and $g(d')$ are both nonzero, then for some $n, n' \in \mathbb{Z}_{>0}$ we have $d/n = d'/n'$. But

$$g(d) = \pm g(d/n) = \pm g(d').$$

Finally, suppose that for some d, d' we have $g(d) = c$ and $g(d') = -c$ for some nonzero c. So we have

$$f(d) + f(-d) - f(d') - f(-d') = 2c$$

which rearranges to become $(f(d) - f(d')) - (f(-d') - f(-d)) = 2c$.

Each of the bracketed terms must be equal to either $f(d - d')$ or $-f(d' - d)$. However, they cannot be equal since c is nonzero, so $g(d - d') = f(d - d') + f(d' - d) = \pm 2c$. This contradicts the assertion that $g(-x) = \pm c$ for all x.

Solution 3. As in Solution 1, we start by establishing Lemma 1 as above, and write $f^{-1}(x) = -f(-x)$ for the inverse of f, and $g(x) = f(x) - f^{-1}(x)$.

We now prove the following.

Lemma 2. If $g(x) \neq g(y)$, then $g(x + y) = \pm(g(x) - g(y))$.

Proof. Assume x and y are such that $g(x) \neq g(y)$. Applying $P(x, f^{-1}(y))$ gives $x + y \sim f(x) + f^{-1}(y)$, and applying $P(f^{-1}(x), y)$ gives $x + y \sim f^{-1}(x) + f(y)$.

Observe that

$$\begin{aligned}(f(x) + f^{-1}(y)) - (f^{-1}(x) + f(y)) &= (f(x) - f^{-1}(x)) - (f(y) - f^{-1}(y)) \\ &= g(x) - g(y).\end{aligned}$$

By assumption, $g(x) \neq g(y)$, and so $f(x) + f^{-1}(y) \neq f^{-1}(x) + f(y)$. Since f is bijective, this means that these two values must be $f(x + y)$ and $f^{-1}(x + y)$ in some order, and so $g(x+y) = f(x+y) - f^{-1}(x+y)$ must be their difference up to sign, which is either $g(x) - g(y)$ or $g(y) - g(x)$. □

Claim. If x and q are rational numbers such that $g(q) = 0$ and n is an integer, then $g(x + nq) = g(x)$.

Proof. If $g(b) = 0$ and $g(a) \neq g(a + b)$, then the lemma tells us that $g(b) = \pm(g(a + b) - g(a))$, which contradicts our assumptions. Therefore, $g(a) = g(a + b)$ whenever $g(b) = 0$.

A simple induction then gives that $g(nb) = 0$ for any positive integer n, and $g(nb) = 0$ for negative n as $g(x) = g(-x)$. The claim follows immediately. □

Lemma 3. There cannot be both positive and negative elements in the range of g.

Proof. Suppose that $g(x) > 0$ and $g(y) < 0$. Let $\mathcal{S}$ be the set of numbers of the form $mx + ny$ for integers m, n. We first show that $g(\mathcal{S})$ has infinitely many elements. Indeed, suppose $g(\mathcal{S})$ is finite, and let $a \in \mathcal{S}$ maximise g and $b \in \mathcal{S}$ maximise $-g$. Then $a + b \in \mathcal{S}$, and $g(a + b) = g(a) - g(b)$ or $g(b) - g(a)$. In the first case $g(a + b) > g(a)$ and in the second case $g(a + b) < g(b)$; in either case we get a contradiction.

Now, we show that there must exist some nonzero rational number q with $g(q) = 0$. Indeed, suppose first that $a + f(a) = 0$ for all a. Then $g(a) = f(a) + f(-a) = 0$ for all a, and so g takes no nonzero value. Otherwise, there is some a with $a + f(a) \neq 0$, and so (1) yields that $f(q) = 0$ for $q = a + f(a) \neq 0$. Noting that $f(-q) = 0$ from Lemma 1 tells us that $g(q) = 0$, as required.

Now, there must exist integers s and s' such that $xs = qs'$ and integers t and t' such that $yt = qt'$. The claim above gives that the value of $g(mx + ny)$ depends only on the values of m mod s and n mod t, so $g(mx + ny)$ can only take finitely many values. □

Finally, suppose that $g(x) = u$ and $g(y) = v$ where $u \neq v$ have the same sign. Assume $u, v > 0$ (the other case is similar) and assume $u > v$ without loss of generality.

$P(f^{-1}(x), f^{-1}(y))$ gives $x - y \sim f^{-1}(x) - f^{-1}(y) = f(x) - f(y) - (u - v)$, and $P(x, y)$ gives $x - y \sim f(x) - f(y)$. $u - v$ is nonzero, so $f(x - y)$ and $f^{-1}(x - y)$ must be $f(x) - f(y) - (u - v)$ and $f(x) - f(y)$ in some order, and since $g(x - y)$ must be nonnegative, we have

$$f(x) - f(y) - (u - v) \to x - y \to f(x) - f(y).$$

Then, $P(x - y, f^{-1}(y))$ tells us that $(x - y) + y \sim (f(x) - f(y)) + (f(y) - v)$, so $x \sim f(x) - v$, contradicting either $v \neq u$ or $v > 0$.

Comment. Lemma 2 also follows from $f(x + d) - f(x) \in \{f(d), -f(-d)\}$ as proven in Solution 2. Indeed, we also have $f(-x) - f(-x - d) \in \{f(d), -f(-d)\}$, and then subtracting the second from the first we get $g(x+d)-g(x) \in \{g(d), -g(d), 0\}$. Replacing $x+d$ and x with x and $-y$ gives the statement of Lemma 2.

Comment. It is possible to prove using Lemma 2 that g must have image of the form $\{0, c, 2c\}$ if it has size greater than 2. Indeed, if $g(x) = c$ and $g(y) = d$ with $0 < c < d$, then $g(x + y) = d - c$ as it must be nonnegative, and $g(y) = g((x + y) + (-x)) = |d - 2c|$ provided that $d \neq 2c$.

However, it is not possible to rule out $\{0, c, 2c\}$ based entirely on the conclusion of Lemma 2; indeed, the function given by

$$g(x) = \begin{cases} 0, & \text{if } x = 2n \text{ for } n \in \mathbb{Z}; \\ 2, & \text{if } x = 2n + 1 \text{ for } n \in \mathbb{Z}; \\ 1, & \text{if } x \notin \mathbb{Z}. \end{cases}$$

satisfies the conclusion of Lemma 2 (even though there is no function f giving this choice of g).

Note. Solution 1 actually implies that the result also holds over $\mathbb{R}$. The proposal was originally submitted and evaluated over $\mathbb{Q}$ as it is presented here, and the Problem Selection Committee believes that this form is more suitable for the competition because it allows for more varied and interesting approaches once Lemma 1 has been established. Even the variant here defined over $\mathbb{Q}$ was found to be fairly challenging.

A8. Let $p \neq q$ be coprime positive integers. Determine all infinite sequences $a_1, a_2, \ldots$ of positive integers such that the following conditions hold for all $n \geqslant 1$:

$$\max(a_n, a_{n+1}, \ldots, a_{n+p}) - \min(a_n, a_{n+1}, \ldots, a_{n+p}) = p \qquad \text{and}$$
$$\max(a_n, a_{n+1}, \ldots, a_{n+q}) - \min(a_n, a_{n+1}, \ldots, a_{n+q}) = q.$$

(Japan)

Answer: The only such sequences are $a_n = n + C$, where C is a nonnegative integer.

Common remarks.

- Denote by $a_{[i,j]}$ the subsequence $a_i, a_{i+1}, \ldots, a_j$.
- Without loss of generality, in each solution we suppose $p < q$. It can be convenient to treat the case where $p = 1$ separately.
- The problem can also be posed for sequences of arbitrary integers (rather than positive). Refer to the comment after Solution 1 for a proof.

Solution 1. Let $k = \lceil \frac{q}{p} \rceil$. Note that $k \geqslant 2$.

Lemma 1. If i, j and m are positive integers such that $|i - j| \leqslant mp$ then $|a_i - a_j| \leqslant mp$.

Proof. By the given condition, if $|i - j| \leqslant p$ then $|a_i - a_j| \leqslant p$. So the lemma follows from induction on m and the triangle inequality. □

Lemma 2. For a fixed n, suppose that a_i is minimal over $i \geqslant n$. Then $i \leqslant n + p - 1$.

Proof. Suppose for contradiction that $i \geqslant n + p$. Then $\min(a_{[i-p,i+q-p]}) = a_i$. Since $q - p \leqslant (k - 1)p$, it follows from Lemma 1 that $\max(a_{[i-p,i+q-p]}) \leqslant a_i + (k - 1)p < a_i + q$, which is a contradiction. □

Lemma 3. For a fixed $n > q$, suppose that a_i is maximal over $i \leqslant n$. Then $i \geqslant n - p + 1$.

Proof. Suppose a_j is minimal over $j \geqslant n - q$. Then by Lemma 2, $j \leqslant n - q + p - 1$. So $\min(a_{[n-q,n]}) = a_j$ and $a_i \geqslant \max(a_{[n-q,n]})$, which implies that $a_i \geqslant a_j + q$.

Lemma 2 also implies that if $j \geqslant n$ then $a_j \geqslant \min(a_{[n,n+p]})$. So if $i < j$, then we have $a_j \geqslant a_i - p$, which contradicts $a_i \geqslant a_j + q$. Hence we must have $i > j$.

The above inequality also gives $|a_i - a_j| \geqslant q > (k - 1)p$, so by Lemma 1 it follows that $|i - j| > (k - 1)p$. Therefore $i > j + (k - 1)p \geqslant n - q + (k - 1)p \geqslant n - p + 1$. □

Let b_n be the minimal value of a_i for $i \geqslant n$. By Lemma 2, $b_{n+p} > b_n$ for all n. Hence $b_n = \min(a_{[n,n+p]}) = \min(a_{[n,n+q]})$. Let c_n be the maximal value of a_i for $i \leqslant n$. By Lemma 3, $c_{n-p} > c_n$ for all $n > q$. Hence $c_n = \max(a_{[n-p,n]}) = \max(a_{[n-q,n]})$ for $n > q$.

So if $n > q$ then $b_n = c_{n+p} - p = c_{n+q} - q$. So for $n > q$ we get $b_{n+q-p} + p = c_{n+q} = b_n + q$, and hence $b_{n+q-p} = b_n + q - p$.

Next note that $b_{n+p} \leqslant a_{n+p} \leqslant b_n + p$. So $b_{n+p} - b_n \leqslant p$ for all $n > q$, and iterating this $(q-p)$ times gives $b_{n+p(q-p)} - b_n \leqslant p(q-p)$. But using $b_{n+q-p} = b_n + q - p$ gives $b_{n+p(q-p)} - b_n = p(q-p)$. Since equality occurs, we must have $b_{n+p} = b_n + p$.

So for $n > q$, $b_{n+p} = b_n + p$ and $b_{n+q-p} = b_n + q - p$. Since p and $q - p$ are coprime, $b_{n+1} = b_n + 1$ for all $n > q$. The only way for b_n and b_{n+1} to be different is if $b_n = a_n$, so we deduce that $a_{n+1} = a_n + 1$ and there is a constant C such that $a_n = n + C$ for all $n > q$.

Finally, suppose $a_n = n + C$ for all $n \geqslant N$. Then $p = \max(a_{N-1}, N + C + p - 1) - \min(a_{N-1}, N + C)$. So $a_{N-1} = N + C + p$ or $N + C - 1$. Similarly, $a_{N-1} = N + C + q$ or $N + C - 1$. Hence $a_{N-1} = N + C - 1$. So, by induction, we have $a_n = n + C$ for all positive integers n. Since $a_1 \geqslant 1$, C is a nonnegative integer.

It is trivial to check that $a_n = n + C$ satisfies the given condition.

Comment. Here is a variant of Solution 1. Proceed up to proving $b_n - n$ is eventually periodic with period $q-p$. Then there is some minimal value of $b_n - n$. Suppose n attains this minimal value. Since $b_{n+p} - n - p \leqslant b_n - n$, $n+p$ also attains this minimal value. And since p and $q-p$ are coprime, all $n \geqslant q$ must attain this minimal value. Hence $b_{n+1} = b_n + 1$ for all $n \geqslant q$. Finish as above.

Comment. It is also possible to solve the problem using a weaker version of Lemma 2 and without Lemma 3. For example, the following lemma plays a similar role.

Lemma 2'. Let $b'_n = \min(a_{[n,n+p]})$. Then $b'_n < b'_{n+p}$.

Comment. To solve the problem for sequences a_n of arbitrary integers, we will use the following lemma.

Lemma 4. The sequence a_n is either bounded above or bounded below.

Proof. Suppose that a_n is unbounded above and below. Then there is some i such that $a_i < a_1 - p$. There is also some j such that $a_j > \max(a_{[1,i]}) + q$. Now let a_l be minimal amongst $a_{[1,j]}$. Since $a_l \leqslant a_i$, $a_l < a_1 - p$ and $a_l < a_j - kp$. By Lemma 1, $1 + p < l < j - kp$. So $\min(a_{[l-p,l+q-p]}) = a_l$. By Lemma 1 again, $\max(a_{[l-p,l+q-p]}) \leqslant a_l + (k-1)p < a_l + q$, which is a contradiction. □

From there, the solution above can be adapted to prove that $a_n = n + C$ for all n or $a_n = -n + C$ for all n, where C can be any constant integer.

Solution 2. For $n, x \geqslant 1$, let the *x-width* of n be $\max(a_{[n,n+x]}) - \min(a_{[n,n+x]})$. We call a positive integer x *good* if the x-width of n is less than or equal to x for all sufficiently large n, and we call x *very good* if the x-width of n is equal to x for sufficiently large n.

Lemma 1. If p' is good and q' is very good with $p' < q' < 2p'$, then $2p' - q'$ is also good.

Proof. Note that $0 < q' - p' < p' < q'$. Let n be a sufficiently large positive integer. Then for $k \in [n + q' - p', n + p']$, we have $a_k \geqslant \max(a_{[n,n+p']}) - p'$ and $a_k \geqslant \max(a_{[n+q'-p',n+q']}) - p'$ since p' is good, which shows $a_k \geqslant \max(a_{[n,n+q']}) - p'$. Similarly we get $a_k \leqslant \min(a_{[n,n+q']}) + p'$.

Therefore, for all $k \in [n + q' - p', n + p']$ we have $a_k \in [\max(a_{[n,n+q']}) - p', \min(a_{[n,n+q']}) + p']$. Thus, the $(2p' - q')$-width of $n + q' - p'$ is at most $(\min(a_{[n,n+q']}) + p') - (\max(a_{[n,n+q']}) - p') = 2p' - q'$. The lemma follows. □

Lemma 2. Let p' be a good number and q' a very good number with $p' < q'$. For sufficiently large n, take $s, t \in [n, n + q']$ such that $\min(a_{[n,n+q']}) = a_s$ and $\max(a_{[n,n+q']}) = a_t$. Then $s \in [n, n + p']$ and $t \in [n + q' - p', n + q']$.

Proof. Lemma 2 and Lemma 3 from Solution 1 hold with p and q replaced by p' and q' by similar arguments. We can deduce the statement about s from Lemma 2 of Solution 1. We can deduce the statement about t from Lemma 3 of Solution 1. □

Lemma 3. If p' is good and q' is very good with $2p' < q'$, then there exists a positive integer r such that for all sufficiently large n, we have $a_{n+r} - a_n \geqslant r$.

Proof. Let $r = q' - 2p'$, and let s and t be as defined in Lemma 2. Then consider the identity

$$(a_t - a_{n+q'-p'}) + (a_{n+p'+r} - a_{n+p'}) + (a_{n+p'} - a_s) = a_t - a_s = q'.$$

By Lemma 2, we have $s \in [n, n + p']$ and $t \in [n + q' - p', n + q']$, so $a_{n+p'} - s \leqslant p'$ and $a_t - a_{n+q'-p'} \leqslant p'$. Combining these, we get $a_{n+p'+r} - a_{n+p'} \geqslant q' - 2p' = r$. This proves that $a_{n+r} - a_n \geqslant r$ for sufficiently large n. □

Lemma 4. Suppose $(p, q) \neq (1, 2)$. Then there exists a good number p' such that $2p' < q$.

Proof. Let p' be the smallest good positive integer. Note that p is good, so p' exists and is less than q.

Suppose for contradiction that $2p' \geqslant q$. If $2p' > q$, then by Lemma 1, $2p' - q$ is a good number strictly less than p', which contradicts minimality of p'. If $2p' = q$, then $p' < p < 2p'$. So we can apply Lemma 1 with $q_0 = p$ to get that $2p' - p$ is a good number that is strictly less than p', which again contradicts minimality. □

If $(p, q) = (1, 2)$ then the problem is easily solved. Otherwise, Lemmas 3 and 4 combined give us some $r > 0$ such that $a_{n+r} - a_n \geqslant r$ for n sufficiently large.

By iterating, we get $a_{n+pr} - a_n \geqslant pr$ for all sufficiently large n, and hence it follows that $a_{n+p} - a_n = p$. Similarly we get $a_{n+q} - a_n = q$. As p and q are coprime, we deduce that $a_{n+1} - a_n = 1$ for sufficiently large n. Thus we get $a_n = n + C$ for sufficiently large n, and we can conclude by the same argument as Solution 1.

Combinatorics

C1. Let n be a positive integer. A class of n students run n races, in each of which they are ranked with no draws. A student is eligible for a rating (a, b) for positive integers a and b if they come in the top b places in at least a of the races. Their final score is the maximum possible value of $a - b$ across all ratings for which they are eligible.

Find the maximum possible sum of all the scores of the n students.

(Australia)

Answer: The maximum possible sum is $\frac{n(n-1)}{2}$.

Solution 1. The answer can be achieved by the students finishing in the same order in every race. To show that this is the maximum, we will apply a series of modifications to the results of the races, each of which does not decrease the total score, such that after k such modifications the first k positions are the same in every race. Say that a student is scored on the bth place if their score is $a - b$ because they came in the top b places in a of the races and b is minimal with this property for that student.

Supposing that the first $k - 1$ positions are the same in every race, look at the students scored on the kth place. If there are no such students, let $\ell > k$ be minimal such that some student S is scored on the ℓth place. Then, in every race where S appears in any place from the kth through the ℓth inclusive (of which there must be at least ℓ, otherwise S would achieve a higher rating of 0 based on the nth place), reorder the students in places k through ℓ so that S finishes in the kth place instead (and otherwise the ordering of those students is arbitrary). Now S is scored on the kth place, their score has gone up by $\ell - k$ and no other scores have gone down (some might have gone up as well).

Now we know that the first $k - 1$ positions are the same in every race and at least one student is scored on the kth place. Pick one such student S. In each race where S finishes behind the kth place, swap them with the student T who finishes in the kth place, leaving the positions of all other students unchanged. Each such swap increases the score of S by 1 and decreases the score of T by at most 1, so such swaps do not decrease the total score. At the end of this process, the first k positions are the same in every race and the total score has not decreased.

Repeating this n times yields the required result.

Comment. The following simpler approach to modifying results of races is tempting: find pairs of students S and T who are scored on places k and ℓ respectively, where $k < \ell$, but where S finishes after T in some race, and swap the positions of those two students in that race so they finish in the same order as the places they are scored on. However, such a swap can decrease the total score; for example, suppose that $k = 1$ and $\ell = 4$, and in some race S finishes 6th and T finishes 3rd; then swapping those students reduces the number of races contributing to T's score without increasing the number contributing to S's score.

Solution 2. The answer can be achieved by having the same ranking for all n races.

Note that taking $a = b = n$ shows each student has a nonnegative score. Consider a student who has race ranks $r_1, r_2, \ldots, r_n$ and a final score of s. We first prove that

$$\sum_i r_i \leqslant n(n - s).$$

Without loss of generality, suppose that $r_1 \leqslant r_2 \leqslant \cdots \leqslant r_n$. There must exist some k with $s + 1 \leqslant k \leqslant n$ and $k - r_k = s$. In order to maximise $\sum_i r_i$ while retaining the score of s, we can

replace each of $r_1, \ldots, r_{k-1}$ by r_k, and replace each of $r_{k+1}, \ldots, r_n$ by n. Then the sum is

$$\sum_i r_i \leqslant kr_k + (n-k)n = n^2 - k(n-r_k) = n^2 - k(n+s-k) \leqslant n^2 - s_n. \tag{1}$$

The final inequality follows from the fact that given $s+1 \leqslant k \leqslant n$, the quantity $k(n+s-k)$ is minimised when $k = n$.

The sum of ranks of all students across all races is $\frac{n^2(n+1)}{2}$. If the total of all student scores is t, then (1) implies

$$\frac{n^2(n+1)}{2} \leqslant n^3 - tn.$$

This rearranges to $t \leqslant \frac{n(n-1)}{2}$, as required.

Solution 3. In each race, assign the student in the kth place a weight of $1 - \frac{k}{n}$. If a student finishes in the top b places in at least a of the races, the total of their weights is at least $a\left(1 - \frac{b}{n}\right) = a - b\left(\frac{a}{n}\right) \geqslant a - b$. Thus the sum of a student's weights across all races is at least their score, and so the sum of all weights for all students across all races is at least the sum of all the scores of all students. The sum of weights in each race is $\frac{n-1}{2}$, so the sum of all weights across all races is $\frac{n(n-1)}{2}$. Equality is achieved if and only if, for each student, the values of b and a determining that student's score have $a = n$ and they finish in exactly the bth place in all n races; that is, if the students are ranked the same in every race.

Solution 4. Given a positive integer $b(S)$ for each student S, define $a_b(S)$ to be the number of races in which S finished in the top $b(S)$ places, and define $\text{score}_b(S) = a_b(S) - b(S)$; for a race r, let $I_b(S, r)$ be 1 if S finished in the top $b(S)$ places in race r and 0 otherwise, so

$$a_b(S) = \sum_r I_b(S, r).$$

Then the problem asks for the maximum across all possible results of the races of

$$\max_b \sum_S \text{score}_b(S) = \max_b \left(\sum_r \sum_S I_b(S, r) - \sum_S b(S) \right).$$

Given b, the sum $\sum_S I_b(S, r)$ is maximised (not necessarily uniquely) for some choice of the rankings in race r, which is the same choice for every race. So the maximum possible sum of the scores of all the students occurs when all students are ranked the same in all races, which yields the given answer.

C2. Let n be a positive integer. The integers 1, 2, 3, ..., n^2 are to be written in the cells of an $n \times n$ board such that each integer is written in exactly one cell and each cell contains exactly one integer. For every integer d with $d \mid n$, the *d-division* of the board is the division of the board into $(n/d)^2$ nonoverlapping sub-boards, each of size $d \times d$, such that each cell is contained in exactly one $d \times d$ sub-board.

We say that n is a *cool number* if the integers can be written on the $n \times n$ board such that, for each integer d with $d \mid n$ and $1 < d < n$, in the d-division of the board, the sum of the integers written in each $d \times d$ sub-board is not a multiple of d.

Determine all even cool numbers.

(Türkiye)

Answer: The even cool numbers are $n = 2^k$ where k is a positive integer.

Solution. We first show by induction that $n = 2^k$ is a cool number. The base case of $n = 2$ is trivial as there is no such d.

For induction, assume that 2^k is a cool number. We construct a numbering of a $2^{k+1} \times 2^{k+1}$ board that satisfies the conditions.

Take the $2^{k+1} \times 2^{k+1}$ board and divide it into four $2^k \times 2^k$ sub-boards. By assumption, there is some numbering P of a $2^k \times 2^k$ board that satisfies the required condition; we write down the numbering P in each sub-board. Next, add 2^{2k} to every number in the second sub-board, add 2×2^{2k} to every number in the third sub-board, and add 3×2^{2k} to every number in the fourth sub-board. Then the numbers in the cells of the $2^{k+1} \times 2^{k+1}$ board are the numbers 1 to $2^{2(k+1)}$.

Now locate 2^{2k} from the first sub-board, and swap it with $2^{2k} + 2^{k-1}$ from the second sub-board. Locate 3×2^{2k} from the third sub-board, and swap it with $3 \times 2^{2k} + 2^{k-1}$ from the fourth sub-board.

We claim that this numbering of the $2^{k+1} \times 2^{k+1}$ board satisfies the required conditions. For any $d = 2^i$ where $i < k$, consider any $2^i \times 2^i$ sub-board. The sum of its cells modulo 2^i is not changed in the addition step or the swapping step, so the sum is congruent modulo 2^i to the sum of the corresponding $2^i \times 2^i$ sub-board in P, which is nonzero, as required.

In the case of $d = 2^k$, we can directly evaluate the sum of the $(b+1)^{\text{th}}$ sub-board for $b \in \{0, 1, 2, 3\}$. The sum is given by

$$2^{2k-1}(1 + 2^{2k}) + b2^{4k} + (-1)^b 2^{k-1} \equiv 2^{k-1} \pmod{2^k}.$$

Therefore all sub-boards satisfy the required conditions and so 2^{k+1} is a cool number, completing the induction.

It remains to show that no other even number is a cool number. Let $n = 2^s m$ where s is a positive integer and m is an odd integer greater than 1. For the sake of contradiction, suppose that there is a numbering of the $n \times n$ board satisfying the required conditions.

Claim. In the 2^i-division of the board, where $1 \leqslant i \leqslant s$, the sum of numbers in each $2^i \times 2^i$ sub-board is congruent to 2^{i-1} modulo 2^i.

Proof. We prove the claim by induction on i. The base case of $i = 1$ holds as the sum of numbers in each 2×2 sub-board must be odd. Next, suppose the claim is true for 2^i. In the 2^{i+1}-division, each $2^{i+1} \times 2^{i+1}$ sub-board is made up of four $2^i \times 2^i$ sub-boards, each with a sum congruent to 2^{i-1} modulo 2^i. Hence the sum of each $2^{i+1} \times 2^{i+1}$ sub-board is a multiple of 2^i. It cannot be a multiple of 2^{i+1} because of the conditions, which means it must be congruent to 2^i modulo 2^{i+1}. This proves the claim. □

Back to the problem, since m is odd, summing up the m^2 sums of $2^s \times 2^s$ sub-boards gives

$$2^{s-1}m^2 \equiv 2^{s-1} \pmod{2^s}.$$

However, the sum of the numbers from 1 to n^2 is

$$\frac{n^2(n^2+1)}{2} = 2^{2s-1}m^2(2^{2s}m^2+1) \equiv 0 \pmod{2^s}.$$

This is a contradiction. Therefore n is not a cool number.

Comment. In the case of odd n, similar arguments show that prime powers are cool numbers.

If the definition of cool numbers additionally requires that all $d \times d$ sub-boards in the d-division have the same nonzero residue modulo d, then the cool numbers are precisely the prime powers.

C3. Let n be a positive integer. There are $2n$ knights sitting at a round table. They consist of n pairs of partners, each pair of which wishes to shake hands. A pair can shake hands only when next to each other. Every minute, one pair of adjacent knights swaps places.

Find the minimum number of exchanges of adjacent knights such that, regardless of the initial arrangement, every knight can meet her partner and shake hands at some time.

(Belarus)

Answer: The minimum number of exchanges is $\frac{n(n-1)}{2}$.

Common remarks. The solution is divided into three lemmas. We provide multiple proofs of each lemma.

Solution. Join each pair of knights with a chord across the table. We'll refer to these chords as *chains*.

First we show that $n(n-1)/2$ exchanges are required for some arrangements.

Lemma 1. If each knight is initially sitting directly opposite her partner, then at least $n(n-1)/2$ exchanges are required for all knights to meet and shake hands with their partners.

Proof 1. In this arrangement any two chains are initially intersecting. For two knights to be adjacent to each other, it is necessary that their chain does not cross any other chain, and thus every pair of chains must be uncrossed at some time. Each exchange of adjacent knights can only uncross a single pair of intersecting chains, and thus the number of exchanges required is at least the number of pairs of chains, which is $n(n-1)/2$. □

Proof 2. In this arrangement the two knights in each pair are initially separated by $n-1$ seats in either direction around the table, and so each pair must move a total of at least $n-1$ steps so as to be adjacent. There are n pairs, and each exchange moves two knights by a single step. Hence at least $n(n-1)/2$ moves are required. □

We will now prove that $n(n-1)/2$ exchanges is sufficient in all cases. We'll prove a stronger version of this bound than is required, namely that every knight can shake hands with her partner at the end, after all exchanges have finished.

Begin by adding a pillar at the centre of the table. For each chain that passes through the centre of the table, we arbitrarily choose one side of the chain and say that the pillar lies on that side of the chain. While the pillar may lie on a chain, we will never move a knight if that causes the pillar to cross to the other side of a chain. Say that a chain passes in front of a knight if it passes between that knight and the pillar, and define the length of a chain to be the number of knights it passes in front of. Then each chain has a length between 0 and $n-1$ inclusive.

Say that a chain C *encloses* another chain C' if C and C' do not cross, and C passes between C' and the pillar. Say that two chains are *intersecting* if they cross on the table; *enclosing* if one chain encloses the other; and *disjoint* otherwise. Let k, l and m denote respectively the number of enclosing, intersecting and disjoint pairs of chains. Then we have

$$k + l + m = \frac{n(n-1)}{2}.$$

Lemma 2. $2k + l$ exchanges are sufficient to reach a position with all pairs of knights sitting adjacent to each other.

Proof 1. We proceed by induction on $2k + l$.

If every chain has length 0, then every pair of knights is adjacent and the statement is trivial.

Otherwise, let A and B be a pair of knights whose chain C_0 has length $q \geqslant 1$. Let $S_0 = A$, and let $S_1, \ldots, S_q$ be the knights which C_0 passes in front of, sitting in that order from A to B. We know that C_0 passes in front of S_1, and there are three cases for the chain C_1 for knight S_1.

If C_1 passes in front of S_0 then C_0 and C_1 are intersecting, and we can make them disjoint by exchanging the positions of S_0 and S_1. This reduces the sum $2k + l$ by 1.

If C_1 passes in front of neither S_0 nor B then C_1 is enclosed by C_0, and we can swap S_0 and S_1 to make C_0 and C_1 an intersecting pair. This increases l by 1 and decreases k by 1, and hence reduces the sum $2k + l$ by 1.

If this C_1 passes in front of B then we cannot immediately find a beneficial exchange.

In the third case, we look instead at the knights S_i and S_{i+1}, for each i in turn. Each time, we will either find a beneficial exchange, or find that the chain C_{i+1} for knight S_{i+1} passes in front of B. Eventually we will either find a beneficial exchange in one of the first two cases above, or we will find that the chain C_q for S_q passes in front of B, in which case C_q and C_0 are intersecting and we can make C_q and C_0 disjoint by swapping S_q and B.

Also note that the only times a chain is increased in length is when it is enclosed by another chain. But this cannot happen for a chain containing the pillar, so no chains ever cross the pillar. □

Proof 2. We begin by ignoring the seats, and let each knight walk freely to a predetermined destination. Each pair of knights will walk around the table to one of the two points on the circumference midway between their initial locations, such that the chain between them passes between the pillar and the destination. If more than one pair of knights would have the same destination point, then we make small adjustments to the destination points so that each pair has a distinct destination point.

We then imagine each knight walking at a constant speed (which may be different for each knight). They all start and stop walking at the same time. We want to count how many times two knights pass (either in opposite directions, or in the same direction but at different speeds). For any two pairs of knights, the number of passes depends on the relation between their two chains.

If their two chains are intersecting then there will be one pass, involving the two knights for whom the other chain passes between them and the pillar.

If their two chains are enclosing then there will be two passes, with one of the knights with the enclosing chain passing both of the knights with the shorter enclosed chain.

If their two chains are disjoint then there will be no passes.

The number of passes is therefore $2k + l$. If multiple pairs of knights would pass at the same time, we can slightly adjust the walking speeds so that the passes happen at distinct times. We can then convert this sequence of passes into a sequence of seat exchanges in the original problem, which shows that $2k + l$ exchanges is sufficient. □

Lemma 3. $k \leqslant m$.

Proof 1. We proceed by induction on n. The base case $n = 2$ is clear.

Consider a chain C of greatest length, and suppose it joins knights A and B. Let x be the number of chains that intersect C, and let y be the number of chains that are enclosed by C. Note that no chain can enclose C. Then C passes in front of one knight from each pair whose chain intersects C, and both knights in any pair whose chain is enclosed by C. Thus the length of C is $x + 2y \leqslant n - 1$. The number of chains that form a disjoint pair with C is then

$$n - 1 - x - y \geqslant (x + 2y) - x - y = y.$$

Now we can remove A and B and use the induction hypothesis. We need to show that the length of each remaining chain is at most $n - 2$ so the chains remain valid. No chain increases in length after removing A and B. If any chain C had length $n - 1$, then the chain between A and B also had length $n - 1$. Then C must have passed in front of exactly one of A or B, and so has length $n - 2$ after removing A and B. □

Proof 2. Let k_C denote the number of chains C' such that C encloses C'.

Note that if C encloses C', then $k_{C'} < k_C$.

First we will show that there at least k_C chains that are disjoint from C. Let x be the length of C, let $\mathcal{S}$ be the set of x knights that C passes in front of, and let $\mathcal{T}$ be the set of x knights sitting directly opposite them. None of the knights in $\mathcal{T}$ can have a chain that encloses or is enclosed by C, and if any knight in $\mathcal{T}$ has a chain that intersects C, then her partner must be a knight in $\mathcal{S}$. So we have that

$$\begin{aligned} 2k_C &= \text{number of knights in } \mathcal{S} \text{ whose chain is enclosed by } C \\ &= x - \text{number of knights in } \mathcal{S} \text{ whose chain intersects } C \\ &\leqslant x - \text{number of knights in } \mathcal{T} \text{ whose chain intersects } C \\ &\leqslant \text{number of knights in } \mathcal{T} \text{ whose chain is disjoint from } C \\ &\leqslant 2 \times \text{number of chains that are disjoint from } C. \end{aligned}$$

Now let m_C denote the number of chains C' with C and C' disjoint, and $k_{C'} < k_C$. We will show that $m_C \geqslant k_C$.

Let $\mathcal{R}$ be a set of k_C chains that are disjoint from C, such that $\sum_{C' \in \mathcal{R}} k_{C'}$ is minimal. If every chain $C' \in \mathcal{R}$ has $k_{C'} < k_C$, then we are done. Otherwise, let consider a chain C' with $k_{C'} \geqslant k_C$. There are then at least k_C chains C'' for which the chain C' passes between C'' and the pillar. Each of these chains must have $k_{C''} < k_{C'}$, and at least one of them is not in $\mathcal{R}$ (otherwise $\mathcal{R}$ would contain C' and at least k_C other chains), so we can swap this chain with C' to obtain a set $\mathcal{R}'$ with $\sum_{C' \in \mathcal{R}'} k_{C'} < \sum_{C' \in \mathcal{R}} k_{C'}$. But this contradicts the minimality of $\mathcal{R}$.

We finish by summing these inequalities over all chains C:

$$k = \sum_C k_C \leqslant \sum_C m_C \leqslant m.$$ □

By Lemma 3, we have that $2k + l \leqslant k + l + m = n(n-1)/2$. Combining this with Lemma 2, we have that $n(n-1)/2$ exchanges is enough to reach an arrangement where every knight is sitting next to her partner, as desired.

Comment. Either proof of Lemma 3 can be adapted to show that the configuration in Lemma 1 is the only one which achieves the bound.

C4. On a board with 2024 rows and 2023 columns, Turbo the snail tries to move from the first row to the last row. On each attempt, he chooses to start on any cell in the first row, then moves one step at a time to an adjacent cell sharing a common side. He wins if he reaches any cell in the last row. However, there are 2022 predetermined, hidden monsters in 2022 of the cells, one in each row except the first and last rows, such that no two monsters share the same column. If Turbo unfortunately reaches a cell with a monster, his attempt ends and he is transported back to the first row to start a new attempt. The monsters do not move.

Suppose Turbo is allowed to take n attempts. Determine the minimum value of n for which he has a strategy that guarantees reaching the last row, regardless of the locations of the monsters.

(Hong Kong)

Comment. One of the main difficulties of solving this question is in determining the correct expression for n. Students may spend a long time attempting to prove bounds for the wrong value for n before finding better strategies.

Students may incorrectly assume that Turbo is not allowed to backtrack to squares he has already visited within a single attempt. Fortunately, making this assumption does not change the answer to the problem, though it may make it slightly harder to find a winning strategy.

Answer: The answer is $n = 3$.

Solution. First we demonstrate that there is no winning strategy if Turbo has 2 attempts.

Suppose that $(2, i)$ is the first cell in the second row that Turbo reaches on his first attempt. There can be a monster in this cell, in which case Turbo must return to the first row immediately, and he cannot have reached any other cells past the first row.

Next, suppose that $(3, j)$ is the first cell in the third row that Turbo reaches on his second attempt. Turbo must have moved to this cell from $(2, j)$, so we know $j \neq i$. So it is possible that there is a monster on $(3, j)$, in which case Turbo also fails on his second attempt. Therefore Turbo cannot guarantee to reach the last row in 2 attempts.

Next, we exhibit a strategy for $n = 3$. On the first attempt, Turbo travels along the path

$$(1, 1) \to (2, 1) \to (2, 2) \to \cdots \to (2, 2023).$$

This path meets every cell in the second row, so Turbo will find the monster in row 2 and his attempt will end.

If the monster in the second row is not on the edge of the board (that is, it is in cell $(2, i)$ with $2 \leqslant i \leqslant 2022$), then Turbo takes the following two paths in his second and third attempts:

$$(1, i-1) \to (2, i-1) \to (3, i-1) \to (3, i) \to (4, i) \to \cdots \to (2024, i).$$
$$(1, i+1) \to (2, i+1) \to (3, i+1) \to (3, i) \to (4, i) \to \cdots \to (2024, i).$$

The only cells that may contain monsters in either of these paths are $(3, i-1)$ and $(3, i+1)$. At most one of these can contain a monster, so at least one of the two paths will be successful.

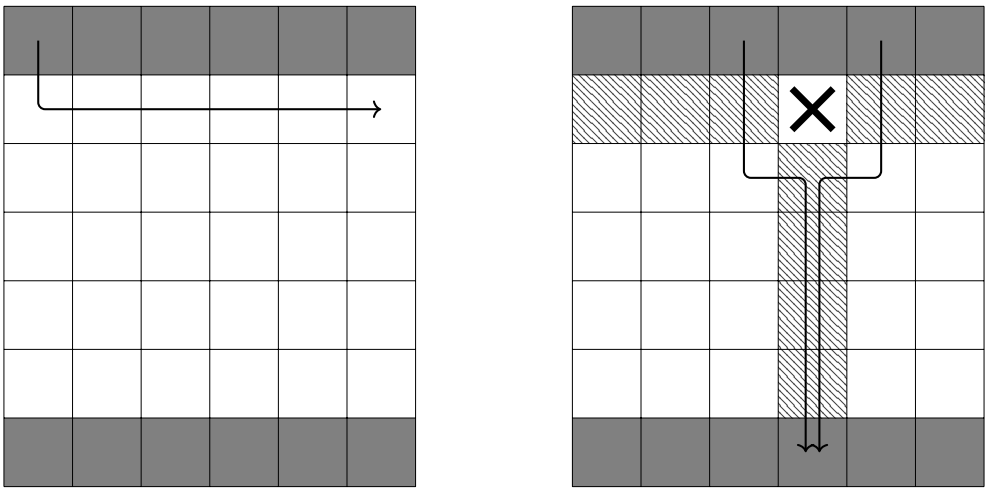


Figure 1: Turbo's first attempt, and his second and third attempts in the case where the monster on the second row is not on the edge. The cross indicates the location of a monster, and the shaded cells are cells guaranteed to not contain a monster.

If the monster in the second row is on the edge of the board, without loss of generality we may assume it is in $(2, 1)$. Then, on the second attempt, Turbo takes the following path:

$$(1, 2) \to (2, 2) \to (2, 3) \to (3, 3) \to \cdots \to (2022, 2023) \to (2023, 2023) \to (2024, 2023).$$

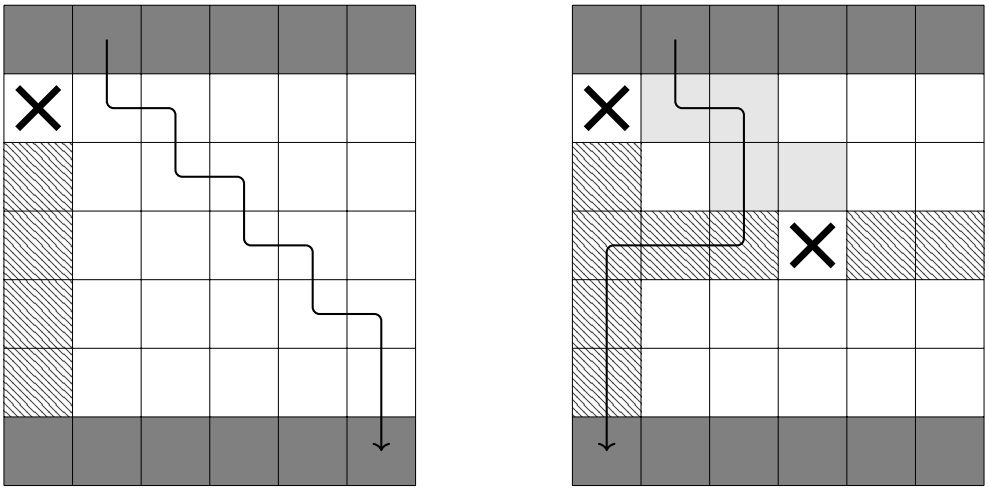


Figure 2: Turbo's second and third attempts in the case where the monster on the second row is on the edge. The light gray cells on the right diagram indicate cells that were visited on the previous attempt. Note that not all safe cells have been shaded.

If there are no monsters on this path, then Turbo wins. Otherwise, let (i, j) be the first cell on which Turbo encounters a monster. We have that $j = i$ or $j = i + 1$. Then, on the third attempt, Turbo takes the following path:

$$\begin{aligned}(1, 2) &\to (2, 2) \to (2, 3) \to (3, 3) \to \cdots \to (i - 2, i - 1) \to (i - 1, i - 1) \\ &\to (i, i - 1) \to (i, i - 2) \to \cdots \to (i, 2) \to (i, 1) \\ &\to (i + 1, 1) \to \cdots \to (2023, 1) \to (2024, 1).\end{aligned}$$

Now note that

- The cells from $(1, 2)$ to $(i - 1, i - 1)$ do not contain monsters because they were reached earlier than (i, j) on the previous attempt.
- The cells (i, k) for $1 \leqslant k \leqslant i - 1$ do not contain monsters because there is only one monster in row i, and it lies in (i, i) or $(i, i + 1)$.
- The cells $(k, 1)$ for $i \leqslant k \leqslant 2024$ do not contain monsters because there is at most one monster in column 1, and it lies in $(2, 1)$.

Therefore Turbo will win on the third attempt.

Comment. A small variation on Turbo's strategy when the monster on the second row is on the edge is possible. On the second attempt, Turbo can instead take the path

$$\begin{aligned}(1, 2023) &\to (2, 2023) \to (2, 2022) \to \cdots \to (2, 3) \to (2, 2) \to (2, 3) \to \cdots \to (2, 2023) \\ &\to (3, 2023) \to (3, 2022) \to \cdots \to (3, 4) \to (3, 3) \to (3, 4) \to \cdots \to (3, 2023) \\ &\to \cdots \\ &\to (2022, 2023) \to (2022, 2022) \to (2022, 2023) \\ &\to (2023, 2023) \\ &\to (2024, 2023).\end{aligned}$$

If there is a monster on this path, say in cell (i, j), then on the third attempt Turbo can travel straight down to the cell just left of the monster instead of following the path traced out in the second attempt.

$$\begin{aligned}(1, j - 1) &\to (2, j - 1) \to \cdots \to (i - 1, j - 1) \to (i, j - 1) \\ &\to (i, j - 2) \to \cdots \to (i, 2) \to (i, 1) \\ &\to (i + 1, 1) \to \cdots \to (2023, 1) \to (2024, 1).\end{aligned}$$

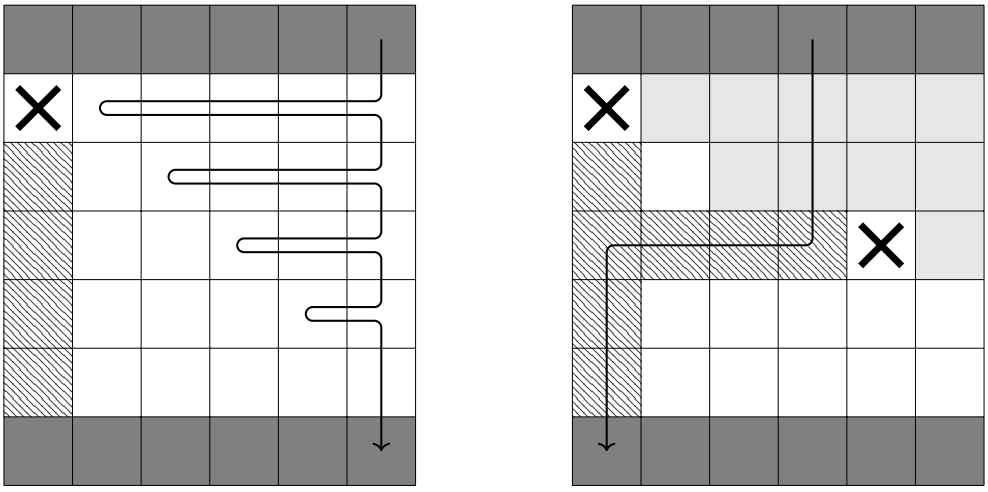


Figure 3: Alternative strategy for Turbo's second and third attempts.

C5. Let N be a positive integer. Geoff and Ceri play a game in which they start by writing the numbers 1, 2, ..., N on a board. They then take turns to make a move, starting with Geoff. Each move consists of choosing a pair of integers (k, n), where $k \geqslant 0$ and n is one of the integers on the board, and then erasing every integer s on the board such that $2^k \mid n - s$. The game continues until the board is empty. The player who erases the last integer on the board loses.

Determine all values of N for which Geoff can ensure that he wins, no matter how Ceri plays.

(Indonesia)

Answer: The answer is that Geoff wins when N is of the form 2^n for n odd or of the form $t2^n$ for n even and $t > 1$ odd.

Common remarks. We will say that a set $\mathcal{S}$ wins if the current player wins given $\mathcal{S}$ as the current set of integers on the board. Otherwise, we will say that $\mathcal{S}$ loses.

We will let $J(\mathcal{S}, \mathcal{T}) = (2\mathcal{S}-1) \cup (2\mathcal{T})$. Note that every subset of $\mathbb{Z}$ can be written as $J(\mathcal{S}, \mathcal{T})$ for some unique pair $(\mathcal{S}, \mathcal{T})$ of subsets of $\mathbb{Z}$.

We will let $[n]$ denote the set $\{1, 2, \ldots, n\}$.

Solution.

Lemma 1. For any set $\mathcal{S}$, $\mathcal{S}$ wins if and only if $J(\mathcal{S}, \varnothing)$ wins. Similarly, $\mathcal{S}$ wins if and only if $J(\varnothing, \mathcal{S})$ wins.

Proof. Let (k, m) be a move on $\mathcal{S}$, and let $\mathcal{T}$ be the result of applying the move. Then we can reduce $J(\mathcal{S}, \varnothing)$ to $J(\mathcal{T}, \varnothing)$ by applying the move $(k + 1, 2m - 1)$.

Conversely, let (k, m) be a move on $J(\mathcal{S}, \varnothing)$. We can express the result of this move as $J(\mathcal{T}, \varnothing)$ for some $\mathcal{T}$. Then we can reduce $\mathcal{S}$ to $\mathcal{T}$ by applying the move $(\max(k-1, 0), (k+1)/2)$.

This gives us a natural bijection between games starting with $\mathcal{S}$ and games starting with $J(\mathcal{S}, \varnothing)$ and thus proves the first part of the lemma. The second part follows by a similar argument. □

Lemma 2. If $\mathcal{S}$ and $\mathcal{T}$ are nonempty and at least one of them loses, then $J(\mathcal{S}, \mathcal{T})$ wins.

Proof. If $\mathcal{S}$ is losing, then we can delete $J(\varnothing, \mathcal{T})$ using the move $(1, t)$ for some $t \in J(\varnothing, \mathcal{T})$, which leaves the losing set $J(\mathcal{S}, \varnothing)$. Similarly, if $\mathcal{T}$ is losing, then we can delete $J(\mathcal{S}, \varnothing)$ using the move $(1, s)$ for some $s \in J(\mathcal{S}, \varnothing)$, leaving the losing set $J(\varnothing, \mathcal{T})$. □

Lemma 3. If $\mathcal{S}$ is nonempty and wins, then $J(\mathcal{S}, \mathcal{S})$ loses.

Proof. From this position, we can convert any sequence of moves into another valid sequence of moves by replacing $(k, 2n-1)$ with $(k, 2n)$, and vice versa. Thus we may assume that the initial move (k, m) has m odd. We want to show that any such move results in a winning position for the other player.

The move $(0, m)$ loses immediately. Otherwise, the move results in the set $J(\mathcal{T}, \mathcal{S})$ for some set $\mathcal{T}$. There are three cases.

If $\mathcal{T}$ is empty then the other player gets the winning set $J(\varnothing, \mathcal{S})$.

If $\mathcal{T}$ is losing then the other player can choose the move $(1, s)$ for some $s \in J(\varnothing, \mathcal{S})$, which leaves the losing set $J(\mathcal{T}, \varnothing)$.

If $\mathcal{T}$ is nonempty winning then the other player can choose the move $(k, m + 1)$, which results in the position $J(\mathcal{T}, \mathcal{T})$. We can then proceed by induction on $|\mathcal{S}|$ to show that this is a losing set. □

Lemma 4. $[2n]$ wins if and only if $[n]$ loses.

Proof. Note that $[2n] = J([n], [n])$. The result then follows directly from the previous two lemmas. □

Lemma 5. For any integer $n \geqslant 1$, $[2n+1]$ wins.

Proof. By Lemma 4, either $[n]$ or $[2n]$ loses. If $[n]$ loses, then by Lemma 2 we have that $[2n+1] = J([n+1], [n])$ wins. Otherwise, $[2n]$ loses, and therefore $[2n+1]$ wins by choosing the move $(k, 2n+1)$ for sufficiently large k so that only $2n+1$ is eliminated. □

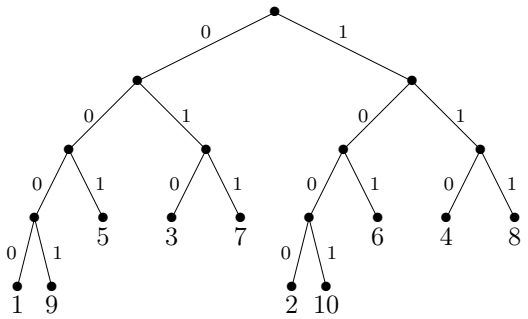
It remains to verify the original answer. We have two cases to consider:

- Suppose $N = 2^n$ for some n. For $N = 1$, every move is an instant loss for Geoff. Then by Lemma 4, Geoff wins for $N = 2^n$ if and only if Geoff loses for $N = 2^{n-1}$, and thus by induction we have that Geoff wins for $N = 2^n$ if and only if n is odd.

- Otherwise, $N = t2^n$, for some n and some $t > 1$ with t odd. By Lemma 5, Geoff wins when $n = 0$. Then by Lemma 4, Geoff wins for $N = t2^n$ if and only if Geoff loses for $N = t2^{n-1}$, and thus by induction on n we have that Geoff wins for $N = t2^n$ if and only if n is even.

Comment. We can represent this game as a game on partial binary trees. This representation could be common in rough working, as it facilitates exploration of small cases. If two sets produce trees which are topologically equivalent, then this equivalence leads to a natural bijection between games starting with the two sets. Such equivalences lead to a significant reduction in the number of distinct cases that need to be considered when exploring the game for small N.

The construction is as follows. First we begin by considering an infinite binary tree. For each positive integer n, we consider the binary representation of $n-1$, starting with the least significant bit and ending with an infinite sequence of leading zeroes. We map this sequence of bits to a path on the binary tree by starting at the root, and then repeatedly choosing the left child if the bit is 0 and the right child if the bit is 1. We can then truncate each path after reaching a sufficient depth to distinguish the path from all other paths in the tree.



Valid moves in this representation of the game consist of selecting a node with two children, and removing either the left child or the right child (and its descendants). Selecting and removing the entire graph is also an allowed move (which loses instantly).

Two trees have equivalent games if they're topologically identical. This equivalence includes swapping the left and right children of any single node, or removing a node with a single child by merging the edges above and below it (and decreasing the depth of its children by one).

Comment. We can also analyse this game using Grundy values (also known as nim-values or nimbers). This requires a slight modification to the rules, wherein any move that would erase all integers on the board is disallowed, and the first player who cannot move loses. This is clearly equivalent to the original game.

Let $g(\mathcal{S})$ denote the Grundy value of the game starting with the set $\mathcal{S}$. Note that the bijection in Lemma 1 shows that

$$g(\mathcal{S}) = g(J(\mathcal{S}, \varnothing)) = g(J(\varnothing, \mathcal{S})).$$

For any set V, let $\operatorname{mex}(V)$ denote the least nonnegative element that is not an element of V. For nonnegative integers x and y, define $j(x, y)$ recursively as

$$j(x, y) = \operatorname{mex}(\{x, y\} \cup \{j(w, y) \mid w < x\} \cup \{j(x, z) \mid z < y\}).$$

The values of $j(x, y)$ for small x and y are:

5	6	7	8	9	1	0
4	5	3	6	2	0	1
3	4	5	1	0	2	9
2	3	4	0	1	6	8
1	2	0	4	5	3	7
0	1	2	3	4	5	6
—	0	1	2	3	4	5

We can show that $g(J(\mathcal{S}, \mathcal{T})) = j(g(\mathcal{S}), g(\mathcal{T}))$ for any nonempty sets $\mathcal{S}$ and $\mathcal{T}$. The remainder of the proof follows a similar structure to the given solution.

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C6. Let n and T be positive integers. James has $4n$ marbles with weights 1, 2, ..., $4n$. He places them on a balance scale, so that both sides have equal weight. Andrew may move a marble from one side of the scale to the other, so that the absolute difference in weights of the two sides remains at most T.

Find, in terms of n, the minimum positive integer T such that Andrew may make a sequence of moves such that each marble ends up on the opposite side of the scale, regardless of how James initially placed the marbles.

(Ghana)

Answer: The minimum value of T is $4n$.

Solution 1. We must have $T \geqslant 4n$, as otherwise we can never move the marble of weight $4n$. We will show that $T = 4n$ by showing that, for any initial configuration, there is a sequence of moves, never increasing the absolute value of the difference above $4n$, that results in every marble ending up on the opposite side of the scale. Because moves are reversible, it suffices to do the following: exhibit at least one configuration C for which this can be achieved, and show that any initial configuration can reach such a configuration C by some sequence of moves.

Consider partitioning the weights into pairs $(t, 4n+1-t)$. Suppose that each side of the balance contains n of those pairs. If one side of the balance contains the pair $(t, 4n+1-t)$ for $1 \leqslant t < 2n$ and the other side contains $(2n, 2n+1)$, then the following sequence of moves swaps those pairs between the sides without ever increasing the absolute value of the difference above $4n$.

$$t, 4n+1-t \mid 2n, 2n+1 \tag{1}$$
$$t, 2n, 4n+1-t \mid 2n+1 \tag{2}$$
$$t, 2n \mid 2n+1, 4n+1-t \tag{3}$$
$$t, 2n, 2n+1 \mid 4n+1-t \tag{4}$$
$$2n, 2n+1 \mid t, 4n+1-t \tag{5}$$

Applying this sequence twice swaps any two pairs $(t, 4n+1-t)$ and $(t', 4n+1-t')$ between the sides. So we can achieve an arbitrary exchange of pairs between the sides, and C can be any configuration where each side of the balance contains n of those pairs.

We now show that any initial configuration can reach one where each side has n of those pairs. Consider a configuration where one side has total weight $A - s$ and the other has total weight $A + s$, for some $0 \leqslant s \leqslant 2n$, and where some pair is split between the two sides. (If no pair is split between the two sides, they must have equal weights and we are done.) Valid moves include moving any weight w with $1 \leqslant w \leqslant 2n + s$ from the $A + s$ side to the $A - s$ side, and moving any weight w with $1 \leqslant w \leqslant 2n - s$ from the $A - s$ side to the $A + s$ side. Suppose the pair $(t, 4n+1-t)$, with $t \leqslant 2n$, is split between the sides. If t is on the $A + s$ side, or on the $A - s$ side and $t \leqslant 2n - s$, it can be moved to the other side. Otherwise, t is on the $A - s$ side and $t \geqslant 2n - s + 1$, so $4n + 1 - t \leqslant 2n + s$ is on the $A + s$ side and can be moved to the other side. So we can unite the two weights from that pair without splitting any other pair, and repeating this we reach a configuration where no pair is split between the sides.

Solution 2. As in Solution 1, $T \geqslant 4n$. Let δ be the weight of the left side minus the weight of the right side. A configuration is called *legal* if $|\delta| \leqslant 4n$, and a move is *legal* if it results in a legal configuration. We will show that if $\delta = 0$ then there is a sequence of legal moves after which every marble is on the opposite side.

We treat the $n = 1$ case separately. The initial configuration has marbles 1, 4 on one side and 2, 3 on the other. So moving marbles 2, 4, 3, 1 in that order is legal and every marble ends on the opposite side. Now assume $n \geqslant 2$.

Marbles of weight at most $2n$ are called *small*. We will make use of the following lemmas:

Lemma 1. If a pair of legal configurations differ only in the locations of small marbles then there is a sequence of legal moves to get from one to the other.

Proof. At first we only move marbles in the wrong position if they are not on the lighter side. (In the case of a tie, neither side is lighter.) Such a move is always legal. Since this reduces the number of marbles in the wrong position, eventually it will no longer be possible to perform such a move.

Then the only marbles in the wrong position are on the lighter side. So moving one marble in the wrong position at a time will always increase $|\delta|$, and $|\delta| \leqslant 4n$ at the end. Hence every move is legal. □

Lemma 2. Let $k \in \mathbb{N}$. A positive integer can be expressed as a sum of distinct positive integers up to k if and only if it is at most $k(k+1)/2$.

Proof. The maximum possible sum of distinct positive integers up to k is $k(k+1)/2$. For the other direction we use induction on k. The case $k = 1$ is trivial. Assume the statement is true for $k-1$. For positive integers up to k we only need a single term. For larger integers, including k in the expression means we are done by the inductive hypothesis. □

Also note that $n(2n+1) \geqslant 4n$ for $n \geqslant 2$.

Let $2n < m \leqslant 4n$. Marbles of weight greater than m are called *big* and marbles from $2n+1$ to m are called *medium*.

Suppose all big marbles are on the correct side (that is, opposite where they started), m is on the incorrect side and the configuration is legal. Then the following steps give a sequence of legal moves after which m is on the correct side and the big marbles were never moved.

Assume m is on the left. In Step 2, we rearrange the small marbles so we can move m. But this is only possible if the weight of big and medium marbles on the right is not too large. So we may need to move some medium marbles from the right first, which we do in Step 1.

Step 1 Skip to Step 2 if the total weight of medium and big marbles on the right side is at most $n(4n+1) + 2n - m$. Since the big marbles are in the correct position and m is in the incorrect position, the big marbles on the right can weigh at most $n(4n+1) - m$. So there must be a medium marble $m' < m$ on the right.

From the first assumption, it is legal to move all small marbles to the left. Then by Lemma 2 we can move some of the small marbles to the right so the right side has weight exactly $n(4n+1) + 2n$. Then moving m' is legal. Repeat this step. Since the total weight of medium marbles on the right decreases, this step will occur a bounded number of times.

Step 2 Let the total weight of the right side be $n(4n+1) + 2n - m + x$ and the weight of small marbles on the right side be y. Note that $y \geqslant x$. If $x \leqslant 0$ then moving m is legal.

Otherwise, by Lemma 2 there is a set of small marbles of weight $y - x$. By Lemma 1, there is a sequence of legal moves of small marbles such that the right side has weight exactly $n(4n+1) + 2n - m$. Now moving m is legal.

Applying the process above for $m = 4n, 4n-1, \ldots, 2n+1$ will move all nonsmall marbles to the opposite side. Then Lemma 1 completes the proof.

C7. Let N be a positive integer and let $a_1, a_2, \ldots$ be an infinite sequence of positive integers. Suppose that, for each $n > N$, a_n is equal to the number of times a_{n-1} appears in the list $a_1, a_2, \ldots, a_{n-1}$.

Prove that at least one of the sequences $a_1, a_3, a_5, \ldots$ and $a_2, a_4, a_6, \ldots$ is eventually periodic.

(Australia)

Solution 1. Let $M > \max(a_1, \ldots, a_N)$. We first prove that some integer appears infinitely many times. If not, then the sequence contains arbitrarily large integers. The first time each integer larger than M appears, it is followed by a 1. So 1 appears infinitely many times, which is a contradiction.

Now we prove that every integer $x \geqslant M$ appears at most $M-1$ times. If not, consider the first time that any $x \geqslant M$ appears for the Mth time. Up to this point, each appearance of x is preceded by an integer which has appeared $x \geqslant M$ times. So there must have been at least M numbers that have already appeared at least M times before x does, which is a contradiction.

Thus there are only finitely many numbers that appear infinitely many times. Let the largest of these be k. Since k appears infinitely many times there must be infinitely many integers greater than M which appear at least k times in the sequence, so each integer $1, 2, \ldots, k-1$ also appears infinitely many times. Since $k+1$ doesn't appear infinitely often there must only be finitely many numbers which appear more than k times. Let the largest such number be $l \geqslant k$. From here on we call an integer x *big* if $x > l$, *medium* if $l \geqslant x > k$ and *small* if $x \leqslant k$. To summarise, each small number appears infinitely many times in the sequence, while each big number appears at most k times in the sequence.

Choose a large enough $N' > N$ such that $a_{N'}$ is small, and in $a_1, \ldots, a_{N'}$:

- every medium number has already made all of its appearances;
- every small number has made more than $\max(k, N)$ appearances.

Since every small number has appeared more than k times, past this point each small number must be followed by a big number. Also, by definition each big number appears at most k times, so it must be followed by a small number. Hence the sequence alternates between big and small numbers after $a_{N'}$.

Lemma 1. Let g be a big number that appears after $a_{N'}$. If g is followed by the small number h, then h equals the amount of small numbers which have appeared at least g times before that point.

Proof. By the definition of N', the small number immediately preceding g has appeared more than $\max(k, N)$ times, so $g > \max(k, N)$. And since $g > N$, the gth appearance of every small number must occur after a_N and hence is followed by g. Since there are k small numbers and g appears at most k times, g must appear exactly k times, always following a small number after a_N. Hence on the hth appearance of g, exactly h small numbers have appeared at least g times before that point. □

Denote by $a_{[i,j]}$ the subsequence $a_i, a_{i+1}, \ldots, a_j$.

Lemma 2. Suppose that i and j satisfy the following conditions:

(a) $j > i > N' + 2$,

(b) a_i is small and $a_i = a_j$,

(c) no small value appears more than once in $a_{[i,j-1]}$.

Then a_{i-2} is equal to some small number in $a_{[i,j-1]}$.

Proof. Let $\mathcal{I}$ be the set of small numbers that appear at least a_{i-1} times in $a_{[1,i-1]}$. By Lemma 1, $a_i = |\mathcal{I}|$. Similarly, let $\mathcal{J}$ be the set of small numbers that appear at least a_{j-1} times in $a_{[1,j-1]}$. Then by Lemma 1, $a_j = |\mathcal{J}|$ and hence by (b), $|\mathcal{I}| = |\mathcal{J}|$. Also by definition, $a_{i-2} \in \mathcal{I}$ and $a_{j-2} \in \mathcal{J}$.

Suppose the small number a_{j-2} is not in $\mathcal{I}$. This means a_{j-2} has appeared less than a_{i-1} times in $a_{[1,i-1]}$. By (c), a_{j-2} has appeared at most a_{i-1} times in $a_{[1,j-1]}$, hence $a_{j-1} \leqslant a_{i-1}$. Combining with $a_{[1,i-1]} \subset a_{[1,j-1]}$, this implies $\mathcal{I} \subseteq \mathcal{J}$. But since $a_{j-2} \in \mathcal{J} \setminus \mathcal{I}$, this contradicts $|\mathcal{I}| = |\mathcal{J}|$. So $a_{j-2} \in \mathcal{I}$, which means it has appeared at least a_{i-1} times in $a_{[1,i-1]}$ and one more time in $a_{[i,j-1]}$. Therefore $a_{j-1} > a_{i-1}$.

By (c), any small number appearing at least a_{j-1} times in $a_{[1,j-1]}$ has also appeared $a_{j-1}-1 \geqslant a_{i-1}$ times in $a_{[1,i-1]}$. So $\mathcal{J} \subseteq \mathcal{I}$ and hence $\mathcal{I} = \mathcal{J}$. Therefore, $a_{i-2} \in \mathcal{J}$, so it must appear at least $a_{j-1} - a_{i-1} = 1$ more time in $a_{[i,j-1]}$. □

For each small number a_n with $n > N' + 2$, let p_n be the smallest number such that $a_{n+p_n} = a_i$ is also small for some i with $n \leqslant i < n + p_n$. In other words, $a_{n+p_n} = a_i$ is the first small number to occur twice after a_{n-1}. If $i > n$, Lemma 2 (with $j = n + p_n$) implies that a_{i-2} appears again before a_{n+p_n}, contradicting the minimality of p_n. So $i = n$. Lemma 2 also implies that $p_n \geqslant p_{n-2}$. So p_n, p_{n+2}, p_{n+4}, ... is a nondecreasing sequence bounded above by $2k$ (as there are only k small numbers). Therefore, p_n, p_{n+2}, p_{n+4}, ... is eventually constant and the subsequence of small numbers is eventually periodic with period at most k.

Note. Since every small number appears infinitely often, Solution 1 additionally proves that the sequence of small numbers has period k. The repeating part of the sequence of small numbers is thus a permutation of the integers from 1 to k. It can be shown that every permutation of the integers from 1 to k is attainable in this way.

Solution 2. We follow Solution 1 until after Lemma 1. For each $n > N'$ we keep track of how many times each of 1, 2, ..., k has appeared in $a_1, \ldots, a_n$. We will record this information in an updating $(k+1)$-tuple

$$(b_1, b_2, \ldots, b_k; j)$$

where each b_i records the number of times i has appeared. The final element j of the $(k+1)$-tuple, also called the *active* element, represents the latest small number that has appeared in $a_1, \ldots, a_n$.

As n increases, the value of $(b_1, b_2, \ldots, b_k; j)$ is updated whenever a_n is small. The $(k+1)$-tuple updates deterministically based on its previous value. In particular, when $a_n = j$ is small, the active element is updated to j and we increment b_j by 1. The next big number is $a_{n+1} = b_j$. By Lemma 1, the next value of the active element, or the next small number a_{n+2}, is given by the number of b terms greater than or equal to the newly updated b_j, or

$$|\{i \mid 1 \leqslant i \leqslant k,\ b_i \geqslant b_j\}|. \tag{1}$$

Each sufficiently large integer which appears $i+1$ times must also appear i times, with both of these appearances occurring after the initial block of N. So there exists a global constant C such that $b_{i+1} - b_i \leqslant C$. Suppose that for some r, $b_{r+1} - b_r$ is unbounded from below. Since the value of $b_{r+1} - b_r$ changes by at most 1 when it is updated, there must be some update where $b_{r+1} - b_r$ decreases and $b_{r+1} - b_r < -(k-1)C$. Combining with the fact that $b_i - b_{i-1} \leqslant C$ for all i, we see that at this particular point, by the triangle inequality

$$\min(b_1, \ldots, b_r) > \max(b_{r+1}, \ldots, b_k). \tag{2}$$

Since $b_{r+1} - b_r$ just decreased, the new active element is r. From this point on, if the new active element is at most r, by (1) and (2), the next element to increase is once again from $b_1, \ldots, b_r$. Thus only $b_1, \ldots, b_r$ will increase from this point onwards, and b_k will no longer increase, contradicting the fact that k must appear infinitely often in the sequence. Therefore $|b_{r+1} - b_r|$ is bounded.

Since $|b_{r+1} - b_r|$ is bounded, it follows that each of $|b_i - b_1|$ is bounded for $i = 1, \ldots, k$. This means that there are only finitely many different states for $(b_1 - b_1, b_2 - b_1, \ldots, b_k - b_1; j)$. Since the next active element is completely determined by the relative sizes of $b_1, b_2, \ldots, b_k$ to each other, and the update of b terms depends on the active element, the active element must be eventually periodic. Therefore the small numbers subsequence, which is either $a_1, a_3, a_5, \ldots$ or $a_2, a_4, a_6, \ldots$, must be eventually periodic.

C8. Let n be a positive integer. Given an $n \times n$ board, the unit cell in the top left corner is initially coloured black, and the other cells are coloured white. We then apply a series of colouring operations to the board. In each operation, we choose a 2×2 square with exactly one cell coloured black and we colour the remaining three cells of that 2×2 square black.

Determine all values of n such that we can colour the whole board black.

(Peru)

Answer: The answer is $n = 2^k$ where k is a nonnegative integer.

Solution 1. We first prove by induction that it is possible the colour the whole board black for $n = 2^k$. The base case of $k = 1$ is trivial. Assume the result holds for $k = m$ and consider the case of $k = m + 1$. Divide the $2^{m+1} \times 2^{m+1}$ board into four $2^m \times 2^m$ sub-boards. Colour the top left $2^m \times 2^m$ sub-board using the inductive hypothesis. Next, colour the centre 2×2 square with a single operation. Finally, each of the remaining $2^m \times 2^m$ sub-board can be completely coloured using the inductive hypothesis, starting from the black square closest to the centre. This concludes the induction.

Now we prove that if such a colouring is possible for n then n must be a power of 2. Suppose it is possible to colour an $n \times n$ board where $n > 1$. Identify the top left corner of the board by $(0, 0)$ and the bottom right corner by (n, n). Whenever an operation takes place in a 2×2 square centred on (i, j), we immediately draw an "X", joining the four cells' centres (see Figure 4). Also, identify this X by (i, j). The first operation implies there's an X at $(1, 1)$. Since the whole board is eventually coloured, every cell centre must be connected to at least one X. The collection of all Xs forms a graph G.

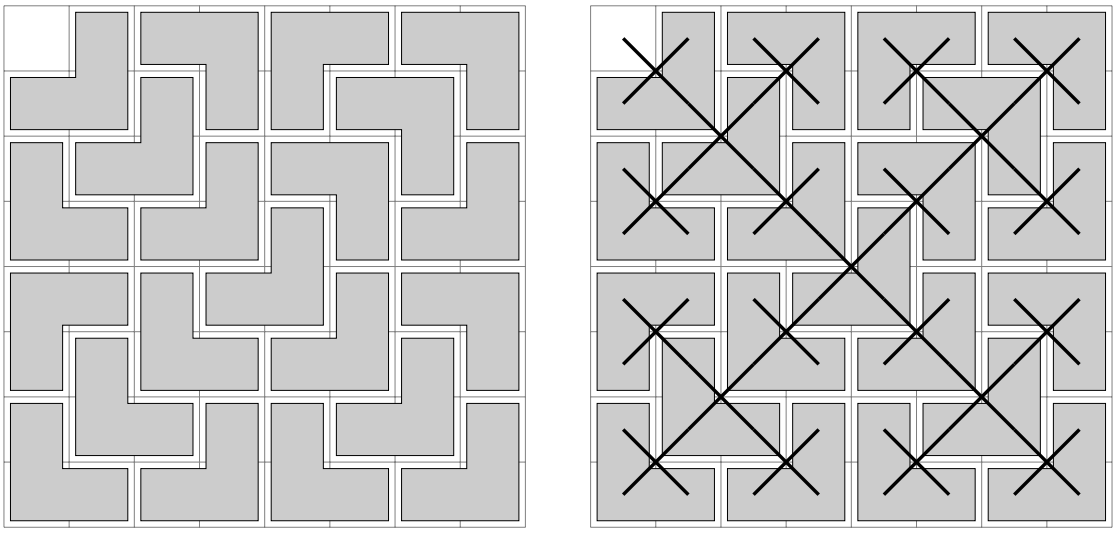


Figure 4: L-trominoes placements corresponding to colouring operations (left) and the corresponding X diagram (right).

Claim 1. The graph G is a tree.

Proof. Since every operation requires a pre-existing black cell, each newly drawn X apart from the first must connect to an existing X. So all Xs are connected to the first X and G must be connected. Now, suppose G has a cycle. Consider the newest X involved in the cycle, it must connect to previous Xs at at least two points. But this implies the corresponding operation will colour at most two cells, which is a contradiction. □

Note that in the following arguments, Claims 2 to 4 only require the condition that G is a tree and every cell is connected to G.

Claim 2. If there's an X at (i, j), then $1 \leqslant i, j \leqslant n-1$ and $i \equiv j \pmod 2$.

Proof. The inequalities $1 \leqslant i, j \leqslant n-1$ are clear. Call an X at (i, j) *good* if $i \equiv j \pmod 2$, or *bad* if $i \not\equiv j \pmod 2$. The first X at $(i, j) = (1, 1)$ is good. Suppose some Xs are bad. Since G is connected, there must exist a good X connecting to a bad X. But this can only occur if they connect at two points, creating a cycle. This is a contradiction, thus all Xs are good. □

Call an X at (i, j) *odd* if $i \equiv j \equiv 1 \pmod 2$, *even* if $i \equiv j \equiv 0 \pmod 2$.

Claim 3. The integer n must be even. Furthermore, there must be $4(n/2-1)$ odd Xs connecting the cells on the perimeter of the board as shown in Figure 5.

Proof. If n is odd, the four corners of the bottom left cell are $(n, 0), (n-1, 0), (n-1, 1)$ and $(n, 1)$, none of which satisfies the conditions of Claim 2. So the bottom left cell cannot connect to any X. If n is even, each cell on the edge of the board has exactly one corner satisfying the conditions of Claim 2, so the X connecting it is uniquely determined. Therefore the cells on the perimeter of the board are connected to Xs according to Figure 5. □

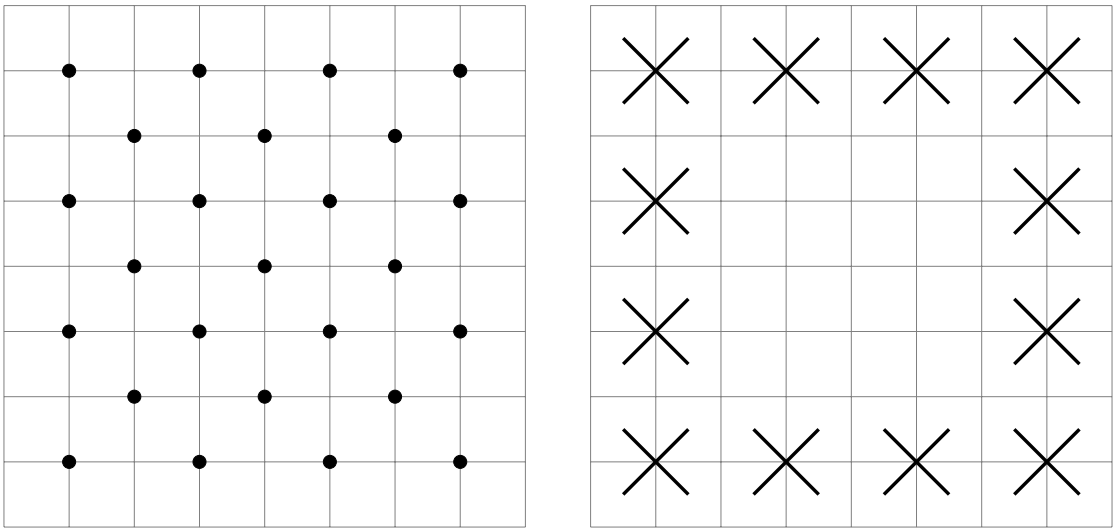


Figure 5: Highlighting the permitted points for Xs (left) and Xs on the perimeter (right).

Divide the $n \times n$ board into $n^2/4$ blocks of 2×2 squares. Call each of these blocks a *big-cell*. We say a big-cell is *filled* if it contains an odd X on its interior, *empty* otherwise. By Claim 3, each big-cell on the perimeter must be filled.

Claim 4. Every big-cell is filled.

Proof. Recall that Xs can only be at (i, j) with $i \equiv j \pmod 2$. Suppose a big-cell centred at (i, j) is empty. Then in order for its four cells to be coloured, there must be four even Xs on $(i-1, j-1)$, $(i+1, j-1)$, $(i-1, j+1)$ and $(i+1, j+1)$, "surrounding" the big-cell (see Figure 6).

By Claim 3, no empty big-cell can be on the perimeter. So if there exist some empty big-cells, the boundary between empty and filled big-cells must consist of a number of closed loops. Each closed loop is made up of several line segments of length 2, each of which separates a filled big-cell from an empty big-cell.

Since every empty big-cell is surrounded by even Xs and every filled big-cell contains an odd X, the two end points of each such line segment must be connected by Xs. Since these line segments form at least one closed loop, it implies the existence of a cycle made up of Xs (see Figure 6). This is a contradiction, thus no big-cell can be empty. □

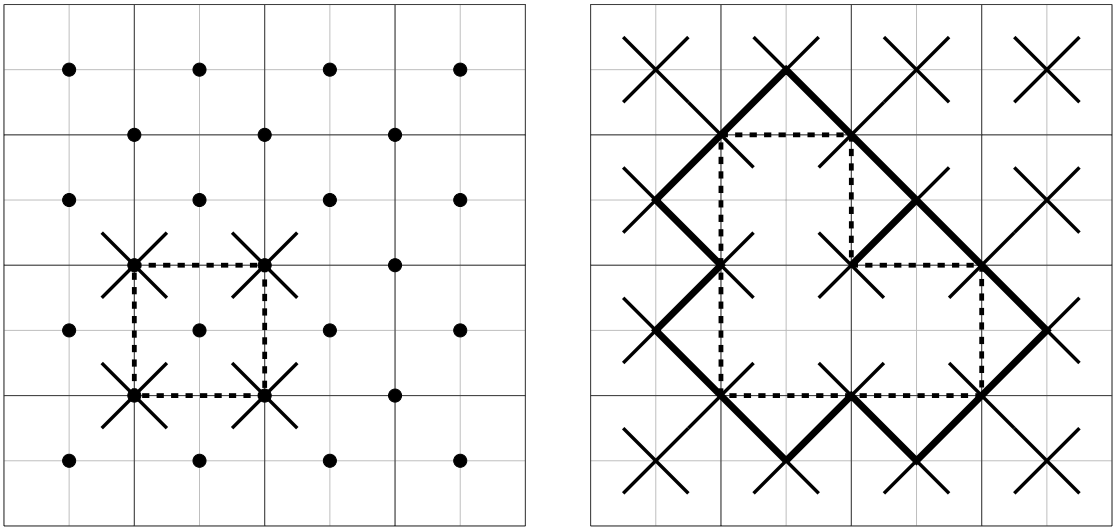


Figure 6: An empty big-cell surrounded by even Xs (left) and the boundary between empty and filled Xs creating a cycle (right).

Therefore every big-cell is filled by an odd X, and the connections between them are provided by even Xs. We can now reduce the $n \times n$ problem to an $n/2 \times n/2$ problem in the following way. Perform a dilation of the board by a factor of $1/2$ with respect to $(0, 0)$. Each big-cell is shrunk to a regular cell. For the Xs, replace each odd X at (i, j) by the point $(i/2, j/2)$, and replace each even X at (i, j) by an X at $(i/2, j/2)$.

We claim the new resulting graph of Xs is a tree that connects all cells of an $n/2 \times n/2$ board. First, two connected Xs in the original $n \times n$ board are still connected after their replacements (noting that some Xs have been replaced by single points). For each cell in the $n/2 \times n/2$ board, its centre corresponds to an odd X from a filled big-cell in the original $n \times n$ board, so it must be connected to the graph. Finally, suppose there exists a cycle in the new graph. The cycle consists of Xs that correspond to even Xs in the original graph connecting big-cells, forming a cycle of big-cells. Since in every big-cell, the four unit squares were connected by an odd X, this implies the existence of a cycle in the original graph, which is a contradiction.

Thus the new graph of Xs must be a tree that connects all cells of an $n/2 \times n/2$ board, which are the required conditions for Claims 2 to 4. Hence we can repeat our argument, halving the dimensions of the board each time, until we reach the base case of a 1×1 board (where the tree is a single point). Therefore n must be a power of 2, completing the solution.

Solution 2. As in Solution 1, it is possible the colour the whole board black for $n = 2^k$.

The colouring operation is equivalent to the placement of L-trominoes. For each L-tromino we place on the board, we draw an arrow and a node as shown in Figure 7. We also draw a node in the top left corner of the board.

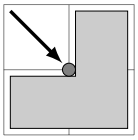


Figure 7: Tromino with corresponding arrow and node drawn.

Claim 1. The arrows and nodes form a directed tree rooted at the top left corner.

Proof. The proof is similar to the proof of Claim 1 in Solution 1, with the additional note that the directions of the arrows inherit the order of the colouring operations, so they must be pointing away from the top left node. □

Note that since all edges of the tree are diagonal, the nodes can only lie on points (i, j) with $i + j \equiv 0 \pmod 2$. This implies that we can only place down L-trominoes of one particular parity: that is, with the centre of the L-tromino on a point with $i + j \equiv 0 \pmod 2$. In the remainder of the proof, we will implicitly use this parity property when determining possible positions of L-trominoes.

Next, we show that certain configurations of edges of the tree are impossible.

Claim 2. There cannot be two edges in a "parallel" configuration (see Figure 8).

Proof. In such a configuration, the two edges can either be directed in the same direction or opposite directions. If they point in the same direction (see Figure 8), then the L-trominoes corresponding to the two edges overlap.

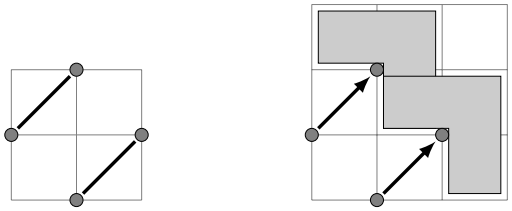


Figure 8: Parallel configuration (left) and two parallel edges, case 1 (right).

If they point in opposite directions, then we get the diagram in Figure 9. The cells marked ($\star$) must lie inside the $n \times n$ board, so they must be covered by L-trominoes. There is only one possible way to cover these with a L-tromino of the right parity. But this makes the arrows form a cycle, which cannot happen. So we have a contradiction. □

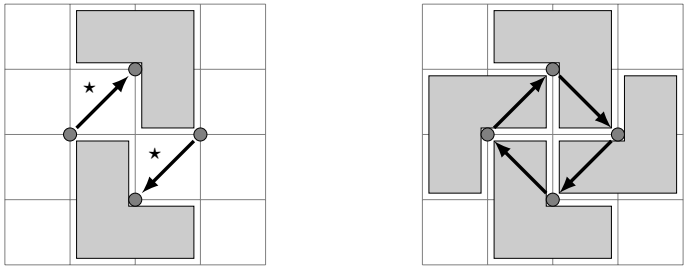


Figure 9: Two parallel edges, case 2.

Claim 3. There cannot be three edges in a "zigzag" configuration, shown in Figure 10.

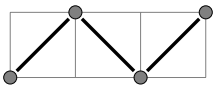


Figure 10: Zigzag configuration.

Proof. Assume for contradiction that there is a zigzag. Then take the zigzag with maximal distance from the root of the tree (measured by distance along the graph from the root to the middle edge of the zigzag).

We may assume without loss of generality that the middle edge is directed down-right. Then the right edge must be directed up-right, since no two arrows can point to the same node. Next, we draw in the corresponding L-trominoes, and consider the cell marked ($\star$). There are two possible ways to cover it with an L-tromino, because of the parity of L-tromino centres.

We could choose the centre of the L-tromino to be the top right corner of the cell (see Figure 11). This immediately gives another zigzag.

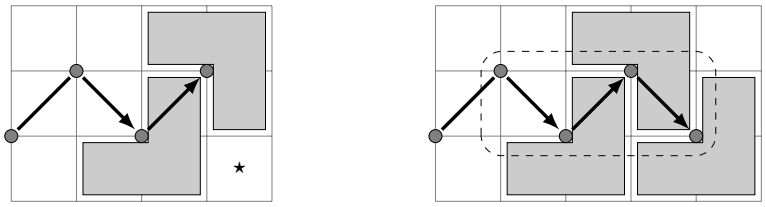


Figure 11: Zigzag configuration, case 1.

The other possibility is if we choose the centre of the L-tromino to be the bottom left corner of the cell (see Figure 12). Then we need to cover the cell marked ($\star\star$) with an L-tromino. If

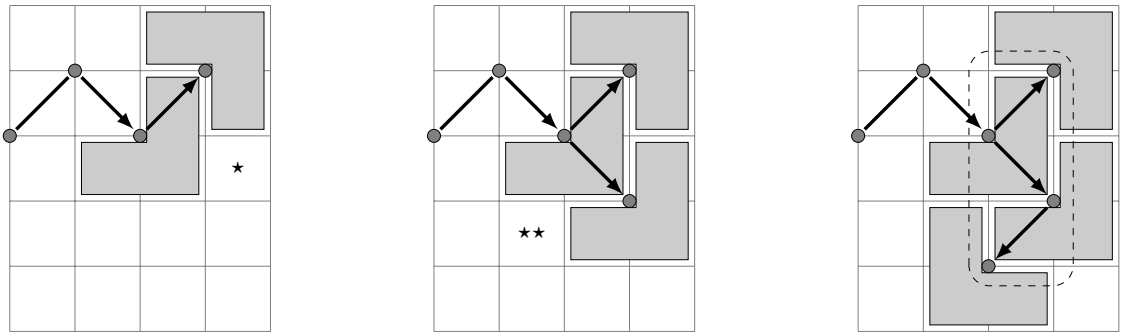


Figure 12: Zigzag configuration, case 2.

we placed the centre of the L-tromino on the top left corner of the cell, this would give two parallel edges, contradicting Claim 2. So we must place the centre of the L-tromino on the bottom right corner of the cell, which gives a zigzag.

In each case, we get another zigzag further away from the root of the tree, which contradicts our assumption of maximality. So there cannot be any zigzags. □

We now colour the nodes of the tree. Colour the root node yellow. For all other nodes, we colour it white if it has an arrow coming out of it in a different direction to the arrow going in, and black otherwise.

Claim 4. Any child of a black node is white.

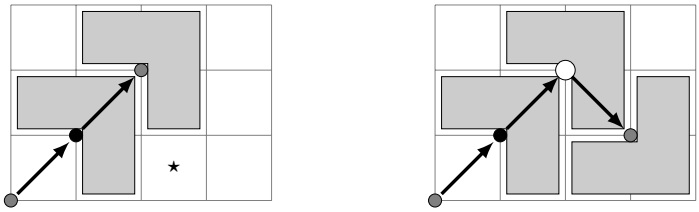


Figure 13: Black node configuration.

Proof. Suppose we have a black node with a child. Then the arrow exiting the black node must be in the same direction as the arrow entering it by the definition of our colouring, giving the left diagram of Figure 13.

The cell marked ($\star$) must be covered by an L-tromino. If the centre of this L-tromino is the bottom left corner, then this would give an arrow leaving the black node in a different direction, which cannot happen. So the centre of the L-tromino must instead be the top right corner, which gives an arrow leaving the upper node in a different direction. Thus the upper node must be white. □

Claim 5. Every white node has three children, all of which are black.

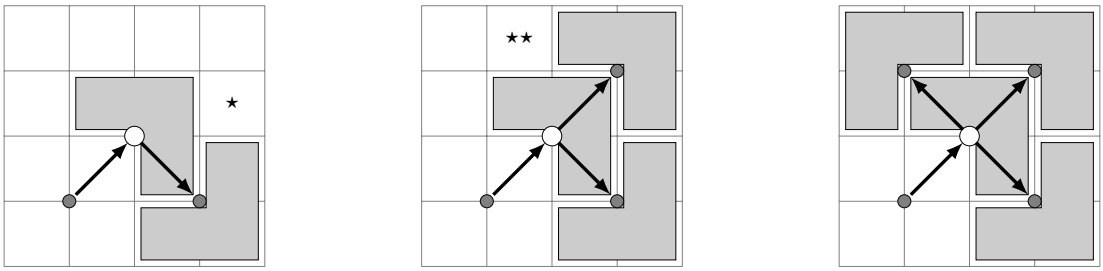


Figure 14: White node configuration.

Proof. Refer to Figure 14. Suppose we have a white node, as in the leftmost diagram. The cell marked ($\star$) must be covered by an L-tromino. If the centre of this L-tromino is the bottom right corner of the cell, then this would form a zigzag, which by Claim 3 is not allowed. So the centre must be the top left corner.

Next, the cell marked ($\star\star$) must be covered by an L-tromino. If the centre of this L-tromino is the top right corner, this would form a zigzag, so the centre must be the bottom left corner instead. Thus we have shown that any white node has three children.

Finally, note that if any of the child nodes had three children of their own, then this would give parallel edges in the diagram, which contradicts Claim 2. Therefore the child nodes of the white node must all be black. □

We now know that the node colours alternate between black and white as you go down the tree, so all white nodes lie on points with coordinates $(2i, 2j)$, and all black nodes lie on points with coordinates $(2i + 1, 2j + 1)$.

Now (assuming $n > 1$) we will construct a new board whose cells are 2×2 squares of our current board. We replace the root node and its child with a single big cell and a big root node,

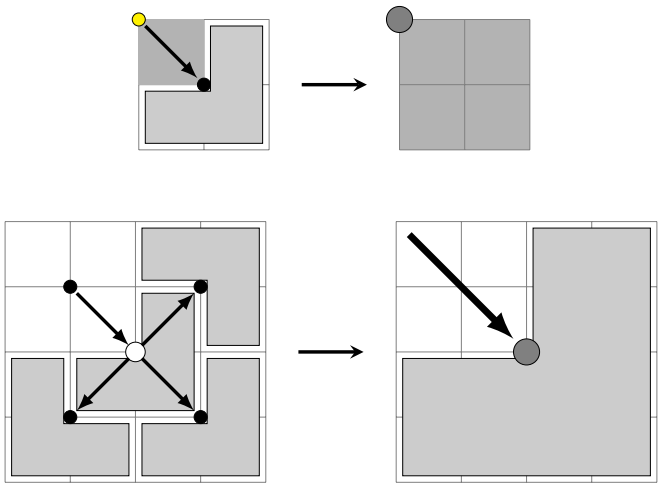


Figure 15: Replacing with larger cells and L-trominoes.

and we replace each white node and its three children with a big L-tromino, big arrow and big node as shown in Figure 15.

Every black node is the child of the root node or a white node, so every L-tromino is involved in exactly one replacement. Also, the parent of any white node is a black node, whose parent, in turn, is a white node or the root. So the starting point of every big arrow will be on a big node. Therefore we obtain an L-tromino tiling forming a tree.

This shows for $n > 1$ that if an $n \times n$ board can be tiled by L-trominoes forming a tree, then n is even, and an $n/2 \times n/2$ board can also be tiled by L-trominoes forming a tree. Since a 1×1 board can trivially be tiled, we conclude that the only values of n for which an $n \times n$ board can be tiled are $n = 2^k$.

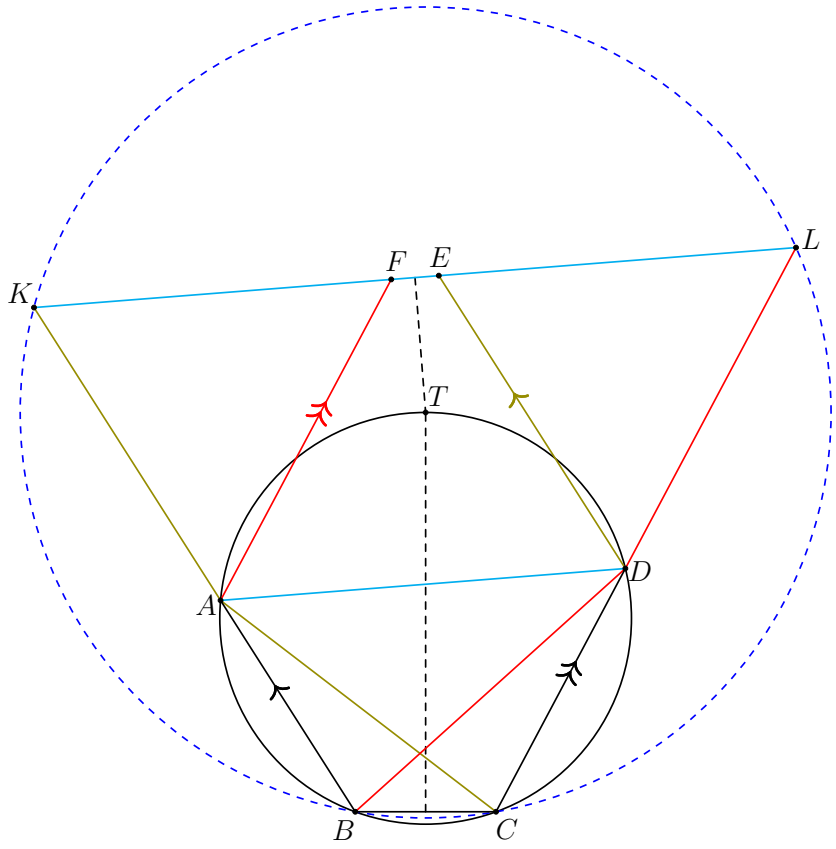
Geometry

G1. Let $ABCD$ be a cyclic quadrilateral such that $AC < BD < AD$ and $\angle DBA < 90°$. Point E lies on the line through D parallel to AB such that E and C lie on opposite sides of line AD, and $AC = DE$. Point F lies on the line through A parallel to CD such that F and C lie on opposite sides of line AD, and $BD = AF$.

Prove that the perpendicular bisectors of segments BC and EF intersect on the circumcircle of $ABCD$.

(Ukraine)

Solution 1. Let T be the midpoint of arc $\overset{\frown}{BAC}$ and let lines BA and CD intersect EF at K and L, respectively. Note that T lies on the perpendicular bisector of segment BC.



Since $ABCD$ is cyclic, $\frac{BD}{\sin\angle BAD} = \frac{AC}{\sin\angle ADC}$. From parallel lines we have $\angle DAF = \angle ADC$ and $\angle BAD = \angle EDA$. Hence,

$$AF \cdot \sin\angle DAF = BD \cdot \sin\angle ADC = AC \cdot \sin\angle BAD = DE \cdot \sin\angle EDA.$$

So F and E are equidistant from the line AD, meaning that EF is parallel to AD.

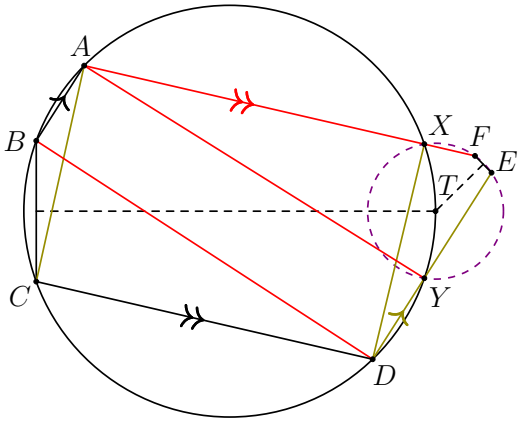
We have that $KADE$ and $FADL$ are parallelograms, hence we get $KA = DE = AC$ and $DL = AF = BD$. Also, $KE = AD = FL$ so it suffices to prove the perpendicular bisector of KL passes through T.

Triangle AKC is isosceles so $\angle BTC = \angle BAC = 2\angle BKC$. Likewise, $\angle BTC = 2\angle BLC$. Since T, K, and L all lie on the same side of BC and T lies on the perpendicular bisector of BC, T is the centre of circle $BKLC$. The result follows.

Solution 2. Let AF and DE meet ω at X and Y, respectively, and let T be as in Solution 1.

As $BD < AD$, $DY \parallel AB$ and $\angle BAY = \angle DBA < 90^\circ$, we have $DY < AB$ and Y lies on the opposite side of line AD to C. Also from $BD < AD$, we have B, C, and D all lie on the same side of the perpendicular bisector of AB which shows $AC > AB$. Combining these, we get $DY < AB < AC = DE$ and, as Y and E both lie on the same side of line AD, Y lies in the interior of segment DE. Similarly, X lies in the interior of segment DF.

Since AB is parallel to DY, we have $YA = BD = FA$. Likewise $XD = AC = ED$.



Claim 1. T is the midpoint of arc $\overset{\frown}{XY}$.

Proof. From $AX \parallel CD$ and $AB \parallel DY$ we have

$$\angle CAX = \angle AXD = \angle AYD = \angle YDB.$$

Since T is the midpoint of arc $\overset{\frown}{BAC}$, we have $\angle BAT = \angle TDC$, so

$$\angle TAX = \angle CAX + \angle BAC - \angle BAT = \angle YDB + \angle BDC - \angle TDC = \angle YDT. \qquad \square$$

Recall from above we have $AB < AC$ and analogously, $DC < DB$, which shows that X, Y and T all lie on the same side of line AD. In particular, T and A lie on opposite sides of XY so T lies on the internal angle bisector of $\angle XAY$. Since $AF = AY$, we have $\triangle ATF \cong \triangle ATY$, giving $TF = TY$.

Likewise, $TE = TX$, so $TE = TF$, meaning that T lies on the perpendicular bisector of segment EF as required.

Comment. The statement remains true without the length and angle conditions on cyclic quadrilateral $ABCD$ however additional care is required to consider different cases based on the ordering of points on lines DE and AF. It is also possible for T to be on the external angle bisector of $\angle XAY$.

Solution 3. From $AF = DB$, $AC = DE$ and

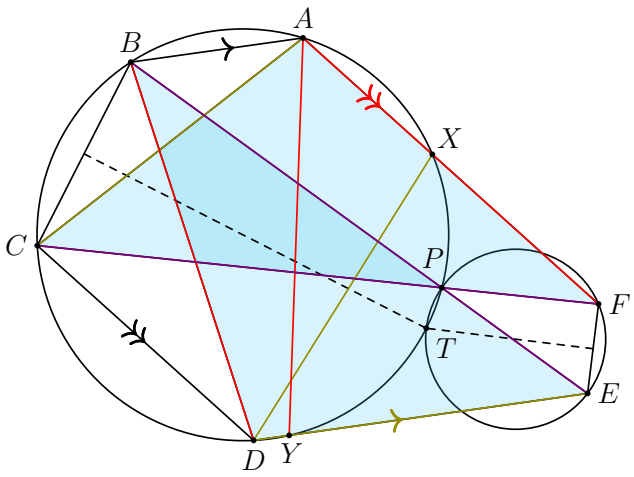
$$\angle(AC, AF) = \angle(AC, CD) = \angle(AB, BD) = \angle(DE, DB),$$

triangles ACF and DEB are congruent, so $CF = BE$.

Let $P = BE \cap CF$. Since

$$\angle(CP, BP) = \angle(CF, BE) = \angle(AF, DB) = \angle(DC, DB),$$

we have that P lies on circle $ABCD$.



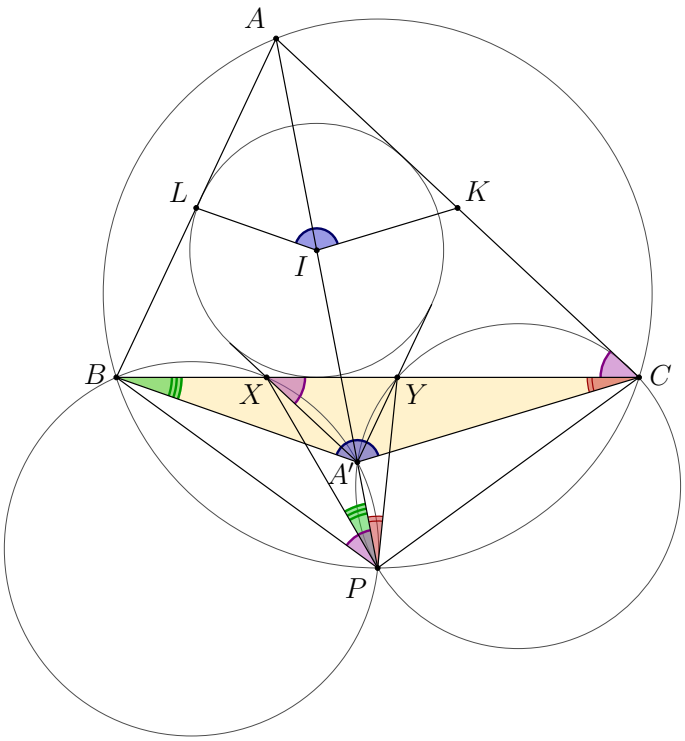
Finally, let T be the Miquel point of the quadrilateral $BCFE$ so T lies on circles EFP and $ABCD$. Note that T is the centre of spiral similarity taking segments BE to CF and since $BE = CF$, this is in fact just a rotation, so $TB = TC$ and $TE = TF$; that is, the perpendicular bisectors of BC and EF meet at T, on circle $ABCD$.

G2. Let ABC be a triangle with $AB < AC < BC$, incentre I and incircle ω. Let X be the point in the interior of side BC such that the line through X parallel to AC is tangent to ω. Similarly, let Y be the point in the interior of side BC such that the line through Y parallel to AB is tangent to ω. Let AI intersect the circumcircle of triangle ABC again at $P \neq A$. Let K and L be the midpoints of AB and AC, respectively.

Prove that $\angle KIL + \angle YPX = 180^\circ$.

(Poland)

Solution 1. Let A' be the reflection of A in I, then A' lies on the angle bisector AP. Lines $A'X$ and $A'Y$ are the reflections of AC and AB in I, respectively, and so they are the tangents to ω from X and Y. As is well-known, $PB = PC = PI$, and since $\angle BAP = \angle PAC > 30^\circ$, $PB = PC$ is greater than the circumradius. Hence $PI > \frac{1}{2}AP > AI$; we conclude that A' lies in the interior of segment AP.

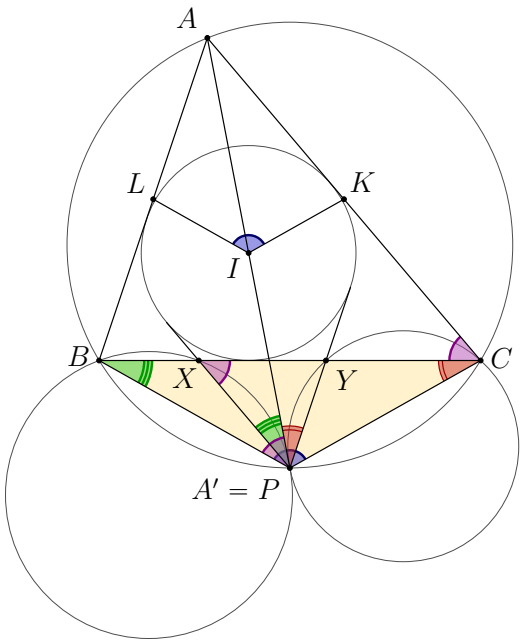


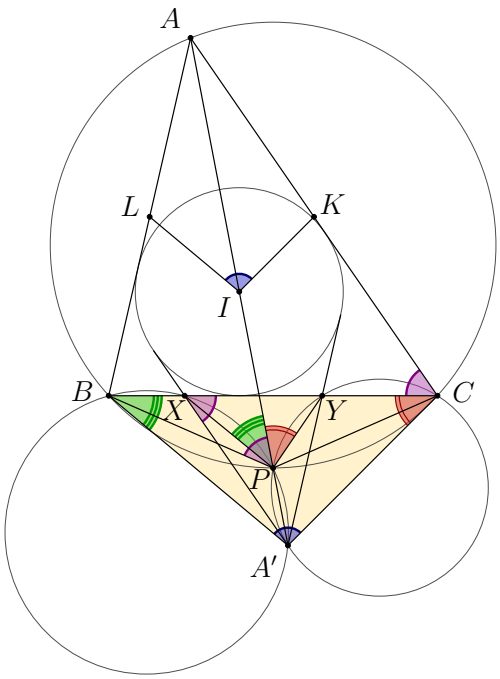
We have $\angle APB = \angle ACB$ in the circumcircle and $\angle ACB = \angle A'XC$ because $A'X \parallel AC$. Hence, $\angle APB = \angle A'XC$, and so quadrilateral $BPA'X$ is cyclic. Similarly, it follows that $CYA'P$ is cyclic.

Now we are ready to transform $\angle KIL + \angle YPX$ to the sum of angles in triangle $A'CB$. By a homothety of factor 2 at A we have $\angle KIL = \angle CA'B$. In circles $BPA'X$ and $CYA'P$ we have $\angle APX = \angle A'BC$ and $\angle YPA = \angle BCA'$, therefore

$$\angle KIL + \angle YPX = \angle CA'B + \big(\angle YPA + \angle APX\big) = \angle CA'B + \angle BCA' + \angle A'BC = 180^\circ.$$

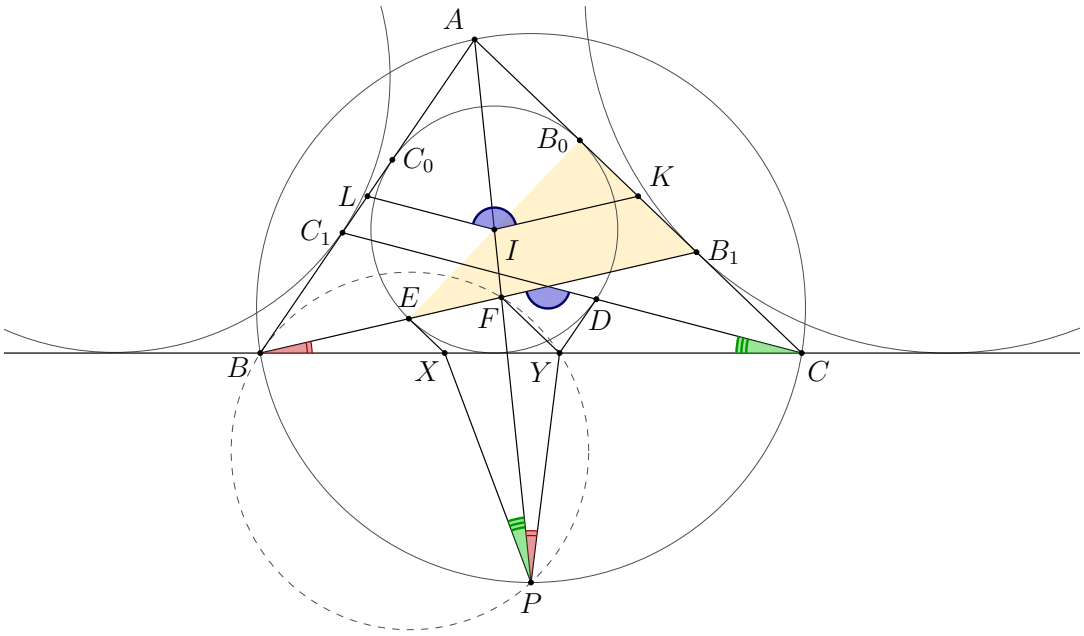
Comment. The constraint $AB < AC < BC$ was added by the Problem Selection Committee in order to reduce case-sensitivity. Without that, there would be two more possible configurations according to the possible orders of points A, P and A', as shown in the pictures below. The solution for these cases is broadly the same, but some extra care is required in the degenerate case when A' coincides with P and line AP is a common tangent to circles BPX and CPY.





Solution 2. Let $BC = a$, $AC = b$, $AB = c$ and $s = \frac{a+b+c}{2}$, and let the radii of the incircle, B-excircle and C-excircle be r, r_b and r_c, respectively. Let the incircle be tangent to AC and AB at B_0 and C_0, respectively; let the B-excircle be tangent to AC at B_1, and let the C-excircle be tangent to AB at C_1. As is well-known, $AB_1 = s - c$ and $\operatorname{area}(\triangle ABC) = rs = r_c(s - c)$.

Let the line through X, parallel to AC be tangent to the incircle at E, and the line through Y, parallel to AB be tangent to the incircle at D. Finally, let AP meet BB_1 at F.



It is well-known that points B, E, and B_1 are collinear by the homothety between the incircle and the B-excircle, and $BE \parallel IK$ because IK is a midline in triangle B_0EB_1. Similarly, it follows that C, D, and C_1 are collinear and $CD \parallel IL$. Hence, the problem reduces to proving $\angle YPA = \angle CBE$ (and its symmetric counterpart $\angle APX = \angle DCB$ with respect to the vertex C), so it suffices to prove that $FYPB$ is cyclic. Since $ACPB$ is cyclic, that is equivalent to $FY \parallel B_1C$ and $\frac{BF}{FB_1} = \frac{BY}{YC}$.

By the angle bisector theorem we have

$$\frac{BF}{FB_1} = \frac{AB}{AB_1} = \frac{c}{s-c}.$$

The homothety at C that maps the incircle to the C-excircle sends Y to B, so

$$\frac{BC}{YC} = \frac{r_c}{r} = \frac{s}{s-c}.$$

So,

$$\frac{BY}{YC} = \frac{BC}{YC} - 1 = \frac{s}{s-c} - 1 = \frac{c}{s-c} = \frac{BF}{FB_1},$$

which completes the solution.

G3. Let $ABCDE$ be a convex pentagon and let M be the midpoint of AB. Suppose that segment AB is tangent to the circumcircle of triangle CME at M and that D lies on the circumcircles of triangles AME and BMC. Lines AD and ME intersect at K, and lines BD and MC intersect at L. Points P and Q lie on line EC so that $\angle PDC = \angle EDQ = \angle ADB$.

Prove that lines KP, LQ, and MD are concurrent.

(Belarus)

Common remarks. Each of solutions we present consists of three separate parts:

(a) proving $KP \parallel MC$ and $LQ \parallel ME$;

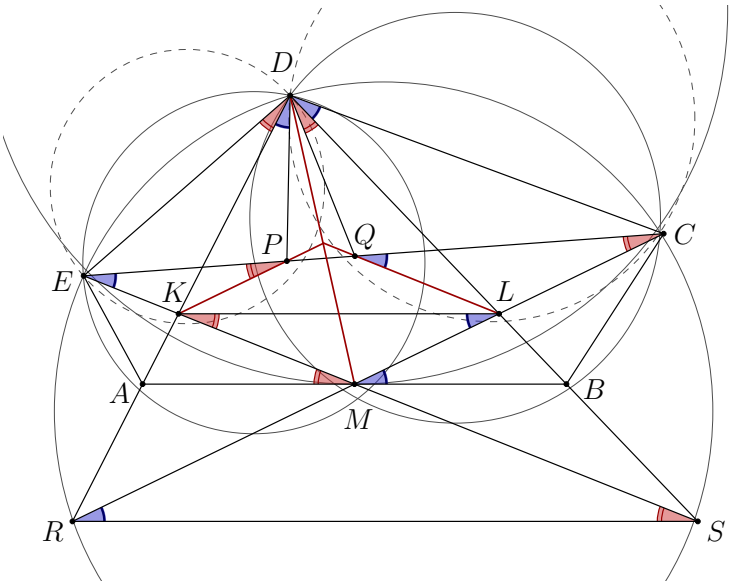
(b) proving $KL \parallel AB$ and, optionally, showing that points C, E, K, and L are concyclic;

(c) completing the solution either using homotheties or the parallelogram enclosed by lines KP, MK, ML and LQ, or radical axes between three circles.

Solution 1.

(a) Notice that the condition $\angle PDC = \angle ADB$ is equivalent to $\angle ADP = \angle BDC$, and $\angle EDQ = \angle ADB$ is equivalent to $\angle EDA = \angle QDB$. From line AB being tangent to circle CME, and circles $AMDE$ and $CDME$ we read $\angle ECM = \angle EMA = \angle EDA = \angle QDB$ and $\angle MEC = \angle BMC = \angle BDC = \angle ADP$.

Using $\angle ADP = \angle MEC$, the points D, E, K, and P are concyclic, which gives that $\angle EPK = \angle EDA = \angle ECM$. From that, we can see that $KP \parallel MC$. It can be shown similarly that C, D, Q, and L are concyclic, $\angle LQC = \angle MEC$ and therefore $LQ \parallel ME$.



(b) Let rays DA and DB intersect circle CDE at R and S, respectively. We now observe that $\angle SEC = \angle SDC = \angle MEC$, so points E, M, and S are collinear. We similarly obtain that C, M, and R are collinear.

From $\angle SRC = \angle SEC = \angle BMC$ we can see that $RS \parallel AB$. Since M bisects AB, it follows that $KL \parallel RS$.

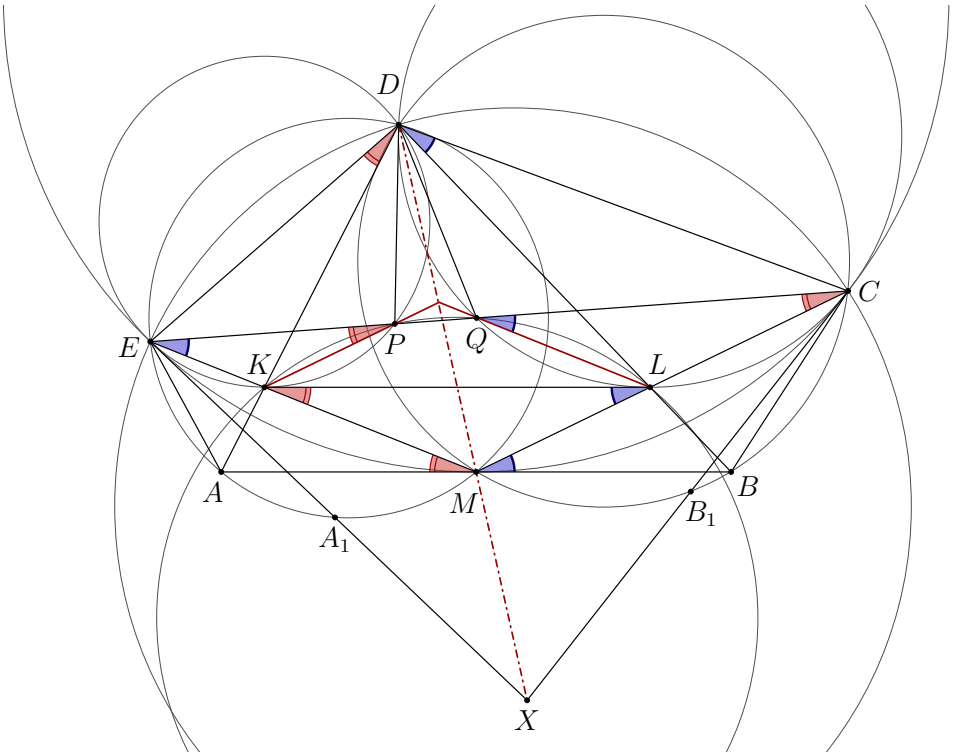
(c) Consider the homothety at D that sends RS to KL. Because $KP \parallel RC$ and $LQ \parallel SE$, that homothety sends the concurrent lines DM, RC, and SE to DM, KP, and LQ, so these lines are also concurrent.

Solution 2.

(a) As in Solution 1, we show the following: $\angle ECM = \angle EMA = \angle EDA = \angle EPK$; $\angle MEC = \angle BMC = \angle BDC = \angle LQC$; the points C, D, Q, and L are concyclic; the points D, E, K, and P are concyclic; $KP \parallel MC$; and $LQ \parallel ME$.

(b) Notice that triangles EKP and EMC are homothetic at E, so their circumcircles CME and $DEKP$ are tangent to each other at E. Similarly, circle $CDQL$ is tangent to circle CME at C.

Suppose that the tangents to circle CME at C and E intersect at point X. (The case when CE is a diameter in circle CME can be considered as a limit case.) Moreover, let EX and CX intersect circles $DEAM$ and $BCDM$ again at $A_1 \neq E$ and $B_1 \neq C$, respectively.



We have $XE = XC$ because they are the tangents from X to circle CME. Moreover, in circle $DEAM$, chords AM and A_1E are tangent to circle CME, so $A_1E = AM$. Similarly, we have $B_1C = BM$, hence $A_1E = AM = BM = B_1C$. We conclude $XA_1 = XB_1$, so the powers of X with respect to circles $DEAM$ and $BCDM$ are equal. Therefore, X lies on the radical axis of these two circles, which is DM.

Now notice that by $XC = XE$, point X has equal powers to circles $CDQL$ and $DEKP$, so DX is the radical axis of these circles. Point M lies on DX, so $ME \cdot MK = MC \cdot ML$; we conclude that C, E, K, and L are concyclic. Hence, by $\angle MKL = \angle ECM = \angle KMA$ we have $KL \parallel AB$.

(c) As $\angle EPK = \angle EMA = \angle QLK$, we have that $KLQP$ is cyclic. The radical axes between circles $DEKP$, $CDQL$ and $KLQP$ are DM, KP and LQ, so they are concurrent at the radical centre of the three circles.

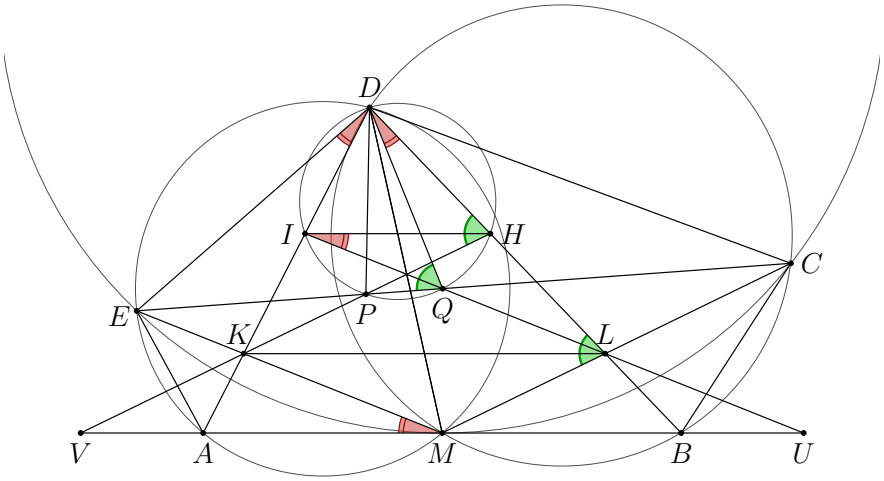
Solution 3.

(b) We present another proof that $KL \parallel AB$.

Let $AD \cap LQ = I$, $BD \cap KP = H$, $AB \cap LQ = U$ and $AB \cap KP = V$. Since

$$\angle DHP = \angle DLM = 180^\circ - \angle CLD = 180^\circ - \angle CQD = \angle DQE,$$

point H lies on circle DPQ. Similarly, we obtain that point I lies on this circle. Hence, $\angle LIH = \angle QDB = \angle EDA = \angle EMA$, and $LQ \parallel ME$ implies that $HI \parallel AB$.



Let $AM = BM = d$, then we have

$$\frac{BU}{IH} = \frac{BL}{LH} = \frac{BM}{MV} = \frac{d}{d + AV} \quad \text{and} \quad \frac{AV}{IH} = \frac{AK}{KI} = \frac{AM}{MU} = \frac{d}{d + BU}.$$

Hence, $BU \cdot (d + AV) = AV \cdot (d + BU)$, so $BU = AV$. Therefore, $\triangle MLU \cong \triangle VKM$ which implies $KL \parallel AB \parallel HI$.

(c) Lines MK, ML, KP and LQ enclose a parallelogram. Line DM passes through the midpoint of KL, which the centre of the parallelogram, and passes through the vertex M. Therefore, DM passes through the opposite vertex, which is the intersection of KP and LQ.

G4. Let $ABCD$ be a quadrilateral with AB parallel to CD and $AB < CD$. Lines AD and BC intersect at a point P. Point $X \neq C$ on the circumcircle of triangle ABC is such that $PC = PX$. Point $Y \neq D$ on the circumcircle of triangle ABD is such that $PD = PY$. Lines AX and BY intersect at Q.

Prove that PQ is parallel to AB.

(Ukraine)

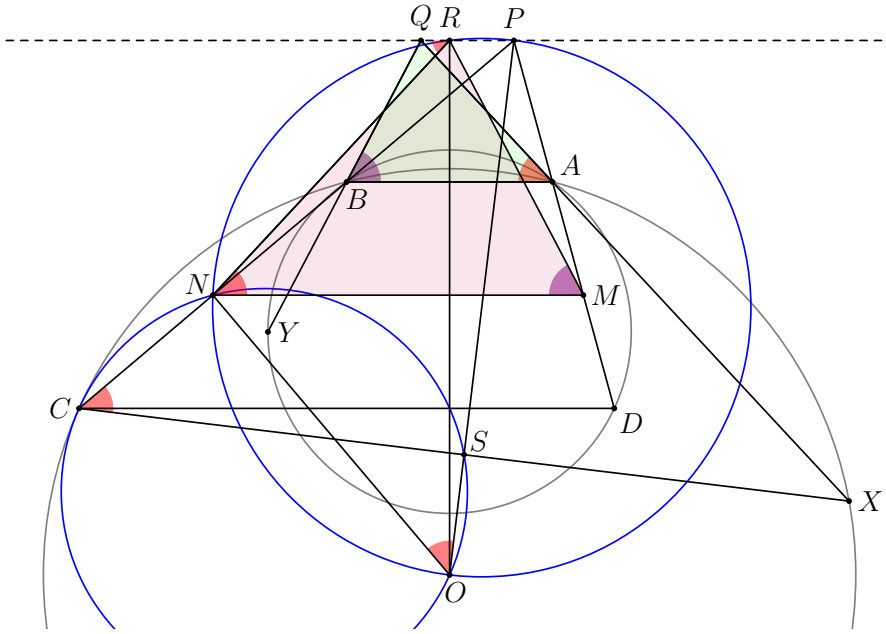
Solution 1. Let M and N be the midpoints of AD and BC, respectively and let the perpendicular bisector of AB intersect the line through P parallel to AB at R.

Lemma. Triangles QAB and RNM are similar.

Proof. Let O be the circumcentre of triangle ABC, and let S be the midpoint of CX. Since N, S, and R are the respective perpendicular feet from O to BC, CX, and PR, we have that quadrilaterals $PRNO$ and $CNSO$ are cyclic. Furthermore, P, S, and O are collinear as $PC = PX$. Since $ABCX$ is also cyclic, we have that

$$\angle QAB = \angle XCB = \angle PON = 180^\circ - \angle NRP = \angle MNR.$$

Analogously, we have that $\angle ABQ = \angle RMN$, so triangles QAB and RNM are similar. □

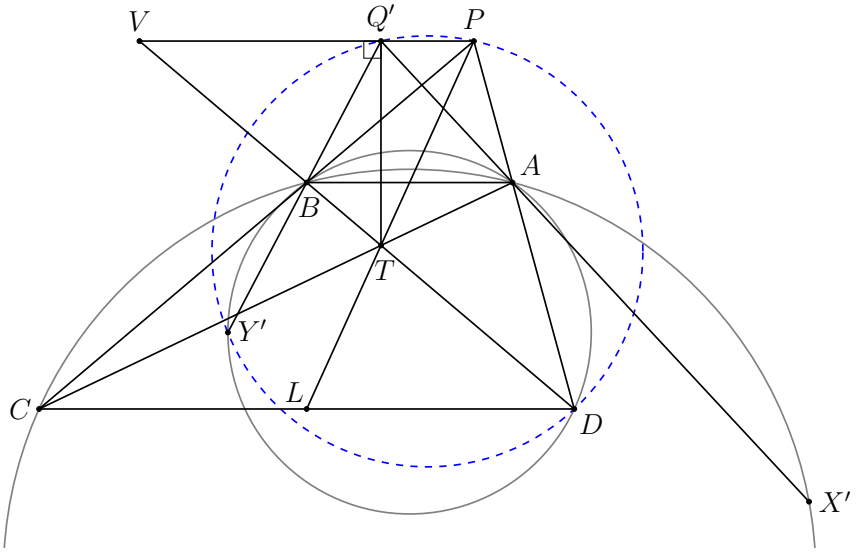


Let $d(Z, \ell)$ denote the perpendicular distance from the point Z to the line ℓ. Using that $PR \parallel AB$ along with the similarities $QAB \sim RNM$ and $PAB \sim PMN$, we have that

$$\frac{d(Q, AB)}{AB} = \frac{d(R, MN)}{MN} = \frac{d(P, MN)}{MN} = \frac{d(P, AB)}{AB},$$

which implies that $PQ \parallel AB$.

Solution 2. Let BD and AC intersect at T and let the line through P parallel to AB intersect BD at V. Next, let Q' be the foot of the perpendicular from T to PV. Finally, let $Q'A$ intersect circle ABC again at X' and $Q'B$ intersect circle ABD again at Y'.



Claim. PQ' bisects $\angle BQ'D$ externally.

Proof. Let PT intersect CD at L. Let ∞_{CD} be the point at infinity on line CD. From the standard Ceva-Menelaus configuration we have $(D, C; L, \infty_{CD})$ is harmonic. Hence projecting through P we have

$$-1 = (D, C; L, \infty_{CD}) = (D, B; T, V).$$

As $(D, B; T, V)$ is harmonic, and also $\angle VQ'T = 90^\circ$ (by construction), the claim follows. □

Now as

$$\angle Q'PD = \angle BAD = 180^\circ - \angle DY'B = 180^\circ - \angle DY'Q'$$

we have $Q'PDY'$ cyclic. By the claim, we have that P is the midpoint of arc $\overset{\frown}{DQ'Y'}$, so $PD = PY'$.

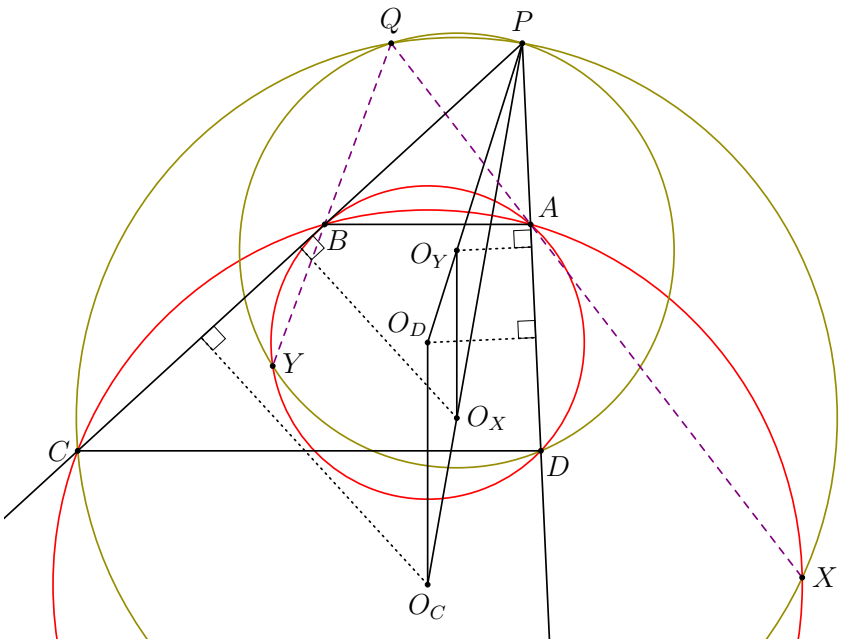
Since Y is the unique point not equal to D on circle ABD satisfying $PD = PY$, we have $Y' = Y$.

Likewise $X' = X$ so $Q' = Q$ and we are done.

Solution 3. Let AX intersect circle PCX for the second time at Q'. Then

$$\angle AQ'P = \angle XQ'P = \angle XCP = \angle XCB = 180^\circ - \angle BAX = \angle Q'AB$$

so PQ' is parallel to AB. Hence, it suffices to show that Q' is equal to Q. To do so, we aim to show the common chord of circles PCX and PDY is parallel to AB, since then by symmetry Q' is also the second intersection of BY and circle PDY.



Let the centres of circles PCX and PDY be O_X and O_Y, respectively. Let the centres of circles ABC and ABD be O_C and O_D, respectively.

Note P, O_X, and O_C are collinear since they all lie on the perpendicular bisector of CX. Likewise P, O_Y, and O_D are collinear on the perpendicular bisector of DY. By considering the projections of O_X and O_C onto BC, and O_Y and O_D onto AD, we have

$$\frac{PO_X}{PO_C} = \frac{\frac{PC}{2}}{\frac{PB+PC}{2}} = \frac{\frac{PD}{2}}{\frac{PA+PD}{2}} = \frac{PO_Y}{PO_D}.$$

Hence O_XO_Y is parallel to O_CO_D, which is perpendicular to AB as desired.

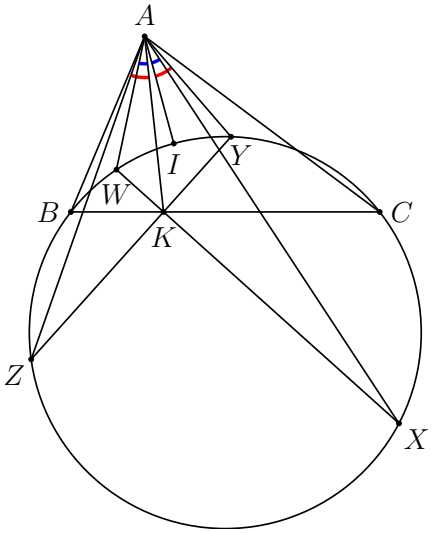
G5. Let ABC be a triangle with incentre I, and let Ω be the circumcircle of triangle BIC. Let K be a point in the interior of segment BC such that $\angle BAK < \angle KAC$. The angle bisector of $\angle BKA$ intersects Ω at points W and X such that A and W lie on the same side of BC, and the angle bisector of $\angle CKA$ intersects Ω at points Y and Z such that A and Y lie on the same side of BC.

Prove that $\angle WAY = \angle ZAX$.

(Uzbekistan)

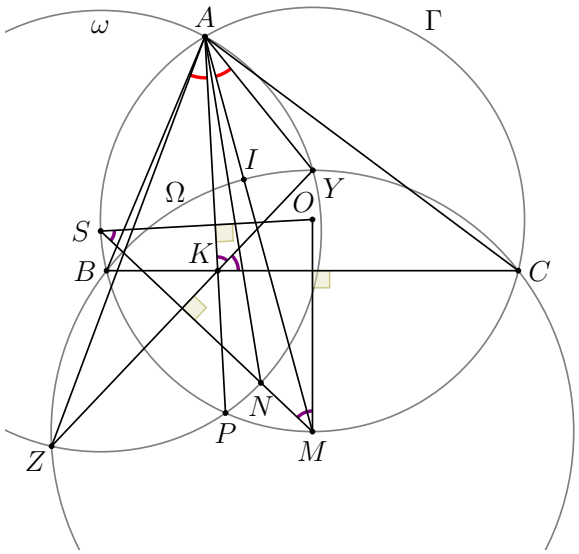
Common remarks. The key step in each solution is to prove that $\angle ZAK = \angle IAY$ and $\angle WAK = \angle IAX$. The problem is implied by these equalities, as we then have that

$$\angle WAY = \angle WAK + \angle KAI + \angle IAY = \angle IAX + \angle KAI + \angle ZAK = \angle ZAX.$$



We now present several proofs that $\angle ZAK = \angle IAY$, with $\angle WAK = \angle IAX$ following in an analogous manner.

Solution 1. Let Γ be circle ABC and ω be circle AYZ. Let O, M, and S be the centres of Γ, Ω, and ω, respectively. Let AK intersect Γ again at P, and let the angle bisector of $\angle ZAY$ intersect ω again at N.

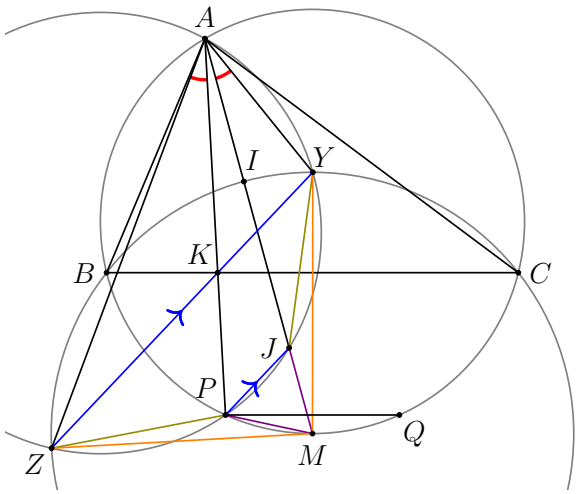


By power of a point from K to Γ and Ω, we have that $KA \cdot KP = KB \cdot KC = KY \cdot KZ$, so P also lies on ω. The pairwise common chords of Γ, Ω, and ω are then $AP \perp OS$, $BC \perp OM$, and $YZ \perp MS$, so we have that $\angle OMS = \angle CKY = \angle YKA = \angle MSO$. As M lies on Γ and $OM = OS$, S also lies on Γ. Note that N lies on MS as $NY = NZ$, so

$$\angle PAN = \frac{1}{2}\angle PSN = \frac{1}{2}\angle PSM = \frac{1}{2}\angle PAM.$$

Thus, AN bisects $\angle PAM$ in addition to $\angle ZAY$, which means that $\angle ZAK = \angle IAY$ as K lies on AP and I lies on AM.

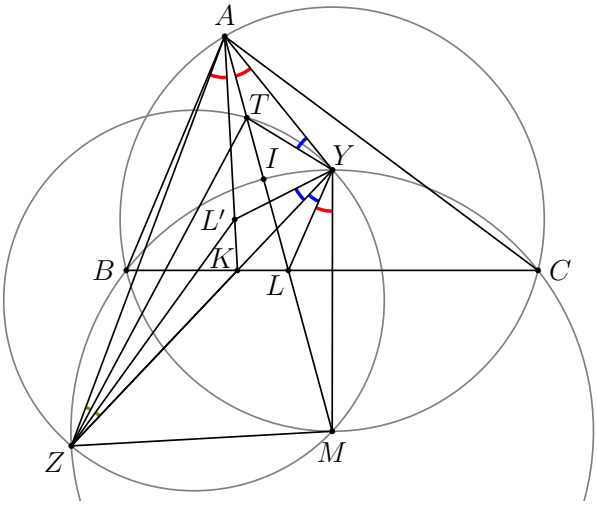
Solution 2. Define M and P as in Solution 1, and recall that $AYPZ$ is cyclic. Let Q be the second intersection of the line parallel to BC through P with circle ABC and let J be the incentre of triangle APQ.



Since PQ is parallel to BC and $\angle BAP < \angle PAC$, the angle bisector of $\angle APQ$ is parallel to the angle bisector of $\angle AKC$. Hence, PJ is parallel to YZ. As M is the midpoint of $\widehat{PQ}$ on circle APQ, we have that $MP = MJ$. Then since segments YZ and PJ are parallel and have a common point M on their perpendicular bisectors, $PJYZ$ is cyclic with $JY = PZ$. It follows that J also lies on circle $AYPZ$ and that $\angle ZAP = \angle JAY = \angle IAY$.

Comment. The proof of the analogous case of $\angle WAK = \angle IAX$ is slightly different. In this case, J should be defined as the A-excentre of APQ so that PJ is the external bisector of $\angle APQ$ and $PJ \parallel WX$. The proof is otherwise exactly the same.

Solution 3. As in the previous solutions, let M be the centre of Ω. Let L be the intersection of AM and BC, and let L' be the reflection of L over YZ. Let the circle MYZ intersect AM again at T.



Note that as M is the midpoint of $\widehat{BC}$ on circle ABC and L is the foot of the bisector of $\angle BAC$, we have that $MA \cdot ML = MI^2 = MY^2$. It follows by power of a point that MY is tangent to circle ALY, so $\angle LAY = \angle LYM$. Using directed angles, we then have that

$$\measuredangle AYT = \measuredangle MTY - \measuredangle MAY = \measuredangle MZY - \measuredangle LYM = \measuredangle ZYM - \measuredangle LYM = \measuredangle ZYL = \measuredangle L'YZ,$$

where we use the fact that $MY = MZ$ and that L and L' are symmetric about YZ. Thus, YT and YL' are isogonal in $\angle AYZ$. Analogously, we have that ZT and ZL' are isogonal in $\angle YZA$. This means that T and L' are isogonal conjugates in triangle AYZ, which allows us to conclude that $\angle ZAK = \angle IAY$ since L' lies on AK and T lies on AI.

Comment. Owing to the condition $\angle BAK < \angle KAC$, points L' and T lie inside triangle AYZ. However, if one tries to write down the same proof for $\angle WAK = \angle IAX$, the analogues L_1' and T_1 of L' and T would lie outside triangle AWX. Thus, the solution has been written using directed angles so that it applies directly to this case as well. It is also possible that L_1' lies on circle AWX and T_1 is a point at infinity. In this case, it is straightforward to interpret the directed angle chase to prove the isogonality, and the isogonality also follows from this scenario being a limit case of other configurations.

Note. The original proposal remarks that this problem is a special case of a more general property:

A convex quadrilateral $ABCD$ is inscribed in a circle ω. The bisectors between AC and BD intersect ω at four points, forming a convex quadrilateral $PQRS$. Then the conditions

$$XA \cdot XC = XB \cdot XD \qquad \text{and} \qquad \measuredangle(XP, XQ) = \measuredangle(XS, XR)$$

on point X are equivalent.

The Problem Selection Committee believes that the proof of this generalisation is beyond the scope of the competition and considers the original problem to be more suitable.

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G6. Let ABC be an acute triangle with $AB < AC$, and let Γ be the circumcircle of ABC. Points X and Y lie on Γ so that XY and BC intersect on the external angle bisector of $\angle BAC$. Suppose that the tangents to Γ at X and Y intersect at a point T on the same side of BC as A, and that TX and TY intersect BC at U and V, respectively. Let J be the centre of the excircle of triangle TUV opposite the vertex T.

Prove that AJ bisects $\angle BAC$.

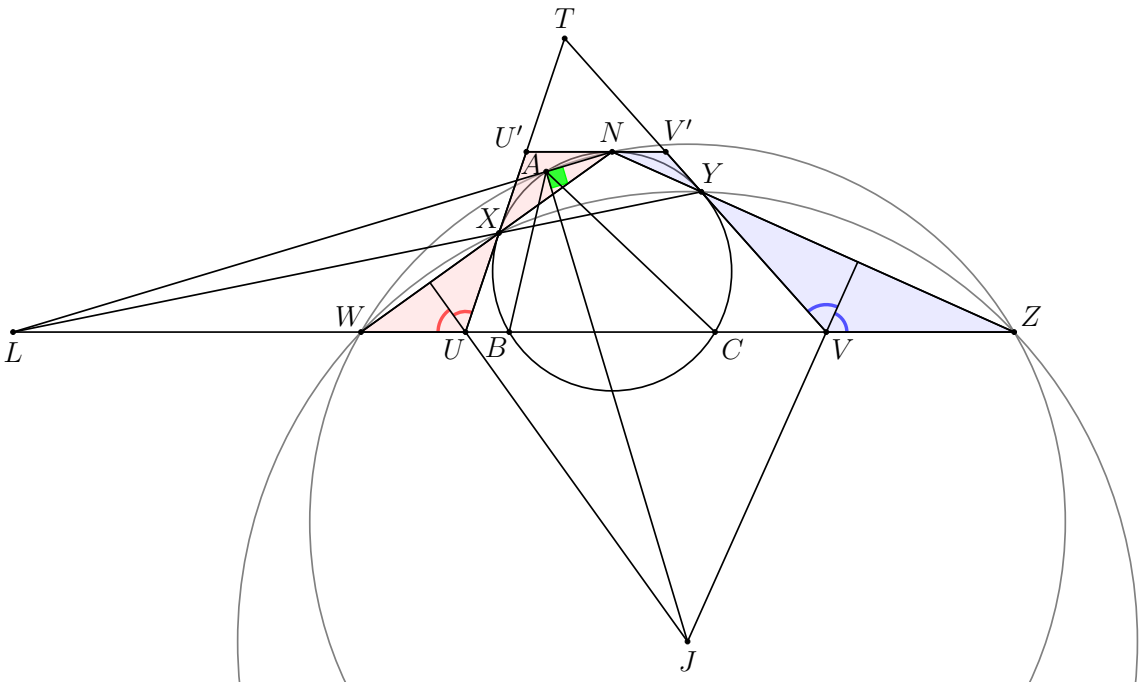
(Poland)

Solution 1. Let N be the midpoint of $\overset{\frown}{BAC}$ on Γ, and let NX and NY intersect BC at W and Z, respectively.

Claim. Quadrilateral $WXYZ$ is cyclic, and its circumcentre is J.

Proof. As N is the midpoint of $\overset{\frown}{BAC}$, W and Z lie on BC, and X and Y are the second intersections of NW and NZ with Γ, we have that $WXYZ$ is cyclic.

Let the parallel to BC through N intersect TU and TV at U' and V', respectively. Then U' is the intersection of the tangents to Γ at N and X, so $U'N = U'X$. As $NU' \parallel BC$, $U'NX$ is similar to UWX, so $UW = UX$ as well. Hence, the perpendicular bisector of WX is the internal bisector of $\angle XUW$, which is the external bisector of $\angle VUT$. Analogously, the perpendicular bisector of YZ is the external bisector of $\angle TVU$. This means that the circumcentre of $WXYZ$ is the intersection of the external bisectors of $\angle VUT$ and $\angle TVU$, which is J. □



Let AN intersect BC at L, so XY passes through L as well. By power of a point from L to Γ and circle $WXYZ$, we have that $LA \cdot LN = LX \cdot LY = LW \cdot LZ$, so $WANZ$ is also cyclic. Thus, A is the Miquel point of quadrilateral $WXYZ$. As $WXYZ$ is cyclic with circumcentre J and its opposite sides WX and YZ intersect at N, we have that $AN \perp AJ$. Since AN is the external bisector of $\angle BAC$, this implies that AJ is the internal bisector of $\angle BAC$.

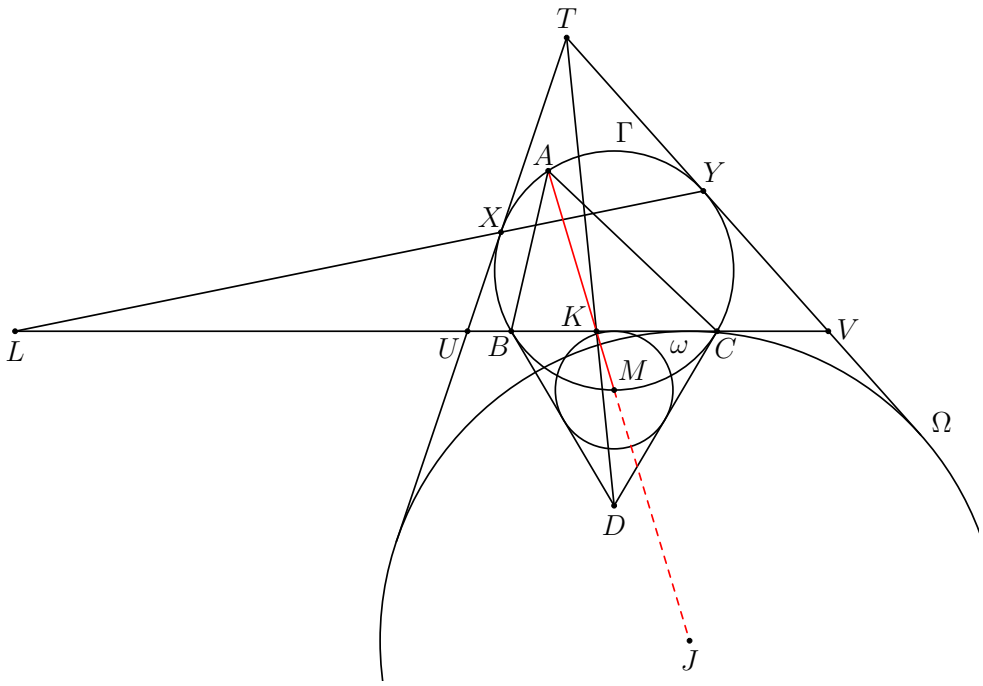
Solution 2. Let the internal and external angle bisectors of $\angle BAC$ intersect BC at K and L, respectively. Let AK intersect circle ABC again at M, and let D be the intersection of the tangents to Γ at B and C. Let Ω be the T-excircle of TUV, and let ω be the incircle of DBC.

Claim. The points T, K, and D are collinear.

Proof. Note that BC and XY are the polars of T and D with respect to Γ. By La Hire's Theorem, TD is the polar of L with respect to Γ. As $(B, C; K, L) = -1$, K also lies on the polar of L, thus proving the collinearity. □

Claim. The incentre of DBC is M.

Proof. We have that $\angle MBC = \angle MAC = \frac{1}{2}\angle BAC = \frac{1}{2}\angle DBC$, so BM bisects $\angle DBC$. Similarly, CM bisects $\angle BCD$, so M is the incentre of DBC. □



Claim. The intersection of the common external tangents of Ω and ω is K.

Proof. Let K' be the intersection of the common external tangents of Ω and ω. As Ω and ω are both tangent to BC and lie on the same side of BC opposite to A, K' lies on BC. As T is the intersection of the common external tangents of Γ and Ω and D is the intersection of the common external tangents of Γ and ω, by Monge's theorem K' lies on TD. As K' lies on both BC and TD, it is the same point as K. □

Hence, K is collinear with the centres of Ω and ω, which are M and J, respectively. As K and M both lie on the bisector of $\angle BAC$, so does J.

Note. It can be shown that circles AUV and ABC are tangent and that the tangents from U and V to circle ABC different from TU and TV intersect at a point W on line TK. Reframing the problem in terms of quadrilateral $TUWV$ using these properties, we obtain the following problem:

Let $ABCD$ be a convex quadrilateral with an incircle ω, and let AC and BD intersect at P. Point E lies on ω such that the circumcircle of ACE is tangent to ω. Prove that if B and E lie on the same side of line AC, then the centre of the excircle of triangle ABC opposite the vertex B lies on line EP.

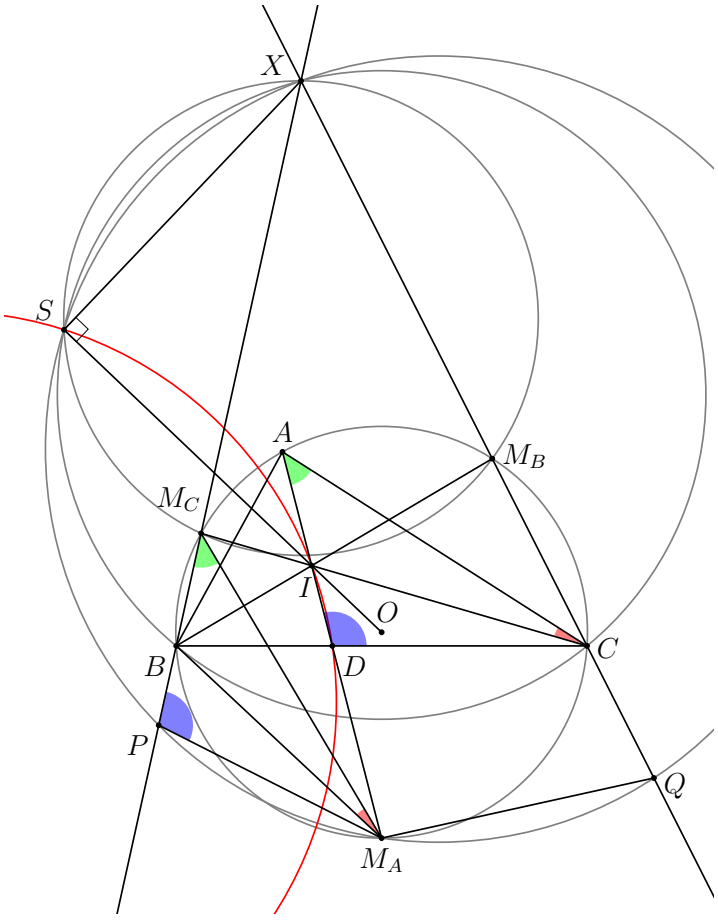
While this is an appealing statement, the Problem Selection Committee is uncertain about its difficulty and whether it has solutions that do not proceed by reducing to the original problem. Thus, it is believed that the original statement is more suitable for the competition.

G7. Let ABC be a triangle with incentre I such that $AB < AC < BC$. The second intersections of AI, BI, and CI with the circumcircle of triangle ABC are M_A, M_B, and M_C, respectively. Lines AI and BC intersect at D and lines BM_C and CM_B intersect at X. Suppose the circumcircles of triangles XM_BM_C and XBC intersect again at $S \neq X$. Lines BX and CX intersect the circumcircle of triangle SXM_A again at $P \neq X$ and $Q \neq X$, respectively.

Prove that the circumcentre of triangle SID lies on PQ.

(Thailand)

Solution 1.



Let O be the circumcentre of $\triangle ABC$. First we note from standard properties of the Miquel point S we have:

- $\triangle SM_CM_B \sim \triangle SBC \sim \triangle SPQ$; $(*)$
- I and S are inverses with respect to circle ABC;
- $\angle OSX = 90^\circ$.

Claim 1. $\angle M_APB = \angle CDA$.

Proof. From the above we have $\triangle OM_AI \sim \triangle OSM_A$ and

$$\angle M_APB = \angle M_APX = \angle M_ASX = 90^\circ + \angle M_ASO = 90^\circ + \angle OM_AI = \angle M_ABA = \angle CDA. \square$$

Claim 2. $\frac{M_CB}{BP} = \frac{M_BC}{CQ} = \frac{AI}{ID}$.

Proof. Observe that $\angle PM_CM_A = \angle BM_CM_A = \angle DAC$ and $\angle M_CM_AB = \angle ICD$. Combining these with Claim 1 shows $M_CPM_AB \sim ADCI$. Therefore, $\frac{M_CB}{BP} = \frac{AI}{ID}$. Similarly, $\frac{M_BC}{CQ} = \frac{AI}{ID}$. □

Claim 3. $\frac{DP}{DQ} = \frac{IB}{IC}$.

Proof. Firstly, observe that $\angle ICB = \angle AM_BM_C$ and $\angle CBI = \angle M_BM_CA$ which gives that $\triangle IBC \sim \triangle AM_CM_B$. This, combined with Claim 2, is enough to show $\triangle DPQ \sim \triangle IBC$ by linearity, proving the claim. □

Claim 4. $\frac{IP}{IQ} = \frac{IB}{IC}$.

Proof. Combining $\triangle IBM_C \sim \triangle ICM_B$ with Claim 2 shows $IBM_CP \sim ICM_BQ$ giving the result. □

Finally, we have that

$$\frac{SP}{SQ} = \frac{SB}{SC} = \frac{BM_C}{CM_B} = \frac{IB}{IC}$$

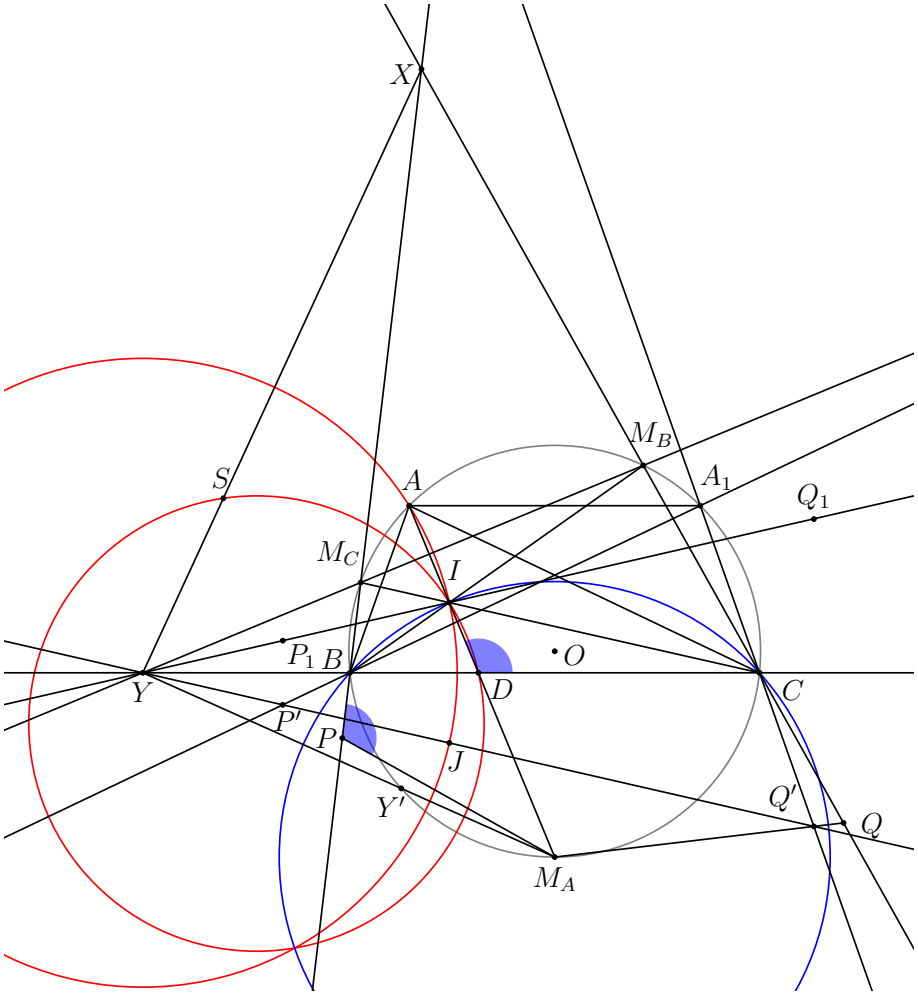
from $(*)$ and $\triangle IBM_C \sim \triangle ICM_B$. Putting this together with Claims 3 and 4, we have that

$$\frac{IB}{IC} = \frac{DP}{DQ} = \frac{IP}{IQ} = \frac{SP}{SQ},$$

which shows that circle SID is an Apollonius circle with respect to P and Q, giving the desired conclusion.

Comment. The condition $AB < AC$ ensures $S \neq X$. We also need to avoid the case $\angle BAC = 60^\circ$ as then $BM_C \parallel CM_B$.

Solution 2. We use Claim 1 from Solution 1. We will show that P and Q are inverses in circle SID which implies the result. Perform an inversion in circle BIC and denote the inverse of a point $\bullet$ by $\bullet'$.



Claim 1. $S' = J$ where J is the reflection of I across BC.

Proof. We have that S and I are inverses in circle ABC. Inverting this assertion in circle BIC shows that S' and I are inverses with respect to line BC, which is just a reflection in line BC. □

Let $Y = M_BM_C \cap BC$. From $\angle IM_CM_B = \angle M_BM_CA$ and $\angle AM_BM_C = \angle M_CM_BI$, we see that A and I are reflections in line M_BM_C so $YA = YI$. We have that circle SID maps to circle AIJ which, from the previous comment, has centre Y. Inverting the conclusion that P and Q are inverses with respect to circle SID in circle BIC, it suffices to show P' and Q' are inverses with respect to circle AIJ or equivalently, that $YP' \cdot YQ' = YA^2$.

Claim 2. Circle XSM_A maps to line YJ under the inversion in circle BIC.

Proof. Since circle BIC has centre M_A, the inverse of this circle is a line. By Claim 1, this line passes through J hence it suffices to prove that circle XSM_A passes through Y'. From inverting line BC in circle BIC, we have that BCM_AY' is cyclic so

$$YS \cdot YX = YB \cdot YC = YY' \cdot YM_A.$$

where we have used that Y, S and X are collinear by a standard property of the Miquel point. Hence Y' lies on circle XSM_A as required. □

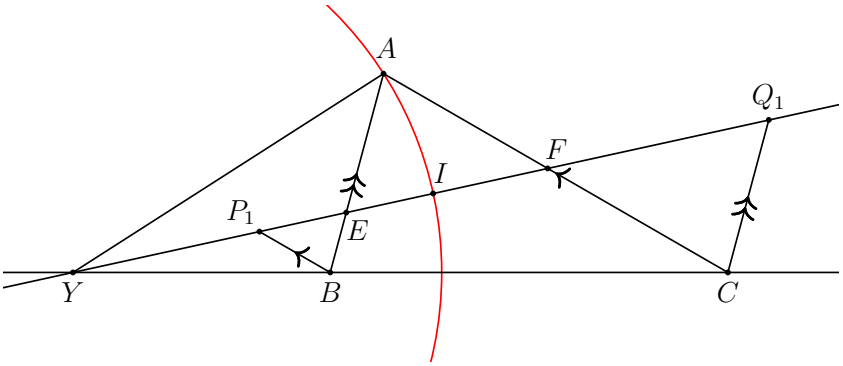
Let A_1 be the reflection of A in the perpendicular bisector of BC. Using Claim 1 from Solution 1,

$$\angle P'BM_A = \angle M_APB = \angle CDA = 180^\circ - \angle ACM_A = 180^\circ - \angle M_ABA_1.$$

Hence, P', B, and A_1 are collinear. Similarly Q', C, and A_1 are collinear. Let P_1 and Q_1 be the reflections of P' and Q' across BC. As P' and Q' lie on line YJ, it follows that P_1 and Q_1 lie on line YI. Also from the previous collinearities, we get $BP_1 \parallel AC$ and $CQ_1 \parallel AB$.

We have now reduced the problem to the following:

Claim 3 (Inverted Problem). Let ABC be a triangle with incentre I. Let Y be the point on BC such that $YA = YI$. Let P_1 and Q_1 be points on YI such that $BP_1 \parallel AC$ and $CQ_1 \parallel AB$. Then $YA^2 = YP_1 \cdot YQ_1$.

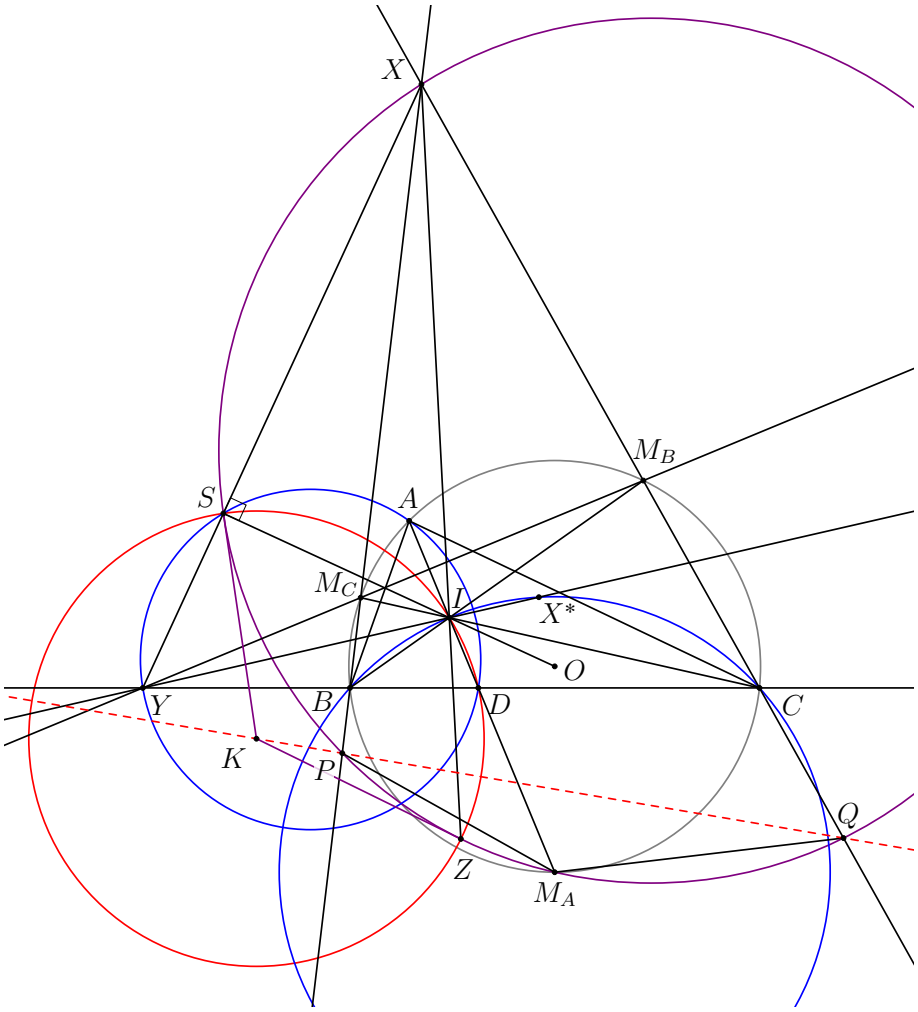


Proof. Let YI intersect AB and AC at E and F, respectively. From the parallel lines, we get that $\triangle BEP_1$ and $\triangle CQ_1F$ are homothetic with centre Y. Thus we have

$$\frac{YE}{YP_1} = \frac{YQ_1}{YF} \implies YP_1 \cdot YQ_1 = YE \cdot YF.$$

Moreover, AI bisects $\angle EAF$ and $YA = YI$ so the circle centred at Y with radius YA is the Apollonius circle of $\triangle AEF$ with respect to the feet of the internal and external angle bisectors at A. This gives $YE \cdot YF = YA^2$. Combining these results proves the claim. □

Solution 3. As in Solution 1, let O be the circumcentre of $\triangle ABC$. Let XI intersect circle XSM_A again at $Z \neq X$ and let $Y = BC \cap M_BM_C$. Let X^* be the inverse of X in circle ABC. We will use the properties of Miquel point S noted at the top of Solution 1 and in addition, that S lies on line XY.



Claim 1. $YSAD$ is cyclic.

Proof. From $OM_A \perp BC$ and $YS \perp OS$ we have $\angle DYS = 180^\circ - \angle SOM_A$. From inverting collinear points A, I and M_A in circle ABC we get ASM_AO is cyclic which gives

$$\angle SOM_A = \angle SAM_A = \angle SAD \implies \angle SAD + \angle DYS = 180^\circ$$

proving the claim. □

Claim 2. X^* lies on circle BIC which has centre M_A.

Proof. This follows immediately from inverting circle $SBCX$ in circle ABC. □

Claim 3. Z lies on circle SID.

Proof. We have that

$$\angle IZS = \angle XM_AS = \angle OM_AS - \angle OM_AX = \angle M_AIO - \angle M_AX^*O = \angle DIO - \angle M_AX^*O$$

where in the penultimate step we inverted in circle ABC to get the angle equalities.

From Brocard's Theorem applied to cyclic quadrilateral BM_CM_BC, we get Y, I, and X^* collinear and $\angle YX^*O = 90^\circ$. This gives that

$$\angle M_AX^*O = 90^\circ - \angle IX^*M_A = 90^\circ - \angle M_AIX^* = 90^\circ - \angle AIY,$$

where the second equality is by Claim 2. We have that A and I are reflections in line M_BM_C. Hence,

$$90^\circ - \angle AIY = 90^\circ - \angle YAD = 90^\circ - \angle YSD = \angle DSO$$

where the second step is by Claim 1, and in the last step we are using $OS \perp YS$. Putting these together,

$$\angle IZS = \angle DIO - \angle DSO = \angle IDS,$$

proving the claim. □

Let the tangents from S and Z to circle XSM_A intersect at K. Observe from the standard Ceva-Menelaus configuration,

$$-1 = (XY, XI; XB, XC) \overset{X}{=} (S, Z; P, Q)\,.$$

This shows that K lies on line PQ. We then have

$$\angle ZKS = 180^\circ - 2\angle SXZ = 2\,(90^\circ - \angle SXI) = 2\,(180^\circ - \angle SIZ)\,,$$

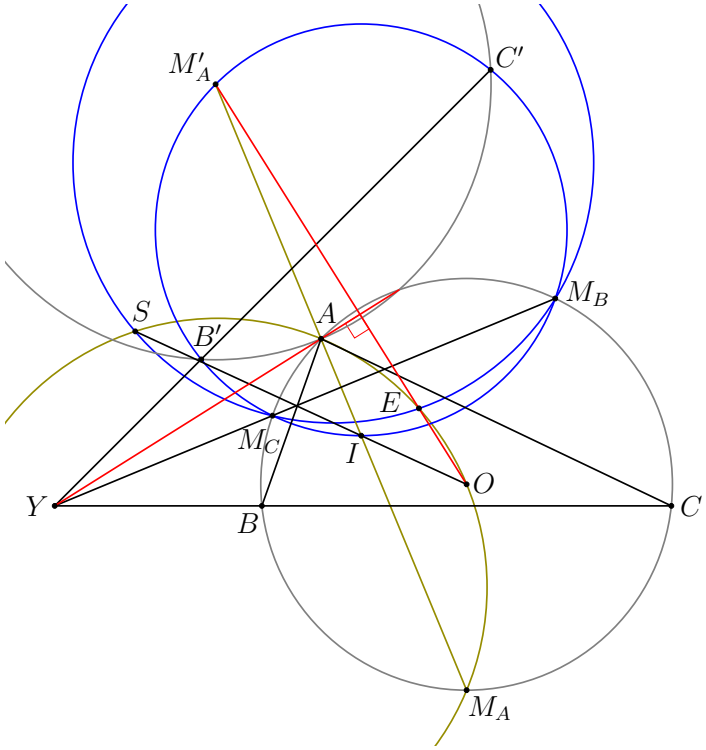
where we are using $\angle ISX = 90^\circ$. As K lies on the perpendicular bisector of SZ, this is enough to show that K is the centre of circle $SIDZ$ completing the proof.

Solution 4. Solution 1 solves the problem by establishing $\frac{SP}{SQ} = \frac{IP}{IQ} = \frac{DP}{DQ}$, which implies that circle SID is an Apollonius circle with respect to P and Q. We demonstrate an alternate approach that only requires us to show *two* of the ratios $\frac{SP}{SQ}$, $\frac{IP}{IQ}$, and $\frac{DP}{DQ}$ to be equal. This can arise from missing some of the observations in Solution 1, for example not proving Claim 3.

Claim. Given we have shown two of the ratios listed above to be equal, it suffices to show that circle SID is orthogonal to circle SXM_A, which the same circle as SPQ.

Proof. Supposing we have shown the orthogonality, if $\frac{SP}{SQ} = \frac{IP}{IQ}$ or $\frac{SP}{SQ} = \frac{DP}{DQ}$, then we immediately have that circle SID is an Apollonius circle with respect to P and Q. If $\frac{IP}{IQ} = \frac{DP}{DQ}$ and S does not lie on the Apollonius circle $\mathcal{C}$ defined by this common ratio, then I and D lie on two distinct circles orthogonal to circle SPQ, namely circle SID and $\mathcal{C}$. This implies that I and D are inverses with respect to circle SPQ, which is a contradiction as both I and D lie inside circle SPQ. □

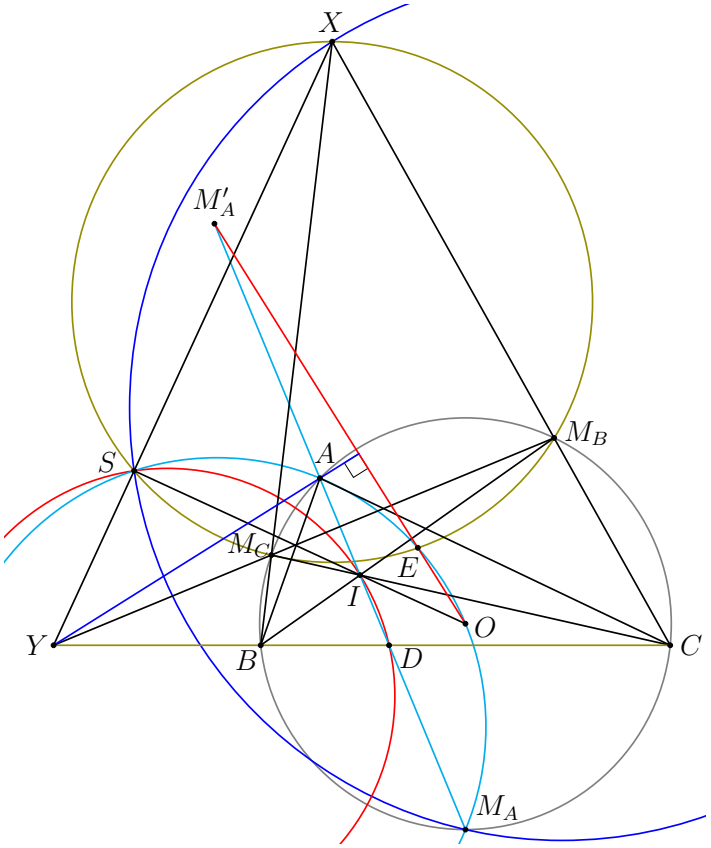
Throughout this solution, we will use the properties of S from the beginning of Solution 1. Define O and Y as in previous solutions, and let E be the second intersection of circles SOM_A and SM_BM_C.



Lemma. We have that $OE \perp AY$.

Proof. Let M'_A, B', and C' be the respective reflections of M_A, B, and C over line M_BM_C. As noted in Solution 3, A and I are reflections across M_BM_C. Because M_A is the centre of circle BIC, it follows that M'_A is the centre of circle $AB'C'$. On the other hand, Y lies on M_BM_C, so we have that $YB \cdot YC = YB' \cdot YC'$. Thus, Y lies on the radical axis of circles ABC and $AB'C'$, so $OM'_A \perp AY$. Finally, note that the inverses of circles SOM_A and SM_BM_C in circle ABC are line IM_A and circle IM_BM_C respectively, so E and M'_A are inverses in circle ABC. Thus, E lies on OM'_A and the lemma follows. □

Let $\mathcal{T}$ denote the composition of an inversion at S with radius $\sqrt{SI \cdot SO}$ with a reflection across line SI. By standard properties of the Miquel point, $\mathcal{T}$ swaps X and Y and any points Z_1 and Z_2 on circle ABC with $I \in Z_1Z_2$. Hence, $\mathcal{T}$ swaps the pairs (A, M_A), (B, M_B), (C, M_C), (O, I), and (X, Y). As $D = AI \cap BC$ and E is the intersection of circles SOM_A and SM_BM_C, we have that $\mathcal{T}(D) = E$. Thus, $\mathcal{T}$ maps circles SID and SXM_A to lines OE and AY, so by the Lemma, circles SID and SXM_A are orthogonal, as required.



G8. Let ABC be a triangle with $AB < AC < BC$, and let D be a point in the interior of segment BC. Let E be a point on the circumcircle of triangle ABC such that A and E lie on opposite sides of line BC and $\angle BAD = \angle EAC$. Let I, I_B, I_C, J_B, and J_C be the incentres of triangles ABC, ABD, ADC, ABE, and AEC, respectively.

Prove that I_B, I_C, J_B, and J_C are concyclic if and only if AI, I_BJ_C, and J_BI_C concur.

(Canada)

Solution 1. Let X be the intersection of I_BJ_C and J_BI_C. We will prove that, provided that $AB < AC < BC$, the following two conditions are equivalent:

(1) AX bisects $\angle BAC$;

(2) I_B, I_C, J_B, and J_C are concyclic.

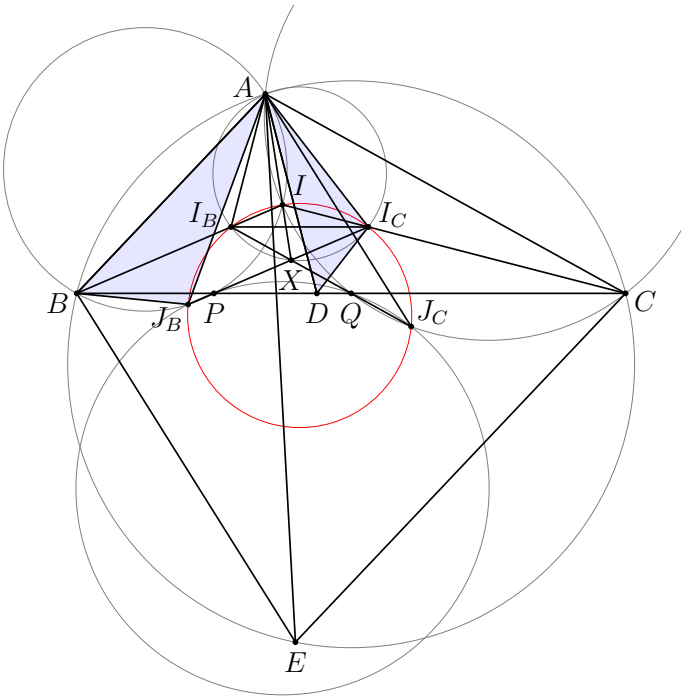
Let circles AIB and AIC meet BC again at P and Q, respectively. Note that $AB = BQ$ and $AC = CP$ because the centres of circles AIB and AIC lie on CI and BI, respectively. Thus, B, P, Q, and C are collinear in this order as $BQ + PC = AB + AC > BC$ by the triangle inequality.

Claim 1. Points P, J_B, and I_C are collinear, and points Q, I_B, and J_C are collinear.

Proof. We have that

$$\angle AJ_BB = 90^\circ + \frac{1}{2}\angle AEB = 90^\circ + \frac{1}{2}\angle ACB = \angle AIB = \angle APB,$$

so ABJ_BP is cyclic. As A is the centre of spiral similarity between ABE and ADC, it is also the centre of spiral similarity between ABJ_B and ADI_C. Hence, A is the Miquel point of self-intersecting quadrilateral BDI_CJ_B, so P lies on J_BI_C. Analogously, we have that Q lies on I_BJ_C. □



Throughout the rest of the solution, we will use directed angles.

Proof of (1) $\implies$ (2). We assume that (1) holds.

Claim 1 and the similarities $ABDI_B \sim AECJ_C$ and $ABEJ_B \sim ADCI_C$ tell us that

$$\measuredangle I_BXI_C = \measuredangle J_CQC + \measuredangle BPJ_B = \measuredangle J_CAC + \measuredangle BAJ_B = \measuredangle I_BAD + \measuredangle DAI_C = \measuredangle I_BAI_C,$$

so AI_BXI_C is cyclic. Also, as $X \in AI$, we have that

$$\measuredangle I_BAX = \measuredangle BAI - \measuredangle BAI_B = \measuredangle I_BAI_C - \measuredangle IB_AD = \measuredangle DAI_C.$$

Using these, we have that

$$\measuredangle I_BI_CP = \measuredangle I_BAX = \measuredangle DAI_C = \measuredangle BAJ_B = \measuredangle BPJ_B,$$

so $I_BI_C \parallel BC$. Hence,

$$\measuredangle I_BI_CJ_B = \measuredangle BPJ_B = \measuredangle BIJ_B = \measuredangle I_BIJ_B,$$

so $II_BJ_BI_C$ is cyclic. Analogously, we have that $II_CJ_CI_B$ is cyclic, so $I_BJ_BJ_CI_C$ is cyclic, thus proving (2). □

Proof of (2) $\implies$ (1). We assume that (2) holds.

Claim 2. Circles IBC, IJ_BI_C, and II_BJ_C are tangent at I.

Proof. Using the cyclic quadrilateral BIJ_BP, we have that

$$\measuredangle IBC = \measuredangle IBP = \measuredangle IJ_BP = \measuredangle IJ_BI_C.$$

As C, I_C, and I are collinear, the tangents to circles IJ_BI_C and IBC at I coincide, so circles IJ_BI_C and IBC are tangent at I. Analogously, circles II_BJ_C and IBC are tangent at I, so all three circles are tangent at I. □

Claim 3. Point I lies on circle $I_BJ_BJ_CI_C$.

Proof. Suppose that I does not lie on circle $I_BJ_BJ_CI_C$. Then the circles II_BJ_C, IJ_BI_C, and $I_BJ_BJ_CI_C$ are distinct. We apply the radical axis theorem to these three circles. By Claim 2, the radical axis of circles II_BJ_C and IJ_BI_C is the tangent to circle IBC at I. As I_BJ_C and J_BI_C intersect at X, IX must be tangent to circle IBC.

However, by Claim 1 we have that X is the intersection of PI_C and QI_B. As D lies on the interior of segment BC, I_B lies on the interior of segment BI and I_C lies on the interior of segment CI. Hence, I_B, P, Q, and I_C all lie on the perimeter of triangle IBC in this order, so X must be in the interior of triangle IBC. This means that IX cannot be tangent to circle BIC, so I must lie on circle $I_BJ_BJ_CI_C$. □

By Claims 2 and 3, circles II_BI_C and IBC are tangent, so $I_BI_C \parallel BC$. Since $I_BJ_BJ_CI_C$ is cyclic, we have that

$$\measuredangle PJ_BJ_C = \measuredangle I_CJ_BJ_C = \measuredangle I_CI_BJ_C = \measuredangle PQI_B = \measuredangle PQJ_C,$$

so PJ_BJ_CQ is cyclic. By the radical axis theorem on circles $AIPJ_B$, $AIQJ_C$, and PJ_BJ_CQ, we have that AI, I_BJ_C, and J_BI_C concur at X, thus proving (1). □

Solution 2. Let X be the intersection of I_BJ_C and J_BI_C. As in Solution 1, we will prove that conditions (1) and (2) are equivalent. To do so, we introduce the new condition:

(3) $I_BI_C \parallel BC$

and show that (3) is equivalent to both (1) and (2), provided that $AB < AC < BC$.

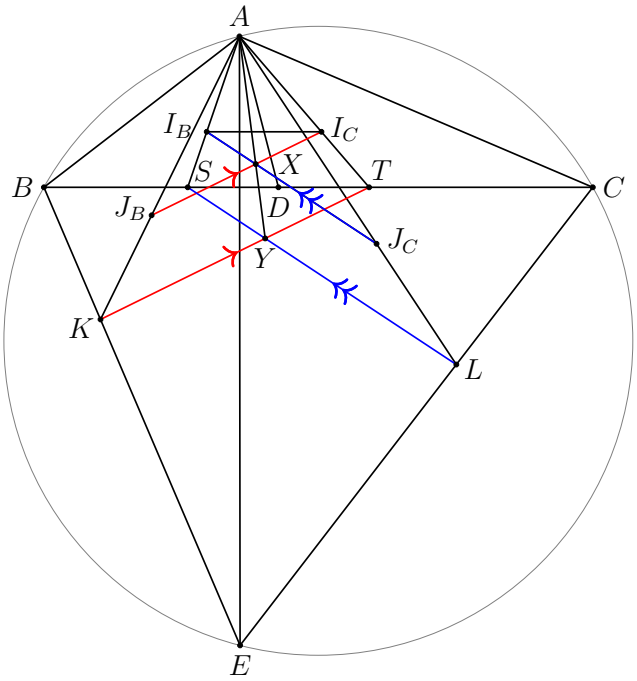
Note that $ABD \overset{+}{\sim} AEC$ and $ABE \overset{+}{\sim} ADC$, where $\overset{+}{\sim}$ denotes positive similarity. We will make use of the following fact.

Fact. For points P, P_1, P_2, P_3, and P_4, the positive similarities

$$PP_1P_2 \overset{+}{\sim} PP_3P_4 \quad \text{and} \quad PP_1P_3 \overset{+}{\sim} PP_2P_4$$

are equivalent.

Proof of (1) $\iff$ (3). Let AI_B and AI_C meet BC at S and T, respectively. Let AJ_B meet BE at K, AJ_C meet CE at L, and KT and SL meet at Y.



Claim 1. Line AY bisects $\angle BAC$.

Proof. Let Y' be the intersection of KT and the bisector of $\angle BAC$. As

$$\angle BAK = \frac{1}{2}\angle BAE = \frac{1}{2}\angle DAC = \angle TAC,$$

AY' also bisects $\angle KAT$. Hence, Y' is the foot of the bisector of $\angle KAT$ in triangle AKT. Using the Fact, we have that

$$\begin{aligned} ABE \overset{+}{\sim} ADC &\implies ABEK \overset{+}{\sim} ADCT \\ &\implies ABD \overset{+}{\sim} AKT \overset{+}{\sim} AEC \\ &\implies ABDS \overset{+}{\sim} AKTY' \overset{+}{\sim} AECL \\ &\implies ABEK \overset{+}{\sim} ASLY' \overset{+}{\sim} ADCT. \end{aligned}$$

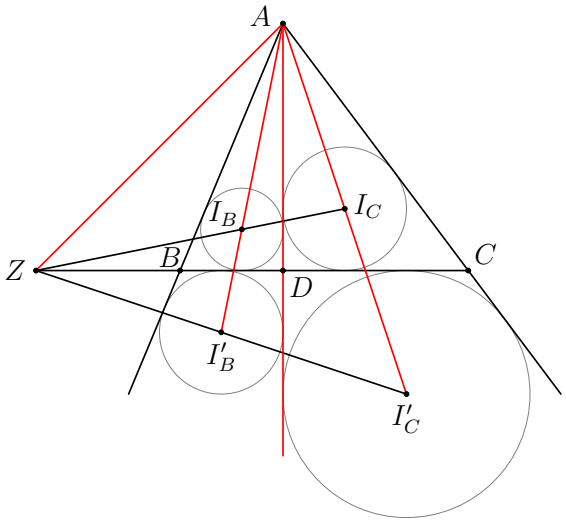
As K lies on BE, we have that Y' lies on SL, so $Y = Y'$ and AY bisects $\angle BAC$. □

We show that X lies on AY if and only if $I_BI_C \parallel BC$, which implies the equivalence of (1) and (3) by Claim 1. Let AY meet I_BJ_C and J_BI_C at X_1 and X_2, respectively. As ABD and AEC are similar, we have that $\frac{AI_B}{AS} = \frac{AJ_C}{AL}$, so $I_BJ_C \parallel SL$. Analogously, we have that $J_BI_C \parallel KT$. Hence, X_1 and X_2 coincide with X if and only if

$$\frac{AI_B}{AS} = \frac{AX_1}{AY} = \frac{AX_2}{AY} = \frac{AI_C}{AT},$$

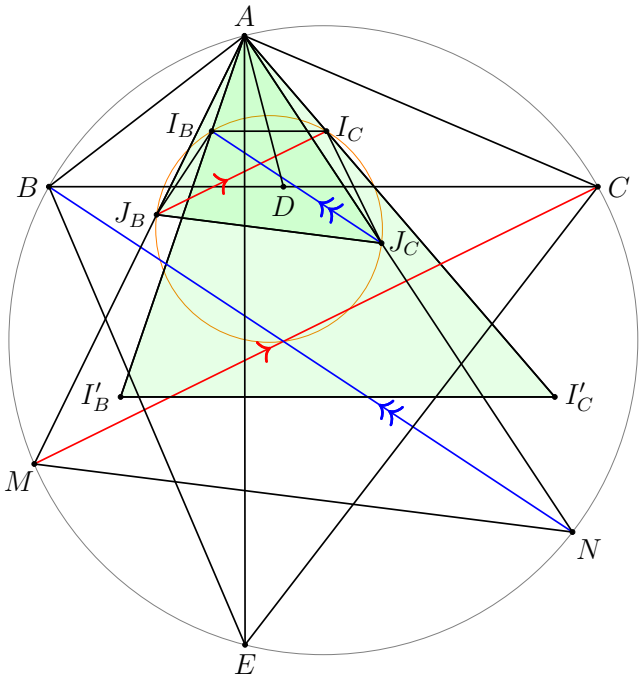
which is equivalent to $I_BI_C \parallel BC$. □

Proof of (2) $\iff$ (3). Let AJ_B and AJ_C meet circle ABC at M and N, respectively, and let I'_B and I'_C be the A-excentres of ABD and ADC, respectively.



Claim 2. Lines I_BI_C, $I'_BI'_C$, and BC are concurrent or pairwise parallel.

Proof. We work in the projective plane. Let I_BI_C and $I'_BI'_C$ meet BC at Z and Z', respectively. Note that Z is the intersection of the external common tangents of the incircles of ABD and ADC and AD is a common internal tangent of the incircles of ABD and ADC, so $(AD, AZ; AI_B, AI_C) = -1$. Applying the same argument to the A-excircles of ABD and ADC gives $(AD, AZ'; AI'_B, AI'_C) = -1$, which means that $Z = Z'$. Thus, I_BI_C, $I'_BI'_C$, and BC concur, possibly at infinity. □



Claim 3. Lines J_BI_C and CM are parallel, and lines I_BJ_C and BN are parallel.

Proof. Using the Fact, we have that

$$ABE \overset{+}{\sim} ADC \implies ABEJ_B \overset{+}{\sim} ADCI_C \implies AJ_BI_C \overset{+}{\sim} ABD.$$

Thus, $\angle(BD, J_BI_C) = \angle BAJ_B = \angle BCM$, so $J_BI_C \parallel CM$. Similarly, we have that $I_BJ_C \parallel BN$. □

Claim 4. The centre of spiral similarity between J_BJ_C and $I'_BI'_C$ is A.

Proof. As I_B and I'_B are respectively the incentre and A-excentre of triangle ABD, we have that $ABI'_B \overset{+}{\sim} AI_BD$. Using the similarity $ABD \overset{+}{\sim} AEC$, this means that $ABI'_B \overset{+}{\sim} AJ_CC$, so $AB \cdot AC = AI'_B \cdot AJ_C$ and $\angle BAI'_B = \angle J_CAC$. Similarly, we have that $AB \cdot AC = AJ_B \cdot AI'_C$ and $\angle BAJ_B = \angle I'_CAC$. Together, these imply that $AI'_B \cdot AJ_C = AJ_B \cdot AI'_C$ and $\angle J_BAJ_C = \angle I'_BAI'_C$, so $AJ_BJ_C \overset{+}{\sim} AI'_BI'_C$. □

We proceed using directed angles. By Claim 3, we have that $I_BJ_BJ_CI_C$ is cyclic if and only if

$$\begin{aligned} \measuredangle I_BI_CJ_B = \measuredangle I_BJ_CJ_B &\iff \measuredangle I_BI_CJ_B + \measuredangle MCB = \measuredangle I_BJ_CJ_B + \measuredangle MNB \\ &\iff \measuredangle(I_BI_C, BC) = \measuredangle(MN, J_BJ_C). \end{aligned}$$

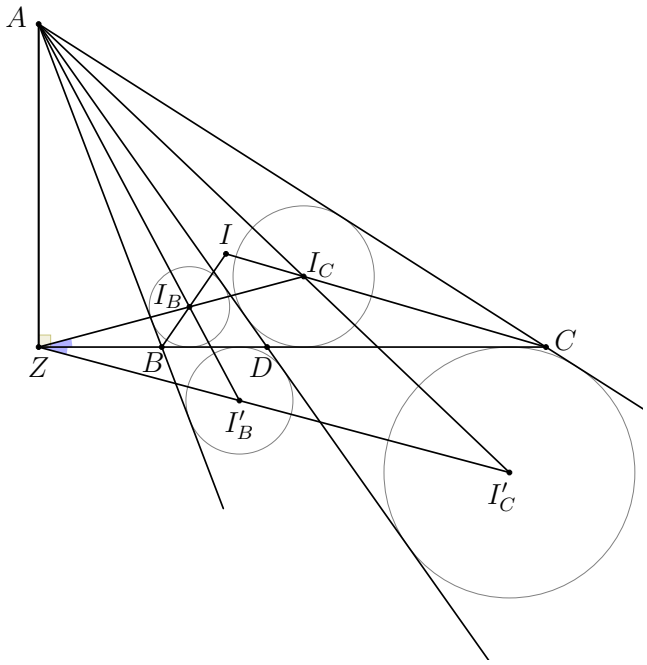
By Claim 4, we have that

$$\begin{aligned} \measuredangle(J_BJ_C, I'_BI'_C) &= \measuredangle J_BAI'_B \\ &= \measuredangle BAI_B + \measuredangle MAB \\ &= \measuredangle EAJ_C + \measuredangle MAB \\ &= \measuredangle NAC + \measuredangle MAB \\ &= \measuredangle(MN, BC), \end{aligned}$$

which is equivalent to $\measuredangle(BC, I'_BI'_C) = \measuredangle(MN, J_BJ_C)$. Thus, $I_BJ_BJ_CI_C$ is cyclic if and only if

$$\measuredangle(I_BI_C, BC) = \measuredangle(BC, I'_BI'_C). \tag{$*$}$$

Suppose that I_BI_C is parallel to BC. By Claim 2, $I'_BI'_C$ is also parallel to BC, so we have that $\measuredangle(I_BI_C, BC) = \measuredangle(BC, I'_BI'_C) = 0°$. Thus, $(*)$ is satisfied, so $I_BJ_BJ_CI_C$ is cyclic.



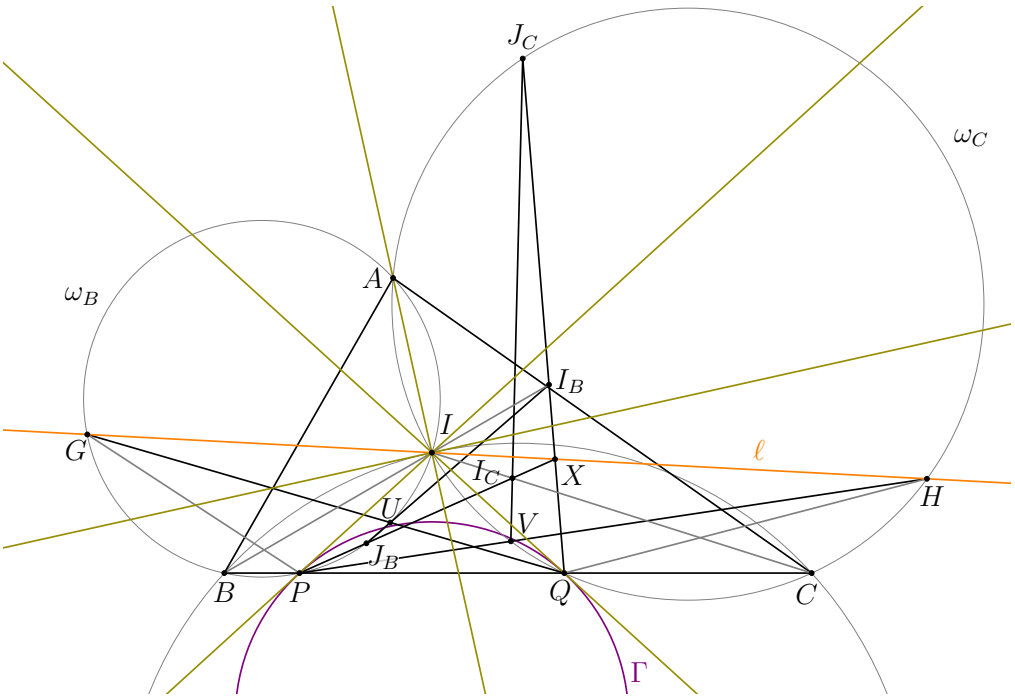
Suppose now that I_BI_C is not parallel to BC while $I_BJ_BJ_CI_C$ is cyclic. By Claim 2, I_BI_C, $I'_BI'_C$, and BC concur at a point Z. As I_B and I_C lie on segments BI and CI, Z must lie outside segment BC. Since A is the intersection of the common external tangents of the incircle and A-excircle of ABD and ZD is a common internal tangent of the incircle and A-excircle of ABD, we have that $(ZA, ZD; ZI_B, ZI'_B) = -1$. By $(*)$, ZD bisects $\angle I_BZI'_B$, so $\angle AZD = 90°$: that is, Z is the foot from A to BC. But this implies that $\angle ABC$ or $\angle BCA$ is obtuse, contradicting the fact that $AB < AC < BC$. □

Comment. While we have written the solution using harmonic bundles for the sake of brevity, there are ways to prove Claim 2 and obtain the final contradiction without the use of projective geometry. Claim 2 can be proven using an application of Menelaus's theorem, and the final contradiction can be obtained using the fact that an excircle of a triangle is always larger than its incircle.

Solution 3. Let ω_B and ω_C denote circles AIB and AIC, respectively. Introduce P, Q and X as in Solution 1 and recall from Claim 1 in Solution 1 that P, J_B and I_C are collinear with J_B lying on ω_B. From this, we can define J_B and I_C in terms of X by $I_C = XP \cap CI$ and $J_B \neq P$ as the second intersection of line XP with ω_B. Similarly, we can define $I_B = XQ \cap BI$ and $J_C \neq Q$ as the second intersection of line XQ with ω_C. Note that this now detaches the definitions of points I_B, I_C, J_B, and J_C from points D and E.

Let ℓ be a line passing through I. We now allow X to vary along ℓ while fixing $\triangle ABC$ and points I, P, and Q. We use the definitions from above to construct I_B, I_C, J_B, and J_C. We will classify all cases where these four points are concyclic. Throughout the rest of the solution we use directed angles and directed lengths.

For nondegeneracy reasons, we exclude cases where $X = I$ and X lies on line BC, which means that I_B, $J_B \neq B$ and I_C, $J_C \neq C$. We also exclude the cases where ℓ is tangent to either ω_B or ω_C. Similar results hold in these cases and they can be treated as limit cases.



Claim 1. Line I_BJ_B passes through a fixed point on ω_B, and line I_CJ_C passes through a fixed point on ω_C as X varies on ℓ.

Proof. Let $U \neq J_B$ be the second intersection of I_BJ_B with ω_B. We have by the law of sines that

$$\frac{\sin\measuredangle IJ_BU}{\sin\measuredangle UJ_BB} = \frac{\sin\measuredangle IJ_BI_B}{\sin\measuredangle I_BJ_BB} = \frac{\sin\measuredangle J_BII_B}{\sin\measuredangle J_BBI_B}\cdot\frac{II_B}{I_BB} = \frac{\sin\measuredangle J_BIB}{\sin\measuredangle J_BBI}\cdot\frac{II_B}{I_BB} = \frac{\sin\measuredangle XPQ}{\sin\measuredangle XPI}\cdot\frac{II_B}{I_BB}.$$

We also have

$$\frac{II_B}{I_BB} = \frac{\sin\measuredangle IQI_B}{\sin\measuredangle I_BQB}\cdot\frac{|IQ|}{|BQ|} = \frac{\sin\measuredangle IQX}{\sin\measuredangle XQP}\cdot\frac{|IQ|}{|BQ|}.$$

Combining these and applying Ceva's Theorem in $\triangle PIQ$ with point X, we get

$$\frac{\sin\measuredangle IJ_BU}{\sin\measuredangle UJ_BB} = \frac{\sin\measuredangle XPQ}{\sin\measuredangle XPI}\cdot\frac{\sin\measuredangle IQX}{\sin\measuredangle XQP}\cdot\frac{|IQ|}{|BQ|} = \frac{\sin\measuredangle XIQ}{\sin\measuredangle XIP}\cdot\frac{|IQ|}{|BQ|} = \frac{\sin\measuredangle(\ell,IQ)}{\sin\measuredangle(\ell,IP)}\cdot\frac{|IQ|}{|BQ|},$$

which is independent of the choice of X on ℓ. As $\measuredangle IJ_BU + \measuredangle UJ_BB = \measuredangle IJ_BB = \measuredangle IAB$ is fixed, this is enough to show point U is fixed on ω_B.

Similarly, if we define $V \neq J_C$ to be the second intersection of I_CJ_C with ω_C, we get that V is fixed on ω_C. □

Let $G \neq X$ and $H \neq X$ be the second intersections of ℓ with ω_B and ω_C, respectively which exist as we are assuming ℓ is not tangent to either ω_B or ω_C.

Claim 2. Points U, G and Q are collinear and points V, H and P are collinear.

Proof. Taking $X = G$, we have $J_B = G$ and $I_B = XQ \cap BI$. Both of these points lie on line QG which, by Claim 1, shows that U, G, Q are collinear. Similarly, V, H, P are collinear. □

Claim 3. Points I_B, I_C, J_B, J_C are concyclic if and only if points P, Q, G, H are concyclic. In particular, this depends only on ℓ, not on the choice of X on ℓ.

Proof. We have that

$$\measuredangle I_CJ_BI_B = \measuredangle PJ_BU = \measuredangle PGU = \measuredangle PGQ$$
$$\measuredangle I_CJ_CI_B = \measuredangle VJ_CQ = \measuredangle VHQ = \measuredangle PHQ.$$

Thus $\measuredangle I_CJ_BI_B = \measuredangle I_CJ_CI_B \iff \measuredangle PGQ = \measuredangle PHQ$ which proves the claim. □

Claim 4. P, Q, G, H are concyclic if and only if $\ell \in \{IA, IP, IQ, t\}$ where t is the tangent to circle BIC at I.

Proof. When $\ell = IA$, we have $G = H = A$ so the cyclic condition from Claim 3 holds. Similarly, when $\ell = IP$ or $\ell = IQ$, $G = P$ or $H = Q$, respectively, so again the cyclic condition holds.

Now, consider the case where $\ell \notin \{IA, IP, IQ\}$. In this case it is straightforward to see that the four points P, Q, G, and H are distinct. We then have that $\measuredangle QPG = \measuredangle BPG = \measuredangle BIG$, so

$$PQGH \text{ concyclic} \iff \measuredangle QHG = \measuredangle QPG \iff \measuredangle QHG = \measuredangle BIG \iff QH \parallel BI.$$

We also have that $\measuredangle CQH = \measuredangle CIH$, so

$$\ell \text{ tangent to circle } BIC \iff \measuredangle CIH = \measuredangle CBI \iff \measuredangle CQH = \measuredangle CBI \iff QH \parallel BI.$$

Hence, in this case P, Q, G, H are concyclic if and only if ℓ is tangent to circle BIC, as claimed. □

We now revert to using points D and E to define points I_B, I_C, J_B, J_C, and X, returning to the original set-up.

Claim 5. Let Γ be the circle passing through P and Q that is tangent to IP and IQ, which exists as $IP = IQ = IA$. Then X lies on Γ. Furthermore, X lies on the same side of BC as A and does not lie on line BC.

Proof. We have that

$$\begin{aligned}\measuredangle XPI &= \measuredangle J_BPI = \measuredangle J_BAI = \measuredangle BAI - \measuredangle BAJ_B = \measuredangle J_BAJ_C - \measuredangle J_BAE \\ &= \measuredangle EAJ_C = \measuredangle J_CAC = \measuredangle J_CQC = \measuredangle XQP,\end{aligned}$$

so circle XPQ is tangent to IP. Similarly, circle XPQ is tangent to IQ, so X lies on Γ.

As D lies in the interior of segment BC, I_C lies in the interior of segment CI. Since X is the second intersection of PI_C with Γ and IP is tangent to Γ, X lies in the interior of $\overset{\frown}{PQ}$ on Γ on the same side of BC as A. This implies the second part of the claim. □

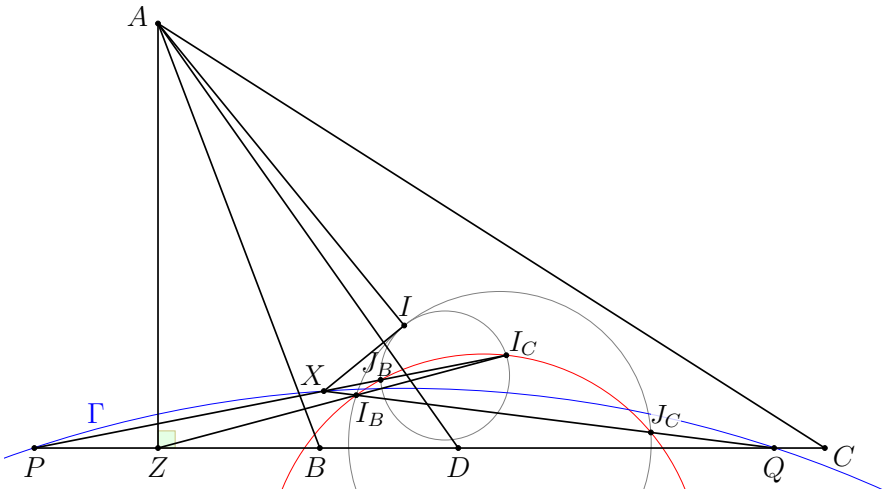
By Claim 5, we cannot have $\ell \in \{IP, IQ\}$ in the original problem. Furthermore, as shown in Claim 2 of Solution 1, we have that X lies inside triangle IBC, which means that $\ell \neq t$. Thus, the only remaining possibility in Claim 4 is $\ell = AI$. We then have

$$I_BI_CJ_BJ_C \text{ concyclic} \overset{\text{Claim 3}}{\Longleftrightarrow} PQGH \text{ concyclic} \overset{\text{Claim 4}}{\Longleftrightarrow} X \text{ lies on } AI,$$

finishing the problem.

Comment. The condition $AB < AC < BC$ is used in an essential way in the solutions. In Solution 1, it is used in the proof of Claim 3 to ensure that X lies in the interior of triangle IBC. In Solution 2, it is used in the final step to ensure that $\angle ABC$ and $\angle BCA$ cannot be obtuse. In Solution 3, it is used to exclude the case $\ell = t$. If the condition is removed, then the problem is no longer true: whenever $\angle ABC$ or $\angle BCA$ is obtuse, there exists a choice of D on BC such that $I_BJ_BJ_CI_C$ is cyclic but AI, I_BJ_C, and J_BI_C do not concur. This counterexample configuration can be constructed using Solution 3 by letting X be the intersection of t with Γ that lies on the same side of BC as A and constructing I_B, I_C, J_B, and J_C as described in the solution, from which we can reconstruct D.

Conversely, the problem holds whenever $\angle ABC$ and $\angle BCA$ are both not obtuse, as can be seen from Solution 2. This is thus the weakest possible condition on triangle ABC that is necessary for the problem to be true.



When X lies on the tangent to circle IBC at I, there is no contradiction in the proof of Claim 3 in Solution 1: circles II_BJ_C and IJ_BI_C are distinct, and X is the radical centre of circles II_BJ_C, IJ_BI_C, and $I_BJ_BI_CJ_C$. There is also no contradiction in the final step of Solution 2, and indeed I_BI_C and BC intersect at the foot of the altitude from A to BC.

There are no configuration issues with the direction (1) $\Longrightarrow$ (2). This implication holds without any constraint on triangle ABC, and the proofs in Solutions 1 and 2 apply without any modification.

Number Theory

N1. Find all positive integers n with the following property: for all positive divisors d of n, we have that $d+1 \mid n$ or $d+1$ is prime.

(Ghana)

Answer: $n \in \{1, 2, 4, 12\}$.

Solution 1. It is easy to verify that $n = 1$, 2, 4, 12 all work. We must show they are the only possibilities. We write $n = 2^k m$, where k is a nonnegative integer and m is odd. Since $m \mid n$, either $m+1$ is prime or $m+1 \mid n$.

In the former case, since $m+1$ is even it must be 2, so $n = 2^k$. If $k \geqslant 3$, we get a contradiction, since $8 \mid n$ but $9 \nmid n$. Hence $k \leqslant 2$, so $n \in \{1, 2, 4\}$.

In the latter case, we have $m+1 \mid 2^k m$ and $m+1$ coprime to m, and hence $m+1 \mid 2^k$. This means that $m+1 = 2^j$ with $2 \leqslant j \leqslant k$ (since $j = 1$ gives $m = 1$, which was considered earlier).

Then we have $2^k + 1 \nmid n$: since $2^k + 1$ is odd, it would have to divide m but is larger than m. Hence, by the condition of the problem, $2^k + 1$ is prime. If $k = 2$, j must be 2 as well, and this gives the solution $n = 12$. Also, $2^{k-1} + 1 \nmid n$ for $k > 2$: since it is odd, it would have to divide m. However, we have no solutions to $2^{k-1} + 1 \mid 2^j - 1$ with $j \leqslant k$: the left-hand side is greater than the right unless $j = k$, when the left-hand side is just over half the right-hand side.

Since we have $2^k \mid n$ and $2^k + 1 \nmid n$, and $2^{k-1} \mid n$ and $2^{k-1} + 1 \nmid n$, we must have $2^k + 1$ and $2^{k-1} + 1$ both prime. However, $2^a + 1$ is a multiple of three if a is odd, so we must have $2^k + 1 = 3$ (impossible as this gives $k = 1$) or $2^{k-1} + 1 = 3$, which gives $j = k = 2$, whence $n = 12$.

Solution 2. We proceed as in Solution 1 as far as determining that $n = 2^k(2^j - 1)$ with $j \leqslant k$.

Now, we have $2^j \mid n$ but $2^j + 1 \nmid n$, as it is odd and does not divide $2^j - 1$. Thus $2^j + 1$ is prime. The theory of Fermat primes tells us we must have $j = 2^h$ with $h > 0$.

Then $2^{2^h} - 1$ is congruent to 3 or 6 (modulo 9) depending on whether h is odd or even, respectively. In particular it is not divisible by 9, so $n = 2^k(2^{2^h} - 1)$ is not divisible by 9; so we must have $k \leqslant 2$, since if $k \geqslant 3$ then $8 \mid n$ but $9 \nmid n$ with 9 not prime.

Solution 3. Let p be the smallest integer not dividing n. Since $p-1$ is a divisor of n, p must be a prime. Let $1 \leqslant r \leqslant p-1$ be the remainder of n modulo p. Since $p - r < p$, we have $p - r \mid n$, so we may consider the divisor $d = \frac{n}{p-r}$.

Since $p \mid n - r$, we have $p \mid n + p - r$, whence $p \mid d + 1$. Thus $d + 1 \nmid n$; so it must be prime. On the other hand, this prime is divisible by p, so we conclude $d + 1 = p$, which means that $n = (p-1)(p-r)$.

Then from $p-2, p-3 \mid n$ we get $(p-2)(p-3) \mid 2(p-r)$, from which we find

$$(p-2)(p-3) \leqslant 2(p-r) \leqslant 2(p-1).$$

Solving this quadratic inequality gives $p \leqslant 5$, which means that $n \in \{1, 2, 4, 8, 12, 16\}$. Of this set, $n = 8$ and $n = 16$ are not solutions.

Solution 4. We suppose that n is not 1 or 2.

Since $n \mid n$ and $n + 1 \nmid n$, we know that $n + 1$ is prime. Thus it is odd, so $2 \mid n$; as $n > 2$, we have $\frac{n}{2} \mid n$ and $\frac{n}{2} + 1 \nmid n$, so $\frac{n}{2} + 1$ is prime. Thus it is also odd, so $4 \mid n$.

We must then have $\frac{n}{4} + 1 \mid n$ or $\frac{n}{4} + 1$ prime.

In the former case, $\frac{n}{4} \mid 4(\frac{n}{4}+1)-n$, so $\frac{n}{4}+1 \mid 4$. This means that $n=4$ or $n=12$.

In the latter case, $\frac{n}{4}+1$ must be odd if $n \neq 4$. Thus we have $n=8m$ where $2m+1$, $4m+1$, $8m+1$ are all prime; $n=8$ does not work, so $3 \mid m$ (otherwise one of those numbers would be divisible by 3). Thus $24 \mid n$, so $25 \mid n$ as 25 is not prime.

Now suppose that p is the least positive integer not dividing n: as in Solution 3 we know that p is prime, and what we have done so far shows that $p \geqslant 7$. If $p^2-1=(p-1)(p+1)$ is the product of coprime integers less than p, it divides n, and p^2 is not prime so also divides n (a contradiction); $p-1$ and $p+1$ are even and have no common factor higher than 2, so all odd prime power divisors of their product are less than p and the only case where p^2-1 is not a product of coprime integers less than p is when one of $p-1$ and $p+1$ is a power of 2, say 2^m (with $m \geqslant 3$). If $p=2^m-1$, then $3p-1=4(3\times 2^{m-2}-1)$ and $3\times 2^{m-2}-1$ is an odd integer less than p, so $3p-1 \mid n$ and so $3p \mid n$. Finally, if $p=2^m+1$, then m is even and $2p-1=2^{m+1}+1$ is a multiple of 3; the only case where it is a power of 3 is when $m=2$, but we have $m \geqslant 3$, so $2p-1$ is a product of coprime integers less than p and again we have a contradiction.

Solution 5. As in Solution 4, we deduce that if $n>2$ then n must be even. We write $n=2\cdot 3^k\cdot r$, where k is a nonnegative integer and $3 \nmid r$.

Since r and $2r$ are both different and nonzero modulo 3, one of them must be congruent to 2 modulo 3. We'll say that it is ar, where $a \in \{1,2\}$.

Since $ar \mid n$, we must have that $ar+1$ is either prime or a factor of n. In the first case, $ar+1=3$ because $3 \mid ar+1$, and so $n=2\cdot 3^k\cdot r$, where $r=2/a$ is 1 or 2. Noting that we must have $k \leqslant 1$ (else $9 \mid n$ but $10 \nmid n$), we can examine cases to deduce that $n \in \{2,4,12\}$ are the only possibilities.

Otherwise, $ar+1 \mid n$. Since $ar+1$ is coprime to r, we must in fact have that $ar+1 \mid 2\cdot 3^k$, and since $3 \mid ar+1$ by assumption we deduce that $k \geqslant 1$. In particular, 3^k+1 is an even number that is at least 4, so is not prime and must divide n. As it is coprime to 3, we must in fact have $3^k+1 \mid 2r$.

Let q_1 and q_2 be such that $q_1(ar+1)=2\cdot 3^k$ and $q_2(3^k+1)=2r$. We have that $q_1 ar<2\cdot 3^k$ and $q_2 3^k<2r$, and multiplying these together gives $q_1q_2a<4$.

If $a=2$ then $q_1=q_2=1$, so $2r+1=2\cdot 3^k$, which is not possible (considering both sides modulo 2).

If $a=1$ then r must be equivalent to 2 modulo 3, so $q_2(3^k+1)=2r$ gives that q_2 is equivalent to 1 modulo 3, whence $q_2=1$. So we deduce that $2r=3^k+1$. Thus, we deduce that $q_1(3^k+3)=4\cdot 3^k$, which rearranges to give $3^{k-1}(4-q_1)=q_1$, whence $3^{k-1}\leqslant q_1<4$ and so $k\leqslant 2$. We can examine cases to deduce that $n=12$ is the only possibility.

N2. Determine all finite, nonempty sets $\mathcal{S}$ of positive integers such that for every $a, b \in \mathcal{S}$ there exists $c \in \mathcal{S}$ with $a \mid b + 2c$.

(Netherlands)

Answer: The possible sets are $\mathcal{S} = \{t\}$ and $\mathcal{S} = \{t, 3t\}$ for any positive integer t.

Solution 1. Without loss of generality, we may divide all elements of $\mathcal{S}$ by any common factor, after which they cannot all be even. As $a \nmid b + 2c$ for a even and b odd, the elements of $\mathcal{S}$ are all odd.

We now divide into three cases:

Case 1: $|\mathcal{S}| = 1$.

The set $\mathcal{S} = \{t\}$ clearly works.

Case 2: $|\mathcal{S}| = 2$.

Say $\mathcal{S} = \{r, s\}$ with $r < s$, so either $s \mid r + 2r$ or $s \mid r + 2s$, and in either case $s \mid 3r$. We cannot have $s = 3r/2$ as we assumed that r is odd, so $s = 3r$ and $\mathcal{S} = \{r, 3r\}$, which clearly works by examining cases for a and b.

Case 3: $|\mathcal{S}| \geqslant 3$.

If all elements of $\mathcal{S}$ are odd then for any b, $c \in \mathcal{S}$, $b + 2c \not\equiv b \pmod 4$. If $a \mid b + 2c$ with $a \equiv b \pmod 4$, this means there exists k with $b + 2c = ka$ and $k \equiv 3 \pmod 4$, so $k \geqslant 3$. If a is the greatest element of $\mathcal{S}$ and $b < a$, we have $b + 2c < 3a$, a contradiction. Thus when a is the greatest element, no $b < a$ has $b \equiv a \pmod 4$ (and thus all elements other than the greatest are congruent modulo 4).

Let d and e be the largest and second largest element of $\mathcal{S}$ respectively. Let $f \neq d, e$ be any other element of $\mathcal{S}$. There is some $c \in \mathcal{S}$ with $e \mid f + 2c$, and $e \not\equiv f + 2c \pmod 4$, so $f + 2c \geqslant 3e$, so $c > e$. Since e is the second largest element of $\mathcal{S}$, $c = d$, so $e \mid f + 2d$, and this holds for all $f \in \mathcal{S}$ with $f < e$, but can only hold for at most one such f. So $|\mathcal{S}| \leqslant 3$.

Hence the elements of $\mathcal{S}$ are $d > e > f$, and by the discussion above without loss of generality we may suppose these elements are all odd, $e \equiv f \pmod 4$ and $d \not\equiv e \pmod 4$. We have above that $e \mid f + 2d$. Furthermore, there exists some $c \in \mathcal{S}$ with $d \mid f + 2c$, and $c \neq d$ as $d > f$ so $d \nmid f$, so $c \leqslant e$; as $f + 2e < 3e$, we have $e > d/3$. Since $f + 2c$ is odd and $f + 2c < 3d$, we have $f + 2c = d$.

Subcase 3.1: $c = f$.

Here $d = 3f$ and $e \mid f + 2d = 7f$. As $e > f$ and $e \equiv f \pmod 4$, we have $e = 7f/3$ and the elements are some multiples of $\{3, 7, 9\}$. But $a = 7$ and $b = 9$ have no corresponding value of c.

Subcase 3.2: $c = e$.

Here $d = f + 2e$ and $e \mid f + 2d = 3f + 4e$ so $e \mid 3f$. But this is not possible with $e > f$ and $e \equiv f \pmod 4$.

Solution 2. As in Solution 1, we reduce to the case where all elements of $\mathcal{S}$ are odd. Since all one-element sets satisfy the given conditions, we show that if $|\mathcal{S}| \geqslant 2$, then $|\mathcal{S}| = 2$ and $\mathcal{S} = \{t, 3t\}$ for some positive integer t.

Let d be the largest element. For any $e \in \mathcal{S}$ with $e \neq d$ there must be a $f \in \mathcal{S}$ such that $d \mid e + 2f$. This implies $2f \equiv -e \pmod d$, hence $2f \equiv d - e \pmod d$. Now $d - e$ is even (because all elements in $\mathcal{S}$ are odd) and d is odd, so $\frac{d-e}{2}$ is an integer and we have $f \equiv \frac{d-e}{2} \pmod d$. Further, $0 < \frac{d-e}{2} < d$, while we must also have $0 < f \leqslant d$, so $f = \frac{d-e}{2}$. We conclude that for any $e \in \mathcal{S}$ with $e \neq d$ the integer $\frac{d-e}{2}$ is also in $\mathcal{S}$ and not equal to d.

Denote by $e_1 < e_2 < \cdots < e_k < d$ the elements of $\mathcal{S}$, where $k \geqslant 1$. Then $\frac{d-e_1}{2} > \frac{d-e_2}{2} > \cdots > \frac{d-e_k}{2}$ are also elements of $\mathcal{S}$, none of them equal to d. Hence we must have $e_1 = \frac{d-e_k}{2}$ and

$e_k = \frac{d-e_1}{2}$, so $2e_1+e_k = d = 2e_k+e_1$. We conclude $e_1 = e_k$, so $k = 1$, and also $d = 2e_k+e_1 = 3e_1$. Hence $\mathcal{S} = \{e_1, 3e_1\}$ for some positive integer e_1.

Solution 3. As in Solution 1, we reduce to the case where all elements of $\mathcal{S}$ are odd. Since all one-element sets satisfy the given conditions, we show that if $|\mathcal{S}| \geqslant 2$, then $|\mathcal{S}| = 2$ and $\mathcal{S} = \{t, 3t\}$ for some positive integer t.

Let d be the largest element, and let $e \in \mathcal{S}$ be any other element. We will say that $x \in \mathcal{S} \pmod d$ if the unique element y in $\{1, \ldots, d\}$ such that $x \equiv y \pmod d$ is an element of $\mathcal{S}$. Note that by the choice of d being the largest element, if $x \neq d$, then $x \notin 0 \pmod d$. The given condition implies that if $b \in \mathcal{S}$, then $-\frac{b}{2} \in \mathcal{S} \pmod d$. Repeating this gives $-\frac{b}{2} \in \mathcal{S} \Rightarrow \frac{b}{4} \in \mathcal{S} \pmod d$, and by iterating, we have $b \in \mathcal{S} \Rightarrow \frac{b}{(-2)^k} \in \mathcal{S} \pmod d$ for all k. Since d is odd, there is some g such that $(-2)^g \equiv 1 \pmod d$, so by setting $k = g - 1$, we get that

$$\text{for all } d \neq e \in \mathcal{S}, -2e \in \mathcal{S} \pmod d.$$

Now, if $e > \frac{d}{2}$, then $-2e \in \mathcal{S} \pmod d$ and $d - 2e < 0$, so $2d - 2e \in \mathcal{S}$, contradicting the lack of even elements. Then $e < \frac{d}{2}$ for any $e \in \mathcal{S} \setminus \{d\}$, so we have $e \in \mathcal{S} \Rightarrow d - 2e \in \mathcal{S}$. Since $d - 2e \neq d$, we must have $d - 2e < \frac{d}{2}$, which rearranges to $e > \frac{d}{4}$.

Let $\lambda \in (0, 1)$ be a positive real number and suppose we have proved that $e > \lambda d$ for any $e \in \mathcal{S} \setminus \{d\}$. Then $d - 2e > \lambda d$, which rearranges to $e < \frac{(1-\lambda)d}{2}$. Then $d - 2e < \frac{(1-\lambda)d}{2}$, which rearranges to $e > \frac{(1+\lambda)d}{4}$. Defining $\lambda_0 = \frac{1}{4}$ and $\lambda_i = \frac{1+\lambda_{i-1}}{4}$ for $i \geqslant 1$, we have shown that for all $e \in \mathcal{S} \setminus \{d\}$ and all λ_i, $e > \lambda_i d$. Now note that the sequence λ_i is increasing and bounded above by $\frac{1}{3}$, so it converges to some limit ℓ, which satisfies $\ell = \frac{1+\ell}{4}$, so $\ell = \frac{1}{3}$. Hence $e \geqslant \frac{d}{3}$, but then $d - 2e \geqslant \frac{d}{3}$ implies $e \leqslant \frac{d}{3}$, so e must be $\frac{d}{3}$, and we are done.

Comment. We can finish Solution 3 alternatively as follows: after showing that if $e \in \mathcal{S} \setminus \{d\}$ then $d - 2e \in \mathcal{S} \setminus \{d\}$, note that

$$(d - 2e) - \frac{d}{3} = \frac{2d}{3} - 2e = -2\left(e - \frac{d}{3}\right).$$

So consider $e \in \mathcal{S} \setminus \{d\}$ maximising $|e - \frac{d}{3}|$. If $e \neq \frac{d}{3}$, them the above shows that $|(d-2e) - \frac{d}{3}| > |e - \frac{d}{3}|$, which is a contradiction. Thus $\mathcal{S} \setminus \{d\}$ is empty or equal to $\{\frac{d}{3}\}$, which completes the proof.

N3. Determine all sequences $a_1, a_2, \ldots$ of positive integers such that, for any pair of positive integers $m \leqslant n$, the arithmetic and geometric means

$$\frac{a_m + a_{m+1} + \cdots + a_n}{n-m+1} \quad \text{and} \quad (a_m a_{m+1} \cdots a_n)^{\frac{1}{n-m+1}}$$

are both integers.

(Singapore)

Answer: The only such sequences are the constant sequences (which clearly work).

Solution 1. We say that an integer sequence $b_1, b_2, \ldots$ is *good* if for any pair of positive integers $m \leqslant n$, the arithmetic mean $\frac{b_m + b_{m+1} + \cdots + b_n}{n-m+1}$ is an integer. Then the condition in the question is equivalent to saying that the sequences (a_i) and $(\nu_p(a_i))$ for all primes p are good.

Claim 1. If (b_i) is a good sequence, then $n - m \mid b_n - b_m$ for all pairs of integers m, n.

Proof. This follows from $n-m$ dividing $b_m + b_{m+1} + \cdots + b_{n-1}$ and $b_{m+1} + b_{m+2} + \cdots + b_n$, and then taking the difference. □

Claim 2. If (b_i) is a good sequence where some integer b occurs infinitely many times, then (b_i) is constant.

Proof. Say $b_{n_1}, b_{n_2}, b_{n_3}, \ldots$ are equal to b. Then for all m, we have that $b - b_m = b_{n_j} - b_m$ is divisible by infinitely many different integers $n_j - m$, so it must be zero. Therefore the sequence is constant. □

Now, for a given prime p, we look at the sequence $(\nu_p(a_i))$. Let $k = \nu_p(a_1)$. Then Claim 1 tells us that $a_1 \equiv a_{np^{k+1}+1} \pmod{p^{k+1}}$ for all n, which implies that $\nu_p(a_{np^{k+1}+1}) = k$ for all n. We now have that k appears infinitely many times in this good sequence, so by Claim 2, the sequence $(\nu_p(a_i))$ is constant. This holds for all primes p, so (a_i) must in fact be constant.

Solution 2. As in Claim 1 of Solution 1, we have that $a_{i+r} \equiv a_i \pmod{r}$, which tells us that the sequence a_i is periodic modulo p with period p. Also, by a similar argument, we have that a_{i+r}/a_i is the rth power of a rational number.

Now suppose that for some $i \not\equiv j \pmod{p}$ we have $a_i, a_j \not\equiv 0 \pmod{p}$. As p and $p-1$ are coprime, we can find some $i' \equiv i \pmod{p}$, $j' \equiv j \pmod{p}$ such that $p - 1 \mid i' - j'$. Then $a_{i'}/a_{j'}$ is a perfect $(p-1)^{\text{th}}$ power, so

$$a_{i'} = tu^{p-1}, \quad a_{j'} = tv^{p-1}$$

for some positive integers t, u, v not divisible by p. By Fermat's little theorem, u^{p-1} and v^{p-1} must be 1 modulo p. So we must have

$$a_i \equiv a_{i'} \equiv t \equiv a_{j'} \equiv a_j \pmod{p}.$$

Thus all values of a_i that are not divisible by p are congruent modulo p.

For the sum of p consecutive values to be divisible by p, this means that all the a_i are congruent modulo p. Since this is true for all primes p, the sequence must therefore be constant.

Solution 3. Fix an arbitrary index m. First, we show that a_m divides a_n for sufficiently large n. Let n be sufficiently large that $n > \nu_p(a_m) + m$ for every prime p. By Claim 1 of Solution 1, we have

$$\nu_p(a_m) \equiv \nu_p(a_n) \pmod{n-m}.$$

Since $\nu_p(a_m) < n-m$, it follows that $\nu_p(a_m) \leqslant \nu_p(a_n)$. This holds for every prime p, so $a_m \mid a_n$.

Next, suppose that there is some index k such that a_m does not divide a_k. By the previous, there is a maximal such k. Then a_{k+1}, a_{k+2}, ... are all divisible by a_m. But now applying the first condition gives

$$a_m \mid a_k + a_{k+1} + \cdots + a_{k+a_m-1},$$

so a_m divides a_k, a contradiction. Therefore every term a_n is divisible by a_m.

As m was arbitrary, we now have $a_m \mid a_n$ and vice versa for all m, n. So the sequence must be constant.

N4. Determine all positive integers a and b such that there exists a positive integer g such that $\gcd(a^n + b, b^n + a) = g$ for all sufficiently large n.

(Indonesia)

Answer: The only solution is $(a, b) = (1, 1)$.

Solution 1. It is clear that we may take $g = 2$ for $(a, b) = (1, 1)$. Supposing that (a, b) satisfies the conditions in the problem, let N be a positive integer such that $\gcd(a^n + b, b^n + a) = g$ for all $n \geqslant N$.

Lemma. We have that $g = \gcd(a, b)$ or $g = 2\gcd(a, b)$.

Proof. Note that both $a^N + b$ and $a^{N+1} + b$ are divisible by g. Hence

$$a(a^N + b) - (a^{N+1} + b) = ab - b = a(b - 1)$$

is divisible by g. Analogously, $b(a - 1)$ is divisible by g. Their difference $a - b$ is then divisible by g, so g also divides $a(b - 1) + a(a - b) = a^2 - a$. All powers of a are then congruent modulo g, so $a + b \equiv a^N + b \equiv 0 \pmod{g}$. Then $2a = (a + b) + (a - b)$ and $2b = (a + b) - (a - b)$ are both divisible by g, so $g \mid 2\gcd(a, b)$. On the other hand, it is clear that $\gcd(a, b) \mid g$, thus proving the Lemma. □

Let $d = \gcd(a, b)$, and write $a = dx$ and $b = dy$ for coprime positive integers x and y. We have that

$$\gcd\left((dx)^n + dy, (dy)^n + dx\right) = d\gcd\left(d^{n-1}x^n + y, d^{n-1}y^n + x\right),$$

so the Lemma tells us that

$$\gcd\left(d^{n-1}x^n + y, d^{n-1}y^n + x\right) \leqslant 2$$

for all $n \geqslant N$. Defining $K = d^2xy + 1$, note that K is coprime to each of d, x, and y. By Euler's theorem, for $n \equiv -1 \pmod{\varphi(K)}$ we have that

$$d^{n-1}x^n + y \equiv d^{-2}x^{-1} + y \equiv d^{-2}x^{-1}(1 + d^2xy) \equiv 0 \pmod{K},$$

so $K \mid d^{n-1}x^n + y$. Analogously, we have that $K \mid d^{n-1}y^n + x$. Taking such an n which also satisfies $n \geqslant N$ gives us that

$$K \mid \gcd(d^{n-1}x^n + y, d^{n-1}y^n + x) \leqslant 2.$$

This is only possible when $d = x = y = 1$, which yields the only solution $(a, b) = (1, 1)$.

Solution 2. After proving the Lemma, one can finish the solution as follows.

For any prime factor p of $ab + 1$, p is coprime to a and b. Take an $n \geqslant N$ such that $n \equiv -1 \pmod{p - 1}$. By Fermat's little theorem, we have that

$$a^n + b \equiv a^{-1} + b = a^{-1}(1 + ab) \equiv 0 \pmod{p},$$
$$b^n + a \equiv b^{-1} + a = b^{-1}(1 + ab) \equiv 0 \pmod{p},$$

then p divides g. By the Lemma, we have that $p \mid 2\gcd(a, b)$, and thus $p = 2$. Therefore, $ab + 1$ is a power of 2, and a and b are both odd numbers.

If $(a, b) \neq (1, 1)$, then $ab + 1$ is divisible by 4, hence $\{a, b\} = \{-1, 1\} \pmod 4$. For odd $n \geqslant N$, we have that

$$a^n + b \equiv b^n + a \equiv (-1) + 1 = 0 \pmod 4,$$

then $4 \mid g$. But by the Lemma, we have that $\nu_2(g) \leqslant \nu_2(2\gcd(a, b)) = 1$, which is a contradiction. So the only solution to the problem is $(a, b) = (1, 1)$.

N5. Let $\mathcal{S}$ be a finite nonempty set of prime numbers. Let $1 = b_1 < b_2 < \cdots$ be the sequence of all positive integers whose prime divisors all belong to $\mathcal{S}$. Prove that, for all but finitely many positive integers n, there exist positive integers $a_1, a_2, \ldots, a_n$ such that

$$\frac{a_1}{b_1} + \frac{a_2}{b_2} + \cdots + \frac{a_n}{b_n} = \left\lceil \frac{1}{b_1} + \frac{1}{b_2} + \cdots + \frac{1}{b_n} \right\rceil.$$

(Croatia)

Solution 1. If $\mathcal{S}$ has only one element p, then $b_i = p^{i-1}$ and we can easily find $a_1, \ldots, a_n$ with $2 = \left\lceil \sum_{i=0}^{n-1} \frac{1}{p^i} \right\rceil = \sum_{i=0}^{n-1} \frac{a_i}{p^{i-1}}$ by taking $a_1 = a_2 = \cdots = a_{n-1} = 1$ and choosing $a_n = p^{n-1} - (p + p^2 + \ldots + p^{n-2})$.

More generally, observe that the sum of $\frac{1}{b_i}$ over all i is

$$\begin{aligned} \sum_i \frac{1}{b_i} &= \prod_i \left(1 + \frac{1}{p_i} + \frac{1}{p_i^2} + \ldots\right) \\ &= \prod_{p \in \mathcal{S}} \frac{p}{p-1}. \end{aligned}$$

In particular, if n is large enough, then

$$\left\lceil \sum_{j=1}^{n} \frac{1}{b_j} \right\rceil = \left\lceil \prod_{p \in \mathcal{S}} \frac{p}{p-1} \right\rceil.$$

For the remainder of the proof, we will only consider n large enough that this equality holds.

Next, we handle the special case $\mathcal{S} = \{2, 3\}$, for which this product is 3. Start by setting

$$a_i = \begin{cases} 1, & \text{if } 2b_i \leqslant b_n; \\ 2, & \text{if } 2b_i > b_n. \end{cases}$$

Then,

$$\sum_{\substack{i \leqslant n \\ \nu_3(b_i) = t}} \frac{a_i}{b_i} = \begin{cases} \frac{2}{3^t}, & \text{if } b_n \geqslant 3^t; \\ 0, & \text{otherwise.} \end{cases}$$

As a result,

$$\begin{aligned} \sum_{i \leqslant n} \frac{a_i}{b_i} &= \sum_{\substack{t \geqslant 0 \\ 3^t \leqslant b_n}} \frac{2}{3^t} \\ &= 3 - \frac{1}{3^T} \end{aligned}$$

where T is the largest $t \geqslant 0$ with $3^t \leqslant b_n$. Thus, increasing a_j by one (where $b_j = 3^T$) gives a sequence of a_i that works.

Otherwise, we may assume that $|\mathcal{S}| > 1$ and $\mathcal{S} \neq \{2, 3\}$, which means that the product $\prod_{p \in \mathcal{S}} \frac{p}{p-1}$ is not an integer. Indeed,

- if $|\mathcal{S}| > 2$ then 2 divides the denominator at least twice and so divides the denominator of the overall fraction;
- if $|\mathcal{S}| = 2$ and $2 \notin \mathcal{S}$ then 2 divides the denominator and not the numerator;

- if $\mathcal{S} = \{2, p\}$ then the product is $2p/(p-1)$ which is not an integer for $p > 3$.

It follows that for some fixed $\alpha > 0$, we have that

$$\left\lceil \prod_{p \in \mathcal{S}} \frac{p}{p-1} \right\rceil = \prod_{p \in \mathcal{S}} \frac{p}{p-1} + \alpha,$$

from which it follows that

$$\left\lceil \sum_{i=1}^{n} \frac{1}{b_i} \right\rceil - \sum_{i=1}^{n} \frac{1}{b_i} > \alpha.$$

It will now suffice to prove the following claim.

Claim. Suppose that n is large enough, and let e_p be the largest nonnegative integer such that $p^{e_p} \leqslant b_n$. Let $M = \prod_{p \in \mathcal{S}} p^{e_p}$. If u is a positive integer such that $u/M > \alpha$, then there exist nonnegative integers a_i such that

$$\sum_i \frac{a_i}{b_i} = \frac{u}{M}.$$

The problem statement follows after replacing a_i with $a_i + 1$ for each i.

To prove this, choose some constant c such that $\sum_{p \in \mathcal{S}} p^{-c} < \alpha$, and suppose n is large enough that $p^c < b_n$ for each $p \in \mathcal{S}$; in particular, $p^c \mid M$ with M defined as above.

For each $p \in \mathcal{S}$, let i_p be such that $b_{i_p} = p^{e_p}$ and choose the smallest nonnegative integer a_{i_p} satisfying

$$p^{e_p - c} \,\Big|\, a_{i_p} \left(\frac{M}{p^{e_p}} \right) - u.$$

Such an a_{i_p} must exist and be at most $p^{e_p - c}$; indeed, $\frac{M}{p^{e_p}}$ is an integer coprime to p, so we can take a_{i_p} to be equal to u times its multiplicative inverse modulo $p^{e_p - c}$. The sum of the contributions to the sum from the a_{i_p} is at most

$$\sum_{p \in \mathcal{S}} \frac{p^{e_p - c}}{p^{e_p}} = \sum_{p \in \mathcal{S}} p^{-c} < \alpha.$$

So, we have

$$\frac{u}{M} = \sum_{p \in \mathcal{S}} \frac{a_{i_p}}{p^{e_p}} + \frac{r}{\prod_{p \in \mathcal{S}} p^c},$$

where r is an integer because of our choice of a_{i_p} and r is nonnegative because of the bound on u. Simply choose $a_i = r$ where $b_i = \prod_{p \in \mathcal{S}} p^c$ to complete the proof.

Solution 2. We reduce to the claim as in Solution 1, and provide an alternative approach for constructing the a_i.

Let $p_0 \in \mathcal{S}$ be the smallest prime in $\mathcal{S}$. Let $z_0 = u/M$. We construct a sequence z_0, z_1, z_2, … and values of a_i by the following iterative process: to construct z_{j+1},

- select the largest prime $p \in \mathcal{S}$ dividing the denominator of z_j, and let μ be the number of times p divides the denominator of z_j;
- choose the largest ν such that $p_0^\nu p^\mu \leqslant b_n$, and let $i \leqslant n$ be such that $b_i = p_0^\nu p^\mu$;
- choose $0 \leqslant a_i < p$ such that the denominator of $z_k - a_i/b_i$ has at most $\mu - 1$ factors of p, and let $z_{k+1} = z_k - a_i/b_i$;
- continue until p_0 is the only prime dividing the denominator of z_k.

Note that we can always choose a_i in step 3; by construction, $z_k b_i$ has no factors of p in its denominator, so must be realised as an element of $\mathbb{Z}_p$.

Each time we do this, $b_i > M/p_0$ by construction, so

$$\frac{a_i}{b_i} < \frac{pp_0}{M} \leqslant \frac{p_0 p_1}{M},$$

where p_1 is the largest prime in $\mathcal{S}$. And the number of times we do this operation is at most

$$\sum_{\substack{p \in \mathcal{S} \\ p > p_0}} e_p \leqslant |\mathcal{S}| \log_2(M),$$

so the sum of the a_i/b_i we have assigned is at most $|\mathcal{S}| p_0 p_1 \log_2(M)/M$.

Choose n large enough that $\log_2(M)/M < \alpha$; after subtracting the above choices of a_i/b_i from u/M, we have a quantity of the form $r/p_0^{e_{p_0}}$, where r is an integer by construction and r is positive by the above bounds. Simply set $a_i = r$ where $b_i = p_0^{e_{p_0}}$ to complete the proof.

Solution 3. As in Solution 1, we may handle $|\mathcal{S}| = 1$ and $\mathcal{S} = \{2, 3\}$ separately; otherwise, we can define α as we did in that solution. Also define e_p to be the largest nonnegative integer such that $p^{e_p} \leqslant b_n$ as we did in Solution 1.

We will show that, for n sufficiently large, we may choose some $j \leqslant n$, and positive integers a_i, such that

$$\sum_{i \neq j} \frac{a_i}{b_i} - \sum_{i \neq j} \frac{1}{b_i} < \alpha.$$

and all $\frac{a_i}{b_i}$ are integer multiples of $\frac{1}{b_j}$. We then set a_j to be the least positive integer such that the sum on the left is an integer, which will obviously have the required value.

Concretely, choose j such that $b_j = \prod_{p \in \mathcal{S}} p^{\lfloor e_p/|\mathcal{S}| \rfloor}$, which is less than b_n by construction. For $i \neq j$, set $a_i = b_i / \gcd(b_i, b_j)$. We have

$$\sum_{i \neq j} \frac{a_i}{b_i} - \sum_{i \neq j} \frac{1}{b_i} < \sum_{\substack{i \neq j \\ a_i > 1}} \frac{a_i}{b_i}.$$

If $a_i > 1$, then there must be some $p \in \mathcal{S}$ for which $p^{\lfloor e_p/|\mathcal{S}| \rfloor + 1} \mid b_i$, and so

$$\frac{a_i}{b_i} = \frac{1}{\gcd(b_i, b_j)} \leqslant \frac{1}{p^{\lfloor e_p/|\mathcal{S}| \rfloor}} < \frac{p}{b_n^{1/|\mathcal{S}|}},$$

where the last inequality follows from the fact that $p^{e_p+1} > b_n$.

Now $n \leqslant \prod_{p \in \mathcal{S}} (\log_p(b_n) + 1) \leqslant (2 \log b_n)^{|\mathcal{S}|}$, so

$$\sum_{\substack{i \neq j \\ a_i > 1}} \frac{a_i}{b_i} \leqslant \frac{(2 \log b_n)^{|\mathcal{S}|}}{b_n^{1/|\mathcal{S}|}},$$

and so we can choose n large enough that this quantity is less than α, as required.

N6. Let n be a positive integer. We say that a polynomial P with integer coefficients is *n-good* if there exists a polynomial Q of degree 2 with integer coefficients such that $Q(k)(P(k)+Q(k))$ is never divisible by n for any integer k.

Determine all integers n such that every polynomial with integer coefficients is an n-good polynomial.

(France)

Answer: The set of such n is any $n > 2$.

Solution 1. First, observe that no polynomial is 1-good (because $Q(X)(P(X)+Q(X))$ always has roots modulo 1) and the polynomial $P(X) = 1$ is not 2-good (because $Q(X)(Q(X)+1)$ is always divisible by 2).

Now, if P is d-good with some Q, then $Q \cdot (P+Q)$ has no roots mod d. Therefore, it certainly has no roots mod n for $d \mid n$, so P must be n-good. Consequently, it suffices to show that all polynomials are n-good whenever n is an odd prime, or $n = 4$.

We start by handling the case $n = 4$. We will construct a Q such that $Q(X)$ is never divisible by 4 and $Q(X) + P(X)$ is always odd; this will clearly show that P is 4-good. Note that any function modulo 2 must be either constant or linear – in other words, there are $a, b \in \{0, 1\}$ such that $P(X) = aX + b \bmod 2$ for all X. If $a = 0$ then set $Q(X) = 4X^2 + b + 1$, and if $a = 1$ then set $Q(X) = X^2 + b + 1$; in all cases, Q will satisfy the required properties.

It remains to prove that any polynomial is p-good, where p is an odd prime. We will prove that for any *function* f defined mod p, there is a quadratic Q with no roots mod p such that $Q(x) \neq f(x) \bmod p$ for all x; the statement about P then follows with f replaced by $-P$. For the remainder of the proof, we will consider all equalities modulo p.

Suppose that a function f not satisfying the above exists; in other words, f has the property that for any quadratic Q with no roots mod p, there is some x such that $Q(x) = f(x)$. Without loss of generality, we may assume that f has no roots mod p. To see why, suppose that $f(u) = 0$ for some u, and let g be the function such that $g(x) = f(x)$ for $x \neq u$ and $g(u) = 1$. For any Q with no roots, we know that there is some $x \neq u$ such that $P(x) = f(x)$, and so $P(x) = g(x)$ for that choice of x. In particular, g is also not p-good.

Now, suppose first that there is some nonzero t such that t is not in the image of f. Then we may take $Q(X) = pX^2 + t$; this quadratic is never equal to f and is never zero. Thus, f must be surjective onto the nonzero residues mod p. There are p choices for X and $p-1$ nonzero residues mod p, so there must be some $x_1 \neq x_2 \bmod p$ such that $f(x_1) = f(x_2)$, and f is a bijection from the set of residues mod p not equal to x_2 to the set of nonzero residues mod p.

Now, note that we may choose any b and c with b nonzero and replace $f(X)$ with $g(X) = f(bX+c)$; if there were some Q with no roots such that $Q(x) \neq g(x)$ for all x, then $Q(X/b-c/b)$ would work for f. Choose b and c such that $bx_1 + c = 1$ and $bx_2 + c = -1$; such b and c must exist (we may take $b = 2/(x_1 - x_2)$ and $c = (x_1 + x_2)/(x_2 - x_1)$). Renaming g to f, we see that we may assume $f(1) = f(-1)$.

Let r' be a quadratic nonresidue mod p. Choose $y \neq 0$ such that $f(y) = (1-r')f(0)$, which must exist as the right hand side is nonzero and $1 - r'$ is not equal to 1. Choose $r = y^2/r'$, which is a quadratic nonresidue.

Consider $\phi(X) = f(X)/(X^2 - r)$. By definition, $\phi(1) = \phi(-1)$ and $\phi(0) = \phi(y)$, so there are no more than $p-2$ values in the image of ϕ. Choose some nonzero a not in the image of ϕ, so $f(X)/(X^2 - r)$ is never equal to a. The quadratic $Q(X) = a(X^2 - r)$ is never zero and also never equal to $f(X)$, which completes the proof.

Comment. In fact, there is no need to pass from polynomials P to functions f, as any function mod p is a polynomial. Concretely, instead of passing from f to g, we would have instead replaced $P(X)$ with $P(X) + 1 - (X - u)^{p-1}$, which is a polynomial that is unchanged except at $X = u$.

Solution 2. Given f a function mod p such that f is surjective onto the nonzero elements of $\mathbb{Z}/p\mathbb{Z}$ and $f(1) = f(-1)$, we provide an alternative approach to construct a nonzero quadratic $Q(X)$ such that $Q(X) \neq f(X)$. Let r be the smallest quadratic nonresidue mod p (so $r - 1$ is a square) and let a vary over the nonzero elements mod p; we will show that it is possible to choose $Q_a(X) = a(X^2 - r)$ for some choice of a. Note that any quadratic of this form will be nowhere zero.

Suppose that no such Q_a works. Then, for each a, there exists x such that $a(x^2 - r) = f(x)$. We may assume that $x \neq -1$, as if the equality holds for $x = -1$ then it also holds for $x = 1$. However, $a(x^2 - r) = f(x)$ implies $a = f(x)/(x^2 - r)$, so $f(x)/(x^2 - r)$ must be a surjection from $\{x \neq -1\}$ to the set of nonzero a, and so this is a bijection. In particular, for each a, there exists a unique x_a such that $f(x_a) = a(x_a^2 - r)$.

We now have

$$\begin{aligned}\prod_{t\neq 0} t &= \prod_{a\neq 0} f(x_a) \\ &= \prod_{a\neq 0} a \prod_{a\neq 0} (x_a^2 - r) \\ &= \prod_{a\neq 0} a \prod_{x\neq -1} (x^2 - r)\end{aligned}$$

where the first equality follows because f is surjective onto the nonzero residues mod p, and the second equality follows from the definition of x_a. The two products cancel, which means that $\prod_{x\neq -1}(x^2 - r) = 1$.

However, we also get

$$\prod_{x\neq -1} (x^2 - r) = (-r)(1 - r) \left(\prod_{x=2}^{(p-1)/2} (x^2 - r) \right)^2.$$

However, this is a contradiction as $-r(1 - r) = r(r - 1)$, which is not a quadratic residue (by our choice of r).

Comment. By Wilson's theorem, we know that the product of the nonzero elements mod p is -1; however, this fact was not necessary for the solution so we chose to present the solution without needing to state it.

Comment. One can in fact show that

$$\prod_{x\neq -1} (x^2 - r) = \frac{-4r}{1 - r}.$$

To do this, note that the polynomial $X^{\frac{p-1}{2}} - 1$ has the $\frac{p-1}{2}$ quadratic residues as roots, so we have

$$\prod_{s \text{ quad. res.}} (X - s) = X^{\frac{p-1}{2}} - 1$$

and so

$$\prod_{x\neq 0} (X - x^2) = (X^{\frac{p-1}{2}} - 1)^2.$$

Since r is a quadratic nonresidue, by Euler's criterion $r^{\frac{p-1}{2}} = -1$, and the result follows.

Therefore, one can replace the condition that r is the smallest quadratic nonresidue with the condition that r is a quadratic nonresidue not equal to $-\frac{1}{3}$ (which is possible for all $p \geqslant 3$).

Solution 3. As in Solution 1, we will reduce to the case of p being an odd prime and f being a function mod p with no roots which is surjective onto the set of nonzero residues mod p, although we make no assumption about the values of x_1 and x_2 with $f(x_1) = f(x_2)$.

We will again consider quadratics of the form $Q_{a,b,c}(X) = aR(bX+c)$, where $R(X) = X^2 - r$ for an arbitrary fixed quadratic nonresidue r, a and b are nonzero mod p, and c is any residue mod p.

For each fixed b and c, there must be n pairs (a, x) such that $aR(bx + c) = f(x)$, because there must be exactly one value of a for each x. If any a appears in no such pair then we are done, so assume otherwise. In other words, there must be exactly one a such that there are two such x, and for all other a there is only one such x.

Thus, for each (b, c), there is exactly one unordered pair $\{x_1, x_2\}$ such that for some a we have $f(x_i) = aR(bx_i + c)$; in other words, there is exactly one unordered pair $\{x_1, x_2\}$ such that $f(x_1)/R(bx_1 + c) = f(x_2)/R(bx_2 + c)$.

Now, we show that for each unordered pair $\{x_1, x_2\}$ there must be at least one pair (b, c) such that $f(x_1)/R(bx_1 + c) = f(x_2)/R(bx_2 + c)$. Indeed, let $t = f(x_1)/f(x_2)$. There must be some x_1', x_2' such that $R(x_1')/R(x_2') = t$; this is because $R(X)$ and $tR(X)$ both take $\frac{p+1}{2}$ nonzero values mod p, so the intersection must be nonempty by the pigeonhole principle. Choosing b and c such that $bx_1 + c = x_1'$ and $bx_2 + c = x_2'$ gives the claim.

Note further that if (b, c) and $\{x_1, x_2\}$ satisfy the relation, then the same is true for $(-b, -c)$ and $\{x_1, x_2\}$ because $R(bx + c) = R(-bx - c)$. Since b is nonzero, this means that each pair $\{x_1, x_2\}$ corresponds to at least two pairs (b, c). However, since there are $p(p - 1)$ pairs (b, c) with b nonzero and $p(p-1)/2$ unordered pairs $\{x_1, x_2\}$, each $\{x_1, x_2\}$ must correspond to exactly two pairs (b, c) and $(-b, -c)$ for some (b, c).

Now, since the image of f has only $p - 1$ elements, there must be some x_1, x_2 such that $f(x_1) = f(x_2)$. Choose any b, c such that $bx_1 + c = -(bx_2 + c)$, so $R(bx_1 + c) = R(bx_2 + c)$ and so $f(x_1)/R(bx_1 + c) = f(x_2)/R(bx_2 + c)$. There is such a pair b, c for any nonzero b, so there are at least $p - 1$ such pairs, and this quantity is greater than 2 for $p \geqslant 5$.

Finally, for the special case that $p = 3$, we observe that there must be at least one allowed value for $Q(x)$ for each x, so there must exist such a quadratic Q by Lagrange interpolation.

Comment. We may also handle the case $p = 3$ as follows. Recall that we may assume f is nonzero and surjective onto $\{1, 2\}$ mod 3, so the image of f must be $(1, 1, 2)$ or $(1, 2, 2)$ in some order. Without loss of generality $f(1) = f(2)$, so we either have $(f(0), f(1), f(2)) = (1, 2, 2)$ or $(2, 1, 1)$. In the first case, take $Q(X) = 2X^2 + 2$, and in the second case take $Q(X) = X^2 + 1$.

In some sense, this is equivalent to the Lagrange interpolation approach, as in each case the polynomial $Q(X)$ can be determined by Lagrange interpolation.

Solution 4. Again, we reduce to the case of p being an odd prime and f being a function mod p; we will show that there is a quadratic which is nowhere zero such that $Q(x) = f(x)$ has no root. We can handle the case of $p = 3$ separately as in Solution 3, so assume that $p \geqslant 5$.

We will prove the following more general statement: let $p \geqslant 5$ be a prime and let $\mathcal{A}_1$, $\mathcal{A}_2$, $\ldots$, $\mathcal{A}_p$ be subsets of $\mathbb{Z}/p\mathbb{Z}$ with $|\mathcal{A}_i| = 2$ for all i. Then there exists a polynomial $Q \in \mathbb{Z}/p\mathbb{Z}[X]$ of degree at most 2 such that $Q(i) \notin \mathcal{A}_i$ for all i. Indeed, applying this statement to the sets $\mathcal{A}_i = \{0, f(i)\}$ (and adding pX^2 if necessary) produces a quadratic Q satisfying the desired property.

Choose the coefficients of Q uniformly at random from $\mathbb{Z}/p\mathbb{Z}$, and let T be the random variable denoting the number of i for which $Q(i) \in \mathcal{A}_i$. Observe that for $k \leqslant 3$, we have

$$\mathbb{E}\left[\binom{T}{k}\right] = 2^k \binom{p}{k} p^{-k}.$$

To see why, let $k \leqslant 3$. If $\mathcal{S} \subseteq \mathbb{Z}/p\mathbb{Z}$ has size k and $(a_i)_{i\in\mathcal{S}}$ is a k-tuple, the probability that $Q(i) = a_i$ on $\mathcal{S}$ is equal to p^{-k}; for $k = 3$ this follows by Lagrange interpolation, and for $k < 3$

it follows from the $k = 3$ case by summing. The expectation is therefore equal to the number of $\mathcal{S} \subseteq \mathbb{Z}/p\mathbb{Z}$ of size k times the probability that $Q(i) \in \mathcal{A}_i$ for each $i \in \mathcal{S}$, which is equal to the right hand side as each $\mathcal{A}_i$ has size 2.

Now, observe that we have the identity $(t-1)(t-3)(t-4) = -12 + 12\binom{t}{1} - 10\binom{t}{2} + 6\binom{t}{3}$, so

$$\begin{aligned}\mathbb{E}[(T-1)(T-3)(T-4)] &= -12 + 12\mathbb{E}\left[\binom{T}{1}\right] - 10\mathbb{E}\left[\binom{T}{2}\right] + 6\mathbb{E}\left[\binom{T}{3}\right] \\ &= -12 + 12 \cdot 2 - 10 \cdot 2\left(1 - \frac{1}{p}\right) + 6 \cdot \frac{4}{3}\left(1 - \frac{1}{p}\right)\left(1 - \frac{2}{p}\right) \\ &= -\frac{4}{p} + \frac{16}{p^2}.\end{aligned}$$

This is negative for $p \geqslant 5$. Because $(t-1)(t-3)(t-4) \geqslant 0$ for all integers $t > 0$, it then follows that $T = 0$ with positive probability, which implies that there must exist some Q with $Q(i) \notin \mathcal{A}_i$ for all i, as desired.

Comment. We do not have much freedom to choose a different polynomial in place of $R(T) = (T-1)(T-3)(T-4)$ in this argument. Indeed, it can be shown (by comparing coefficients of $\binom{T}{K}$) that if R has degree at most 3, then the expected value of $R(T)$ tends to $\frac{1}{3}(R(4) + 2R(1))$ as p tends to infinity, so R must have both 1 and 4 as roots. In particular, R must be of the form $R(T) = (T-1)(T-4)(T-d)$ for some $d \geqslant 3$, and if $d < 4$ then the argument works for any p with $p > 4/(4-d)$.

N7. Let $\mathbb{Z}_{>0}$ denote the set of positive integers. Let $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ be a function satisfying the following property: for m, $n \in \mathbb{Z}_{>0}$, the equation

$$f(mn)^2 = f(m^2)f(f(n))f(mf(n))$$

holds if and only if m and n are coprime.

For each positive integer n, determine all the possible values of $f(n)$.

(Japan)

Answer: All numbers with the same set of prime factors as n.

Common remarks. We refer to the given property as $P(m,n)$. We use the notation $\mathrm{rad}(n)$ for the radical of n: the product of the distinct primes dividing n.

Solution 1. We start with a series of straightforward deductions:

- From $P(1,1)$, we have $f(1)^2 = f(1)f(f(1))^2$, so $f(1) = f(f(1))^2$.
- From $P(1,f(1))$, we have $f(f(1))^2 = f(1)f(f(f(1)))f(f(f(1)))$, so $f(f(f(1))) = 1$.
- From $P(1,f(f(1)))$, we have $f(f(f(1)))^2 = f(1)f(f(f(f(1))))f(f(f(f(1))))$, which simplifies to $1 = f(1)^3$, so $f(1) = 1$.
- From $P(1,n)$ we deduce $f(n) = f(f(n))$ for all n.
- From $P(m,1)$ we deduce $f(m) = f(m^2)$ for all m.
- Simplifying $P(m,n)$, we have that

  $$f(mn)^2 = f(m)f(n)f(mf(n))$$

  if and only if m and n are coprime; refer to this as $Q(m,n)$.
- From $Q(m,f(n))$, we have that $f(mf(n)) = f(m)f(n)$ if and only if m and $f(n)$ are coprime; refer to this as $R(m,n)$.

Claim. If $f(a) = 1$, then $a = 1$.

Proof. If $a \neq 1$, then $Q(a,a)$ gives $f(a)^2 \neq f(a)^2 f(af(a))$. If $f(a) = 1$, then both sides simplify to 1, a contradiction. □

Claim. If $n \neq 1$ then $\gcd(n, f(n)) \neq 1$.

Proof. If $\gcd(n,f(n)) = 1$, then $Q(f(n),n)$ gives $f(nf(n))^2 = f(n)^3$, and $Q(n,f(n))$ gives $f(nf(n))^2 = f(n)^2 f(nf(n))$, which together yield $f(n) = 1$ for a contradiction. □

Claim. For all n we have $\mathrm{rad}(n) \mid f(n)$.

Proof. For any prime $p \mid n$, write $n = p^v n'$ with $p \nmid n'$. From $Q(p^v, n')$ we have $f(n)^2 = f(p^v)f(n')f(p^v f(n'))$. Since $\gcd(p^v, f(p^v)) \neq 1$, it follows that $p \mid f(p^v)$, so $p \mid f(n)$, and thus $\mathrm{rad}(n) \mid f(n)$. □

Claim. If n is coprime to $f(k)$, then $f(n)$ is coprime to $f(k)$.

Proof. From $Q(f(k),n)$ we have $f(nf(k))^2 = f(k)f(n)f(f(k)f(n))$; applying $R(n,k)$ to the LHS, we conclude that $f(k)f(n) = f(f(k)f(n))$. Applying $R(f(n),k)$ we deduce that $f(n)$ is coprime to $f(k)$, as required. □

Claim. If p is prime then $f(p)$ is a power of p.

Proof. Suppose otherwise. We know that $p \mid f(p)$; let $q \neq p$ be another prime with $q \mid f(p)$.

If, for some positive integer N, we have $p \nmid f(N)$, then $f(p)$ is coprime to $f(N)$, so $q \nmid f(N)$, so $q \nmid N$; thus, if $q \mid N$, then $p \mid f(N)$ (and in particular, $p \mid f(q)$, by taking $N = q$).

Similarly, if $q \nmid f(N)$ then $f(q)$ is coprime to $f(N)$; as $p \mid f(q)$, this means $p \nmid f(N)$, so $p \nmid N$. So if $p \mid N$, then $q \mid f(N)$.

Together with $\operatorname{rad}(n) \mid f(n)$, this means that for any n not coprime to pq, we have $pq \mid f(n)$.

Let $m = \min\{\nu_p(f(x)) \mid x \text{ is not coprime to } pq\}$, and let X be a positive integer not coprime to pq such that $\nu_p(f(X)) = m$. The argument above shows $m \geqslant 1$. We can write $f(X) = p^m q^y X'$, where $y \geqslant 1$, $p \nmid X'$ and $q \nmid X'$. Since $f(f(X)) = f(X)$ we have $f(p^m q^y X') = p^m q^y X'$. Applying $Q(p^m, q^y X')$ gives $(p^m q^y X')^2 = f(p^m) f(q^y X') f(p^m f(q^y X'))$. The RHS is divisible by p^{3m} but the LHS is only divisible by p^{2m}, yielding a contradiction. □

Claim. For any integer n, $\operatorname{rad}(f(n)) = \operatorname{rad}(n)$.

Proof. We already have that $\operatorname{rad}(n) \mid f(n)$, so it remains only to show that no other primes divide $f(n)$. If p is prime and $p \nmid n$, the previous Claim shows that n is coprime to $f(p)$, and thus $f(n)$ is coprime to $f(p)$; that is, $p \nmid f(n)$. So exactly the same primes divide $f(n)$ as divide n. □

It remains only to exhibit functions that show all values of $f(n)$ with $\operatorname{rad}(f(n)) = \operatorname{rad}(n)$ are possible. Given $e(p) \geqslant 1$ for each prime p, take

$$f(n) = \prod_{p \mid n} p^{e(p)}$$

and we verify by examining exponents of each prime that this satisfies the conditions of the problem.

Comment. A quicker but less straightforward proof that $f(1) = 1$ is to let $f(n) = M$ be the least value that f takes; then $P(1, n)$ gives $M^2 = f(n)^2 = f(1) f(f(n))^2 \geqslant M^3$ so $M = 1$ and $f(1) = 1$.

Solution 2. As in Solution 1, we see that there are indeed functions f satisfying the given condition and producing all the given values of $f(n)$, and we follow Solution 1 to show the following facts:

- $f(1) = 1$.
- $f(m) = f(m^2)$ for all m.
- $f(n) = f(f(n))$ for all n.
- $f(mn)^2 = f(m) f(n) f(m f(n))$ if and only if m and n are coprime; refer to this as $Q(m, n)$.

Taking $Q(m, n)$ together with $Q(n, m)$ gives that $f(m f(n)) = f(n f(m))$ if m and n are coprime.

Suppose now that m is coprime to both n and $f(n)$. We have $f(mn)^2 = f(m) f(n) f(m f(n))$ and squaring both sides gives

$$\begin{aligned} f(mn)^4 &= f(m)^2 f(n)^2 f(m f(n))^2 \\ &= f(m)^2 f(n)^2 f(m) f(f(n)) f(m f(f(n))) \\ &= f(m)^3 f(n)^3 f(m f(n)). \end{aligned}$$

Thus $f(m f(n)) = f(m) f(n)$, so $f(mn)^2 = f(m)^2 f(n)^2$, so $f(mn) = f(m) f(n) = f(m f(n)) = f(n f(m))$.

If m is coprime to both n and $f(n)$ but however n is not coprime to $f(m)$, we have

$$\begin{aligned} f(nf(m))^2 &\neq f(n)f(f(m))f(nf(f(m))) \\ &= f(n)f(m)f(nf(m)) \\ &= f(nf(m))^2, \end{aligned}$$

a contradiction. Thus, given that m and n are coprime, we know that m is coprime to $f(n)$ if and only if n is coprime to $f(m)$. In particular, if p and q are different primes, then $p \mid f(q)$ if and only if $q \mid f(p)$, and likewise, for any positive integer k, $p \mid f(q^k)$ if and only if $q \mid f(p)$. More generally, if $p \nmid n$, then $p \mid f(n)$ if and only if n is not coprime to $f(p)$.

Now form a graph whose vertices are the primes, and where there is an edge between primes $p \neq q$ if and only if $p \mid f(q)$ (and so $q \mid f(p)$); every vertex has finite degree. For any integer n, the primes dividing $f(n)$ are all the primes that are neighbours of any prime $q \mid n$, together possibly with some further primes $p \mid n$.

If p and q are different primes, we have $f(pf(q)) = f(qf(p))$. The LHS is divisible by all primes that (in the graph) are neighbours of p or neighbours of neighbours of q, and possibly also by p and by some primes that are neighbours of q, and a corresponding statement with p and q swapped applies to the RHS. Thus any prime that is a neighbour of a neighbour of q must be one of: p, q, distance 1 from q, or distance 1 or 2 from p. For any prime r that is distance 2 from q, there are only finitely many primes p that it is distance 2 or less from, so by choosing a suitable prime p (depending on q) we conclude that every prime that is a neighbour of a neighbour of q is actually q itself or a neighbour of q.

So the connected components of the graph are (finite) complete graphs. If m is divisible only by primes in one component, and n is divisible only by primes in another component, then $f(mn) = f(m)f(n)$. If n is divisible by more than one prime from a component, considering the expression for $f(mn)^2$ as applied with successive prime power divisors of n shows that $f(n)$ is divisible by all the primes in that component. However, while $f(p^k)$ is divisible by all the primes in the component of p except possibly for p itself, we do not yet know that $p \mid f(p^k)$. We now consider cases for the order of a component.

For any prime p, we cannot have $f(p^k) = 1$, because $Q(p^k, p^k)$ gives

$$f(p^{2k})^2 \neq f(p^k)f(p^k)f(p^k f(p^k)),$$

and simplifying using $f(m^2) = f(m)$ results in $1 \neq 1$. So for a component of order 1, $f(p^k)$ is a positive power of p, so has the same set of prime factors as p, as required.

Now consider a component of order at least 2. Since $f(f(n)) = f(n)$, if the component has order at least 3, then for any $n \neq 1$ whose prime divisors are in that component, $f(n)$ is divisible by all the primes in that component. If the component has order 2, we saw above that this is true except possibly for $n = p^k$. However, if the primes in the component are p and q, and $f(p^k) = q^\ell$, then $f(q^\ell) = f(f(p^k)) = f(p^k) = q^\ell$, which contradicts $p \mid f(q^\ell)$. So for any component of order at least 2, and any $n \neq 1$ whose prime divisors are in that component, $f(n)$ is divisible by all the primes in that component.

In a component of order at least 2, let m be the product of all the primes in that component, and let t be maximal such that $m^t \mid f(n)$ for all $n \neq 1$ whose prime divisors are in that component; we have seen that $t \geqslant 1$. If m and n are coprime numbers greater than 1, all of whose prime divisors are in that component, then $Q(m, n)$ tells us that $m^{3t/2} \mid f(mn)$. For any $n' \neq 1$, all of whose prime divisors are in that component, $f(n')$ is divisible by all the primes in that component, so can be expressed as such a product, so $m^{3t/2} \mid f(f(n')) = f(n')$. But this means $t \geqslant 3t/2$, a contradiction, so all components have order 1, and we are done.

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Shortlist 2023

with solutions



64th International Mathematical Olympiad
Chiba, Japan, 2nd–13th July 2023

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SHORTLISTED PROBLEMS
WITH SOLUTIONS

Note of Confidentiality

> **The Shortlist has to be kept strictly confidential until the conclusion of the following International Mathematical Olympiad.**
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Contributing Countries

The Organising Committee and the Problem Selection Committee of IMO 2023 thank the following 52 countries for contributing 167 problem proposals:

Armenia, Australia, Austria, Azerbaijan, Bangladesh, Belarus, Belgium, Bulgaria, Brazil, Canada, China, Colombia, Croatia, Cyprus, Czech Republic, Estonia, Georgia, Germany, Greece, Hong Kong, Hungary, India, Iran, Israel, Latvia, Liechtenstein, Lithuania, Malaysia, Mexico, Mongolia, Morocco, Netherlands, New Zealand, North Macedonia, Poland, Portugal, Serbia, Singapore, Slovakia, Slovenia, South Africa, South Korea, Spain, Syria, Sweden, Taiwan, Tajikistan, Thailand, Turkey, Ukraine, United Kingdom, U.S.A., Uzbekistan.

Problem Selection Committee



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Problems

Algebra

A1. Professor Oak is feeding his 100 Pokémon. Each Pokémon has a bowl whose capacity is a positive real number of kilograms. These capacities are known to Professor Oak. The total capacity of all the bowls is 100 kilograms. Professor Oak distributes 100 kilograms of food in such a way that each Pokémon receives a non-negative integer number of kilograms of food (which may be larger than the capacity of their bowl). The *dissatisfaction level* of a Pokémon who received N kilograms of food and whose bowl has a capacity of C kilograms is equal to $|N - C|$.

Find the smallest real number D such that, regardless of the capacities of the bowls, Professor Oak can distribute the food in a way that the sum of the dissatisfaction levels over all the 100 Pokémon is at most D.

(Ukraine)

A2. Let $\mathbb{R}$ be the set of real numbers. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a function such that

$$f(x+y)f(x-y) \geqslant f(x)^2 - f(y)^2$$

for every $x, y \in \mathbb{R}$. Assume that the inequality is strict for some $x_0, y_0 \in \mathbb{R}$.

Prove that $f(x) \geqslant 0$ for every $x \in \mathbb{R}$ or $f(x) \leqslant 0$ for every $x \in \mathbb{R}$.

(Malaysia)

A3. Let $x_1, x_2, \ldots, x_{2023}$ be *distinct* real positive numbers such that

$$a_n = \sqrt{(x_1 + x_2 + \cdots + x_n)\left(\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}\right)}$$

is an integer for every $n = 1, 2, \ldots, 2023$. Prove that $a_{2023} \geqslant 3034$.

(Netherlands)

A4. Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f\colon \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that

$$x\big(f(x) + f(y)\big) \geqslant \big(f(f(x)) + y\big)f(y)$$

for every $x, y \in \mathbb{R}_{>0}$.

(Belgium)

A5. Let $a_1, a_2, \ldots, a_{2023}$ be positive integers such that

- $a_1, a_2, \ldots, a_{2023}$ is a permutation of $1, 2, \ldots, 2023$, and
- $|a_1 - a_2|, |a_2 - a_3|, \ldots, |a_{2022} - a_{2023}|$ is a permutation of $1, 2, \ldots, 2022$.

Prove that $\max\big(a_1, a_{2023}\big) \geqslant 507$.

(Australia)

A6. Let $k \geqslant 2$ be an integer. Determine all sequences of positive integers $a_1, a_2, \ldots$ for which there exists a monic polynomial P of degree k with non-negative integer coefficients such that

$$P(a_n) = a_{n+1}a_{n+2}\cdots a_{n+k}$$

for every integer $n \geqslant 1$.

(Malaysia)

A7. Let N be a positive integer. Prove that there exist three permutations $a_1, a_2, \ldots, a_N$; $b_1, b_2, \ldots, b_N$; and $c_1, c_2, \ldots, c_N$ of $1, 2, \ldots, N$ such that

$$\left|\sqrt{a_k} + \sqrt{b_k} + \sqrt{c_k} - 2\sqrt{N}\right| < 2023$$

for every $k = 1, 2, \ldots, N$.

(China)

Combinatorics

C1. Let m and n be positive integers greater than 1. In each unit square of an $m \times n$ grid lies a coin with its tail-side up. A *move* consists of the following steps:

1. select a 2×2 square in the grid;
2. flip the coins in the top-left and bottom-right unit squares;
3. flip the coin in either the top-right or bottom-left unit square.

Determine all pairs (m, n) for which it is possible that every coin shows head-side up after a finite number of moves.

(Thailand)

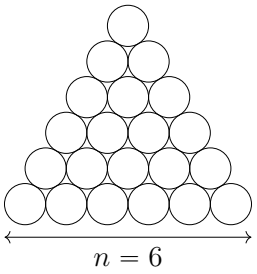
C2. Determine the maximal length L of a sequence $a_1, \ldots, a_L$ of positive integers satisfying both the following properties:

- every term in the sequence is less than or equal to 2^{2023}, and
- there does not exist a consecutive subsequence $a_i, a_{i+1}, \ldots, a_j$ (where $1 \leqslant i \leqslant j \leqslant L$) with a choice of signs $s_i, s_{i+1}, \ldots, s_j \in \{1, -1\}$ for which

$$s_i a_i + s_{i+1} a_{i+1} + \cdots + s_j a_j = 0.$$

(Czech Republic)

C3. Let n be a positive integer. We arrange $1 + 2 + \cdots + n$ circles in a triangle with n rows, such that the ith row contains exactly i circles. The following figure shows the case $n = 6$.



In this triangle, a *ninja-path* is a sequence of circles obtained by repeatedly going from a circle to one of the two circles directly below it. In terms of n, find the largest value of k such that if one circle from every row is coloured red, we can always find a ninja-path in which at least k of the circles are red.

(Netherlands)

C4. Let $n \geqslant 2$ be a positive integer. Paul has a $1 \times n^2$ rectangular strip consisting of n^2 unit squares, where the ith square is labelled with i for all $1 \leqslant i \leqslant n^2$. He wishes to cut the strip into several pieces, where each piece consists of a number of consecutive unit squares, and then *translate* (without rotating or flipping) the pieces to obtain an $n \times n$ square satisfying the following property: if the unit square in the ith row and j^{th} column is labelled with a_{ij}, then $a_{ij} - (i + j - 1)$ is divisible by n.

Determine the smallest number of pieces Paul needs to make in order to accomplish this.

(U.S.A.)

C5. Elisa has 2023 treasure chests, all of which are unlocked and empty at first. Each day, Elisa adds a new gem to one of the unlocked chests of her choice, and afterwards, a fairy acts according to the following rules:

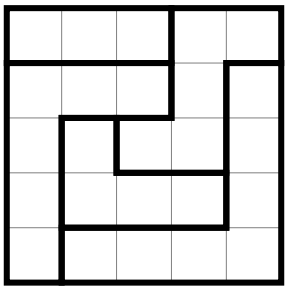
- if more than one chests are unlocked, it locks one of them, or
- if there is only one unlocked chest, it unlocks all the chests.

Given that this process goes on forever, prove that there is a constant C with the following property: Elisa can ensure that the difference between the numbers of gems in any two chests never exceeds C, regardless of how the fairy chooses the chests to lock.

(Israel)

C6. Let N be a positive integer, and consider an $N \times N$ grid. A *right-down path* is a sequence of grid cells such that each cell is either one cell to the right of or one cell below the previous cell in the sequence. A *right-up path* is a sequence of grid cells such that each cell is either one cell to the right of or one cell above the previous cell in the sequence.

Prove that the cells of the $N \times N$ grid cannot be partitioned into less than N right-down or right-up paths. For example, the following partition of the 5×5 grid uses 5 paths.



(Canada)

C7. The Imomi archipelago consists of $n \geqslant 2$ islands. Between each pair of distinct islands is a unique ferry line that runs in both directions, and each ferry line is operated by one of k companies. It is known that if any one of the k companies closes all its ferry lines, then it becomes impossible for a traveller, no matter where the traveller starts at, to visit all the islands exactly once (in particular, not returning to the island the traveller started at).

Determine the maximal possible value of k in terms of n.

(Ukraine)

Geometry

G1. Let $ABCDE$ be a convex pentagon such that $\angle ABC = \angle AED = 90^\circ$. Suppose that the midpoint of CD is the circumcentre of triangle ABE. Let O be the circumcentre of triangle ACD.

Prove that line AO passes through the midpoint of segment BE.

(Slovakia)

G2. Let ABC be a triangle with $AC > BC$. Let ω be the circumcircle of triangle ABC and let r be the radius of ω. Point P lies on segment AC such that $BC = CP$ and point S is the foot of the perpendicular from P to line AB. Let ray BP intersect ω again at D and let Q lie on line SP such that $PQ = r$ and S, P, Q lie on the line in that order. Finally, let the line perpendicular to CQ from A intersect the line perpendicular to DQ from B at E.

Prove that E lies on ω.

(Iran)

G3. Let $ABCD$ be a cyclic quadrilateral with $\angle BAD < \angle ADC$. Let M be the midpoint of the arc CD not containing A. Suppose there is a point P inside $ABCD$ such that $\angle ADB = \angle CPD$ and $\angle ADP = \angle PCB$.

Prove that lines AD, PM, BC are concurrent.

(Slovakia)

G4. Let ABC be an acute-angled triangle with $AB < AC$. Denote its circumcircle by Ω and denote the midpoint of arc CAB by S. Let the perpendicular from A to BC meet BS and Ω at D and $E \neq A$ respectively. Let the line through D parallel to BC meet line BE at L and denote the circumcircle of triangle BDL by ω. Let ω meet Ω again at $P \neq B$.

Prove that the line tangent to ω at P, and line BS intersect on the internal bisector of $\angle BAC$.

(Portugal)

G5. Let ABC be an acute-angled triangle with circumcircle ω and circumcentre O. Points $D \neq B$ and $E \neq C$ lie on ω such that $BD \perp AC$ and $CE \perp AB$. Let CO meet AB at X, and BO meet AC at Y.

Prove that the circumcircles of triangles BXD and CYE have an intersection on line AO.

(Malaysia)

G6. Let ABC be an acute-angled triangle with circumcircle ω. A circle Γ is internally tangent to ω at A and also tangent to BC at D. Let AB and AC intersect Γ at P and Q respectively. Let M and N be points on line BC such that B is the midpoint of DM and C is the midpoint of DN. Lines MP and NQ meet at K and intersect Γ again at I and J respectively. The ray KA meets the circumcircle of triangle IJK at $X \neq K$.

Prove that $\angle BXP = \angle CXQ$.

(United Kingdom)

G7. Let ABC be an acute, scalene triangle with orthocentre H. Let ℓ_a be the line through the reflection of B with respect to CH and the reflection of C with respect to BH. Lines ℓ_b and ℓ_c are defined similarly. Suppose lines ℓ_a, ℓ_b, and ℓ_c determine a triangle $\mathcal{T}$.

Prove that the orthocentre of $\mathcal{T}$, the circumcentre of $\mathcal{T}$ and H are collinear.

(Ukraine)

G8. Let ABC be an equilateral triangle. Points A_1, B_1, C_1 lie inside triangle ABC such that triangle $A_1B_1C_1$ is scalene, $BA_1 = A_1C$, $CB_1 = B_1A$, $AC_1 = C_1B$ and

$$\angle BA_1C + \angle CB_1A + \angle AC_1B = 480^\circ.$$

Lines BC_1 and CB_1 intersect at A_2; lines CA_1 and AC_1 intersect at B_2; and lines AB_1 and BA_1 intersect at C_2.

Prove that the circumcircles of triangles AA_1A_2, BB_1B_2, CC_1C_2 have two common points.

(U.S.A.)

Number Theory

N1. Determine all positive, composite integers n that satisfy the following property: if the positive divisors of n are $1 = d_1 < d_2 < \cdots < d_k = n$, then d_i divides $d_{i+1} + d_{i+2}$ for every $1 \leqslant i \leqslant k-2$.

(Colombia)

N2. Determine all pairs (a, p) of positive integers with p prime such that $p^a + a^4$ is a perfect square.

(Bangladesh)

N3. For positive integers n and $k \geqslant 2$ define $E_k(n)$ as the greatest exponent r such that k^r divides $n!$. Prove that there are infinitely many n such that $E_{10}(n) > E_9(n)$ and infinitely many m such that $E_{10}(m) < E_9(m)$.

(Brazil)

N4. Let $a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_n$ be $2n$ positive integers such that the $n+1$ products

$$\begin{gathered} a_1 a_2 a_3 \cdots a_n, \\ b_1 a_2 a_3 \cdots a_n, \\ b_1 b_2 a_3 \cdots a_n, \\ \vdots \\ b_1 b_2 b_3 \cdots b_n \end{gathered}$$

form a strictly increasing arithmetic progression in that order. Determine the smallest positive integer that could be the common difference of such an arithmetic progression.

(Canada)

N5. Let $a_1 < a_2 < a_3 < \cdots$ be positive integers such that a_{k+1} divides $2(a_1 + a_2 + \cdots + a_k)$ for every $k \geqslant 1$. Suppose that for infinitely many primes p, there exists k such that p divides a_k. Prove that for every positive integer n, there exists k such that n divides a_k.

(Netherlands)

N6. A sequence of integers $a_0, a_1, a_2, \ldots$ is called *kawaii*, if $a_0 = 0, a_1 = 1$, and, for any positive integer n, we have

$$(a_{n+1} - 3a_n + 2a_{n-1})(a_{n+1} - 4a_n + 3a_{n-1}) = 0.$$

An integer is called *kawaii* if it belongs to a kawaii sequence.

Suppose that two consecutive positive integers m and $m+1$ are both kawaii (not necessarily belonging to the same kawaii sequence). Prove that 3 divides m, and that $m/3$ is kawaii.

(China)

N7. Let a, b, c, d be positive integers satisfying

$$\frac{ab}{a+b} + \frac{cd}{c+d} = \frac{(a+b)(c+d)}{a+b+c+d}.$$

Determine all possible values of $a + b + c + d$.

(Netherlands)

N8. Let $\mathbb{Z}_{>0}$ be the set of positive integers. Determine all functions $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that

$$f^{bf(a)}(a+1) = (a+1)f(b)$$

holds for all $a, b \in \mathbb{Z}_{>0}$, where $f^k(n) = f(f(\cdots f(n)\cdots))$ denotes the composition of f with itself k times.

(Taiwan)

Solutions

Algebra

A1. Professor Oak is feeding his 100 Pokémon. Each Pokémon has a bowl whose capacity is a positive real number of kilograms. These capacities are known to Professor Oak. The total capacity of all the bowls is 100 kilograms. Professor Oak distributes 100 kilograms of food in such a way that each Pokémon receives a non-negative integer number of kilograms of food (which may be larger than the capacity of their bowl). The *dissatisfaction level* of a Pokémon who received N kilograms of food and whose bowl has a capacity of C kilograms is equal to $|N - C|$.

Find the smallest real number D such that, regardless of the capacities of the bowls, Professor Oak can distribute the food in a way that the sum of the dissatisfaction levels over all the 100 Pokémon is at most D.

(Ukraine)

Answer: The answer is $D = 50$.

Solution 1. First, consider the situation where 99 bowls have a capacity of 0.5 kilograms and the last bowl has a capacity of 50.5 kilograms. No matter how Professor Oak distributes the food, the dissatisfaction level of every Pokémon will be at least 0.5. This amounts to a total dissatisfaction level of at least 50, proving that $D \geqslant 50$.

Now we prove that no matter what the capacities of the bowls are, Professor Oak can always distribute food in a way that the total dissatisfaction level is at most 50. We start by fixing some notation. We number the Pokémon from 1 to 100. Let $C_i > 0$ be the capacity of the bowl of the ith Pokémon. By assumption, we have $C_1 + C_2 + \cdots + C_{100} = 100$. We write $F_i := C_i - \lfloor C_i \rfloor$ for the fractional part of C_i. Without loss of generality, we may assume that $F_1 \leqslant F_2 \leqslant \cdots \leqslant F_{100}$.

Here is a strategy: Professor Oak starts by giving $\lfloor C_i \rfloor$ kilograms of food to the ith Pokémon. Let

$$R := 100 - \lfloor C_1 \rfloor - \lfloor C_2 \rfloor - \cdots - \lfloor C_{100} \rfloor = F_1 + F_2 + \cdots + F_{100} \geqslant 0$$

be the amount of food left. He continues by giving an extra kilogram of food to the R Pokémon numbered $100 - R + 1, 100 - R + 2, \ldots, 100$, i.e. the Pokémon with the R largest values of F_i. By doing so, Professor Oak distributed 100 kilograms of food. The total dissatisfaction level with this strategy is

$$d := F_1 + \cdots + F_{100-R} + (1 - F_{100-R+1}) + \cdots + (1 - F_{100}).$$

We can rewrite

$$\begin{aligned} d &= 2(F_1 + \cdots + F_{100-R}) + R - (F_1 + \cdots + F_{100}) \\ &= 2(F_1 + \cdots + F_{100-R}). \end{aligned}$$

Now, observe that the arithmetic mean of $F_1, F_2, \ldots, F_{100-R}$ is not greater than the arithmetic mean of $F_1, F_2, \ldots, F_{100}$, because we assumed $F_1 \leqslant F_2 \leqslant \cdots \leqslant F_{100}$. Therefore

$$d \leqslant 2(100 - R) \cdot \frac{F_1 + \cdots + F_{100}}{100} = 2 \cdot \frac{R(100 - R)}{100}.$$

Finally, we use the AM-GM inequality to see that $R(100 - R) \leqslant \frac{100^2}{2^2}$ which implies $d \leqslant 50$. We conclude that there is always a distribution for which the total dissatisfaction level is at most 50, proving that $D \leqslant 50$.

Solution 2. We adopt the same notation as in Solution 1. Let $C_i > 0$ be the capacity of the bowl of the ith Pokémon. By assumption, we have $C_1 + C_2 + \cdots + C_{100} = 100$. We write $F_i := C_i - \lfloor C_i \rfloor$ for the fractional part of C_i, and $R = F_1 + F_2 + \cdots + F_{100}$. Note that $R = 100 - \lfloor C_1 \rfloor - \cdots - \lfloor C_{100} \rfloor$ is an integer.

This solution uses the probabilistic method. We consider all distributions in which each Pokémon receives $\lfloor C_i \rfloor + \varepsilon_i$ kilograms of food, where $\varepsilon_i \in \{0, 1\}$ and $\varepsilon_1 + \varepsilon_2 + \cdots + \varepsilon_{100} = R$. There are $\binom{100}{R}$ such distributions. Suppose each of them occurs in an equal probability. In other words,

$$\varepsilon_i = \begin{cases} 0 & \text{with probability } \frac{100-R}{100}, \\ 1 & \text{with probability } \frac{R}{100}. \end{cases}$$

The expected value of the dissatisfaction level of the ith Pokémon is

$$\frac{100-R}{100}\big(C_i - \lfloor C_i \rfloor\big) + \frac{R}{100}\big(\lfloor C_i \rfloor + 1 - C_i\big) = \frac{100-R}{100}F_i + \frac{R}{100}(1 - F_i).$$

Hence, the expected value of the total dissatisfaction level is

$$\begin{aligned} \sum_{i=1}^{100}\left(\frac{100-R}{100}F_i + \frac{R}{100}(1-F_i)\right) &= \frac{100-R}{100}\sum_{i=1}^{100}F_i + \frac{R}{100}\sum_{i=1}^{100}(1-F_i) \\ &= \frac{100-R}{100}\cdot R + \frac{R}{100}\cdot(100-R) \\ &= 2\cdot\frac{R(100-R)}{100}. \end{aligned}$$

As in Solution 1, this is at most 50. We conclude that there is at least one distribution for which the total dissatisfaction level is at most 50.

A2. Let $\mathbb{R}$ be the set of real numbers. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a function such that

$$f(x+y)f(x-y) \geqslant f(x)^2 - f(y)^2$$

for every $x, y \in \mathbb{R}$. Assume that the inequality is strict for some $x_0, y_0 \in \mathbb{R}$.

Prove that $f(x) \geqslant 0$ for every $x \in \mathbb{R}$ or $f(x) \leqslant 0$ for every $x \in \mathbb{R}$.

(Malaysia)

Common remarks. We will say that f *has constant sign*, if f satisfies the conclusion of the problem.

Solution 1. We introduce the new variables $s := x + y$ and $t := x - y$. Equivalently, $x = \frac{s+t}{2}$ and $y = \frac{s-t}{2}$. The inequality becomes

$$f(s)f(t) \geqslant f\left(\frac{s+t}{2}\right)^2 - f\left(\frac{s-t}{2}\right)^2$$

for every $s, t \in \mathbb{R}$. We replace t by $-t$ to obtain

$$f(s)f(-t) \geqslant f\left(\frac{s-t}{2}\right)^2 - f\left(\frac{s+t}{2}\right)^2.$$

Summing the previous two inequalities gives

$$f(s)\big(f(t) + f(-t)\big) \geqslant 0$$

for every $s, t \in \mathbb{R}$. This inequality is strict for $s = x_0 + y_0$ and $t = x_0 - y_0$ by assumption. In particular, there exists some $t_0 = x_0 - y_0$ for which $f(t_0) + f(-t_0) \neq 0$. Since $f(s)\big(f(t_0) + f(-t_0)\big) \geqslant 0$ for every $s \in \mathbb{R}$, we conclude that $f(s)$ must have constant sign.

Solution 2. We do the same change of variables as in Solution 1 to obtain

$$f(s)f(t) \geqslant f\left(\frac{s+t}{2}\right)^2 - f\left(\frac{s-t}{2}\right)^2. \tag{1}$$

In this solution, we replace s by $-s$ (instead of t by $-t$). This gives

$$f(-s)f(t) \geqslant f\left(\frac{-s+t}{2}\right)^2 - f\left(\frac{-s-t}{2}\right)^2. \tag{2}$$

We now go back to the original inequality. Substituting $x = y$ gives $f(2x)f(0) \geqslant 0$ for every $x \in \mathbb{R}$. If $f(0) \neq 0$, then we conclude that f indeed has constant sign. From now on, we will assume that

$$f(0) = 0.$$

Substituting $x = -y$ gives $f(-x)^2 \geqslant f(x)^2$. By permuting x and $-x$, we conclude that

$$f(-x)^2 = f(x)^2$$

for every $x \in \mathbb{R}$.

Using the relation $f(x)^2 = f(-x)^2$, we can rewrite (2) as

$$f(-s)f(t) \geqslant f\left(\frac{s-t}{2}\right)^2 - f\left(\frac{s+t}{2}\right)^2.$$

Summing this inequality with (1), we obtain

$$\big(f(s) + f(-s)\big)f(t) \geqslant 0$$

for every $s, t \in \mathbb{R}$ and we can conclude as in Solution 1.

Solution 3. We prove the contrapositive of the problem statement. Assume that there exist $a, b \in \mathbb{R}$ such that $f(a) < 0$ and $f(b) > 0$. We want to prove that the inequality is actually an equality, i.e. it is never strict.

Lemma 1. The function f is odd, i.e. $f(x) + f(-x) = 0$ for every $x \in \mathbb{R}$.

Proof. We plug in $x = \frac{a+u}{2}$ and $y = \frac{a-u}{2}$ in the original inequality, where u is a free variable. We obtain

$$f(a)f(u) \geqslant f\left(\frac{a+u}{2}\right)^2 - f\left(\frac{a-u}{2}\right)^2.$$

Replacing u with $-u$ and summing the two inequalities as in the previous solutions, we get

$$f(a)\big(f(u) + f(-u)\big) \geqslant 0$$

for every $u \in \mathbb{R}$. Since $f(a) < 0$ by assumption, we conclude that $f(u) + f(-u) \leqslant 0$ for every $u \in \mathbb{R}$.

We can repeat the above argument with b instead of a. Since $f(b) > 0$ by assumption, we conclude that $f(u) + f(-u) \geqslant 0$ for every $u \in \mathbb{R}$. This implies that $f(u) + f(-u) = 0$ for every $u \in \mathbb{R}$. □

Now, using that f is odd, we can write the following chain of inequalities

$$\begin{aligned} f(x)^2 - f(y)^2 &\leqslant f(x+y)f(x-y) \\ &= -f(y+x)f(y-x) \\ &\leqslant -\big(f(y)^2 - f(x)^2\big) \\ &= f(x)^2 - f(y)^2. \end{aligned}$$

We conclude that every inequality above is actually an inequality, so

$$f(x+y)f(x-y) = f(x)^2 - f(y)^2$$

for every $x, y \in \mathbb{R}$.

Solution 4. As in Solution 3, we prove the contrapositive of the statement. Assume that there exist $a, b \in \mathbb{R}$ such that $f(a)f(b) < 0$. We want to prove that the inequality is actually an equality, i.e. it is never strict.

In this solution, we construct an argument by multiplying inequalities, rather than adding them as in Solutions 1-3.

Lemma 2. $f(b)f(-b) < 0$.

Proof. Let $x_1 := \frac{a+b}{2}$ and $y_1 := \frac{a-b}{2}$ so that $a = x_1 + y_1$ and $b = x_1 - y_1$. Plugging in $x = x_1$ and $y = y_1$, we obtain

$$0 > f(a)f(b) = f(x_1 + y_1)f(x_1 - y_1) \geqslant f(x_1)^2 - f(y_1)^2$$

which implies $f(x_1)^2 - f(y_1)^2 < 0$. Similarly, by plugging in $x = y_1$ and $y = x_1$, we get

$$f(a)f(-b) = f(y_1 + x_1)f(y_1 - x_1) \geqslant f(y_1)^2 - f(x_1)^2.$$

Using $f(x_1)^2 - f(y_1)^2 < 0$, we conclude $f(a)f(-b) > 0$. If we multiply the two inequalities $f(a)f(b) < 0$ and $f(a)f(-b) > 0$, we get $f(a)^2 f(b)f(-b) < 0$ and hence

$$f(b)f(-b) < 0.$$

□

Lemma 3. $f(x)f(-x) \leqslant 0$ for every $x \in \mathbb{R}$.

Proof. As in Solution 2, we prove that $f(x)^2 = f(-x)^2$ for every $x \in \mathbb{R}$ and we rewrite the original inequality as

$$f(s)f(t) \geqslant f\left(\frac{s+t}{2}\right)^2 - f\left(\frac{s-t}{2}\right)^2.$$

We replace s by $-s$ and t by $-t$, and use the relation $f(x)^2 = f(-x)^2$, to get

$$\begin{aligned} f(-s)f(-t) &\geqslant f\left(\frac{-s-t}{2}\right)^2 - f\left(\frac{-s+t}{2}\right)^2 \\ &= f\left(\frac{s+t}{2}\right)^2 - f\left(\frac{s-t}{2}\right)^2. \end{aligned}$$

Up to replacing t by $-t$, we can assume that $f\left(\frac{s+t}{2}\right)^2 - f\left(\frac{s-t}{2}\right)^2 \geqslant 0$. Multiplying the two previous inequalities leads to

$$f(s)f(-s)f(t)f(-t) \geqslant 0$$

for every $s, t \in \mathbb{R}$. This shows that $f(s)f(-s)$ (as a function of s) has constant sign. Since $f(b)f(-b) < 0$, we conclude that

$$f(x)f(-x) \leqslant 0$$

for every $x \in \mathbb{R}$. □

Lemma 3, combined with the relation $f(x)^2 = f(-x)^2$, implies $f(x) + f(-x) = 0$ for every $x \in \mathbb{R}$, i.e. f is odd. We conclude with the same argument as in Solution 3.

Comment. The presence of squares on the right-hand side of the inequality is not crucial as Solution 1 illustrates very well. However, it allows non-constant functions such as $f(x) = |x|$ to satisfy the conditions of the problem statement.

A3. Let $x_1, x_2, \ldots, x_{2023}$ be *distinct* real positive numbers such that

$$a_n = \sqrt{(x_1 + x_2 + \cdots + x_n)\left(\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}\right)}$$

is an integer for every $n = 1, 2, \ldots, 2023$. Prove that $a_{2023} \geqslant 3034$.

(Netherlands)

Solution 1. We start with some basic observations. First note that the sequence $a_1, a_2, \ldots, a_{2023}$ is increasing and thus, since all elements are integers, $a_{n+1} - a_n \geqslant 1$. We also observe that $a_1 = 1$ and

$$a_2 = \sqrt{(x_1 + x_2)\left(\frac{1}{x_1} + \frac{1}{x_2}\right)} > 2$$

by Cauchy-Schwarz inequality and using $x_1 \neq x_2$. So, $a_2 \geqslant 3$.

Now, we proceed to the main part of the argument. We observe that 3034 is about three halves of 2023. Motivated by this observation, we will prove the following.

Claim. If $a_{n+1} - a_n = 1$, then $a_{n+2} - a_{n+1} \geqslant 2$.

In other words, the sequence has to increase by at least 2 at least half of the times. Assuming the claim is true, since $a_1 = 1$, we would be done since

$$\begin{aligned} a_{2023} &= (a_{2023} - a_{2022}) + (a_{2022} - a_{2021}) + \cdots + (a_2 - a_1) + a_1 \\ &\geqslant (2 + 1) \cdot 1011 + 1 \\ &= 3034. \end{aligned}$$

We now prove the claim. We start by observing that

$$\begin{aligned} a_{n+1}^2 &= (x_1 + \cdots + x_{n+1})\left(\frac{1}{x_1} + \cdots + \frac{1}{x_{n+1}}\right) \\ &= (x_1 + \cdots + x_n)\left(\frac{1}{x_1} + \cdots + \frac{1}{x_n}\right) + 1 \\ &\quad + \frac{1}{x_{n+1}}(x_1 + \cdots + x_n) + x_{n+1}\left(\frac{1}{x_1} + \cdots + \frac{1}{x_n}\right) \\ &\geqslant a_n^2 + 1 + 2\sqrt{\frac{1}{x_{n+1}}(x_1 + \cdots + x_n) \cdot x_{n+1}\left(\frac{1}{x_1} + \cdots + \frac{1}{x_n}\right)} \\ &= a_n^2 + 1 + 2a_n \\ &= (a_n + 1)^2, \end{aligned}$$

where we used AM-GM to obtain the inequality. In particular, if $a_{n+1} = a_n + 1$, then

$$\frac{1}{x_{n+1}}(x_1 + \cdots + x_n) = x_{n+1}\left(\frac{1}{x_1} + \cdots + \frac{1}{x_n}\right). \tag{1}$$

Now, assume for the sake of contradiction that both $a_{n+1} = a_n + 1$ and $a_{n+2} = a_{n+1} + 1$ hold. In this case, (1) gives

$$\frac{1}{x_{n+2}}(x_1 + \cdots + x_{n+1}) = x_{n+2}\left(\frac{1}{x_1} + \cdots + \frac{1}{x_{n+1}}\right).$$

We can rewrite this relation as

$$\frac{x_{n+1}}{x_{n+2}}\left(\frac{1}{x_{n+1}}(x_1 + \cdots + x_n) + 1\right) = \frac{x_{n+2}}{x_{n+1}}\left(x_{n+1}\left(\frac{1}{x_1} + \cdots + \frac{1}{x_n}\right) + 1\right).$$

From (1) again, we conclude that $x_{n+1} = x_{n+2}$ which is a contradiction.

Solution 2. The trick is to compare a_{n+2} and a_n. Observe that

$$\begin{aligned}
a_{n+2}^2 &= (x_1+\cdots+x_{n+2})\left(\frac{1}{x_1}+\cdots+\frac{1}{x_{n+2}}\right)\\
&= (x_1+\cdots+x_n)\left(\frac{1}{x_1}+\cdots+\frac{1}{x_n}\right)+(x_{n+1}+x_{n+2})\left(\frac{1}{x_{n+1}}+\frac{1}{x_{n+2}}\right)\\
&\qquad+(x_1+\cdots+x_n)\left(\frac{1}{x_{n+1}}+\frac{1}{x_{n+2}}\right)+(x_{n+1}+x_{n+2})\left(\frac{1}{x_1}+\cdots+\frac{1}{x_n}\right)\\
&\geqslant a_n^2+(x_{n+1}+x_{n+2})\left(\frac{1}{x_{n+1}}+\frac{1}{x_{n+2}}\right)\\
&\qquad+2\sqrt{(x_{n+1}+x_{n+2})\left(\frac{1}{x_{n+1}}+\frac{1}{x_{n+2}}\right)(x_1+\cdots+x_n)\left(\frac{1}{x_1}+\cdots+\frac{1}{x_n}\right)}\\
&= a_n^2+(x_{n+1}+x_{n+2})\left(\frac{1}{x_{n+1}}+\frac{1}{x_{n+2}}\right)+2a_n\sqrt{(x_{n+1}+x_{n+2})\left(\frac{1}{x_{n+1}}+\frac{1}{x_{n+2}}\right)},
\end{aligned}$$

where we used AM-GM to obtain the inequality. Furthermore, we have

$$(x_{n+1}+x_{n+2})\left(\frac{1}{x_{n+1}}+\frac{1}{x_{n+2}}\right)>4$$

because $x_{n+1}\neq x_{n+2}$ by assumption. Therefore, it follows that

$$a_{n+2}^2>a_n^2+4+4a_n=(a_n+2)^2.$$

Because a_{n+2} and a_n are both positive integers, we conclude that

$$a_{n+2}\geqslant a_n+3.$$

A simple induction gives $a_{2k+1}\geqslant 3k+a_1$ for every $k\geqslant 0$. Since $a_1=1$, it follows that $a_{2k+1}\geqslant 3k+1$. We get the desired conclusion for $k=1011$. □

Comment 1. A similar argument as in Solution 2 shows that $a_2\geqslant 3$ and $a_{2k}\geqslant 3k$ for every $k\geqslant 1$. Actually, these lower bounds on a_n are sharp (at least for $n\leqslant 2023$). In other words, there exists a sequence of distinct values $x_1,\ldots,x_{2023}>0$ for which

$$a_n=\begin{cases}\frac{3n-1}{2} & \text{if } n \text{ is odd,}\\ \frac{3n}{2} & \text{if } n \text{ is even,}\end{cases}$$

for $n=1,\ldots,2023$. The value of x_1 can be chosen arbitrarily. The next values can be obtained inductively by solving the quadratic equation

$$a_{n+1}^2=a_n^2+1+\left(\sum_{i=1}^n x_i\right)\frac{1}{x_{n+1}}+\left(\sum_{i=1}^n\frac{1}{x_i}\right)x_{n+1}$$

for x_{n+1}. Computation gives, for $n\geqslant 1$,

$$x_{n+1}=\begin{cases}\frac{3n}{2\left(\sum_{i=1}^n\frac{1}{x_i}\right)} & \text{if } n \text{ is even,}\\ \frac{6n+1\pm 3\sqrt{n(3n+2)}}{2\left(\sum_{i=1}^n\frac{1}{x_i}\right)} & \text{if } n \text{ is odd.}\end{cases}$$

One can check (with the help of a computer), that the values $x_1,\ldots,x_{2023}$ obtained by choosing $x_1=1$ and "+" every time in the odd case are indeed distinct.

It is interesting to note that the discriminant always vanishes in the even case. This is a consequence of $a_{n+1}=a_n+1$ being achieved as an equality case of AM-GM. Another cute observation is that the ratio x_2/x_1 is equal to the fourth power of the golden ratio.

Comment 2. The estimations in Solutions 1 and 2 can be made more efficiently if one applies the following form of the Cauchy–Schwarz inequality instead:

$$\sqrt{(a+b)(c+d)} \geqslant \sqrt{ac} + \sqrt{bd} \tag{2}$$

for arbitrary nonnegative numbers a, b, c, d. Equality occurs if and only if $a : c = b : d = (a+b) : (c+d)$.

For instance, by applying (2) to $a = x_1 + \cdots + x_n$, $b = x_{n+1}$, $c = \frac{1}{x_1} + \cdots + \frac{1}{x_n}$ and $d = \frac{1}{x_{n+1}}$ we get

$$\begin{aligned} a_{n+1} &= \sqrt{\left(x_1 + \cdots + x_n + x_{n+1}\right)\left(\frac{1}{x_1} + \cdots + \frac{1}{x_n} + \frac{1}{x_{n+1}}\right)} \\ &\geqslant \sqrt{\left(x_1 + \cdots + x_n\right)\left(\frac{1}{x_1} + \cdots + \frac{1}{x_n}\right)} + \sqrt{x_{n+1} \cdot \frac{1}{x_{n+1}}} = a_n + 1. \end{aligned}$$

A study of equality cases show that equality cannot occur twice in a row, as in Solution 1. Suppose that $a_{n+1} = a_n + 1$ and $a_{n+2} = a_{n+1} + 1$ for some index n. By the equality case in (2) we have

$$\frac{\left(x_1 + \cdots + x_n\right) + x_{n+1}}{\left(\frac{1}{x_1} + \cdots + \frac{1}{x_n}\right) + \frac{1}{x_{n+1}}} = \frac{x_{n+1}}{1/x_{n+1}} = x_{n+1}^2 \quad \text{because } a_{n+1} = a_n + 1,$$

and

$$\frac{x_1 + \cdots + x_n + x_{n+1}}{\frac{1}{x_1} + \cdots + \frac{1}{x_n} + \frac{1}{x_{n+1}}} = \frac{x_{n+2}}{1/x_{n+2}} = x_{n+2}^2 \quad \text{because } a_{n+2} = a_{n+1} + 1.$$

The left-hand sides are the same, so $x_{n+1} = x_{n+2}$, but this violates the condition that x_{n+1} and x_{n+2} are distinct.

The same trick applies to Solution 2. We can compare a_n and a_{n+2} directly as

$$\begin{aligned} a_{n+2} &= \sqrt{\left(x_1 + \cdots + x_n + x_{n+1} + x_{n+2}\right)\left(\frac{1}{x_1} + \cdots + \frac{1}{x_n} + \frac{1}{x_{n+1}} + \frac{1}{x_{n+2}}\right)} \\ &\geqslant \sqrt{\left(x_1 + \cdots + x_n\right)\left(\frac{1}{x_1} + \cdots + \frac{1}{x_n}\right)} + \sqrt{\left(x_{n+1} + x_{n+2}\right) \cdot \left(\frac{1}{x_{n+1}} + \frac{1}{x_{n+2}}\right)} \\ &= a_n + \sqrt{\left(x_{n+1} + x_{n+2}\right) \cdot \left(\frac{1}{x_{n+1}} + \frac{1}{x_{n+2}}\right)} \\ &\geqslant a_n + 2. \end{aligned}$$

In the last estimate, equality is not possible because x_{n+1} and x_{n+2} are distinct, so $a_{n+2} > a_n + 2$ and therefore $a_{n+2} \geqslant a_n + 3$.

A4. Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f\colon \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that

$$x\big(f(x)+f(y)\big) \geqslant \big(f(f(x))+y\big)f(y)$$

for every $x, y \in \mathbb{R}_{>0}$.

(Belgium)

Answer: All functions $f(x) = \frac{c}{x}$ for some $c > 0$.

Solution 1. Let $f\colon \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ be a function that satisfies the inequality of the problem statement. We will write $f^k(x) = f(f(\cdots f(x)\cdots))$ for the composition of f with itself k times, with the convention that $f^0(x) = x$. Substituting $y = x$ gives

$$x \geqslant f^2(x).$$

Substituting $x = f(y)$ instead leads to $f(y) + f^2(y) \geqslant y + f^3(y)$, or equivalently

$$f(y) - f^3(y) \geqslant y - f^2(y).$$

We can generalise this inequality. If we replace y by $f^{n-1}(y)$ in the above inequality, we get

$$f^n(y) - f^{n+2}(y) \geqslant f^{n-1}(y) - f^{n+1}(y),$$

for every $y \in \mathbb{R}_{>0}$ and for every integer $n \geqslant 1$. In particular, $f^n(y) - f^{n+2}(y) \geqslant y - f^2(y) \geqslant 0$ for every $n \geqslant 1$. Hereafter consider even integers $n = 2m$. Observe that

$$y - f^{2m}(y) = \sum_{i=0}^{m-1} \big(f^{2i}(y) - f^{2i+2}(y)\big) \geqslant m\big(y - f^2(y)\big).$$

Since f takes positive values, it holds that $y - f^{2m}(y) < y$ for every $m \geqslant 1$. So, we have proved that $y > m(y - f^2(y))$ for every $y \in \mathbb{R}_{>0}$ and every $m \geqslant 1$. Since $y - f^2(y) \geqslant 0$, this holds if only if

$$f^2(y) = y$$

for every $y \in \mathbb{R}_{>0}$. The original inequality becomes

$$xf(x) \geqslant yf(y)$$

for every $x, y \in \mathbb{R}_{>0}$. Hence, $xf(x)$ is constant. We conclude that $f(x) = c/x$ for some $c > 0$.

We now check that all the functions of the form $f(x) = c/x$ are indeed solutions of the original problem. First, note that all these functions satisfy $f(f(x)) = c/(c/x) = x$. So it's sufficient to check that $xf(x) \geqslant yf(y)$, which is true since $c \geqslant c$.

Solution 2. Let $f\colon \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ be a function that satisfies the inequality of the problem statement. As in Solution 1, we prove that

$$f^n(y) \geqslant f^{n+2}(y)$$

for every $y \in \mathbb{R}_{>0}$ and every $n \geqslant 0$. Since f takes positive values, this implies that

$$yf(y) \geqslant f(y)f^2(y) \geqslant f^2(y)f^3(y) \geqslant \cdots.$$

In other words, $yf(y) \geqslant f^n(y)f^{n+1}(y)$ for every $y \in \mathbb{R}_{>0}$ and every $n \geqslant 1$.

We replace x by $f^n(x)$ in the original inequality and get

$$f^n(x) - f^{n+2}(x) \geqslant \frac{yf(y) - f^n(x)f^{n+1}(x)}{f(y)}.$$

Using that $xf(x) \geqslant f^n(x)f^{n+1}(x)$, we obtain

$$f^n(x) - f^{n+2}(x) \geqslant \frac{yf(y) - xf(x)}{f(y)}$$

for every $n \geqslant 0$. The same trick as in Solution 1 gives

$$x > x - f^{2m}(x) = \sum_{i=0}^{m-1} (f^{2i}(x) - f^{2i+2}(x)) \geqslant m \cdot \frac{yf(y) - xf(x)}{f(y)}$$

for every $x, y \in \mathbb{R}_{>0}$ and every $m \geqslant 1$. Possibly permuting x and y, we may assume that $yf(y) - xf(x) \geqslant 0$ then the above inequality implies $xf(x) = yf(y)$. We conclude as in Solution 1.

A5. Let $a_1, a_2, \ldots, a_{2023}$ be positive integers such that

- $a_1, a_2, \ldots, a_{2023}$ is a permutation of $1, 2, \ldots, 2023$, and
- $|a_1 - a_2|, |a_2 - a_3|, \ldots, |a_{2022} - a_{2023}|$ is a permutation of $1, 2, \ldots, 2022$.

Prove that $\max\big(a_1, a_{2023}\big) \geqslant 507$.

(Australia)

Solution. For the sake of clarity, we consider and prove the following generalisation of the original problem (which is the case $N = 1012$):

> Let N be a positive integer and $a_1, a_2, \ldots, a_{2N-1}$ be positive integers such that
>
> - $a_1, a_2, \ldots, a_{2N-1}$ is a permutation of $1, 2, \ldots, 2N-1$, and
> - $|a_1 - a_2|, |a_2 - a_3|, \ldots, |a_{2N-2} - a_{2N-1}|$ is a permutation of $1, 2, \ldots, 2N-2$.
>
> Then $a_1 + a_{2N-1} \geqslant N + 1$ and hence $\max\big(a_1, a_{2N-1}\big) \geqslant \left\lceil \frac{N+1}{2} \right\rceil$.

Now we proceed to the proof of the generalised statement. We introduce the notion of *score* of a number $a \in \{1, 2, \ldots, 2N-1\}$. The score of a is defined to be

$$s(a) := |a - N|.$$

Note that, by the triangle inequality,

$$|a - b| \leqslant |a - N| + |N - b| = s(a) + s(b).$$

Considering the sum $|a_1 - a_2| + |a_2 - a_3| + \cdots + |a_{2N-2} - a_{2N-1}|$, we find that

$$\begin{aligned}(N-1)(2N-1) &= |a_1 - a_2| + |a_2 - a_3| + \cdots + |a_{2N-2} - a_{2N-1}| \\ &\leqslant 2\big(s(a_1) + s(a_2) + \cdots + s(a_{2N-1})\big) - \big(s(a_1) + s(a_{2N-1})\big) \\ &= 2N(N-1) - \big(s(a_1) + s(a_{2N-1})\big).\end{aligned}$$

For the last equality we used that the numbers $s(a_1), s(a_2), \ldots, s(a_{2N-1})$ are a permutation of $0, 1, 1, 2, 2, \ldots, N-1, N-1$.

Hence, $s(a_1) + s(a_{2N-1}) \leqslant 2N(N-1) - (N-1)(2N-1) = N - 1$. We conclude that

$$(N - a_1) + (N - a_{2N-1}) \leqslant s(a_1) + s(a_{2N-1}) \leqslant N - 1,$$

which implies $a_1 + a_{2N-1} \geqslant N + 1$.

Comment 1. In the case $N = 1012$, such a sequence with $\max\big(a_1, a_{2023}\big) = 507$ indeed exists:

$$507, 1517, 508, 1516, \ldots, 1011, 1013, 1012, 2023, 1, 2022, 2, \ldots, 1518, 506.$$

For a general even number N, a sequence with $\max\big(a_1, a_{2N-1}\big) = \left\lceil \frac{N+1}{2} \right\rceil$ can be obtained similarly. If $N \geqslant 3$ is odd, the inequality is not sharp, because $\max(a_1, a_{2N-1}) = \frac{N+1}{2}$ and $a_1 + a_{2N-1} \geqslant N + 1$ together imply $a_1 = a_{2N-1} = \frac{N+1}{2}$, a contradiction.

Comment 2. The formulation of the author's submission was slightly different:

> **Author's formulation.** Consider a sequence of positive integers $a_1, a_2, a_3, \ldots$ such that the following conditions hold for all positive integers m and n:
>
> - $a_{n+2023} = a_n + 2023$,
> - If $|a_{n+1} - a_n| = |a_{m+1} - a_m|$, then $2023 \mid (n - m)$, and
> - The sequence contains every positive integer.
>
> Prove that $a_1 \geqslant 507$.

The two formulations are equivalent up to relatively trivial arguments. Suppose (a_n) is a sequence satisfying the author's formulation. From the first and third conditions, we see that $a_1, \ldots, a_{2023}$ is a permutation of $1, \ldots, 2023$. Moreover, the sequence $|a_i - a_{i+1}|$ for $i = 1, 2, \ldots, 2022$ consists of positive integers $\leqslant 2022$ and has pairwise distinct elements by the second condition. Hence, it is a permutation of $1, \ldots, 2022$. It also holds that $a_1 > a_{2023}$, since if $a_1 < a_{2023}$ then $|a_{2024} - a_{2023}| = |2023 + a_1 - a_{2023}| \leqslant 2022$, which should be equal to $|a_i - a_{i+1}|$ for some $1 \leqslant i \leqslant 2022$, contradicting the second condition. This reduces the problem to the Shortlist formulation.

Conversely, if the numbers $a_1, \ldots, a_{2023}$ satisfy the conditions of the Shortlist formulation, then, after possibly reversing the sequence to ensure $a_1 > a_{2023}$, the sequence can be extended to an infinite sequence satisfying the conditions of the author's formulation.

A6. Let $k \geqslant 2$ be an integer. Determine all sequences of positive integers $a_1, a_2, \ldots$ for which there exists a monic polynomial P of degree k with non-negative integer coefficients such that

$$P(a_n) = a_{n+1}a_{n+2}\cdots a_{n+k}$$

for every integer $n \geqslant 1$.

(Malaysia)

Answer: The sequence (a_n) must be an arithmetic progression consisting of positive integers with common difference $d \geqslant 0$, and $P(x) = (x+d)\cdots(x+kd)$.

Common remarks. The following arguments and observations are implicit in the solutions given below.

Suppose the sequence (a_n) is an arithmetic progression with common difference $d \geqslant 0$. Then it satisfies the condition with

$$P(x) = (x+d)\cdots\big(x+kd\big).$$

This settles one direction. Now suppose (a_n) is a sequence satisfying the condition. We will show that it is a non-decreasing arithmetic progression.

Since $P(x)$ has non-negative integer coefficients, it is strictly increasing on the positive real line. In particular, it holds that, for any positive integer x, y,

$$P(x) < P(y) \quad \Longleftrightarrow \quad x < y.$$

Furthermore, if the sequence (a_n) is eventually constant, then $P(x) = x^k$ and the sequence (a_n) is actually constant. Indeed, if $P(x)$ were not the polynomial x^k, then $P(a_n) = a_{n+1}\cdots a_{n+k}$ cannot be satisfied for n such that $a_n = \cdots = a_{n+k}$. By a descending induction, we conclude that (a_n) is constant. Thus we can restrict to the case (a_n) is not eventually constant.

Solution 1. We assume that (a_n) is not eventually constant.

Step 1. The first goal is to show that the sequence must be *increasing*, i.e. $a_n < a_{n+1}$ for all $n \geqslant 1$.

First, by comparing the two equalities

$$P(a_n) = a_{n+1}a_{n+2}\cdots a_{n+k},$$
$$P(a_{n+1}) = a_{n+2}\cdots a_{n+k}a_{n+k+1},$$

we observe that

$$a_n < a_{n+1} \quad \Longleftrightarrow \quad P(a_n) < P(a_{n+1}) \quad \Longleftrightarrow \quad a_{n+1} < a_{n+k+1}, \tag{1}$$
$$a_n > a_{n+1} \quad \Longleftrightarrow \quad P(a_n) > P(a_{n+1}) \quad \Longleftrightarrow \quad a_{n+1} > a_{n+k+1}, \tag{2}$$
$$a_n = a_{n+1} \quad \Longleftrightarrow \quad P(a_n) = P(a_{n+1}) \quad \Longleftrightarrow \quad a_{n+1} = a_{n+k+1}. \tag{3}$$

Claim 1. $a_n \leqslant a_{n+1}$ for all $n \geqslant 1$.

Proof. Suppose, to the contrary, that $a_{n(0)-1} > a_{n(0)}$ for some $n(0) \geqslant 2$. We will give an infinite sequence of positive integers $n(0) < n(1) < \cdots$ satisfying

$$a_{n(i)-1} > a_{n(i)} \text{ and } a_{n(i)} > a_{n(i+1)}.$$

Then $a_{n(0)}, a_{n(1)}, a_{n(2)}, \ldots$ is an infinite decreasing sequence of positive integers, which is absurd.

We construct such a sequence inductively. If we have chosen $n(i)$, then we let $n(i+1)$ be the smallest index larger than $n(i)$ such that $a_{n(i)} > a_{n(i+1)}$. Note that such an index always exists and satisfies $n(i)+1 \leqslant n(i+1) \leqslant n(i)+k$ because $a_{n(i)} > a_{n(i)+k}$ by (2). We need to check that $a_{n(i+1)-1} > a_{n(i+1)}$. This is immediate if $n(i+1) = n(i)+1$ by construction. If $n(i+1) \geqslant n(i)+2$, then $a_{n(i+1)-1} \geqslant a_{n(i)}$ by minimality of $n(i+1)$, and so $a_{n(i+1)-1} \geqslant a_{n(i)} > a_{n(i+1)}$. □

We are now ready to prove that the sequence a_n is increasing. Suppose $a_n = a_{n+1}$ for some $n \geqslant 1$. Then we also have $a_{n+1} = a_{n+k+1}$ by (3), and since the sequence is non-decreasing we have $a_n = a_{n+1} = a_{n+2} = \cdots = a_{n+k+1}$. We repeat the argument for $a_{n+k} = a_{n+k+1}$ and get that the sequence is eventually constant, which contradicts our assumption. Hence

$$a_n < a_{n+1} \text{ for all } n \geqslant 1.$$

Step 2. The next and final goal is to prove that the sequence a_n is an *arithmetic progression*. Observe that we can make differences of terms appear as follows

$$\begin{aligned} P(a_n) &= a_{n+1}a_{n+2}\cdots a_{n+k} \\ &= \big(a_n + (a_{n+1} - a_n)\big)\big(a_n + (a_{n+2} - a_n)\big)\cdots\big(a_n + (a_{n+k} - a_n)\big). \end{aligned}$$

We will prove that, for n large enough, the sum

$$(a_{n+1} - a_n) + (a_{n+2} - a_n) + \cdots + (a_{n+k} - a_n)$$

is equal to the coefficient b of the term x^{k-1} in P. The argument is based on the following claim.

Claim 2. There exists a bound A with the following properties:

1. If $(c_1, \ldots, c_k)$ is a k-tuple of positive integers with $c_1 + \cdots + c_k > b$, then for every $x \geqslant A$ we have $P(x) < (x + c_1)(x + c_2)\cdots(x + c_k)$.
2. If $(c_1, \ldots, c_k)$ is a k-tuple of positive integers with $c_1 + \cdots + c_k < b$, then for every $x \geqslant A$ we have $P(x) > (x + c_1)(x + c_2)\cdots(x + c_k)$.

Proof. It suffices to show parts 1 and 2 separately, because then we can take the maximum of two bounds.

We first show part 1. For each single $(c_1, \ldots, c_k)$ such a bound A exists since

$$P(x) - (x + c_1)(x + c_2)\cdots(x + c_k) = (b - (c_1 + \cdots + c_k))x^{k-1} + (\text{terms of degree} \leqslant k - 2)$$

has negative leading coefficient and hence takes negative values for x large enough.

Suppose A is a common bound for all tuples $c = (c_1, \ldots, c_k)$ satisfying $c_1 + \cdots + c_k = b + 1$ (note that there are only finitely many such tuples). Then, for any tuple $c' = (c'_1, \ldots, c'_k)$ with $c'_1 + \cdots + c'_k > b$, there exists a tuple $c = (c_1, \ldots, c_k)$ with $c_1 + \cdots + c_k = b + 1$ and $c'_i \geqslant c_i$, and then the inequality for c' follows from the inequality for c.

We can show part 2 either in a similar way, or by using that there are only finitely many such tuples. □

Take A satisfying the assertion of Claim 2, and take N such that $n \geqslant N$ implies $a_n \geqslant A$. Then for each $n \geqslant N$, we have

$$(a_{n+1} - a_n) + \cdots + (a_{n+k} - a_n) = b.$$

By taking the difference of this equality and the equality for $n + 1$, we obtain

$$a_{n+k+1} - a_{n+1} = k(a_{n+1} - a_n)$$

for every $n \geqslant N$.

We conclude using an extremal principle. Let $d = \min\{a_{n+1} - a_n \mid n \geqslant N\}$, and suppose it is attained at some index $n \geqslant N$. Since

$$kd = k(a_{n+1} - a_n) = a_{n+k+1} - a_{n+1} = \sum_{i=1}^{k}(a_{n+i+1} - a_{n+i})$$

and each summand is at least d, we conclude that d is also attained at $n + 1, \ldots, n + k$, and inductively at all $n' \geqslant n$. We see that the equation $P(x) = (x + d)(x + 2d)\cdots(x + kd)$ is true for infinitely many values of x (all $a_{n'}$ for $n' \geqslant n$), hence this is an equality of polynomials. Finally we use (backward) induction to show that $a_{n+1} - a_n = d$ for every $n \geqslant 1$.

Solution 2. We assume that (a_n) is not eventually constant. In this solution, we first prove an alternative version of Claim 1.

Claim 3. There exist infinitely many $n \geqslant 1$ with

$$a_n \leqslant \min\{a_{n+1}, \ldots, a_{n+k}\}.$$

Proof. Suppose not, then for all but finitely many $n \geqslant 1$, it holds that $a_n > \min\{a_{n+1}, \ldots, a_{n+k}\}$. Hence for all large enough n, there always exist some $1 \leqslant l \leqslant k$ such that $a_n > a_{n+l}$. This induces an infinite decreasing sequence $a_n > a_{n+l_1} > a_{n+l_2} > \cdots$ of positive integers, which is absurd. □

We use Claim 3 to quickly settle the case $P(x) = x^k$. In that case, for every n with $a_n \leqslant \min\{a_{n+1}, \ldots, a_{n+k}\}$, since $a_{n+1} \cdots a_{n+k} = a_n^k$, it implies $a_n = a_{n+1} = \cdots = a_{n+k}$. This shows that the sequence is eventually constant, which contradicts our assumption.

From now on, assume

$$P(x) > x^k \text{ for all } x > 0.$$

Claim 4. For every $M > 0$, there exists some $N > 0$ such that $a_n > M$ for all $n > N$.

Proof. Suppose there exists some $M > 0$, such that $a_n \leqslant M$ for infinitely many n. For each i with $a_i \leqslant M$, we consider the k-tuple $(a_{i+1}, \ldots, a_{i+k})$. Then each of the terms in the k-tuple is bounded from above by $P(a_i)$, and hence by $P(M)$ too. Since the number of such k-tuples is bounded by $P(M)^k$, we deduce by the Pigeonhole Principle that there exist some indices $i < j$ such that $(a_{i+1}, \ldots, a_{i+k}) = (a_{j+1}, \ldots, a_{j+k})$. Since a_n is uniquely determined by the k terms before it, we conclude that $a_{i+k+1} = a_{j+k+1}$ must hold, and similarly $a_{i+l} = a_{j+l}$ for all $l \geqslant 0$, so the sequence is eventually periodic, for some period $p = j - i$.

Take K such that $a_n = a_{n+p}$ for every $n \geqslant K$. Then, by taking the products of the inequalities

$$a_n^k < P(a_n) = a_{n+1} \cdots a_{n+k}$$

for $K \leqslant n \leqslant K + p - 1$, we obtain

$$\begin{aligned}
\prod_{n=K}^{K+p-1} a_n^k &< \prod_{n=K}^{K+p-1} a_{n+1} \cdots a_{n+k} \\
&= a_{K+1} a_{K+2}^2 \cdots a_{K+k-1}^{k-1} \left(\prod_{n=K+k}^{K+p} a_n \right)^k a_{K+p+1}^{k-1} \cdots a_{K+p+k-2}^2 a_{K+p+k-1} \\
&= \left(\prod_{n=K}^{K+p-1} a_n \right)^k \qquad \text{(by periodicity)},
\end{aligned}$$

which is a contradiction. □

Write $P(x) = x^k + bx^{k-1} + Q(x)$, where $Q(x)$ is of degree at most $k-2$. Take M such that $x > M$ implies $x^{k-1} > Q(x)$.

Claim 5. There exist non-negative integers $b_1, \cdots, b_k$ such that $P(x) = (x + b_1) \cdots (x + b_k)$, and such that, for infinitely many $n \geqslant 1$, we have $a_{n+i} = a_n + b_i$ for every $1 \leqslant i \leqslant k$.

Proof. By Claims 3 and 4, there are infinitely many n such that

$$a_n > M \text{ and } a_n \leqslant \min\{a_{n+1}, \ldots, a_{n+k}\}.$$

Call such indices n to be *good*. We claim that if n is a good index then

$$\max\{a_{n+1}, \ldots, a_{n+k}\} \leqslant a_n + b.$$

Indeed, if $a_{n+i} \geqslant a_n + b + 1$, then together with $a_n \leqslant \min\{a_{n+1}, \ldots, a_{n+k}\}$ and $a_n^{k-1} > Q(a_n)$, we have

$$a_n^k + (b+1)a_n^{k-1} > a_n^k + ba_n^{k-1} + Q(a_n) = P(a_n) \geqslant (a_n + b + 1)a_n^{k-1},$$

a contradiction.

Hence for each good index n, we may write $a_{n+i} = a_n + b_i$ for all $1 \leqslant i \leqslant k$ for some choices of $(b_1, \ldots, b_k)$ (which may depend on n) and $0 \leqslant b_i \leqslant b$. Again by Pigeonhole Principle, some k-tuple $(b_1, \ldots, b_k)$ must be chosen for infinitely such good indices n. This means that the equation $P(a_n) = (a_n + b_1) \cdots (a_n + b_k)$ is satisfied by infinitely many good indices n. By Claim 4, a_n is unbounded among these a_n's, hence $P(x) = (x + b_1) \cdots (x + b_k)$ must hold identically. □

Claim 6. We have $b_i = ib_1$ for all $1 \leqslant i \leqslant k$.

Proof. Call an index n *excellent* if $a_{n+i} = a_n + b_i$ for every $1 \leqslant i \leqslant k$. From Claim 5 we know there are infinitely many excellent n.

We first show that for any pair $1 \leqslant i < j \leqslant k$ there is $1 \leqslant l \leqslant k$ such that $b_j = b_i + b_l$. Indeed, for such i and j and for excellent n, $a_n + b_j$ (which is equal to a_{n+j}) divides $P(a_{n+i}) = \prod_{l=1}^{k}(a_n + b_i + b_l)$, and hence divides $\prod_{l=1}^{k}(b_i + b_l - b_j)$. Since $a_n + b_j$ is unbounded among excellent n, we have $\prod_{l=1}^{k}(b_i + b_l - b_j) = 0$, hence there is l such that $b_j = b_i + b_l$.

In particular, $b_j = b_i + b_l \geqslant b_i$, i.e. $(b_1, \ldots, b_k)$ is non-decreasing.

Suppose $b_1 = 0$ and n is an excellent number. In particular, it holds that $a_n = a_{n+1}$. Moreover, since

$$a_{n+k+1}P(a_n) = a_{n+1} \cdots a_{n+k+1} = a_{n+1}P(a_{n+1}),$$

we have $a_n = a_{n+1} = a_{n+k+1}$, which divides $P(a_{n+i}) = \prod_{l=1}^{k}(a_n + b_i + b_l)$ for each $1 \leqslant i \leqslant k$. Hence a_n divides $\prod_{l=1}^{k}(b_i + b_l)$. By the same reasoning, we have $b_i + b_l = 0$ for some l, but since $b_i, b_l \geqslant 0$ we obtain $b_i = 0$ for each $1 \leqslant i \leqslant k$.

Now suppose $b_1 \geqslant 1$. Then, for each $1 \leqslant i < j \leqslant k$, we have $b_j - b_i = b_l \geqslant b_1 \geqslant 1$, hence $(b_1, \ldots, b_k)$ is strictly increasing. Therefore, the $k - 1$ elements $b_2 < b_3 < \cdots < b_k$ are exactly equal to $b_1 + b_1 < \cdots < b_1 + b_{k-1}$, since they cannot be equal to $b_1 + b_k$. This gives $b_i = ib_1$ for all $1 \leqslant i \leqslant k$ as desired. □

Claim 6 implies $P(x) = (x + d)(x + 2d) \cdots (x + kd)$ for some $d \geqslant 1$, and there are infinitely many indices n with $a_{n+i} = a_n + id$ for $1 \leqslant i \leqslant k$. By backwards induction, $P(a_{n-1}) = a_n \cdots a_{n+k-1}$ implies $a_{n-1} = a_n - d$, and so on. Thus $a_1, \ldots, a_n$ forms an arithmetic progression with common difference d. Since n can be arbitrarily large, the whole sequence is an arithmetic progression too, as desired.

Comment 1. A typical solution would first show some kind of increasing property (assuming a_n is not constant), and then use that property to deduce informations on the numbers $a_{n+i} - a_n$ $(1 \leqslant i \leqslant k)$ and/or on the polynomial P.

Solution 1 shows a strict one: $a_n < a_{n+1}$ (arguments after Claim 1), which makes the latter part easier. Solution 2 (Claims 3 and 4) shows only weaker increasing properties, which require more complicated/tricky arguments in the latter part but still can solve the problem.

Comment 2. It would be interesting to sort out the case when P can take negative integer coefficients, or (a_n) is just an integer sequence. Then a decreasing arithmetic progression is possible too, yet that is not the only possibility. There exist bounded examples such as $1, -1, 1, -1, \ldots$ with $P(x) = -x^2$, or $0, 1, -1, 0, 1, -1, \ldots$ with $P(x) = x^2 - 1$. If furthermore P is allowed to be non-monic, then the situation is even more unclear. For instance, the sequence $1, 2, 4, 8, \ldots$ works for the polynomial $P(x) = 8x^2$.

A7. Let N be a positive integer. Prove that there exist three permutations $a_1, a_2, \ldots, a_N$; $b_1, b_2, \ldots, b_N$; and $c_1, c_2, \ldots, c_N$ of $1, 2, \ldots, N$ such that

$$\left|\sqrt{a_k} + \sqrt{b_k} + \sqrt{c_k} - 2\sqrt{N}\right| < 2023$$

for every $k = 1, 2, \ldots, N$.

(China)

Solution 1. The idea is to approximate the numbers $\sqrt{1}, \sqrt{2}, \ldots, \sqrt{N}$ by the nearest integer with errors < 0.5. This gives the following sequence

$$1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 4, \ldots.$$

More precisely, for each $k \geqslant 1$, we round $\sqrt{k^2 - k + 1}, \ldots, \sqrt{k^2 + k}$ to k, so that there are $2k$ copies of k.

Step 1. We first consider the easier case when N has the form

$$N = m(m+1).$$

In this case, the numbers $\sqrt{1}, \sqrt{2}, \ldots, \sqrt{N}$ are approximated by the elements of the multiset $\{1_{\times 2}, 2_{\times 4}, 3_{\times 6}, \ldots, m_{\times 2m}\}$. Let T_m denote "half of" the multiset, i.e.

$$T_m := \{1_{\times 1}, 2_{\times 2}, 3_{\times 3}, \ldots, m_{\times m}\}.$$

We will prove by induction that there exists three permutations (u_k), (v_k), and (w_k) of the elements in the multiset T_m such that $u_k + v_k + w_k = 2m+1$ is constant for $k = 1, 2, \ldots, \frac{m(m+1)}{2}$.

When $m = 1$, take $1 + 1 + 1 = 3$. When $m = 2$, take $(1, 2, 2) + (2, 1, 2) + (2, 2, 1) = (5, 5, 5)$. Suppose that we have constructed three permutations (u_k), (v_k), and (w_k) of T_{m-1} satisfying $u_k + v_k + w_k = 2m - 1$ for every $k = 1, 2, \ldots, \frac{m(m-1)}{2}$. For T_m, we note that

$$T_m = T_{m-1} \sqcup \{m_{\times m}\},$$

and also

$$T_m = (T_{m-1} + 1) \sqcup \{1, 2, \ldots, m\}. \tag{1}$$

Here $T_{m-1} + 1$ means to add 1 to all elements in T_{m-1}. We construct the permutations (u'_k), (v'_k), and (w'_k) of T_m as follows:

- For $k = 1, 2, \ldots, \frac{m(m-1)}{2}$, we set $u'_k = u_k$, $v'_k = v_k + 1$, $w'_k = w_k + 1$.
- For $k = \frac{m(m-1)}{2} + r$ with $r = 1, 2, \ldots, m$, we set $u'_k = m$, $v'_k = r$, $w'_k = m + 1 - r$.

It is clear from (1) that (u'_k), (v'_k), and (w'_k) give three permutations of T_m, and that they satisfy $u'_k + v'_k + w'_k = 2m + 1$ for every $k = 1, 2, \ldots, \frac{m(m+1)}{2}$.

The inductive construction can be visualised by the $3 \times \frac{m(m+1)}{2}$ matrix

$$\begin{bmatrix} u_1 & \ldots & u_{m(m-1)/2} & m & \ldots & m \\ v_1 + 1 & \ldots & v_{m(m-1)/2} + 1 & 1 & \ldots & m \\ w_1 + 1 & \ldots & w_{m(m-1)/2} + 1 & m & \ldots & 1 \end{bmatrix},$$

in which the three rows represent the permutations $(u'_k), (v'_k), (w'_k)$, and the sum of the three entries of each column is $2m + 1$.

Thus, when $N = m^2 + m$, we can construct permutations (a_k), (b_k), and (c_k) of $1, 2, \ldots, N$ such that

$$2m + 1 - 1.5 < \sqrt{a_k} + \sqrt{b_k} + \sqrt{c_k} < 2m + 1 + 1.5. \tag{2}$$

This gives

$$\left|\sqrt{a_k} + \sqrt{b_k} + \sqrt{c_k} - 2\sqrt{N}\right| < 2.5 < 2023,$$

where we used that $-1 < 2m - 2\sqrt{m^2 + m} < 0$ for positive m.

Step 2. We now proceed to the general case. Let m be such that

$$m(m + 1) \leqslant N < (m + 1)(m + 2).$$

Write $N = m(m + 1) + t$ for some $t \in \{0, 1, \ldots, 2m + 1\}$ and let

$$L := \left\lfloor \frac{4}{9}N \right\rfloor.$$

We will make use of the following inequalities below:

$$N > m^2, \quad N < (m + 2)^2, \quad t \leqslant 2m + 1, \quad L + 1 > 4N/9, \quad L \leqslant 4N/9.$$

As above, we construct three permutations (a_k), (b_k), and (c_k) of $1, 2, \ldots, m(m+1)$ satisfying (2). Now we construct the three required permutations (A_k), (B_k), and (C_k) of $1, 2, \ldots, N$ as follows:

For $k = 1, 2, \ldots, m(m + 1)$, if $a_k \leqslant L$, take $A_k = a_k$, and if $a_k > L$, take $A_k = a_k + t$. For $k = m(m + 1) + r$ with $r = 1, 2, \ldots, t$, set $A_k = L + r$. Define the permutations (B_k) and (C_k) similarly. Now for $k = 1, 2, \ldots, m(m + 1)$, we show $0 \leqslant \sqrt{A_k} - \sqrt{a_k} \leqslant 2$. The lower bound is obvious. If $m \leqslant 1$, then $N \leqslant 5$ and hence $\sqrt{A_k} - \sqrt{a_k} \leqslant \sqrt{5} - \sqrt{1} \leqslant 2$. If $m \geqslant 2$, then

$$\sqrt{A_k} - \sqrt{a_k} = \frac{A_k - a_k}{\sqrt{A_k} + \sqrt{a_k}} \leqslant \frac{t}{2\sqrt{L + 1}} \leqslant \frac{2m + 1}{\frac{4}{3}m} \leqslant 2.$$

We have similar inequalities for (B_k) and (C_k). Thus

$$2\sqrt{N} - 4.5 < 2m + 1 - 1.5 \leqslant \sqrt{A_k} + \sqrt{B_k} + \sqrt{C_k} \leqslant 2m + 1 + 1.5 + 6 < 2\sqrt{N} + 8.5.$$

For $k = m^2 + m + 1, \ldots, m^2 + m + t$, we have

$$2\sqrt{N} < 3\sqrt{L + 1} \leqslant \sqrt{A_k} + \sqrt{B_k} + \sqrt{C_k} \leqslant 3\sqrt{L + t} \leqslant \sqrt{4N + 9t} < 2\sqrt{N} + 8.5.$$

To sum up, we have defined three permutations (A_k), (B_k), and (C_k) of $1, 2, \ldots, N$, such that

$$\left|\sqrt{A_k} + \sqrt{B_k} + \sqrt{C_k} - 2\sqrt{N}\right| < 8.5 < 2023.$$

holds for every $k = 1, 2, \ldots, N$. □

Solution 2. This is a variation of Solution 1 that uses induction for Step 2.

Let n be an integer satisfying $0 \leqslant n \leqslant m + 1$ and define the multiset $T_{m,n}$ by

$$T_{m,n} := \{1_{\times 1}, 2_{\times 2}, 3_{\times 3}, \ldots, m_{\times m}, (m + 1)_{\times n}\}.$$

In other words, $T_{m,0} = T_m$, $T_{m,n} = T_m \sqcup \{(m + 1)_{\times n}\}$ and $T_{m,m+1} = T_{m+1}$, where T_m is the set defined in Solution 1.

Claim. There exist three permutations (u_k), (v_k), (w_k) of $T_{m,n}$ such that

$$\begin{cases} u_k+v_k+w_k &= 2m+1 & (n=0),\\ u_k+v_k+w_k &\in \{2m+1,2m+2,2m+3\} & (1\leqslant n\leqslant m),\\ u_k+v_k+w_k &= 2m+3 & (n=m+1). \end{cases}$$

Proof. We proceed by induction on m. If $n=0$ or $n=m+1$, the assertion can be proved as in Solution 1. If $1\leqslant n\leqslant m$, we note that

$$T_{m,n}=T_{m-1,n}\sqcup\{m_{\times(m-n)},(m+1)_{\times n}\}=(T_{m-1,n}+1)\sqcup\{1,2,\dots,m\}.$$

From the hypothesis of induction, it follows that we have three permutations (u_k), (v_k), (w_k) of $T_{m-1,n}$ satisfying $u_k+v_k+w_k\in\{2m-1,2m,2m+1\}$ for every k. We construct the permutations (u'_k), (v'_k), and (w'_k) of $T_{m,n}$ as follows:

- For $k=1,2,\dots,\frac{m(m-1)}{2}+n$, we set $u'_k=u_k$, $v'_k=v_k+1$, and $w'_k=w_k+1$.
- For $k=\frac{m(m-1)}{2}+n+r$ with $r=1,2,\dots,m$, we set $u'_k=m$ if $1\leqslant r\leqslant m-n$ while $u'_k=m+1$ if $m-n+1\leqslant r\leqslant m$, $v'_k=r$, and $w'_k=m+1-r$.

It is clear from the construction that (u'_k), (v'_k), and (w'_k) give three permutations of $T_{m,n}$, and they satisfy $u'_k+v'_k+w'_k\in\{2m+1,2m+2,2m+3\}$ for every $k=1,2,\dots,\frac{m(m+1)}{2}+n$.

Again, we can visualise the construction using the matrix

$$\begin{bmatrix} u_1 & \dots & u_{m(m-1)/2+n} & m & \dots & m & m+1 & \dots & m+1\\ v_1+1 & \dots & v_{m(m-1)/2+n}+1 & 1 & \dots & \dots & \dots & \dots & m\\ w_1+1 & \dots & w_{m(m-1)/2+n}+1 & m & \dots & \dots & \dots & \dots & 1 \end{bmatrix}. \qquad \square$$

In general, we have $m(m+1)\leqslant N<(m+1)(m+2)$ for some $m\geqslant 0$. Set $N=m(m+1)+t$ for some $t\in\{0,1,\dots,2m+1\}$. Then the approximation of $\{\sqrt{1},\sqrt{2},\dots,\sqrt{N}\}$ by the nearest integer with errors <0.5 is a multiset

$$\{1_{\times 2},2_{\times 4},\dots,m_{\times 2m},(m+1)_{\times t}\}=T_{m,n_1}\sqcup T_{m,n_2}$$

with $n_1=\lfloor t/2\rfloor$ and $n_2=\lceil t/2\rceil$.

Since $0\leqslant n_1\leqslant n_2\leqslant m+1$, by using the Claim we can construct permutations (a_k), (b_k), and (c_k) to satisfy the following inequality:

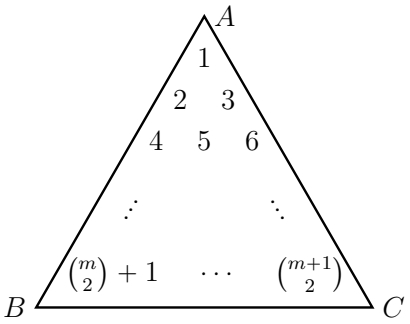
$$2m+1-1.5<\sqrt{a_k}+\sqrt{b_k}+\sqrt{c_k}<2m+3+1.5.$$

Since $m<\sqrt{N}<m+2$, it follows that

$$2\sqrt{N}-4.5<2m+1-1.5<\sqrt{a_k}+\sqrt{b_k}+\sqrt{c_k}<2m+3+1.5<2\sqrt{N}+4.5,$$

and so

$$\left|\sqrt{A_k}+\sqrt{B_k}+\sqrt{C_k}-2\sqrt{N}\right|<4.5<2023. \qquad \square$$



Solution 3. This solution is based on the geometrical insight of equilateral triangles.

Step 1. We first consider the easier case of triangle numbers

$$N = \frac{m(m+1)}{2}.$$

As shown in the following picture, consider the triangular shaped lattice points inside an equilateral triangle ABC with a total of N points. The lattice is built in a way that the ℓth row has exactly ℓ points for each $\ell = 1, 2, \ldots, m$. Rows are numbered in three different ways, one for each vertex.

Each point P_k in the triangular lattice is labelled with a triple of integers (a_k, b_k, c_k) as follows. The first coordinate is called the *A-coordinate*, and so on for B, C. To define the A-coordinate, denoted $W_a(\bullet)$, first label the lattice points by $1, 2, 3, \ldots$ starting with the point closest to A and then going down the rows with the rule that within a row, the labelling is from left to right (see right picture). The B-coordinate, denoted $W_b(\bullet)$, is defined by rotating the A-coordinate counterclockwise by 120°. The C-coordinate, denoted $W_c(\bullet)$, similarly, by rotating the A-coordinate counterclockwise by 240°.

Assume that a point P lies in the ${\ell_a}^{\text{th}}$ row from the vertex A, in the ${\ell_b}^{\text{th}}$ row from the vertex B, and in the ${\ell_c}^{\text{th}}$ row from the vertex C. Note that ℓ_a is proportional to the height of A in the triangle, minus the height of P. Since inside an equilateral triangle, the sum of the lengths of the heights from a point to the three sides is independent of the point, we must have

$$\ell_a + \ell_b + \ell_c = 2m + 1 = \sqrt{8N+1}.$$

Since there are exactly $1 + 2 + \cdots + \ell = \frac{\ell(\ell+1)}{2}$ points in the first ℓ rows, the A-labeling $W_a(P)$ of the point P satisfies

$$\frac{\ell_a(\ell_a - 1)}{2} + 1 \leqslant W_a(P) \leqslant \frac{\ell_a(\ell_a + 1)}{2}.$$

In paticular,

$$\left(\ell_a - \frac{1}{2}\right)^2 < 2W_a(P) < \left(\ell_a + \frac{1}{2}\right)^2.$$

Taking the cyclic sum gives

$$\left|\sqrt{2W_a(P)} + \sqrt{2W_b(P)} + \sqrt{2W_c(P)} - (\ell_a + \ell_b + \ell_c)\right| < \frac{3}{2}$$

and thus

$$\left|\sqrt{W_a(P)} + \sqrt{W_b(P)} + \sqrt{W_c(P)} - 2\sqrt{N + \frac{1}{8}}\right| < \frac{3}{2} \cdot \frac{1}{\sqrt{2}} = \frac{3\sqrt{2}}{4}.$$

Step 2. Now, for a general positive integer N, there exists a positive integer m such that

$$\frac{m(m-1)}{2} + 1 \leqslant N \leqslant \frac{m(m+1)}{2}.$$

Write $N = \frac{m(m+1)}{2} - t$ with $t \in \{0, 1, \ldots, m-1\}$. We modify the above construction for $\frac{m(m+1)}{2}$ points into a construction for N points as follows. We remove t arbitrary points from the mth row (namely the bottom row) of the triangular lattice. The remaining triangular lattice has $\frac{m(m+1)}{2} - t = N$ points, and we assign their A-, B-, and C-coordinates as before (in the same order, yet skipping over the points that are removed so that the coordinates exactly form permutations of $1, 2, \ldots, N$).

For each point P in the triangular lattice (that was not removed earlier), suppose that it is in the ${\ell_a}^{\text{th}}$, ${\ell_b}^{\text{th}}$, and ${\ell_c}^{\text{th}}$ row when viewed from A, B, and C, respectively. Now the A-coordinates $W_a(P)$ still satisfies

$$\frac{\ell_a(\ell_a-1)}{2} + 1 \leqslant W_a(P) \leqslant \frac{\ell_a(\ell_a+1)}{2}.$$

The B-coordinate $W_b(P)$ satisfies

$$\frac{(\ell_b-1)(\ell_b-2)}{2} + 1 \leqslant W_b(P) \leqslant \frac{\ell_b(\ell_b+1)}{2},$$

because, viewing from point B, we have removed either 0 or 1 point from each row, and the first $\ell_b - 1$ rows have at least $0 + 1 + \cdots + (\ell_b - 2) = \frac{(\ell_b-1)(\ell_b-2)}{2}$ points left. For the same reason, the C-labeling $W_c(P)$ satisfies

$$\frac{(\ell_c-1)(\ell_c-2)}{2} + 1 \leqslant W_c(P) \leqslant \frac{\ell_c(\ell_c+1)}{2}.$$

From this, we deduce that

$$\begin{aligned}
\ell_a - \frac{1}{2} &< \sqrt{2W_a(P)} < \ell_a + \frac{1}{2}, \\
\ell_b - \frac{3}{2} &< \sqrt{2W_b(P)} < \ell_b + \frac{1}{2}, \\
\ell_c - \frac{3}{2} &< \sqrt{2W_c(P)} < \ell_c + \frac{1}{2}.
\end{aligned}$$

Combining all above with the inequalities $2m-1 < 2\sqrt{2N} < 2m+1$ and $\ell_a+\ell_b+\ell_c = 2m+1$, we deduce that

$$\begin{aligned}
2\sqrt{2N} - \frac{7}{2} < (2m+1) - \frac{7}{2} < \sqrt{2W_a(P)} + \sqrt{2W_b(P)} + \sqrt{2W_c(P)} \\
< (2m+1) + \frac{3}{2} < 2\sqrt{2N} + \frac{7}{2}.
\end{aligned}$$

Therefore, for each point P, we have

$$\left|\sqrt{W_a(P)} + \sqrt{W_b(P)} + \sqrt{W_c(P)} - 2\sqrt{N}\right| < \frac{7}{2} \cdot \frac{1}{\sqrt{2}} < 2.5 < 2013.$$

We may finally order of the N points in an arbitrary way. Then the A-labelings $W_a(\bullet)$ give the permutation $a_1, \ldots, a_N$, the B-labelings $W_b(\bullet)$ give $b_1, \ldots, b_N$, and the C-labelings $W_c(\bullet)$ give $c_1, \ldots, c_N$.

For each $k = 1, 2, \ldots, N$, we have

$$\left|\sqrt{a_k} + \sqrt{b_k} + \sqrt{c_k} - 2\sqrt{N}\right| < 2.5 < 2023. \qquad \square$$

Comment. We can make the same argument as in Solution 3 without using geometry or diagram instead using barycentric coordinates in integers and lexicographic order.

For $N = \frac{m(m+1)}{2}$, consider a set

$$X = \{(x, y, z) \in \mathbb{Z}^3 \mid 0 \leqslant x, y, z \leqslant m-1,\ x+y+z = m-1\}.$$

and the lexicographic order of X, i.e.

$$(x, y, z) > (x', y', z') \Longleftrightarrow \begin{cases} x > x' \text{ or} \\ x = x' \text{ and } y > y' \text{ or} \\ x = x' \text{ and } y = y' \text{ and } z > z'. \end{cases}$$

Then for an element $Q_k = (x_k, y_k, z_k)$

- Define $W_a(Q_k)$ so that (x_k, y_k, z_k) is the $W_a(Q_k)^{\text{th}}$ biggest element in X.
- Define $W_b(Q_k)$ so that (y_k, z_k, x_k) is the $W_b(Q_k)^{\text{th}}$ biggest element in $X' = \{(y, z, x) \mid (x, y, z) \in X\}$.
- Define $W_c(Q_k)$ so that (z_k, x_k, y_k) is the $W_c(Q_k)^{\text{th}}$ biggest element in $X'' = \{(z, x, y) \mid (x, y, z) \in X\}$.

The same argument as in Solution 2 then holds.

Observe that for an element $Q_k = (x_k, y_k, z_k)$, it holds that $\ell_a = m - x_k$, $\ell_b = m - y_k$, and $\ell_c = m - z_k$.

Combinatorics

C1. Let m and n be positive integers greater than 1. In each unit square of an $m \times n$ grid lies a coin with its tail-side up. A *move* consists of the following steps:

1. select a 2×2 square in the grid;
2. flip the coins in the top-left and bottom-right unit squares;
3. flip the coin in either the top-right or bottom-left unit square.

Determine all pairs (m, n) for which it is possible that every coin shows head-side up after a finite number of moves.

(Thailand)

Answer: The answer is all pairs (m, n) satisfying $3 \mid mn$.

Solution 1. Let us denote by (i, j)-square the unit square in the ith row and the jth column.

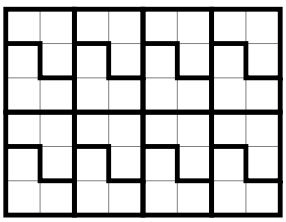
We first prove that when $3 \mid mn$, it is possible to make all the coins show head-side up. For integers $1 \leqslant i \leqslant m-1$ and $1 \leqslant j \leqslant n-1$, denote by $A(i, j)$ the move that flips the coin in the (i, j)-square, the $(i+1, j+1)$-square and the $(i, j+1)$-square. Similarly, denote by $B(i, j)$ the move that flips the coin in the (i, j)-square, $(i+1, j+1)$-square, and the $(i+1, j)$-square. Without loss of generality, we may assume that $3 \mid m$.

Case 1: n is even.

We apply the moves

- $A(3k-2, 2l-1)$ for all $1 \leqslant k \leqslant \frac{m}{3}$ and $1 \leqslant l \leqslant \frac{n}{2}$,
- $B(3k-1, 2l-1)$ for all $1 \leqslant k \leqslant \frac{m}{3}$ and $1 \leqslant l \leqslant \frac{n}{2}$.

This process will flip each coin exactly once, hence all the coins will face head-side up afterwards.



Case 2: n is odd.

We start by applying

- $A(3k-2, 2l-1)$ for all $1 \leqslant k \leqslant \frac{m}{3}$ and $1 \leqslant l \leqslant \frac{n-1}{2}$,
- $B(3k-1, 2l-1)$ for all $1 \leqslant k \leqslant \frac{m}{3}$ and $1 \leqslant l \leqslant \frac{n-1}{2}$

as in the previous case. At this point, the coins on the rightmost column have tail-side up and the rest of the coins have head-side up. We now apply the moves

- $A(3k-2, n-1)$, $A(3k-1, n-1)$ and $B(3k-2, n-1)$ for every $1 \leqslant k \leqslant \frac{m}{3}$.

For each k, the three moves flip precisely the coins in the $(3k-2,n)$-square, the $(3k-1,n)$-square, and the $(3k,n)$-square. Hence after this process, every coin will face head-side up.

We next prove that mn being divisible by 3 is a necessary condition. We first label the (i,j)-square by the remainder of $i+j-2$ when divided by 3, as shown in the figure.

0	1	2	0	$\cdots$
1	2	0	1	$\cdots$
2	0	1	2	$\cdots$
0	1	2	0	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Let $T(c)$ be the number of coins facing head-side up in those squares whose label is c. The main observation is that each move does not change the parity of both $T(0)-T(1)$ and $T(1)-T(2)$, since a move flips exactly one coin in a square with each label. Initially, all coins face tail-side up at the beginning, thus all of $T(0),T(1),T(2)$ are equal to 0. Hence it follows that any configuration that can be achieved from the initial state must satisfy the parity condition of

$$T(0) \equiv T(1) \equiv T(2) \pmod 2.$$

We now calculate the values of T for the configuration in which all coins are facing head-side up.

- When $m \equiv n \equiv 1 \pmod 3$, we have $T(0)-1 = T(1) = T(2) = \frac{mn-1}{3}$.
- When $m \equiv 1 \pmod 3$ and $n \equiv 2 \pmod 3$, or $m \equiv 2 \pmod 3$ and $n \equiv 1 \pmod 3$, we have $T(0)-1 = T(1)-1 = T(2) = \frac{mn-2}{3}$.
- When $m \equiv n \equiv 2 \pmod 3$, we have $T(0) = T(1)-1 = T(2) = \frac{mn-1}{3}$.
- When $m \equiv 0 \pmod 3$ or $n \equiv 0 \pmod 3$, we have $T(0) = T(1) = T(2) = \frac{mn}{3}$.

From this calculation, we see that $T(0)$, $T(1)$ and $T(2)$ has the same parity only when mn is divisible by 3.

Comment 1. The original proposal of the problem also included the following question as part (b):

> For each pair (m,n) of integers greater than 1, how many configurations can be obtained by applying a finite number of moves?

An explicit construction of a sequence of moves shows that $T(0)$, $T(1)$, and $T(2)$ having the same parity is a necessary and sufficient condition for a configuration to obtainable after a finite sequence of moves, and this shows that the answer is 2^{mn-2}.

Comment 2. A significantly more difficult problem is to ask the following question: for pairs (m,n) such that the task is possible (i.e. $3 \mid mn$), what is the smallest number of moves required to complete this task? The answer is:

- $\frac{mn}{3}$ if mn is even;
- $\frac{mn}{3}+2$ if mn is odd.

To show this, we observe that we can flip all coins in any 2×3 (or 3×2) by using a minimum of two moves. Furthermore, when mn is odd with $3 \mid mn$, it is impossible to tile an $m\times n$ table with one type of L-tromino and its 180°-rotated L-tromino (disallowing rotations and reflections). The only known proof of the latter claim is lengthy and difficult, and it requires some group-theoretic arguments by studying the title homotopy group given by these two L-tromino tiles. This technique was developed by J. H. Conway and J. C. Lagarias in *Tiling with Polyominoes and Combinatorial Group Theory*, Journal of Combinatorial Group Theory, Series A **53**, 183-208 (1990).

Comment 3. Here is neat way of defining the invariant. Consider a finite field $\mathbb{F}_4 = \{0, 1, \omega, \omega + 1\}$, where $1 + 1 = \omega^2 + \omega + 1 = 0$ in $\mathbb{F}_4$. Consider the set

$$H = \{(i, j) | 1 \leqslant i \leqslant m,\ 1 \leqslant j \leqslant n, \text{ the coin in the } (i, j)\text{-square is head-side up}\}.$$

and the invariant

$$I(H) = \sum_{(i,j)\in H} \omega^{i+j} \in \mathbb{F}_4.$$

Then the value of $I(H)$ does not change under applying moves, and when all coins are tail-side up, it holds that $I(H) = 0$. On the other hand, its value when all coins are head-side up can be computed as

$$I(H) = \sum_{i=1}^{m} \sum_{j=1}^{n} \omega^{i+j} = \Big(\sum_{i=1}^{m} \omega^i\Big)\Big(\sum_{j=1}^{n} \omega^j\Big).$$

This is equal to $0 \in \mathbb{F}_4$ if and only if $3 \mid mn$.

C2. Determine the maximal length L of a sequence $a_1, \ldots, a_L$ of positive integers satisfying both the following properties:

- every term in the sequence is less than or equal to 2^{2023}, and
- there does not exist a consecutive subsequence $a_i, a_{i+1}, \ldots, a_j$ (where $1 \leqslant i \leqslant j \leqslant L$) with a choice of signs $s_i, s_{i+1}, \ldots, s_j \in \{1, -1\}$ for which

$$s_i a_i + s_{i+1} a_{i+1} + \cdots + s_j a_j = 0.$$

(Czech Republic)

Answer: The answer is $L = 2^{2024} - 1$.

Solution. We prove more generally that the answer is $2^{k+1} - 1$ when 2^{2023} is replaced by 2^k for an arbitrary positive integer k. Write $n = 2^k$.

We first show that there exists a sequence of length $L = 2n-1$ satisfying the properties. For a positive integer x, denote by $v_2(x)$ the maximal nonnegative integer v such that 2^v divides x. Consider the sequence $a_1, \ldots, a_{2n-1}$ defined as

$$a_i = 2^{k - v_2(i)}.$$

For example, when $k = 2$ and $n = 4$, the sequence is

$$4, 2, 4, 1, 4, 2, 4.$$

This indeed consists of positive integers less than or equal to $n = 2^k$, because $0 \leqslant v_2(i) \leqslant k$ for $1 \leqslant i \leqslant 2^{k+1} - 1$.

Claim 1. This sequence $a_1, \ldots, a_{2n-1}$ does not have a consecutive subsequence with a choice of signs such that the signed sum equals zero.

Proof. Let $1 \leqslant i \leqslant j \leqslant 2n - 1$ be integers. The main observation is that amongst the integers

$$i, i+1, \ldots, j-1, j,$$

there exists a unique integer x with the maximal value of $v_2(x)$. To see this, write $v = \max(v_2(i), \ldots, v_2(j))$. If there exist at least two multiples of 2^v amongst $i, i+1, \ldots, j$, then one of them must be a multiple of 2^{v+1}, which is a contradiction.

Therefore there is exactly one $i \leqslant x \leqslant j$ with $v_2(x) = v$, which implies that all terms except for $a_x = 2^{k-v}$ in the sequence

$$a_i, a_{i+1}, \ldots, a_j$$

are a multiple of 2^{k-v+1}. The same holds for the terms $s_i a_i, s_{i+1} a_{i+1}, \ldots, s_j a_j$, hence the sum cannot be equal to zero. □

We now prove that there does not exist a sequence of length $L \geqslant 2n$ satisfying the conditions of the problem. Let $a_1, \ldots, a_L$ be an arbitrary sequence consisting of positive integers less than or equal to n. Define a sequence $s_1, \ldots, s_L$ of signs recursively as follows:

- when $s_1 a_1 + \cdots + s_{i-1} a_{i-1} \leqslant 0$, set $s_i = +1$,
- when $s_1 a_1 + \cdots + s_{i-1} a_{i-1} \geqslant 1$, set $s_i = -1$.

Write

$$b_i = \sum_{j=1}^{i} s_i a_i = s_1 a_1 + \cdots + s_i a_i,$$

and consider the sequence

$$0 = b_0, b_1, b_2, \ldots, b_L.$$

Claim 2. All terms b_i of the sequence satisfy $-n+1 \leqslant b_i \leqslant n$.

Proof. We prove this by induction on i. It is clear that $b_0 = 0$ satisfies $-n+1 \leqslant 0 \leqslant n$. We now assume $-n+1 \leqslant b_{i-1} \leqslant n$ and show that $-n+1 \leqslant b_i \leqslant n$.

Case 1: $-n+1 \leqslant b_{i-1} \leqslant 0$.

Then $b_i = b_{i-1} + a_i$ from the definition of s_i, and hence

$$-n+1 \leqslant b_{i-1} < b_{i-1} + a_i \leqslant b_{i-1} + n \leqslant n.$$

Case 2: $1 \leqslant b_{i-1} \leqslant n$.

Then $b_i = b_{i-1} - a_i$ from the definition of s_i, and hence

$$-n+1 \leqslant b_{i-1} - n \leqslant b_{i-1} - a_i < b_{i-1} \leqslant n.$$

This finishes the proof. □

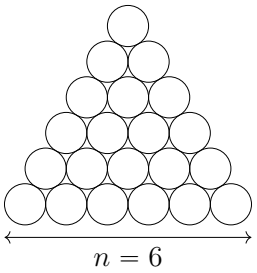
Because there are $2n$ integers in the closed interval $[-n+1, n]$ and at least $2n+1$ terms in the sequence $b_0, b_1, \ldots, b_L$ (as $L+1 \geqslant 2n+1$ by assumption), the pigeonhole principle implies that two distinct terms b_{i-1}, b_j (where $1 \leqslant i \leqslant j \leqslant L$) must be equal. Subtracting one from another, we obtain

$$s_i a_i + \cdots + s_j a_j = b_j - b_{i-1} = 0$$

as desired.

Comment. The same argument gives a bound $L \leqslant 2n-1$ that works for all n, but this bound is not necessarily sharp when n is not a power of 2. For instance, when $n = 3$, the longest sequence has length $L = 3$.

C3. Let n be a positive integer. We arrange $1 + 2 + \cdots + n$ circles in a triangle with n rows, such that the ith row contains exactly i circles. The following figure shows the case $n = 6$.



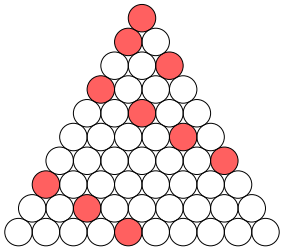
In this triangle, a *ninja-path* is a sequence of circles obtained by repeatedly going from a circle to one of the two circles directly below it. In terms of n, find the largest value of k such that if one circle from every row is coloured red, we can always find a ninja-path in which at least k of the circles are red.

(Netherlands)

Answer: The maximum value is $k = 1 + \lfloor \log_2 n \rfloor$.

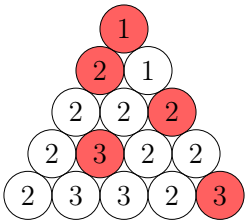
Solution 1. Write $N = \lfloor \log_2 n \rfloor$ so that we have $2^N \leqslant n \leqslant 2^{N+1} - 1$.

We first provide a construction where every ninja-path passes through at most $N + 1$ red circles. For the row $i = 2^a + b$ for $0 \leqslant a \leqslant N$ and $0 \leqslant b < 2^a$, we colour the $(2b + 1)^{\text{th}}$ circle.



Then every ninja-path passes through at most one red circle in each of the rows $2^a, 2^a + 1, \ldots, 2^{a+1} - 1$ for each $0 \leqslant a \leqslant N$. It follows that every ninja-path passes through at most $N + 1$ red circles.

We now prove that for every colouring, there exists a ninja-path going through at least $N+1$ red circles. For each circle C, we assign the maximum number of red circles in a ninja-path that starts at the top of the triangle and ends at C.



Note that

- if C is not red, then the number assigned to C is the maximum of the number assigned to the one or two circles above C, and
- if C is red, then the number assigned to C is one plus the above maximum.

Write $v_1, \ldots, v_i$ for the numbers in row i, and let v_m be the maximum among these numbers. Then the numbers in row $i+1$ will be at least

$$v_1, \ldots, v_{m-1}, v_m, v_m, v_{m+1}, \ldots, v_i,$$

not taking into account the fact that one of the circles in row $i+1$ is red. On the other hand, for the red circle in row $i+1$, the lower bound on the assigned number can be increased by 1. Therefore the sum of the numbers in row $i+1$ is at least

$$(v_1 + \cdots + v_i) + v_m + 1.$$

Using this observation, we prove the following claim.

Claim 1. Let σ_k be the sum of the numbers assigned to circles in row k. Then for $0 \leqslant j \leqslant N$, we have $\sigma_{2^j} \geqslant j \cdot 2^j + 1$.

Proof. We use induction on j. This is clear for $j = 0$, since the number in the first row is always 1. For the induction step, suppose that $\sigma_{2^j} \geqslant j \cdot 2^j + 1$. Then the maximum value assigned to a circle in row 2^j is at least $j+1$. As a consequence, for every $k \geqslant 2^j$, there is a circle on row k with number at least $j+1$. Then by our observation above, we have

$$\sigma_{k+1} \geqslant \sigma_k + (j+1) + 1 = \sigma_k + (j+2).$$

Then we get

$$\sigma_{2^{j+1}} \geqslant \sigma_{2^j} + 2^j(j+2) \geqslant j \cdot 2^j + 1 + 2^j(j+2) = (j+j+2)2^j + 1 = (j+1)2^{j+1} + 1.$$

This completes the inductive step. □

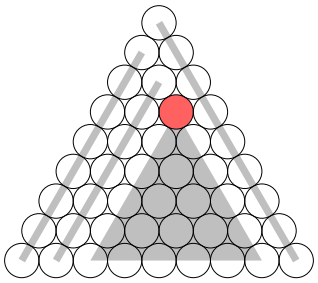
For $j = N$, this immediately implies that some circle in row 2^N has number at least $N+1$. This shows that there is a ninja-path passing through at least $N+1$ red circles.

Solution 2. We give an alternative proof that there exists a ninja-path passing through at least $N+1$ red circles. Assign numbers to circles as in the previous solution, but we only focus on the numbers assigned to red circles.

For each positive integer i, denote by e_i the number of red circles with number i.

Claim 2. If the red circle on row l has number i, then $e_i \leqslant l$.

Proof. Note that if two circles C and C' are both assigned the same number i, then there cannot be a ninja-path joining the two circles. We partition the triangle into a smaller triangle with the red circle in row l at its top along with $l-1$ lines that together cover all other circles.



In each set, there can be at most one red circle with number i, and therefore $e_i \leqslant l$. □

We observe that if there exists a red circle C with number $i \geqslant 2$, then there also exists a red circle with number $i-1$ in some row that is above the row containing C. This is because the second last red circle in the ninja-path ending at C has number $i-1$.

Claim 3. We have $e_i \leqslant 2^{i-1}$ for every positive integer i.

Proof. We prove by induction on i. The base case $i = 1$ is clear, since the only red circle with number 1 is the one at the top of the triangle . We now assume that the statement is true for $1 \leqslant i \leqslant j-1$ and prove the statement for $i = j$. If $e_j = 0$, there is nothing to prove. Otherwise, let l be minimal such that the red circle on row l has number j. Then all the red circles on row $1, \ldots, l-1$ must have number less than j. This shows that

$$l - 1 \leqslant e_1 + e_2 + \cdots + e_{j-1} \leqslant 1 + 2 + \cdots + 2^{j-2} = 2^{j-1} - 1.$$

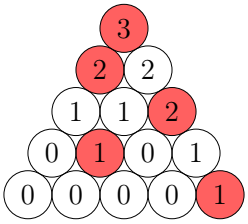
This proves that $l \leqslant 2^{j-1}$, and by Claim 2, we also have $e_j \leqslant l$. Therefore $e_j \leqslant 2^{j-1}$. □

We now see that

$$e_1 + e_2 + \cdots + e_N \leqslant 1 + \cdots + 2^{N-1} = 2^N - 1 < n.$$

Therefore there exists a red circle with number at least $N + 1$, which means that there exists a ninja-path passing through at least $N + 1$ red circles.

Solution 3. We provide yet another proof that there exists a ninja-path passing through at least $N + 1$ red circles. In this solution, we assign to a circle C the maximum number of red circles on a ninja-path *starting* at C (including C itself).



Denote by f_i the number of red circles with number i. Note that if a red circle C has number i, and there is a ninja-path from C to another red circle C', then the number assigned to C' must be less than i.

Claim 4. If the red circle on row l has number less than or equal to i, then $f_i \leqslant l$.

Proof. This proof is same as the proof of Claim 2. The additional input is that if the red circle on row l has number strictly less than i, then the smaller triangle cannot have a red circle with number i. □

Claim 5. We have

$$f_1 + f_2 + \cdots + f_i \leqslant n - \left\lfloor \frac{n}{2^i} \right\rfloor$$

for all $0 \leqslant i \leqslant N$.

Proof. We use induction on i. The base case $i = 0$ is clear as the left hand side is the empty sum and the right hand side is zero. For the induction step, we assume that $i \geqslant 1$ and that the statement is true for $i - 1$. Let l be minimal such that the red circle on row l has number less than or equal to i. Then all the red circles with number less than or equal to i lie on rows $l, l + 1, \ldots, n$, and therefore

$$f_1 + f_2 + \cdots + f_i \leqslant n - l + 1.$$

On the other hand, the induction hypothesis together with the fact that $f_i \leqslant l$ shows that

$$f_1 + \cdots + f_{i-1} + f_i \leqslant n - \left\lfloor \frac{n}{2^{i-1}} \right\rfloor + l.$$

Averaging the two inequalities gives

$$f_1 + \cdots + f_i \leqslant n - \frac{1}{2}\left\lfloor \frac{n}{2^{i-1}} \right\rfloor + \frac{1}{2}.$$

Since the left hand side is an integer, we conclude that

$$f_1 + \cdots + f_i \leqslant n - \left\lfloor \frac{1}{2} \left\lfloor \frac{n}{2^{i-1}} \right\rfloor \right\rfloor = n - \left\lfloor \frac{n}{2^i} \right\rfloor.$$

This completes the induction step. □

Taking $i = N$, we obtain

$$f_1 + f_2 + \cdots + f_N \leqslant n - \left\lfloor \frac{n}{2^N} \right\rfloor < n.$$

This implies that there exists a ninja-path passing through at least $N + 1$ red circles.

Comment. Using essentially the same argument, one may inductively prove

$$e_a + e_{a+1} + \cdots + e_{a+i-1} \leqslant n - \left\lfloor \frac{n}{2^i} \right\rfloor.$$

instead. Taking $a = 1$ and $i = N$ gives the desired statement.

C4. Let $n \geqslant 2$ be a positive integer. Paul has a $1 \times n^2$ rectangular strip consisting of n^2 unit squares, where the ith square is labelled with i for all $1 \leqslant i \leqslant n^2$. He wishes to cut the strip into several pieces, where each piece consists of a number of consecutive unit squares, and then *translate* (without rotating or flipping) the pieces to obtain an $n \times n$ square satisfying the following property: if the unit square in the ith row and j^{th} column is labelled with a_{ij}, then $a_{ij} - (i + j - 1)$ is divisible by n.

Determine the smallest number of pieces Paul needs to make in order to accomplish this.

(U.S.A.)

Answer: The minimum number of pieces is $2n - 1$.

Solution 1. For the entirety of the solution, we shall view the labels as taking values in $\mathbb{Z}/n\mathbb{Z}$, as only their values modulo n play a role.

Here are two possible constructions consisting of $2n - 1$ pieces.

1. Cut into pieces of sizes $n, 1, n, 1, \ldots, n, 1, 1$, and glue the pieces of size 1 to obtain the last row.

2. Cut into pieces of sizes $n, 1, n-1, 2, n-2, \ldots, n-1, 1$, and switch the pairs of consecutive strips that add up to size n.

We now prove that using $2n - 1$ pieces is optimal. It will be more helpful to think of the reverse process: start with n pieces of size $1 \times n$, where the kth piece has squares labelled $k, k + 1, \ldots, k + n - 1$. The goal is to restore the original $1 \times n^2$ strip.

Note that each piece, after cutting at appropriate places, is of the form $a, a + 1, \ldots, b - 1$. Construct an (undirected but not necessarily simple) graph Γ with vertices labelled by $1, \ldots, n$, where a piece of the form $a, a + 1, \ldots, b - 1$ corresponds to an edge from a to b. We make the following observations.

- The cut pieces came from the kth initial piece $k, k + 1, \ldots, k + n - 1$ corresponds to a cycle γ_k (possibly of length 1) containing the vertex k.

- Since it is possible to rearrange the pieces into one single $1 \times n^2$ strip, the graph Γ has an Eulerian cycle.

- The number of edges of Γ is equal to the total number of cut pieces.

The goal is to prove that Γ has at least $2n - 1$ edges. Since Γ has an Eulerian cycle, it is connected. For every $1 \leqslant k \leqslant n$, pick one edge from γ_k, delete it from Γ to obtain a new graph Γ'. Since no two cycles γ_i and γ_j share a common edge, removing one edge from each cycle does not affect the connectivity of the graph. This shows that the new graph Γ' must also be connected. Therefore Γ' has at least $n - 1$ edges, which means that Γ has at least $2n - 1$ edges.

Solution 2. We provide an alternative proof that at least $2n - 1$ pieces are needed. Instead of having a linear strip, we work with a number of *circular strips*, each having length a multiple of n and labelled as

$$1, 2, \ldots, n, 1, 2, \ldots, n, \ldots, 1, 2, \ldots, n,$$

where there are n^2 cells in total across all circular strips. The goal is still to create the $n \times n$ square by cutting and translating. Here, when we say "translating" the strips, we imagine that each cell has a number written on it and the final $n \times n$ square is required to have every number written in the same upright, non-mirrored orientation.

Note that the number of cuts will be equal to the number of pieces, because performing $l \geqslant 1$ cuts on a single circular strip results in l pieces.

Consider any "seam" in the interior of the final square, between two squares S and T, so that S and T belongs to two separate pieces. We are interested in the positions of these two squares in the original circular strips, with the aim of removing the seam.

- If the two squares S and T come from the same circular strip and are adjacent, then the cut was unnecessary and we can simply remove the seam and reduce the number of required cuts by 1. The circular strips are not affected.
- If these two squares S and T were not adjacent, then they are next to two different cuts (either from the same circular strip or two different circular strips). Denote the two cuts by $(S|Y)$ and $(X|T)$. We perform these two cuts and then glue the pieces back according to $(S|T)$ and $(X|Y)$. Performing this move would either split one circular strip into two or merge two circular strips into one, changing the number of circular strips by at most one. Afterwards, we may eliminate cut $(S|T)$ since it is no longer needed, which also removes the corresponding seam from the final square.

By iterating this process, eventually we reach a state where there are some number of circular strips, but the final $n \times n$ square no longer has any interior seams.

Since no two rows of the square can be glued together while maintaining the consecutive numbering, the only possibility is to have exactly n circular strips, each with length n. In this state at least n cuts are required to reassemble the square. Recall that each seam removal operation changed the number of circular strips by at most one. So if we started with only one initial circular strip, then at least $n-1$ seams were removed. Hence in total, at least $n + (n-1) = 2n-1$ cuts are required to transform one initial circular strip into the final square. Hence at least $2n-1$ pieces are required to achieve the desired outcome.

Solution 3. As with the previous solution, we again work with *circular strips*. In particular, we start out with k *circular strips*, each having length a multiple of n and labelled as

$$1, 2, \ldots, n, 1, 2, \ldots, n, \ldots, 1, 2, \ldots, n,$$

where there are n^2 cells in total across all k circular strips. The goal is still to create the $n \times n$ square by cutting and translating the circular strips.

Claim. Constructing the $n \times n$ square requires at least $2n-k$ cuts (or alternatively, $2n-k$ pieces).

Proof. We prove by induction on n. The base case $n = 1$ is clear, because we can only have $k = 1$ and the only way of producing a 1×1 square from a 1×1 circular strip is by making a single cut. We now assume that $n \geqslant 2$ and the statement is true for $n-1$.

Each cut is a cut between a cell of label i on the left and a cell of label $i+1$ on the right side, for a unique $1 \leqslant i \leqslant n$. Let a_i be the number of such cuts, so that $a_1 + a_2 + \cdots + a_n$ is the total number of cuts. Since all the left and right edges of the $n \times n$ square at the end must be cut, we have $a_i \geqslant 1$ for all $1 \leqslant i \leqslant n$.

If $a_i \geqslant 2$ for all i, then

$$a_1 + a_2 + \cdots + a_n \geqslant 2n > 2n - k$$

and hence there is nothing to prove. We therefore assume that there exist some $1 \leqslant m \leqslant n$ for which $a_m = 1$. This unique cut must form the two ends of the linear strip

$$m+1, m+2, \ldots, m-1+n, m+n$$

from the final product. There are two cases.

Case 1: The strip is a single connected piece.

In this case, the strip must have come from a single circular strip of length exactly n. We now remove this circular strip from of the cutting and pasting process. By definition of m, none of the edges between m and $m+1$ are cut. Therefore we may pretend that all the adjacent pairs of cells labelled m and $m+1$ are single cells. The induction hypothesis then implies that

$$a_1 + \cdots + a_{m-1} + a_{m+1} + \cdots + a_n \geqslant 2(n-1) - (k-1).$$

Adding back in a_m, we obtain

$$a_1 + \cdots + a_n \geqslant 2(n-1) - (k-1) + 1 = 2n - k.$$

Case 2: The strip is not a single connected piece.

Say the linear strip $m+1, \ldots, m+n$ is composed of $l \geqslant 2$ pieces $C_1, \ldots, C_l$. We claim that if we cut the initial circular strips along both the left and right end points of the pieces $C_1, \ldots, C_l$, and then remove them, the remaining part consists of at most $k+l-2$ connected pieces (where some of them may be circular and some of them may be linear). This is because C_l, C_1 form a consecutive block of cells on the circular strip, and removing $l-1$ consecutive blocks from k circular strips results in at most $k + (l-1) - 1$ connected pieces.

Once we have the connected pieces that form the complement of $C_1, \ldots, C_l$, we may glue them back at appropriate endpoints to form circular strips. Say we get k' circular strips after this procedure. As we are gluing back from at most $k+l-2$ connected pieces, we see that

$$k' \leqslant k + l - 2.$$

We again observe that to get from the new circular strips to the $n-1$ strips of size $1 \times n$, we never have to cut along the cell boundary between labels m and $m+1$. Therefore the induction hypothesis applies, and we conclude that the total number of pieces is bounded below by

$$l + (2(n-1) - k') \geqslant l + 2(n-1) - (k+l-2) = 2n - k.$$

This finishes the induction step, and therefore the statement holds for all n. □

Taking $k = 1$ in the claim, we see that to obtain a $n \times n$ square from a circular $1 \times n^2$ strip, we need at least $2n-1$ connected pieces. This shows that constructing the $n \times n$ square out of a linear $1 \times n^2$ strip also requires at least $2n-1$ pieces.

C5. Elisa has 2023 treasure chests, all of which are unlocked and empty at first. Each day, Elisa adds a new gem to one of the unlocked chests of her choice, and afterwards, a fairy acts according to the following rules:

- if more than one chests are unlocked, it locks one of them, or
- if there is only one unlocked chest, it unlocks all the chests.

Given that this process goes on forever, prove that there is a constant C with the following property: Elisa can ensure that the difference between the numbers of gems in any two chests never exceeds C, regardless of how the fairy chooses the chests to lock.

(Israel)

Solution 1. We will prove that such a constant C exists when there are n chests for n an *odd* positive integer. In fact we can take $C = n - 1$. Elisa's strategy is simple: place a gem in the chest with the fewest gems (in case there are more than one such chests, pick one arbitrarily).

For each integer $t \geqslant 0$, let $a_1^t \leqslant a_2^t \leqslant \cdots \leqslant a_n^t$ be the numbers of gems in the n chests at the end of the tth day. In particular, $a_1^0 = \cdots = a_n^0 = 0$ and

$$a_1^t + a_2^t + \cdots + a_n^t = t.$$

For each $t \geqslant 0$, there is a unique index $m = m(t)$ for which $a_m^{t+1} = a_m^t + 1$. We know that $a_j^t > a_{m(t)}^t$ for all $j > m(t)$, since $a_{m(t)}^t < a_{m(t)}^{t+1} \leqslant a_j^{t+1} = a_j^t$. Elisa's strategy also guarantees that if an index j is greater than the remainder of t when divided by n (i.e. the number of locked chests at the end of the tth day), then $a_j^t \geqslant a_{m(t)}^t$, because some chest with at most a_j^t gems must still be unlocked at the end of the tth day.

Recall that a sequence $x_1 \leqslant x_2 \leqslant \cdots \leqslant x_n$ of real numbers is said to *majorise* another sequence $y_1 \leqslant y_2 \leqslant \cdots \leqslant y_n$ of real numbers when for all $1 \leqslant k \leqslant n$ we have

$$x_1 + x_2 + \cdots + x_k \leqslant y_1 + y_2 + \cdots + y_k$$

and

$$x_1 + x_2 + \cdots + x_n = y_1 + y_2 + \cdots + y_n.$$

Our strategy for proving $a_n^t - a_1^t \leqslant n - 1$ is to inductively show that the sequence (a_i^t) is majorised by some other sequence (b_i^t).

We define this other sequence (b_i^t) as follows. Let $b_k^0 = k - \frac{n+1}{2}$ for $1 \leqslant k \leqslant n$. As n is odd, this is a strictly increasing sequence of integers, and the sum of its terms is 0. Now define $b_i^t = b_i^0 + \left\lfloor \frac{t-i}{n} \right\rfloor + 1$ for $t \geqslant 1$ and $1 \leqslant i \leqslant n$. Thus for $t \geqslant 0$,

$$b_i^{t+1} = \begin{cases} b_i^t & \text{if } t+1 \not\equiv i \pmod{n}, \\ b_i^t + 1 & \text{if } t+1 \equiv i \pmod{n}. \end{cases}$$

From these properties it is easy to see that

- $b_1^t + b_2^t + \cdots + b_n^t = t$ for all $t \geqslant 0$, and
- $b_i^t \leqslant b_{i+1}^t$ for all $t \geqslant 0$ and $1 \leqslant i \leqslant n-1$, with the inequality being strict if $t \not\equiv i \pmod{n}$.

Claim 1. For each $t \geqslant 0$, the sequence of integers $b_1^t, b_2^t, \ldots, b_n^t$ majorises the sequence of integers $a_1^t, a_2^t, \ldots, a_n^t$.

Proof. We use induction on t. The base case $t = 0$ is trivial. Assume $t \geqslant 0$ and that (b_i^t) majorises (a_i^t). We want to prove the same holds for $t + 1$.

First note that the two sequences (b_i^{t+1}) and (a_i^{t+1}) both sum up to $t + 1$. Next, we wish to show that for $1 \leqslant k < n$, we have

$$b_1^{t+1} + b_2^{t+1} + \cdots + b_k^{t+1} \leqslant a_1^{t+1} + a_2^{t+1} + \cdots + a_k^{t+1}.$$

When $t+1$ is replaced by t, the above inequality holds by the induction hypothesis. For the sake of contradiction, suppose k is the smallest index such that the inequality for $t + 1$ fails. Since the left hand side increases by at most 1 during the transition from t to $t + 1$, the inequality for $t + 1$ can fail only if all of the following occur:

- $b_1^t + b_2^t + \cdots + b_k^t = a_1^t + a_2^t + \cdots + a_k^t$,
- $t + 1 \equiv j \pmod{n}$ for some $1 \leqslant j \leqslant k$ (so that $b_j^{t+1} = b_j^t + 1$),
- $m(t) > k$ (so that $a_i^{t+1} = a_i^t$ for $1 \leqslant i \leqslant k$).

The first point and the minimality of k tell us that $b_1^t, \ldots, b_k^t$ majorises $a_1^t, \ldots, a_k^t$ as well (again using the induction hypothesis), and in particular $b_k^t \geqslant a_k^t$.

The second point tells us that the remainder of t when divided by n is at most $k - 1$, so $a_k^t \geqslant a_{m(t)}^t$ (by Elisa's strategy). But by the third point ($m(t) \geqslant k + 1$) and the non-decreasing property of a_i^t, we must have the equalities $a_k^t = a_{k+1}^t = a_{m(t)}^t$. On the other hand, $a_k^t \leqslant b_k^t < b_{k+1}^t$, with the second inequality being strict because $t \not\equiv k \pmod{n}$. We conclude that

$$b_1^t + b_2^t + \cdots + b_{k+1}^t > a_1^t + a_2^t + \cdots + a_{k+1}^t,$$

a contradiction to the induction hypothesis. □

This completes the proof as it implies

$$a_n^t - a_1^t \leqslant b_n^t - b_1^t \leqslant b_n^0 - b_1^0 = n - 1.$$

Comment 1. The statement is true even when n is even. In this case, we instead use the initial state

$$b_k^0 = \begin{cases} k - \frac{n}{2} - 1 & k \leqslant \frac{n}{2}, \\ k - \frac{n}{2} & k > \frac{n}{2}. \end{cases}$$

The same argument shows that $C = n$ works.

Comment 2. The constants $C = n - 1$ for odd n and $C = n$ for even n are in fact optimal. To see this, we will assume that the fairy always locks a chest with the minimal number of gems. Then at every point, if a chest is locked, any other chest with fewer gems will also be locked. Thus $m(t)$ is always greater than the remainder of t when divided by n. This implies that the quantities

$$I_k = a_1^t + \cdots + a_k^t - b_1^t - \cdots - b_k^t$$

for each $0 \leqslant k \leqslant n$, do not increase regardless of how Elisa acts. If Elisa succeeds in keeping $a_n^t - a_1^t$ bounded, then these quantities must also be bounded; thus they are eventually constant, say for $t \geqslant t_0$. This implies that for all $t \geqslant t_0$, the number $m(t)$ is equal to 1 plus the remainder of t when divided by n.

Claim 2. For $T \geqslant t_0$ divisible by n, we have

$$a_1^T < a_2^T < \cdots < a_n^T.$$

Proof. Suppose otherwise, and let j be an index for which $a_j^T = a_{j+1}^T$. We have $m(T+k-1) = k$ for all $1 \leqslant k \leqslant n$. Then $a_j^{T+j} > a_{j+1}^{T+j}$, which gives a contradiction. □

This implies $a_n^T - a_1^T \geqslant n-1$, which already proves optimality of $C = n-1$ for odd n. For even n, note that the sequence (a_i^T) has sum divisible by n, so it cannot consist of n consecutive integers. Thus $a_n^T - a_1^T \geqslant n$ for n even.

Solution 2. We solve the problem when 2023 is replaced with an arbitrary integer n. We assume that Elisa uses the following strategy:

> At the beginning of the $(nt+1)^{\text{th}}$ day, Elisa first labels her chests as $C_1^t, \ldots, C_n^t$ so that before she adds in the gem, the number of gems in C_i^t is less than or equal C_j^t for all $1 \leqslant i < j \leqslant n$. Then for days $nt+1, nt+2, \ldots, nt+n$, she adds a gem to chest C_i^t, where i is chosen to be minimal such that C_i^t is unlocked.

Denote by c_i^t the number of gems in chest C_i^t at the beginning of the $(nt+1)^{\text{th}}$ day, so that

$$c_1^t \leqslant c_2^t \leqslant \cdots \leqslant c_n^t$$

by construction. Also, denote by δ_i^t the total number of gems added to chest C_i^t during days $nt+1, \ldots, nt+n$. We make the following observations.

- We have $c_1^0 = c_2^0 = \cdots = c_n^0 = 0$.
- We have $c_1^t + \cdots + c_n^t = nt$, since n gems are added every n days.
- The sequence (c_i^{t+1}) is a permutation of the sequence $(c_i^t + \delta_i^t)$ for all $t \geqslant 0$.
- We have $\delta_1^t + \cdots + \delta_n^t = n$ for all $t \geqslant 0$.
- Since Elisa adds a gem to an unlocked chest C_i^t with i minimal, we have

  $$\delta_1^t + \delta_2^t + \cdots + \delta_k^t \geqslant k$$

  for every $1 \leqslant k \leqslant n$ and $t \geqslant 0$.

We now define another sequence of sequences of integers as follows.

$$d_i^0 = 3n(i - \tfrac{n+1}{2}), \quad d_i^t = d_i^0 + t.$$

We observe that

$$d_1^t + \cdots + d_n^t = c_1^t + \cdots + c_n^t = nt.$$

Claim 3. For each $t \geqslant 0$, the sequence (d_i^t) majorises the sequence (c_i^t).

Proof. We induct on t. For $t = 0$, this is clear as all the terms in the sequence (c_i^t) are equal. For the induction step, we assume that (d_i^t) majorises (c_i^t). Given $1 \leqslant k \leqslant n-1$, we wish to show that

$$d_1^{t+1} + \cdots + d_k^{t+1} \leqslant c_1^{t+1} + \cdots + c_k^{t+1}.$$

Case 1: $c_1^{t+1}, \ldots, c_k^{t+1}$ is a permutation of $c_1^t + \delta_1^t, \ldots, c_k^t + \delta_k^t$.

Since $d_1^t + \cdots + d_k^t \leqslant c_1^t + \cdots + c_k^t$ by the induction hypothesis, we have

$$\sum_{i=1}^k d_i^{t+1} = k + \sum_{i=1}^k d_i^t \leqslant k + \sum_{i=1}^k c_i^t \leqslant \sum_{i=1}^k (c_i^t + \delta_i^t) = \sum_{i=1}^k c_i^{t+1}.$$

Case 2: $c_1^{t+1}, \ldots, c_k^{t+1}$ *is not a permutation of* $c_1^t + \delta_1^t, \ldots, c_k^t + \delta_k^t$.

In this case, we have $c_i^t + \delta_i^t > c_j^t + \delta_j^t$ for some $i \leqslant k < j$. It follows that

$$c_k^t + n \geqslant c_i^t + n \geqslant c_i^t + \delta_i^t > c_j^t + \delta_j^t \geqslant c_j^t \geqslant c_{k+1}^t.$$

Using $d_k^t + 3n = d_{k+1}^t$ and the induction hypothesis, we obtain

$$\begin{aligned}\sum_{i=1}^{k} c_i^{t+1} &\geqslant \sum_{i=1}^{k} c_i^t > c_1^t + \cdots + c_{k-1}^t + \frac{1}{2}c_k^t + \frac{1}{2}c_{k+1}^t - \frac{n}{2} = \frac{1}{2}\sum_{i=1}^{k-1} c_i^t + \frac{1}{2}\sum_{i=1}^{k+1} c_i^t - \frac{n}{2} \\ &\geqslant \frac{1}{2}\sum_{i=1}^{k-1} d_i^t + \frac{1}{2}\sum_{i=1}^{k+1} d_i^t - \frac{n}{2} = n + \sum_{i=1}^{k} d_i^t \geqslant k + \sum_{i=1}^{k} d_i^t = \sum_{i=1}^{k} d_i^{t+1},\end{aligned}$$

This finishes the induction step. □

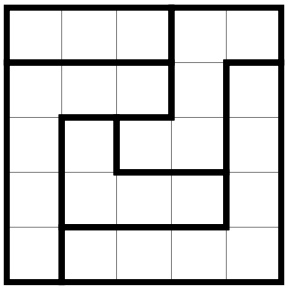
It follows that

$$c_n^t - c_1^t \leqslant d_n^t - d_1^t = 3n(n-1).$$

From day $nt+1$ to day $n(t+1)+1$, Elisa adds n gems, and therefore the difference may increase by at most n. This shows that the difference of the number of gems in two chests never exceeds $C = 3n(n-1) + n$.

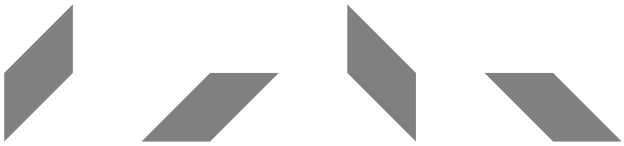
C6. Let N be a positive integer, and consider an $N \times N$ grid. A *right-down path* is a sequence of grid cells such that each cell is either one cell to the right of or one cell below the previous cell in the sequence. A *right-up path* is a sequence of grid cells such that each cell is either one cell to the right of or one cell above the previous cell in the sequence.

Prove that the cells of the $N \times N$ grid cannot be partitioned into less than N right-down or right-up paths. For example, the following partition of the 5×5 grid uses 5 paths.

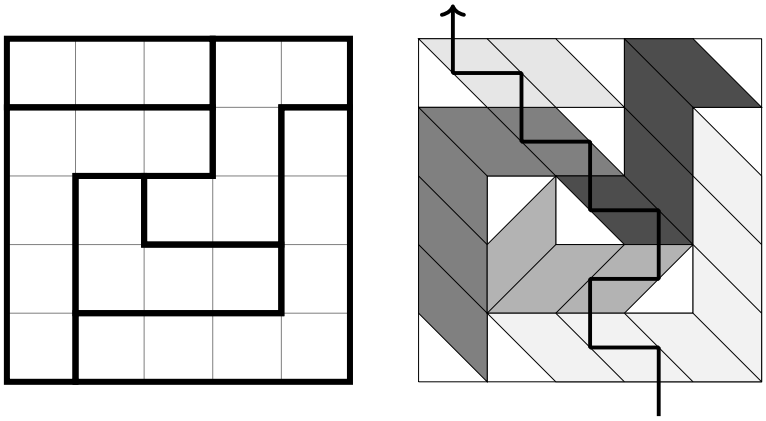


(Canada)

Solution 1. We define a *good parallelogram* to be a parallelogram composed of two isosceles right-angled triangles glued together as shown below.



Given any partition into k right-down or right-up paths, we can find a corresponding packing of good parallelograms that leaves an area of k empty. Thus, it suffices to prove that we must leave an area of at least N empty when we pack good parallelograms into an $N \times N$ grid. This is actually equivalent to the original problem since we can uniquely recover the partition into right-down or right-up paths from the corresponding packing of good parallelograms.



We draw one of the diagonals in each cell so that it does not intersect any of the good parallelograms. Now, view these segments as mirrors, and consider a laser entering each of the $4N$ boundary edges (with starting direction being perpendicular to the edge), bouncing along these mirrors until it exits at some other edge. When a laser passes through a good parallelogram, its direction goes back to the original one after bouncing two times. Thus, if the final direction of a laser is perpendicular to its initial direction, it must pass through at least

one empty triangle. Similarly, if the final direction of a laser is opposite to its initial direction, it must pass though at least two empty triangles. Using this, we will estimate the number of empty triangles in the $N \times N$ grid.

We associate the starting edge of a laser with the edge it exits at. Then, the boundary edges are divided into $2N$ pairs. These pairs can be classified into three types:

(1) a pair of a vertical and a horizontal boundary edge,

(2) a pair of boundary edges from the same side, and

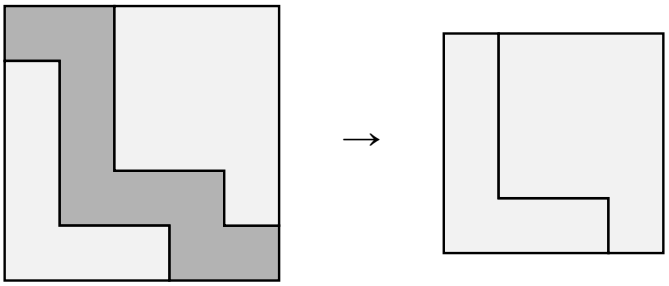
(3) a pair of boundary edges from opposite sides.

Since the beams do not intersect, we cannot have one type (3) pair from vertical boundary edges and another type (3) pair from horizontal boundary edges. Without loss of generality, we may assume that we have t pairs of type (3) and they are all from vertical boundary edges. Then, out of the remaining boundary edges, there are $2N$ horizontal boundary edges and $2N-2t$ vertical boundary edges. It follows that there must be at least t pairs of type (2) from horizontal boundary edges. We know that a laser corresponding to a pair of type (1) passes through at least one empty triangle, and a laser corresponding to a pair of type (2) passes through at least two empty triangles. Thus, as the beams do not intersect, we have at least $(2N-2t)+2\cdot t = 2N$ empty triangles in the grid, leaving an area of at least N empty as required.

Solution 2. We apply an induction on N. The base case $N = 1$ is trivial. Suppose that the claim holds for $N-1$ and prove it for $N \geqslant 2$.

Let us denote the path containing the upper left corner by P. If P is right-up, then every cell in P is in the top row or in the leftmost column. By the induction hypothesis, there are at least $N-1$ paths passing through the lower right $(N-1) \times (N-1)$ subgrid. Since P is not amongst them, we have at least N paths.

Next, assume that P is right-down. If P contains the lower right corner, then we get an $(N-1) \times (N-1)$ grid by removing P and glueing the remaining two parts together. The main idea is to extend P so that it contains the lower right corner and the above procedure gives a valid partition of an $(N-1) \times (N-1)$ grid.



We inductively construct Q, which denotes an extension of P as a right-down path. Initially, $Q = P$. Let A be the last cell of Q, B be the cell below A, and C be the cell to the right of A (if they exist). Suppose that A is not the lower right corner, and that $(*)$ both B and C do not belong to the same path as A. Then, we can extend Q as follows (in case we have two or more options, we can choose any one of them to extend Q).

1. If B belongs to a right-down path R, then we add the part of R, from B to its end, to Q.
2. If C belongs to a right-down path R, then we add the part of R, from C to its end, to Q.
3. If B belongs to a right-up path R which ends at B, then we add the part of R in the same column as B to Q.

4. If C belongs to a right-up path R which starts at C, then we add the part of R in the same row as C to Q.

5. Otherwise, B and C must belong to the same right-up path R. In this case, we add B and the cell to the right of B to Q.

Note that if B does not exist, then case (4) must hold. If C does not exist, then case (3) must hold.

It is easily seen that such an extension also satisfies the hypothesis $(*)$, so we can repeat this construction to get an extension of P containing the lower right corner, denoted by Q. We show that this is a desired extension, i.e. the partition of an $(N-1)\times(N-1)$ grid obtained by removing Q and glueing the remaining two parts together consists of right-down or right-up paths.

Take a path R in the partition of the $N\times N$ grid intersecting Q. If the intersection of Q and R occurs in case (1) or case (2), then there exists a cell D in R such that the intersection of Q and R is the part of R from D to its end, so R remains a right-down path after removal of Q. Similarly, if the intersection of Q and R occurs in case (3) or case (4), then R remains a right-up path after removal of Q. If the intersection of Q and R occurs in case (5), then this intersection has exactly two adjacent cells. After the removal of these two cells (as we remove Q), R is divided into two parts that are glued into a right-up path.

Thus, we may apply the induction hypothesis to the resulting partition of an $(N-1)\times(N-1)$ grid, to find that it must contain at least $N-1$ paths. Since P is contained in Q and is not amongst these paths, the original partition must contain at least N paths.

C7. The Imomi archipelago consists of $n \geqslant 2$ islands. Between each pair of distinct islands is a unique ferry line that runs in both directions, and each ferry line is operated by one of k companies. It is known that if any one of the k companies closes all its ferry lines, then it becomes impossible for a traveller, no matter where the traveller starts at, to visit all the islands exactly once (in particular, not returning to the island the traveller started at).

Determine the maximal possible value of k in terms of n.

(Ukraine)

Answer: The largest k is $k = \lfloor \log_2 n \rfloor$.

Solution. We reformulate the problem using graph theory. We have a complete graph K_n on n nodes (corresponding to islands), and we want to colour the edges (corresponding to ferry lines) with k colours (corresponding to companies), so that every Hamiltonian path contains all k different colours. For a fixed set of k colours, we say that an edge colouring of K_n is *good* if every Hamiltonian path contains an edge of each one of these k colours.

We first construct a good colouring of K_n using $k = \lfloor \log_2 n \rfloor$ colours.

Claim 1. Take $k = \lfloor \log_2 n \rfloor$. Consider the complete graph K_n in which the nodes are labelled by $1, 2, \ldots, n$. Colour node i with colour $\min(\lfloor \log_2 i \rfloor + 1, k)$ (so the colours of the first nodes are $1, 2, 2, 3, 3, 3, 3, 4, \ldots$ and the last $n - 2^{k-1} + 1$ nodes have colour k), and for $1 \leqslant i < j \leqslant n$, colour the edge ij with the colour of the node i. Then the resulting edge colouring of K_n is good.

Proof. We need to check that every Hamiltonian path contains edges of every single colour. We first observe that the number of nodes assigned colour k is $n - 2^{k-1} + 1$. Since $n \geqslant 2^k$, we have

$$n - 2^{k-1} + 1 \geqslant \frac{n}{2} + 1.$$

This implies that in any Hamiltonian path, there exists an edge between two nodes with colour k. Then that edge must have colour k.

We next show that for each $1 \leqslant i < k$, every Hamiltonian path contains an edge of colour i. Suppose the contrary, that some Hamiltonian path does not contain an edge of colour i. Then nodes with colour i can only be adjacent to nodes with colour less than i inside the Hamiltonian path. Since there are 2^{i-1} nodes with colour i and $2^{i-1} - 1$ nodes with colour less than i, the Hamiltonian path must take the form

$$(i) \leftrightarrow (< i) \leftrightarrow (i) \leftrightarrow (< i) \leftrightarrow \cdots \leftrightarrow (< i) \leftrightarrow (i),$$

where (i) denotes a node with colour i, $(< i)$ denotes a node with colour less than i, and $\leftrightarrow$ denotes an edge. But this is impossible, as the Hamiltonian path would not have any nodes with colours greater than i. □

Fix a set of k colours, we now prove that if there exists a good colouring of K_n, then $k \leqslant \lfloor \log_2 n \rfloor$. For $n = 2$, this is trivial, so we assume $n \geqslant 3$. For any node v of K_n and $1 \leqslant i \leqslant k$, we denote by $d_i(v)$ the number of edges with colour i incident with the node v.

Lemma 1. Consider a good colouring of K_n, and let AB be an arbitrary edge with colour i. If $d_i(A) + d_i(B) \leqslant n - 1$, then the colouring will remain good after recolouring edge AB with any other colour.

Proof. Suppose there exists a good colouring together with an edge AB of colour i, such that if AB is recoloured with another colour, the colouring will no longer be good. The failure of the new colouring being good will come from colour i, and thus there exists a Hamiltonian path containing edge AB such that initially (i.e. before recolouring) AB is the only edge of colour i in this path. Writing $A = A_0$ and $B = B_0$, denote this Hamiltonian path by

$$A_s \leftrightarrow A_{s-1} \leftrightarrow \cdots \leftrightarrow A_1 \leftrightarrow A_0 \leftrightarrow B_0 \leftrightarrow B_1 \leftrightarrow \cdots \leftrightarrow B_{t-1} \leftrightarrow B_t,$$

where $s, t \geqslant 0$ and $s + t + 2 = n$.

In the initial colouring, we observe the following.

- The edge B_0A_s must have colour i, since otherwise the path
$$A_0 \leftrightarrow A_1 \leftrightarrow \cdots \leftrightarrow A_{s-1} \leftrightarrow A_s \leftrightarrow B_0 \leftrightarrow B_1 \leftrightarrow \cdots \leftrightarrow B_{t-1} \leftrightarrow B_t$$
has no edges of colour i.
- Similarly, the edge A_0B_t must have colour i.
- For each $0 \leqslant p < s$, at least one of the edges B_0A_p and A_0A_{p+1} must have colour i, since otherwise the path
$$A_s \leftrightarrow \cdots \leftrightarrow A_{p+2} \leftrightarrow A_{p+1} \leftrightarrow A_0 \leftrightarrow A_1 \leftrightarrow \cdots \leftrightarrow A_{p-1} \leftrightarrow A_p \leftrightarrow B_0 \leftrightarrow B_1 \leftrightarrow \cdots \leftrightarrow B_t$$
has no edges of colour i.
- Similarly, for each $0 \leqslant q < t$, at least one of the edges A_0B_q and B_0B_{q+1} must have colour i.

In the above list, each edge A_0X appears exactly once and also each edge B_0X appears exactly once (where A_0B_0 and B_0A_0 are counted separately). Adding up the contributions to $d_i(A) + d_i(B)$, we obtain
$$d_i(A) + d_i(B) \geqslant (s + 1) + (t + 1) = n.$$
This contradicts our assumption that $d_i(A) + d_i(B) \leqslant n - 1$. □

Our strategy now is to repeatedly recolour the edges using Lemma 1 until the colouring has a simple structure. For a node v, we define $m(v)$ to be the largest value of $d_i(v)$ over all colours i.

Lemma 2. Assume we have a good colouring of K_n. Let A, B be two distinct nodes, and let j be the colour of edge AB where $1 \leqslant j \leqslant k$. If

- $m(A) \geqslant m(B)$ and
- $m(A) = d_i(A)$ for some $i \neq j$,

then after recolouring edge AB with colour i, the colouring remains good.

Proof. Note that
$$d_j(A) + d_j(B) \leqslant (n - 1 - m(A)) + m(B) \leqslant n - 1,$$
and so we may apply Lemma 1. □

Lemma 3. Assume we have a good colouring of K_n. Let S be a nonempty set of nodes. Let $A \in S$ be a node such that $m(A) \geqslant m(B)$ for all $B \in S$, and choose $1 \leqslant i \leqslant k$ for which $d_i(A) = m(A)$. Then after recolouring the edge AB with colour i for all $B \in S$ distinct from A, the colouring remains good.

Proof. We repeatedly perform the following operation until all edges AB with $B \in S$ have colour i:

> choose an edge AB with $B \in S$ that does not have colour i, and recolour it with colour i.

By Lemma 2, the colouring remains good after one operation. Moreover, $m(A)$ increase by 1 during an operation, and all other $m(B)$ may increase by at most 1. This shows that $m(A)$ will remain maximal amongst $m(B)$ for $B \in S$. We will also have $d_i(A) = m(A)$ after the operation, since both sides increase by 1. Therefore the operation can be performed repeatedly, and the colouring remains good. □

We first apply Lemma 3 to the set of all n nodes in K_n. After recolouring, there exists a node A_1 such that every edge incident with A_1 has colour c_1. We then apply Lemma 3 to the set of nodes excluding A_1, and we obtain a colouring where

- every edge incident with A_1 has colour c_1,
- every edge incident with A_2 except for A_1A_2 has colour c_2.

Repeating this process, we arrive at the following configuration:

- the n nodes of K_n are labelled $A_1, A_2, \ldots, A_n$,
- the node A_i has a corresponding colour c_i (as a convention, we also colour A_i with c_i),
- for all $1 \leqslant u < v \leqslant n$, the edge between A_u and A_v has colour c_u,
- this colouring is good.

Claim 2. For every colour i, there exists a $1 \leqslant p \leqslant n$ such that the number of nodes of colour i amongst $A_1, \ldots, A_p$ is greater than $p/2$.

Proof. Suppose the contrary, that for every $1 \leqslant p \leqslant n$, there are at most $\lfloor p/2 \rfloor$ nodes of colour i. We then construct a Hamiltonian path not containing any edge of colour i. Let $A_{x_1}, \ldots, A_{x_t}$ be the nodes with colour i, where $x_1 < x_2 < \cdots < x_t$, and let $A_{y_1}, A_{y_2}, \ldots, A_{y_s}$ be the nodes with colour different from i, where $y_1 < y_2 < \cdots < y_s$. We have $s + t = n$ and $t \leqslant \lfloor n/2 \rfloor$, so $t \leqslant s$. We also see that $y_j < x_j$ for all $1 \leqslant j \leqslant t$, because otherwise, $A_1, A_2, \ldots, A_{x_j}$ will have j nodes of colour i and less than j nodes of colour different from i. Then we can construct a Hamiltonian path

$$A_{x_1} \leftrightarrow A_{y_1} \leftrightarrow A_{x_2} \leftrightarrow A_{y_2} \leftrightarrow A_{x_3} \leftrightarrow \cdots \leftrightarrow A_{x_t} \leftrightarrow A_{y_t} \leftrightarrow A_{y_{t+1}} \leftrightarrow \cdots \leftrightarrow A_{y_s}$$

that does not contain an edge with colour i. This contradicts that the colouring is good. □

So for every colour i, there has to be an integer p_i with $1 \leqslant p_i \leqslant n$ such that there are more than $p_i/2$ nodes assigned colour i amongst $A_1, \ldots, A_{p_i}$. Choose the smallest such p_i for every i, and without loss of generality assume

$$p_1 < p_2 < \cdots < p_k.$$

Note that the inequalities are strict by the definition of p_i.

Then amongst the nodes $A_1, \ldots, A_{p_i}$, there are at least $\lceil (p_j + 1)/2 \rceil$ nodes of colour j for all $1 \leqslant j \leqslant i$. Then

$$p_i \geqslant \left\lceil \frac{p_1 + 1}{2} \right\rceil + \left\lceil \frac{p_2 + 1}{2} \right\rceil + \cdots + \left\lceil \frac{p_i + 1}{2} \right\rceil.$$

This inductively shows that

$$p_i \geqslant 2^i - 1$$

for all $1 \leqslant i \leqslant k$, and this already proves $n \geqslant 2^k - 1$.

It remains to show that $n = 2^k - 1$ is not possible. If $n = 2^k - 1$, then all inequalities have to be equalities, so $p_i = 2^i - 1$ and there must be exactly 2^{i-1} nodes of colour i. Moreover, there cannot be a node of colour i amongst $A_1, A_2, \ldots, A_{p_{i-1}}$, and so the set of nodes of colour i must precisely be

$$A_{2^{i-1}}, A_{2^{i-1}+1}, \ldots, A_{2^i-1}.$$

Then we can form a Hamiltonian path

$$A_{2^{k-1}} \leftrightarrow A_1 \leftrightarrow A_{2^{k-1}+1} \leftrightarrow A_2 \leftrightarrow A_{2^{k-1}+2} \leftrightarrow A_3 \leftrightarrow \ldots \leftrightarrow A_n,$$

which does not contain an edge of colour k. This is a contradiction, and therefore $n \geqslant 2^k$.

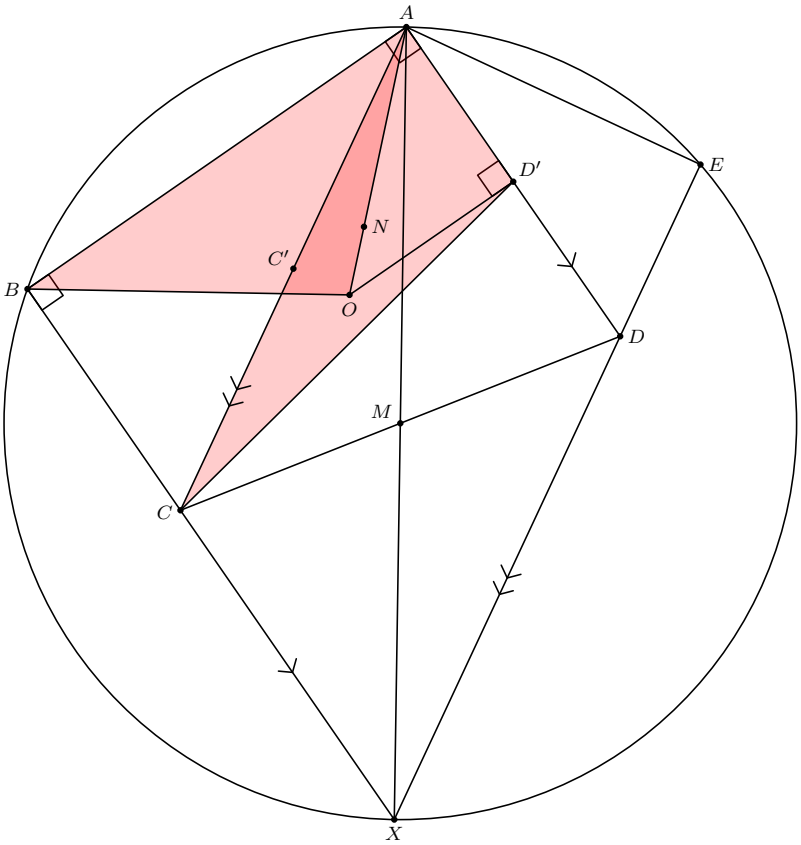
Geometry

G1. Let $ABCDE$ be a convex pentagon such that $\angle ABC = \angle AED = 90^\circ$. Suppose that the midpoint of CD is the circumcentre of triangle ABE. Let O be the circumcentre of triangle ACD.

Prove that line AO passes through the midpoint of segment BE.

(Slovakia)

Solution 1 (Area Ratio).



Let M be the midpoint of CD and $X = BC \cap ED$. Since $\angle ABX = \angle AEX = 90^\circ$, AX is a diameter of the circumcircle of $\triangle ABE$ so the midpoint of AX is the circumcentre of $\triangle ABE$. Therefore, the midpoint of AX coincides with M. This means $ACXD$ is a parallelogram and in particular, $AD \parallel BC$ and $AC \parallel ED$.

We denote the area of $\triangle P_1P_2P_3$ by $[P_1P_2P_3]$. To prove that line AO bisects BE, it suffices to show $[OAB] = [OAE]$.

Let C', D' be the midpoints of AC, AD respectively. Since $OD' \perp AD$, $AD \parallel BC$, and $BC \perp AB$, we have $AB \parallel OD'$, so $[OAB] = [D'AB]$. Using $AD \parallel BC$ again, we have $[D'AB] = [D'AC]$. Therefore

$$[OAB] = [D'AB] = [D'AC] = \frac{1}{2}[ACD].$$

Similarly

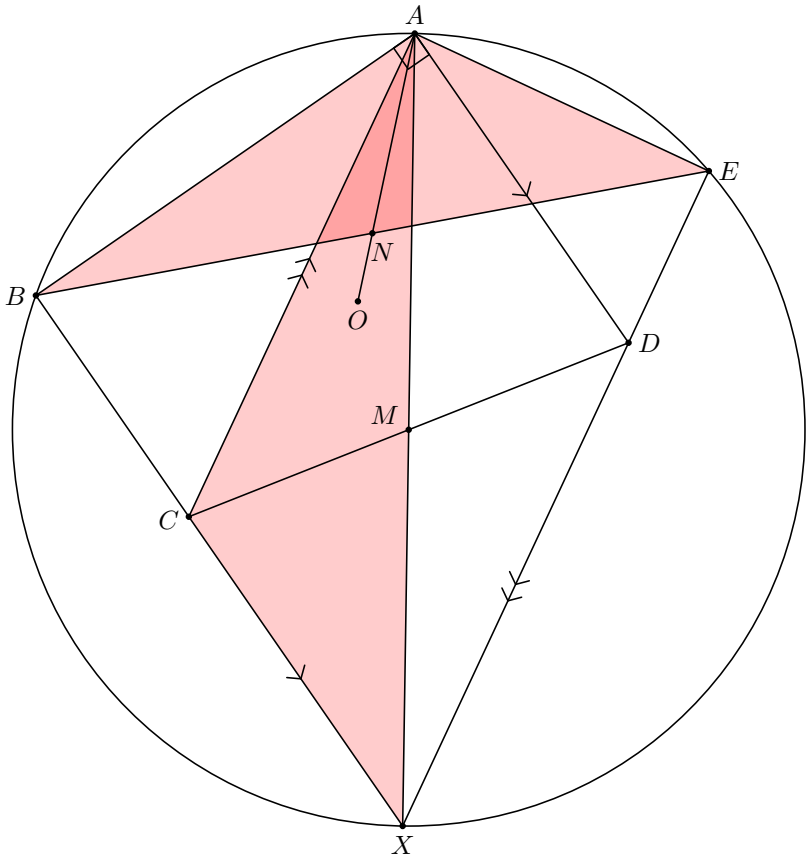
$$[OAE] = [C'AE] = [C'AD] = \frac{1}{2}[ACD].$$

Combining these gives $[OAB] = [OAE]$.

Comment 1. The following is another way to prove $AD \parallel BC$ and $AC \parallel ED$.

Let ℓ be the perpendicular bisector of AB. Since the midpoint of CD is the circumcentre of $\triangle ABE$, it must lie on ℓ. Also, since $\angle ABC = 90^\circ$, the midpoint of AC is on ℓ. Therefore, we get $AD \parallel \ell \parallel BC$ and similarly $AC \parallel ED$.

Solution 2 (Similar Triangles).



Let M be the midpoint of CD and $X = BC \cap ED$. As in Solution 1, M is the midpoint of AX and $ACXD$ is a parallelogram. Since $AD \parallel BC$ and $\angle ABC = 90^\circ$, we have $\angle DAB = 90^\circ$.

Let N be the midpoint of BE. It is enough to show that $\angle NAB = \angle OAB$. Since $ABXE$ is cyclic, we have

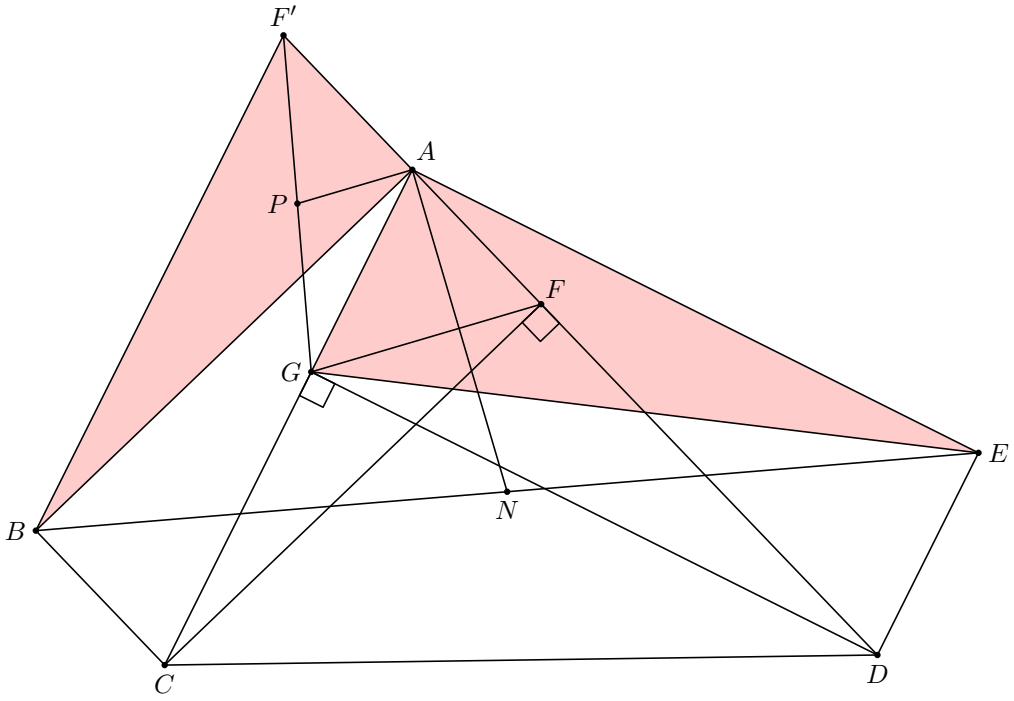
$$\angle ABE = \angle AXE = \angle XAC \quad \text{and} \quad \angle BEA = \angle CXA.$$

Therefore, $\triangle ABE \sim \triangle CAX$, and N corresponds to M under this similarity. In particular, $\angle NAB = \angle ACM$.

Also, we have

$$\angle OAB = 90^\circ - \angle DAO = \angle ACM = \angle NAB.$$

Solution 3 (Reflection).



Let N be the midpoint of BE, and let F, G be the projections of C, D onto AD, AC respectively. $CGFD$ is cyclic so

$$\angle AGF = \angle CDF = \angle CDA = 90^\circ - \angle OAC$$

giving $AO \perp FG$. Therefore it's enough to show that $AN \perp FG$.

As in Solution 1, $AD \parallel BC$ and $AC \parallel ED$ so $\angle EAG = \angle FAB = 90^\circ$ and in fact $AEDG$ and $AFCB$ are rectangles. From this we get

$$\angle AGE = \angle DAG = \angle FAC = \angle BFA$$

so $\triangle GAE \sim \triangle FAB$.

Let F' be the reflection of F in A, then $\triangle F'AB \sim \triangle FAB \sim \triangle GAE$. Thus A is the centre of the spiral symmetry taking $F'B \to GE$.

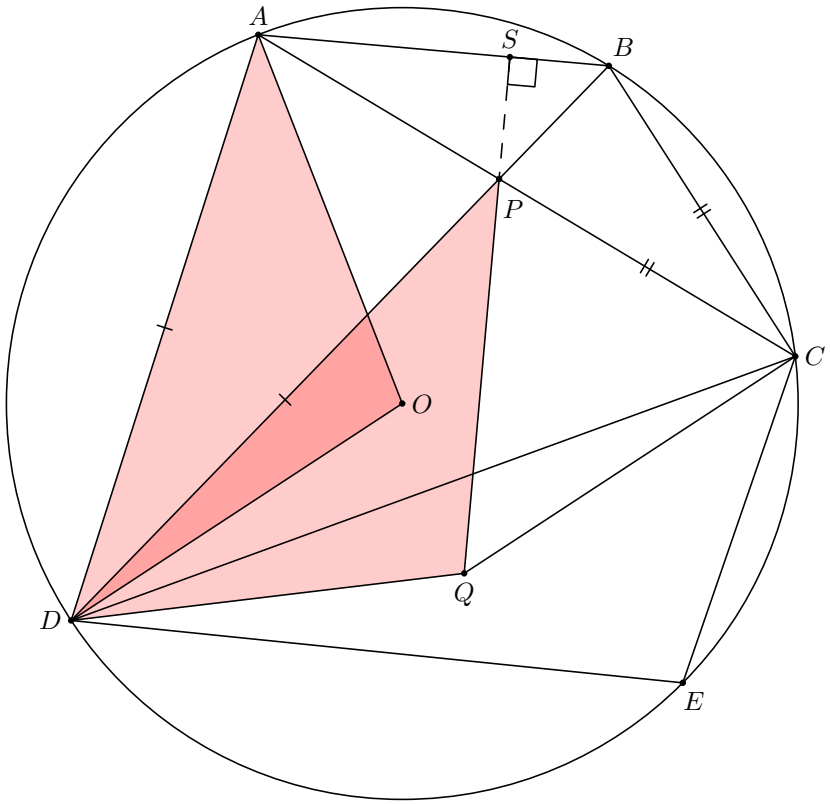
Let P be the midpoint of $F'G$ then by the spiral similarity, we have $\triangle APN \sim \triangle AGE$ which implies $\angle NAP = 90^\circ$. From A being the midpoint of FF' we have $AP \parallel FG$. Combining the results gives $AN \perp FG$.

G2. Let ABC be a triangle with $AC > BC$. Let ω be the circumcircle of triangle ABC and let r be the radius of ω. Point P lies on segment AC such that $BC = CP$ and point S is the foot of the perpendicular from P to line AB. Let ray BP intersect ω again at D and let Q lie on line SP such that $PQ = r$ and S, P, Q lie on the line in that order. Finally, let the line perpendicular to CQ from A intersect the line perpendicular to DQ from B at E.

Prove that E lies on ω.

(Iran)

Solution 1 (Similar Triangles).



First observe that

$$\angle DPA = \angle BPC \overset{CP=CB}{=} \angle CBP = \angle CBD = \angle CAD = \angle PAD$$

so $DP = DA$. Thus there is a symmetry in the problem statement swapping $(A, D) \leftrightarrow (B, C)$.

Let O be the centre of ω and let E be the reflection of P in CD which, by

$$\angle CED = \angle DPC = 180^\circ - \angle CPB \overset{CP=CB}{=} 180^\circ - \angle PBC = 180^\circ - \angle DBC$$

lies on ω. We claim the two lines concur at E. By the symmetry noted above, it suffices to prove that $BE \perp DQ$ and then $AE \perp CQ$ will follow by symmetry.

We have $AO = PQ$, $AD = DP$ and

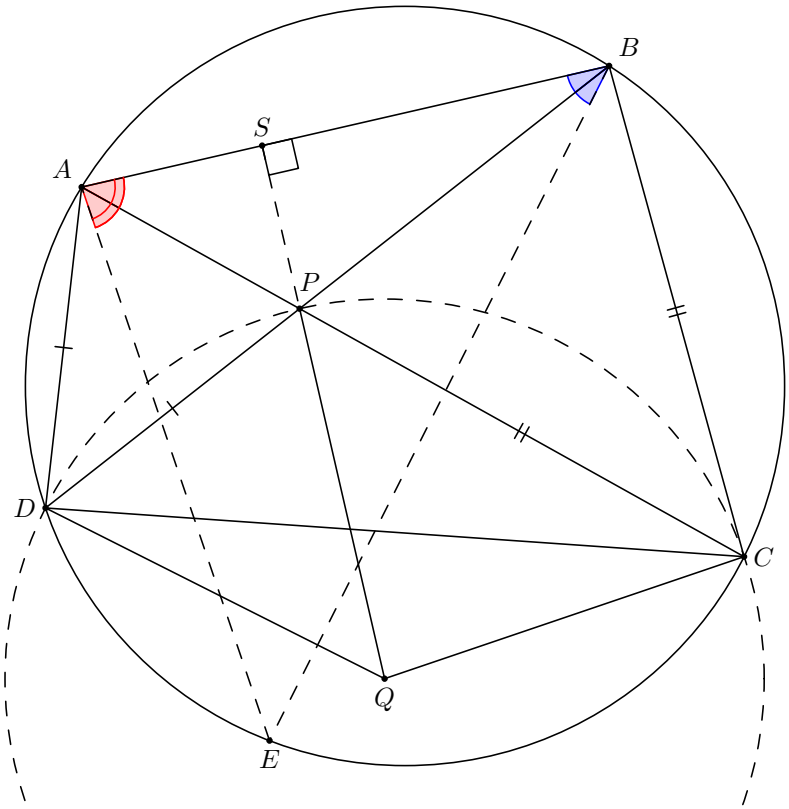
$$\angle DAO = 90^\circ - \angle ABD \overset{PQ \perp AB}{=} \angle DPQ.$$

Hence $\triangle AOD \cong \triangle PQD$. Thus

$$\angle QDB + \angle DBE = \angle ODA + \angle DAE \overset{DE=DA}{=} \angle ODA + \angle AED = (90^\circ - \cancel{\angle AED}) + \cancel{\angle AED} = 90^\circ$$

giving $BE \perp DQ$ as required.

Solution 2 (Second Circle).



As in Solution 1, we prove that $DA = DP$ and note the symmetry in the problem statement swapping $(A, D) \leftrightarrow (B, C)$.

Let Γ be the circumcircle of $\triangle PCD$. Since $DP = DA$ and $\angle ACD = \angle PCD$, the radius of Γ is equal to that of ω. We have that

$$\angle DPQ = \angle BPS = 90^\circ - \angle ABD = 90^\circ - \angle PCD.$$

This, combined with PQ being equal to the common circumradius of Γ and ω, means that Q is the circumcentre of Γ.

Let the perpendiculars to CQ, DQ from A, B intersect at E then we have

$$\angle EAC = 90^\circ - \angle ACQ \overset{QC=QP}{=} 90^\circ - \angle QPC = 90^\circ - \angle SPA = \angle CAB \implies \angle EAB = 2\angle PAB$$
$$\angle DBE = 90^\circ - \angle QDP \overset{QD=QP}{=} 90^\circ - \angle DPQ = 90^\circ - \angle BPS = \angle ABD \implies \angle ABE = 2\angle ABP.$$

Combining these

$$\angle BEA = 180^\circ - 2\left(\angle PAB + \angle ABP\right) = 180^\circ - 2\angle APD \overset{DA=DP}{=} \angle BDA$$

which gives that E lies on ω.

Comment 1. An alternative final angle chase is

$$\angle BEA = 180^\circ - \angle CQD \overset{\Gamma}{=} 180^\circ - 2\left(180^\circ - \angle DPC\right) = 180^\circ - 2\angle APD \overset{DA=DP}{=} \angle PDA = \angle BDA.$$

Comment 2. An alternative formulation of the problem in terms of a cyclic quadrilateral is given below:

> Let $ABCD$ be a cyclic quadrilateral with circumcircle ω and circumradius r. The diagonals AC and BD intersect at P. Suppose that $AD = DP$. Let S be the foot of the perpendicular from P to the line AB. Point Q lies on line SP such that $PQ = r$ and S, P, Q lie on the line in that order. Let the line perpendicular to CQ from A intersect the line perpendicular to DQ from B at E.
>
> Prove that E lies on ω.

G3. Let $ABCD$ be a cyclic quadrilateral with $\angle BAD < \angle ADC$. Let M be the midpoint of the arc CD not containing A. Suppose there is a point P inside $ABCD$ such that $\angle ADB = \angle CPD$ and $\angle ADP = \angle PCB$.

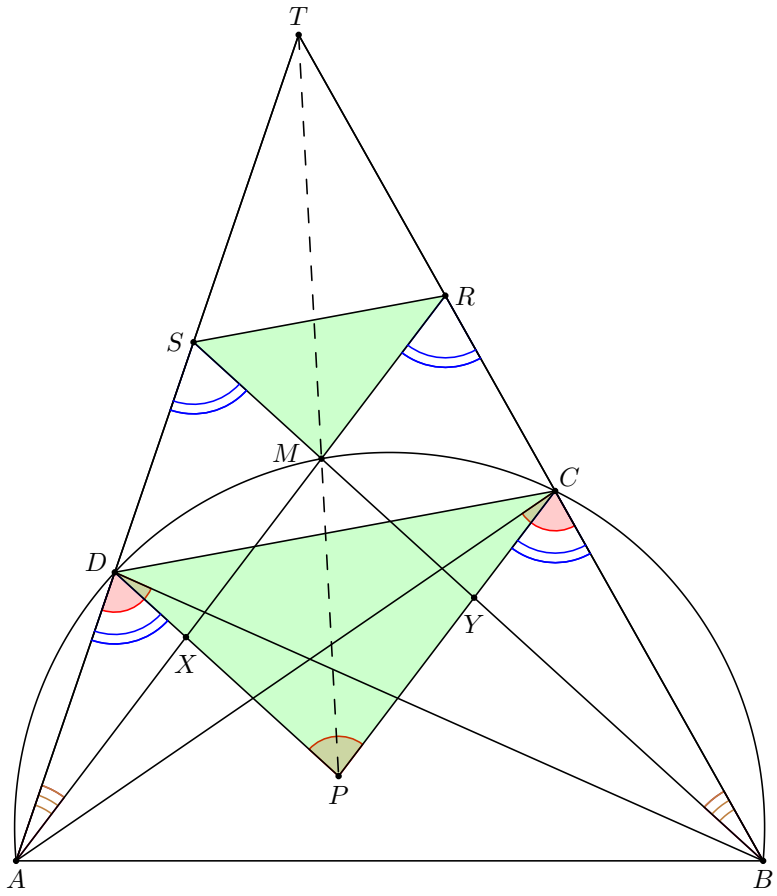
Prove that lines AD, PM, BC are concurrent.

(Slovakia)

Solution 1. Let X and Y be the intersection points of AM and BM with PD and PC respectively. Since $ABCMD$ is cyclic and $CM = MD$, we have

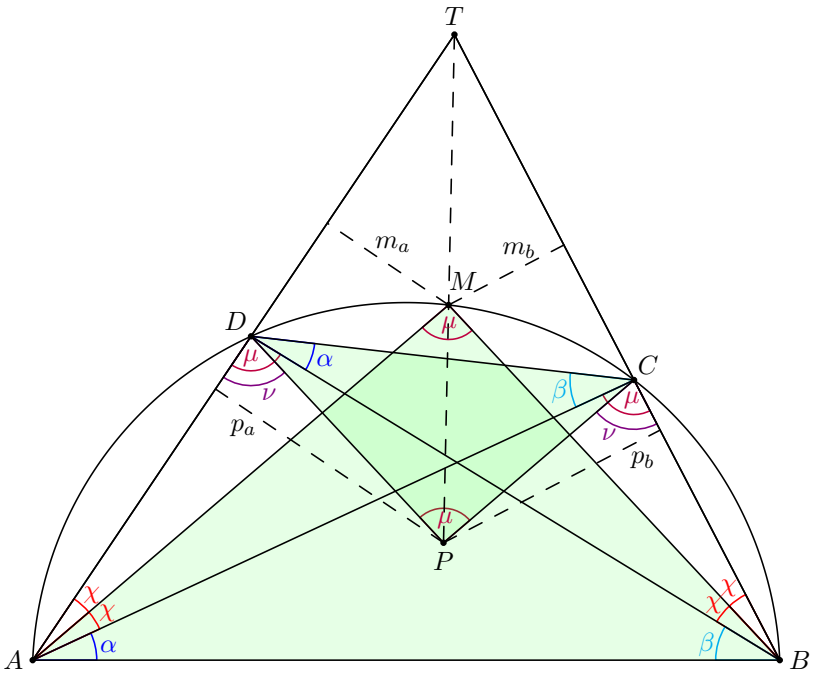
$$\angle XAD = \angle MAD = \angle CBM = \angle CBY.$$

Combining this with $\angle ADX = \angle YCB$, we get $\angle DXA = \angle BYC$, and so $\angle PXM = \angle MYP$. Moreover, $\angle YPX = \angle CPD = \angle ADB = \angle AMB$. The quadrilateral $MXPY$ therefore has equal opposite angles and so is a parallelogram.



Let R and S be the intersection points of AM and BM with BC and AD respectively. Due to $AM \parallel PC$ and $BM \parallel PD$, we have $\angle ASB = \angle ADP = \angle PCB = \angle ARB$ and so the quadrilateral $ABRS$ is cyclic. We then have $\angle SRB = 180^\circ - \angle BAS = \angle DCB$ and so $SR \parallel CD$. In triangles PCD and MRS, the corresponding sides are parallel so they are homothetic meaning lines DS, PM, CR concur at the centre of this homothety.

Solution 2. Let AD and BC meet at T. Denote by p_a, p_b, m_a and m_b the distances between line TA and P, TB and P, TA and M and between TB and M respectively. Our goal is to prove $p_a : p_b = m_a : m_b$ which is equivalent to the collinearity of T, P and M.



Let $\angle BAC = \angle BDC = \alpha$, $\angle DBA = \angle DCA = \beta$, $\angle ADB = \angle AMB = \angle ACB = \angle CPD = \mu$, $\angle ADP = \angle PCB = \nu$ and $\angle MAD = \angle CAM = \angle MBD = \angle CBM = \chi$.

From $\angle ADP = \angle PCB = \nu$ and $\angle MAD = \angle CBM = \chi$ we get

$$\frac{p_a}{p_b} = \frac{PD \cdot \sin\nu}{PC \cdot \sin\nu} = \frac{PD}{PC} \quad \text{and} \quad \frac{m_a}{m_b} = \frac{MA \cdot \sin\chi}{MB \cdot \sin\chi} = \frac{MA}{MB}.$$

Hence $p_a : p_b = m_a : m_b$ is equivalent to $PD : PC = MA : MB$, and since $\angle CPD = \angle AMB = \mu$, this means we have to show that triangles PDC and MAB are similar.

In triangle PDC we have

$$\angle PDC + \angle DCP = 180^\circ - \angle CPD = 180^\circ - \mu,$$
$$\angle PDC - \angle DCP = (\alpha + \mu - \nu) - (\beta + \mu - \nu) = \alpha - \beta.$$

Similarly, in triangle MAB we have

$$\angle BAM + \angle MBA = 180^\circ - \angle AMB = 180^\circ - \mu,$$
$$\angle BAM - \angle MBA = (\alpha + \chi) - (\beta + \chi) = \alpha - \beta.$$

Therefore, $(\angle BAM, \angle MBA)$ and $(\angle PDC, \angle DCP)$ satisfy the same system of linear equations. The common solution is

$$\angle BAM = \angle PDC = \frac{180^\circ - \mu + \alpha - \beta}{2} \quad \text{and} \quad \angle MBA = \angle DCP = \frac{180^\circ - \mu - \alpha + \beta}{2}.$$

Hence triangles PDC and MAB have equal angles and so are similar. This completes the proof.

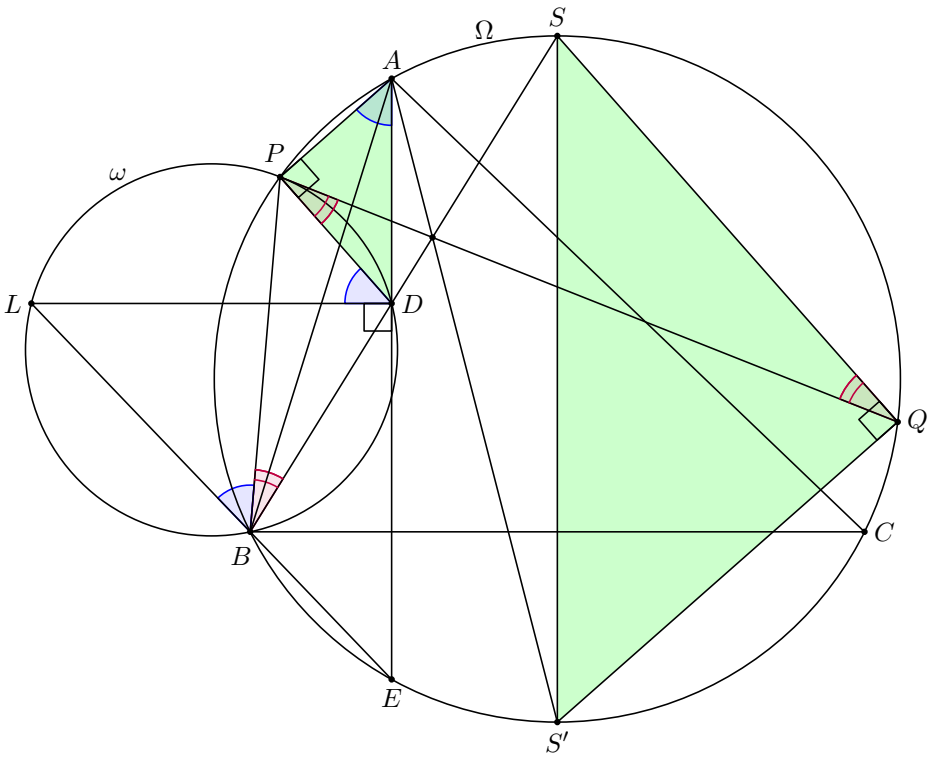
G4. Let ABC be an acute-angled triangle with $AB < AC$. Denote its circumcircle by Ω and denote the midpoint of arc CAB by S. Let the perpendicular from A to BC meet BS and Ω at D and $E \neq A$ respectively. Let the line through D parallel to BC meet line BE at L and denote the circumcircle of triangle BDL by ω. Let ω meet Ω again at $P \neq B$.

Prove that the line tangent to ω at P, and line BS intersect on the internal bisector of $\angle BAC$.

(Portugal)

Solution 1 (Triangles in Perspective). Let S' be the midpoint of arc BC of Ω, diametrically opposite to S so SS' is a diameter in Ω and AS' is the angle bisector of $\angle BAC$. Let the tangent of ω at P meet Ω again at $Q \neq P$, then we have $\angle SQS' = 90^\circ$.

We will show that triangles APD and $S'QS$ are similar and their corresponding sides are parallel. Then it will follow that the lines connecting the corresponding vertices, namely line AS', that is the angle bisector of $\angle BAC$, line PQ, that is the tangent to ω at P, and DS are concurrent. Note that the sides AD and $S'S$ have opposite directions, so the three lines cannot be parallel.



First we show that $AP \perp DP$. Indeed, from cyclic quadrilaterals $APBE$ and $DPLB$ we can see that

$$\angle PAD = \angle PAE = 180^\circ - \angle EBP = \angle PBL = \angle PDL = 90^\circ - \angle ADP.$$

Then, in triangle APD we have $\angle DPA = 180^\circ - \angle PAD - \angle ADP = 90^\circ$.

Now we can see that:

- Both lines ADE and SS' are perpendicular to BC, so $AD \parallel S'S$.
- Line PQ is tangent to circle ω at P so $\angle DPQ = \angle DBP = \angle SBP = \angle SQP$; it follows that $PD \parallel QS$.

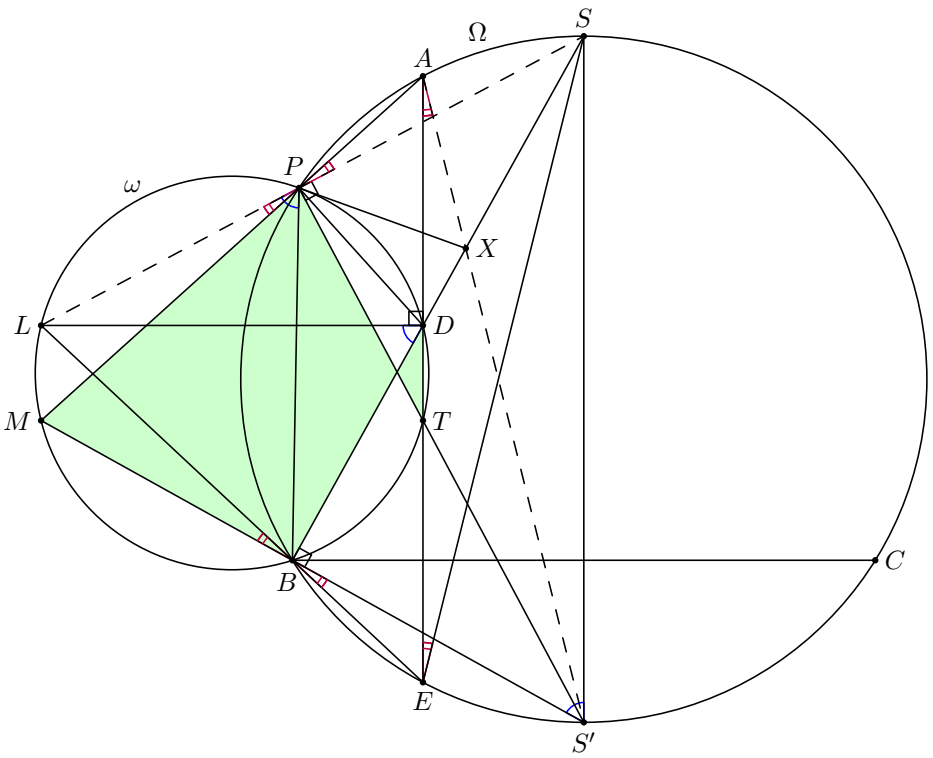
- Finally, since $AP \perp PD \parallel QS \perp S'Q$, we have $AP \parallel S'Q$ as well.

Hence the corresponding sides of triangles APD and $S'QS$ are parallel completing the solution.

Solution 2 (Pascal). Again, let S' be the midpoint of arc BC, diametrically opposite to S, so $AES'S$ is an isosceles trapezoid, and $\angle S'BS = \angle S'PS = 90^\circ$. Let lines AE and PS' meet at T and let AP and $S'B$ meet at point M.

We will need that points L, P, S are collinear, and points T and M lie on circle ω.

- From $\angle LPB = \angle LDB = 90^\circ - \angle BDE = 90^\circ - \angle BSS' = \angle SS'B = 180^\circ - \angle BPS$ we get $\angle LPB + \angle BPS = 180^\circ$, so L, P and S are indeed collinear.
- Since SS' is a diameter in Ω, lines LPS and PTS' are perpendicular. We also have $LD \parallel BC \perp AE$ hence $\angle LDT = \angle LPT = 90^\circ$ and therefore $T \in \omega$.
- By $\angle LPM = \angle SPA = \angle SEA = \angle EAS' = \angle EBS' = \angle LBM$, point M is concyclic with B, P, L so $M \in \omega$.



Now let X be the intersection of line BDS with the tangent of ω at P and apply Pascal's theorem to the degenerate cyclic hexagon $PPMBDT$. This gives points $PP \cap BD = X$, $PM \cap DT = A$ and $MB \cap TP = S'$ are collinear so X lies on line AS', that is the bisector of $\angle BAC$.

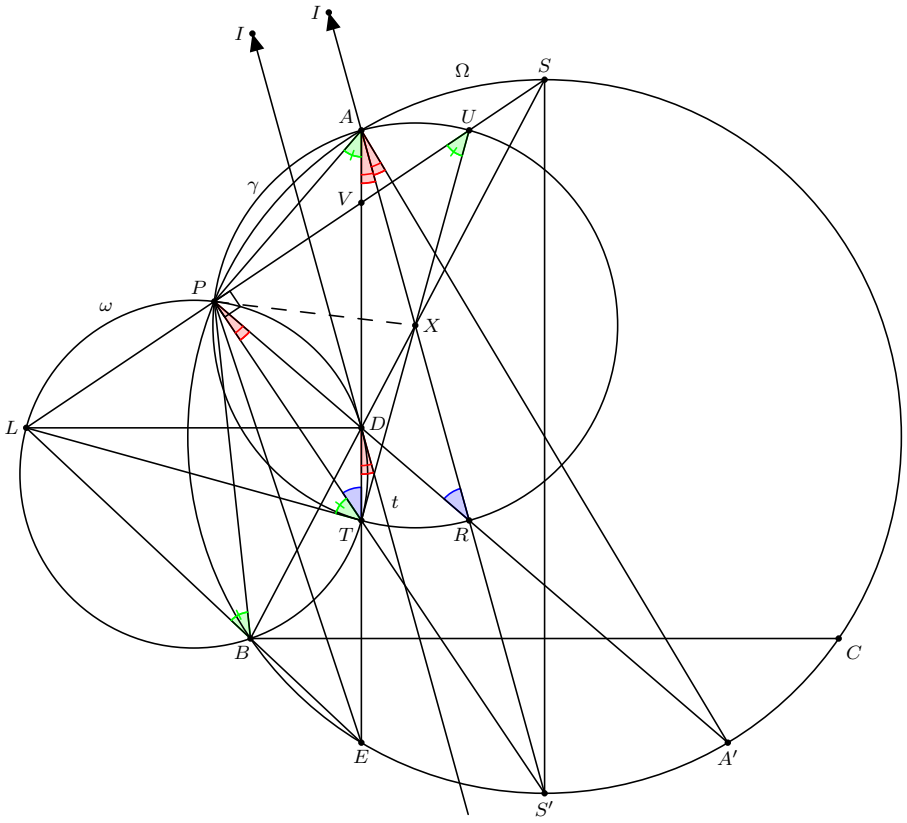
Comment. It is easy to see that $LMTD$ is a rectangle, but we did not need this information for the solution.

Solution 3 (Projections). Let A' and S' be the points of Ω diametrically opposite to A and S respectively. It is well-known that E and A' are reflections with respect to SS' so AS' is the angle bisector of $\angle EAA'$. Define point T to be the intersection of AE and PS'. As in the previous two solutions, we have: $\angle DPA = 90^\circ$ so PD passes through A'; points L, P, S are collinear; and $T \in \omega$.

Let lines AS' and PDA' meet at R. From the angles of triangles PRS' and PTE we get

$$\angle ARP = \angle AS'P + \angle S'PA' = \angle AEP + \angle EPS' = \angle ATP$$

so points A, P, T, R are concyclic. Denote their circle by γ. Due to $\angle RPA = \angle DPA = 90^\circ$, segment AR is a diameter in γ.



We claim that circles ω and γ are perpendicular. Let line LPS meet γ again at $U \neq P$, and consider triangles PLT and PTU. By $\angle LPT = \angle TPU = 90^\circ$ and

$$\angle PTL = \angle PBL = 180^\circ - \angle EBP = \angle PAE = \angle PAT = \angle PUT,$$

triangles PLT and PTU are similar. It follows that the spiral similarity that takes PLT to PTU, maps ω to γ and the angle of this similarity is 90°, so circles ω and γ are indeed perpendicular.

Finally, let lines BDS and ARS' meet at X. We claim that X bisects AR, so point X is the centre of γ and, as ω and γ are perpendicular, PX is tangent to ω.

Let t be the tangent of ω at D. From $\angle(DT, t) = \angle TPD = \angle S'PA' = \angle EAS'$ it can be seen that $t \parallel AS'$. Let I be the common point at infinity of t and AS'. Moreover, let lines LPS and $ADTE$ meet at V. By projecting line AS' to circle ω through D, then projecting ω to line AE through L, finally projecting AE to Ω through P, we find

$$\frac{AX}{RX} = (A, R; X, I) \overset{D}{=} (T, P; B, D) \overset{L}{=} (T, V; E, D) \overset{P}{=} (S', S; E, A') = -1,$$

so X is the midpoint of AR.

G5. Let ABC be an acute-angled triangle with circumcircle ω and circumcentre O. Points $D \neq B$ and $E \neq C$ lie on ω such that $BD \perp AC$ and $CE \perp AB$. Let CO meet AB at X, and BO meet AC at Y.

Prove that the circumcircles of triangles BXD and CYE have an intersection on line AO.
(Malaysia)

Solution 1 (Reflections).

Note that $AO = OC$ implies the lines AO, XO are reflections of each other about the line parallel to AC through O, which is the perpendicular bisector of BD. Call this line ℓ.

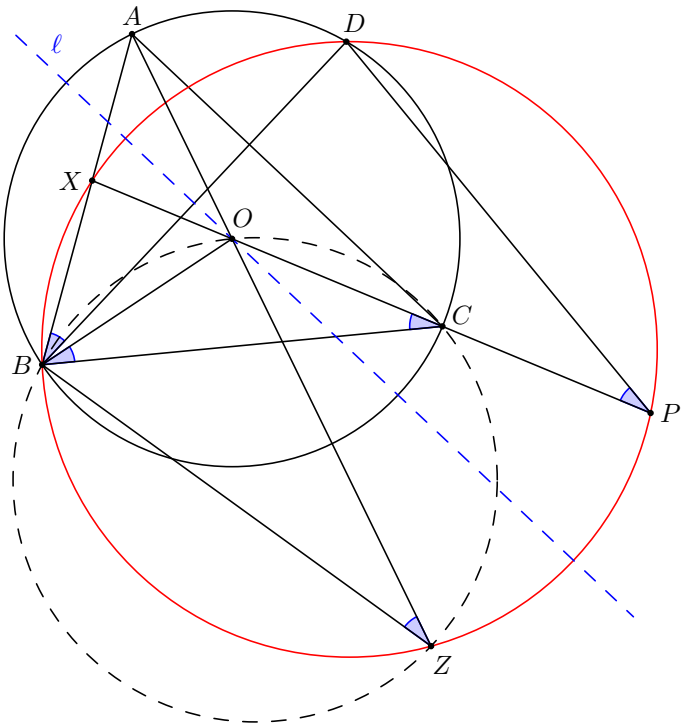
Let $P \neq X$ be the second intersection of circle $\odot BXD$ with line XO, and let Z be the intersection of circle $\odot BXD$ with line AO furthest from A.

Consider a reflection across ℓ. This maps B to D, AO to XO, and circle $\odot BXD$ to itself so the transformation must map P, the intersection of XO and circle $\odot BXD$, to the intersection of AO and $\odot BXD$ furthest from A i.e. Z. Thus we have

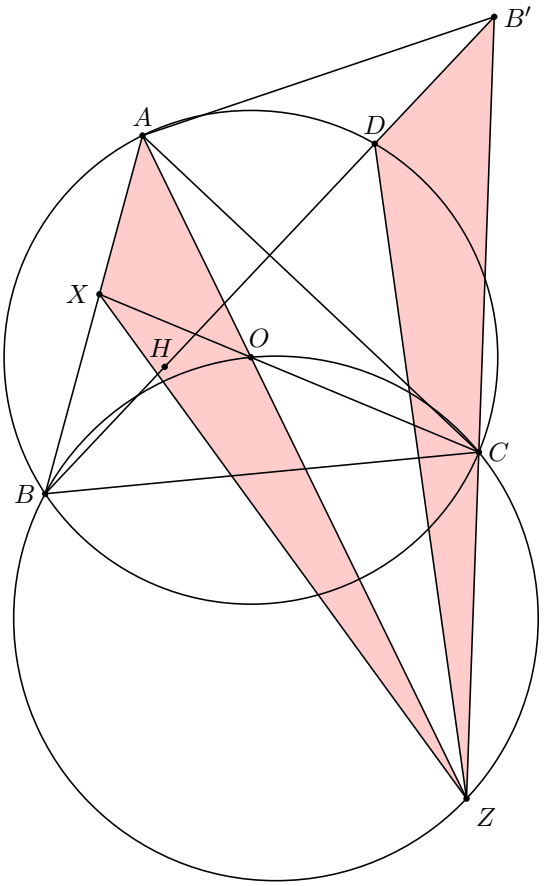
$$\angle OZB = \angle DPO = \angle DPX = \angle DBX = 90^\circ - \angle BAC = \angle OCB$$

which implies $BOCZ$ is cyclic.

Therefore the second intersection of circle $\odot BOC$ with line AO lies on circle $\odot BXD$. Similarly, Z lies on circle $\odot CYE$ so the two circles have common point Z on AO.



Solution 2 (Similar Triangles).



Let B' be the reflection of B in AC and let AO intersect circle $\odot OBC$ again at $Z \neq O$. Observe that

$$\angle B'CA+\angle ACZ = 2\angle ACB+\angle BCZ = 2\angle ACB+\angle BOZ = 2\angle ACB+(180^\circ - \angle AOB) = 180^\circ$$

so Z, C, B' are collinear.

Claim. Triangles ZXA and ZDB' are similar.
Proof. We have

$$\angle XAZ = \angle BAO = 90^\circ - \angle ACB = \angle CBB' = \angle BB'C = \angle DB'Z.$$

So it suffices to prove that $\frac{B'Z}{B'D} = \frac{AZ}{AX}$. To do this, first observe

$$\angle B'ZA = \angle CZO = \angle CBO = \angle XCB \quad \text{and} \quad \angle AB'Z = \angle AB'C = \angle CBA = \angle CBX.$$

Hence triangles ZAB' and CXB are similar so

$$\frac{B'Z}{AZ} = \frac{BC}{CX}.$$

Note that the orthocentre H of triangle ABC is the reflection of D in AC. Applying sine rule to triangles ACX and BHC gives

$$\frac{AX}{CX} = \frac{\sin\angle ACX}{\sin\angle XAC} = \frac{\sin(90^\circ - \angle CBA)}{\sin\angle BAC} = \frac{\sin(90^\circ - \angle CBA)}{\sin(180^\circ - \angle BAC)} = \frac{\sin\angle HCB}{\sin\angle BHC} = \frac{BH}{BC} = \frac{B'D}{BC}.$$

Multiplying the two results gives

$$\frac{B'Z}{AZ} \cdot \frac{AX}{CX} = \frac{\cancel{BC}}{CX} \cdot \frac{B'D}{\cancel{BC}} = \frac{B'D}{CX}$$

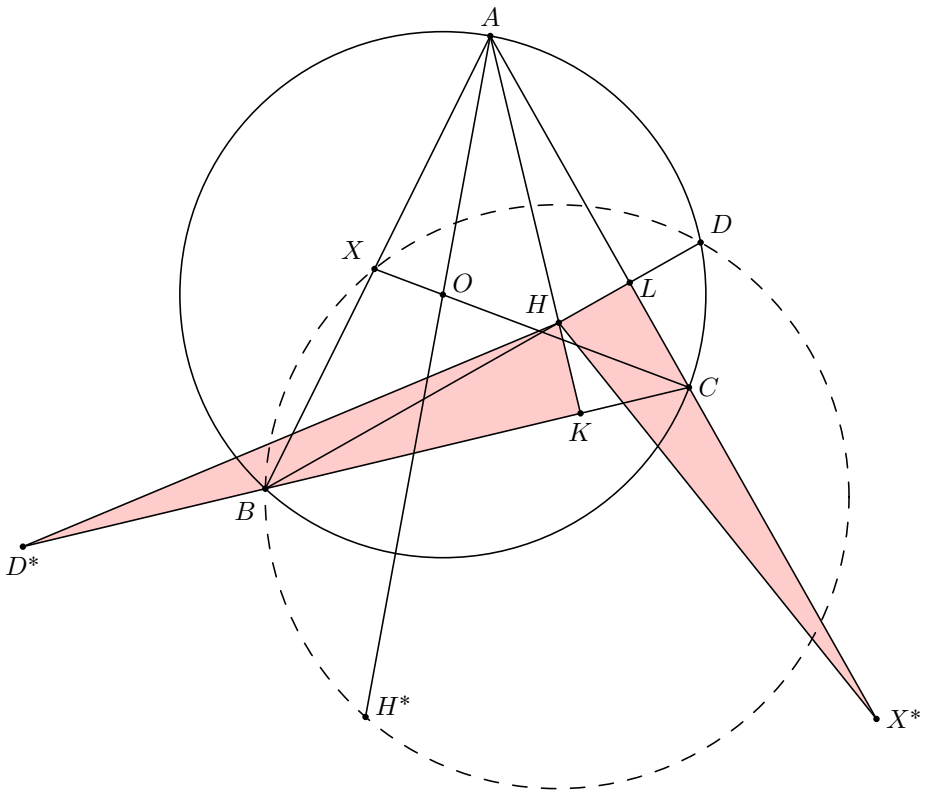
which implies $\frac{B'Z}{B'D} = \frac{AZ}{AX}$, as required. □

From the claim

$$\angle BDZ = 180^\circ - \angle ZDB' = 180^\circ - \angle ZXA = \angle BXZ$$

which means Z lies on circle $\odot BXD$. Similarly, Z lies on circle $\odot CYE$ completing the proof.

Solution 3 (Inversion at A).



Consider the composition of the inversion at A with radius $\sqrt{AB \times AC}$ and reflection in the angle bisector of $\angle BAC$, and use P^* to denote the image of a point P under this transformation. Let H be the orthocentre of triangle ABC and let K, L be the feet of the perpendicular from A, B to BC, CA respectively. Denote $A = \angle BAC$, $B = \angle CBA$ and $C = \angle ACB$.

We have

$$\angle D^*AK = \angle OAD = 90^\circ - \angle DBA = 90^\circ - (90^\circ - A) = A.$$

Hence, using right-angled $\triangle AKD^*$

$$\begin{cases} D^*K = AK \tan A = 2R \sin B \sin C \tan A \\ HK = 2R \cos B \cos C \end{cases} \implies \frac{D^*K}{HK} = \tan A \tan B \tan C$$

We also have

$$\angle AX^*B = \angle ACX = \angle ACO = 90^\circ - B.$$

Hence, using right-angled $\triangle BLX^*$

$$\begin{cases} X^*L = BL\tan B = 2R\sin A\sin C\tan B \\ HL = 2R\cos A\cos C \end{cases} \implies \frac{X^*L}{HL} = \tan A\tan B\tan C.$$

Thus $\frac{D^*K}{HK} = \frac{X^*L}{HL}$ and as $\angle HLX^* = \angle HKD^* = 90^\circ$, this means that triangle D^*HK and X^*HL are similar and in particular

$$\angle CD^*H = \angle KD^*H = \angle LX^*H = \angle CX^*H$$

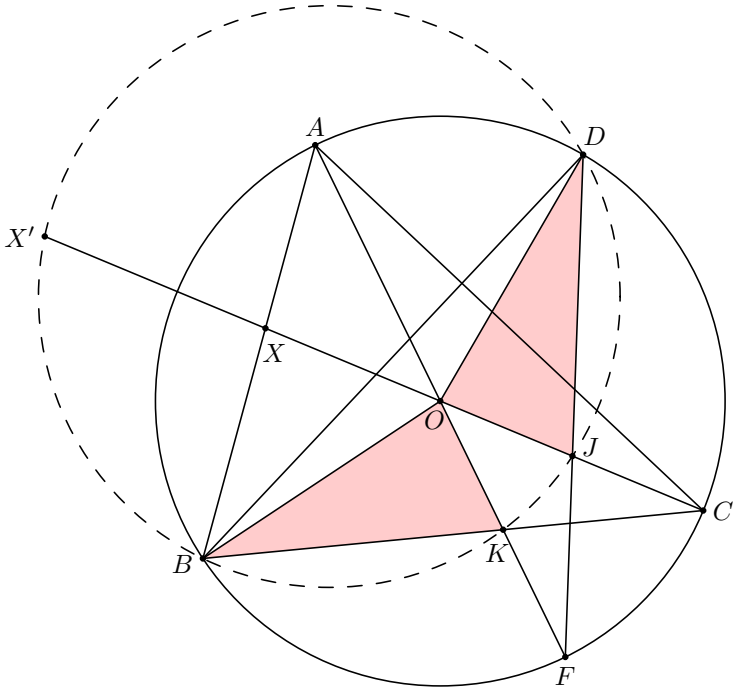
so D^*HCX^* is cyclic.

Inverting back, this gives DH^*BX cyclic so H^* lies on circle $\odot BXD$. Similarly, H^* lies on circle $\odot CYE$.

Since AO and AH are isogonal in $\angle BAC$, H^* lies on line AO completing the proof.

Solution 4 (Inversion at O). Let F be the point on ω such that AF is a diameter of ω, and J be the intersection of DF with CO.

Consider the inversion with respect to ω and use P' to denote the image of a point P.



X' lies on line CO and we have

$$\angle BX'J = \angle BX'O = \angle OBX = \angle OBA = \angle BAO = \angle BAF = \angle BDF = \angle BDJ$$

so $BX'DJ$ is cyclic.

Let K be the intersection of AF with BC. Then we have $OB = OD$ and

$$\angle KBO = 90^\circ - A = \angle DBA = \angle DFA = \angle DFO = \angle ODJ$$
$$\angle BOK = 2\angle OBA = 2\angle CBD = \angle COD = \angle JOD$$

Hence triangle BOK and DOJ are congruent. In particular $BK = DJ$ and

$$\angle KBD = \angle KBO + \angle OBD = \angle ODJ + \angle BDO = \angle BDJ.$$

Thus $BDJK$ is an isosceles trapezoid and $BX'DJK$ is cyclic.

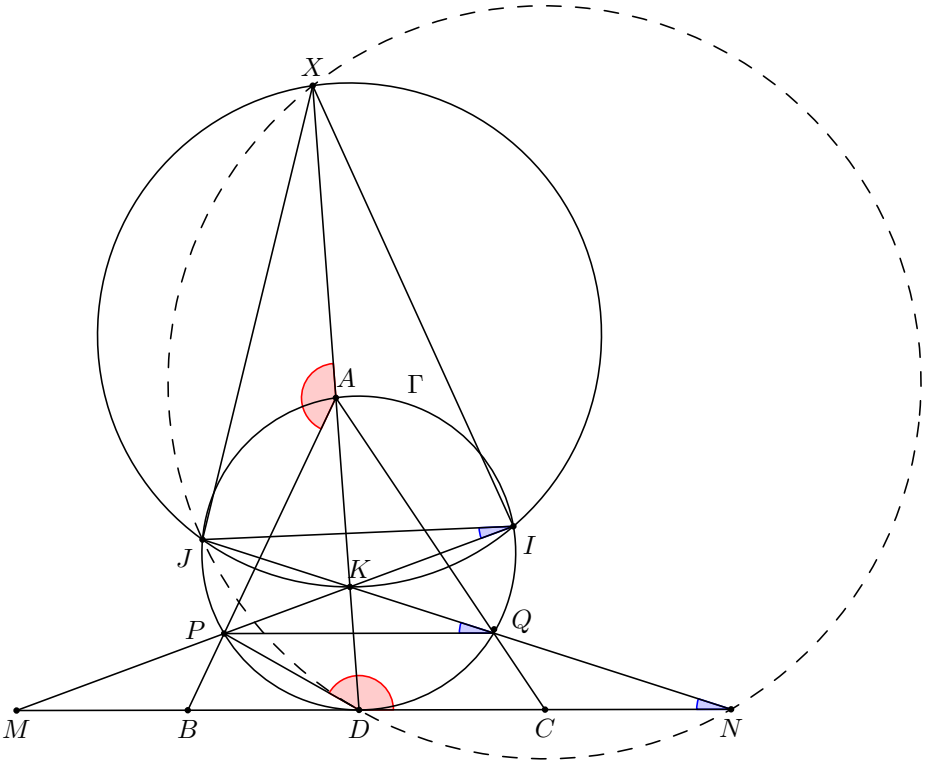
Inverting back this gives that $BXDK'$ is cyclic. Similarly $CYEK'$ is cyclic. Since K lies on AO, K' also lies on AO completing the proof.

G6. Let ABC be an acute-angled triangle with circumcircle ω. A circle Γ is internally tangent to ω at A and also tangent to BC at D. Let AB and AC intersect Γ at P and Q respectively. Let M and N be points on line BC such that B is the midpoint of DM and C is the midpoint of DN. Lines MP and NQ meet at K and intersect Γ again at I and J respectively. The ray KA meets the circumcircle of triangle IJK at $X \neq K$.

Prove that $\angle BXP = \angle CXQ$.

(United Kingdom)

Solution 1 (Similar Triangles).



Let MP and NQ intersect AD at K_1 and K_2 respectively. By applying Menelaus' theorem to triangle ABD and line MPK_1, we have

$$\frac{AK_1}{K_1D} = \frac{AP}{PB} \cdot \frac{BM}{MD} = \frac{AP}{2PB}$$

and similarly $\frac{AK_2}{K_2D} = \frac{AQ}{2QC}$. A homothety at A takes $\Gamma \to \omega$ and D to the midpoint of arc BC not containing A, so $PQ \parallel BC$ and AD bisects $\angle BAC$. Thus

$$\frac{AK_1}{K_1D} = \frac{AP}{2PB} = \frac{AQ}{2QC} = \frac{AK_2}{K_2D}$$

which implies $K_1 \equiv K_2$, and K lies on AD.

Then we obtain

$$\angle JXD = \angle JXK = \angle JIK = \angle JIP = \angle JQP = \angle JND$$

where the last equality follows from $PQ \parallel BC$. This shows $JXND$ is cyclic and hence

$$\angle DXN = \angle DJN = \angle DJQ = \angle DAQ = \angle DAC$$

which shows $AC \parallel XN$. As C is the midpoint of DN, A is the midpoint of XD.

Now observe that $\angle ADP = \angle AQP = \angle ACB$ and $\angle PAD = \angle DAC = \frac{\angle A}{2}$ so triangle APD and ADC are similar. Therefore we have

$$\frac{CD}{DP} = \frac{AD}{AP} = \frac{XA}{AP}$$

and also have

$$\angle CDP = 180^\circ - \angle PDB = 180^\circ - \angle PAD = \angle XAP.$$

Combining the two results gives triangles PDC and PAX are similar, which shows P is the centre of spiral similarity taking $CD \to XA$. Hence also triangles PXC and PAD are similar which shows $\angle PXC = \angle PAD = \frac{\angle A}{2}$. This gives

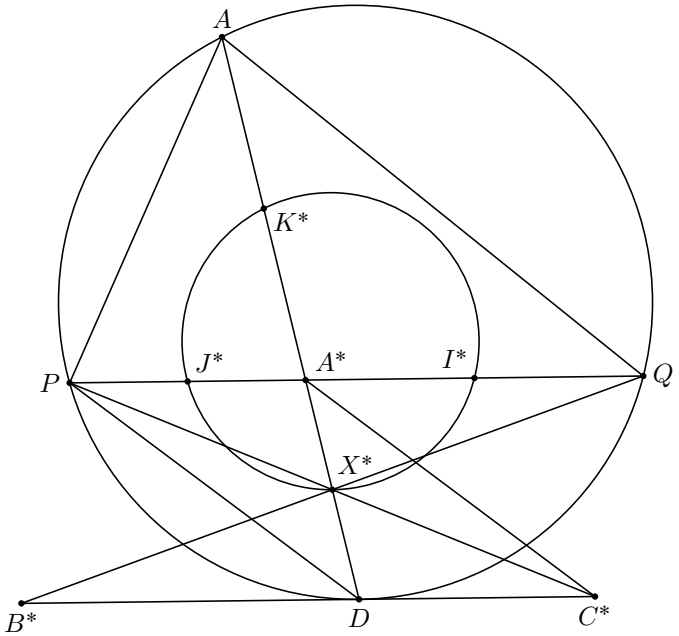
$$\angle BXP = \angle BXC - \angle PXC = \angle BXC - \frac{\angle A}{2}$$

which is symmetric in B, C giving the result.

Solution 2 (Inversion). As in the first solution we show that K lies on AD. From C being the midpoint of DN and $BC \parallel PQ$ we get

$$-1 = (C, \infty_{BC}; N, D) \overset{Q}{=} (A, PQ \cap AD; K, D) \overset{Q}{=} (A, P; J, D)\,.$$

Similarly we get $(A, Q; I, D) = -1$.



Now invert about D with radius $DP = DQ$ denoting the inverse of a point Z by Z^*. Since $\odot APQ$ and line PQ swap we have $A^* = PQ \cap AD$. Thus we have:

$$-1 = (A, A^*; K, D) = (A, P; J, D) = (A, Q; I, D)$$

As inversion preserves cross ratio and D inverts to the point at infinity, it follows I^*, J^*, K^* are the midpoints of A^*Q, A^*P, A^*A respectively. We know $XIKJ$ cyclic so X is the second intersection of circle $(I^*J^*K^*)$ with AD. Homothety of factor 2 at A^* takes circle $(I^*J^*K^*)$ to circle $\odot APQ \equiv \Gamma$ hence in fact X^* is the midpoint of A^*D.

Then we have

$$\angle PDB = \angle BAD = \angle DAC = \angle DC^*A^*$$

so $DP \parallel C^*A^*$. Also A^* lies on PQ so as $PQ \parallel BC$ we get $A^*P \parallel DC^*$, which gives PA^*C^*D is a parallelogram. Similarly QA^*B^*D is also a parallelogram. As X^* is the midpoint of A^*D this shows that X^* lies on lines B^*Q and C^*P.

By applying standard properties of angles under inversion, we have

$$\begin{aligned}
\angle BXP - \angle CXQ &= (\angle BXD - \angle PXD) - (\angle DXC - \angle DXQ) \\
&= (\angle DB^*X^* - \angle DPX^*) - (\angle X^*C^*D - \angle X^*QD) \\
&= (\angle DB^*X^* + \angle X^*PQ) - (\angle X^*C^*D + \angle PQX^*) \\
&\qquad\qquad (\text{as } \angle DPQ = \angle PQD) \\
&= \underbrace{(\angle DB^*Q - \angle PQB^*)}_{=0} - \underbrace{(\angle PC^*D - \angle C^*PQ)}_{=0} \\
&= 0 \qquad\qquad (\text{as } PQ \parallel BC)
\end{aligned}$$

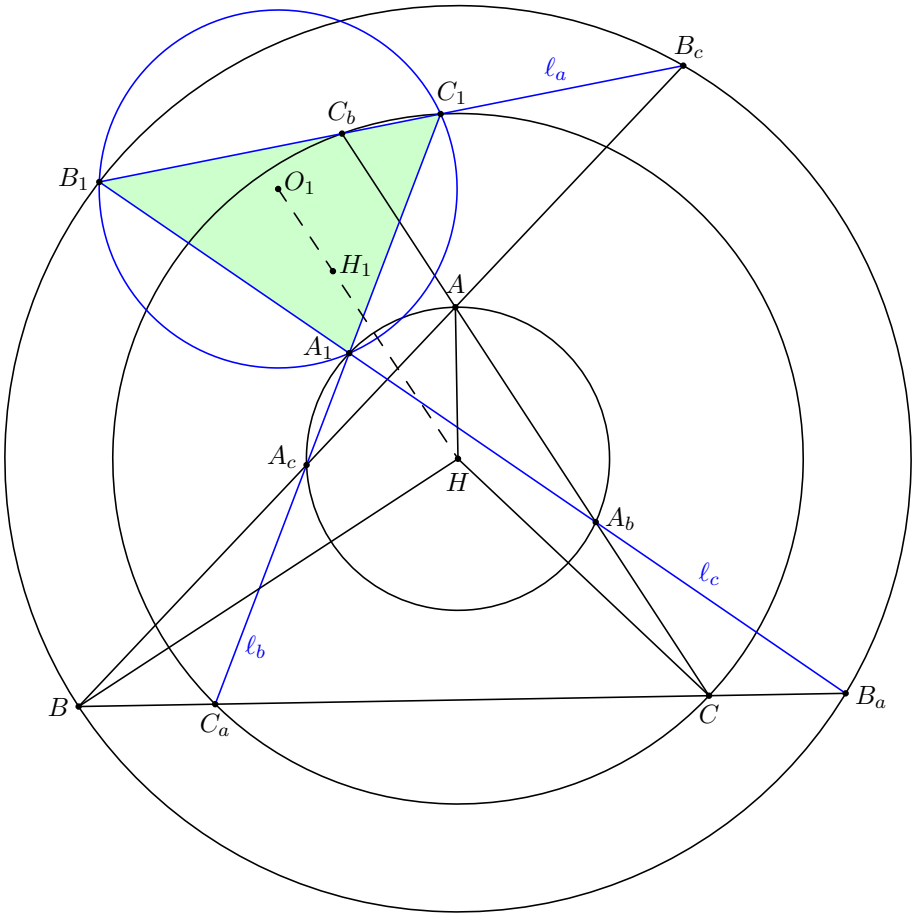
which gives the result.

G7. Let ABC be an acute, scalene triangle with orthocentre H. Let ℓ_a be the line through the reflection of B with respect to CH and the reflection of C with respect to BH. Lines ℓ_b and ℓ_c are defined similarly. Suppose lines ℓ_a, ℓ_b, and ℓ_c determine a triangle $\mathcal{T}$.

Prove that the orthocentre of $\mathcal{T}$, the circumcentre of $\mathcal{T}$ and H are collinear.

(Ukraine)

Solution 1.



We write $\triangle P_1P_2P_3 \overset{+}{\sim} \triangle Q_1Q_2Q_3$ (resp. $\triangle P_1P_2P_3 \overset{-}{\sim} \triangle Q_1Q_2Q_3$) to indicate that two triangles are directly (resp. oppositely) similar. We use directed angles throughout denoted with $\measuredangle$.

Denote by A_b, A_c the reflections of A in BH and CH respectively. B_c, B_a and C_a, C_b are defined similarly. By definition, $\ell_a = B_cC_b$, $\ell_b = C_aA_c$, $\ell_c = A_bB_a$. Let $A_1 = \ell_b \cap \ell_c$, $B_1 = \ell_c \cap \ell_a$, $C_1 = \ell_a \cap \ell_b$ and let O_1, H_1 be the orthocentre and circumcentre of $\mathcal{T} \equiv \triangle A_1B_1C_1$ respectively.

Claim 1. $\triangle AA_bA_c \overset{-}{\sim} \triangle ABC$.

Proof. Let $P = BH \cap AC$, $Q = CH \cap AB$, then it is well known that $\triangle APQ \overset{-}{\sim} \triangle ABC$. By the dilation with factor 2 centred at A, $\triangle APQ$ is sent to $\triangle AA_bA_c$, so we have $\triangle AA_bA_c \overset{-}{\sim} \triangle ABC$. □

Claim 2. $\triangle AA_bA_c \overset{+}{\sim} \triangle AB_aC_a$ and A_1 lies on the circumcircle of $\triangle AA_bA_c$ which is centred at H.

Proof. Since B_a, C_a are reflections of B, C in AH, we have $\triangle AB_aC_a \overset{-}{\sim} \triangle ABC$. Combining this with *Claim 1*, we have $\triangle AA_bA_c \overset{+}{\sim} \triangle AB_aC_a$, where A is the centre of this similarity. Therefore, $\measuredangle A_cA_1A_b = \measuredangle A_cAA_b$ meaning A_1 lies on $\odot AA_bA_c$. By symmetry, $HA_b = HA = HA_c$, so H is centre of this circle. □

Claim 3. $\triangle A_1B_1C_1 \overset{-}{\sim} \triangle ABC$.

Proof. From *Claim 2* we have

$$\measuredangle C_1A_1B_1 = \measuredangle A_cA_1A_b \overset{Claim\ 2}{=} \measuredangle A_cAA_b = -\measuredangle CAB$$

and similarly $\measuredangle A_1B_1C_1 = -\measuredangle ABC$, $\measuredangle B_1C_1A_1 = -\measuredangle BCA$, which imply $\triangle A_1B_1C_1 \overset{-}{\sim} \triangle ABC$. □

Denote the ratio of similitude of $\triangle A_1B_1C_1$ and $\triangle ABC$ by λ $(= \frac{B_1C_1}{BC})$, then

$$\lambda = \frac{H_1A_1}{HA} = \frac{H_1B_1}{HB} = \frac{H_1C_1}{HC}.$$

Since $HA = HA_1$ and similarly $HB = HB_1$, $HC = HC_1$ from *Claim 2*, we get

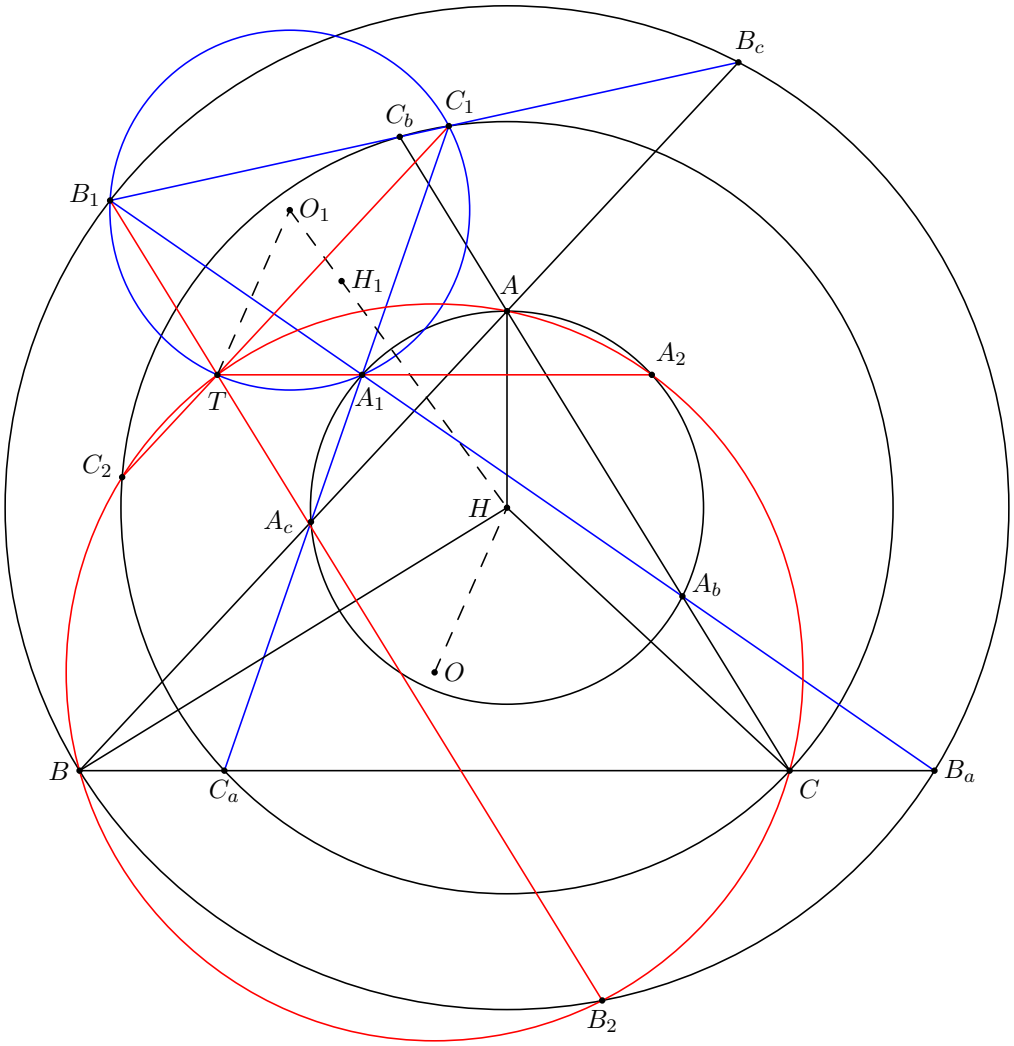
$$\lambda = \frac{H_1A_1}{HA_1} = \frac{H_1B_1}{HB_1} = \frac{H_1C_1}{HC_1}.$$

Therefore, the circle $A_1B_1C_1$ is the Apollonian circle of the segment HH_1 with ratio λ so the line HH_1 passes through O_1.

Solution 2. We use the same notation $A_b, A_c, B_c, B_a, C_a, C_b$ and A_1, B_1, C_1, O_1, H_1 as in Solution 1, and also show *Claim 1*, *Claim 2* and *Claim 3* in the same way.

Let O be the circumcentre of $\triangle ABC$ and A_2 be the reflection of A_1 in AH. As $\odot AA_bA_c$ is centred at H, A_2 also lies on this circle.

By *Claim 2*, $\measuredangle B_aA_1C_a = \measuredangle A_bAA_c = \measuredangle B_aAC_a$, so A_1 lies on $\odot AB_aC_a$. Reflecting this in AH gives that A_2 lies on $\odot ABC$. We now have circles centred at O and H passing through A and A_2 so these points are symmetric with respect to OH. Define B_2 and C_2 similarly then $\triangle ABC$ and $\triangle A_2B_2C_2$ are symmetric with respect to OH and also $\odot ABC = \odot A_2B_2C_2$.



Claim 4. A_1A_2, B_1B_2 and C_1C_2 have an intersection on $\odot ABC$ which we denote by T.

Proof. Let $T = A_1A_2 \cap B_1B_2$. Since $A_1A_2 \parallel BC$ and $B_1B_2 \parallel AC$, we have

$$\measuredangle A_2TB_2 = \measuredangle BCA = -\measuredangle B_2C_2A_2 = \measuredangle A_2C_2B_2.$$

So T lies on $\odot A_2B_2C_2 = \odot ABC$. Similarly the intersection of A_1A_2 and C_1C_2 lies on $\odot ABC$, so C_1C_2 also passes through T. □

Claim 5. T also lies on $\odot A_1B_1C_1$ and T corresponds to T itself under the similarity $\triangle A_1B_1C_1 \overset{-}{\sim} \triangle ABC$.

Proof. We know $\triangle A_1B_1C_1 \overset{-}{\sim} \triangle ABC$ by *Claim 3*. We also have

$$\measuredangle B_1TC_1 = \measuredangle B_2TC_2 \quad = \quad \measuredangle B_2A_2C_2 = -\measuredangle BAC \overset{\text{Claim 3}}{=} \measuredangle B_1A_1C_1,$$

so T lies on $\odot A_1B_1C_1$. The remaining part is concluded by the following angle chase:

$$\measuredangle A_1B_1T = \measuredangle A_1B_1B_2 \overset{B_1B_2 \parallel AC}{=} \measuredangle A_1A_bA = \measuredangle A_1A_2A = -\measuredangle AA_2T = -\measuredangle ABT. \qquad \square$$

Claim 6. The circumradius of $\triangle A_1B_1C_1$ is equal to HO.

Proof. Two circles centred at H intersect ℓ_c at A_1, A_b and B_1, B_a, so A_1A_b and B_1B_a have the same midpoint and thus $A_1B_1 = A_bB_a$. Consider the spiral symmetry $\triangle AA_bA_c \overset{+}{\sim} \triangle AB_aC_a$. This takes H, the circumcentre of $\triangle AA_bA_c$, to the circumcentre of $\triangle AB_aC_a$, denoted by O_a, which is symmetric to O in AH. Hence $\triangle AA_bB_a \overset{+}{\sim} \triangle AHO_a$, so

$$\frac{AA_b}{B_aA_b} = \frac{AH}{HO_a} = \frac{AH}{HO} \implies \frac{AH}{AA_b} = \frac{HO}{A_bB_a}.$$

Also since $\triangle A_1O_1B_1 \overset{+}{\sim} \triangle AHA_b$ (both of them are $\overset{-}{\sim} \triangle AOB$), we have

$$\frac{A_1O_1}{A_1B_1} = \frac{AH}{AA_b} = \frac{HO}{A_bB_a} = \frac{HO}{A_1B_1} \implies A_1O_1 = HO$$

as desired. $\square$

Since

$$\measuredangle (TA_1, TO_1) \overset{\text{Claim 5}}{=} \measuredangle (TO, TA) = 90^\circ + \measuredangle (TA_1, AA_2) \overset{OH \perp AA_2}{=} \measuredangle (TA_1, OH),$$

we have $O_1T \parallel HO$. Combined with $O_1T = HO$, O_1TOH is a parallelogram. Therefore, using this and *Claim 5*, we have $\measuredangle H_1O_1T = \measuredangle TOH = \measuredangle HO_1T$, which imply that O_1, H_1 and H are collinear as desired.

G8. Let ABC be an equilateral triangle. Points A_1, B_1, C_1 lie inside triangle ABC such that triangle $A_1B_1C_1$ is scalene, $BA_1 = A_1C$, $CB_1 = B_1A$, $AC_1 = C_1B$ and

$$\angle BA_1C + \angle CB_1A + \angle AC_1B = 480^\circ.$$

Lines BC_1 and CB_1 intersect at A_2; lines CA_1 and AC_1 intersect at B_2; and lines AB_1 and BA_1 intersect at C_2.

Prove that the circumcircles of triangles AA_1A_2, BB_1B_2, CC_1C_2 have two common points.
(U.S.A.)

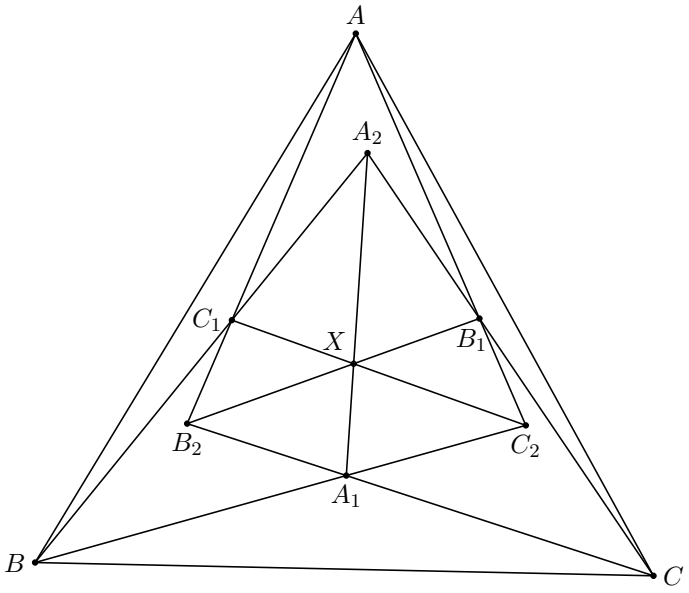
Solution. Let $\delta_A, \delta_B, \delta_C$ be the circumcircles of $\triangle AA_1A_2$, $\triangle BB_1B_2$, $\triangle CC_1C_2$. The general strategy of the solution is to find two different points having equal power with respect to $\delta_A, \delta_B, \delta_C$.

Claim. A_1 is the circumcentre of A_2BC and cyclic variations.

Proof. Since A_1 lies on the perpendicular bisector of BC and inside $\triangle BA_2C$, it suffices to prove $\angle BA_1C = 2\angle BA_2C$. This follows from

$$\begin{aligned}\angle BA_2C &= \angle A_2BA + \angle BAC + \angle ACA_2 \\ &= \frac{1}{2}\left((180^\circ - \angle AC_1B) + (180^\circ - \angle CB_1A)\right) + 60^\circ \\ &= 240^\circ - \frac{1}{2}\left(480^\circ - \angle BA_1C\right) \\ &= \frac{1}{2}\angle BA_1C\end{aligned}$$

□



The circumcentres above give

$$\angle B_1B_2C_1 = \angle B_1B_2A = \angle B_2AB_1 = \angle C_1AC_2 = \angle AC_2C_1 = \angle B_1C_2C_1$$

and so $B_1C_1B_2C_2$ is cyclic. Likewise $C_1A_1C_2A_2$ and $A_1B_1A_2B_2$ are cyclic. Note that hexagon $A_1B_2C_1A_2B_1C_2$ is not cyclic since

$$\angle C_2A_1B_2 + \angle B_2C_1A_2 + \angle A_2B_1C_2 = 480^\circ \neq 360^\circ.$$

Thus we can apply radical axis theorem to the three circles to show that A_1A_2, B_1B_2, C_1, C_2 concur at a point X and this point has equal power with respect to $\delta_A, \delta_B, \delta_C$.

Let the circumcircle of $\triangle A_2BC$ meet δ_A at $A_3 \neq A_2$. Define B_3 and C_3 similarly.

Claim. BCB_3C_3 cyclic.

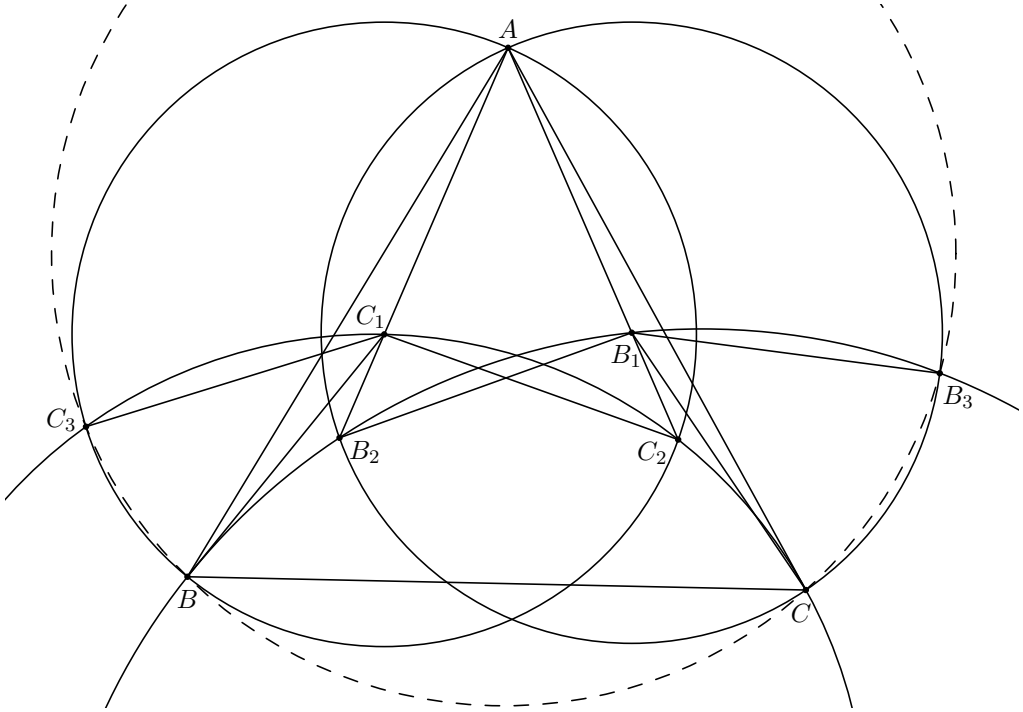
Proof. Using directed angles

$$\begin{aligned} \measuredangle BC_3C &= \measuredangle BC_3C_2 + \measuredangle C_2C_3C \\ &= \measuredangle BAC_2 + \measuredangle C_2C_1C \\ &= 90^\circ + \measuredangle(C_1C, AC_2) + \measuredangle C_2C_1C \qquad (CC_1 \perp AB) \\ &= 90^\circ + \measuredangle C_1C_2B_1. \end{aligned}$$

Similarly $\measuredangle CB_3B = 90^\circ + \measuredangle B_1B_2C_1$. Hence, using $B_1C_1B_2C_2$ cyclic

$$\measuredangle BB_3C = 90^\circ + \measuredangle C_1B_2B_1 = 90^\circ + \measuredangle C_1C_2B_1 = \measuredangle BC_3C$$

as required. □



Similarly CAC_3A_3 and ABA_3B_3 are cyclic. $AC_3BA_3CB_3$ is not cyclic because then AB_2CB_3 cyclic would mean B_2 lies on $\odot ABC$ which is impossible since B_2 lies inside $\triangle ABC$. Thus we can apply radical axis theorem to the three circles to get AA_3, BB_3, CC_3 concur at a point Y which has equal power with respect to $\delta_A, \delta_B, \delta_C$.

We now make some technical observations before finishing.

- Let O be the centre of $\triangle ABC$. We have that

$$\angle BA_1C = 480^\circ - \angle CB_1A - \angle AC_1B > 480^\circ - 180^\circ - 180^\circ = 120^\circ.$$

so A_1 lies inside $\triangle BOC$. We have similar results for B_1, C_1 and thus $\triangle BA_1C$, $\triangle CB_1A$, $\triangle AC_1B$ have disjoint interiors. It follows that $A_1B_2C_1A_2B_1C_2$ is a convex hexagon thus X lies on segment A_1A_2 and therefore is inside δ_A.

- Since A_1 is the centre of A_2BC we have that $A_1A_2 = A_1A_3$ so, from cyclic quadrilateral $AA_2A_1A_3$ we get that lines AA_2 and $AA_3 \equiv AY$ are reflections in line AA_1. As X lies on segment A_1A_2, the only way $X \equiv Y$ is if A_1 and A_2 both lie on the perpendicular bisector of BC. But this forces B_1 and C_1 to also be reflections in this line meaning $A_1B_1 = A_1C_1$ contradicting the scalene condition.

Summarising, we have distinct points X, Y with equal power with respect to $\delta_A, \delta_B, \delta_C$ thus these circles have a common radical axis. As X lies inside δ_A (and similarly δ_B, δ_C), this radical axis intersects the circles at two points and so $\delta_A, \delta_B, \delta_C$ have two points in common.

Comment. An alternative construction for Y comes by observing that

$$\frac{\sin\angle BAA_2}{\sin\angle A_2AC} = \frac{\frac{A_2B}{AA_2}\sin\angle A_2BA}{\frac{A_2C}{AA_2}\sin\angle ACA_2} = \frac{A_2B}{A_2C}\cdot\frac{\sin\angle C_1BA}{\sin\angle ACB_1} = \frac{\sin\angle B_1CB}{\sin\angle CBC_1}\cdot\frac{\sin\angle C_1BA}{\sin\angle ACB_1}$$

and hence

$$\frac{\sin\angle BAA_2}{\sin\angle A_2AC}\cdot\frac{\sin\angle CBB_2}{\sin\angle B_2BA}\cdot\frac{\sin\angle ACC_2}{\sin\angle C_2CB} = 1$$

so by Ceva's theorem, AA_2, BB_2, CC_2 concur and thus we can construct the isogonal conjugate of this point of concurrency which turns out to be Y.

Number Theory

N1. Determine all positive, composite integers n that satisfy the following property: if the positive divisors of n are $1 = d_1 < d_2 < \cdots < d_k = n$, then d_i divides $d_{i+1} + d_{i+2}$ for every $1 \leqslant i \leqslant k-2$.

(Colombia)

Answer: $n = p^r$ is a prime power for some $r \geqslant 2$.

Solution 1. It is easy to see that such an $n = p^r$ with $r \geqslant 2$ satisfies the condition as $d_i = p^{i-1}$ with $1 \geqslant i \geqslant k = r+1$ and clearly

$$p^{i-1} \mid p^i + p^{i+1}.$$

Now, let us suppose that there is a positive integer n that satisfies the divisibility condition of the problem and that has two different prime divisors p and q. Without lost of generality, we assume $p < q$ and that they are the two smallest prime divisors of n. Then there is a positive integer j such that

$$d_1 = 1,\, d_2 = p, \ldots, d_j = p^{j-1}, d_{j+1} = p^j, d_{j+2} = q,$$

and it follows that

$$d_{k-j-1} = \frac{n}{q}, d_{k-j} = \frac{n}{p^j},\, d_{k-j+1} = \frac{n}{p^{j-1}}, \ldots, d_{k-1} = \frac{n}{p},\, d_k = n.$$

Thus

$$d_{k-j-1} = \frac{n}{q} \mid d_{k-j} + d_{k-j+1} = \frac{n}{p^j} + \frac{n}{p^{j-1}} = \frac{n}{p^j}(p+1). \tag{1}$$

This gives $p^j \mid q(p+1)$, which is a contradiction since $\gcd(p, p+1) = 1$ and $p \neq q$.

Solution 2. Since $d_i d_{k+1-i} = n$, we have the equivalence:

$$d_{k-i-1} \mid d_{k-i} + d_{k-i+1} \iff \frac{n}{d_{i+2}} \mid \frac{n}{d_{i+1}} + \frac{n}{d_i}.$$

We multiply both sides by $d_i d_{i+1} d_{i+2}$ and cancel the n's to get

$$d_i d_{i+1} \mid d_i d_{i+2} + d_{i+1} d_{i+2}.$$

Hence,

$$d_i \mid d_{i+1} d_{i+2}. \tag{2}$$

Moreover, by the condition of the problem,

$$d_i \mid d_{i+1}(d_{i+1} + d_{i+2}) = d_{i+1}^2 + d_{i+1} d_{i+2}.$$

Combining this with (2) we get that $d_i \mid d_{i+1}^2$ for all $1 \leqslant i \leqslant k-2$.

Let $d_2 = p$ be the smallest prime divisor of n. By induction on i we prove that $p \mid d_i$ for all $2 \leqslant i \leqslant k-1$. The base case $d_2 = p$ is obvious. Let us suppose that $p \mid d_j$ for some $2 \leqslant j \leqslant k-2$. Then we have that

$$p \mid d_j \mid d_{j+1}^2 \implies p \mid d_{j+1}$$

as p is prime, which completes the induction. This implies that n has to be a prime power, as otherwise there would be another prime q that divides n and we would get that $p \mid q$ which is obviously false.

We finally check that the powers of p satisfy the condition in the statement of the problem as in Solution 1.

Solution 3. We start by proving the following claim:

Claim. $d_i \mid d_{i+1}$ for every $1 \leqslant i \leqslant k-1$.

Proof. We prove the Claim by induction on i; it is trivial for $i = 1$ because $d_1 = 1$. Suppose that $2 \leqslant i \leqslant k-1$ and the Claim is true for $i-1$, i.e. $d_{i-1} \mid d_i$. By the induction hypothesis and the problem condition, $d_{i-1} \mid d_i$ and $d_{i-1} \mid d_i + d_{i+1}$, so $d_{i-1} \mid d_{i+1}$.

Now consider the divisors $d_{k-i} = \dfrac{n}{d_{i+1}}$, $d_{k-i+1} = \dfrac{n}{d_i}$, $d_{k-i+2} = \dfrac{n}{d_{i-1}}$. By the problem condition,

$$\frac{d_{k-i+1} + d_{k-i+2}}{d_{k-i}} = \frac{\dfrac{n}{d_i} + \dfrac{n}{d_{i-1}}}{\dfrac{n}{d_{i+1}}} = \frac{d_{i+1}}{d_i} + \frac{d_{i+1}}{d_{i-1}}$$

is an integer. We conclude that $\dfrac{d_{i+1}}{d_i}$ is an integer, so $d_i \mid d_{i+1}$. □

By the Claim, n cannot have two different prime divisors because the smallest one would divide the other one. Hence, n must be a power of a prime, and powers of primes satisfy the condition of the problem as we saw in Solution 1.

Solution 4. We present here a more technical way of finishing Solution 1 after obtaining (1). We let $v_p(m)$ denote the p-adic valuation of m. Notice that $v_p(n/q) = v_p(n)$ as $\gcd(p, q) = 1$ and that

$$v_p\left(\frac{n}{p^j}(p+1)\right) = v_p(n) - j$$

as $\gcd(p, p+1) = 1$. But (1) implies

$$v_p(n) = v_p(n/q) \leqslant v_p\left(\frac{n}{p^j}(p+1)\right) = v_p(n) - j$$

which is a contradiction. Thus n has only one prime divisor as desired.

N2. Determine all pairs (a, p) of positive integers with p prime such that $p^a + a^4$ is a perfect square.

(Bangladesh)

Answer: $(a, p) = (1, 3),\ (2, 3),\ (6, 3),\ (9, 3)$ are all the possible solutions.

Solution. Let $p^a + a^4 = b^2$ for some positive integer b. Then we have

$$p^a = b^2 - a^4 = (b + a^2)(b - a^2).$$

Hence both $b + a^2$ and $b - a^2$ are powers of p.

Let $b - a^2 = p^x$ for some integer x. Then $b + a^2 = p^{a-x}$ and $a - x > x$. Therefore, we have

$$2a^2 = (b + a^2) - (b - a^2) = p^{a-x} - p^x = p^x(p^{a-2x} - 1). \tag{1}$$

We shall consider two cases according to whether $p = 2$ or $p \neq 2$. We let $v_p(m)$ denote the p-adic valuation of m.

Case 1 ($p = 2$): In this case,

$$a^2 = 2^{x-1}(2^{a-2x} - 1) = 2^{2v_2(a)}(2^{a-2x} - 1),$$

where the first equality comes from (1) and the second one from $\gcd(2, 2^{a-2x} - 1) = 1$. So, $2^{a-2x} - 1$ is a square.

If $v_2(a) > 0$, then 2^{a-2x} is also a square. So, $2^{a-2x} - 1 = 0$, and $a = 0$ which is a contradiction.

If $v_2(a) = 0$, then $x = 1$, and $a^2 = 2^{a-2} - 1$. If $a \geqslant 4$, the right hand side is congruent to 3 modulo 4, thus cannot be a square. It is easy to see that $a = 1, 2, 3$ do not satisfy this condition.

Therefore, we do not get any solutions in this case.

Case 2 ($p \neq 2$): In this case, we have $2v_p(a) = x$. Let $m = v_p(a)$. Then we have $a^2 = p^{2m} \cdot n^2$ for some integer $n \geqslant 1$. So, $2n^2 = p^{a-2x} - 1 = p^{p^m \cdot n - 4m} - 1$.

We consider two subcases.

Subcase 2-1 ($p \geqslant 5$): By induction, one can easily prove that $p^m \geqslant 5^m > 4m$ for all m. Then we have

$$2n^2 + 1 = p^{p^m \cdot n - 4m} > p^{p^m \cdot n - p^m} \geqslant 5^{5^m \cdot (n-1)} \geqslant 5^{n-1}.$$

But, by induction, one can easily prove that $5^{n-1} > 2n^2 + 1$ for all $n \geqslant 3$. Therefore, we conclude that $n = 1$ or 2. If $n = 1$ or 2, then $p = 3$, which is a contradiction. So there are no solutions in this subcase.

Subcase 2-2 ($p = 3$): Then we have $2n^2 + 1 = 3^{3^m \cdot n - 4m}$. If $m \geqslant 2$, one can easily prove by induction that $3^m > 4m$. Then we have

$$2n^2 + 1 = 3^{3^m \cdot n - 4m} > 3^{3^m \cdot n - 3^m} = 3^{3^m \cdot (n-1)} \geqslant 3^{9(n-1)}.$$

Again, by induction, one can easily prove that $3^{9(n-1)} > 2n^2 + 1$ for all $n \geqslant 2$. Therefore, we conclude that $n = 1$. Then we have $2 \cdot 1^2 + 1 = 3^{3^m - 4m}$ hence $3 = 3^{3^m - 4m}$. Consequently, we have $3^m - 4m = 1$. The only solution of this equation is $m = 2$ in which case we have $a = 3^m \cdot n = 3^2 \cdot 1 = 9$.

If $m \leqslant 1$, then there are two possible cases: $m = 0$ or $m = 1$.

- If $m = 1$, then we have $2n^2 + 1 = 3^{3n-4}$. Again, by induction, one can easily prove that $3^{3n-4} > 2n^2 + 1$ for all $n \geqslant 3$. By checking $n = 1, 2$, we only get $n = 2$ as a solution. This gives $a = 3^m \cdot n = 3^1 \cdot 2 = 6$.
- If $m = 0$, then we have $2n^2 + 1 = 3^n$. By induction, one can easily prove that $3^n > 2n^2 + 1$ for all $n \geqslant 3$. By checking $n = 1, 2$, we find the solutions $a = 3^0 \cdot 1 = 1$ and $a = 3^0 \cdot 2 = 2$.

Therefore, $(a, p) = (1, 3),\ (2, 3),\ (6, 3),\ (9, 3)$ are all the possible solutions.

N3. For positive integers n and $k \geqslant 2$ define $E_k(n)$ as the greatest exponent r such that k^r divides $n!$. Prove that there are infinitely many n such that $E_{10}(n) > E_9(n)$ and infinitely many m such that $E_{10}(m) < E_9(m)$.

(Brazil)

Solution 1. We let $v_p(m)$ denote the p-adic valuation of m. By Legendre's Formula we know, for p prime, that $v_p(n!) = \lfloor n/p \rfloor + \lfloor n/p^2 \rfloor + \cdots$. We can see that $E_9(n) = \lfloor \frac{v_3(n!)}{2} \rfloor$. Since $v_5(n!) \leqslant v_2(n!)$ and $E_{10}(n) = \min(v_5(n!), v_2(n!))$, we have $E_{10}(n) = v_5(n!)$.

Let l be a positive integer. Set $n = 5^{2l-1}$. Then we have

$$E_{10}(n) = v_5(n!) = 5^{2l-2} + 5^{2l-3} + \cdots + 5 + 1 = \frac{5^{2l-1} - 1}{4} = \frac{n-1}{4}.$$

Since $n = 5^{2l-1} \equiv 2 \pmod 3$, we have $\left\lfloor \frac{n}{3} \right\rfloor = \frac{n-2}{3}$ and it implying

$$v_3(n!) = \left\lfloor \frac{n}{3} \right\rfloor + \left\lfloor \frac{n}{3^2} \right\rfloor + \left\lfloor \frac{n}{3^3} \right\rfloor + \cdots < \frac{n-2}{3} + \frac{n}{3^2} + \frac{n}{3^3} + \cdots = \frac{n}{2} - \frac{2}{3}.$$

From this we obtain

$$E_9(n) = \left\lfloor \frac{v_3(n!)}{2} \right\rfloor \leqslant \frac{v_3(n!)}{2} \leqslant \frac{n}{4} - \frac{1}{3} < \frac{n}{4} - \frac{1}{4} = E_{10}(n).$$

In a similar way, we set now $m = 3^{4l-2}$. Then we have

$$v_3(m!) = 3^{4l-3} + 3^{4l-4} + \cdots + 3 + 1 = \frac{3^{4l-2} - 1}{2} = \frac{m-1}{2}.$$

Note that $m = 3^{4l-2} \equiv 1 \pmod 4$ and hence $E_9(m) = \lfloor \frac{v_3(m!)}{2} \rfloor = \lfloor \frac{m-1}{4} \rfloor = \frac{m-1}{4}$. We also have $m = 3^{4l-2} \equiv 4 \pmod 5$ implying $\lfloor \frac{m}{5} \rfloor = \frac{m-4}{5}$. Therefore we obtain

$$E_{10}(m) = v_5(m!) = \left\lfloor \frac{m}{5} \right\rfloor + \left\lfloor \frac{m}{5^2} \right\rfloor + \cdots < \frac{m-4}{5} + \frac{m}{5^2} + \cdots = \frac{m}{4} - \frac{4}{5} < \frac{m}{4} - \frac{1}{4} = E_9(m).$$

We can take infinitely many $n = 5^{2l-1}$ and $m = 3^{4l-2}$ completing the proof.

Solution 2. In the setting of Solution 1, we consider two subsequences:

First, we take $n = 5^{3^{b-1}}$ with $b \geqslant 2$. Because 5 is not a square modulo 3 and $\varphi(3^b) = 2 \cdot 3^{b-1}$, we have $n \equiv -1 \pmod{3^b}$. Hence,

$$v_3(n!) = \left\lfloor \frac{n}{3} \right\rfloor + \left\lfloor \frac{n}{3^2} \right\rfloor + \cdots < \frac{n-2}{3} + \frac{n-8}{9} + \cdots + \frac{n-(3^b-1)}{3^b} + \frac{n}{3^{b+1}} + \cdots < \frac{n}{2} - b + \frac{1}{2},$$

and $E_{10}(n) = \frac{n-1}{4} > \frac{n+1-2b}{4} > E_9(n)$.

In the same way, for $m = 3^{2 \cdot 5^{b-1}} \equiv -1 \pmod{5^b}$ with $b \geqslant 2$,

$$E_{10}(m) = \left\lfloor \frac{m}{5} \right\rfloor + \left\lfloor \frac{m}{5^2} \right\rfloor + \cdots < \frac{m-4}{5} + \frac{m-24}{25} + \cdots + \frac{m-(5^b-1)}{5^b} + \frac{m}{5^{b+1}} + \cdots < \frac{m}{4} - b + \frac{1}{4},$$

and $E_9(m) = \frac{m-1}{4} > E_{10}(m)$ holds.

Comment. From Solution 2 we can see that for any positive real B, there exist infinitely many positive integers m and n such that $E_{10}(n) - E_9(n) > B$ and $E_{10}(m) - E_9(m) < -B$.

N4. Let $a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_n$ be $2n$ positive integers such that the $n+1$ products

$$\begin{gathered} a_1a_2a_3\cdots a_n, \\ b_1a_2a_3\cdots a_n, \\ b_1b_2a_3\cdots a_n, \\ \vdots \\ b_1b_2b_3\cdots b_n \end{gathered}$$

form a strictly increasing arithmetic progression in that order. Determine the smallest positive integer that could be the common difference of such an arithmetic progression.

(*Canada*)

Answer: The smallest common difference is $n!$.

Solution 1. The condition in the problem is equivalent to

$$D = (b_1 - a_1)a_2a_3\cdots a_n = b_1(b_2 - a_2)a_3a_4\cdots a_n = \cdots = b_1b_2\cdots b_{n-1}(b_n - a_n),$$

where D is the common difference. Since the progression is strictly increasing, $D > 0$, hence $b_i > a_i$ for every $1 \leqslant i \leqslant n$. Individually, these equalities simplify to

$$(b_i - a_i)a_{i+1} = b_i(b_{i+1} - a_{i+1}) \text{ for every } 1 \leqslant i \leqslant n-1. \tag{1}$$

If $g_i := \gcd(a_i, b_i) > 1$ for some $1 \leqslant i \leqslant n$, then we can replace a_i with $\frac{a_i}{g_i}$ and b_i with $\frac{b_i}{g_i}$ to get a smaller common difference. Hence we may assume $\gcd(a_i, b_i) = 1$ for every $1 \leqslant i \leqslant n$.

Then, we have $\gcd(b_i - a_i, b_i) = \gcd(a_i, b_i) = 1$ and $\gcd(a_{i+1}, b_{i+1} - a_{i+1}) = \gcd(a_{i+1}, b_{i+1}) = 1$ for every $1 \leqslant i \leqslant n-1$. The equality (1) implies $a_{i+1} = b_i$ and $b_i - a_i = b_{i+1} - a_{i+1}$. Thus,

$$a_1, \quad b_1 = a_2, \quad b_2 = a_3, \quad \ldots, \quad b_{n-1} = a_n, \quad b_n$$

is an arithmetic progression with positive common difference. Since $a_1 \geqslant 1$, we have $a_i \geqslant i$ for every $1 \leqslant i \leqslant n$, so

$$D = (b_1 - a_1)a_2a_3\cdots a_n \geqslant 1 \cdot 2 \cdot 3 \cdots n = n!$$

Equality is achieved when $b_i - a_i = 1$ for $1 \leqslant i \leqslant n$ and $a_1 = 1$, i.e. $a_i = i$ and $b_i = i + 1$ for every $1 \leqslant i \leqslant n$. Indeed, it is straightforward to check that these integers produce an arithmetic progression with common difference $n!$.

Solution 2 (Variant of Solution 1). Similarly to Solution 1, we may assume $\gcd(a_i, b_i) = 1$ for every $1 \leqslant i \leqslant n$.

Denote by $p_1, p_2, \ldots, p_{n+1}$ the sequence obtained as the product in the problem statement. Then we have $\frac{p_{i+1}}{p_i} = \frac{b_i}{a_i} > 1$, so $b_i > a_i$. Since $p_1, p_2, \ldots, p_{n+1}$ is an arithmetic progression, we have $p_{i+2} = 2p_{i+1} - p_i$ hence

$$2 - \frac{a_i}{b_i} = \frac{2b_i - a_i}{b_i} = \frac{2p_{i+1} - p_i}{p_{i+1}} = \frac{p_{i+2}}{p_{i+1}} = \frac{b_{i+1}}{a_{i+1}}.$$

But since the fractions on the left-hand side and the right-hand side are both irreducible, we conclude that $b_i = a_{i+1}$, so $2 - \frac{a_i}{a_{i+1}} = \frac{a_{i+2}}{a_{i+1}}$. Then we have $a_i + a_{i+2} = 2a_{i+1}$, which means that $a_1, a_2, \ldots, a_n$ is an arithmetic progression with positive common difference.

We conclude as in Solution 1.

Solution 3. (The following solution is purely algebraic: it does not involve considerations on greatest common divisors.)

We retake Solution 1 from (1). Then we have

$$\frac{a_{i+1}}{b_{i+1}-a_{i+1}} = \frac{b_i}{b_i-a_i} = 1 + \frac{a_i}{b_i-a_i}.$$

So, for $1 \leqslant i \leqslant n$,

$$\frac{a_i}{b_i-a_i} = \frac{a_1}{b_1-a_1} + (i-1).$$

Then

$$a_i \geqslant \frac{a_i}{b_i-a_i} = \frac{a_1}{b_1-a_1} + (i-1) > i-1.$$

since $b_i - a_i \geqslant 1$ and $b_1 - a_1 > 0$. As a_i is an integer, we have $a_i \geqslant i$.

We again conclude as in Solution 1.

N5. Let $a_1 < a_2 < a_3 < \cdots$ be positive integers such that a_{k+1} divides $2(a_1 + a_2 + \cdots + a_k)$ for every $k \geqslant 1$. Suppose that for infinitely many primes p, there exists k such that p divides a_k. Prove that for every positive integer n, there exists k such that n divides a_k.

(Netherlands)

Solution. For every $k \geqslant 2$ define the quotient $b_k = 2(a_1 + \cdots + a_{k-1})/a_k$, which must be a positive integer. We first prove the following properties of the sequence (b_k):

Claim 1. We have $b_{k+1} \leqslant b_k + 1$ for all $k \geqslant 2$.

Proof. By subtracting $b_k a_k = 2(a_1 + \cdots + a_{k-1})$ from $b_{k+1}a_{k+1} = 2(a_1 + \cdots + a_k)$, we find that $b_{k+1}a_{k+1} = b_k a_k + 2a_k = (b_k + 2)a_k$. From $a_k < a_{k+1}$ it follows that $b_k + 2 > b_{k+1}$. □

Claim 2. The sequence (b_k) is unbounded.

Proof. We start by rewriting $b_{k+1}a_{k+1} = (b_k + 2)a_k$ as

$$a_{k+1} = a_k \cdot \frac{b_k + 2}{b_{k+1}} \implies a_{k+1} \mid a_k (b_k + 2).$$

If the sequence (b_k) were bounded, say by some positive integer B, then the prime factors of the terms of the sequence (a_k) could only be primes less than or equal to $B + 2$ or those dividing a_1 or a_2, which contradicts the property in the statement of the problem. □

Consider now an arbitrary positive integer n. We assume $n > b_2$, otherwise we replace n by an arbitrary multiple of n that is bigger than b_2. By Claim 2, there exists k such that $b_{k+1} \geqslant n$. Consider the smallest such k. From Claim 1, it follows that we must have $b_k = n - 1$ and $b_{k+1} = n$ (we assumed $n > b_2$ to ensure that $k \geqslant 2$). We now find that

$$a_{k+1} = a_k \cdot \frac{b_k + 2}{b_{k+1}} = a_k \cdot \frac{n + 1}{n}.$$

Because n and $n + 1$ are coprime, this immediately implies that a_k is divisible by n.

Comment. For c a positive integer, the sequence $a_k = ck$ satisfies the conditions of the problem. Another example is

$$a_1 = 1, \quad a_2 = 2, \quad a_k = 3(k - 1) \text{ for } k \geqslant 3.$$

N6. A sequence of integers $a_0, a_1, a_2, \ldots$ is called *kawaii*, if $a_0 = 0, a_1 = 1$, and, for any positive integer n, we have

$$(a_{n+1} - 3a_n + 2a_{n-1})(a_{n+1} - 4a_n + 3a_{n-1}) = 0.$$

An integer is called *kawaii* if it belongs to a kawaii sequence.

Suppose that two consecutive positive integers m and $m+1$ are both kawaii (not necessarily belonging to the same kawaii sequence). Prove that 3 divides m, and that $m/3$ is kawaii.

(China)

Solution 1. We start by rewriting the condition in the problem as:

$$a_{n+1} = 3a_n - 2a_{n-1}, \text{ or } a_{n+1} = 4a_n - 3a_{n-1}.$$

We have $a_{n+1} \equiv a_n$ or $a_{n-1} \pmod 2$ and $a_{n+1} \equiv a_{n-1}$ or $a_n \pmod 3$ for all $n \geqslant 1$. Now, since $a_0 = 0$ and $a_1 = 1$, we have that $a_n \equiv 0, 1 \bmod 3$ for all $n \geqslant 0$. Since m and $m+1$ are kawaii integers, then necessarily $m \equiv 0 \bmod 3$.

We also observe that $a_2 = 3$ or $a_2 = 4$. Moreover,

(1) If $a_2 = 3$, then $a_n \equiv 1 \pmod 2$ for all $n \geqslant 1$ since $a_1 \equiv a_2 \equiv 1 \pmod 2$.

(2) If $a_2 = 4$, then $a_n \equiv 1 \pmod 3$ for all $n \geqslant 1$ since $a_1 \equiv a_2 \equiv 1 \pmod 3$.

Since $m \equiv 0 \pmod 3$, any kawaii sequence containing m does not satisfy (2), so it must satisfy (1). Hence, m is odd and $m+1$ is even.

Take a kawaii sequence (a_n) containing $m+1$. Let $t \geqslant 2$ be such that $a_t = m+1$. As (a_n) does not satisfy (1), it must satisfy (2). Then $a_n \equiv 1 \pmod 3$ for all $n \geqslant 1$. We define the sequence $a'_n = (a_{n+1} - 1)/3$. This is a kawaii sequence: $a'_0 = 0$, $a'_1 = 1$ and for all $n \geqslant 1$,

$$(a'_{n+1} - 3a'_n + 2a'_{n-1})(a'_{n+1} - 4a'_n + 3a'_{n-1}) = (a_{n+2} - 3a_{n+1} + 2a_n)(a_{n+2} - 4a_{n+1} + 3a_n)/9 = 0.$$

Finally, we notice that the term $a'_{t-1} = m/3$ which implies that $m/3$ is kawaii.

Solution 2. We start by proving the following:

Claim 1. We have $a_n \equiv 0, 1 \bmod 3$ for all $n \geqslant 0$.

Proof. We have $a_{n+1} = 3a_n - 2a_{n-1} = 3(a_n - a_{n-1}) + a_{n-1}$ or $a_{n+1} = 4a_n - 3a_{n-1} = 3(a_n - a_{n-1}) + a_n$, so $a_{n+1} \equiv a_n$ or $a_{n-1} \bmod 3$, and since $a_0 = 0$ and $a_1 = 1$ the result follows. □

Hence if m and $m+1$ are kawaii, then necessarily $m \equiv 0 \bmod 3$.

Claim 2. An integer $\geqslant 2$ is kawaii if and only if it can be written as $1 + b_2 + \cdots + b_n$ for some $n \geqslant 2$ with $b_i = 2^{r_i}3^{s_i}$ satisfying $r_i + s_i = i - 1$ for $i = 2, \ldots, n$ and $b_i \mid b_{i+1}$ for all $i = 2, \ldots, n-1$.

Proof. For a kawaii sequence (a_n), we can write $a_{n+1} = 3a_n - 2a_{n-1} = a_n + 2(a_n - a_{n-1})$ or $a_{n+1} = 4a_n - 3a_{n-1} = a_n + 3(a_n - a_{n-1})$, so $a_{n+1} - a_n = 2(a_n - a_{n-1})$ or $3(a_n - a_{n-1})$. Hence, $a_n = 1 + b_2 + \cdots + b_n$ where $b_2 = 2$ or 3 and $b_{i+1} = 2b_i$ or $3b_i$.

Conversely, given a number that can be written in that way, we consider any sequence given by $a_0 = 0$, $a_1 = 1$ and $a_i = 1 + b_2 + \cdots + b_i$ for $2 \leqslant i \leqslant n$ and a_i given by the kawaii condition for $i \geqslant n+1$. This defines a kawaii sequence containing the given number as a_n. □

Let us suppose that m and $m+1$ are kawaii, then they belong to some kawaii sequences and we can write them as in Claim 2 as $m = 1+2+\cdots+2^\ell+2^\ell\cdot 3\cdot A$ and $m+1 = 1+2+\cdots+2^{\ell'}+2^{\ell'}\cdot 3\cdot A'$ where ℓ is odd and ℓ' is even because of modulo 3 reasons. Since $m+1 \equiv m \pmod{2^{\min(\ell,\ell')}}$, we have $\min(\ell, \ell') = 0$, so $\ell' = 0$.

Then $m+1 = 1 + b_2 + \cdots + b_j$ for some b_i's as in Claim 2 with $b_2 = 3$ and $b_i \mid b_{i+1}$: so with $3 \mid b_i$ for all $i = 2, \ldots, j$. Then $\dfrac{m}{3} = 1 + b'_1 + \cdots + b'_{j-1}$ with $b'_i = \dfrac{b_{i+1}}{3}$ as in Claim 2 and $\dfrac{m}{3}$ is a kawaii integer.

Solution 3. (This solution is just a different combination of the ideas in Solutions 1 and 2) We first prove that, in a kawaii sequence $a_0, a_1, a_2, \ldots$, every term a_t with $t \geqslant 0$ is congruent to 0 or 1 modulo 3.

For $n \geqslant 1$, put $b_n = a_n - a_{n-1}$. We have

$$a_t = a_0 + \sum_{k=1}^{t} (a_k - a_{k-1}) = \sum_{k=1}^{t} b_k. \qquad (*)$$

Note that

$$a_{n+1} - 3a_n + 2a_{n-1} = b_{n+1} - 2b_n \quad \text{and} \quad a_{n+1} - 4a_n + 3a_{n-1} = b_{n+1} - 3b_n.$$

The conditions on the b_i's for defining a kawaii sequence are

$$b_1 = a_1 - a_0 = 1, \quad \text{and} \quad \frac{b_{n+1}}{b_n} \in \{2, 3\} \quad \text{for} \quad n \geqslant 1.$$

1. If we have $\dfrac{b_{n+1}}{b_n} = 2$ for any n with $1 \leqslant n \leqslant t-1$, then $(*)$ implies that

$$a_t = \sum_{k=1}^{t} 2^{k-1} = 2^t - 1 \equiv 0, 1 \pmod 3.$$

2. If there exists some integer s with $2 \leqslant s \leqslant t-1$ such that

$$\frac{b_2}{b_1} = \frac{b_3}{b_2} = \cdots = \frac{b_s}{b_{s-1}} = 2, \quad \frac{b_{s+1}}{b_s} = 3,$$

it implies that $3 \mid b_n$ for any $n \geqslant s+1$. Similarly to the argument in (1), we obtain $a_t \equiv \sum_{k=1}^{s} b_k \equiv 0, 1 \pmod 3$.

3. If $\dfrac{b_2}{b_1} = b_2 = 3$, we have $3 \mid b_n$ for any $n \geqslant 2$, and hence $a_t \equiv 1 \pmod 3$.

Combining these, we have proved that $a_t \equiv 0, 1 \pmod 3$.

We next prove that no positive kawaii integer is divisible by both 2 and 3. If $b_2 = 2$ for some kawaii sequence, then $2 \mid b_n$ and $a_n \equiv 1 \pmod 2$ for all $n \geqslant 2$ in it. If $b_2 = 3$ in some kawaii sequence, then $3 \mid b_n$ and $a_n \equiv 1 \pmod 3$ for all $n \geqslant 2$ in it.

Now, consider the original problem. Since m and $m+1$ are both kawaii integer, it means

$$m \equiv 0, 1 \pmod 3, \quad \text{and} \quad m+1 \equiv 0, 1 \pmod 3,$$

and hence we easily obtain $3 \mid m$. Since a kawaii integer m is divisible by 3, m must be odd, and hence $m+1$ is even. Take a kawaii sequence $a_0, a_1, a_2, \ldots$ containing $m+1$ as a_t. The fact that $m+1$ is even implies that $b_2 = 3$ and so $3 \mid b_n$ for all $n \geqslant 2$ in this sequence. Set $b'_n = \dfrac{b_{n+1}}{3}$ for $n \geqslant 1$. Thus $b'_1 = \dfrac{b_2}{3} = 1$, and $\dfrac{b'_{n+1}}{b'_n} = \dfrac{b_{n+2}}{b_{n+1}} \in \{2, 3\}$ for all $n \geqslant 1$. Define $a'_0 = 0$ and $a'_n = \sum_{k=1}^{n} b'_k$ for $n \geqslant 1$, then $a'_0, a'_1, a'_2, \ldots$ is a kawaii sequence. Now,

$$a'_{t-1} = \sum_{k=1}^{t-1} b'_k = \frac{1}{3} \sum_{k=2}^{t} b_k = \frac{1}{3} \left(-b_1 + \sum_{k=1}^{t} b_k \right) = \frac{1}{3} (-1 + a_t) = \frac{m}{3}.$$

This means that $\dfrac{m}{3}$ is a kawaii integer.

Comment. There are infinitely many positive integers m such that $m, m+1, m/3$ are kawaii. To see this, let $k \geqslant 1$ be a kawaii integer. Then $2k+1$ and $3k+1$ are kawaii by Claim 2 in Solution 2, and $3(2k+1)+1 = 6k+4$ and $2(3k+1)+1 = 6k+3$ are also kawaii.

N7. Let a, b, c, d be positive integers satisfying

$$\frac{ab}{a+b} + \frac{cd}{c+d} = \frac{(a+b)(c+d)}{a+b+c+d}.$$

Determine all possible values of $a+b+c+d$.

(Netherlands)

Answer: The possible values are the positive integers that are not square-free.

Solution.

First, note that if we take $a = \ell$, $b = k\ell$, $c = k\ell$, $d = k^2\ell$ for some positive integers k and ℓ, then we have

$$\frac{ab}{a+b} + \frac{cd}{c+d} = \frac{k\ell^2}{\ell + k\ell} + \frac{k^3\ell^2}{k\ell + k^2\ell} = \frac{k\ell}{k+1} + \frac{k^2\ell}{k+1} = k\ell$$

and

$$\frac{(a+b)(c+d)}{a+b+c+d} = \frac{(\ell + k\ell)(k\ell + k^2\ell)}{\ell + k\ell + k\ell + k^2\ell} = \frac{k(k+1)^2\ell^2}{\ell(k+1)^2} = k\ell,$$

so that

$$\frac{ab}{a+b} + \frac{cd}{c+d} = k\ell = \frac{(a+b)(c+d)}{a+b+c+d}.$$

This means that $a+b+c+d = \ell(1+2k+k^2) = \ell(k+1)^2$ can be attained. We conclude that all non-square-free positive integers can be attained.

Now, we will show that if

$$\frac{ab}{a+b} + \frac{cd}{c+d} = \frac{(a+b)(c+d)}{a+b+c+d}$$

then $a+b+c+d$ is not square-free. We argue by contradiction. Suppose that $a+b+c+d$ is square-free, and note that after multiplying by $(a+b)(c+d)(a+b+c+d)$, we obtain

$$(ab(c+d) + cd(a+b))(a+b+c+d) = (a+b)^2(c+d)^2. \tag{1}$$

A prime factor of $a+b+c+d$ must divide $a+b$ or $c+d$, and therefore divides both $a+b$ and $c+d$. Because $a+b+c+d$ is square-free, the fact that every prime factor of $a+b+c+d$ divides $a+b$ implies that $a+b+c+d$ itself divides $a+b$. Because $a+b < a+b+c+d$, this is impossible. So $a+b+c+d$ cannot be square-free.

Comment 1. Another way to conclude after obtaining (1) is by observing that

$$(a+b)^2(c+d)^2 \equiv (a+b)^4 \pmod{a+b+c+d}$$

Hence $a+b+c+d \mid (a+b)^4$. But if $a+b+c+d$ is square-free, this forces $a+b+c+d \mid a+b$, which is clearly a contradiction.

Comment 2. It seems difficult to characterise all quadruples (a, b, c, d) that satisfy the equality in the problem. Many of them, including those used in the solution, are of the general form $(a, b, c, d) = (xy^2, xyz, xyz, xz^2)$ for some positive integers x, y, and z, but there are more solutions than that, such as $(a, b, c, d) = (2, 7, 8, 10)$ or $(a, b, c, d) = (13, 14, 16, 38)$.

N8. Let $\mathbb{Z}_{>0}$ be the set of positive integers. Determine all functions $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that

$$f^{bf(a)}(a+1) = (a+1)f(b)$$

holds for all $a, b \in \mathbb{Z}_{>0}$, where $f^k(n) = f(f(\cdots f(n)\cdots))$ denotes the composition of f with itself k times.

(Taiwan)

Answer: The only function satisfying the condition is $f(n) = n+1$ for all $n \in \mathbb{Z}_{>0}$.

Let $P(a,b)$ be the equality in the statement.

Solution 1. We divide the solution into 5 steps.

Step 1. (f is injective)

Claim 1. For any $a \geqslant 2$, the set $\{f^n(a) \mid n \in \mathbb{Z}_{>0}\}$ is infinite.

Proof. First, we have $f^{f(a)}(a+1) \overset{P(a,1)}{=} (a+1)f(1)$. Varying a, we see that $f(\mathbb{Z}_{>0})$ is infinite. Next, we have $f^{bf(a-1)}(a) \overset{P(a-1,b)}{=} af(b)$. So, varying b, $f^{bf(a-1)}(a)$ takes infinitely many values. □

Claim 2. For any $a \geqslant 2$ and $n \in \mathbb{Z}_{>0}$, we have $f^n(a) \neq a$.

Proof. Otherwise we would get a contradiction with Claim 1. □

Assume $f(b) = f(c)$ for some $b < c$. Then we have

$$\begin{aligned}(a+1)f(c) &\overset{P(a,c)}{=} f^{cf(a)}(a+1)\\ &= f^{(c-b)f(a)}\big(f^{bf(a)}(a+1)\big)\\ &\overset{P(a,b)}{=} f^{(c-b)f(a)}\big((a+1)f(b)\big)\\ &= f^{(c-b)f(a)}\big((a+1)f(c)\big),\end{aligned}$$

which contradicts Claim 2. So, f is injective.

Step 2. ($f(\mathbb{Z}_{>0}) = \mathbb{Z}_{\geqslant 2}$)

Claim 3. 1 is not in the range of f.

Proof. If $f(b) = 1$, then $f^{f(a)}(a+1) = a+1$ by $P(a,1)$, which contradicts Claim 2. □

We say that a is a *descendant* of b if $f^n(b) = a$ for some $n \in \mathbb{Z}_{>0}$.

Claim 4. For any $a, b \geqslant 1$, both of the following cannot happen at the same time:

- a is a descendant of b;
- b is a descendant of a.

Proof. If both of the above hold, then $a = f^m(b)$ and $b = f^n(a)$ for some $m, n \in \mathbb{Z}_{>0}$. Then $a = f^{m+n}(a)$, which contradicts Claim 2. □

Claim 5. For any $a, b \geqslant 2$, exactly one of the following holds:

- a is a descendant of b;
- b is a descendant of a;
- $a = b$.

Proof. For any $c \geqslant 2$, taking $m = f^{cf(a-1)-1}(a)$ and $n = f^{cf(b-1)-1}(b)$, we have

$$f(m) = f^{cf(a-1)}(a) \overset{P(a-1,c)}{=} af(c) \text{ and } f(n) = f^{cf(b-1)}(b) \overset{P(b-1,c)}{=} bf(c).$$

Hence

$$f^{nf(a-1)}(a) \overset{P(a-1,n)}{=} af(n) = abf(c) = bf(m) \overset{P(a-1,m)}{=} f^{mf(b-1)}(b).$$

The assertion then follows from the injectivity of f and Claim 2. □

Now, we show that any $a \geqslant 2$ is in the range of f. Let $b = f(1)$. If $a = b$, then a is in the range of f. If $a \neq b$, either a is a descendant of b, or b is a descendant of a by Claim 5. If b is a descendant of a, then $b = f^n(a)$ for some $n \in \mathbb{Z}_{>0}$, so $1 = f^{n-1}(a)$. Then, by Claim 3, we have $n = 1$, so $1 = a$, which is absurd. So, a is a descendant of b. In particular, a is in the range of f. Thus, $f(\mathbb{Z}_{>0}) = \mathbb{Z}_{\geqslant 2}$.

Step 3. ($f(1) = 2$)

Claim 6. Let $a, n \geqslant 2$, then na is a descendant of a.

Proof. We write $n = f(m)$ by Step 2. We have $na = f(m)a \overset{P(a-1,m)}{=} f^{mf(a-1)}(a)$, which shows na is a descendant of a. □

By Claim 6, all even integers $\geqslant 4$ are descendants of 2. Hence $2 = f(2k+1)$ for some $k \geqslant 0$.

Next, we show $f(2k+1) \geqslant f(1)$, which implies $f(1) = 2$. It trivially holds if $k = 0$. If $k \geqslant 1$, let n be the integer such that $f^n(2) = 2k + 2$. For any $b > n/f(1)$, we have

$$f^{bf(1)-n}(2k+2) = f^{bf(1)}(2) \overset{P(1,b)}{=} 2f(b) \text{ and } f^{bf(2k+1)}(2k+2) \overset{P(2k+1,b)}{=} (2k+2)f(b).$$

By Claim 6, $(2k+2)f(b)$ is a descendant of $2f(b)$. By Claim 2, we have $bf(2k+1) > bf(1) - n$. By taking b large enough, we conclude $f(2k+1) \geqslant f(1)$.

Step 4. ($f(2) = 3$ and $f(3) = 4$) From $f(1) = 2$ and $P(1, b)$, we have $f^{2b}(2) = 2f(b)$. So taking $b = 1$, we obtain $f^2(2) = 2f(1) = 4$; and taking $b = f(2)$, we have $f^{2f(2)}(2) = 2f^2(2) = 8$. Hence, $f^{2f(2)-2}(4) = f^{2f(2)}(2) = 8$ and $f^{f(3)}(4) \overset{P(3,1)}{=} 8$ give $f(3) = 2f(2) - 2$.

Claim 7. For any $m, n \in \mathbb{Z}_{>0}$, if $f(m)$ divides $f(n)$, then $m \leqslant n$.

Proof. If $f(m) = f(n)$, the assertion follows from the injectivity of f. If $f(m) < f(n)$, by $P(a, m)$, $P(a, n)$ and Claim 6, we have that $f^{nf(a)}(a+1)$ is a descendant of $f^{mf(a)}(a+1)$ for any $a \in \mathbb{Z}_{>0}$. So $mf(a) < nf(a)$, and $m < n$. □

By Claim 7, every possible divisor of $f(2)$ is in $\{1, f(1) = 2, f(2)\}$. Thus $f(2)$ is an odd prime or $f(2) = 4$. Since $f^2(2) = 4$, we have $f(2) \neq 4$, and hence $f(2)$ is an odd prime. We set $p = f(2)$.

Now, $f(3) = 2f(2) - 2 = 2(p-1)$. Since $p - 1$ divides $f(3)$, we have $p - 1 \in \{1, f(1) = 2, f(2) = p\}$ by Claim 7, so $p - 1 = 2$. Thus, $f(2) = p = 3$ and $f(3) = 2(p-1) = 4$.

Step 5. ($f(n) = n + 1$)

Claim 8. For any $b \geqslant 1$, $f(2f(b) - 1) = 2b + 2$.

Proof. Since $f^2(2) = 4$, we have $f^{2b-2}(4) = f^{2b}(2) = 2f(b)$, so

$$f^{f(2f(b)-1)+2b-2}(4) = f^{f(2f(b)-1)}(2f(b)) \overset{P(2f(b)-1,1)}{=} 4f(b) \overset{P(3,b)}{=} f^{4b}(4),$$

which gives us $f(2f(b) - 1) = 2b + 2$. □

Finally, we prove $f(n) = n+1$ by induction on n. Suppose $f(n) = n+1$ for all $1 \leqslant n \leqslant 2b+1$. Replace b by $b+1$ in $f(2f(b)-1) = 2b+2$ to get

$$f(2b+3) = f(2f(b+1)-1) = 2(b+1)+2 = 2b+4.$$

By induction hypothesis, we have $f^b(b+2) = 2b+2$. Hence

$$f(f(2b+2)) = f^{b+2}(b+2) = f^{f(b+1)}(b+2) \overset{P(b+1,1)}{=} 2(b+2) = f(2b+3).$$

By injectivity, $f(2b+2) = 2b+3$. Then $f(n) = n+1$ for all $n \in \mathbb{Z}_{>0}$, which is indeed a solution.

Solution 2. In the same way as Steps 1-2 of Solution 1, we have that f is injective and $f(\mathbb{Z}_{>0}) = \mathbb{Z}_{\geqslant 2}$.

We first note that Claim 2 in Solution 1 is also true for $a = 1$.

Claim 2'. For any $a, n \in \mathbb{Z}_{>0}$, we have $f^n(a) \neq a$.

Proof. If $a \geqslant 2$, the assertion was proved in Claim 2 in Solution 1. If $a = 1$, we have that 1 is not in the range of f by Claim 3 in Solution 1. So, $f^n(1) \neq 1$ for every $n \in \mathbb{Z}_{>0}$. □

For any $a, b \in \mathbb{Z}_{>0}$, we have

$$f^{bf(f(a)-1)+1}(a) = f^{bf(f(a)-1)}(f(a)) \overset{P(f(a)-1,b)}{=} f(a)f(b).$$

Since the right-hand side is symmetric in a, b, we have

$$f^{bf(f(a)-1)+1}(a) = f(a)f(b) = f^{af(f(b)-1)+1}(b).$$

Since f is injective, we have $f^{bf(f(a)-1)}(a) = f^{af(f(b)-1)}(b)$. We set $g(n) = f(f(n)-1)$. Then we have $f^{bg(a)}(a) = f^{ag(b)}(b)$ for any $a, b \in \mathbb{Z}_{>0}$. We set $n_{a,b} = bg(a) - ag(b)$. Then, for sufficiently large n, we have $f^{n+n_{a,b}}(a) = f^n(b)$. For any $a, b, c \in \mathbb{Z}_{>0}$ and sufficiently large n, we have

$$f^{n+n_{a,b}+n_{b,c}+n_{c,a}}(a) = f^n(a).$$

By Claim 2' above, we have $n_{a,b} + n_{b,c} + n_{c,a} = 0$, so

$$(a-b)g(c) + (b-c)g(a) + (c-a)g(b) = 0.$$

Taking $(a, b, c) = (n, n+1, n+2)$, we have $g(n+1) - g(n) = g(n+2) - g(n+1)$. So, $\{g(n)\}_{n \geqslant 1}$ is an arithmetic progression.

There exist $C, D \in \mathbb{Z}$ such that $g(n) = f(f(n)-1) = Cn + D$ for all $n \in \mathbb{Z}_{>0}$. By Step 2 of Solution 1, we have $f(\mathbb{Z}_{>0}) = \mathbb{Z}_{\geqslant 2}$, so $C = 1$. Since $2 = \min_{n \in \mathbb{Z}_{>0}}\{f(f(n)-1)\}$, we have $D = 1$.

Thus, $g(n) = f(f(n)-1) = n+1$ for all $n \geqslant 1$. For any $a, b \in \mathbb{Z}_{>0}$, we have $f^{b(a+1)}(a) = f^{a(b+1)}(b)$. By the injectivity of f, we have $f^b(a) = f^a(b)$. For any $n \in \mathbb{Z}_{>0}$, taking $(a, b) = (1, n)$, we have $f^n(1) = f(n)$, so $f^{n-1}(1) = n$ again by the injectivity of f. For any $n \geqslant 1$, we have $f(n) = f(f^{n-1}(1)) = f^n(1) = n+1$.

Solution 3. The following is another way of finishing Solution 2 after Claim 2' and having introduced $g(n) = f(f(n)-1)$. For $a, b \in \mathbb{Z}_{>0}$ satisfying $b = f^k(a)$, we have

$$\begin{aligned} f^{bg(a)}(f(a)) &= f^{bf(f(a)-1)}(f(a)) \overset{P(f(a)-1,b)}{=} f(a)f(b) \overset{P(f(b)-1,a)}{=} f^{af(f(b)-1)}(f(b)) \\ &= f^{ag(b)+k}(f(a)). \end{aligned}$$

By Claim 2' in Solution 2, we have $bg(a) = ag(b)+k$, so $f^k(a)\cdot g(a) = a\cdot g(f^k(a))+k$. Therefore, for any $n \geqslant 0$, we put $a_n = f^n(1)$. We have

$$\begin{cases} a_{n+1}\cdot g(a_n) = a_n \cdot g(a_{n+1}) + 1 \\ a_{n+2}\cdot g(a_{n+1}) = a_{n+1} \cdot g(a_{n+2}) + 1 \\ a_{n+2}\cdot g(a_n) = a_n \cdot g(a_{n+2}) + 2. \end{cases}$$

Then we have

$$\begin{aligned} a_n \cdot a_{n+1} \cdot g(a_{n+2}) + 2a_{n+1} &= a_{n+1}\cdot a_{n+2}\cdot g(a_n) \\ &= a_{n+2}\big(a_n \cdot g(a_{n+1}) + 1\big) \\ &= a_n \cdot a_{n+2} \cdot g(a_{n+1}) + a_{n+2} \\ &= a_n\big(a_{n+1} \cdot g(a_{n+2}) + 1\big) + a_{n+2} \\ &= a_n \cdot a_{n+1} \cdot g(a_{n+2}) + a_n + a_{n+2}. \end{aligned}$$

From these, we have $2a_{n+1} = a_n + a_{n+2}$. Thus (a_n) is an arithmetic progression, and we have $a_n = f^n(1) = Cn + D$ for some $C, D \in \mathbb{Z}$.

By Step 2 in Solution 1, any $a \in \mathbb{Z}_{>0}$ is a descendant of 1, and $f(\mathbb{Z}_{>0}) = \mathbb{Z}_{\geqslant 2}$. Hence $D = 1$ and $C = 1$, and so $f^n(1) = n+1$. For any $n \geqslant 1$, we have $f^{n-1}(1) = n$, so $f(n) = f(f^{n-1}(1)) = f^n(1) = n+1$.

Solution 4.

We provide yet another (more technical) solution assuming Step 1 and Step 2 of Solution 1.

By Claim 5 in Solution 1, every $a \geqslant 2$ is a descendant of 1. Let g and h be the functions on $\mathbb{Z}_{\geqslant 2}$ such that $f^{g(a)}(1) = a$ and $h(a) = f(a-1)$. Then, $g\colon \mathbb{Z}_{\geqslant 2} \to \mathbb{Z}_{\geqslant 1}$ and $h\colon \mathbb{Z}_{\geqslant 2} \to \mathbb{Z}_{\geqslant 2}$ are bijections. The equation $P(a,b)$ can be rewritten as

$$g(ah(b)) = g(a) + (b-1)h(a).$$

Consider the set $S_a = g(a\cdot \mathbb{Z}_{>0})$. Since h is a bijection onto $\mathbb{Z}_{\geqslant 2}$, we have

$$S_a = g(a) + h(a)\cdot \mathbb{Z}_{\geqslant 0}.$$

Consider the intersection $S_a \cap S_b = S_{\mathrm{lcm}(a,b)}$. If we put $c = \mathrm{lcm}(a,b)$, this gives

$$(g(a) + h(a)\cdot \mathbb{Z}_{\geqslant 0}) \cap (g(b) + h(b)\cdot \mathbb{Z}_{\geqslant 0}) = g(c) + h(c)\cdot \mathbb{Z}_{\geqslant 0}.$$

Then we have $h(c) = \mathrm{lcm}(h(a), h(b))$ since the left hand side must be of the form $m + \mathrm{lcm}(h(a),h(b))\cdot \mathbb{Z}_{\geqslant 0}$ for some m.

If b is a multiple of a, then $\mathrm{lcm}(a,b) = b$, so $h(b) = \mathrm{lcm}(h(a),h(b))$, and hence $h(b)$ is a multiple of $h(a)$. Conversely, if $h(b)$ is a multiple of $h(a)$, then $h(b) = \mathrm{lcm}(h(a),h(b))$. On the other hand, we have $h(c) = \mathrm{lcm}(h(a),h(b))$. Since h is injective, we have $c = b$, so b is a multiple of a.

We apply the following claim for $H = h$.

Claim. Suppose $H\colon \mathbb{Z}_{\geqslant 2} \to \mathbb{Z}_{\geqslant 2}$ is a bijection such that a divides b if and only if $H(a)$ divides $H(b)$. Then:

1. $H(p)$ is prime if and only if p is prime;
2. $H(\prod_{i=1}^m p_i^{e_i}) = \prod_{i=1}^m H(p_i)^{e_i}$ i.e. H is completely multiplicative;
3. H preserves gcd and lcm.

Proof. We define $H(1) = 1$, and consider the bijection $H\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$. By the conditions on H, for any $n \in \mathbb{Z}_{\geqslant 2}$, n and $H(n)$ have the same number of divisors. Hence $H(p)$ is prime if and only if p is prime.

Since the only prime dividing $H(p^r)$ is $H(p)$, we have $H(p^r) = H(p)^s$ for some $s \geqslant 1$. Counting the number of divisors, we have $s = r$, so $H(p^r) = H(p)^r$ for any prime p and $r \geqslant 1$.

For $a, b \in \mathbb{Z}_{>0}$, recall that $\gcd(a, b)$ is a unique positive integer satisfying the following condition: for any $c \in \mathbb{Z}_{>0}$, c divides $\gcd(a, b)$ if and only if c divides both a and b. By the condition on H, for any $c \in \mathbb{Z}_{>0}$, $H(c)$ divides $H(\gcd(a, b))$ if and only if $H(c)$ divides both $H(a)$ and $H(b)$. Hence we have $H(\gcd(a, b)) = \gcd(H(a), H(b))$.

Similarly, we have $H(\operatorname{lcm}(a, b)) = \operatorname{lcm}(H(a), H(b))$. Hence we have

$$\begin{aligned} H\left(\prod_{i=1}^{m} p_i^{e_i}\right) &= H\big(\operatorname{lcm}(p_1^{e_1}, \ldots, p_r^{e_r})\big) = \operatorname{lcm}\big(H(p_1^{e_1}), \ldots, H(p_r^{e_r})\big) = \operatorname{lcm}\big(H(p_1)^{e_1}, \ldots, H(p_r)^{e_r}\big) \\ &= \prod_{i=1}^{m} H(p_i)^{e_i} \end{aligned}$$

since $H(p_i)$ and $H(p_j)$ are different primes for $i \neq j$. □

Take two primes $p \neq q$, and let x, y be positive integers such that

$$g(p) + (x-1)h(p) = g(q) + (y-1)h(q).$$

This is possible as $h(p)$ and $h(q)$ are two distinct primes. For every $k \geqslant 0$, by $P(p, x + kh(q))$ and $P(q, y + kh(p))$, we have

$$\begin{cases} g\big(p \cdot h(x + kh(q))\big) = g(p) + \big(x + kh(q) - 1\big)h(p), \\ g\big(q \cdot h(y + kh(p))\big) = g(q) + \big(y + kh(p) - 1\big)h(q), \end{cases}$$

where the right hand sides are equal. By the injectivity of g, we have

$$p \cdot h(x + kh(q)) = q \cdot h(y + kh(p)).$$

So, $h(y + kh(p))$ is divisible by p for all $k \geqslant 0$. By the above Claim, h preserves gcd, so

$$h(\gcd(y, h(p))) = \gcd(h(y), h(y + h(p)))$$

is divisible by p. Since $h(p)$ is a prime, y must be divisible by $h(p)$. Moreover, $h(h(p))$ is also a prime, so we have $h(h(p)) = p$. The function $h \circ h$ is completely multiplicative, so we have $h(h(n)) = n$ for every $n \geqslant 2$.

By $P(a, h(b))$ and $P(b, h(a))$, we have

$$\begin{cases} g(ab) = g\big(a \cdot h(h(b))\big) \overset{P(a,h(b))}{=} g(a) + \big(h(b) - 1\big)h(a), \\ g(ba) = g\big(b \cdot h(h(a))\big) \overset{P(b,h(a))}{=} g(b) + \big(h(a) - 1\big)h(b), \end{cases}$$

so

$$g(a) + h(a)\big(h(b) - 1\big) = g(ab) = g(b) + h(b)\big(h(a) - 1\big).$$

Hence $g(a) - h(a) = g(b) - h(b)$ for any $a, b \geqslant 2$, so $g - h$ is a constant function. By comparing the images of g and h, the difference is -1, i.e. $g(a) - h(a) = 1$ for any $a \geqslant 2$.

So, we have $g(h(a)) = h(h(a)) - 1 = a - 1$. By definition,

$$f(a-1) = h(a) = f^{g(h(a))}(1) = f^{a-1}(1).$$

By the injectivity of f, we have $f^{a-2}(1) = a - 1$ for every $a \geqslant 2$. From this, we can deduce inductively that $f(a) = a + 1$ for every $a \geqslant 1$.





OSLO2022

63rd International Mathematical Olympiad

SHORTLISTED PROBLEMS WITH SOLUTIONS

63rd International Mathematical Olympiad
Oslo, Norway, 6th–16th July 2022





SHORTLISTED PROBLEMS
WITH SOLUTIONS

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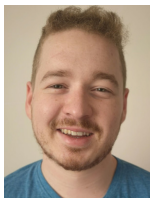
The Organising Committee and the Problem Selection Committee of IMO 2022 thank the following 50 countries for contributing 193 problem proposals (43 A, 59 C, 52 G, 39 N):

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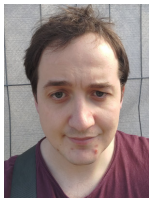

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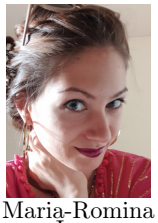

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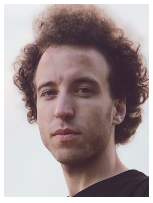

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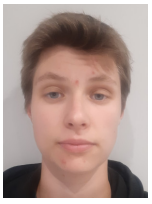

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Problems

Algebra

A1. Let $(a_n)_{n\geqslant 1}$ be a sequence of positive real numbers with the property that

$$(a_{n+1})^2 + a_n a_{n+2} \leqslant a_n + a_{n+2}$$

for all positive integers n. Show that $a_{2022} \leqslant 1$.

(Nigeria)

A2. Let $k \geqslant 2$ be an integer. Find the smallest integer $n \geqslant k+1$ with the property that there exists a set of n distinct real numbers such that each of its elements can be written as a sum of k other distinct elements of the set.

(Slovakia)

A3. Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Find all functions $f\colon \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that, for every $x \in \mathbb{R}_{>0}$, there exists a unique $y \in \mathbb{R}_{>0}$ satisfying

$$xf(y) + yf(x) \leqslant 2.$$

(Netherlands)

A4. Let $n \geqslant 3$ be an integer, and let $x_1, x_2, ..., x_n$ be real numbers in the interval $[0, 1]$. Let $s = x_1 + x_2 + \ldots + x_n$, and assume that $s \geqslant 3$. Prove that there exist integers i and j with $1 \leqslant i < j \leqslant n$ such that

$$2^{j-i} x_i x_j > 2^{s-3}.$$

(Trinidad and Tobago)

A5. Find all positive integers $n \geqslant 2$ for which there exist n real numbers $a_1 < \cdots < a_n$ and a real number $r > 0$ such that the $\frac{1}{2}n(n-1)$ differences $a_j - a_i$ for $1 \leqslant i < j \leqslant n$ are equal, in some order, to the numbers $r^1, r^2, \ldots, r^{\frac{1}{2}n(n-1)}$.

(Czech Republic)

A6. Let $\mathbb{R}$ be the set of real numbers. We denote by $\mathcal{F}$ the set of all functions $f\colon \mathbb{R} \to \mathbb{R}$ such that

$$f(x + f(y)) = f(x) + f(y)$$

for every $x, y \in \mathbb{R}$. Find all rational numbers q such that for every function $f \in \mathcal{F}$, there exists some $z \in \mathbb{R}$ satisfying $f(z) = qz$.

(Indonesia)

A7. For a positive integer n we denote by $s(n)$ the sum of the digits of n. Let $P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial, where $n \geqslant 2$ and a_i is a positive integer for all $0 \leqslant i \leqslant n-1$. Could it be the case that, for all positive integers k, $s(k)$ and $s(P(k))$ have the same parity?

(Belarus)

A8. For a positive integer n, an *n-sequence* is a sequence $(a_0, \ldots, a_n)$ of non-negative integers satisfying the following condition: if i and j are non-negative integers with $i + j \leqslant n$, then $a_i + a_j \leqslant n$ and $a_{a_i + a_j} = a_{i+j}$.

Let $f(n)$ be the number of n-sequences. Prove that there exist positive real numbers c_1, c_2 and λ such that

$$c_1 \lambda^n < f(n) < c_2 \lambda^n$$

for all positive integers n.

(Canada)

Combinatorics

C1. A ± 1-*sequence* is a sequence of 2022 numbers $a_1, \ldots, a_{2022}$, each equal to either $+1$ or -1. Determine the largest C so that, for any ± 1-sequence, there exists an integer k and indices $1 \leqslant t_1 < \ldots < t_k \leqslant 2022$ so that $t_{i+1} - t_i \leqslant 2$ for all i, and

$$\left| \sum_{i=1}^{k} a_{t_i} \right| \geqslant C.$$

(Czech Republic)

C2. The Bank of Oslo issues coins made out of two types of metal: aluminium (denoted A) and copper (denoted C). Morgane has n aluminium coins, and n copper coins, and arranges her $2n$ coins in a row in some arbitrary initial order. Given a fixed positive integer $k \leqslant 2n$, she repeatedly performs the following operation: identify the largest subsequence containing the k-th coin from the left which consists of consecutive coins made of the same metal, and move all coins in that subsequence to the left end of the row. For example, if $n = 4$ and $k = 4$, the process starting from the configuration $AACCCACA$ would be

$$AACCCACA \to CCCAAACA \to AAACCCCA \to CCCCAAAA \to \cdots .$$

Find all pairs (n, k) with $1 \leqslant k \leqslant 2n$ such that for every initial configuration, at some point of the process there will be at most one aluminium coin adjacent to a copper coin.

(France)

C3. In each square of a garden shaped like a 2022×2022 board, there is initially a tree of height 0. A gardener and a lumberjack alternate turns playing the following game, with the gardener taking the first turn:

- The gardener chooses a square in the garden. Each tree on that square and all the surrounding squares (of which there are at most eight) then becomes one unit taller.
- The lumberjack then chooses four different squares on the board. Each tree of positive height on those squares then becomes one unit shorter.

We say that a tree is *majestic* if its height is at least 10^6. Determine the largest number K such that the gardener can ensure there are eventually K majestic trees on the board, no matter how the lumberjack plays.

(Colombia)

C4.

Let $n > 3$ be a positive integer. Suppose that n children are arranged in a circle, and n coins are distributed between them (some children may have no coins). At every step, a child with at least 2 coins may give 1 coin to each of their immediate neighbours on the right and left. Determine all initial distributions of coins from which it is possible that, after a finite number of steps, each child has exactly one coin.

(Israel)

C5. Let $m, n \geqslant 2$ be integers, let X be a set with n elements, and let $X_1, X_2, \ldots, X_m$ be pairwise distinct non-empty, not necessary disjoint subsets of X. A function $f : X \to \{1, 2, \ldots, n+1\}$ is called *nice* if there exists an index k such that

$$\sum_{x \in X_k} f(x) > \sum_{x \in X_i} f(x) \quad \text{for all } i \neq k.$$

Prove that the number of nice functions is at least n^n.

(Germany)

C6. Let n be a positive integer. We start with n piles of pebbles, each initially containing a single pebble. One can perform moves of the following form: choose two piles, take an equal number of pebbles from each pile and form a new pile out of these pebbles. For each positive integer n, find the smallest number of non-empty piles that one can obtain by performing a finite sequence of moves of this form.

(Croatia)

C7. Lucy starts by writing s integer-valued 2022-tuples on a blackboard. After doing that, she can take any two (not necessarily distinct) tuples $\mathbf{v} = (v_1, \ldots, v_{2022})$ and $\mathbf{w} = (w_1, \ldots, w_{2022})$ that she has already written, and apply one of the following operations to obtain a new tuple:

$$\mathbf{v} + \mathbf{w} = (v_1 + w_1, \ldots, v_{2022} + w_{2022})$$
$$\mathbf{v} \vee \mathbf{w} = (\max(v_1, w_1), \ldots, \max(v_{2022}, w_{2022}))$$

and then write this tuple on the blackboard.

It turns out that, in this way, Lucy can write any integer-valued 2022-tuple on the blackboard after finitely many steps. What is the smallest possible number s of tuples that she initially wrote?

(Czech Republic)

C8.

Alice fills the fields of an $n \times n$ board with numbers from 1 to n^2, each number being used exactly once. She then counts the total number of good paths on the board. *A good path* is a sequence of fields of arbitrary length (including 1) such that:

(i) The first field in the sequence is one that is only adjacent to fields with larger numbers,

(ii) Each subsequent field in the sequence is adjacent to the previous field,

(iii) The numbers written on the fields in the sequence are in increasing order.

Two fields are considered adjacent if they share a common side. Find the smallest possible number of good paths Alice can obtain, as a function of n.

(Serbia)

C9. Let $\mathbb{Z}_{\geqslant 0}$ be the set of non-negative integers, and let $f : \mathbb{Z}_{\geqslant 0} \times \mathbb{Z}_{\geqslant 0} \to \mathbb{Z}_{\geqslant 0}$ be a bijection such that whenever $f(x_1, y_1) > f(x_2, y_2)$, we have $f(x_1 + 1, y_1) > f(x_2 + 1, y_2)$ and $f(x_1, y_1 + 1) > f(x_2, y_2 + 1)$.

Let N be the number of pairs of integers (x, y), with $0 \leqslant x, y < 100$, such that $f(x, y)$ is odd. Find the smallest and largest possible value of N.

(U.S.A.)

Geometry

G1. Let $ABCDE$ be a convex pentagon such that $BC = DE$. Assume there is a point T inside $ABCDE$ with $TB = TD$, $TC = TE$ and $\angle TBA = \angle AET$. Let lines CD and CT intersect line AB at points P and Q, respectively, and let lines CD and DT intersect line AE at points R and S, respectively. Assume that points P, B, A, Q and R, E, A, S respectively, are collinear and occur on their lines in this order. Prove that the points P, S, Q, R are concyclic.

(Slovakia)

G2. In the acute-angled triangle ABC, the point F is the foot of the altitude from A, and P is a point on the segment AF. The lines through P parallel to AC and AB meet BC at D and E, respectively. Points $X \neq A$ and $Y \neq A$ lie on the circles ABD and ACE, respectively, such that $DA = DX$ and $EA = EY$.

Prove that B, C, X and Y are concyclic.

(Netherlands)

G3. Let $ABCD$ be a cyclic quadrilateral. Assume that the points Q, A, B, P are collinear in this order, in such a way that the line AC is tangent to the circle ADQ, and the line BD is tangent to the circle BCP. Let M and N be the midpoints of BC and AD, respectively. Prove that the following three lines are concurrent: line CD, the tangent of circle ANQ at point A, and the tangent to circle BMP at point B.

(Slovakia)

G4. Let ABC be an acute-angled triangle with $AC > AB$, let O be its circumcentre, and let D be a point on the segment BC. The line through D perpendicular to BC intersects the lines AO, AC and AB at W, X and Y, respectively. The circumcircles of triangles AXY and ABC intersect again at $Z \neq A$.

Prove that if $OW = OD$, then DZ is tangent to the circle AXY.

(United Kingdom)

G5. Let ABC be a triangle, and let ℓ_1 and ℓ_2 be two parallel lines. For $i = 1, 2$, let ℓ_i meet the lines BC, CA, and AB at X_i, Y_i, and Z_i, respectively. Suppose that the line through X_i perpendicular to BC, the line through Y_i perpendicular to CA, and finally the line through Z_i perpendicular to AB, determine a non-degenerate triangle Δ_i.

Show that the circumcircles of Δ_1 and Δ_2 are tangent to each other.

(Vietnam)

G6. In an acute-angled triangle ABC, point H is the foot of the altitude from A. Let P be a moving point such that the bisectors k and ℓ of angles PBC and PCB, respectively, intersect each other on the line segment AH. Let k and AC meet at E, let ℓ and AB meet at F, and let EF and AH meet at Q. Prove that, as P varies, the line PQ passes through a fixed point.

(Iran)

G7. Let ABC and $A'B'C'$ be two triangles having the same circumcircle ω, and the same orthocentre H. Let Ω be the circumcircle of the triangle determined by the lines AA', BB' and CC'. Prove that H, the centre of ω, and the centre of Ω are collinear.

(Denmark)

G8. Let $AA'BCC'B'$ be a convex cyclic hexagon such that AC is tangent to the incircle of the triangle $A'B'C'$, and $A'C'$ is tangent to the incircle of the triangle ABC. Let the lines AB and $A'B'$ meet at X and let the lines BC and $B'C'$ meet at Y.

Prove that if $XBYB'$ is a convex quadrilateral, then it has an incircle.

(Australia)

Number Theory

N1. A number is called *Norwegian* if it has three distinct positive divisors whose sum is equal to 2022. Determine the smallest Norwegian number.

(Note: The total number of positive divisors of a Norwegian number is allowed to be larger than 3.)

(Cyprus)

N2. Find all positive integers $n > 2$ such that

$$n! \Bigg| \prod_{\substack{p<q\leqslant n,\\ p,q \text{ primes}}} (p+q).$$

(Nigeria)

N3. Let $a > 1$ be a positive integer, and let $d > 1$ be a positive integer coprime to a. Let $x_1 = 1$ and, for $k \geqslant 1$, define

$$x_{k+1} = \begin{cases} x_k + d & \text{if } a \text{ doesn't divide } x_k, \\ x_k/a & \text{if } a \text{ divides } x_k. \end{cases}$$

Find the greatest positive integer n for which there exists an index k such that x_k is divisible by a^n.

(Croatia)

N4. Find all triples of positive integers (a, b, p) with p prime and

$$a^p = b! + p.$$

(Belgium)

N5. For each $1 \leqslant i \leqslant 9$ and $T \in \mathbb{N}$, define $d_i(T)$ to be the total number of times the digit i appears when all the multiples of 1829 between 1 and T inclusive are written out in base 10.

Show that there are infinitely many $T \in \mathbb{N}$ such that there are precisely two distinct values among $d_1(T), d_2(T), \ldots, d_9(T)$.

(United Kingdom)

N6. Let Q be a set of prime numbers, not necessarily finite. For a positive integer n consider its prime factorisation; define $p(n)$ to be the sum of all the exponents and $q(n)$ to be the sum of the exponents corresponding only to primes in Q. A positive integer n is called *special* if $p(n) + p(n+1)$ and $q(n) + q(n+1)$ are both even integers. Prove that there is a constant $c > 0$ independent of the set Q such that for any positive integer $N > 100$, the number of special integers in $[1, N]$ is at least cN.

(For example, if $Q = \{3, 7\}$, then $p(42) = 3$, $q(42) = 2$, $p(63) = 3$, $q(63) = 3$, $p(2022) = 3$, $q(2022) = 1$.)

(Costa Rica)

N7. Let k be a positive integer and let S be a finite set of odd prime numbers. Prove that there is at most one way (modulo rotation and reflection) to place the elements of S around a circle such that the product of any two neighbors is of the form $x^2 + x + k$ for some positive integer x.

(U.S.A.)

N8. Prove that $5^n - 3^n$ is not divisible by $2^n + 65$ for any positive integer n.

(Belgium)

Solutions

Algebra

A1. Let $(a_n)_{n\geqslant 1}$ be a sequence of positive real numbers with the property that

$$(a_{n+1})^2 + a_n a_{n+2} \leqslant a_n + a_{n+2}$$

for all positive integers n. Show that $a_{2022} \leqslant 1$.

(Nigeria)

Solution. We begin by observing that $(a_{n+1})^2 - 1 \leqslant a_n + a_{n+2} - a_n a_{n+2} - 1$, which is equivalent to

$$(a_{n+1})^2 - 1 \leqslant (1 - a_n)(a_{n+2} - 1).$$

Suppose now that there exists a positive integer n such that $a_{n+1} > 1$ and $a_{n+2} > 1$. Since $(a_{n+1})^2 - 1 \leqslant (1 - a_n)(a_{n+2} - 1)$, we deduce that $0 < 1 - a_n < 1 < 1 + a_{n+2}$, thus $(a_{n+1})^2 - 1 < (a_{n+2} + 1)(a_{n+2} - 1) = (a_{n+2})^2 - 1$.

On the other hand, $(a_{n+2})^2 - 1 \leqslant (1 - a_{n+3})(a_{n+1} - 1) < (1 + a_{n+1})(a_{n+1} - 1) = (a_{n+1})^2 - 1$, a contradiction. We have shown that we cannot have two consecutive terms, except maybe a_1 and a_2, strictly greater than 1.

Finally, suppose $a_{2022} > 1$. This implies that $a_{2021} \leqslant 1$ and $a_{2023} \leqslant 1$. Therefore $0 < (a_{2022})^2 - 1 \leqslant (1 - a_{2021})(a_{2023} - 1) \leqslant 0$, a contradiction. We conclude that $a_{2022} \leqslant 1$.

A2. Let $k \geqslant 2$ be an integer. Find the smallest integer $n \geqslant k+1$ with the property that there exists a set of n distinct real numbers such that each of its elements can be written as a sum of k other distinct elements of the set.

(Slovakia)

Answer: $n = k + 4$.

Solution. First we show that $n \geqslant k+4$. Suppose that there exists such a set with n numbers and denote them by $a_1 < a_2 < \cdots < a_n$.

Note that in order to express a_1 as a sum of k distinct elements of the set, we must have $a_1 \geqslant a_2 + \cdots + a_{k+1}$ and, similarly for a_n, we must have $a_{n-k} + \cdots + a_{n-1} \geqslant a_n$. We also know that $n \geqslant k+1$.

If $n = k+1$ then we have $a_1 \geqslant a_2 + \cdots + a_{k+1} > a_1 + \cdots + a_k \geqslant a_{k+1}$, which gives a contradiction.

If $n = k+2$ then we have $a_1 \geqslant a_2 + \cdots + a_{k+1} \geqslant a_{k+2}$, that again gives a contradiction.

If $n = k+3$ then we have $a_1 \geqslant a_2 + \cdots + a_{k+1}$ and $a_3 + \cdots + a_{k+2} \geqslant a_{k+3}$. Adding the two inequalities we get $a_1 + a_{k+2} \geqslant a_2 + a_{k+3}$, again a contradiction.

It remains to give an example of a set with $k+4$ elements satisfying the condition of the problem. We start with the case when $k = 2l$ and $l \geqslant 1$. In that case, denote by $A_i = \{-i, i\}$ and take the set $A_1 \cup \cdots \cup A_{l+2}$, which has exactly $k+4 = 2l+4$ elements. We are left to show that this set satisfies the required condition.

Note that if a number i can be expressed in the desired way, then so can $-i$ by negating the expression. Therefore, we consider only $1 \leqslant i \leqslant l+2$.

If $i < l+2$, we sum the numbers from some $l-1$ sets A_j with $j \neq 1, i+1$, and the numbers $i+1$ and -1.

For $i = l+2$, we sum the numbers from some $l-1$ sets A_j with $j \neq 1, l+1$, and the numbers $l+1$ and 1.

It remains to a give a construction for odd $k = 2l+1$ with $l \geqslant 1$ (since $k \geqslant 2$). To that end, we modify the construction for $k = 2l$ by adding 0 to the previous set.

This is a valid set as 0 can be added to each constructed expression, and 0 can be expressed as follows: take the numbers $1, 2, -3$ and all the numbers from the remaining $l-1$ sets $A_4, A_5, \cdots, A_{l+2}$.

A3. Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Find all functions $f\colon \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that, for every $x \in \mathbb{R}_{>0}$, there exists a unique $y \in \mathbb{R}_{>0}$ satisfying

$$xf(y) + yf(x) \leqslant 2.$$

(Netherlands)

Answer: The function $f(x) = 1/x$ is the only solution.

Solution 1. First we prove that the function $f(x) = 1/x$ satisfies the condition of the problem statement. The AM-GM inequality gives

$$\frac{x}{y} + \frac{y}{x} \geqslant 2$$

for every $x, y > 0$, with equality if and only if $x = y$. This means that, for every $x > 0$, there exists a unique $y > 0$ such that

$$\frac{x}{y} + \frac{y}{x} \leqslant 2,$$

namely $y = x$.

Let now $f\colon \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ be a function that satisfies the condition of the problem statement. We say that a pair of positive real numbers (x, y) is *good* if $xf(y) + yf(x) \leqslant 2$. Observe that if (x, y) is good, then so is (y, x).

Lemma 1.0. If (x, y) is good, then $x = y$.

Proof. Assume that there exist positive real numbers $x \neq y$ such that (x, y) is good. The uniqueness assumption says that y is the unique positive real number such that (x, y) is good. In particular, (x, x) is not a good pair. This means that

$$xf(x) + xf(x) > 2$$

and thus $xf(x) > 1$. Similarly, (y, x) is a good pair , so (y, y) is not a good pair, which implies $yf(y) > 1$. We apply the AM-GM inequality to obtain

$$xf(y) + yf(x) \geqslant 2\sqrt{xf(y) \cdot yf(x)} = 2\sqrt{xf(x) \cdot yf(y)} > 2.$$

This is a contradiction, since (x, y) is a good pair.

By assumption, for any $x > 0$, there always exists a good pair containing x, however Lemma 1 implies that the only good pair that can contain x is (x, x), so

$$xf(x) \leqslant 1 \quad \Longleftrightarrow \quad f(x) \leqslant \frac{1}{x},$$

for every $x > 0$.

In particular, with $x = 1/f(t)$ for $t > 0$, we obtain

$$\frac{1}{f(t)} \cdot f\left(\frac{1}{f(t)}\right) \leqslant 1.$$

Hence

$$t \cdot f\left(\frac{1}{f(t)}\right) \leqslant tf(t) \leqslant 1.$$

We claim that $(t, 1/f(t))$ is a good pair for every $t > 0$. Indeed,

$$t \cdot f\left(\frac{1}{f(t)}\right) + \frac{1}{f(t)} f(t) = t \cdot f\left(\frac{1}{f(t)}\right) + 1 \leqslant 2.$$

Lemma 1 implies that $t = 1/f(t) \Longleftrightarrow f(t) = 1/t$ for every $t > 0$.

Solution 1.1. We give an alternative way to prove that $f(x) = 1/x$ assuming $f(x) \leqslant 1/x$ for every $x > 0$.

Indeed, if $f(x) < 1/x$ then for every $a > 0$ with $f(x) < 1/a < 1/x$ (and there are at least two of them), we have

$$af(x) + xf(a) < 1 + \frac{x}{a} < 2.$$

Hence (x, a) is a good pair for every such a, a contradiction. We conclude that $f(x) = 1/x$.

Solution 1.2. We can also conclude from Lemma 1 and $f(x) \leqslant 1/x$ as follows.

Lemma 2. The function f is decreasing.

Proof. Let $y > x > 0$. Lemma 1 says that (x, y) is not a good pair, but (y, y) is. Hence

$$xf(y) + yf(x) > 2 \geqslant 2yf(y) > yf(y) + xf(y),$$

where we used $y > x$ (and $f(y) > 0$) in the last inequality. This implies that $f(x) > f(y)$, showing that f is decreasing.

We now prove that $f(x) = 1/x$ for all x. Fix a value of x and note that for $y > x$ we must have $xf(x) + yf(x) > xf(y) + yf(x) > 2$ (using that f is decreasing for the first step), hence $f(x) > \frac{2}{x+y}$. The last inequality is true for every $y > x > 0$. If we fix x and look for the supremum of the expression $\frac{2}{x+y}$ over all $y > x$, we get

$$f(x) \geqslant \frac{2}{x + x} = \frac{1}{x}.$$

Since we already know that $f(x) \leqslant 1/x$, we conclude that $f(x) = 1/x$.

Solution 2.0. As in the first solution, we note that $f(x) = 1/x$ is a solution, and we set out to prove that it is the only one. We write $g(x)$ for the unique positive real number such that $(x, g(x))$ is a good pair. In this solution, we prove Lemma 2 without assuming Lemma 1.

Lemma 2. The function f is decreasing.

Proof. Consider $x < y$. It holds that $yf(g(y)) + g(y)f(y) \leqslant 2$. Moreover, because y is the only positive real number such that $(g(y), y)$ is a good pair and $x \neq y$, we have $xf(g(y)) + g(y)f(x) > 2$. Combining these two inequalities yields

$$xf(g(y)) + g(y)f(x) > 2 \geqslant yf(g(y)) + g(y)f(y),$$

or $f(g(y))(x - y) > g(y)(f(y) - f(x))$. Because $g(y)$ and $f(g(y))$ are both positive while $x - y$ is negative, it follows that $f(y) < f(x)$, showing that f is decreasing.

We now prove Lemma 1 using Lemma 2. Suppose that $x \neq y$ but $xf(y) + yf(x) \leqslant 2$. As in the first solution, we get $xf(x) + xf(x) > 2$ and $yf(y) + yf(y) > 2$, which implies $xf(x) + yf(y) > 2$. Now

$$xf(x) + yf(y) > 2 \geqslant xf(y) + yf(x)$$

implies $(x - y)(f(x) - f(y)) > 0$, which contradicts the fact that f is decreasing. So $y = x$ is the unique y such that (x, y) is a good pair, and in particular we have $f(x) \leqslant 1/x$.

We can now conclude the proof as in any of the Solutions 1.x.

Solution 3.0. As in the other solutions we verify that the function $f(x) = 1/x$ is a solution. We first want to prove the following lemma:

Lemma 3. For all $x \in \mathbb{R}_{>0}$ we actually have $xf(g(x)) + g(x)f(x) = 2$ (that is: the inequality is actually an equality).

Proof. We proceed by contradiction: Assume there exists some number $x > 0$ such that for $y = g(x)$ we have $xf(y) + yf(x) < 2$. Then for any $0 < \epsilon < \frac{2-xf(y)-yf(x)}{2f(x)}$ we have, by uniqueness of y, that $xf(y+\epsilon) + (y+\epsilon)f(x) > 2$. Therefore

$$\begin{aligned} f(y+\epsilon) &> \frac{2-(y+\epsilon)f(x)}{x} = \frac{2-yf(x)-\epsilon f(x)}{x} \\ &> \frac{2-yf(x)-\frac{2-xf(y)-yf(x)}{2}}{x} \\ &= \frac{2-xf(y)-yf(x)}{2x} + f(y) > f(y). \end{aligned} \tag{1}$$

Furthermore, for every such ϵ we have $g(y+\epsilon)f(y+\epsilon) + (y+\epsilon)f(g(y+\epsilon)) \leqslant 2$ and $g(y+\epsilon)f(y) + yf(g(y+\epsilon)) > 2$ (since $y \neq y+\epsilon = g(g(y+\epsilon))$). This gives us the two inequalities

$$f(g(y+\epsilon)) \leqslant \frac{2-g(y+\epsilon)f(y+\epsilon)}{y+\epsilon} \qquad \text{and} \qquad f(g(y+\epsilon)) > \frac{2-g(y+\epsilon)f(y)}{y}.$$

Combining these two inequalities and rearranging the terms leads to the inequality

$$2\epsilon < g(y+\epsilon)[(y+\epsilon)f(y) - yf(y+\epsilon)].$$

Moreover combining with the inequality (1) we obtain

$$2\epsilon < g(y+\epsilon)\left[(y+\epsilon)f(y) - y\left(\frac{2-xf(y)-yf(x)}{2x} + f(y)\right)\right] = g(y+\epsilon)\left[\epsilon f(y) - y\frac{2-xf(y)-yf(x)}{2x}\right].$$

We now reach the desired contradiction, since for ϵ sufficiently small we have that the left hand side is positive while the right hand side is negative.

With this lemma it then follows that for all $x, y \in \mathbb{R}_{>0}$ we have

$$xf(y) + yf(x) \geqslant 2,$$

since for $y = g(x)$ we have equality and by uniqueness for $y \neq g(x)$ the inequality is strict.

In particular for every $x \in \mathbb{R}_{>0}$ and for $y = x$ we have $2xf(x) \geqslant 2$, or equivalently $f(x) \geqslant 1/x$ for all $x \in \mathbb{R}_{>0}$. With this inequality we obtain for all $x \in \mathbb{R}_{>0}$

$$2 \geqslant xf(g(x)) + g(x)f(x) \geqslant \frac{x}{g(x)} + \frac{g(x)}{x} \geqslant 2,$$

where the first inequality comes from the problem statement. Consequently each of these inequalities must actually be an equality, and in particular we obtain $f(x) = 1/x$ for all $x \in \mathbb{R}_{>0}$.

Solution 4. Again, let us prove that $f(x) = 1/x$ is the only solution. Let again $g(x)$ be the unique positive real number such that $(x, g(x))$ is a good pair.

Lemma 4. The function f is strictly convex.

Proof. Consider the function $q_s(x) = f(x) + sx$ for some real number s. If f is not strictly convex, then there exist $u < v$ and $t \in (0,1)$ such that

$$f(tu + (1-t)v) \geqslant tf(u) + (1-t)f(v).$$

Hence

$$\begin{aligned} q_s(tu + (1-t)v) &\geqslant tf(u) + (1-t)f(v) + s(tu + (1-t)v) \\ &= tq_s(u) + (1-t)q_s(v). \end{aligned}$$

Let $w = tu+(1-t)v$ and consider the case $s = f(g(w))/g(w)$. For that particular choice of s, the function $q_s(x)$ has a unique minimum at $x = w$. However, since $q_s(w) \geqslant tq_s(u)+(1-t)q_s(v)$, it must hold $q_s(u) \leqslant q_s(w)$ or $q_s(v) \leqslant q_s(w)$, a contradiction.

Lemma 5. The function f is continuous.

Proof. Since f is strictly convex and defined on an open interval, it is also continuous.

As in Solution 1, we can prove that $f(x) \leqslant 1/x$. If $f(x) < 1/x$, then we consider the function $h(y) = xf(y) + yf(x)$ which is continuous. Since $h(x) < 2$, there exist at least two distinct $z \neq x$ such that $h(z) < 2$ giving that (x, z) is good pair for both values of z, a contradiction. We conclude that $f(x) = 1/x$ as desired.

Comment. Lemma 5 implies Lemma 3, using an argument similar as in the end of Solution 4.

A4. Let $n \geqslant 3$ be an integer, and let $x_1, x_2, ..., x_n$ be real numbers in the interval $[0, 1]$. Let $s = x_1 + x_2 + \ldots + x_n$, and assume that $s \geqslant 3$. Prove that there exist integers i and j with $1 \leqslant i < j \leqslant n$ such that

$$2^{j-i} x_i x_j > 2^{s-3}.$$

(Trinidad and Tobago)

Solution.

Let $1 \leqslant a < b \leqslant n$ be such that $2^{b-a} x_a x_b$ is maximal. This choice of a and b implies that $x_{a+t} \leqslant 2^t x_a$ for all $1 - a \leqslant t \leqslant b - a - 1$, and similarly $x_{b-t} \leqslant 2^t x_b$ for all $b - n \leqslant t \leqslant b - a + 1$. Now, suppose that $x_a \in \left(\frac{1}{2^{u+1}}, \frac{1}{2^u}\right]$ and $x_b \in \left(\frac{1}{2^{v+1}}, \frac{1}{2^v}\right]$, and write $x_a = 2^{-\alpha}$, $x_b = 2^{-\beta}$. Then

$$\sum_{i=1}^{a+u-1} x_i \leqslant 2^u x_a \left(\frac{1}{2} + \frac{1}{4} + \ldots + \frac{1}{2^{a+u-1}}\right) < 2^u x_a \leqslant 1,$$

and similarly,

$$\sum_{i=b-v+1}^{n} x_i \leqslant 2^v x_b \left(\frac{1}{2} + \frac{1}{4} + \ldots + \frac{1}{2^{n-b+v}}\right) < 2^v x_b \leqslant 1,$$

In other words, the sum of the x_i's for i outside of the interval $[a + u, b - v]$ is strictly less than 2. Since the total sum is at least 3, and each term is at most 1, it follows that this interval must have at least two integers. i.e., $a + u < b - v$. Thus, by bounding the sum of the x_i for $i \in [1, a + u] \cup [b - v, n]$ like above, and trivially bounding each $x_i \in (a + u, b - v)$ by 1, we obtain

$$s < 2^{u+1} x_a + 2^{v+1} x_b + ((b - v) - (a + u) - 1) = b - a + (2^{u+1-\alpha} + 2^{v+1-\beta} - (u + v + 1)).$$

Now recall $\alpha \in (u, u + 1]$ and $\beta \in (v, v + 1]$, so applying Bernoulli's inequality yields

$$2^{u+1-\alpha} + 2^{v+1-\beta} - u - v - 1 \leqslant (1 + (u + 1 - \alpha))) + (1 + (v + 1 - \beta)) - u - v - 1 = 3 - \alpha - \beta.$$

It follows that $s - 3 < b - a - \alpha - \beta$, and so

$$2^{s-3} < 2^{b-a-\alpha-\beta} = 2^{b-a} x_a x_b.$$

Comment 1. It is not hard to see that the inequality is tight. Suppose that $n = 2k + 1$, and we set $x_{k+1} = 1$, $x_k = x_{k+2} = \frac{1}{2} + \frac{1}{2^k}$, and $x_{k+1-t} = x_{k+1+t} = \frac{1}{2^t}$. Then $s = 3$ (so the right hand side is 1), and

$$\max_{i<j} 2^{j-i} x_i x_j = 2^2 \left(\frac{1}{2} + \frac{1}{2^k}\right)^2,$$

which can be made arbitrarily close to 1. Note that s can be made larger by putting in more 1-s in the middle (and this accommodates even n as well).

Comment 2. An alternative formulation of the problem is to show that there exists a positive constant c (or find the best such c) such that

$$\max_{i<j} 2^{j-i} x_i x_j > c 2^s.$$

The above shows that $c = 1/8$ is the best possible. A somewhat simpler ending to the proof can be given for $c = 1/32$.

End of solution for $c = 1/32$.

As in the original solution, we arrive at

$$s < 2^{u+1} x_a + 2^{v+1} x_b + ((b - v) - (a + u) - 1) = b - a + (2^{u+1-\alpha} + 2^{v+1-\beta} - (u + v + 1)).$$

Now $2^{b-a} x_a x_b \geqslant 2^{b-a} 2^{-u-1} 2^{-v-1}$, so it is enough to show $s - 5 < b - a - u - v - 2$, or $s < b - a - u - v + 3$. The fact that $u + 1 - \alpha < 1$ and $v + 1 - \beta < 1$ implies $2^{u+1-\alpha} + 2^{v+1-\beta} < 4$, and so $s < b - a + (2 + 2 - u - v - 1) = b - a - u - v - 3$.

A5. Find all positive integers $n \geqslant 2$ for which there exist n real numbers $a_1 < \cdots < a_n$ and a real number $r > 0$ such that the $\frac{1}{2}n(n-1)$ differences $a_j - a_i$ for $1 \leqslant i < j \leqslant n$ are equal, in some order, to the numbers $r^1, r^2, \ldots, r^{\frac{1}{2}n(n-1)}$.

(Czech Republic)

Answer: $n \in \{2, 3, 4\}$.

Solution. We first show a solution for each $n \in \{2, 3, 4\}$. We will later show the impossibility of finding such a solution for $n \geqslant 5$.

For $n = 2$, take for example $(a_1, a_2) = (1, 3)$ and $r = 2$.

For $n = 3$, take the root $r > 1$ of $x^2 - x - 1 = 0$ (the golden ratio) and set $(a_1, a_2, a_3) = (0, r, r + r^2)$. Then

$$(a_2 - a_1, a_3 - a_2, a_3 - a_1) = (r, r^2, r + r^2 = r^3).$$

For $n = 4$, take the root $r \in (1, 2)$ of $x^3 - x - 1 = 0$ (such a root exists because $1^3 - 1 - 1 < 0$ and $2^3 - 2 - 1 > 0$) and set $(a_1, a_2, a_3, a_4) = (0, r, r + r^2, r + r^2 + r^3)$. Then

$$(a_2 - a_1, a_3 - a_2, a_4 - a_3, a_3 - a_1, a_4 - a_2, a_4 - a_1) = (r, r^2, r^3, r^4, r^5, r^6).$$

For $n \geqslant 5$, we will proceed by contradiction. Suppose there exist numbers $a_1 < \cdots < a_n$ and $r > 1$ satisfying the conditions of the problem. We start with a lemma:

Lemma. We have $r^{n-1} > 2$.

Proof. There are only $n - 1$ differences $a_j - a_i$ with $j = i + 1$, so there exists an exponent $e \leqslant n$ and a difference $a_j - a_i$ with $j \geqslant i + 2$ such that $a_j - a_i = r^e$. This implies

$$r^n \geqslant r^e = a_j - a_i = (a_j - a_{j-1}) + (a_{j-1} - a_i) > r + r = 2r,$$

thus $r^{n-1} > 2$ as desired.

To illustrate the general approach, we first briefly sketch the idea behind the argument in the special case $n = 5$. In this case, we clearly have $a_5 - a_1 = r^{10}$. Note that there are 3 ways to rewrite $a_5 - a_1$ as a sum of two differences, namely

$$(a_5 - a_4) + (a_4 - a_1),\ (a_5 - a_3) + (a_3 - a_1),\ (a_5 - a_2) + (a_2 - a_1).$$

Using the lemma above and convexity of the function $f(n) = r^n$, we argue that those three ways must be $r^{10} = r^9 + r^1 = r^8 + r^4 = r^7 + r^6$. That is, the "large" exponents keep dropping by 1, while the "small" exponents keep increasing by $n - 2$, $n - 3$, ..., 2. Comparing any two such equations, we then get a contradiction unless $n \leqslant 4$.

Now we go back to the full proof for any $n \geqslant 5$. Denote $b = \frac{1}{2}n(n-1)$. Clearly, we have $a_n - a_1 = r^b$. Consider the $n - 2$ equations of the form:

$$a_n - a_1 = (a_n - a_i) + (a_i - a_1) \text{ for } i \in \{2, \ldots, n-1\}.$$

In each equation, one of the two terms on the right-hand side must be at least $\frac{1}{2}(a_n - a_1)$. But from the lemma we have $r^{b-(n-1)} = r^b / r^{n-1} < \frac{1}{2}(a_n - a_1)$, so there are at most $n - 2$ sufficiently large elements in $\{r^k \mid 1 \leqslant k < b\}$, namely $r^{b-1}, \ldots, r^{b-(n-2)}$ (note that r^b is already used for $a_n - a_1$). Thus, the "large" terms must be, in some order, precisely equal to elements in

$$L = \{r^{b-1}, \ldots, r^{b-(n-2)}\}.$$

Next we claim that the "small" terms in the $n - 2$ equations must be equal to the elements in

$$S = \{r^{b-(n-2)-\frac{1}{2}i(i+1)} \mid 1 \leqslant i \leqslant n-2\},$$

in the corresponding order (the largest "large" term with the smallest "small" term, etc.). Indeed, suppose that

$$r^b = a_n - a_1 = r^{b-i} + r^{\alpha_i} \text{ for } i \in \{1, \ldots, n-2\},$$

where $1 \leqslant \alpha_1 < \cdots < \alpha_{n-2} \leqslant b-(n-1)$. Since $r > 1$ and $f(r) = r^n$ is convex, we have

$$r^{b-1} - r^{b-2} > r^{b-2} - r^{b-3} > \ldots > r^{b-(n-3)} - r^{b-(n-2)},$$

implying

$$r^{\alpha_2} - r^{\alpha_1} > r^{\alpha_3} - r^{\alpha_2} > \ldots > r^{\alpha_{n-2}} - r^{\alpha_{n-3}}.$$

Convexity of $f(r) = r^n$ further implies

$$\alpha_2 - \alpha_1 > \alpha_3 - \alpha_2 > \ldots > \alpha_{n-2} - \alpha_{n-3}.$$

Note that $\alpha_{n-2} - \alpha_{n-3} \geqslant 2$: Otherwise we would have $\alpha_{n-2} - \alpha_{n-3} = 1$ and thus

$$r^{\alpha_{n-3}} \cdot (r-1) = r^{\alpha_{n-2}} - r^{\alpha_{n-3}} = r^{b-(n-3)} - r^{b-(n-2)} = r^{b-(n-2)} \cdot (r-1),$$

implying that $\alpha_{n-3} = b-(n-2)$, a contradiction. Therefore, we have

$$\begin{aligned} \alpha_{n-2} - \alpha_1 &= (\alpha_{n-2} - \alpha_{n-3}) + \cdots + (\alpha_2 - \alpha_1) \\ &\geqslant 2 + 3 + \cdots + (n-2) \\ &= \frac{1}{2}(n-2)(n-1) - 1 = \frac{1}{2}n(n-3). \end{aligned}$$

On the other hand, from $\alpha_{n-2} \leqslant b-(n-1)$ and $\alpha_1 \geqslant 1$ we get

$$\alpha_{n-2} - \alpha_1 \leqslant b - n = \frac{1}{2}n(n-1) - n = \frac{1}{2}n(n-3),$$

implying that equalities must occur everywhere and the claim about the small terms follows.

Now, assuming $n-2 \geqslant 2$, we have the two different equations:

$$r^b = r^{b-(n-2)} + r^{b-(n-2)-1} \text{ and } r^b = r^{b-(n-3)} + r^{b-(n-2)-3},$$

which can be rewritten as

$$r^{n-1} = r+1 \quad \text{and} \quad r^{n+1} = r^4 + 1. \tag{1}$$

Simple algebra now gives

$$r^4 + 1 = r^{n+1} = r^{n-1} \cdot r^2 = r^3 + r^2 \implies (r-1)(r^3 - r - 1) = 0.$$

Since $r \neq 1$, using Equation (1) we conclude $r^3 = r+1 = r^{n-1}$, thus $n = 4$, which gives a contradiction.

A6. Let $\mathbb{R}$ be the set of real numbers. We denote by $\mathcal{F}$ the set of all functions $f\colon \mathbb{R} \to \mathbb{R}$ such that

$$f(x + f(y)) = f(x) + f(y)$$

for every $x, y \in \mathbb{R}$. Find all rational numbers q such that for every function $f \in \mathcal{F}$, there exists some $z \in \mathbb{R}$ satisfying $f(z) = qz$.

(Indonesia)

Answer: The desired set of rational numbers is $\left\{\frac{n+1}{n} : n \in \mathbb{Z}, n \neq 0\right\}$.

Solution. Let Z be the set of all rational numbers q such that for every function $f \in \mathcal{F}$, there exists some $z \in \mathbb{R}$ satisfying $f(z) = qz$. Let further

$$S = \left\{\frac{n+1}{n} : n \in \mathbb{Z}, n \neq 0\right\}.$$

We prove that $Z = S$ by showing the two inclusions: $S \subseteq Z$ and $Z \subseteq S$.

We first prove that $S \subseteq Z$. Let $f \in \mathcal{F}$ and let $P(x, y)$ be the relation $f(x + f(y)) = f(x) + f(y)$. First note that $P(0, 0)$ gives $f(f(0)) = 2f(0)$. Then, $P(0, f(0))$ gives $f(2f(0)) = 3f(0)$. We claim that

$$f(kf(0)) = (k + 1)f(0)$$

for every integer $k \geqslant 1$. The claim can be proved by induction. The cases $k = 1$ and $k = 2$ have already been established. Assume that $f(kf(0)) = (k + 1)f(0)$ and consider $P(0, kf(0))$ which gives

$$f\big((k + 1)f(0)\big) = f(0) + f(kf(0)) = (k + 2)f(0).$$

This proves the claim. We conclude that $\frac{k+1}{k} \in Z$ for every integer $k \geqslant 1$. Note that $P(-f(0), 0)$ gives $f(-f(0)) = 0$. We now claim that

$$f(-kf(0)) = (-k + 1)f(0)$$

for every integer $k \geqslant 1$. The proof by induction is similar to the one above. We conclude that $\frac{-k+1}{-k} \in Z$ for every integer $k \geqslant 1$. This shows that $S \subseteq Z$.

We now prove that $Z \subseteq S$. Let p be a rational number outside the set S. We want to prove that p does not belong to Z. To that end, we construct a function $f \in \mathcal{F}$ such that $f(z) \neq pz$ for every $z \in \mathbb{R}$. The strategy is to first construct a function

$$g\colon [0, 1) \to \mathbb{Z}$$

and then define f as $f(x) = g(\{x\}) + \lfloor x \rfloor$. This function f belongs to $\mathcal{F}$. Indeed,

$$\begin{aligned} f(x + f(y)) &= g(\{x + f(y)\}) + \lfloor x + f(y) \rfloor \\ &= g(\{x + g(\{y\}) + \lfloor y \rfloor\}) + \lfloor x + g(\{y\}) + \lfloor y \rfloor \rfloor \\ &= g(\{x\}) + \lfloor x \rfloor + g(\{y\}) + \lfloor y \rfloor \\ &= f(x) + f(y), \end{aligned}$$

where we used that g only takes integer values.

Lemma 1. For every $\alpha \in [0, 1)$, there exists $m \in \mathbb{Z}$ such that

$$m + n \neq p(\alpha + n)$$

for every $n \in \mathbb{Z}$.

Proof. Note that if $p = 1$ the claim is trivial. If $p \neq 1$, then the claim is equivalent to the existence of an integer m such that

$$\frac{m - p\alpha}{p - 1}$$

is never an integer. Assume the contrary. That would mean that both

$$\frac{m - p\alpha}{p - 1} \quad \text{and} \quad \frac{(m + 1) - p\alpha}{p - 1}$$

are integers, and so is their difference. The latter is equal to

$$\frac{1}{p - 1}.$$

Since we assumed $p \notin S$, $1/(p - 1)$ is never an integer. This is a contradiction.

Define $g\colon [0, 1) \to \mathbb{Z}$ by $g(\alpha) = m$ for any integer m that satisfies the conclusion of Lemma 1. Note that $f(z) \neq pz$ if and and only if

$$g(\{z\}) + \lfloor z \rfloor \neq p(\{z\} + \lfloor z \rfloor)$$

The latter is guaranteed by the construction of the function g. We conclude that $p \notin Z$ as desired. This shows that $Z \subset S$.

A7. For a positive integer n we denote by $s(n)$ the sum of the digits of n. Let $P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$ be a polynomial, where $n \geqslant 2$ and a_i is a positive integer for all $0 \leqslant i \leqslant n-1$. Could it be the case that, for all positive integers k, $s(k)$ and $s(P(k))$ have the same parity?

(Belarus)

Answer: No. For any such polynomial there exists a positive integer k such that $s(k)$ and $s(P(k))$ have different parities.

Solution. With the notation above, we begin by choosing a positive integer t such that

$$10^t > \max\left\{\frac{100^{n-1}a_{n-1}}{(10^{\frac{1}{n-1}} - 9^{\frac{1}{n-1}})^{n-1}}, \frac{a_{n-1}}{9}10^{n-1}, \frac{a_{n-1}}{9}(10a_{n-1})^{n-1}, \ldots, \frac{a_{n-1}}{9}(10a_0)^{n-1}\right\}.$$

As a direct consequence of 10^t being bigger than the first quantity listed in the above set, we get that the interval

$$I = \left[\left(\frac{9}{a_{n-1}}10^t\right)^{\frac{1}{n-1}}, \left(\frac{1}{a_{n-1}}10^{t+1}\right)^{\frac{1}{n-1}}\right)$$

contains at least 100 consecutive positive integers.

Let X be a positive integer in I such that X is congruent to 1 mod 100. Since $X \in I$ we have

$$9 \cdot 10^t \leqslant a_{n-1}X^{n-1} < 10^{t+1},$$

thus the first digit (from the left) of $a_{n-1}X^{n-1}$ must be 9.

Next, we observe that $a_{n-1}(10a_i)^{n-1} < 9 \cdot 10^t \leqslant a_{n-1}X^{n-1}$, thus $10a_i < X$ for all i, which immediately implies that $a_0 < a_1X < \cdots < a_nX^n$, and the number of digits of this strictly increasing sequence forms a strictly increasing sequence too. In other words, if $i < j$, the number of digits of a_iX^i is less than the number of digits of a_jX^j.

Let α be the number of digits of $a_{n-1}X^{n-1}$, thus $10^{\alpha-1} \leqslant a_{n-1}X^{n-1} < 10^\alpha$. We are now going to look at $P(10^\alpha X)$ and $P(10^{\alpha-1}X)$ and prove that the sum of their digits has different parities. This will finish the proof since $s(10^\alpha X) = s(10^{\alpha-1}X) = s(X)$.

We have $P(10^\alpha X) = 10^{\alpha n}X^n + a_{n-1}10^{\alpha(n-1)}X^{n-1} + \cdots + a_0$, and since $10^{\alpha(i+1)} > 10^{\alpha i}a_{n-1}X^{n-1} > 10^{\alpha i}a_iX^i$, the terms $a_i10^{\alpha i}X^i$ do not interact when added; in particular, there is no carryover caused by addition. Thus we have $s(P(10^\alpha X)) = s(X^n) + s(a_{n-1}X^{n-1}) + \cdots + s(a_0)$.

We now look at $P(10^{\alpha-1}X) = 10^{(\alpha-1)n}X^n + a_{n-1}10^{(\alpha-1)(n-1)}X^{n-1} + \cdots + a_0$. Firstly, if $i < n-1$, then $a_{n-1}X^{n-1}$ has more digits than a_iX^i and $a_{n-1}X^{n-1} \geqslant 10a_iX^i$. It now follows that $10^{(\alpha-1)(i+1)+1} > 10^{(\alpha-1)i}a_{n-1}X^{n-1} \geqslant 10^{(\alpha-1)i+1}a_iX^i$, thus all terms $10^{(\alpha-1)i}a_iX^i$ for $0 \leqslant i \leqslant n-1$ come in 'blocks', exactly as in the previous case.

Finally, $10^{(\alpha-1)n+1} > 10^{(\alpha-1)(n-1)}a_{n-1}X^{n-1} \geqslant 10^{(\alpha-1)n}$, thus $10^{(\alpha-1)(n-1)}a_{n-1}X^{n-1}$ has exactly $(\alpha-1)n+1$ digits, and its first digit is 9, as established above. On the other hand, $10^{(\alpha-1)n}X^n$ has exactly $(\alpha-1)n$ zeros, followed by 01 (as X is 1 mod 100). Therefore, when we add the terms, the 9 and 1 turn into 0, the 0 turns into 1, and nothing else is affected.

Putting everything together, we obtain

$$s(P(10^{\alpha-1}X)) = s(X^n) + s(a_{n-1}X^{n-1}) + \cdots + s(a_0) - 9 = s(P(10^\alpha X)) - 9,$$

thus $s(P(10^\alpha X))$ and $s(P(10^{\alpha-1}X))$ have different parities, as claimed.

A8. For a positive integer n, an *n-sequence* is a sequence $(a_0, \ldots, a_n)$ of non-negative integers satisfying the following condition: if i and j are non-negative integers with $i + j \leqslant n$, then $a_i + a_j \leqslant n$ and $a_{a_i+a_j} = a_{i+j}$.

Let $f(n)$ be the number of n-sequences. Prove that there exist positive real numbers c_1, c_2 and λ such that

$$c_1\lambda^n < f(n) < c_2\lambda^n$$

for all positive integers n.

(Canada)

Answer: Such constants exist with $\lambda = 3^{1/6}$; we will discuss appropriate values of c_1 and c_2 in the solution below.

Solution. In order to solve this, we will give a complete classification of n-sequences.

Let $k = \lfloor n/2 \rfloor$. We will say that an n-sequence is *large* if $a_i > k$ for some i, and *small* if no such i exists. For now we will assume that (a_i) is not the identity sequence (in other words, that $a_i \neq i$ for some i).

Lemma 1. If $a_r = a_s$ and $r, s < n$, then $a_{r+1} = a_{s+1}$.

Proof. We have $a_{r+1} = a_{a_r+a_1} = a_{a_s+a_1} = a_{s+1}$.

Lemma 2. If $i \leqslant k$, then $a_i \leqslant k$.

Proof. We have $i + i \leqslant n$, so $a_i + a_i \leqslant n$ whence $a_i \leqslant k$.

Lemma 3. There exist r, s such that $a_r = a_s$ and $r \neq s$.

Proof. If $a_0 \neq 0$ then $a_{2a_0} = a_0$. Otherwise, $a_{a_i} = a_i$ for all i, so take i such that $a_i \neq i$ (which we can do by our earlier assumption).

Lemma 4. Let r be the smallest index such that $a_s = a_r$ for some $s > r$, and let d be the minimum positive integer such that $a_{r+d} = a_r$. Then

1. The subsequence $(a_r, a_{r+1}, \ldots, a_n)$ is periodic with minimal period d. That is, for $u < v$, we have $a_u = a_v$ if and only if $u, v \geqslant r$ and $d \mid v - u$.

2. $a_i = i$ for $i < r$ and $a_i \geqslant r$ for $i \geqslant r$.

In this case we say (a_i) has *period* d and *offset* r.

Proof. We prove each in turn:

1. The "if" implication is clear from Lemma 1. For the reverse direction, suppose $a_u = a_v$. Then there are integers $r \leqslant u_0, v_0 < r + d$ such that $d \mid u - u_0, v - v_0$, so $a_{u_0} = a_u = a_v = a_{v_0}$. If $u_0 < v_0$ then $a_{r+d+u_0-v_0} = a_{r+d} = a_r$, contradicting the minimality of d. There is a similar contradiction when $u_0 > v_0$. Thus $u_0 = v_0$, so $d \mid u - v$.

2. If $r = 0$ there is nothing to prove. Otherwise $a_0 = a_{2a_0}$ so $2a_0 = 0$. Then we have $a_{a_i} = a_i$ for all i, so $a_i = i$ for $i < r$.

Lemma 5. Either

1. $d \mid a_i - i$ for all i, or

2. $r = 0$ and $d \mid a_i - i - d/2$ for all i.

Proof. Note that Lemma 4 tells us that if $a_u = a_v$ then $d \mid u - v$. Since $a_{a_i+a_0} = a_i$ for all i, we have $d \mid a_i - i + a_0$. For $i = 0$, this means that $d \mid 2a_0$. If $d \mid a_0$ then that means that $d \mid a_i - i$ for all i. Otherwise, if $d \nmid a_0$, then $d \mid a_0 - d/2$ and thus $d \mid a_i - i - d/2$ for all i. In addition, part 2 of Lemma 4 says that we must have $r = 0$ in this case.

Lemma 6. If d is even and $d \mid a_0 - d/2$, then (a_i) is small. (Note that we must have $r = 0$ in this case.)

Proof. Note that if $d \leqslant k + 1$, then by Lemma 2, $(a_0, \ldots, a_{d-1})$ is a period for the sequence consisting of elements at most k, so (a_i) must be small. Now suppose $d > k + 1$. We show that $a_i \leqslant k$ for all i by induction. Note that Lemma 2 already establishes this for $i \leqslant k$. We must have $d \mid a_{d/2}$ and $a_{d/2} \leqslant k < d$ so $a_{d/2} = 0$. Thus, for $i > k$, if $a_j \leqslant k$ for $j < i$, then $a_{i-d/2} \leqslant k$, so $a_i = a_{(i-d/2)+d/2} = a_{a_{i-d/2}} \leqslant k$.

Lemma 7. If (a_i) is small, then $r + d \leqslant k + 1$.

Proof. Since (a_i) is small, there exists $u, v \leqslant k + 1$ such that $u < v$ and $a_u = a_v$. Thus $u \leqslant r$ and $d \mid v - u$, so $r + d \leqslant v \leqslant k + 1$.

Lemma 8. If (a_i) is large, then $r + d > k + 1$ and $a_i = i$ for all $0 \leqslant i < r + d$.

Proof. Since (a_i) is large and has period d and offset r, the period $(a_r, \ldots, a_{r+d-1})$ must have an element that is larger than k, so by Lemma 2 we must have $r + d - 1 > k$.

We already have $a_i = i$ for $i < r$. Now we show that $a_i = i$ for $r \leqslant i \leqslant k$. By Lemma 6 we have $d \mid a_i - i$ but $r \leqslant i \leqslant k$. Since $k - r + 1 > d$, this means that $a_i = i$ for $i \leqslant k$.

Finally, one can show inductively that $a_i = i$ for $k < i < r + d$. Indeed, if $a_j = j$ for all $j < i$, then $a_i \geqslant i$ (otherwise $a_i = a_j$ for some $j < i$, but then $r \leqslant j$ and $i < r + d$ means that $d \nmid j - i$.) However, $a_i + (n - i) = a_i + a_{n-i} \leqslant n$, so $a_i = i$.

Thus large sequences are determined by r and d. It is not hard to check that all sequences of the form $a_i = i$ for $i < r + d$ and with period d and offset r are n-sequences. There are $(n-k-1)(n+k+2)/2$ possible choices of (r, d) where $0 \leqslant r < n$, $d \geqslant 1$, and $k+1 < r+d \leqslant n$.

For small sequences, for a given period d and offset r, we need to choose the period $(a_r, \ldots, a_{r+d-1})$ satisfying $r \leqslant aj \leqslant k$ and $d \mid a_j - j$ for $r \leqslant j < r + d$. There are $g(k + 1 - r, d)$ such choices, where we define $g(x, d)$ to be $(p + 1)^q p^{d-q}$ with $p = \lfloor x/d \rfloor$ and $q = x - dp$.

Furthermore, if d is even then there are $g(k + 1, d)$ choices for the period $(a_0, \ldots, a_{d-1})$ satisfying $d \mid a_j - j - d/2$ for $j < d$. Again it is not hard to check that, once these choices are made, then the resulting sequence is an n-sequence.

Thus the total number of n-sequences is

$$f(n) = 1 + \frac{(n-k-1)(n+k+2)}{2} + \sum_{d'=1}^{\lfloor (k+1)/2 \rfloor} g(k+1, 2d') + \sum_{r=0}^{k} \sum_{d=1}^{k+1-r} g(k+1-r, d). \qquad (1)$$

Now, to show that $f(n) > c_1 \lambda^n$ for some c_1, we note that

$$f(n) > g\left(k+1, \left\lfloor \frac{k+1}{3} \right\rfloor\right) \geqslant 3^{\lfloor (k+1)/3 \rfloor} > 3^{n/6} - 1.$$

To show that $f(n) < c_2 \lambda^n$ for some c_2, it actually suffices to show that there is a positive real number c_3 such that for all positive integers x,

$$\sum_{d=1}^{x} g(x, d) \leqslant c_3 3^{x/3}.$$

In fact, the following lemma suffices, as it bounds the left hand side of the above inequality by a pair of geometric series with initial term $3^{x/3}$:

Lemma. For positive d, x, we have:

$$g(x,d) \leqslant \begin{cases} 3^{x/3}\left(\frac{64}{81}\right)^{x/3-d}, & \text{if } d \leqslant x/3; \\ 3^{x/3}\left(\frac{8}{9}\right)^{d-x/3}, & \text{if } d \geqslant x/3. \end{cases}$$

Proof. There are a few key observations needed, all of which are immediate from the definition:

- (x,d) is the maximum product of a sequence of d integers that sums to x.
- For any positive integer k, we have $g(kx,kd) = g(x,d)^k$.
- If $2d \leqslant x \leqslant 3d$, then $g(x,d) = 2^{3d-x}3^{x-2d}$. Likewise, if $3d \leqslant x \leqslant 4d$ then $g(x,d) = 3^{4d-x}4^{x-3d}$.

With these observations, if $d \leqslant x/3$, then

$$3^{4(x-3d)}g(3x,3d) \leqslant g(3x+12(x-3d), 3d+4(x-3d)) = g(15x-36d, 4x-9d)$$

To calculate $g(15x-36d, 4x-9d)$, note that

$$\begin{aligned} 3(4x-9d) &= 12x-27d \leqslant 15x-36d, \\ 4(4x-9d) &= 16x-36d \geqslant 15x-36d, \end{aligned}$$

so

$$g(15x-36d, 4x-9d) = 3^{4(4x-9d)-(15x-36d)}4^{(15x-36d)-3(4x-9d)} = 3^x 4^{3(x-3d)}.$$

Thus

$$g(x,d)^3 = g(3x,3d) = \frac{3^{4(x-3d)}g(3x,3d)}{3^{4(x-3d)}} \leqslant \frac{3^x 4^{3(x-3d)}}{3^{4(x-3d)}} = 3^x\left(\frac{64}{81}\right)^{x-3d},$$

which completes the proof of the first claim. Likewise, if $d \geqslant x/3$,

$$3^{2(3d-x)}g(3x,3d)3 \leqslant g(3x+6(3d-x), 3d+2(3d-x)) = g(18d-3x, 9d-2x).$$

Again we have

$$2(9d-2x) \leqslant 18-3x \leqslant 18-3x+3(3d-x) = 3(9d-2x),$$

so

$$g(18d-3x, 9d-2x) = 2^{3(9d-2x)-(18d-3x)}3^{(18d-3x)-2(9d-2x)} = 2^{3(3d-x)}3^x.$$

Thus

$$g(x,d)^3 = g(3x,3d) = \frac{2^{3(3d-x)}g(3x,3d)}{3^{2(3d-x)}} \leqslant \frac{2^{3(3d-x)3^x}}{3^{2(3d-x)}} = 3^x\left(\frac{8}{9}\right)^{3d-x},$$

from which the second claim follows.

Common remarks. The best possible value of c_1 is constrained by $n = 1$: in this case, $f(1) = 2$, which means that $c_1 > 2 \cdot 3^{-1/6} \approx 1.66537$.

With a careful analysis one can show that the best possible value of c_2 is $\frac{236567}{4930}3^{1/3} \approx 69.20662$.

Combinatorics

C1. A ± 1-*sequence* is a sequence of 2022 numbers $a_1, \ldots, a_{2022}$, each equal to either $+1$ or -1. Determine the largest C so that, for any ± 1-sequence, there exists an integer k and indices $1 \leqslant t_1 < \ldots < t_k \leqslant 2022$ so that $t_{i+1} - t_i \leqslant 2$ for all i, and

$$\left| \sum_{i=1}^{k} a_{t_i} \right| \geqslant C.$$

(*Czech Republic*)

Answer: The answer is $C = 506$.

Solution. First, we prove that this can always be achieved. Without loss of generality, suppose at least $\frac{2022}{2} = 1011$ terms of the ± 1-sequence are $+1$. Define a subsequence as follows: starting at $t = 0$, if $a_t = +1$ we always include a_t in the subsequence. Otherwise, we skip a_t if we can (i.e. if we included a_{t-1} in the subsequence), otherwise we include it out of necessity, and go to the next t. Clearly, this subsequence will include all $+1$s. Also, for each -1 included in the sequence, a -1 must have been skipped, so at most $\lfloor \frac{1011}{2} \rfloor = 505$ can be included. Hence the sum is at least $1011 - 505 = 506$, as desired.

Next, we prove that, for the ± 1-sequence

$$(\{-1\}, \{+1, +1\}, \{-1, -1\}, \{+1, +1\}, \ldots, \{+1, +1\}, \{-1, -1\}, \{+1\}),$$

each admissible subsequence a_{t_i} has $-506 \leqslant \sum_i a_{t_i} \leqslant 506$. We say that the terms inside each curly bracket is a *block*. In total, there are 1012 blocks - 506 of them hold $+1$-s, and 506 of them hold -1s. (The two blocks at each end hold 1 number each, each other block holds 2.)

Suppose an admissible subsequence includes terms from k blocks holding $+1$-s. Then, in each -1-pair in between the $+1$-pairs, the subsequence must also include at least one -1. There can be at most two $+1$s included from each $+1$-block, and at least one -1 must be included from each -1-block, so the sum is at most $2k - (k-1) = k + 1$.

For $k < 506$, this is at most 506. If $k = 506$, one of the $+1$-blocks must be the one at the end, meaning it can only include one $+1$, so that the maximum in this case is only k, not $k+1$, so in this case the sum is also at most 506.

Hence we have shown that for any admissible subsequence, $\sum_i a_{t_i} \leqslant 506$. Analogously we can show that $-506 \leqslant \sum_i a_{t_i}$, meaning that $C \leqslant 506$ as desired.

Comment. A possible reformulation of the problem is the following.

> 2022 buckets of water are arranged in a row, each coloured either red or blue. Sally the salmon plays a game in the following way: first, she chooses any bucket she likes to start in. Then, any number of times she may jump either to the next bucket in the row, or across it to land in the bucket after that. (She may not jump across more than one bucket.) At any point, she may finish the game. At that time, her score is the absolute value of the difference between the number of red and blue buckets she visited during the game. Determine the largest C so that no matter how the buckets are coloured, Sally can achieve a score of at least C.

C2. The Bank of Oslo issues coins made out of two types of metal: aluminium (denoted A) and copper (denoted C). Morgane has n aluminium coins, and n copper coins, and arranges her $2n$ coins in a row in some arbitrary initial order. Given a fixed positive integer $k \leqslant 2n$, she repeatedly performs the following operation: identify the largest subsequence containing the k-th coin from the left which consists of consecutive coins made of the same metal, and move all coins in that subsequence to the left end of the row. For example, if $n = 4$ and $k = 4$, the process starting from the configuration $AACCCACA$ would be

$$AACCCACA \to CCCAAACA \to AAACCCCA \to CCCCAAAA \to \cdots .$$

Find all pairs (n, k) with $1 \leqslant k \leqslant 2n$ such that for every initial configuration, at some point of the process there will be at most one aluminium coin adjacent to a copper coin.

(France)

Answer: All pairs (n, k) such that $n \leqslant k \leqslant \frac{3n+1}{2}$.

Solution. Define a *block* to be a maximal subsequence of consecutive coins made out of the same metal, and let M^b denote a block of b coins of metal M. The property that there is at most one aluminium coin adjacent to a copper coin is clearly equivalent to the configuration having two blocks, one consisting of all A-s and one consisting of all C-s.

First, notice that if $k < n$, the sequence $A^{n-1}C^{n-1}AC$ remains fixed under the operation, and will therefore always have 4 blocks. Next, if $k > \frac{3n+1}{2}$, let $a = k - n - 1, b = 2n - k + 1$. Then $k > 2a + b$, $k > 2b + a$, so the configuration $A^aC^bA^bC^a$ will always have four blocks:

$$A^aC^bA^bC^a \to C^aA^aC^bA^b \to A^bC^aA^aC^b \to C^bA^bC^aA^a \to A^aC^bA^bC^a \to \ldots$$

Therefore a pair (n, k) can have the desired property only if $n \leqslant k \leqslant \frac{3n+1}{2}$. We claim that all such pairs in fact do have the desired property. Clearly, the number of blocks in a configuration cannot increase, so whenever the operation is applied, it either decreases or remains constant. We show that unless there are only two blocks, after a finite amount of steps the number of blocks will decrease.

Consider an arbitrary configuration with $c \geqslant 3$ blocks. We note that as $k \geqslant n$, the leftmost block cannot be moved, because in this case all n coins of one type are in the leftmost block, meaning there are only two blocks. If a block which is not the leftmost or rightmost block is moved, its neighbor blocks will be merged, causing the number of blocks to decrease.

Hence the only case in which the number of blocks does not decrease in the next step is if the rightmost block is moved. If c is odd, the leftmost and the rightmost blocks are made of the same metal, so this would merge two blocks. Hence $c \geqslant 4$ must be even. Suppose there is a configuration of c blocks with the i-th block having size a_i so that the operation always moves the rightmost block:

$$A^{a_1} \ldots A^{a_{c-1}}C^{a_c} \to C^{a_c}A^{a_1} \ldots A^{a_{c-1}} \to A^{a_{c-1}}C^{a_c}A^{a_1} \ldots C^{a_{c-2}} \to \ldots$$

Because the rightmost block is always moved, $k \geqslant 2n + 1 - a_i$ for all i. Because $\sum a_i = 2n$, summing this over all i we get $ck \geqslant 2cn + c - \sum a_i = 2cn + c - 2n$, so $k \geqslant 2n + 1 - \frac{2n}{c} \geqslant \frac{3n}{2} + 1$. But this contradicts $k \leqslant \frac{3n+1}{2}$. Hence at some point the operation will not move the rightmost block, meaning that the number of blocks will decrease, as desired.

C3. In each square of a garden shaped like a 2022×2022 board, there is initially a tree of height 0. A gardener and a lumberjack alternate turns playing the following game, with the gardener taking the first turn:

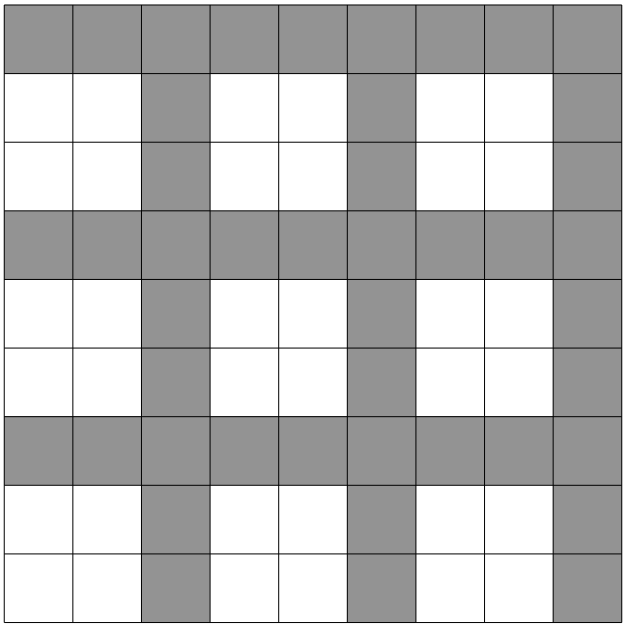
- The gardener chooses a square in the garden. Each tree on that square and all the surrounding squares (of which there are at most eight) then becomes one unit taller.
- The lumberjack then chooses four different squares on the board. Each tree of positive height on those squares then becomes one unit shorter.

We say that a tree is *majestic* if its height is at least 10^6. Determine the largest number K such that the gardener can ensure there are eventually K majestic trees on the board, no matter how the lumberjack plays.

(*Colombia*)

Answer: $K = 5 \cdot \frac{2022^2}{9} = 2271380$. In general, for a $3N \times 3N$ board, $K = 5N^2$.

Solution. We solve the problem for a general $3N \times 3N$ board. First, we prove that the lumberjack has a strategy to ensure there are never more than $5N^2$ majestic trees. Giving the squares of the board coordinates in the natural manner, colour each square where at least one of its coordinates are divisible by 3, shown below for a 9×9 board:



Then, as each 3×3 square on the board contains exactly 5 coloured squares, each move of the gardener will cause at most 4 trees on non-coloured squares to grow. The lumberjack may therefore cut those trees, ensuring no tree on a non-coloured square has positive height after his turn. Hence there cannot ever be more majestic trees than coloured squares, which is $5N^2$.

Next, we prove the gardener may ensure there are $5N^2$ majestic trees. In fact, we prove this statement in a modified game which is more difficult for the gardener: on the lumberjack's turn in the modified game, he may decrement the height of *all* trees on the board except those the gardener did not just grow, in addition to four of the trees the gardener just grew. Clearly, a sequence of moves for the gardener which ensures that there are K majestic trees in the modified game also ensures this in the original game.

Let $M = \binom{9}{5}$; we say that a *map* is one of the M possible ways to mark 5 squares on a 3×3 board. In the modified game, after the gardener chooses a 3×3 subboard on the board, the lumberjack chooses a map in this subboard, and the total result of the two moves is that each tree marked on the map increases its height by 1, each tree in the subboard which is not in the map remains unchanged, and each tree outside the subboard decreases its height by 1. Also note that if the gardener chooses a 3×3 subboard Ml times, the lumberjack will have to choose some map at least l times, so there will be at least 5 trees which each have height $\geqslant l$.

The strategy for the gardener will be to divide the board into N^2 disjoint 3×3 subboards, number them $0, \ldots, N^2 - 1$ in some order. Then, for $b = N^2 - 1, \ldots, 0$ in order, he plays $10^6 M(M+1)^b$ times on subboard number b. Hence, on subboard number b, the moves on that subboard will first ensure 5 of its trees grows by at least $10^6(M+1)^b$, and then each move after that will decrease their heights by 1. (As the trees on subboard b had height 0 before the gardener started playing there, no move made on subboards $\geqslant b$ decreased their heights.) As the gardener makes $10^6 M(M+1)^{b-1} + \ldots = 10^6((M+1)^b - 1)$ moves after he finishes playing on subboard b, this means that on subboard b, there will be 5 trees of height at least $10^6(M+1)^b - 10^6((M+1)^b - 1) = 10^6$, hence each of the subboard has 5 majestic trees, which was what we wanted.

C4.

Let $n > 3$ be a positive integer. Suppose that n children are arranged in a circle, and n coins are distributed between them (some children may have no coins). At every step, a child with at least 2 coins may give 1 coin to each of their immediate neighbours on the right and left. Determine all initial distributions of coins from which it is possible that, after a finite number of steps, each child has exactly one coin.

(Israel)

Answer: All distributions where $\sum_{i=1}^{n} ic_i = \frac{n(n+1)}{2} \pmod{n}$, where c_i denotes the number of coins the i-th child starts with.

Solution 1.

Number the children $1, \ldots, n$, and denote the number of coins the i-th child has by c_i. A step of this process consists of reducing some c_i by 2, and increasing c_{i-1}, c_{i+1} by 1. (Indices are considered $\pmod{n}$.) Because $(i-1) - 2i + (i+1) = 0$, the quantity $\sum_{i=1}^{n} ic_i \pmod{n}$ will be invariant under this process. Hence a necessary condition for the children to end up with an uniform distribution of coins is that

$$\sum_{i=1}^{n} ic_i = \frac{n(n+1)}{2} \pmod{n}.$$

We will show that this condition is also sufficient. Consider an arbitrary initial distribution of coins. First, whenever child i has more than one coin and $i \neq n$, have child i pass coins to its neighbors. (Child i does nothing.) Then, after some amount of such steps, it must eventually become impossible to do any more steps because no child except perhaps child i has more than 1 coin. (To see this, consider e.g. the quantity $\sum_{i=1}^{n-1} i^2 c_i$, which (as $(i-1)^2 + (i+1)^2 > 2i^2$) increases at each step.)

Hence we can reach a state of the form $(z_1, \ldots, z_{n-1}, M)$, where $z_i = 0$ or 1. Call such states *semi-uniform states of irregularity* M.

Lemma. If there is a string of children having coins $a, \overbrace{1, \ldots, 1}^{k \text{ ones}}, b, \overbrace{1, \ldots, 1}^{k \text{ ones}}, c$, with $b \geqslant 2$, after some sequence of steps we may reach the state $a+1, \overbrace{1, \ldots, 1}^{k \text{ ones}}, b-2, \overbrace{1, \ldots, 1}^{k \text{ ones}}, c+1$. We call performing this sequence of steps *long-passing coins*.

Proof. This is simply repeated application of the operation. We prove the lemma by induction on k. For $k = 0$, this is just the operation of the problem. If $k = 1$, have the child with b coins pass coins, then both of their neighbors pass coins, then the child with b coins pass coins again. For $k \geqslant 2$, first, have the child with b coins pass coins, then have both their neigbors send coins, giving the state

$$a, \overbrace{1, \ldots, 1}^{k-2 \text{ ones}}, 2, 0, b, 0, 2, \overbrace{1, \ldots, 1}^{k-2 \text{ ones}}, c.$$

Now set aside the children with a, b and c coins, and have each child with 2 coins give them to their neighbors until there are no such children remaining. This results in the state

$$a+1, 0, \overbrace{1, \ldots, 1}^{k-2 \text{ ones}}, b, \overbrace{1, \ldots, 1}^{k-2 \text{ ones}}, 0, c+1.$$

By the induction hypothesis, we can have the child with b coins may pass a coin to each of the children with 0 coins, proving the lemma. □

Claim. We can reach a semi-uniform state of irregularity $M \leqslant 2$.

Proof. If $M > 3$, because there are only n coins in total, there must be at least two children with 0 coins. Consider the arc of the circle spanned by the two such children closest to the child with M coins. It has the form

$$0, \overbrace{1, \ldots, 1}^{a \text{ ones}}, M, \overbrace{1, \ldots, 1}^{b \text{ ones}}, 0$$

If $a = b$, applying the previous lemma we can have the child with M coins long-pass a coin to each of the children with 0 coins, which yields a semi-uniform state with lower M. Otherwise, WLOG $a > b$, so we can have the child with M coins long-pass a coin to each of the children at distance b from it, reaching a state of the form $(\alpha := a - b - 1, \beta := b)$

$$0, \overbrace{1, \ldots, 1}^{\alpha \text{ ones}}, 2, \overbrace{1, \ldots, 1}^{\beta \text{ ones}}, M - 2, \overbrace{1, \ldots, 1}^{c \text{ ones}}$$

The children in the rightmost string of ones need make no further moves, so consider only the leftmost string. If $\alpha < \beta$, have the child with 2 coins long-pass coins to the child with 0 coins to its left and some child with 1 coin to its right, reaching a new state of the form $0, \overbrace{1, \ldots, 1}^{\alpha \text{ ones}}, 2, \overbrace{1, \ldots, 1}^{\beta \text{ ones}}, M - 2$ with a smaller β. As β cannot decrease indefinitely, eventually $\alpha \geqslant \beta$. If $\alpha = \beta$, have the child with 2 coins long-pass to the child with M coins and the child with 0 coins, reaching a semi-uniform state of irregularity $M - 1$ as desired. Otherwise, $\alpha < \beta$, so have the child with 2 coins long-pass to the child with M coins and a child with 1 coin, reaching a state of the form

$$0, \overbrace{1, \ldots, 1}^{x \text{ ones}}, 2, \overbrace{1, \ldots, 1}^{y \text{ ones}}, 0, \overbrace{1, \ldots, 1}^{z \text{ ones}}, M - 1$$

Now, consider only the substring between the two children with 0 coins, which has the form $0, \overbrace{1, \ldots, 1}^{x \text{ ones}}, 2, \overbrace{1, \ldots, 1}^{y \text{ ones}}, 0$. Repeatedly have the child in this substring with 2 coins long-pass to the closest child with 0 coins and some other child. If the other child has 1 coin, we have a new strictly shorter substring of the form $0, \overbrace{1, \ldots, 1}^{x \text{ ones}}, 2, \overbrace{1, \ldots, 1}^{y \text{ ones}}, 0$. Hence eventually it must happen that the other child also has 0 coins, at which point we reach a semi-uniform state of irregularity $M - 1$, proving the claim. □

We have now shown that we can reach a semi-regular state of irregularity $M \leqslant 2$, If $M = 1$, each child must have one coin, as desired. Otherwise, we must have $M = 2$, so there is one child with 0 coins, one child with 2 coins, and the remaining children all have 1 coin. Recall that the state we started with satisfied the invariant

$$\sum_{i=1}^{n} ic_i = \frac{n(n+1)}{2} \pmod{n}$$

Because each step preserves this invariant, this must also be true of the current state. Number the children so that the child with M coins is child number n, and suppose the child with 0 coins is child k. Then $\frac{n(n+1)}{2} = \sum_{i=1}^{n} ic_i = \left(\sum_{i=1}^{n} 1 \cdot c_i\right) - k = \frac{n(n+1)}{2} - k \pmod{n}$, so $k = 0 \pmod{n}$. But this is impossible, as no child except the child with M coins has an index divisible by n. Hence we cannot end up in a semi-regular state of irregularity 2, so we are done.

Solution 2. Encode the sequence c_i as a polynomial $p(x) = \sum_i a_i x_i$. The cyclic nature of the problem makes it natural to work modulo $x^n - 1$. Child i performing a step is equivalent to adding $x^i(x-1)^2$ to the polynomial, and we want to reach the polynomial $q(x) = 1 + x + \ldots + x^{n-1}$. Since we only add multiples of $(x-1)^2$, this is only possible if $p(x) = q(x)$ modulo the ideal generated by $x^n - 1$ and $(x-1)^2$, i.e.

$$(x^n - 1, (x-1)^2) = (x-1)\left(\frac{x^n - 1}{x-1}, x - 1\right) = (x-1) \cdot (n, (x-1))$$

This is equivalent to $p(1) = q(1)$ (which simply translates to the condition that there are n coins) and $p'(1) = q'(1) \pmod{n}$, which translates to the invariant described in solution 1.

C5. Let $m, n \geqslant 2$ be integers, let X be a set with n elements, and let $X_1, X_2, \ldots, X_m$ be pairwise distinct non-empty, not necessary disjoint subsets of X. A function $f : X \to \{1, 2, \ldots, n+1\}$ is called *nice* if there exists an index k such that

$$\sum_{x \in X_k} f(x) > \sum_{x \in X_i} f(x) \quad \text{for all } i \neq k.$$

Prove that the number of nice functions is at least n^n.

(Germany)

Solution. For a subset $Y \subseteq X$, we write $f(Y)$ for $\sum_{y \in Y} f(y)$. Note that a function $f : X \to \{1, \ldots, n+1\}$ is nice, if and only if $f(X_i)$ is maximized by a unique index $i \in \{1, \ldots, m\}$.

We will first investigate the set $\mathcal{F}$ of functions $f : X \to \{1, \ldots, n\}$; note that $|\mathcal{F}| = n^n$. For every function $f \in \mathcal{F}$, define a corresponding function $f^+ : X \to \{1, 2, \ldots, n+1\}$ in the following way: Pick some set X_l that maximizes the value $f(X_l)$.

- For all $x \in X_l$, define $f^+(x) = f(x) + 1$.
- For all $x \in X \backslash X_l$, define $f^+(x) = f(x)$.

Claim. The resulting function f^+ is nice.

Proof. Note that $f^+(X_i) = f(X_i) + |X_i \cap X_l|$ holds for all X_i. We show that $f^+(X_i)$ is maximized at the unique index $i = l$. Hence consider some arbitrary index $j \neq l$. Then $X_l \subset X_j$ is impossible, as this would imply $f(X_j) > f(X_l)$ and thereby contradict the choice of set X_l; this in particular yields $|X_l| > |X_j \cap X_l|$.

$$f^+(X_l) = f(X_l) + |X_l| \geqslant f(X_j) + |X_l| > f(X_j) + |X_j \cap X_l| = f^+(X_j)$$

The first inequality follows since X_l was chosen to maximize the value $f(X_l)$. The second (strict) inequality follows from $|X_l| > |X_j \cap X_l|$ as observed above. This completes the proof of the claim. □

Next observe that function f can be uniquely reconstructed from f^+ : the claim yields that f^+ has a unique maximizer X_l, and by decreasing the value of f^+ on X_l by 1, we get we can fully determine the values of f. As each of the n^n functions $f \in \mathcal{F}$ yields a (unique) corresponding nice function $f^+ : X \to \{1, 2, \ldots, n+1\}$, the proof is complete.

C6. Let n be a positive integer. We start with n piles of pebbles, each initially containing a single pebble. One can perform moves of the following form: choose two piles, take an equal number of pebbles from each pile and form a new pile out of these pebbles. For each positive integer n, find the smallest number of non-empty piles that one can obtain by performing a finite sequence of moves of this form.

(*Croatia*)

Answer: 1 if n is a power of two, and 2 otherwise.

Solution 1. The solution we describe is simple, but not the most effective one.

We can combine two piles of 2^{k-1} pebbles to make one pile of 2^k pebbles. In particular, given 2^k piles of one pebble, we may combine them as follows:

$$\begin{array}{lcl} 2^k \text{ piles of 1 pebble} & \to & 2^{k-1} \text{ piles of 2 pebbles} \\ 2^{k-1} \text{ piles of 2 pebbles} & \to & 2^{k-2} \text{ piles of 4 pebbles} \\ 2^{k-2} \text{ piles of 4 pebbles} & \to & 2^{k-3} \text{ piles of 8 pebbles} \\ & \vdots & \\ 2 \text{ piles of } 2^{k-1} \text{ pebbles} & \to & 1 \text{ pile of } 2^k \text{ pebbles} \end{array}$$

This proves the desired result in the case when n is a power of 2.

If n is *not* a power of 2, choose N such that $2^N < n < 2^{N+1}$. Let $m = n - 2^N$. Then $0 < m < 2^N$. Make a pile of 2^N pebbles and call it the *large* pile. (Alternatively, one can be more efficient and make a pile of 2^M pebbles where $m \leqslant 2^M$.) Since n is not a power of two, there is at least one other pile with pebbles. All other piles have a single pebble (initial condition).

Choose one single pebble pile and remove the pebble and one pebble from the *large* pile and form a pile of 2 pebbles. If $m < 2^N - 1$, remove another pebble from the *large* pile and one pebble from the 2-pile and form a new pile of 2 pebbles. Repeat until the *large* pile contains exactly m pebbles.

At this point we have one pile of m pebbles, one pile of 2 pebbles, and the rest are single pebble piles. There must be $n - m - 2 = 2^N - 2$ single piles. Combine two and two into piles of two pebbles. Then there are 2^{N-1} piles of two pebbles, which we can make into one pile of 2^N pebbles. We are left with exactly two piles of pebbles.

Lastly we will prove that it is not possible to form a single pile with all pebbles when n is not a power of two. A move consists of choosing two piles of say a and b pebbles, then removing $c \leqslant \min(a, b)$ pebbles from both piles, and forming a new pile with $2c$ pebbles. If we include piles of zero pebbles, then this move changes the number of pebbles in three piles as follows (and leaves all other piles unchanged):

$$\begin{array}{lcl} a & \to & a - c \\ b & \to & b - c \\ 0 & \to & 2c \end{array}$$

Assume that after the move the number of pebbles in each pile is divisible by an odd integer m. In particular, $m \mid 2c$, $m \mid a - c$ and $m \mid b - c$. Since m is odd, it follows that $m \mid c$, and then that $m \mid a$ and $m \mid b$. Hence also before the move the number of pebbles in each pile is divisible by the integer m.

If n is *not* a power of 2, then n is divisible by an odd integer $m > 1$. In order to make a single pile of n pebbles, we would have to start with a distribution in which the number of pebbles in each pile is divisible by the integer m. This is impossible when we start with all piles containing a single pebble.

Remarks on starting configurations From any starting configuration that is not a single pile, if there are at least two piles with at least two pebbles, we can remove one pebble from two such piles, and form a new pile with 2 pebbles. We can repeat this until we have one pile of 2 pebbles and the rest are single pebble piles, and then proceed as in the solution. Hence, if n is a power of two, we can make a single pile from any starting configuration.

If n is of the form $n = 2^k m$ where $m > 1$ is odd, then we can make a single pile from any starting configuration in which the number of pebbles in each pile is divisible by the integer m, otherwise two piles is the best we can do. Half of this is proven already. For the other half, assume we start with a configuration in which the number of pebbles in each pile is divisible by the integer m. Replace each pile of tm pebbles with a pile of t boulders. We now have a total of 2^k boulders, hence we can make them into one pile of 2^k boulders. Replacing the boulders with pebbles again, we are done.

Solution 2. We show an alternative strategy if n is not a power of 2. Write n in binary form: $n = 2^{i_1} + 2^{i_2} + \cdots + 2^{i_k}$, where $i_1 > i_2 > \cdots > i_k$. Now we make piles of sizes $2^{i_1}, 2^{i_2}, \ldots, 2^{i_k}$. We call the pile with 2^{i_1} the *large pile*, all the others are the *small piles*.

The strategy is the following: take the two smallest *small pile*. If they have the same size of 2^a, we make a pile of size 2^{a+1}. If they have different sizes, we double the smaller pile using the *large pile* (we allow the *large pile* to have a negative number of pebbles: later we prove that it is not possible). We call all the new piles *small*. When we have only one *small pile*, we terminate the process: we have at most 2 piles.

After each move we have one less number of piles, and all the piles have cardinality power of 2. The number of pebbles is decreasing, and at the end of the process, it has a size of $n - 2^{i_2+1} \geqslant n - 2^{i_1} > 0$, thus we can manage to have two piles.

Solution 3. Throughout the solution, we will consider the moves in reverse order. Namely, imagine we have some piles of pebbles, and we are allowed to perform moves as follows: take a pile with an even number of pebbles, split it into two equal halves and add the pebbles from each half to a different pile, possibly forming new piles (we may assume for convenience that there are infinitely many empty piles at any given moment). Given a configuration of piles $\mathcal{C}$, we will use $|\mathcal{C}|$ to denote the number of non-empty piles in $\mathcal{C}$. Given two configurations $\mathcal{C}_1$, $\mathcal{C}_2$, we will say that $\mathcal{C}_2$ is reachable from $\mathcal{C}_1$ if $\mathcal{C}_2$ can be obtained by performing a finite sequence of moves starting from $\mathcal{C}_1$. Call a configuration of piles $\mathcal{C}$

- *simple* if each (non-empty) pile in $\mathcal{C}$ consists of a single pebble;
- *good* if at least one (non-empty) pile in $\mathcal{C}$ has an even number of pebbles and the numbers of pebbles on the piles in $\mathcal{C}$ have no non-trivial odd common divisor ($\mathcal{C}$ has the *odd divisor property*);
- *solvable* if there exists a simple configuration which is reachable from $\mathcal{C}$.

The problem asks to find the smallest number of non-empty piles in a solvable configuration consisting of n pebbles. We begin the process of answering this question by making the following observation:

Lemma 1. Let $\mathcal{C}$ be a configuration of piles. Let $\mathcal{C}'$ be a configuration obtained by applying a single move to $\mathcal{C}$. Then

(i) if $\mathcal{C}'$ has the odd divisor property, then so does $\mathcal{C}$;

(ii) the converse to (i) holds if $|\mathcal{C}'| \geqslant |\mathcal{C}|$.

Proof. Suppose that the move consists of splitting a pile of size $2a$ and adding a pebbles to each of two piles of sizes b and c. Here, a is a positive integer and b, c are non-negative integers. Thus, $\mathcal{C}'$ can be obtained from $\mathcal{C}$ by replacing the piles of sizes $2a$, b, c by two piles of sizes $a+b$ and $a+c$. Note that the extra assumption $|\mathcal{C}'| \geqslant |\mathcal{C}|$ of part (ii) holds if and only if at least one of b, c is zero.

(i) Suppose $\mathcal{C}$ doesn't have the odd divisor property, i.e. there exists an odd integer $d > 1$ such that the size of each pile in $\mathcal{C}$ is divisible by d. In particular, $2a$, b, c are multiples of d, so since d is odd, it follows that a, b, c are all divisible by d. Thus, $a+b$ and $a+c$ are also divisible by d, so d divides the size of each pile in $\mathcal{C}'$. In conclusion, $\mathcal{C}'$ doesn't have the odd divisor property.

(ii) If $\mathcal{C}'$ doesn't have the odd divisor property and at least one of b, c is zero, then there exists an odd integer $d > 1$ such that the size of each pile in $\mathcal{C}'$ is divisible by d. In particular, d divides $a+b$ and $a+c$, so since at least one of these numbers is equal to a, it follows that d divides a. But then d must divide all three of a, b and c, and hence it certainly divides $2a$, b and c. Thus, $\mathcal{C}$ doesn't have the odd divisor property, as desired. □

Lemma 2. If $\mathcal{C}_2$ is reachable from $\mathcal{C}_1$ and $\mathcal{C}_2$ has the odd divisor property, then so does $\mathcal{C}_1$. In particular, any solvable configuration has the odd divisor property.

Proof. The first statement follows by inductively applying part (i) of Lemma 1. The second statement follows from the first because every simple configuration has the odd divisor property. □

The main claim is the following:

Lemma 3. Let $\mathcal{C}$ be a good configuration. Then there exists a configuration $\mathcal{C}'$ with the following properties:

- $\mathcal{C}'$ is reachable from $\mathcal{C}$ and $|\mathcal{C}'| > |\mathcal{C}|$;
- $\mathcal{C}'$ is either simple or good.

Proof. Call a configuration *terminal* if it is a counterexample to the claim. The following claim is at the heart of the proof:

Claim. Let $a_1, \ldots, a_k$ be the numbers of pebbles on the non-empty piles of a terminal configuration $\mathcal{C}$. Then there exists a unique $i \in [k]$ such that a_i is even. Moreover, for all $t \geqslant 1$ we have $a_j \equiv \frac{a_i}{2} (\mod 2^t)$ for all $j \neq i$.

Proof of Claim. Since the configuration is good, there must exist $i \in [k]$ such that a_i is even. Moreover, by assumption, if we split the pile with a_i pebbles into two equal halves, the resulting configuration will not be good. By part (ii) of Lemma 2,the only way this can happen is that $\frac{a_i}{2}$ and a_j for all $j \neq i$ are odd. To prove the second assertion, we proceed by induction on t, with the case $t = 1$ already being established. If $t \geqslant 2$, then split the pile with a_i pebbles into two equal halves and move one half to the pile with a_j pebbles. Since $\frac{a_i}{2}$ and a_j are both odd, $a_j + \frac{a_i}{2}$ is even, so by part (ii) of Lemma 2, the resulting configuration $\mathcal{C}'$ is good. Thus, $\mathcal{C}'$ is terminal, so by the induction hypothesis, we have $\frac{a_i}{2} \equiv \frac{1}{2}(a_j + \frac{a_i}{2}) (\mod 2^{t-1})$, whence $a_j \equiv \frac{a_i}{2} (\mod 2^t)$, as desired. □

Suppose for contradiction that there exists a configuration as in the Claim. It follows that there exists $i \in [k]$ and an odd integer x such that $a_i = 2x$ and $a_j = x$ for all $j \neq i$. Thus, x is an odd common divisor of $a_1, \ldots, a_k$, so by the odd divisor property, we must have $x = 1$. But then we can obtain a simple configuration by splitting the only pile with two pebbles into two piles each consisting of a single pebble, which is a contradiction. □

With the aid of Lemmas 2 and 3, it is not hard to characterise all solvable configurations:

Lemma 4. A configuration of piles $\mathcal{C}$ is solvable if and only if it is simple or good.

Proof. ($\Longrightarrow$) Suppose $\mathcal{C}$ is not simple. Then since we have to be able to perform at least one move, there must be at least one non-empty pile in $\mathcal{C}$ with an even number of pebbles. Moreover, by Lemma 2, $\mathcal{C}$ has the odd divisor property, so it must be good.

($\Longleftarrow$) This follows by repeatedly applying Lemma 3 until we reach a simple configuration. Note that the process must stop eventually since the number of non-empty piles is bounded from above. □

Finally, the answer to the problem is implied by the following corollary of Lemma 4:

Lemma 5. Let n be a positive integer. Then

(i) a configuration consisting of a single pile of n pebbles is solvable if and only if n is a power of two;

(ii) if $n \geqslant 2$, then the configuration consisting of piles of sizes 2 and $n-2$ is good.

Proof. (i) By Lemma 4, this configuration is solvable if and only if either $n = 1$ or n is even and has no non-trivial odd divisor, so the conclusion follows.

(ii) Since 2 is even and has no non-trivial odd divisor, this configuration is certainly good, so the conclusion follows by Lemma 4. □

Common remarks. Instead of choosing pebbles from two piles, one could allow choosing an equal number of pebbles from each of k piles, where $k \geqslant 2$ is a fixed (prime) integer. However, this seems to yield a much harder problem – if $k = 3$, numerical evidence suggests the same answer as in the case $k = 2$ (with powers of two replaced by powers of three), but the case $k = 5$ is already unclear.

C7. Lucy starts by writing s integer-valued 2022-tuples on a blackboard. After doing that, she can take any two (not necessarily distinct) tuples $\mathbf{v} = (v_1, \ldots, v_{2022})$ and $\mathbf{w} = (w_1, \ldots, w_{2022})$ that she has already written, and apply one of the following operations to obtain a new tuple:

$$\mathbf{v} + \mathbf{w} = (v_1 + w_1, \ldots, v_{2022} + w_{2022})$$
$$\mathbf{v} \vee \mathbf{w} = (\max(v_1, w_1), \ldots, \max(v_{2022}, w_{2022}))$$

and then write this tuple on the blackboard.

It turns out that, in this way, Lucy can write any integer-valued 2022-tuple on the blackboard after finitely many steps. What is the smallest possible number s of tuples that she initially wrote?

(Czech Republic)

Answer: The smallest possible number is $s = 3$.

Solution. We solve the problem for n-tuples for any $n \geqslant 3$: we will show that the answer is $s = 3$, regardless of the value of n.

First, let us briefly introduce some notation. For an n-tuple $\mathbf{v}$, we will write $\mathbf{v}_i$ for its i-th coordinate (where $1 \leqslant i \leqslant n$). For a positive integer n and a tuple $\mathbf{v}$ we will denote by $n \cdot \mathbf{v}$ the tuple obtained by applying addition on $\mathbf{v}$ with itself n times. Furthermore, we denote by $\mathbf{e}(i)$ the tuple which has i-th coordinate equal to one and all the other coordinates equal to zero. We say that a tuple is *positive* if all its coordinates are positive, and *negative* if all its coordinates are negative.

We will show that three tuples suffice, and then that two tuples do not suffice.

Three tuples suffice. Write $\mathbf{c}$ for the constant-valued tuple $\mathbf{c} = (-1, \ldots, -1)$.

We note that it is enough for Lucy to be able to make the tuples $\mathbf{e}(1), \ldots, \mathbf{e}(n), \mathbf{c}$; from those any other tuple $\mathbf{v}$ can be made as follows. First we choose some positive integer k such that $k + \mathbf{v}_i > 0$ for all i. Then, by adding a positive number of copies of $\mathbf{c}, \mathbf{e}(1), \ldots, \mathbf{e}(n)$, she can make

$$k\mathbf{c} + (k + \mathbf{v}_1) \cdot \mathbf{e}(1) + \cdots + (k + \mathbf{v}_n) \cdot \mathbf{e}(n),$$

which we claim is equal to $\mathbf{v}$. Indeed, this can be checked by comparing coordinates: the i-th coordinate of the right-hand side is $-k + (k + \mathbf{v}_i) = \mathbf{v}_i$ as needed.

Lucy can take her three starting tuples to be $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, such that $\mathbf{a}_i = -i^2$, $\mathbf{b}_i = i$ and $\mathbf{c} = -1$ (as above).

For any $1 \leqslant j \leqslant n$, write $\mathbf{d}(j)$ for the tuple $2 \cdot \mathbf{a} + 4j \cdot \mathbf{b} + (2j^2 - 1) \cdot \mathbf{c}$, which Lucy can make by adding together $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ repeatedly. This has ith term

$$\begin{aligned} \mathbf{d}(j)_i &= 2\mathbf{a}_i + 4j\mathbf{b}_i + (2j^2 - 1)\mathbf{c}_i \\ &= -2i^2 + 4ij - (2j^2 - 1) \\ &= 1 - 2(i - j)^2. \end{aligned}$$

This is 1 if $j = i$, and at most -1 otherwise. Hence Lucy can produce the tuple $\mathbf{1} = (1, \ldots, 1)$ as $\mathbf{d}(1) \vee \cdots \vee \mathbf{d}(n)$.

She can then produce the constant tuple $\mathbf{0} = (0, \ldots, 0)$ as $\mathbf{1} + \mathbf{c}$, and for any $1 \leqslant j \leqslant n$ she can then produce the tuple $\mathbf{e}(j)$ as $\mathbf{d}(j) \vee \mathbf{0}$. Since she can now produce $\mathbf{e}(1), \ldots, \mathbf{e}(n)$ and already had $\mathbf{c}$, she can (as we argued earlier) produce any integer-valued tuple.

Two tuples do not suffice. We start with an observation: Let a be a non-negative real number and suppose that two tuples $\mathbf{v}$ and $\mathbf{w}$ satisfy $\mathbf{v}_j \geqslant a\mathbf{v}_k$ and $\mathbf{w}_j \geqslant a\mathbf{w}_k$ for some $1 \leqslant j, k \leqslant n$. Then we claim that the same inequality holds for $\mathbf{v} + \mathbf{w}$ and $\mathbf{v} \vee \mathbf{w}$: Indeed, the property for the sum is verified by an easy computation:

$$(\mathbf{v} + \mathbf{w})_j = \mathbf{v}_j + \mathbf{w}_j \geqslant a\mathbf{v}_k + a\mathbf{w}_k = a(\mathbf{v} + \mathbf{w})_k.$$

For the second operation, we denote by $\mathbf{m}$ the tuple $\mathbf{v} \vee \mathbf{w}$. Then $\mathbf{m}_j \geqslant \mathbf{v}_j \geqslant a\mathbf{v}_k$ and $\mathbf{m}_j \geqslant \mathbf{w}_j \geqslant a\mathbf{w}_k$. Since $\mathbf{m}_k = \mathbf{v}_k$ or $\mathbf{m}_k = \mathbf{w}_k$, the observation follows.

As a consequence of this observation we have that if all starting tuples satisfy such an inequality, then all generated tuples will also satisfy it, and so we would not be able to obtain every integer-valued tuple.

Let us now prove that Lucy needs at least three starting tuples. For contradiction, let us suppose that Lucy started with only two tuples $\mathbf{v}$ and $\mathbf{w}$. We are going to distinguish two cases. In the first case, suppose we can find a coordinate i such that $\mathbf{v}_i, \mathbf{w}_i \geqslant 0$. Both operations preserve the sign, thus we can not generate any tuple that has negative i-th coordinate. Similarly for $\mathbf{v}_i, \mathbf{w}_i \leqslant 0$.

Suppose the opposite, i.e., for every i we have either $\mathbf{v}_i > 0 > \mathbf{w}_i$, or $\mathbf{v}_i < 0 < \mathbf{w}_i$. Since we assumed that our tuples have at least three coordinates, by pigeonhole principle there exist two coordinates $j \neq k$ such that $\mathbf{v}_j$ has the same sign as $\mathbf{v}_k$ and $\mathbf{w}_j$ has the same sign as $\mathbf{w}_k$ (because there are only two possible combinations of signs).

Without loss of generality assume that $\mathbf{v}_j, \mathbf{v}_k > 0$ and $\mathbf{w}_j, \mathbf{w}_k < 0$. Let us denote the positive real number $\mathbf{v}_j/\mathbf{v}_k$ by a. If $\mathbf{w}_j/\mathbf{w}_k \leqslant a$, then both inequalities $\mathbf{v}_j \geqslant a\mathbf{v}_k$ and $\mathbf{w}_j \geqslant a\mathbf{w}_k$ are satisfied. On the other hand, if $\mathbf{w}_j/\mathbf{w}_k \leqslant a$, then both $\mathbf{v}_k \geqslant (1/a)\mathbf{v}_j$ and $\mathbf{w}_k \geqslant (1/a)\mathbf{w}_j$ are satisfied. In either case, we have found the desired inequality satisfied by both starting tuples, a contradiction with the observation above.

Common remarks.

1. For $n \in \{1, 2\}$, two starting n-tuples are necessary and sufficient.

2. The operations $+$, $\vee$ used in this problem are studied in the area of tropical geometry. However, as far as we know, familiarity with tropical geometry does not help when solving the problem.

C8.

Alice fills the fields of an $n \times n$ board with numbers from 1 to n^2, each number being used exactly once. She then counts the total number of good paths on the board. *A good path* is a sequence of fields of arbitrary length (including 1) such that:

(i) The first field in the sequence is one that is only adjacent to fields with larger numbers,

(ii) Each subsequent field in the sequence is adjacent to the previous field,

(iii) The numbers written on the fields in the sequence are in increasing order.

Two fields are considered adjacent if they share a common side. Find the smallest possible number of good paths Alice can obtain, as a function of n.

(Serbia)

Answer: $2n^2 - 2n + 1$.

Solution.

We will call any field that is only adjacent to fields with larger numbers a well. Other fields will be called non-wells. Let us make a second $n \times n$ board B where in each field we will write the number of good sequences which end on the corresponding field in the original board A. We will thus look for the minimal possible value of the sum of all entries in B.

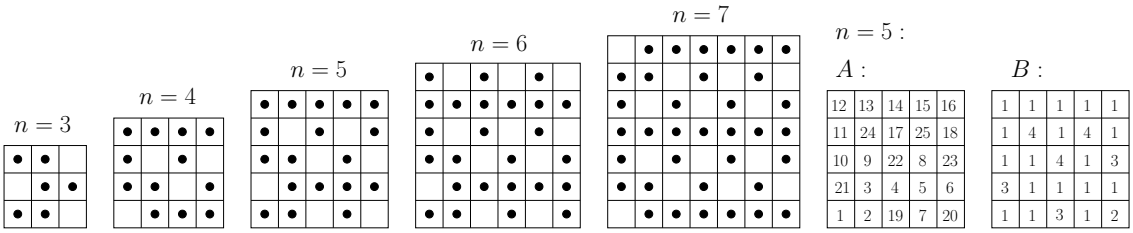
We note that any well has just one good path ending in it, consisting of just the well, and that any other field has the number of good paths ending in it equal to the sum of this quantity for all the adjacent fields with smaller values, since a good path can only come into the field from a field of lower value. Therefore, if we fill in the fields in B in increasing order with respect to their values in A, it follows that each field not adjacent to any already filled field will receive a 1, while each field adjacent to already filled fields will receive the sum of the numbers already written on these adjacent fields.

We note that there is at least one well in A, that corresponding with the field with the entry 1 in A. Hence, the sum of values of fields in B corresponding to wells in A is at least 1. We will now try to minimize the sum of the non-well entries, i.e. of the entries in B corresponding to the non-wells in A. We note that we can ascribe to each pair of adjacent fields the value of the lower assigned number and that the sum of non-well entries will then equal to the sum of the ascribed numbers. Since the lower number is still at least 1, the sum of non-well entries will at least equal the number of pairs of adjacent fields, which is $2n(n-1)$. Hence, the total minimum sum of entries in B is at least $2n(n-1)+1 = 2n^2-2n+1$. The necessary conditions for the minimum to be achieved is for there to be only one well and for no two entries in B larger than 1 to be adjacent to each other.

We will now prove that the lower limit of $2n^2-2n+1$ entries can be achieved. This amounts to finding a way of marking a certain set of squares, those that have a value of 1 in B, such that no two unmarked squares are adjacent and that the marked squares form a connected tree with respect to adjacency.

For $n = 1$ and $n = 2$ the markings are respectively the lone field and the L-trimino. Now, for $n > 2$, let $s = 2$ for $n \equiv 0, 2 \mod 3$ and $s = 1$ for $n \equiv 1 \mod 3$. We will take indices k and l to be arbitrary non-negative integers. For $n \geqslant 3$ we will construct a path of marked squares in the first two columns consisting of all squares of the form $(1, i)$ where i is not of the form $6k + s$ and $(2, j)$ where j is of the form $6k + s - 1$, $6k + s$ or $6 + s + 1$. Obviously, this path is connected. Now, let us consider the fields $(2, 6k + s)$ and $(1, 6k + s + 3)$. For each considered field (i, j) we will mark all squares of the form (l, j) for $l > i$ and $(i + 2k, j \pm 1)$. One can easily see that no set of marked fields will produce a cycle, that the only fields of the unmarked form $(1, 6k+s)$, $(2+2l+1, 6k+s\pm1)$ and $(2+2l, 6k+s+3\pm1)$ and that no two are adjacent, since

the consecutive considered fields are in columns of opposite parity. Examples of markings are given for $n = 3, 4, 5, 6, 7$, and the corresponding constructions for A and B are given for $n = 5$.



12	13	14	15	16
11	24	17	25	18
10	9	22	8	23
21	3	4	5	6
1	2	19	7	20

1	1	1	1	1
1	4	1	4	1
1	1	4	1	3
3	1	1	1	1
1	1	3	1	2

Common remarks.

- The construction can be achieved in different ways. For example, it can also be done recursively; we can complete any construction for n to a construction for $n + 1$.
- It is a natural idea to change the direction of the path: that way it can start anywhere, but only can end in a well, which exactly means that we cannot extend the path. This is just a reformulation of the problem, but can give some intuitions.

C9. Let $\mathbb{Z}_{\geqslant 0}$ be the set of non-negative integers, and let $f : \mathbb{Z}_{\geqslant 0} \times \mathbb{Z}_{\geqslant 0} \to \mathbb{Z}_{\geqslant 0}$ be a bijection such that whenever $f(x_1, y_1) > f(x_2, y_2)$, we have $f(x_1 + 1, y_1) > f(x_2 + 1, y_2)$ and $f(x_1, y_1 + 1) > f(x_2, y_2 + 1)$.

Let N be the number of pairs of integers (x, y), with $0 \leqslant x, y < 100$, such that $f(x, y)$ is odd. Find the smallest and largest possible value of N.

(U.S.A.)

Answer: The optimal bounds are $2500 \leqslant N \leqslant 7500$.

Solution. We defer the constructions to the end of the solution. Instead, we begin by characterizing all such functions f, prove a formula and key property for such functions, and then solve the problem, providing constructions.

Characterization Suppose f satisfies the given relation. The condition can be written more strongly as

$$\begin{aligned} f(x_1, y_1) > f(x_2, y_2) &\iff f(x_1 + 1, y_1) > f(x_2 + 1, y_2) \\ &\iff f(x_1, y_1 + 1) > f(x_2, y_2 + 1). \end{aligned}$$

In particular, this means for any $(k, l) \in \mathbb{Z}^2$, $f(x + k, y + l) - f(x, y)$ has the same sign for all x and y.

Call a non-zero vector $(k, l) \in \mathbb{Z}_{\geqslant 0} \times \mathbb{Z}_{\geqslant 0}$ a *needle* if $f(x + k, y) - f(x, y + l) > 0$ for all x and y. It is not hard to see that needles and non-needles are both closed under addition, and thus under scalar division (whenever the quotient lives in $\mathbb{Z}^2$).

In addition, call a positive rational number $\frac{k}{l}$ a *grade* if the vector (k, l) is a needle. (Since needles are closed under scalar multiples and quotients, this is well-defined.)

Claim. Grades are closed upwards.

Proof. Consider positive rationals $k_1/l_1 < k_2/l_2$ with k_1/l_1 a grade. Then:

- (k_1, l_1) is a needle
- so $(k_1 l_2, l_1 l_2)$ is a needle,
- so $(k_2 l_1, l_1 l_2)$ is a needle (as $k_2 l_1 - k_1 l_2 > 0$ and $(1, 0)$ is a needle).

Thus (k_2, l_2) is a needle, as wanted. □

Claim. A grade exists.

Proof. If no positive integer n is a grade, then $f(1, 0) > f(0, n)$ for all n which is impossible. □

Similarly, there is an n such that $f(0, 1) < f(n, 0)$, thus $1/n$ is not a grade for some large n. That means that small positive rational values are not grades, then there is a switch, and after that all values are grades. Call the place of that switch α. Here α is the infimum of the grades.

Claim (Key property). If $x_1 + y_1\alpha > x_2 + y_2\alpha$ then $f(x_1, y_1) > f(x_2, y_2)$.

Proof. If both $x_1 \geqslant x_2$ and $y_1 \geqslant y_2$ this is clear.
Suppose $x_1 \geqslant x_2$ and $y_1 < y_2$. Then $\frac{x_1 - x_2}{y_2 - y_1} > \alpha$ is a grade. This gives $f(x_1, y_1) > f(x_2, y_2)$.
Suppose $x_1 < x_2$ and $y_1 \geqslant y_2$. Then $\frac{x_2 - x_1}{u_1 - u_2} < \alpha$ is not a grade. This gives $f(x_2, y_2) < f(x_1, y_1)$.

From those observations we get the following claim.

Claim. The function f orders pairs (x, y) based on the value of $x + y\alpha$. If α is rational, tiebreaking is done by larger x- or y-coordinate (depending on whether α is a grade).

We can imagine this the following way: take a line with slope $-\frac{1}{\alpha}$ under the first quadrant of the plane. And we start to move this line upward (but it stays parallel to the original line). First it hits $(0,0)$, so $f(0,0)=0$. And each time the line hits a point p, $f(p)$ is the number of points hit before. If $\alpha \in \mathbb{Q}$, it is possible that the line hits multiple points. Then those points are ordered the same way as their x or y coordinates, depending on whether α is a grade.

We understood the behaviour of f, now we need to focus on the region of $A = \{(x,y) \in \mathbb{Z}_{\geqslant 0} \times \mathbb{Z}_{\geqslant 0} | x < 100,\ y < 100\}$. First, we can assume that α is irrational. If we change it a little bit in the right direction, the behaviour and values of the f function does not change in A.

Claim.

$$f(x,y) + f(x+1,y+1) = f(x+1,y) + f(x,y+1) + 1.$$

Proof.

$$\begin{gathered} f(x+1,y+1) - f(x,y+1) = \\ \#\{(a,b) \in \mathbb{Z}_{\geqslant 0} \times \mathbb{Z}_{\geqslant 0} |\ x + (y+1)\alpha \leqslant a + b\alpha < (x+1) + (y+1)\alpha\} = \\ \#\{(a,b) \in \mathbb{Z}_{\geqslant 0} \times \mathbb{Z}_{>0} |\ x + (y+1)\alpha \leqslant a + b\alpha < (x+1) + (y+1)\alpha\} + \\ \#\{(a,0) \in \mathbb{Z}_{\geqslant 0} \times \mathbb{Z}_{\geqslant 0} |\ (x+1) + y\alpha \leqslant a < (x+1) + (y+1)\alpha\} = \\ \#\{(a,b) \in \mathbb{Z}_{\geqslant 0} \times \mathbb{Z}_{\geqslant 0} |\ x + y\alpha \leqslant a + b\alpha < (x+1) + y\alpha\} + 1 = f(x+1,y) - f(x,y). \end{gathered}$$ □

From this claim we immediately get that $2500 \leqslant N \leqslant 7500$; now we show that those bounds are indeed sharp.

Remember that if α is irrational then

$$f(a,b) = \#\left\{(x,y) \in \mathbb{Z}_{\geqslant 0} \times \mathbb{Z}_{\geqslant 0} \mid\ x + y\alpha < a + b\alpha\right\}.$$

Construction for 7500 Select $\alpha \approx 199.999$.

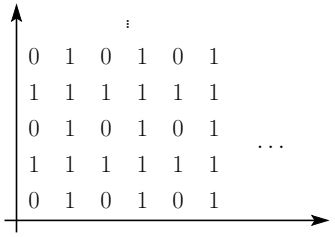
Claim.

1. $f(n,0) = n$ for $0 \leqslant n \leqslant 100$.
2. $f(0,k) \equiv k \mod 2$ for $0 \leqslant k \leqslant 100$.

Proof.

1. $f(n,0) = \#\{(x,y) |\ x + y\alpha < n\} = \#\{(x,y) |\ x + 199y < n\} = n$.
2. $$\begin{aligned} f(0,k) &= \#\{(x,y) |\ x + y\alpha < k\alpha\} = \sum_{l=0}^{k-1} \#\{(x,l) |\ x + l\alpha < k\alpha\} \\ &= \sum_{l=0}^{k-1} \#\{x |\ x < (k-l)\alpha\} = \sum_{l=0}^{k-1} 200(k-l) - 1 = 200A - k \end{aligned}$$ for some integer A.

From this claim, using the equality $f(x,y) + f(x+1,y+1) = f(x+1,y) + f(x,y+1) + 1$, we can prove that mod 2 the region A looks like the following: in the rows $(-, 2y)$ the remainders modulo 2 alternate, while the rows $(-, 2y+1)$ contain only odd numbers.



The numbers mod 2 in the construction for 7500.

Construction for 2500 Select $\alpha \approx 200.001$.

Claim.

1. $f(n, 0) = n$ for $0 \leqslant n \leqslant 100$.
2. $f(0, k) \equiv 0 \mod 2$ for $0 \leqslant k \leqslant 100$.

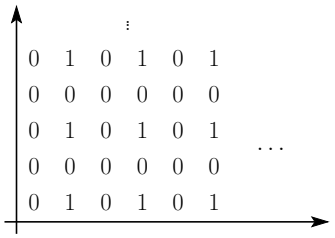
Proof.

1. As above.
2. Similarly to the above:

$$f(0,k) = \#\{(x,y) |\ x + y\alpha < k\alpha\} = \sum_{l=0}^{k-1} \#\{(x,l) |\ x + l\alpha < k\alpha\}$$
$$= \sum_{l=0}^{k-1} \#\{x |\ x < (k-l)\alpha\} = \sum_{l=0}^{k-1} 200(k-l) = 200A$$

for some integer A.

Similarly to the above, we can prove that mod 2 the region A looks like the following: in the rows $(-, 2y)$ the remainder modulo 2 alternate, while the rows $(-, 2y+1)$ contain only even numbers.



The numbers mod 2 in the construction for 7500.

Common remarks.

- In the proposer's solution, an exact formula is proved in the case where α is irrational. Let $\mu = \frac{1}{\alpha}$ and $\nu = \alpha$. Then

$$f(a,b) = ab + (\lceil 1\mu \rceil + \cdots + \lceil \mu \rceil) + (\lceil 1\nu \rceil + \cdots + \lceil \nu \rceil)\,.$$

  They even suggested that this could be the source of a fun olympiad problem: it is not easy to see that f is a bijection.

- (On the choice of statement, by the proposer). As seen in the solution, such functions have various properties, and it was not clear what statement to choose.

  I decided to go with the present statement since I believe it requires a good understanding of the setup and the functions involved. The key property is used in order to establish the bounds, and the formula is used to develop constructions and easily verify they work.

  While choosing a different statement (e.g. just prove the key property) may make the problem less bulky, I believe that doing so runs the risk of introducing unintended solutions which don't "see the full picture." In fact, the present statement probably runs admits such solutions as well, likely through proving the key property via simply examining valid locations for $n+1$ given the locations of $0, \ldots, n$, and using induction.

  Unfortunately, I'm not sure how else to improve the statement.

  One person I've shown this problem to suggests removing one half of the problem, by only asking for the smallest or only the largest possible value of N. Their rationale is that the two parts are mostly the same. I sympathize with them, but am reluctant to break the symmetry.

- (PSC) The real difficulty of this problem is that one should understand the whole picture, and in particular how the function f works, and how that quadrant can behave. In this solution we characterised all such functions f, before even talking about parity, while the problem only asks the number of odd numbers.

Geometry

G1. Let $ABCDE$ be a convex pentagon such that $BC = DE$. Assume there is a point T inside $ABCDE$ with $TB = TD$, $TC = TE$ and $\angle TBA = \angle AET$. Let lines CD and CT intersect line AB at points P and Q, respectively, and let lines CD and DT intersect line AE at points R and S, respectively. Assume that points P, B, A, Q and R, E, A, S respectively, are collinear and occur on their lines in this order. Prove that the points P, S, Q, R are concyclic.
(Slovakia)

Solution 1. By the conditions we have $BC = DE$, $CT = ET$ and $TB = TD$, so the triangles TBC and TDE are congruent, in particular $\angle BTC = \angle DTE$.

In triangles TBQ and TES we have $\angle TBQ = \angle SET$ and $\angle QTB = 180^\circ - \angle BTC = 180^\circ - \angle DTE = \angle ETS$, so these triangles are similar to each other. It follows that $\angle TSE = \angle BQT$ and

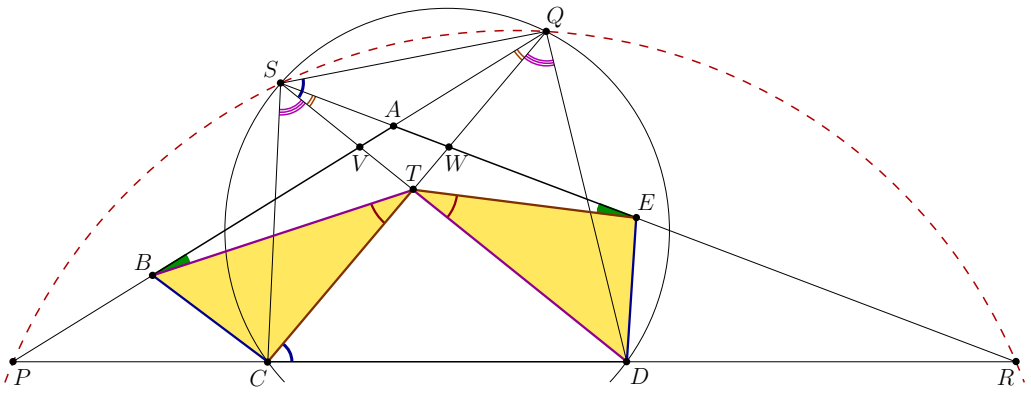
$$\frac{TD}{TQ} = \frac{TB}{TQ} = \frac{TE}{TS} = \frac{TC}{TS}.$$

By rearranging this relation we get $TD \cdot TS = TC \cdot TQ$, so C, D, Q and S are concyclic. (Alternatively, we can get $\angle CQD = \angle CSD$ from the similar triangles TCS and TDQ.) Hence, $\angle DCQ = \angle DSQ$.

Finally, from the angles of triangle CQP we get

$$\angle RPQ = \angle RCQ - \angle PQC = \angle DSQ - \angle DSR = \angle RSQ,$$

which proves that P, Q, R and S are concyclic.



Solution 2. As in the previous solution, we note that triangles TBC and TDE are congruent. Denote the intersection point of DT and BA by V, and the intersection point of CT and EA by W. From triangles BCQ and DES we then have

$$\begin{aligned}\angle VSW = \angle DSE &= 180^\circ - \angle SED - \angle EDS = 180^\circ - \angle AET - \angle TED - \angle EDT \\ &= 180^\circ - \angle TBA - \angle TCB - \angle CBT = 180^\circ - \angle QCB - \angle CBQ = \angle BQC = \angle VQW,\end{aligned}$$

meaning that $VSQW$ is cyclic, and in particular $\angle WVQ = \angle WSQ$. Since

$$\angle VTB = 180^\circ - \angle BTC - \angle CTD = 180^\circ - \angle CTD - \angle DTE = \angle ETW,$$

and $\angle TBV = \angle WET$ by assumption, we have that the triangles VTB and WTE are similar, hence

$$\frac{VT}{WT} = \frac{BT}{ET} = \frac{DT}{CT}.$$

Thus $CD \parallel VW$, and angle chasing yields

$$\angle RPQ = \angle WVQ = \angle WSQ = \angle RSQ,$$

concluding the proof.

G2. In the acute-angled triangle ABC, the point F is the foot of the altitude from A, and P is a point on the segment AF. The lines through P parallel to AC and AB meet BC at D and E, respectively. Points $X \neq A$ and $Y \neq A$ lie on the circles ABD and ACE, respectively, such that $DA = DX$ and $EA = EY$.

Prove that B, C, X and Y are concyclic.

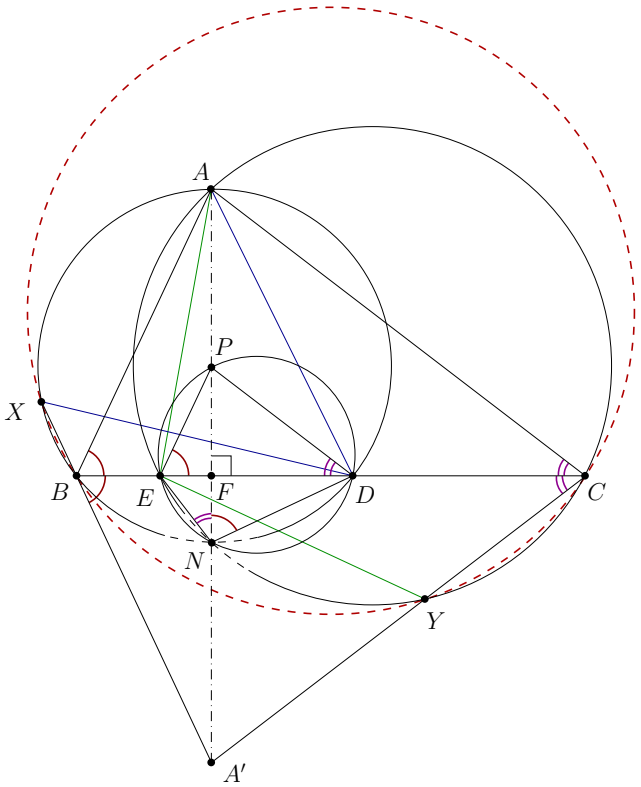
(Netherlands)

Solution 1. Let A' be the intersection of lines BX and CY. By power of a point, it suffices to prove that $A'B \cdot A'X = A'C \cdot A'Y$, or, equivalently, that A' lies on the radical axis of the circles $ABDX$ and $ACEY$.

From $DA = DX$ it follows that in circle $ABDX$, point D bisects of one of the arcs AX. Therefore, depending on the order of points, the line BC is either the internal or external bisector of $\angle ABX$. In both cases, line BX is the reflection of BA in line BDC. Analogously, line CY is the reflection of CA in line BC; we can see that A' is the reflection of A in line BC, so A, F and A' are collinear.

By $PD \parallel AC$ and $PE \parallel AB$ we have $\dfrac{FD}{FC} = \dfrac{FP}{FA} = \dfrac{FE}{FB}$, hence $FD \cdot FB = FE \cdot FC$. So, point F has equal powers with respect to circles $ABDX$ and $ACEY$.

Point A, being a common point of the two circles, is another point with equal powers, so the radical axis of circles $ABDX$ and $ACEY$ is the altitude AF that passes through A'.



Solution 2. We present another way to prove that line APA' is the radical axis of the circles ABD and ACE. It suffices to show that the second intersection point of ABD and ACE lies on AP.

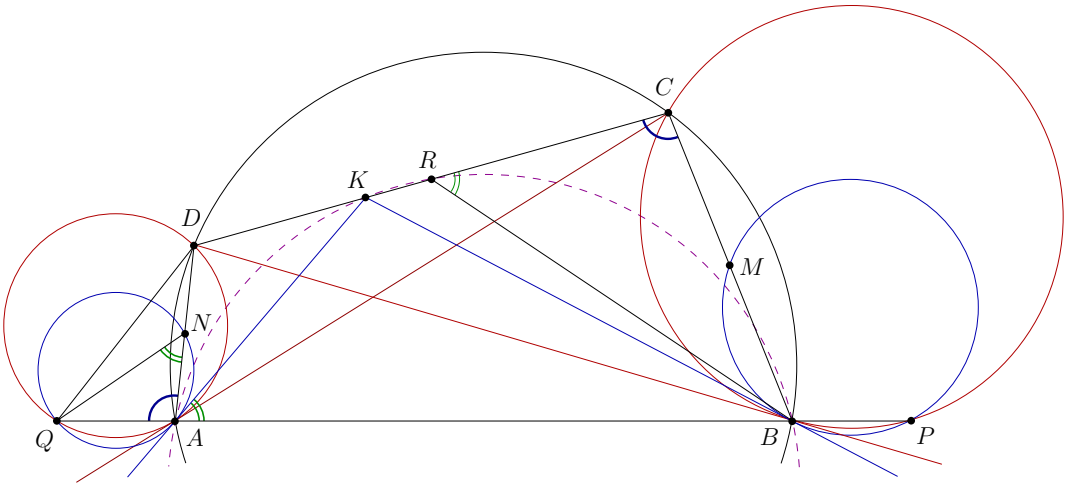
Define N to be the second intersection of circle PDE and AP. From $\angle DNA = \angle DNP = \angle DEP = \angle DBA$ it follows that N lies on circle ABD; analogously, we can show that N lies on circle ACE.

G3. Let $ABCD$ be a cyclic quadrilateral. Assume that the points Q, A, B, P are collinear in this order, in such a way that the line AC is tangent to the circle ADQ, and the line BD is tangent to the circle BCP. Let M and N be the midpoints of BC and AD, respectively. Prove that the following three lines are concurrent: line CD, the tangent of circle ANQ at point A, and the tangent to circle BMP at point B.

(*Slovakia*)

Solution 1. We first prove that triangles ADQ and CDB are similar. Since $ABCD$ is cyclic, we have $\angle DAQ = \angle DCB$. By the tangency of AC to the circle AQD we also have $\angle CBD = \angle CAD = \angle AQD$. The claimed similarity is proven.

Let R be the midpoint of CD. Points N and R correspond in the proven similarity, and so $\angle QNA = \angle BRC$.

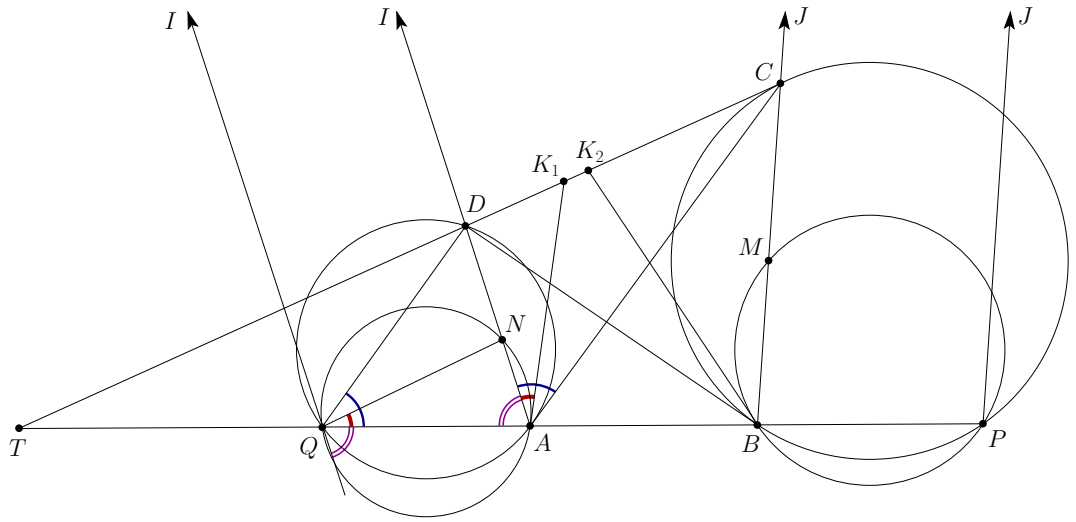


Let K be the second common point of line CD with circle ABR (i.e., if CD intersects circle ABR, then $K \neq R$ is the other intersection; otherwise, if CD is tangent to CD, then $K = R$). In both cases, we have $\angle BAK = \angle BRC = \angle QNA$; that indicates that AK is tangent to circle ANQ. It can be showed analogously that BK is tangent to circle BMP.

Comment. Note that M and N can be any points on lines BC and AD such that $BM : MC = DN : NA$, as we then simply choose R to be such that $DR : RC$ is that same ratio, and the rest of the proof remains unchanged.

Solution 2. We present a second solution, without using the condition that $ABCD$ is cyclic. Again, M and N can be any points on lines BC and AD such that $BM : MC = DN : NA$.

Let AB and CD meet at T (if $AB \parallel CD$ then T is their common ideal point). Let CD meet the tangent to the circle ANQ at A, and the tangent to the circle BMP at B at points K_1 and K_2, respectively.



Let I and J be the ideal points of AD and BC, respectively. Notice that the pencils (AD, AC, AT, AK_1) and (QA, QD, QI, QN) of lines are congruent, because $\angle K_1AD = \angle AQN$, $\angle CAD = \angle AQD$ and $\angle IAT = \angle IQT$. Hence,

$$(D, C; T, K_1) = (AD, AC; AT, AK_1) = (QA, QD; QI, QN) = (A, D; I, N) = \frac{DN}{NA}.$$

It can be obtained analogously that

$$(D, C; T, K_2) = (BD, BC; BT, BK_2) = (PC, PB; PJ, PM) = (C, B; I, N) = \frac{BM}{MC}.$$

From $BM : MC = DN : DA$ we get $(D, C; T, K_1) = (D, C; T, K_2)$ and hence $K_1 = K_2$.

G4. Let ABC be an acute-angled triangle with $AC > AB$, let O be its circumcentre, and let D be a point on the segment BC. The line through D perpendicular to BC intersects the lines AO, AC and AB at W, X and Y, respectively. The circumcircles of triangles AXY and ABC intersect again at $Z \neq A$.

Prove that if $OW = OD$, then DZ is tangent to the circle AXY.

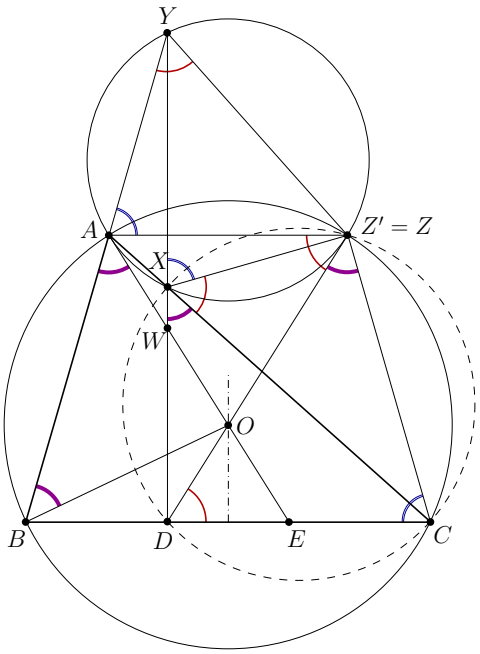
(*United Kingdom*)

Solution 1. Let AO intersect BC at E. As EDW is a right-angled triangle and O is on WE, the condition $OW = OD$ means O is the circumcentre of this triangle. So $OD = OE$ which establishes that D, E are reflections in the perpendicular bisector of BC.

Now observe:

$$180^\circ - \angle DXZ = \angle ZXY = \angle ZAY = \angle ZCD,$$

which shows $CDXZ$ is cyclic.



We next show that $AZ \parallel BC$. To do this, introduce point Z' on circle ABC such that $AZ' \parallel BC$. By the previous result, it suffices to prove that $CDXZ'$ is cyclic. Notice that triangles BAE and $CZ'D$ are reflections in the perpendicular bisector of BC. Using this and that A, O, E are collinear:

$$\angle DZ'C = \angle BAE = \angle BAO = 90^\circ - \tfrac{1}{2}\angle AOB = 90^\circ - \angle C = \angle DXC,$$

so $DXZ'C$ is cyclic, giving $Z \equiv Z'$ as desired.

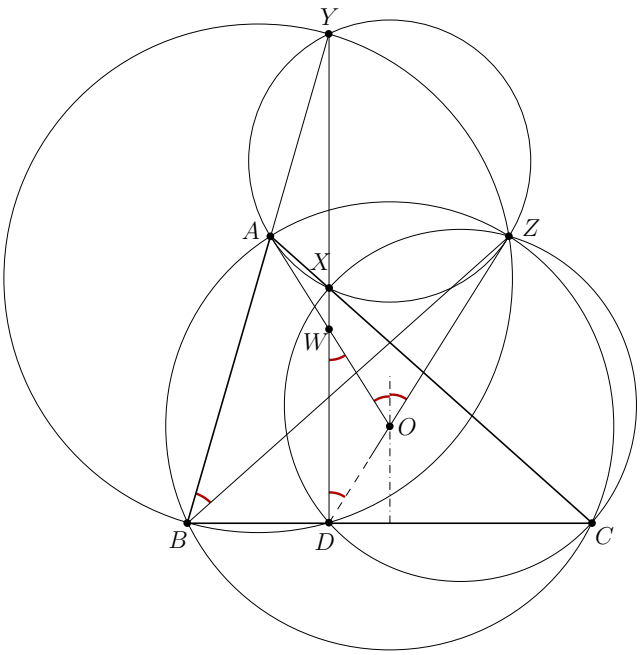
Using $AZ \parallel BC$ and $CDXZ$ cyclic we get:

$$\angle AZD = \angle CDZ = \angle CXZ = \angle AYZ,$$

which by the converse of alternate segment theorem shows DZ is tangent to circle AXY.

Solution 2. Notice that point Z is the Miquel-point of lines AC, BC, BA and DY; then B, D, Z, Y and C, D, X, Y are concyclic. Moreover, Z is the centre of the spiral similarity that maps BC to YX.

By $BC \perp YX$, the angle of that similarity is 90°; hence the circles $ABCZ$ and $AXYZ$ are perpendicular, therefore the radius OZ in circle $ABCZ$ is tangent to circle $AXYZ$.



By $OW = OD$, the triangle OWD is isosceles, and

$$\angle ZOA = 2\angle ZBA = 2\angle ZBY = 2\angle ZDY = \angle ODW + \angle DWO,$$

so D lies on line ZO that is tangent to circle AXY.

G5. Let ABC be a triangle, and let ℓ_1 and ℓ_2 be two parallel lines. For $i = 1, 2$, let ℓ_i meet the lines BC, CA, and AB at X_i, Y_i, and Z_i, respectively. Suppose that the line through X_i perpendicular to BC, the line through Y_i perpendicular to CA, and finally the line through Z_i perpendicular to AB, determine a non-degenerate triangle Δ_i.

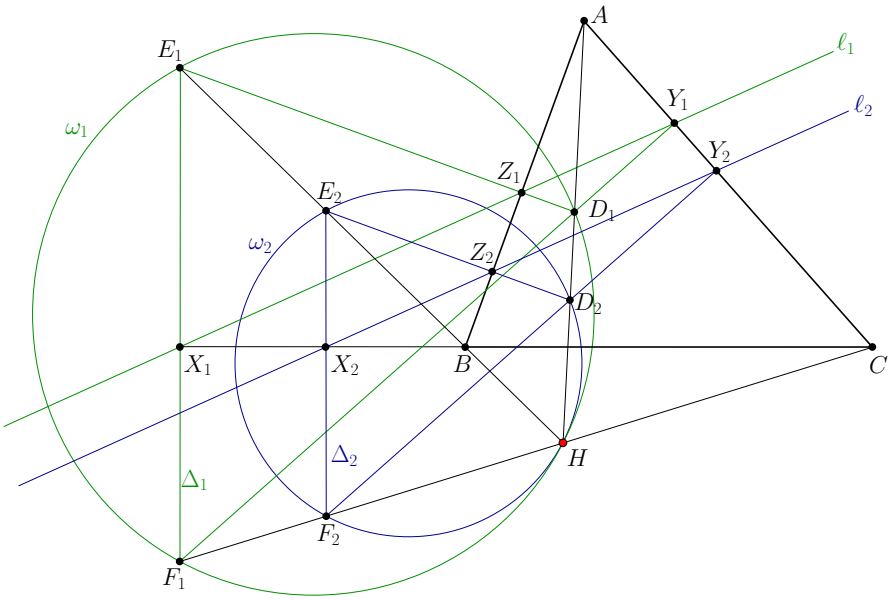
Show that the circumcircles of Δ_1 and Δ_2 are tangent to each other.

(Vietnam)

Solution 1. Throughout the solutions, $\measuredangle(p, q)$ will denote the directed angle between lines p and q, taken modulo 180°.

Let the vertices of Δ_i be D_i, E_i, F_i, such that lines E_iF_i, F_iD_i and D_iE_i are the perpendiculars through X, Y and Z, respectively, and denote the circumcircle of Δ_i by ω_i.

In triangles $D_1Y_1Z_1$ and $D_2Y_2Z_2$ we have $Y_1Z_1 \parallel Y_2Z_2$ because they are parts of ℓ_1 and ℓ_2. Moreover, $D_1Y_1 \parallel D_2Y_2$ are perpendicular to AC and $D_1Z_1 \parallel D_2Z_2$ are perpendicular to AB, so the two triangles are homothetic and their homothetic centre is $Y_1Y_2 \cap Z_1Z_2 = A$. Hence, line D_1D_2 passes through A. Analogously, line E_1E_2 passes through B and F_1F_2 passes through C.



The corresponding sides of Δ_1 and Δ_2 are parallel, because they are perpendicular to the respective sides of triangle ABC. Hence, Δ_1 and Δ_2 are either homothetic, or they can be translated to each other. Using that B, X_2, Z_2 and E_2 are concyclic, C, X_2, Y_2 and F_2 are concyclic, $Z_2E_2 \perp AB$ and $Y_2, F_2 \perp AC$ we can calculate

$$\begin{aligned}\measuredangle(E_1E_2, F_1F_2) &= \measuredangle(E_1E_2, X_1X_2) + \measuredangle(X_1X_2, F_1F_2) = \measuredangle(BE_2, BX_2) + \measuredangle(CX_2, CF_2)\\ &= \measuredangle(Z_2E_2, Z_2X_2) + \measuredangle(Y_2X_2, Y_2F_2) = \measuredangle(Z_2E_2, \ell_2) + \measuredangle(\ell_2, Y_2F_2)\\ &= \measuredangle(Z_2E_2, Y_2F_2) = \measuredangle(AB, AC) \not\equiv 0, \qquad (1)\end{aligned}$$

and conclude that lines E_1E_2 and F_1F_2 are not parallel. Hence, Δ_1 and Δ_2 are homothetic; the lines D_1D_2, E_1E_2, and F_1F_2 are concurrent at the homothetic centre of the two triangles. Denote this homothetic centre by H.

For $i = 1, 2$, using (1), and that A, Y_i, Z_i and D_i are concyclic,

$$\begin{aligned}\measuredangle(HE_i, HF_i) &= \measuredangle(E_1E_2, F_1F_2) = \measuredangle(AB, AC)\\ &= \measuredangle(AZ_i, AY_i) = \measuredangle(D_iZ_i, D_iY_i) = \measuredangle(D_iE_i, D_iF_i),\end{aligned}$$

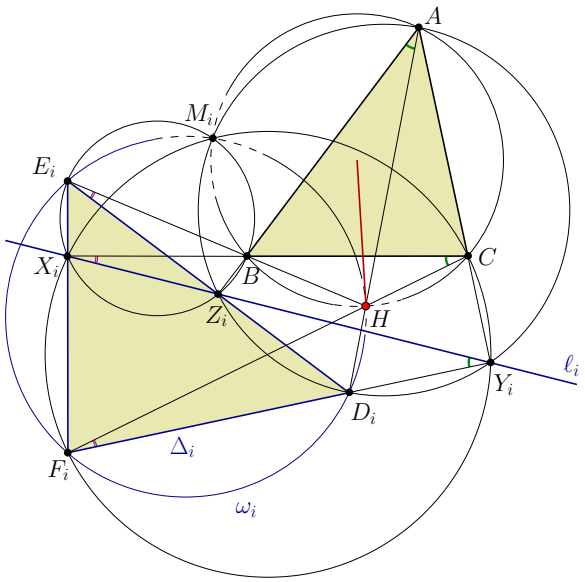
so H lies on circle ω_i.

The same homothety that maps Δ_1 to Δ_2, sends ω_1 to ω_2 as well. Point H, that is the centre of the homothety, is a common point of the two circles, That finishes proving that ω_1 and ω_2 are tangent to each other.

Solution 2. As in the first solution, let the vertices of Δ_i be D_i, E_i, F_i, such that E_iF_i, F_iD_i and D_iE_i are the perpendiculars through X_i, Y_i and Z_i, respectively. In the same way we conclude that (A, D_1, D_2), (B, E_1, E_2) and (C, F_1, F_2) are collinear.

The corresponding sides of triangles ABC and $D_iE_iF_i$ are perpendicular to each other. Hence, there is a spiral similarity with rotation $\pm 90^\circ$ that maps ABC to $D_iE_iF_i$; let M_i be the centre of that similarity. Hence, $\measuredangle(M_iA, M_iD_i) = \measuredangle(M_iB, M_iE_i) = \measuredangle(M_iC, M_iF_i) = 90^\circ$. The circle with diameter AD_i passes through M_i, Y_i, Z_i, so M_i, A, Y_i, Z_i, D_i are concyclic; analogously (M_i, B, X_i, Z_i, E_i) and (M_i, C, X_i, Y_i, F_i) are concyclic.

By applying Desargues' theorem to triangles ABC and $D_iE_iF_i$ we conclude that the lines AD_i, BE_i and BF_i are concurrent; let their intersection be H. Since $(A, D_1.D_2)$, $(B, E_1.E_2)$ and $(C, F_1.F_2)$ are collinear, we obtain the same point H for $i = 1$ and $i = 2$.



By $\measuredangle(CB, CH) = \measuredangle(CX_i, CF_i) = \measuredangle(Y_iX_i, Y_iF_i) = \measuredangle(Y_iZ_i, Y_iD_i) = \measuredangle(AZ_i, AD_i) = \measuredangle(AB, AH)$, point H lies on circle ABC.

Analogously, from $\measuredangle(F_iD_i, F_iH) = \measuredangle(F_iY_i, F_iC) = \measuredangle(X_iY_i, X_iC) = \measuredangle(X_iZ_i, X_iB) = \measuredangle(E_iZ_i, E_iB) = \measuredangle(E_iD_i, E_iH)$, we can see that point H lies on circle $D_iE_iF_i$ as well. Therefore, circles ABC and $D_iE_iF_i$ intersect at point H.

The spiral similarity moves the circle ABC to circle $D_iE_iF_i$, so the two circles are perpendicular. Hence, both circles $D_1E_1F_1$ and $D_2E_2F_2$ are tangent to the radius of circle ABC at H.

Comment 1. As the last picture suggests, the circles ABC and $D_iE_iF_i$ pass through M_i. In fact, point M_i, being the second intersection of circles $D_iE_iF_i$ and $D_iY_iZ_i$, the Miquel point of the lines AY_i, AZ_i, CX_i and X_iY_i, so it is concyclic with A, B, C. Similarly, M_i the Miquel point of lines D_iE_i, E_iF_i, F_iY_i and X_iY_i, so it is concyclic with D_i, E_i, D_i.

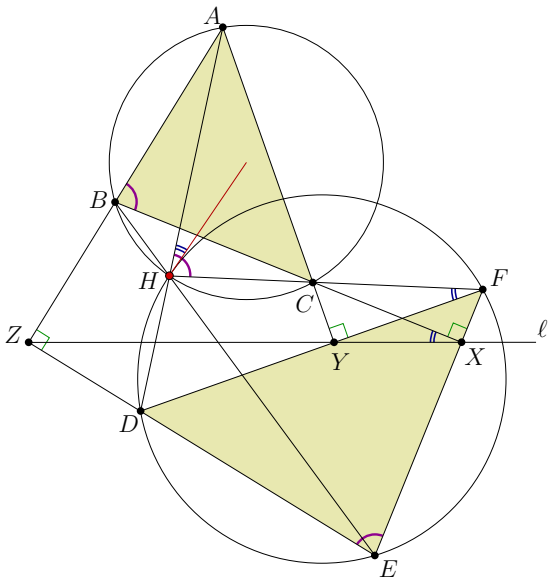
Comment 2. Instead of two lines ℓ_1 and ℓ_2, it is possible to reformulate the problem with a single, varying line ℓ:

Let ABC be a triangle, and let ℓ be a varying line whose direction is fixed. Let ℓ intersect lines BC, CA, and AB at X, Y, and Z, respectively. Suppose that the line through X, perpendicular to BC, the line through Y, perpendicular to CA, and the line through Z, perpendicular to AB, determine a non-degenerate triangle Δ.

Show that as ℓ varies, the circumcircles of the obtained triangles Δ pass through a fixed point, and these circles are tangent to each other.

A reasonable approach is finding the position of line ℓ when the triangle DEF degenerates to a single point. That happens when the line XYZ is the Simson line respect to point $D = E = F$ on the circumcircle ABC. Based on this observation a possible variant of the solution is as follows.

Let H be the second intersection of circles ABC and line AD. Like in the solutions above, we can find that the line AD is fixed, so H is independent of the position of line ℓ.



From $\measuredangle(HF, HD) = \measuredangle(HC, HA) = \measuredangle(BC, BA) = \measuredangle(BX, BZ) = \measuredangle(EX, EZ) = \measuredangle(EF, ED)$ we can see that circle Δ passes through H. Hence, all circles DEF passes through a fixed point.

The corresponding sides of triangles ABC and DEF are perpendicular, so their circumcircle are perpendicular; that proves that circle DEF is tangent to the radius of circle ABC at H.

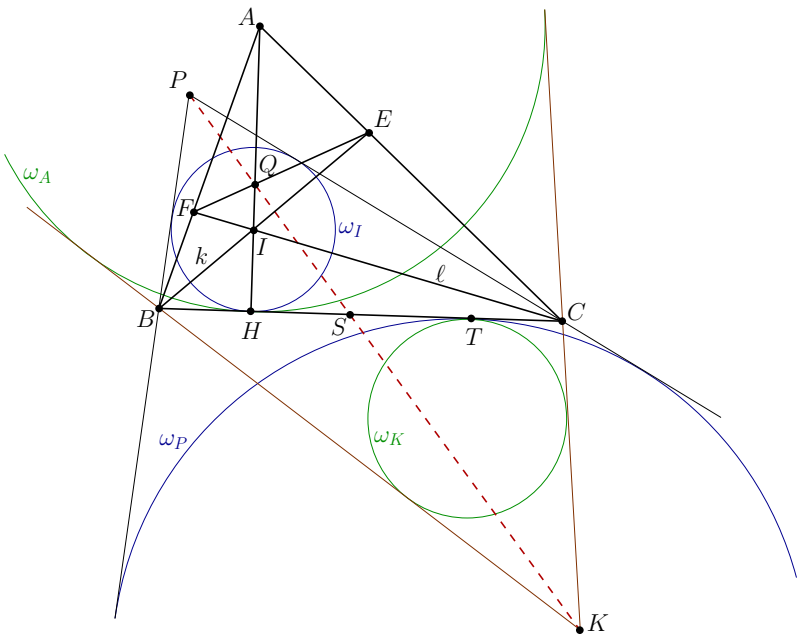
G6. In an acute-angled triangle ABC, point H is the foot of the altitude from A. Let P be a moving point such that the bisectors k and ℓ of angles PBC and PCB, respectively, intersect each other on the line segment AH. Let k and AC meet at E, let ℓ and AB meet at F, and let EF and AH meet at Q. Prove that, as P varies, the line PQ passes through a fixed point.

(Iran)

Solution 1. Let the reflections of the line BC with respect to the lines AB and AC intersect at point K. We will prove that P, Q and K are collinear, so K is the common point of the varying line PQ.

Let lines BE and CF intersect at I. For every point O and $d > 0$, denote by (O, d) the circle centred at O with radius d, and define $\omega_I = (I, IH)$ and $\omega_A = (A, AH)$. Let ω_K and ω_P be the incircle of triangle KBC and the P-excircle of triangle PBC, respectively.

Since $IH \perp BC$ and $AH \perp BC$, the circles ω_A and ω_I are tangent to each other at H. So, H is the external homothetic centre of ω_A and ω_I. From the complete quadrangle $BCEF$ we have $(A, I; Q, H) = -1$, therefore Q is the internal homothetic centre of ω_A and ω_I. Since BA and CA are the external bisectors of angles $\angle KBC$ and $\angle KCB$, circle ω_A is the K-excircle in triangle BKC. Hence, K is the external homothetic centre of ω_A and ω_K. Also it is clear that P is the external homothetic centre of ω_I and ω_P. Let point T be the tangency point of ω_P and BC, and let T' be the tangency point of ω_K and BC. Since ω_I is the incircle and ω_P is the P-excircle of PBC, $TC = BH$ and since ω_K is the incircle and ω_A is the K-excircle of KBC, $T'C = BH$. Therefore $TC = T'C$ and $T \equiv T'$. It yields that ω_K and ω_P are tangent to each other at T.



Let point S be the internal homothetic centre of ω_A and ω_P, and let S' be the internal homothetic centre of ω_I and ω_K. It's obvious that S and S' lie on BC. We claim that $S \equiv S'$. To prove our claim, let r_A, r_I, r_P, and r_K be the radii of ω_A, ω_I, ω_P and ω_k, respectively.

It is well known that if the sides of a triangle are a, b, c, its semiperimeter is $s = (a+b+c)/2$, and the radii of the incircle and the a-excircle are r and r_a, respectively, then $r \cdot r_a = (s-b)(s-c)$. Applying this fact to triangle PBC we get $r_I \cdot r_P = BH \cdot CH$. The same fact in triangle KCB

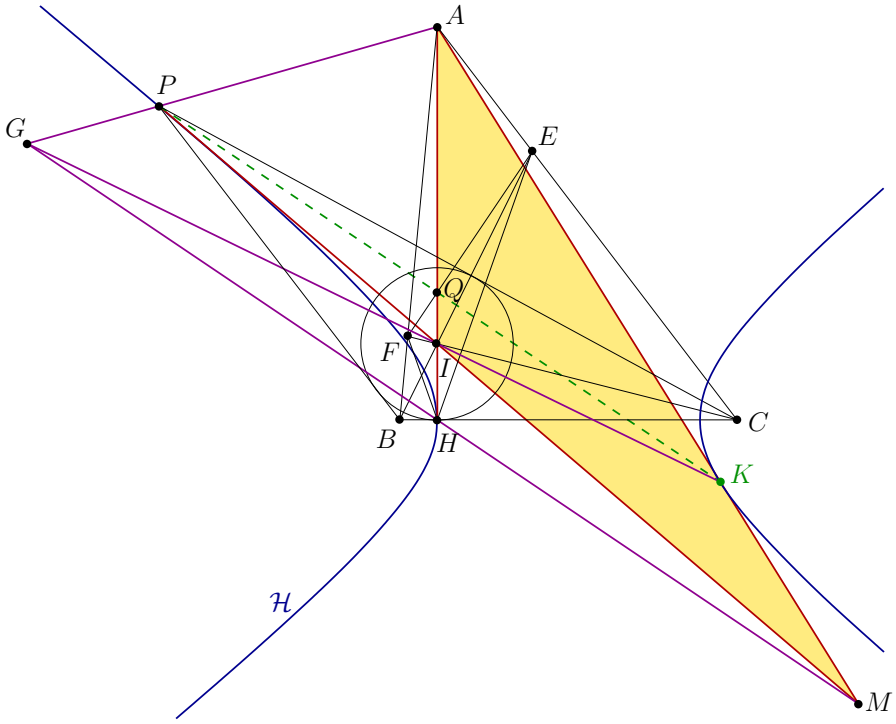
yields $r_K \cdot r_A = CT \cdot BT$. Since $BH = CT$ and $BT = CH$, from these two we get

$$\frac{HS}{ST} = \frac{r_A}{r_P} = \frac{r_I}{r_K} = \frac{HS'}{S'T},$$

so $S = S'$ indeed.

Finally, by applying the generalised Monge's theorem to the circles ω_A, ω_I, and ω_K (with two pairs of internal and one pair of external common tangents), we can see that points Q, S, and K are collinear. Similarly one can show that Q, S and P are collinear, and the result follows.

Solution 2. Again, let BE and CF meet at I, that is the incentre in triangle BCP; then PI is the third angle bisector. From the tangent segments of the incircle we have $BP - CP = BH - CH$; hence, the possible points P lie on a branch of a hyperbola $\mathcal{H}$ with foci B, C, and H is a vertex of $\mathcal{H}$. Since PI bisects the angle between the radii BP and CP of the hyperbola, line PI is tangent to $\mathcal{H}$.



Let K be the second intersection of PQ and $\mathcal{H}$, we will show that AK is tangent to $\mathcal{H}$ at K; this property determines K.

Let $G = KI \cap AP$ and $M = PI \cap AK$. From the complete quadrangle $BCEF$ we can see that $(H, Q; I, A)$ is harmonic, so in the complete quadrangle $APIK$, point H lies on line GM.

Consider triangle AIM. Its side AI is tangent to $\mathcal{H}$ at H, the side IM is tangent to $\mathcal{H}$ at P, and K is a common point of the third side AM and the hyperbola such that the lines AP, IK and MH are concurrent at the generalised Gergonne-point G. It follows that the third side, AM is also tangent to $\mathcal{H}$ at K.

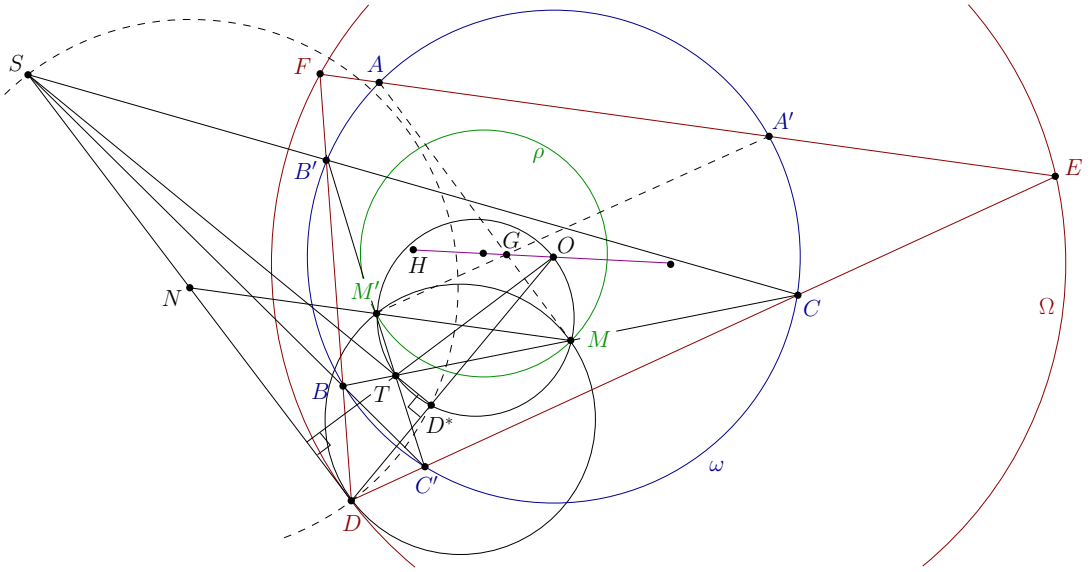
(Alternatively, in the last step we can apply the converse of Brianchon's theorem to the degenerate hexagon $AHIPMK$. By the theorem there is a conic section $\mathcal{H}'$ such that lines AI, IM and MA are tangent to $\mathcal{H}'$ at H, P and K, respectively. But the three points H, K and P, together with the tangents at H and P uniquely determine $\mathcal{H}'$, so indeed $\mathcal{H}' = \mathcal{H}$.)

G7. Let ABC and $A'B'C'$ be two triangles having the same circumcircle ω, and the same orthocentre H. Let Ω be the circumcircle of the triangle determined by the lines AA', BB' and CC'. Prove that H, the centre of ω, and the centre of Ω are collinear.

(Denmark)

Solution. In what follows, $\measuredangle(p, q)$ will denote the directed angle between lines p and q, taken modulo $180°$. Denote by O the centre of ω. In any triangle, the homothety with ratio $-\frac{1}{2}$ centred at the centroid of the triangle takes the vertices to the midpoints of the opposite sides and it takes the orthocentre to the circumcentre. Therefore the triangles ABC and $A'B'C'$ share the same centroid G and the midpoints of their sides lie on a circle ρ with centre on OH. We will prove that ω, Ω, and ρ are coaxial, so in particular it follows that their centres are collinear on OH.

Let $D = BB' \cap CC'$, $E = CC' \cap AA'$, $F = AA' \cap BB'$, $S = BC' \cap B'C$, and $T = BC \cap B'C'$. Since D, S, and T are the intersections of opposite sides and of the diagonals in the quadrilateral $BB'CC'$ inscribed in ω, by Brocard's theorem triangle DST is self-polar with respect to ω, i.e. each vertex is the pole of the opposite side. We apply this in two ways.



First, from D being the pole of ST it follows that the inverse D^* of D with respect to ω is the projection of D onto ST. In particular, D^* lies on the circle with diameter SD. If N denotes the midpoint of SD and R the radius of ω, then the power of O with respect to this circle is $ON^2 - ND^2 = OD \cdot OD^* = R^2$. By rearranging, we see that ND^2 is the power of N with respect to ω.

Second, from T being the pole of SD it follows that OT is perpendicular to SD. Let M and M' denote the midpoints of BC and $B'C'$. Then since $OM \perp BC$ and $OM' \perp B'C'$ it follows that $OMM'T$ is cyclic and

$$\measuredangle(SD, BC) = \measuredangle(OT, OM) = \measuredangle(B'C', MM').$$

From $BB'CC'$ being cyclic we also have $\measuredangle(BC, BB') = \measuredangle(CC', B'C')$, hence we obtain

$$\begin{aligned}\measuredangle(SD, BB') &= \measuredangle(SD, BC) + \measuredangle(BC, BB') \\ &= \measuredangle(B'C', MM') + \measuredangle(CC', B'C') = \measuredangle(CC', MM').\end{aligned}$$

Now from the homothety mentioned in the beginning, we know that MM' is parallel to AA', hence the above implies that $\measuredangle(SD, BB') = \measuredangle(CC', AA')$, which shows that Ω is tangent to SD at D. In particular, ND^2 is also the power of N with respect to Ω.

Additionally, from $BB'CC'$ being cyclic it follows that triangles DBC and $DC'B'$ are inversely similar, so $\measuredangle(BB', DM') = \measuredangle(DM, CC')$. This yields

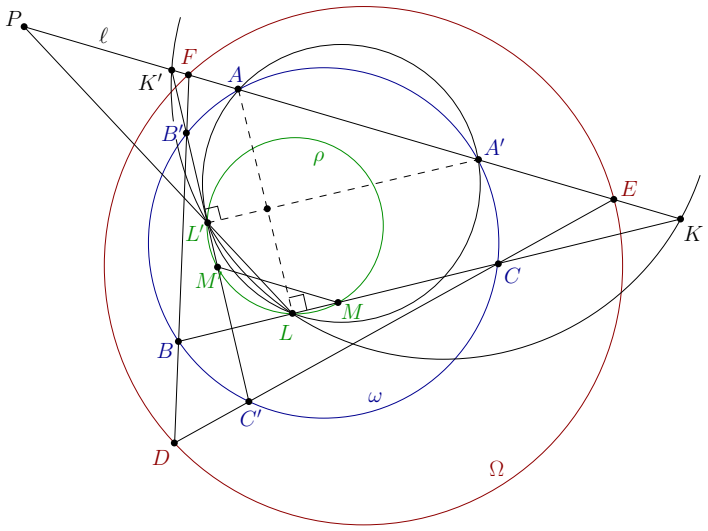
$$\begin{aligned}\measuredangle(SD, DM') &= \measuredangle(SD, BB') + \measuredangle(BB', DM') \\ &= \measuredangle(CC', MM') + \measuredangle(DM, CC') = \measuredangle(DM, MM'),\end{aligned}$$

which shows that the circle DMM' is also tangent to SD. Since N, M, and M' are collinear on the Newton–Gauss line of the complete quadrilateral determined by the lines BB', CC', BC', and $B'C$, it follows that $ND^2 = NM \cdot NM'$. Hence N has the same power with respect to ω, Ω, and ρ.

By the same arguments there exist points on the tangents to Ω at E and F which have the same power with respect to ω, Ω, and ρ. The tangents to a given circle at three distinct points cannot be concurrent, hence we obtain at least two distinct points with the same power with respect to ω, Ω, and ρ. Hence the three circles are coaxial, as desired.

Comment 1. Instead of invoking the Newton–Gauss line, one can also use a nice symmetry argument: If from the beginning we swapped the labels of B' and C', then in the proof above the labels of D and S would be swapped while the labels of M and M' do not change. The consequence is that the circle SMM' is also tangent to SD. Since N is the midpoint of SD it then has the same power with respect to circles DMM' and SMM', so it lies on their radical axis MM'.

Comment 2. There exists another triple of points on the common radical axis of ω, Ω, and ρ which can be used to solve the problem. We outline one such solution.



Let L and L' denote the feet of the altitudes from A and A' in triangle ABC and $A'B'C'$, respectively. Since ρ is the nine-point circle of the two triangles it contains both L and L' . Furthermore, $HA \cdot HL$ and $HA' \cdot HL'$ both equal twice the power of H with respect to ρ, so A, A', L, L' are concyclic as well.

Now let $\ell = AA'$ and denote $P = LL' \cap \ell$, $K = BC \cap \ell$, and $K' = B'C' \cap \ell$. As $MM' \parallel \ell$ (shown in the previous solution) and $LL'MM'$ is cyclic

$$\measuredangle(BC, \ell) = \measuredangle(BC, MM') = \measuredangle(LL', B'C')$$

so K, K', L, and L' are also concyclic. From the cyclic quadrilaterals $AA'LL'$ and $KK'LL'$ we get $PA \cdot PA' = PL \cdot PL' = PK \cdot PK'$. This implies that P is the centre of the (unique) involution σ on ℓ that swaps A, A' and K, K'. On the other hand, by Desargues' involution theorem applied to the line ℓ, the quadrilateral $BB'CC'$, and its circumcircle ω, the involution σ also swaps E and F. Hence

$$PA \cdot PA' = PL \cdot PL' = PE \cdot PF.$$

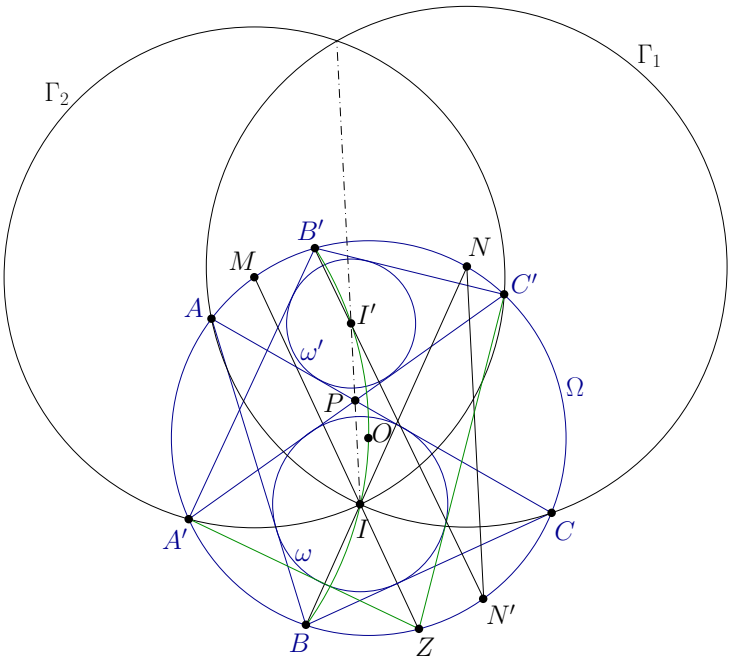
However, this means that P has the same power with respect to ω, Ω, and ρ, and by the same arguments there exist points on BB' and CC' with this property.

G8. Let $AA'BCC'B'$ be a convex cyclic hexagon such that AC is tangent to the incircle of the triangle $A'B'C'$, and $A'C'$ is tangent to the incircle of the triangle ABC. Let the lines AB and $A'B'$ meet at X and let the lines BC and $B'C'$ meet at Y.

Prove that if $XBYB'$ is a convex quadrilateral, then it has an incircle.

(*Australia*)

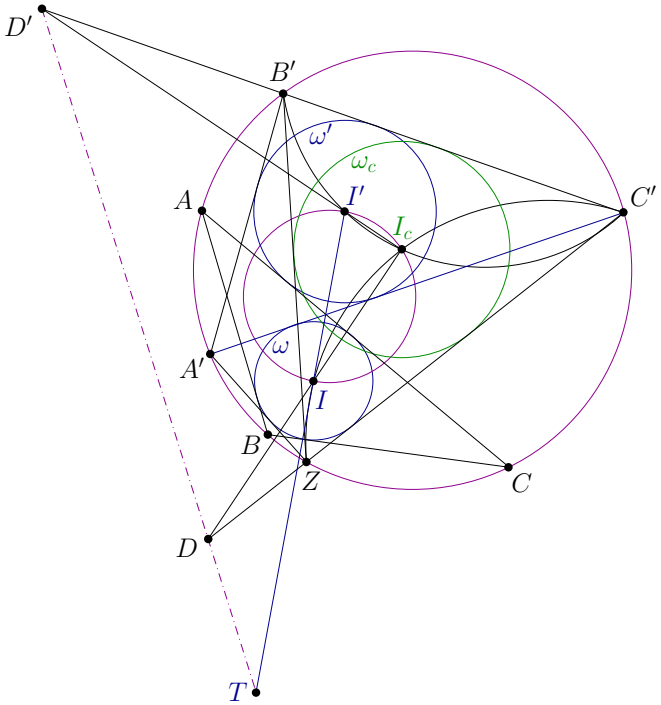
Solution. Denote by ω and ω' the incircles of $\triangle ABC$ and $\triangle A'B'C'$ and let I and I' be the centres of these circles. Let N and N' be the second intersections of BI and $B'I'$ with Ω, the circumcircle of $A'BCC'B'A$, and let O be the centre of Ω. Note that $ON \perp AC$, $ON' \perp A'C'$ and $ON = ON'$ so NN' is parallel to the angle bisector II' of AC and $A'C'$. Thus $II' \parallel NN'$ which is antiparallel to BB' with respect to BI and $B'I'$. Therefore B, I, I', B' are concyclic.



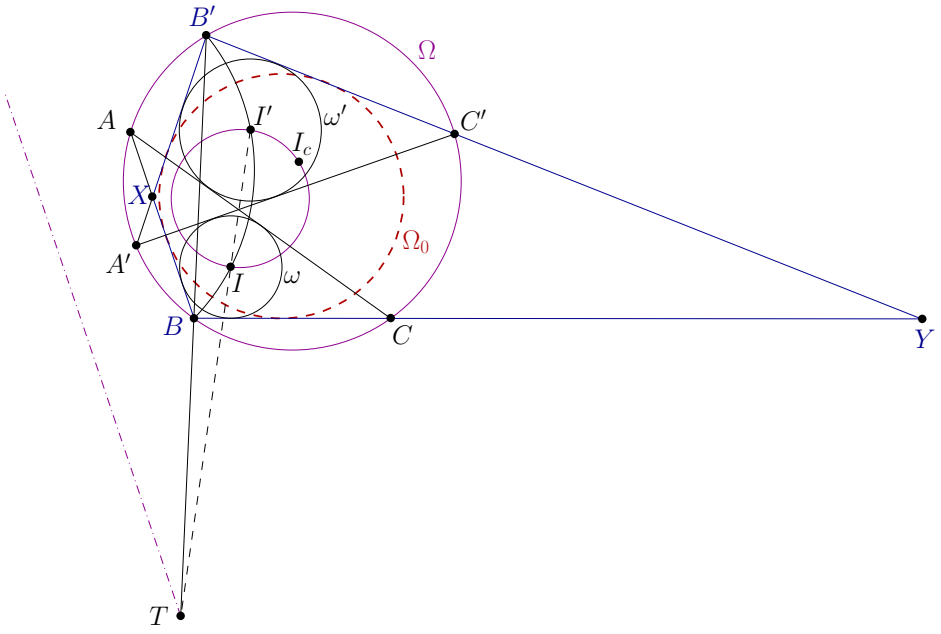
Further define P as the intersection of AC and $A'C'$ and M as the antipode of N' in Ω. Consider the circle Γ_1 with centre N and radius $NA = NC$ and the circle Γ_2 with centre M and radius $MA' = MC'$. Their radical axis passes through P and is perpendicular to $MN \perp NN' \parallel IP$, so I lies on their radical axis. Therefore, since I lies on Γ_1, it must also lie on Γ_2. Thus, if we define Z as the second intersection of MI with Ω, we have that I is the incentre of triangle $ZA'C'$. (Note that the point Z can also be constructed directly via Poncelet's porism.)

Consider the incircle ω_c with centre I_c of triangle $C'B'Z$. Note that $\angle ZIC' = 90^\circ + \frac{1}{2}\angle ZA'C' = 90^\circ + \frac{1}{2}\angle ZB'C' = \angle ZI_cC'$, so Z, I, I_c, C' are concyclic. Similarly B', I', I_c, C' are concyclic.

The external centre of dilation from ω to ω_c is the intersection of II_c and $C'Z$ (D in the picture), that is the radical centre of circles Ω, $C'I_cIZ$ and $II'I_c$. Similarly, the external centre of dilation from ω' to ω_c is the intersection of $I'I_c$ and $B'C'$ (D' in the picture), that is the radical centre of circles Ω, $B'I'I_cC'$ and $II'I_c$. Therefore the Monge line of ω, ω' and ω_c is line DD', and the radical axis of Ω and circle $II'I_c$ coincide. Hence the external centre T of dilation from ω to ω' is also on the radical axis of Ω and circle $II'I_c$.



Now since B, I, I', B' are concyclic, the intersection T' of BB' and II' is on the radical axis of Ω and circle $II'I_c$. Thus $T' = T$ and T lies on line BB'. Finally, construct a circle Ω_0 tangent to $A'B'$, $B'C'$, AB on the same side of these lines as ω'. The centre of dilation from ω' to Ω_0 is B', so by Monge's theorem the external centre of dilation from Ω_0 to ω must be on the line TBB'. However, it is on line AB, so it must be B and BC must be tangent to Ω_0 as desired.



Number Theory

N1. A number is called *Norwegian* if it has three distinct positive divisors whose sum is equal to 2022. Determine the smallest Norwegian number.

(Note: The total number of positive divisors of a Norwegian number is allowed to be larger than 3.)

(*Cyprus*)

Answer: 1344

Solution. Observe that 1344 is a Norwegian number as 6, 672 and 1344 are three distinct divisors of 1344 and $6 + 672 + 1344 = 2022$. It remains to show that this is the smallest such number.

Assume for contradiction that $N < 1344$ is Norwegian and let N/a, N/b and N/c be the three distinct divisors of N, with $a < b < c$. Then

$$2022 = N\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) < 1344\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$$

and so

$$\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) > \frac{2022}{1344} = \frac{337}{224} = \frac{3}{2} + \frac{1}{224}.$$

If $a > 1$ then

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leqslant \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{13}{12} < \frac{3}{2},$$

so it must be the case that $a = 1$. Similarly, it must hold that $b < 4$ since otherwise

$$1 + \frac{1}{b} + \frac{1}{c} \leqslant 1 + \frac{1}{4} + \frac{1}{5} < \frac{3}{2}.$$

This leaves two cases to check, $b = 2$ and $b = 3$.

Case $b = 3$. Then

$$\frac{1}{c} > \frac{3}{2} + \frac{1}{224} - 1 - \frac{1}{3} > \frac{1}{6},$$

so $c = 4$ or $c = 5$. If $c = 4$ then

$$2022 = N\left(1 + \frac{1}{3} + \frac{1}{4}\right) = \frac{19}{12}N,$$

but this is impossible as $19 \nmid 2022$. If $c = 5$ then

$$2022 = N\left(1 + \frac{1}{3} + \frac{1}{5}\right) = \frac{23}{15}N,$$

which again is impossible, as $23 \nmid 2022$.

Case $b = 2$. Note that $c < 224$ since

$$\frac{1}{c} > \frac{3}{2} + \frac{1}{224} - 1 - \frac{1}{2} = \frac{1}{224}.$$

It holds that

$$2022 = N\left(1 + \frac{1}{2} + \frac{1}{c}\right) = \frac{3c+2}{2c}N \Rightarrow (3c+2)N = 4044c.$$

Since $(c, 3c-2) = (c, 2) \in \{1, 2\}$, then $3c+2 \mid 8088 = 2^3 \cdot 3 \cdot 337$ which implies that $3c+2 \mid 2^3 \cdot 337$. But since $3c + 2 \geqslant 3 \cdot 3 + 2 > 8 = 2^3$ and $3c + 2 \neq 337$, then it must hold that $3c + 2 \geqslant 2 \cdot 337$, contradicting $c < 224$.

N2. Find all positive integers $n > 2$ such that

$$n! \Bigg| \prod_{\substack{p<q\leqslant n,\\ p,q \text{ primes}}} (p+q).$$

(Nigeria)

Answer: This only holds for $n = 7$.

Solution. Assume that n satisfies $n! \mid \prod_{p<q\leqslant n}(p+q)$ and let $2 = p_1 < p_2 < \cdots < p_m \leqslant n$ be the primes in $\{1, 2, \ldots, n\}$. Each such prime divides $n!$. In particular, $p_m \mid p_i + p_j$ for some $p_i < p_j \leqslant n$. But

$$0 < \frac{p_i + p_j}{p_m} < \frac{p_m + p_m}{p_m} = 2,$$

so $p_m = p_i + p_j$ which implies $m \geqslant 3$, $p_i = 2$ and $p_m = 2 + p_j = 2 + p_{m-1}$.

Similarly, $p_{m-1} \mid p_k + p_l$ for some $p_k < p_l \leqslant n$. But

$$0 < \frac{p_l + p_k}{p_{m-1}} \leqslant \frac{p_m + p_{m-1}}{p_{m-1}} = \frac{2p_{m-1} + 2}{p_{m-1}} < 3,$$

so either $p_{m-1} = p_l + p_k$ or $2p_{m-1} = p_l + p_k$. As above, the former case gives $p_{m-1} = 2 + p_{m-2}$. If $2p_{m-1} = p_l + p_k$, then $p_{m-1} < p_k$, so $k = m$ and

$$2p_{m-1} = p_l + p_{m-1} + 2 \Rightarrow p_{m-1} = p_l + 2 = p_{m-2} + 2.$$

Either way, $p_{m-2} > 2$ and 3 divides one of p_{m-2}, $p_{m-1} = p_{m-2} + 2$ and $p_m = p_{m-2} + 4$. This implies $p_{m-2} = 3$ and thus $p_m = 7$, giving $7 \leqslant n < 11$.

Finally, a quick computation shows that $7! \mid \prod_{p<q\leqslant 7}(p+q)$ but $8! \nmid \prod_{p<q\leqslant 7}(p+q)$, so neither does $9!$ and $10!$.

N3. Let $a > 1$ be a positive integer, and let $d > 1$ be a positive integer coprime to a. Let $x_1 = 1$ and, for $k \geqslant 1$, define

$$x_{k+1} = \begin{cases} x_k + d & \text{if } a \text{ doesn't divide } x_k, \\ x_k/a & \text{if } a \text{ divides } x_k. \end{cases}$$

Find the greatest positive integer n for which there exists an index k such that x_k is divisible by a^n.

(Croatia)

Answer: n is the exponent with $d < a^n < ad$.

Solution 1. By trivial induction, x_k is coprime to d.

By induction and the fact that there can be at most $a - 1$ consecutive increasing terms in the sequence, it also holds that $x_k < da$ if $x_k = x_{k-1} + d$ and that $x_k < d$ if $x_k = \frac{x_{k-1}}{a}$ or $k = 1$. This gives the upper bound on the exponent.

This implies that the sequence is (eventually) periodic, and that both increasing and decreasing steps happen infinitely many times. Let a^{-k} be the multiplicative inverse of a^k modulo d. The sequence contains elements congruent to $1, a^{-1}, a^{-2}, \ldots$ modulo d.

Let x_{k_0} the first element such that $x_{k_0} \equiv a^{-n} \pmod{d}$. We have either $k_0 = 1$ or $x_{k_0} = x_{k_0-1}/a$; in both cases $x_{k_0} < d < a^n < da$ and therefore

$$x_{k_0} \in \{a^n - d, a^n - 2d, \ldots, a^n - (a-1)d\}.$$

In this set no element is divisible by a, so therefore the sequence will visit the value a^n in the next $a - 1$ steps.

Solution 2. Like in the first solution, x_k is relatively prime to d and $x_k < ad$ for all k.

Let

$$S = \{x \in \mathbb{Z}_{>0} : \ 0 < x < ad, \ \gcd(x, d) = 1\} \quad \text{and} \quad f : S \to S, \ f(x) = \begin{cases} x + d & \text{if } a \nmid x, \\ x/a & \text{if } a \mid x. \end{cases}$$

So, we have $x_1 = 1$ and $x_{k+1} = f(x_k)$.

Notice that the recurrence formula is invertible. Suppose that $f(x) = y$ for some $x, y \in S$. If $y > d$ then $y - d \in S$ but $a \cdot y \notin S$; otherwise, if $y < d$ then $a \cdot y \in S$ but $y - d \notin S$. So, the function f is a permutation on S, and its inverse is

$$f^{-1}(y) = \begin{cases} y - d & \text{if } y > d, \\ a \cdot y & \text{if } y < d \end{cases} \quad \text{for all } y \in S.$$

It follows that the sequence is periodic, and therefore attains the initial value 1 infinitely many times. Let $k_1 > 1$ be an index such that $x_{k_1} = 1$. Then

$$x_{k_1} = 1, \quad x_{k_1-1} = f^{-1}(1) = a, x_{k_1-2} = f^{-1}(a) = a^2, \ldots, \quad x_{k_1-n} = a^n.$$

Solution 3. Like in the first solution, x_k is relatively prime to d and $x_k < ad$ for all k.

We wish to prove that there are n consecutive decreasing indices. Let $m_0 = 0$ and m_k be the k-th smallest decreasing index (an index k is decreasing if $x_k = x_{k-1}/a$) and define a sequence $(y_k)_k = (x_{m_k})_k$, i.e. the subsequence consisting only of elements with decreasing indices. For each $k \geqslant 1$ we can write

$$y_{k+1} = \frac{y_k + z_k d}{a}$$

for some $z_k \in \{0, 1, \ldots, a-1\}$, where $z_k = 0$ if and only if y_k and y_{k+1} are consecutive elements in the original sequence. Then

$$y_k = \frac{1 + (z_0 + z_1 a + \cdots + z_{k-1} a^{k-1})d}{a^k},$$

because y_k is bounded the sequence is (eventually) periodic. Consider some $y_u = y_{u+v}$ where $v \geqslant 1$. We can write

$$y_u = \frac{1 + (z_0 + z_1 a + \cdots + z_{u-1} a^{u-1})d}{a^u}$$

and

$$y_{u+v} = \frac{1 + (z_0 + z_1 a + \cdots + z_{u+v-1} a^{u+v-1})d}{a^{u+v}}$$

and so

$$\frac{1 + (z_0 + z_1 a + \cdots + z_{u+v-1} a^{u+v-1})d}{a^{u+v}} = \frac{1 + (z_0 + z_1 a + \cdots + z_{u-1} a^{u-1})d}{a^u}.$$

Rearranging gives

$$\frac{a^v - 1}{d} = z_0 + z_1 a + \cdots + z_{v-1} a^{v-1} + (z_v - z_0) a^v + \cdots + (z_{u+v-1} - z_{u-1}) a^{u+v-1}.$$

Since $d > a^n/a = a^{n-1}$ the LHS is strictly less than a^{v-n}. This implies that on the RHS, the coefficients of $a^{v-n}, a^{v-n+1}, \ldots$ must all be zero, i.e. $z_{v-n} = z_{v-n+1} = \cdots = z_{v-1} = 0$. This implies that there are n consecutive decreasing indices in the original sequence.

N4. Find all triples of positive integers (a, b, p) with p prime and

$$a^p = b! + p.$$

(Belgium)

Answer: $(2, 2, 2)$ and $(3, 4, 3)$.

Solution 1. Clearly, $a > 1$. We consider three cases.

Case 1: We have $a < p$. Then we either have $a \leqslant b$ which implies $a \mid a^p - b! = p$ leading to a contradiction, or $a > b$ which is also impossible since in this case we have $b! \leqslant a! < a^p - p$, where the last inequality is true for any $p > a > 1$.

Case 2: We have $a > p$. In this case $b! = a^p - p > p^p - p \geqslant p!$ so $b > p$ which means that $a^p = b! + p$ is divisible by p. Hence, a is divisible by p and $b! = a^p - p$ is not divisible by p^2. This means that $b < 2p$. If $a < p^2$ then $a/p < p$ divides both a^p and $b!$ and hence it also divides $p = a^p - b!$ which is impossible. On the other hand, the case $a \geqslant p^2$ is also impossible since then $a^p \geqslant (p^2)^p > (2p-1)! + p \geqslant b! + p$.

Comment. The inequality $p^{2p} > (2p-1)! + p$ can be shown e.g. by using

$$(2p-1)! = [1 \cdot (2p-1)] \cdot [2 \cdot (2p-2)] \cdot \cdots \cdot [(p-1)(p+1)] \cdot p < \left(\left(\frac{2p}{2}\right)^2\right)^{p-1} \cdot p = p^{2p-1},$$

where the inequality comes from applying AM-GM to each of the terms in square brackets.

Case 3: We have $a = p$. In this case $b! = p^p - p$. One can check that the values $p = 2, 3$ lead to the claimed solutions and $p = 5$ does not lead to a solution. So we now assume that $p \geqslant 7$. We have $b! = p^p - p > p!$ and so $b \geqslant p + 1$ which implies that

$$v_2\big((p+1)!\big) \leqslant v_2(b!) = v_2(p^{p-1} - 1) \overset{LTE}{=} 2v_2(p-1) + v_2(p+1) - 1 = v_2\left(\frac{p-1}{2} \cdot (p-1) \cdot (p+1)\right),$$

where in the middle we used lifting-the-exponent lemma. On the RHS we have three factors of $(p+1)!$. But, due to $p + 1 \geqslant 8$, there are at least 4 even numbers among $1, 2, \ldots, p + 1$, so this case is not possible.

Solution 2. The cases $a \neq p$ are covered as in solution 1, as are $p = 2, 3$. For $p \geqslant 5$ we have $b! = p(p^{p-1} - 1)$. By Zsigmondy's Theorem there exists some prime q that divides $p^{p-1} - 1$ but does not divide $p^k - 1$ for $k < p - 1$. It follows that $\mathrm{ord}_q(p) = p - 1$, and hence $q \equiv 1 \mod (p-1)$. Note that $p \neq q$. But then we must have $q \geqslant 2p - 1$, giving

$$b! \geqslant (2p-1)! = [1 \cdot (2p-1)] \cdot [2 \cdot (2p-2)] \cdot \cdots \cdot [(p-1) \cdot (p+1)] \cdot p > (2p-1)^{p-1} p > p^p > p^p - p,$$

a contradiction.

Solution 3. The cases $a \neq p$ are covered as in solution 1, as are $p = 2, 3$. Also $b > p$, as $p^p > p! + p$ for $p > 2$. The cases $p = 5, 7, 11$ are also checked manually, so assume $p \geqslant 13$.
Let $q | p + 1$ be an odd prime. By LTE

$$v_q(p^p - p) = v_q\left((p^2)^{\frac{p-1}{2}} - 1\right) = v_q(p^2 - 1) + v_q\left(\frac{p-1}{2}\right) = v_q(p+1).$$

But $b \geqslant p + 1$, so then $v_q(b!) > v_q(p + 1)$, since $q < p + 1$, a contradiction. This means that $p + 1$ has no odd prime divisor, i.e. $p + 1 = 2^k$ for some k.

Now let $q|p-1$ be an odd prime. By LTE

$$v_q(p^p-p) = 2v_q(p-1).$$

Let $d = v_q(p-1)$. Then $p \geqslant 1+q^d$, so

$$v_q(b!) \geqslant v_q(p!) \geqslant v_q(q^d!) > q^{d-1} \geqslant 2d$$

provided $d \geqslant 2$ and $q > 3$, or $d \geqslant 3$.
If $q = 3$, $d = 2$ and $p \geqslant 13$ then $v_q(b!) \geqslant v_q(p!) \geqslant v_q(13!) = 5 > 2d$. Either way, $d \leqslant 1$.
If $p > 2q+1$ (so $p > 3q$, as $q|p-1$) then

$$v_q(b!) \geqslant v_q((3q)!) = 3,$$

so we must have $q \geqslant \frac{p}{2}$, in other words, $p-1 = 2q$. This implies that $p = 2^k-1$ and $q = 2^{k-1}-1$ are both prime, but it is not possible to have two consecutive Mersenne primes.

Solution 4. Let $a = p$, $b > p$ and $p \geqslant 5$ (the remaining cases are dealt with as in solution 3). Modulo $(p+1)^2$ it holds that

$$p^p-p = (p+1-1)^p-p \equiv \binom{p}{1}(p+1)(-1)^{p-1}+(-1)^p-p = p(p+1)-1-p = p^2-1 \not\equiv 0 \mod \left((p+1)^2\right).$$

Since $p \geqslant 5$, the numbers 2 and $\frac{p+1}{2}$ are distinct and less than or equal to p. Therefore, $p+1|p!$, and so $(p+1)^2|(p+1)!$.
But $b \geqslant p+1$, so $b! \equiv 0 \not\equiv p^p-p \mod (p+1)^2$, a contradiction.

N5. For each $1 \leqslant i \leqslant 9$ and $T \in \mathbb{N}$, define $d_i(T)$ to be the total number of times the digit i appears when all the multiples of 1829 between 1 and T inclusive are written out in base 10.

Show that there are infinitely many $T \in \mathbb{N}$ such that there are precisely two distinct values among $d_1(T), d_2(T), \ldots, d_9(T)$.

(United Kingdom)

Solution. Let $n := 1829$. First, we choose some k such that $n \mid 10^k - 1$. For instance, any multiple of $\varphi(n)$ would work since n is coprime to 10. We will show that either $T = 10^k - 1$ or $T = 10^k - 2$ has the desired property, which completes the proof since k can be taken to be arbitrary large.

For this it suffices to show that $\#\{d_i\left(10^k - 1\right) : 1 \leqslant i \leqslant 9\} \leqslant 2$. Indeed, if

$$\#\{d_i\left(10^k - 1\right) : 1 \leqslant i \leqslant 9\} = 1$$

then, since $10^k - 1$ which consists of all nines is a multiple of n, we have

$$d_i\left(10^k - 2\right) = d_i\left(10^k - 1\right) \text{ for } i \in \{1, \ldots, 8\}, \text{ and } d_9\left(10^k - 2\right) < d_9\left(10^k - 1\right).$$

This means that $\#\{d_i\left(10^k - 2\right) : 1 \leqslant i \leqslant 9\} = 2$.

To prove that $\#\{d_i\left(10^k - 1\right)\} \leqslant 2$ we need an observation. Let $\overline{a_{k-1}a_{k-2}\ldots a_0} \in \left\{1, \ldots, 10^k - 1\right\}$ be the decimal expansion of an arbitrary number, possibly with leading zeroes. Then $\overline{a_{k-1}a_{k-2}\ldots a_0}$ is divisible by n if and only if $\overline{a_{k-2}\ldots a_0a_{k-1}}$ is divisible by n. Indeed, this follows from the fact that

$$10 \cdot \overline{a_{k-1}a_{k-2}\ldots a_0} - \overline{a_{k-2}\ldots a_0a_{k-1}} = \left(10^k - 1\right) \cdot a_{k-1}$$

is divisible by n.

This observation shows that the set of multiples of n between 1 and $10^k - 1$ is invariant under simultaneous cyclic permutation of digits when numbers are written with leading zeroes. Hence, for each $i \in \{1, \ldots, 9\}$ the number $d_i\left(10^k - 1\right)$ is k times larger than the number of k digit numbers which start from the digit i and are divisible by n. Since the latter number is either $\lfloor 10^{k-1}/n \rfloor$ or $1 + \lfloor 10^{k-1}/n \rfloor$, we conclude that $\#\{d_i\left(10^k - 1\right)\} \leqslant 2$.

Comment. More careful analysis shows that $\#\{d_i\left(10^k - 1\right) : 1 \leqslant i \leqslant 9\} = 1$ if and only if $n \equiv 1 \pmod{10}$, which is not the case for $n = 1829$.

N6. Let Q be a set of prime numbers, not necessarily finite. For a positive integer n consider its prime factorisation; define $p(n)$ to be the sum of all the exponents and $q(n)$ to be the sum of the exponents corresponding only to primes in Q. A positive integer n is called *special* if $p(n) + p(n+1)$ and $q(n) + q(n+1)$ are both even integers. Prove that there is a constant $c > 0$ independent of the set Q such that for any positive integer $N > 100$, the number of special integers in $[1, N]$ is at least cN.

(For example, if $Q = \{3, 7\}$, then $p(42) = 3$, $q(42) = 2$, $p(63) = 3$, $q(63) = 3$, $p(2022) = 3$, $q(2022) = 1$.)

(*Costa Rica*)

Solution. Let us call two positive integers m, n *friends* if $p(m) + p(n)$ and $q(m) + q(n)$ are both even integers. We start by noting that the pairs $(p(k), q(k))$ modulo 2 can take at most 4 different values; thus, among any five different positive integers there are two which are friends.

In addition, both functions p and q satisfy $f(ab) = f(a) + f(b)$ for any a, b. Therefore, if m and n are divisible by d, then both p and q satisfy the equality $f(m) + f(n) = f(m/d) + f(n/d) + 2f(d)$. This implies that m, n are friends if and only if $m/d, n/d$ are friends.

Let us call a set of integers $\{n_1, n_2, \ldots, n_5\}$ an *interesting* set if for any indexes i, j, the difference $d_{ij} = |n_i - n_j|$ divides both n_i and n_j. We claim that if elements of an interesting set are all positive, then we can obtain a special integer. Indeed, if we were able to construct such a set, then there would be a pair of integers $\{n_i, n_j\}$ which are friends, according to the first observation. Additionally, the second observation yields that the quotients $n_i/d_{ij}, n_j/d_{ij}$ form a pair of friends, which happen to be consecutive integers, thus giving a special integer as desired.

In order to construct a family of interesting sets, we can start by observing that the set $\{0, 6, 8, 9, 12\}$ is an interesting set. Using that $72 = 2^3 \cdot 3^2$ is the least common multiple of all pairwise differences in this set, we obtain a family of interesting sets by considering

$$\{72k, 72k + 6, 72k + 8, 72k + 9, 72k + 12\},$$

for any $k \geqslant 1$. If we consider the quotients (of these numbers by the appropriate differences), then we obtain that the set

$$S_k = \{6k, 8k, 9k, 12k, 12k + 1, 18k + 2, 24k + 2, 24k + 3, 36k + 3, 72k + 8\},$$

has at least one special integer. In particular, the interval $[1, 100k]$ contains the sets $S_1, S_2, ..., S_k$, each of which has a special number. Any special number can be contained in at most ten sets S_k, from where we conclude that the number of special integers in $[1, 100k]$ is at least $k/10$.

Finally, let $N = 100k + r$, with $k \geqslant 1$ and $0 \leqslant r < 100$, so that we have $N < 100(k + 1) \leqslant 200k$. Then the number of special integers in $[1, N]$ is at least $k/10 > N/2000$, as we wanted to prove.

Comment 1. The statement is also true for $N \geqslant 15$ as at least one of the numbers $7, 14, 15$ is special.

Comment 2. Another approach would be to note that if $p(2n), p(2n + 1), p(2n + 2)$ all have the same parity then one of the numbers $n, 2n, 2n + 1$ is special. Indeed, if $q(n) + q(n + 1)$ is even then n is special since $p(n) + p(n + 1) \equiv p(2n) + p(2n + 2) \equiv 0 \pmod 2$. Otherwise, if $q(n) + q(n + 1)$ is odd, so is $q(2n) + q(2n + 2)$ which implies that exactly one of the numbers $2n, 2n + 1$ is special.

Unfortunately, it seems hard to show that the set of such n has positive density: see a recent paper `https://arxiv.org/abs/1509.01545` for the proof that all eight patterns of the parities of $p(n), p(n + 1), p(n + 2)$ appear for a positive proportion of positive integers.

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N7. Let k be a positive integer and let S be a finite set of odd prime numbers. Prove that there is at most one way (modulo rotation and reflection) to place the elements of S around a circle such that the product of any two neighbors is of the form $x^2 + x + k$ for some positive integer x.

(U.S.A.)

Solution. Let us allow the value $x = 0$ as well; we prove the same statement under this more general constraint. Obviously that implies the statement with the original conditions.

Call a pair $\{p, q\}$ of primes with $p \neq q$ *special* if $pq = x^2 + x + k$ for some nonnegative integer x. The following claim is the key mechanism of the problem:

Claim.

(a) For every prime r, there are at most two primes less than r forming a special pair with r.

(b) If such p and q exist, then $\{p, q\}$ is itself special.

We present two proofs of the claim.

Proof 1. We are interested in integers $1 \leqslant x < r$ satsfying

$$x^2 + x + k \equiv 0 \pmod{r}. \tag{1}$$

Since there are at most two residues modulo r that can satisfy that quadratic congruence, there are at most two possible values of x. That proves (a).

Now suppose there are primes p, q with $p < q < r$ and nonnegative integers x, y such that

$$x^2 + x + k = pr$$
$$y^2 + y + k = qr.$$

From $p < q < r$ we can see that $0 \leqslant x < y \leqslant r - 1$. The numbers x, y are the two solutions of (1); by Vieta's formulas, we should have $x + y \equiv -1 \pmod{r}$, so $x + y = r - 1$.

Letting $K = 4k - 1$, $X = 2x + 1$, and $Y = 2y + 1$, we obtain

$$4pr = X^2 + K,$$
$$4qr = Y^2 + K$$

with $X + Y = 2r$. Multiplying the two above equations,

$$\begin{aligned} 16pqr^2 &= (X^2 + K)(Y^2 + K) \\ &= (XY - K)^2 + K(X + Y)^2 \\ &= (XY - K)^2 + 4Kr^2, \\ 4pq &= \left(\frac{XY - K}{2r}\right)^2 + K. \end{aligned}$$

In particular, the number $Z = \frac{XY-K}{2r}$ should be an integer, and so $4pq = Z^2 + K$. By parity, Z is odd, and thus

$$pq = z^2 + z + k \quad \text{where } z = \frac{Z - 1}{2},$$

so $\{p, q\}$ is special. □

Proof 2. As before, we suppose that

$$x^2 + x + k = pr$$
$$y^2 + y + k = qr.$$

Subtracting, we have

$$(x + y + 1)(x - y) = r(p - q).$$

As before, we have $x + y = r - 1$, so $x - y = p - q$, and

$$x = \tfrac{1}{2}(r + p - q - 1)$$
$$y = \tfrac{1}{2}(r + q - p - 1).$$

Then,

$$\begin{aligned} k = pr - x^2 - x &= \tfrac{1}{4}(4pr - (r + p - q - 1)^2 - 2(r + p - q - 1)) \\ &= \tfrac{1}{4}(4pr - (r + p - q)^2 + 1) \\ &= \tfrac{1}{4}(2pq + 2pr + 2qr - p^2 - q^2 - r^2 + 1), \end{aligned}$$

which is symmetric in p, q, r, so

$$pq = z^2 + z + k \quad \text{where } z = \tfrac{1}{2}(p + q - r - 1),$$

and $\{p, q\}$ is special. □

Now we settle the problem by induction on $|S|$, with $|S| \leqslant 3$ clear.

Suppose we have proven it for $|S| = n$ and consider $|S| = n + 1$. Let r be the largest prime in S; the claim tells us that in any valid cycle of primes:

- the neighbors of r are uniquely determined, and
- removing r from the cycle results in a smaller valid cycle.

It follows that there is at most one valid cycle, completing the inductive step.

Comment. The statement is not as inapplicable as it might seem. For example, for $k = 41$, the following 385 primes form a valid cycle of primes:

53, 4357, 104173, 65921, 36383, 99527, 193789, 2089123, 1010357, 2465263, 319169, 15559, 3449, 2647, 1951, 152297, 542189, 119773, 91151, 66431, 222137, 1336799, 469069, 45613, 1047941, 656291, 355867, 146669, 874879, 2213327, 305119, 3336209, 1623467, 520963, 794201, 1124833, 28697, 15683, 42557, 6571, 39607, 1238833, 835421, 2653681, 5494387, 9357539, 511223, 1515317, 8868173, 114079681, 59334071, 22324807, 3051889, 5120939, 7722467, 266239, 693809, 3931783, 1322317, 100469, 13913, 74419, 23977, 1361, 62983, 935021, 512657, 1394849, 216259, 45827, 31393, 100787, 1193989, 600979, 209543, 357661, 545141, 19681, 10691, 28867, 165089, 2118023, 6271891, 12626693, 21182429, 1100467, 413089, 772867, 1244423, 1827757, 55889, 1558873, 5110711, 1024427, 601759, 290869, 91757, 951109, 452033, 136471, 190031, 4423, 9239, 15809, 24133, 115811, 275911, 34211, 877, 6653, 88001, 46261, 317741, 121523, 232439, 379009, 17827, 2699, 15937, 497729, 335539, 205223, 106781, 1394413, 4140947, 8346383, 43984757, 14010721, 21133961, 729451, 4997297, 1908223, 278051, 529747, 40213, 768107, 456821, 1325351, 225961, 1501921, 562763, 75527, 5519, 9337, 14153, 499, 1399, 2753, 14401, 94583, 245107, 35171, 397093, 195907, 2505623, 34680911, 18542791, 7415917, 144797293, 455529251, 86675291, 252704911, 43385123, 109207907, 204884269, 330414209, 14926789, 1300289, 486769, 2723989, 907757, 1458871, 65063, 4561, 124427, 81343, 252887, 2980139, 1496779, 3779057, 519193, 47381, 135283, 268267, 446333, 669481, 22541, 54167, 99439, 158357, 6823, 32497, 1390709, 998029, 670343, 5180017, 13936673, 2123491, 4391941, 407651, 209953, 77249, 867653, 427117, 141079, 9539, 227, 1439, 18679, 9749, 25453, 3697, 42139, 122327, 712303, 244261, 20873, 52051, 589997, 4310569, 1711069, 291563, 3731527, 11045429, 129098443, 64620427, 162661963, 22233269, 37295047, 1936969, 5033449, 725537, 1353973, 6964457, 2176871, 97231, 7001, 11351, 55673, 16747, 169003, 1218571, 479957, 2779783, 949609, 4975787, 1577959, 2365007, 3310753, 79349, 23189, 107209, 688907, 252583, 30677, 523, 941, 25981, 205103, 85087, 1011233, 509659, 178259, 950479, 6262847, 2333693, 305497, 3199319, 9148267, 1527563, 466801, 17033, 9967, 323003, 4724099, 14278309, 2576557, 1075021, 6462593, 2266021, 63922471, 209814503, 42117791, 131659867, 270892249, 24845153, 12104557, 3896003, 219491, 135913, 406397, 72269, 191689, 2197697, 1091273, 2727311, 368227, 1911661, 601883, 892657, 28559, 4783, 60497, 31259, 80909, 457697, 153733, 11587, 1481, 26161, 15193, 7187, 2143, 21517, 10079, 207643, 1604381, 657661, 126227, 372313, 2176331, 748337, 64969, 844867, 2507291, 29317943, 14677801, 36952793, 69332267, 111816223, 5052241, 8479717, 441263, 3020431, 1152751, 13179611, 38280013, 6536771, 16319657, 91442699, 30501409, 49082027, 72061511, 2199433, 167597, 317963, 23869, 2927, 3833, 17327, 110879, 285517, 40543, 4861, 21683, 50527, 565319, 277829, 687917, 3846023, 25542677, 174261149, 66370753, 9565711, 1280791, 91393, 6011, 7283, 31859, 8677, 10193, 43987, 11831, 13591, 127843, 358229, 58067, 15473, 65839, 17477, 74099, 19603, 82847, 21851, 61.

N8. Prove that $5^n - 3^n$ is not divisible by $2^n + 65$ for any positive integer n.

(Belgium)

Solution 1. Let n be a positive integer, and let $m = 2^n + 65$. For the sake of contradiction, suppose that $m \mid 5^n - 3^n$, so $5^n \equiv 3^n \pmod m$.

Notice that if n is even, then $3 \mid m$, but $3 \nmid 5^n - 3^n$, contradiction. So, from now on we assume that n is odd, $n = 2k + 1$. Obviously $n = 1$ is not possible, so $n \geqslant 3$. Notice that m is coprime to 2, 3 and 5.

> *Let m_1 be the smallest positive multiple of m that can be written in the form of either $|5a^2 - 3b^2|$ or $|a^2 - 15b^2|$ with some integers a and b.*

Note that $5^n - 3^n = 5\left(5^k\right)^2 - 3\left(3^k\right)^2$ is a multiple of m, so the set of such multiples is non-empty, and therefore m_1 is well-defined.

I. First we show that $m_1 \leqslant 5m$. Consider the numbers

$$5^{k+1}x + 3^{k+1}y, \quad 0 \leqslant x, y \leqslant \sqrt{m}.$$

There are $\lfloor \sqrt{m} \rfloor + 1 > \sqrt{m}$ choices for x and y, so there are more than m possible pairs (x, y). Hence, two of these sums are congruent modulo m: $5^{k+1}x_1 + 3^{k+1}y_1 \equiv 5^{k+1}x_2 + 3^{k+1}y_2 \pmod m$.

Now choose $a = x_1 - x_2$ and $b = y_1 - y_2$; at least one of a, b is nonzero, and

$$5^{k+1}a + 3^{k+1}b \equiv 0 \pmod m, \quad |a|, |b| \leqslant \sqrt{m}.$$

From

$$0 \equiv (5^{k+1}a)^2 - (3^{k+1}b)^2 = 5^{n+1}a^2 - 3^{n+1}b^2 \equiv 5 \cdot 3^n a^2 - 3^{n+1}b^2 = 3^n(5a^2 - 3b^2) \pmod m$$

we can see that $|5a^2 - 3b^2|$ is a multiple of m. Since at least one of a and b is nonzero, $5a^2 \neq 3b^2$. Hence, by the choice of a, b, we have $0 < |5a^2 - 3b^2| \leqslant \max(5a^2, 3b^2) \leqslant 5m$. That shows that $m_1 \leqslant 5m$.

II. Next, we show that m_1 cannot be divisible by 2, 3 and 5. Since m_1 equals either $|5a^2 - 3b^2|$ or $|a^2 - 15b^2|$ with some integers a, b, we have six cases to check. In all six cases, we will get a contradiction by presenting another multiple of m, smaller than m_1.

- If $5 \mid m_1$ and $m_1 = |5a^2 - 3b^2|$, then $5 \mid b$ and $\left|a^2 - 15\left(\frac{b}{5}\right)^2\right| = \frac{m_1}{5} < m_1$.
- If $5 \mid m_1$ and $m_1 = |a^2 - 15b^2|$, then $5 \mid a$ and $\left|5\left(\frac{a}{5}\right)^2 - 3b^2\right| = \frac{m_1}{5} < m_1$.
- If $3 \mid m_1$ and $m_1 = |5a^2 - 3b^2|$, then $3 \mid a$ and $\left|b^2 - 15\left(\frac{a}{3}\right)^2\right| = \frac{m_1}{3} < m_1$.
- If $3 \mid m_1$ and $m_1 = |a^2 - 15b^2|$, then $3 \mid a$ and $\left|5b^2 - 3\left(\frac{a}{3}\right)^2\right| = \frac{m_1}{3} < m_1$.
- If $2 \mid m_1$ and $m_1 = |5a^2 - 3b^2|$, then $\left|\left(\frac{5a-3b}{2}\right)^2 - 15\left(\frac{a-b}{2}\right)^2\right| = \frac{m_1}{2} < m_1$.
- If $2 \mid m_1$ and $m_1 = |a^2 - 15b^2|$, then $\left|5\left(\frac{a-3b}{2}\right)^2 - 3\left(\frac{a-5b}{2}\right)^2\right| = \frac{m_1}{2} < m_1$.

(The last two expressions can be obtained from $(\sqrt{5}a + \sqrt{3}b)(\sqrt{5} - \sqrt{3}) = (5a - 3b) + \sqrt{15}(b - a)$ and $(a + \sqrt{15}b)(\sqrt{5} - \sqrt{3}) = \sqrt{5}(a - 3b) + \sqrt{3}(5b - a)$.)

In all six cases, we found that either $\frac{m_1}{2}$, $\frac{m_1}{3}$ or $\frac{m_1}{5}$ is of the form $|5x^2 - 3y^2|$ or $|x^2 - 15y^2|$. Since m is coprime to 2, 3 and 5, the presented number is a multiple of m, but this contradicts the minimality of m_1.

III. The last remaining case is $m_1 = m$, so either $m = |5a^2 - 3b^2|$ or $m = |a^2 - 15b^2|$. We will get a contradiction by considering the two sides modulo 3, 4 and 5.

- $2^n + 65 = 5a^2 - 3b^2$ is not possible, because $2^n + 65 \equiv 1 \pmod 3$, but $5a^2 - 3b^2 \not\equiv 1 \pmod 3$.
- $2^n + 65 = 3b^2 - 5a^2$ is not possible, because $2^n + 65 \equiv 1 \pmod 4$, but $3b^2 - 5a^2 \not\equiv 1 \pmod 4$.
- $2^n + 65 = a^2 - 15b^2$ is not possible, because $2^n + 65 \equiv \pm 2 \pmod 5$, but $a^2 - 15b^2 \not\equiv \pm 2 \pmod 5$.
- $2^n + 65 = 15b^2 - a^2$ is not possible, because $2^n + 65 \equiv 1 \pmod 4$, but $15b^2 - a^2 \not\equiv 1 \pmod 4$.

We found a contradiction in all cases, that completes the solution.

Comment 1. Part I is a standard application of Thue's lemma:

> *Lemma (Thue).* Suppose that $m > 1$ and c are integers, and X, Y are positive integers such that $X \leqslant m < XY$. Then there exist some integers x, y with $|x| < X$ and $0 < y < Y$, such that $x \equiv cy \pmod m$.

It is a well-known corollary that if c, d are coprime to m, the congruence $cx^2 \equiv dy^2 \pmod m$ has a solution such that x, y are coprime to m, and X, Y are positive integers such that $XY > m$, then $cx^2 \equiv dy^2 \pmod m$ has a solution such that at least one of x, y is nonzero, $|x| < X$ and $|y| < Y$.

In the solution we applied this corollary with $c = 5$, $d = 3$, $X = Y = \lfloor \sqrt{m} \rfloor + 1$.

Comment 2. In fact, we proved that a positive integer m with $m \equiv 13$ or $37 \pmod{60}$ cannot divide any nonzero integer of the form $5a^2 - 3b^2$ or $a^2 - 15b^2$ with coprime integers a, b. In other words, if $m \equiv 13$ or $37 \pmod{60}$, then 15 is not a quadratic residue modulo m. Using the tools of quadratic reciprocity, the solution can be significantly shortened.

Suppose that a and b are coprime. For every prime divisor $p > 5$ of $5a^2 - 3b^2$ or $a^2 - 15b^2$, we have

$$1 = \left(\frac{15}{p}\right) = \left(\frac{3}{p}\right)\left(\frac{5}{p}\right) = (-1)^{\frac{p-1}{2}}\left(\frac{p}{3}\right)\left(\frac{p}{5}\right), \tag{1}$$

where $\left(\frac{a}{p}\right)$ stands for the Legendre symbol. Considering the remainders of p when divided by 4, 3 and 5, (1) leads to

$$p \equiv \pm 1, \pm 7, \pm 11 \text{ or } \pm 17 \mod 60.$$

These remainders form a subgroup of the reduced remainders modulo 60. Since 13 and 37 are not elements in this subgroup, the number $m = 2^n + 65$ cannot be a product of such primes.

Instead of handling the prime divisors of m separately, we can use Jacobi symbols for further simplification, as shown in the next solution.

Solution 2. Suppose again that $5^n \equiv 3^n \pmod{m = 2^n + 65}$. Like in the first solution, we conclude that n must be odd, and $n \geqslant 3$, so $8 \mid 2^n$.

Using Jacobi symbols,

$$-1 = \left(\frac{2^n + 65}{5}\right) = \left(\frac{5}{2^n + 65}\right) = \left(\frac{5^n}{2^n + 65}\right) = \left(\frac{3^n}{2^n + 65}\right) = \left(\frac{3}{2^n + 65}\right) = \left(\frac{2^n + 65}{3}\right) = 1,$$

contradiction.

